

Variation, acquisition and change

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Open Germanic Linguistics ??

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Chapter 1

Introduction

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This volume explores issues relating to language variation, acquisition, and change. It consists of 27 chapters that illuminate these issues from a range of different methodological, empirical, and theoretical perspectives. Many chapters also combine two or even three of the themes, illustrating that they are linked and that such combinations increase our understanding of a particular phenomenon.

Keywords: acquisition, change, variation, Norwegian

Humans are seemingly unique in the animal kingdom in that we have language; systems enabling communication about any topic, be it past, present or future. In fact, humans are not limited to one language but can acquire several under the right conditions. Although undoubtedly a cultural object, language is also a fundamental part of the mind/brain of a human being, the constitution of which we refer to as a mental grammar. Changes in states of linguistic knowledge, whether in development (*acquisition*), over the lifespan or across generations (*change*) examined across diverse populations and language combinations (*variation*), provide crucial insights about the organization of mental grammars.

This volume, entitled *Variation, Acquisition and Change*, brings together research on language acquisition in different populations, language change in various languages, and linguistic variation from an empirical and a theoretical perspective. 27 different chapters illuminate these issues from a range of different methodological, empirical, and theoretical perspectives. Importantly, many chapters also combine two or three of the themes, illustrating that they are linked and that such combinations increase our understanding of a given phenomenon. Still, in what follows, we have decided to present the chapters under the three headings.

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1 Variation

At the heart of the volume is linguistic variation. In many ways, this is an old problem in the field of linguistics: What is the scope of linguistic variation across the world's languages and varieties? To better understand this, careful investigations of as many related and unrelated languages as possible is desirable. In this volume, Corbett and Fedden explore verb agreement in Mian, a Papuan language of Papua New Guinea. In Mian, all finite verbs obligatorily mark subject agreement; however, only a minority of transitive verbs show object agreement. This may appear to pose a learnability problem, but Corbett and Fedden demonstrate that learners have several robust cues in the language. They also situate Mian in a typological context, which brings out parallels with other restricted and non-canonical agreement systems elsewhere.

Typological variation can also be studied within one language family, which we see in the chapter by Vangsnes. His starting point is the item *som* in Norwegian, which has a range of uses. Sometimes it corresponds to English *as*, sometimes to *that*, and some uses do not have a translational equivalent in English. Vangsnes discusses the Norwegian variation against other Germanic varieties and proposes a way to formally model the patterns of grammaticalization that these varieties display. He employs the framework of nanosyntax, where the variation can be understood as a gradual growth regarding the specification of what sub-trees *som* may lexicalize.

When studying linguistic variation, the issue of theoretical models often comes up. Several chapters in this volume are concerned with better understanding how to model linguistic data from different varieties. In his chapter, Svenonius returns to an old problem; namely, how to understand variation in verb placement in English. He proposes a novel approach which contains a functional lexicon organized into a finite state automaton. On such an approach, properties of the functional lexicon are distributed over the states and transitions. The finite state automaton defines an extended projection.

Another chapter concerned with theoretical architecture is the one by Áfarli. He is concerned with how to best model argument structure flexibility and lexical creativity, based on data from English and especially Norwegian. Áfarli argues that an exoskeletal approach to grammar is best suited to capture these data, as opposed to an endoskeletal/lexicalist approach.

Fábregas and González Rodríguez build on previous work where they argue for the existence of negative events as objects of natural languages. In the present chapter, they focus on a prediction that this analysis makes, namely that there

should be lexical verbs that already encode negative events. Based on Spanish data, they argue that this prediction is borne out.

In the chapter by Nesset and Janda, the authors discuss agreement properties with certain collective nouns in Norwegian. Based on corpus data, they show that these collective nouns can combine with predicative adjectives in both the singular and the plural. Their distributions are related to the ‘agency hypothesis’ (Nesset et al. 2022), whereby plural agreement is favored when predicate adjectives imply carrying out an action, and the ‘individuation hypothesis’ (Nesset & Janda 2023), which holds that plural agreement is favored when a collective noun is conceptualized as a group of individuals. Nesset and Janda also argue that their data support Westergaard’s (2014) Micro-cue Model.

2 Acquisition

In the past few decades, research on the grammar of multilingual individuals has provided new avenues for exploring the range of linguistic variation. From a multilingual perspective, a core question is how the different varieties that an individual masters interact in their mind/brain. This question is often addressed from the point of view of language acquisition, and specifically how different varieties affect each other in the course of acquisition.

Anderssen & Bentzen extend the Micro-cue Model (Westergaard 2009) to simultaneous bilingual first language acquisition. They do this by considering data on verb placement in English from four bilingual Norwegian/English girls. An important finding is that they display four different patterns of crosslinguistic influence, a type of variation that fits perfectly with the Micro-cue Model.

In her paper, Bohnacker investigates clausal subordination in oral narratives elicited from monolingual Swedish and bilingual Turkish/Swedish children age 4–7. A major finding is that there is a clear increase in the frequency and variety of subordinate clauses with age for both groups. Subordinators are also very rarely omitted. Monolinguals produce higher proportions of subordinate clauses than their bilingual age-peers, and this difference is especially evident for relative clauses. Bohnacker also discusses implications for language assessment in clinical practice.

Verb placement in L3 French and German is the topic of the chapter written by Dahl, Listhaug and Busterud. They consider the written production of L1 Norwegian L2 English speakers in upper secondary schools, which fills an understudied gap in the literature. Their overall findings suggest that the relevant structures are part of the learners’ linguistic repertoire, and furthermore that these findings

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are similar to existing findings based on other methodologies. In terms of theory, their data support an analysis where general mechanisms, such as principles of economy, play a role in the acquisition of verb movement in L3 varieties.

Falkum and Helganger explore a quite different topic in their investigation of how production data can be used to study pragmatic development. They survey a set of studies using production data, both spontaneous and elicited, and argue that these provide a valuable source of evidence, in particular for studies of early pragmatic development and phenomena where controlled comprehension experiments may not be feasible for a given age group.

Gagarina and Alexiadou study the acquisition of case in Russian, which poses a challenge due to its variability in paths and timing. They investigate data from simultaneous bilinguals and compare these data to monolinguals. They demonstrate comparable timing and path of case acquisition across both groups, illustrating that bilingual children's overall developmental trajectory remains comparable to that of their monolingual peers.

Crosslinguistic influence in L3 Norwegian by L1 Ukrainian-L2 English speakers is the topic of the chapter by Kolb, Natvig & van Osch. The authors conduct a gap-filling task in order to scrutinize adverb placement based on similarities and differences between the languages. They probe whether crosslinguistic influence emerges from the L1, the L2, or both. In brief, they find that L3 learners acquire surface-level Norwegian word order in main clauses and that they overgeneralize the same pattern to subordinate clauses.

In their chapter, Krämer, Kupisch and Castro study Brazilian and European Portuguese bilectals. Their main finding is that European Portuguese-speaking Brazilians who acquired European Portuguese late in life (late bilectals) were rated as more Brazilian-sounding than speakers who acquired European Portuguese early in life (early bilectals). Both groups also show traces of the other variety in each speaking mode, which sets them apart from monolinguals. Theoretically, they suggest that the data are best accounted for in a model of indexed violable constraints, and they also employ the Micro-cue Model (Westergaard 2009, 2014) in their analysis.

Lundquist and Ringstad focus on Norwegian and a pattern whereby children and teenagers differ from adults. The former occasionally place verb particles in front of subjects in non-subject initial clauses, in contrast to the latter group. Lundquist and Ringstad consider both spontaneous production and elicited examples, and they observe a development in which verb particle placement becomes increasingly target-like. They also discuss how to best understand this change across the lifespan.

1 Introduction

Dramatic societal changes can lead to new contexts for acquisition. Mitrofanova, Długosz and Hrebenshchikova investigate how Ukrainian refugee children acquire grammatical gender in their new language, Polish. In particular, they are studying morphophonological gender cues based on elicited production. They find that bilingual children tend to use feminine agreement with palatalized consonants (an ambiguous gender cue) more often than their monolingual peers. This tendency increased with age, suggesting stronger cross-linguistic influence from the L1 in older children.

In addition to studying language acquisition from the point of view of production, it is also important to investigate processing. The chapter by Mitrofanova, Minor and Sekerina study the processing of locative prepositional phrases in young speakers of Russian. They do so through eye-tracking and the Visual World Paradigm where they compare the comprehension of locative prepositions in 2- and 3-year-olds. Overall, they find that there is a clear difference between the two age groups, a finding that mirrors a previously established contrast when it comes to the production of prepositions.

Rodina investigates word order patterns in semi-spontaneous spoken narratives produced by Russian-speaking children living in Norway. She argues that her data support the existence of two distinct grammars in these children: One grammar consistent with Russian word order, and another one consistent with Norwegian word order. Put differently, these speakers appear to keep these two grammars apart.

The chapter by Sauermann and Gagarina explores how topic-first contexts, pronominal topics, and language proficiency influence offline OVS sentence comprehension in Russian-German bilinguals. They find that the topic-first context did not influence OVS accuracy in either language, but when the object was realized as a pronoun in OVS sentences, accuracy improved in Russian but not in German. Theoretically, the authors explore the Micro-cue Model (Westergaard 2009, 2014) where children start with specific structures that are later generalized.

Schlenter and Castle focus on syntactic and morphosyntactic processing in bilinguals when they are compared to each other (bilingual-bilingual comparison) or when the two languages of bilinguals are compared (between-language comparison). They study these comparison groups in order to identify whether or not there is crosslinguistic influence. The core argument is that there is ample evidence for crosslinguistic influence from an L1 to an L2, but it is less clear whether it may go in the opposite direction.

Sekerina investigates the effectiveness of cues in gender processing by Russian-speaking children. She considers eye movements of 27 Russian-speaking

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5-to-6-year-olds, and she finds that children use gender agreement cues only in a facilitatory manner. This result can best be explained by the complexity of the design combined with the less efficient gender agreement cues in Russian.

Urek's paper reports on the findings from a large experimental study which investigates the role of probabilistic rules in gender assignment to novel nouns with ambiguous cues by adult monolingual speakers of Russian. Her main results show that speakers can reliably access probabilistic rules to classify novel nouns into gender classes based on their phonological properties, and the information relevant for gender assignment is generalized at the level of segment classes captured by binary phonological features.

In their chapter, Velnić, Slabakova and Dahl investigate the effect of previously acquired languages on the acquisition of article use in generic and specific NPs in Norwegian as a third language. In particular, the authors investigate how multilinguals acquire the form-to-meaning mapping between the universal meaning of genericity and its expressions by plural noun phrases. Based on a truth-value judgment task across three learner groups, they find that all learner groups had acquired the Norwegian form-to-meaning mappings, although there were differences between the groups as well.

The chapter by Wrembel gives a comprehensive review of state-of-the-art approaches to research on L3 phonological acquisition. She considers both methodological and theoretical considerations, including various practical questions that emerge when studying the production, perception and processing of speech in L3 speakers and beyond.

3 Change

Variation can also be understood from a diachronic point of view. Eide investigates why Modern English modals differ from other Germanic modals and modals in earlier stages of English. She links the transition of English modals to the larger grammatical restructuring of Middle English, where several morphosyntactic developments created what she labels a 'linguistic perfect storm'. Concretely, she argues that the following three developments are all related to morphosyntactic finiteness: i) verb movement becoming restricted to modals and auxiliaries, ii) reduction of verbal morphology, and iii) the rise of *do*-support.

Eik, van Baal, Solbakken and Lohndal consider an ongoing change in the grammatical gender system of Norwegian. In their study, elicitation data play a vital role. In their chapter, they discuss methodological challenges that emerge when trying to capture a changing grammatical gender system using elicitation tasks.

They find that speakers produce more feminine markers as the experiment session progresses, and they evaluate some factors that may explain this result.

In her chapter, van Kemenade brings together recent work on the loss of Verb Second in English with insights from the literature on the rise of *do*-support and on the auxiliation of English modal auxiliaries. She also argues that this complex of changes relates to an increasing scale of dialect and language contact, which she considers to be a type of multilingualism.

Putnam, Adeyeye, Wensel and Wilson raise the question of how micro-cues (Westergaard 2009, 2014) are best modeled. They argue that micro-cues are inherently spans; that is, sub-segments of larger hierarchical structures, consisting of contiguous heads and features. Empirically, they draw on L1 acquisition of Norwegian and changes in Verb Second in historical English (i.e., Old to Middle English).

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References

- Nessel, Tore, Hans-Olav Enger & Laura A. Janda. 2022. Er politiet sikker eller sikre? Adjektivkongruens ved kollektiver i norsk. *Arkiv för nordisk filologi* 137. 123–144.

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- Nesset, Tore & Laura A. Janda. 2023. Norwegian agreement clashes on the football field. In T.I. Davidjuk, I.I. Isaev, Ju.B. Mazurova, S.G. Tatevosov & O.V. Fedorova (eds.), *Jazyk kak on est': Sbornik statej k 60-letiju Andreja Aleksandroviča Kibrika*, 293–298. Moscow: Buki Vedi.
- Westergaard, Marit. 2009. *The acquisition of word order: Micro-cues, information structure, and economy* (Linguistik Aktuell/Linguistics Today 145). Amsterdam: John Benjamins.
- Westergaard, Marit. 2014. Linguistic variation and micro-cues in first language acquisition. *Linguistic Variation* 14(1). 26–45. DOI: [10.1075/lv.14.1.02wes](https://doi.org/10.1075/lv.14.1.02wes).

Chapter 2

An exoskeletal grammar approach to linguistic flexibility

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This paper shows how a simple and robust version of an exoskeletal grammar model (dubbed the Exoskeletal Frame Model) is able to account for two ubiquitous types of linguistic flexibility, namely argument structure flexibility and lexical creativity. In both areas, it is concluded that an exoskeletal model should be preferred to a traditional lexicalist or endoskeletal model. The tenor of the paper is somewhat programmatic seeking to make clear the principled differences between endoskeletism on the one hand and exoskeletism on the other, and simultaneously trying to point out the essential motivation for adopting the exoskeletal approach.

1 Introduction

During the early decades of generative grammar, many aberrant and unwieldy types of linguistic data were brushed under the carpet and ignored. This was reasonable to delimit the area of investigation and try to attain descriptive and explanatory adequacy in a situation where the researchers were faced with what appeared to be overwhelming expanses of impenetrable empirical jungles. Thus, Chomsky (1965: 3) adopted the following famous idealization: «Linguistic theory is concerned primarily with an ideal speaker-listener, in a completely homogeneous speech community [...]» This idealization made it possible to concentrate on just the well-behaved and regular data in order to construct corresponding streamlined and efficient grammars. However, it was understood all the time that many kinds of more «mis-behaved» data were put aside for the time being,

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and Chomsky (1981) made that explicit when the *core-periphery* distinction was introduced, see also Chomsky (1991).

During the decades that generative grammar has been in vogue, roughly since the 1960ies, many types of data that were previously relegated to the periphery have been brought to the frontier of research. At the same time, it has become gradually evident that the traditional regular and streamlined grammar models, those that were developed to handle the idealized core data, are no longer fully tenable and should probably be revised. Indeed, models and analyses at different levels of sophistication have been put forth to include “new” types of data.

In this paper, I explore an exoskeletal or constructivist-type model that I call the Exoskeletal Frame Model (EFM). The exoskeletal approach is most forcefully advocated by Hagit Borer in a series of publications (Borer 2003, 2005a,b, 2013).¹ An early proponent of a related analysis was van Hout (1996). It is also related to work in Construction Grammar (Goldberg 1995) and Distributed Morphology (Embick 2015). The exoskeletal approach exists in various technical incarnations. For instance, Borer (2003, 2005a,b, 2013) proposed templates or frames and an elaborated analysis of the role they play in grammar. Åfarli (2007) proposed an exoskeletal frame theory, see also Åfarli & Subbarao (2019). Lohndal (2014) proposed an implementation of the exoskeletal approach, seeking to relate a syntactic exoskeletal analysis with a neo-Davidsonian semantics. Borer (2005b: 10) points out that the general exoskeletal ideas are independent of “any specific functional structure I will propose,” i.e. independent of any technical implementation of the exoskeletal insights. The EFM explored in the present paper is one such technical implementation. Like other exoskeletal models, the EFM is a general model of grammar that also naturally handles various kinds of linguistic flexibility data. As we will see, these are kinds of data that are often perceived as unusual, incongruous, or even weird and funny.

An exoskeletal model should be seen in contrast to traditional lexicalist or endoskeletal models. Such traditional generative models crucially assume that the instructions for the construction of the syntactically displayed argument structure of the clause are inherent in the conceptual or lexico-semantic properties (theta-grid) of the main verb. One could say that such models assume that the basic syntactic structure or skeleton is projected from the *inside* (*endo-*) of the verb. On the other hand, the main property of the EFM and exoskeletal models generally is that the syntactic structure of a clause is assumed to be generated independently of the words that come to fill it, and crucially that the main verb acquires both its word class category and an argument structure by being inserted

¹Notice that the term “exoskeletal” is primarily associated with Hagit Borer’s work, see Borer (2003), where the terms “exoskeletal” and “endoskeletal” are introduced.

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into the independently generated syntactic structure (called template or frame). Thus, the basic syntactic structure or skeleton along with argument structure and word class category is determined from the *outside* (*exo-*) of the verb. Marantz (2013) contains a nice exploration of the difference between the two approaches. In discussing the EFM below, I will not go into theoretical “casuistry”, but instead concentrate on the more basic traits of the model. Thus, the EFM analyses I put forth are simple and robust. For more elaborate discussions of the theoretical foundations and technical aspects of an EFM-type model as well as its relation to other exoskeletal or constructivist-type models, see, e.g., Grimstad et al. (2018), Riksem et al. (2019), Åfarli & Subbarao (2019), or Åfarli et al. (2022).

I will test the EFM against two types of data, namely what I call argument structure flexibility and lexical creativity. I will try to show how data from these two domains present problems for the traditional endoskeletal approach, and how the same phenomena are handled and predicted to exist on an exoskeletal analysis using the EFM. A third interesting domain of linguistic flexibility where exoskeletal analysis proves useful is language mixing. There is no space for dealing with this third domain here. For works using exoskeletal analysis in dealing with language mixing, see for instance Alexiadou et al. (2015); Grimstad (2018); Riksem (2018); Grimstad et al. (2018); Riksem et al. (2019); Åfarli et al. (2022).

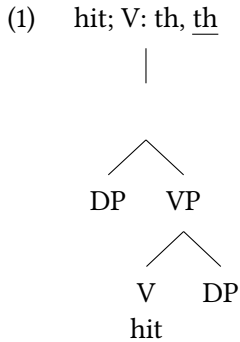
The paper is organized as follows.² Section 2 discusses argument structure flexibility and shows how an exoskeletal analysis fares better in explaining such phenomena than a traditional endoskeletal analysis. Section 3 goes on to discuss lexical creativity and shows again why an exoskeletal analysis fares better than a traditional analysis. Then, Section 4 goes a little deeper into some theoretical aspects and motivation for the EFM, discussing in detail five basic clausal frames, first introduced in Åfarli (2007), that I suggest are relevant for Norwegian. Section 5 concludes the paper.

2 Argument structure flexibility

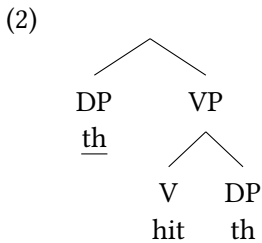
In a traditional endoskeletal approach, the word class category of the main verb and the argument structure properties of the clause are determined by the inherent lexico-semantic properties (theta-grid) of the verb, as indicated in (1).

²Notice that the present paper is a development of Åfarli (2007). Apart from some differences as to theoretical focus and argumentation, in the present paper (a) the analysis is better situated in the history of generative grammar, (b) all examples except one are new, (c) the references are considerably updated, not least regarding research done by MA and PhD students, and (d) the theoretical analysis (in Section 4) concerning constraints on the generation of frame types is new.

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In an exoskeletal model like the EFM, argument structure properties of the clause are generated by the syntactic frame, and the verb is inserted as a modifier of that frame and thus acquires both its argument structure and its word class category from the frame, see (2).



Notice that the representation in (2) is displayed here to emphasize the contrast to the more traditional structure in (1). The EFM structure will be developed in Section 4, retaining the relevant basic traits indicated in (2).³

What motivates an exoskeletal approach? One answer is that the exoskeletal approach is motivated by the existence of what may be seen as unexpected argument structure patterns. Consider first an expected argument structure.

(3) He drank beer.

Drinking is an act that conceptually involves a drinker and something drunk, and in (3) the subject refers to the drinker and the direct object refers to the liquid that is drunk. This argument structure is expected since there is a perfect

³Notice that the verb is not really a verb until it is inserted under V in the frame. I assume that what is inserted is a sound – meaning package, see, e.g., the references given in the penultimate paragraph in Section 1 for discussion of this point, the technical aspects of which will not be further pursued in this paper.

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match between the argument structure of the clause and the conceptual argument structure of the verbal concept involved. Hence, the traditional idea that the argument structure displayed at the clausal level is projected from the lexico-semantic argument structure inherent in the verb.

Now, consider an unexpected argument structure:

- (4) He drank everybody under the table.

Here the conceptual argument structure inherent in the verb appears to be (partly) coerced, and we are faced with a mismatch between the argument structure inherent in the verb and the argument structure displayed by the clause. Specifically, the liquid that is drunk is not expressed in the clause. Instead, the complement of the verb denotes a result of the drinker's drinking activity, namely that everybody else ends up being located under the table (metaphorically speaking). This fact is clearly unexpected on the assumption that the argument structure of the clause is projected from the conceptual argument structure inherent in the verb itself. On the other hand (and as we will see below), it is not unexpected on an exoskeletal analysis since on such an analysis a syntactic resultative argument structure frame is available for a verb like *drink* to be inserted into, independently of the verb's inherent conceptual properties.

Exoskeletal approaches typically assume that there is a limited number of syntactic argument structure frames available for a given language. Åfarli (2007) observed that there are five recurring argument structure patterns in Norwegian and took that observation as motivation for adopting the existence of only five syntactic argument structure frames in that language. Furthermore, it was hypothesized that any verb may be inserted into any frame. As shown in Åfarli (2007) and as seen in the examples below, this prediction is borne out. (5) shows how a Norwegian verb like *sparke* 'kick' can occur with five different argument structures depending on which of the five frames it is inserted into. In Åfarli (2007) the verb *danse* 'dance' was used to make the same point.⁴

- (5) a. Per sparkar. [intransitive frame]
 Per kicks
 b. Per sparkar ballen. [transitive frame]
 Per kicks the ball

⁴Notice that the Norwegian examples shown in this paper are rendered using *nynorsk* spelling, except (6) and (8) below where *bokmål* is used since those examples are taken from *bokmål* sources. The *nynorsk*/*bokmål* differences are irrelevant for the points being made.

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- c. Per sparkar Kari ballen. [ditransitive frame]
Per kicks Kari the ball
- d. Per sparkar ballen flat. [resultative frame]⁵
Per kicks the ball flat
- e. Per sparkar Kari ballen i ryggen. [ditransitive-resultative frame]
Per kicks Kari the ball in the back
'Per kicks the ball into Kari's back.'

As for English, the first four of the frames exemplified in (5) are used. Thus, the English glosses in (5a-d) correspond to sentences that are ok, which is not the case for (5e).⁶ The translation of (5e) indicates that the resultative frame is used in English with what is the goal indirect object in the Norwegian version rendered as a possessive DP that constitutes a part of the small clause predicate PP in the English translation. Notice in that connection that the goal indirect object in the Norwegian version must be interpreted as inalienably possessed by the DP complement of the small clause PP, and that is generally the case regarding the ditransitive-resultative frame. Thus, that frame has a very restricted semantics.⁷

This brief comparison between Norwegian and English suggests that even though the number of frames available to a language seems to be very small, the frames actually employed may still vary somewhat from language to language.

Consider how a traditional endoskeletal analysis as opposed to an exoskeletal analysis would handle an argument structure flexibility case like that illustrated in (5). Given an endoskeletal approach, the verb must be assumed to have a new lexico-semantic argument structure specification (theta-grid) for each syntactic structure that it occurs in. Clearly, that would be a circular non-explanation since the criterion for ascribing a certain lexico-semantic theta-grid to the verb would be the syntactic environment the verb can occur in, which is what the theta-grid properties of the verb are supposed to explain in the first place. On the

⁵The nametags of the frames refer to typical instances and do not necessarily amount to an exhaustive descriptive tag for all instances comprised by a given frame. This remark is particularly relevant regarding the "resultative" frame which includes both resultative and non-resultative small clauses, see Næss (2018) for discussion.

⁶Thanks to Andrew Weir for sharing his intuitions of *kick* with me. As for (5a), he remarks that it is a bit unusual for *kick* to have an absolute/intransitive use, but it is not ungrammatical, for instance if a pregnant woman says *the baby is kicking*. The English versions of (5b, c, d) are deemed to be fine without hesitation.

⁷It must be mentioned that ditransitive-resultative examples are not that frequent in Norwegian as compared to the four other structures, probably not surprising given their restricted semantics. Also, such examples may sound a bit archaic in contemporary Norwegian. See Eik (2014, 2024) for detailed discussion of the ditransitive-resultative frame as employed in Norwegian.

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other hand, given an exoskeletal approach, the verb will acquire the argument structure properties of the syntactic frame that it is inserted into, so in that case the explanation is not circular since the syntactic frames are generated independently according to syntactic principles, and they are therefore independent of the inherent conceptual or lexico-semantic properties of the verb itself. In other words, the actually existing argument structure flexibility illustrated in (5) is explained. Notice also that the exoskeletal analysis is a very constrained analysis since the number of available frames is very limited.⁸

The examples in (5) are reasonably idiomatic with the same verb (*sparke* ‘kick’) used in all five frames. However, that is not always the case. In fact, very often a given verb sounds awkward in one or two or even more of the frames. I assume that this is due to the fact that there are as a main rule different degrees of harmony between the frame and the verb (see also Åfarli 2007: 14). The reason for this is that both the verb and the frame have conceptual or semantic content independently of each other. The frame provides a kind of basic prototypical conceptual-semantic template or schema. This is perhaps best seen in the ditransitive frame, which typically signals a transactional situation with an initiator, a goal/benefactor, and something transferred. On the other hand, a verb also has its inherent conceptual or semantic content independently of any frame it may occur in. For instance, *pisse* ‘piss’ denotes an activity of urinating. Since an urinating or pissing situation typically involves only one participant, namely the one who urinates, this verb is most appropriate in the intransitive frame, which has only one argument, namely the subject, which then denotes the one who is pissing. Thus, in that case there is full harmony between the semantics of the intransitive frame and of the verb, which denotes the activity of a single participant.

Now, consider (6) provided in Brøseth (1997: 30), from the Norwegian author Tor Åge Bringsværd. The relevant part is glossed.

- (6) Slik har det seg at Disney og andre økonomiske ulver stadig glefser etter våre lommebøker

⁸The argument structure flexibility in Norwegian that is facilitated by the five frames exemplified in (5) has been investigated in detail in several master theses submitted in Scandinavian linguistics at NTNU, among them the following (all are written in Norwegian): Frengstad (2016) on verb creativity/flexibility in Norwegian in general, Eik (2014) on the ditransitive-resultative frame in particular, Jordet (2016) on verb frames and Norwegian presentational constructions, Karlsen (2016) on the role of verb frames in Norwegian child language acquisition, and Næss (2018) on verb frames and small clause complements. Beside these master theses, the exoskeletal approach has been exploited in three recent books by NTNU authors: Riksem (2016) on language planning, Nygård (2018) on discourse ellipsis in Norwegian, and Aa (2020) on verb particles in Norwegian.

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og pisser oss dårlig samvittighet opp etter ryggen.
 ... and piss-PRES US bad conscience up after back-DEF
 ‘Thus it is the case that Disney and other economic wolves constantly
 snap at our purses and piss bad conscience up our backs.’

In (6), the verb *pisse* ‘piss’ is used in a non-idiomatic way, but the example is still ok, and it accomplishes an unexpected and striking effect communication-wise. The verb itself still denotes just an activity, albeit understood metaphorically, but here it is inserted into a ditransitive-resultative frame. That is, a frame which conceptually or semantically denotes a prototypical situation which involves an (understood) initiator (*Disney og andre økonomiske ulver*), a goal of a transaction (*oss*), and a result of the transaction (*dårlig samvittighet opp etter ryggen*), a semantics that is clearly not part of the verbal concept of *pisse* ‘piss’ at the outset. What happens here, is that the semantics of the verb is coerced and augmented by the semantics of the frame. In cases where there is non-harmony between the frame and the verb, the frame always wins! The cost is cases of more or less unnatural verbal expressions, but they are not ungrammatical, but only signals different degrees of harmony.⁹

As an example of a quite extreme case of apparent word salad and dis-harmony, consider the following example from Borer (2005a: 9).

(7) The red under fived lunch.

At first reading, (7) seems incomprehensible, and in fact, according to Borer, this example was mentioned in a manuscript by Heidi Harley and Rolf Noyer as an example illustrating the limits of coercion. Still, Borer points out that speakers of English are «perfectly capable of assigning an interpretation to it.» For instance, «some creature [...], for example, a bottom-dweller, which is red, ate lunch five times [...].» Furthermore, (7) «has an understood actor, *the red under*, who acted in some manner pertaining to *five* on some target, *lunch*.» In other words, even examples that at first sight look like gibberish, can be made sense of given an appropriate context or appropriate background assumptions. This suggests that such examples are not ungrammatical, only unusual. They are forced to receive an interpretation that can somehow be made sense of by the meaning of the frame.

⁹Notice that individuals or speech communities may have internalized different normative intuitions regarding what is to count as “acceptable” usage, and that the degree of acceptance of non-harmony may differ for prescriptive reasons.

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Notice that the frame structures are not always as transparently visible as they have been in the examples shown above. Very often the frame structure is obscured by the addition of different kinds of adjuncts and/or auxiliary verbs, as well as by movements of different kinds. Also, the phrasal complexity of the arguments of the frame may contribute to obscuring the frame structure underlying the string. Nevertheless, upon further analysis we always get back to a limited set of basic sentence structures as provided by the frames.

3 Lexical creativity

Conversion (or zero-derivation) is a word-formation phenomenon where the «same» word is used as belonging to different word classes, e.g., *work* used as a noun or a verb. Such cases are of course common and they are amply recorded in dictionaries (see the contributions in [Bauer & Valera 2005](#) for discussions and analyses of conversion generally). However, more radical cases of conversion are also quite common. These are typically uttered spontaneously for instance to spice up language or create an unexpected or striking effect, and they are not found in dictionaries. They therefore bear witness to a resilient lexical creativity. Once you start looking, examples of such lexical creativity abound in languages, even though they have not figured prominently in generative grammar (but see [Clark & Clark 1979](#) and Hagit Borer's work). The analysis sketched below applies equally well to common dictionary-type conversion as to the more radical type that I concentrate on here.

Consider the English example below.



Figure 1: pls provide a caption

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This example is from a protest against a possible military attack on Syria in 2013 by the US Obama administration. The proper name *Iraq* is used as a transitive verb. The protest concerns a possible course of events that can be paraphrased as «Obama Iraqs Syria» or «Obama attacks Syria». The motivation for using *Iraq* as a verb in this context is of course that the US Bush administration started a massive military attack on Iraq in 2003.

How can a proper name like *Iraq* be used as a transitive verb? Given a traditional endoskeletal approach, that is unexpected to say the least: The proper name *Iraq* must be listed in the lexicon as a transitive verb with the appropriate inherent lexico-semantic argument structure specification (theta-grid). In addition, it must of course be listed as a proper name (with no argument structure specification). Not only that, since any proper name or common noun may in principle be used as a verb, any noun must be assumed to have a parallel existence as a verb in the lexicon. The traditional analysis does not explain how such creative use of language is possible; it just states the fact that it is possible. Thus, this approach is clearly circular and non-explanatory.

On the other hand, given an exoskeletal analysis, the use as a verb of what we normally recognize as a noun is not only possible, but also directly predicted. Such a conversion takes place as a result of the word in question being inserted into the verb slot in one of the five verbal frames. Thus, the noun properties that we otherwise associate with the inserted element are nullified by the frame so that the element not only acquires an argument structure, but also an unexpected word class categorization. Given this analysis, the difference between ordinary dictionary-type conversion and the radical type is that only the former is conventionalized.

Like the ordinary cases, the radical cases are ubiquitous once we start looking. Some examples taken from [Frenstad \(2016\)](#) are shown below; the page numbers refer to the page numbers in Frenstad's thesis.

- (8) a. Jeg tindrer ikke nå som jeg er i et forhold. [p. 23]
 I Tinder not now that I am in a relationship
 'I do not use Tinder now that I am in a relationship.'
- b. Han hitler debatten. [p. 24]
 he Hitlers the debate
 'He destroys the debate by acting like Hitler.'
- c. Internett kroppspresser oss. [p. 26]
 internet bodypressures us
 'Internet exerts body image pressure on us.'

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- d. [...] slik at han kan bosmanne seg til Europa. [p. 29]
 so that he can Bosman himself to Europe
 ‘so that he can move to Europe in a manner like [the footballplayer]
 Bosman.’
- e. [...] hun photoshopper seg både tynnere og strammere [...]. [p. 29]
 she photoshops herself both thinner and tighter
 ‘she uses photoshop to manipulate pictures so that she looks thinner
 and tighter.’

Any of the would-be nouns that are turned into verbs in (8) can in turn be employed in any of the five frames exemplified in (5) (with different degrees of resulting harmony). For instance, *kroppspresse* ‘exert body image pressure’, which is used in the transitive frame in (8c), can be used in any of the other frames as well. This is shown in (9) where (9b) is (8c) repeated (with *nynorsk* spelling); my constructed examples.

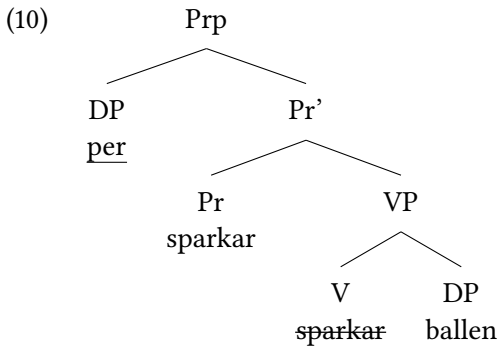
- (9) a. Internett kroppspressar. [intransitive frame]
 internet bodypressures
 ‘Internet exerts body image pressure.’
- b. Internett kroppspressar oss. [transitive frame]
 internet bodypressures us
 ‘Internet exerts body image pressure on us.’
- c. Internett kroppspressar oss dårlege sjølvbilde. [ditransitive frame]
 internet bodypressures us bad selfpictures
 ‘Internet gives us bad self-images by exerting body image pressure.’
- d. Internett kroppspressar oss sjuke. [resultative frame]
 internet bodypressures us ill-PL
 ‘Internet makes us ill by exerting body image pressure.’
- e. Internett kroppspressar oss dårlege sjølvbilde inn i hovudet.
 internet bodypressures us bad-PL selfpictures into in head-DEF.-SG
 [ditrans.-res. frame]
 ‘Internet gives us bad self-images in our heads by exerting body
 image pressure.’

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4 Some theoretical points concerning the EFM

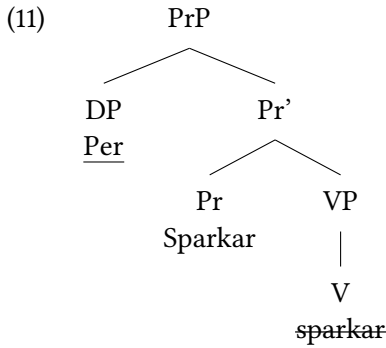
In (2), I showed a simple preliminary representation of the basic exoskeletal clause. In this section, I will present the EFM representations proposed in Åfarli (2007) of the five frames. In that paper, these representations were based on a (development of a) PrP-analysis of the clause (Bowers 1993), and I will not repeat the motivation for that. Instead, I will stress a couple of important theoretical points having to do with restrictions on the generations of the frames that were not included in the previous paper.

Consider first a simple clause with a transitive main verb. (10) shows the EFM PrP structure of the transitive frame example (5b), meaning ‘Per kicks the ball’, after insertion of verb and argument DPs, and after obligatory V to Pr raising of the verb.

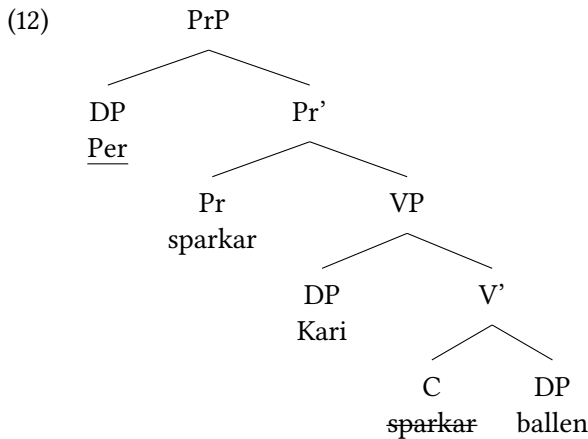


However, the simplest type of clausal PrP is one without a direct object. On the other hand, the subject is obligatory, since its existence follows from the predicational nature of PrP, see Eide & Åfarli (1999), Åfarli & Eide (2000), or Åfarli (2017). Thus, the simplest PrP is exemplified by (5a), meaning ‘Per kicks’, instantiating the intransitive frame type. Its structure is shown in (11) (after insertion and verb raising).

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Given that phrases can have both a specifier and a complement, the VP can have a specifier, which by assumption is the position for indirect objects. Thus the third type of PrP, the ditransitive type, is exemplified by (5c), meaning ‘Per kicks Kari the ball’. Its structure is shown in (12).

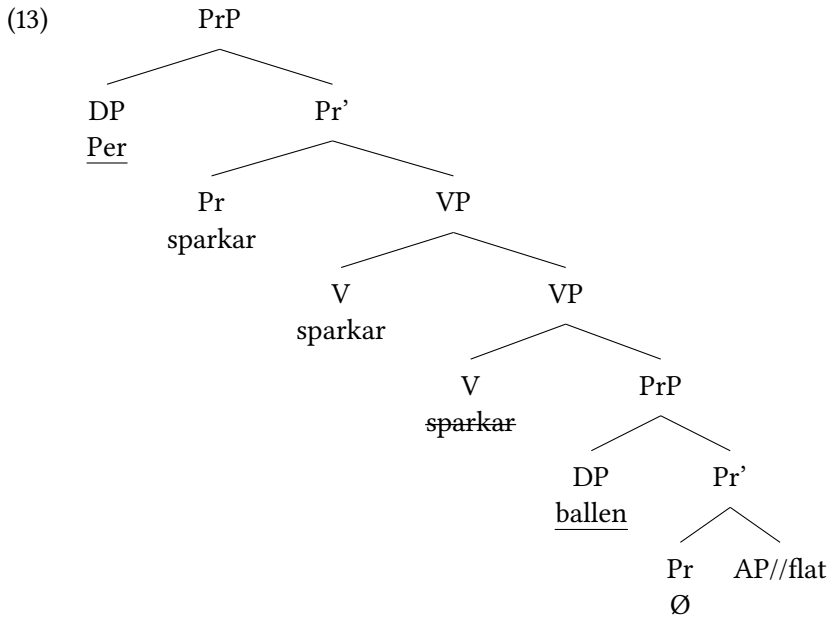


Next, consider the possibility that a clausal PrP may be embedded as a small clause inside the VP that is selected by a superordinate PrP. Can such a small clause PrP occupy both the specifier and complement position in the VP? We know that specifier positions are generally more restricted than complement positions, see, e.g., Eide & Åfarli (1999: 172). For instance, extraction from within specifier positions (subject and indirect object positions) is disallowed, but allowed from within complement (direct object) positions. Also, small clauses are never found in subject or indirect object positions, which suggests that a small clause PrP cannot function as a specifier of the VP. Still, small clause PrPs are common in direct object positions. Therefore, there is reason to believe that independent properties of grammar dictates that small clause PrPs must be restricted

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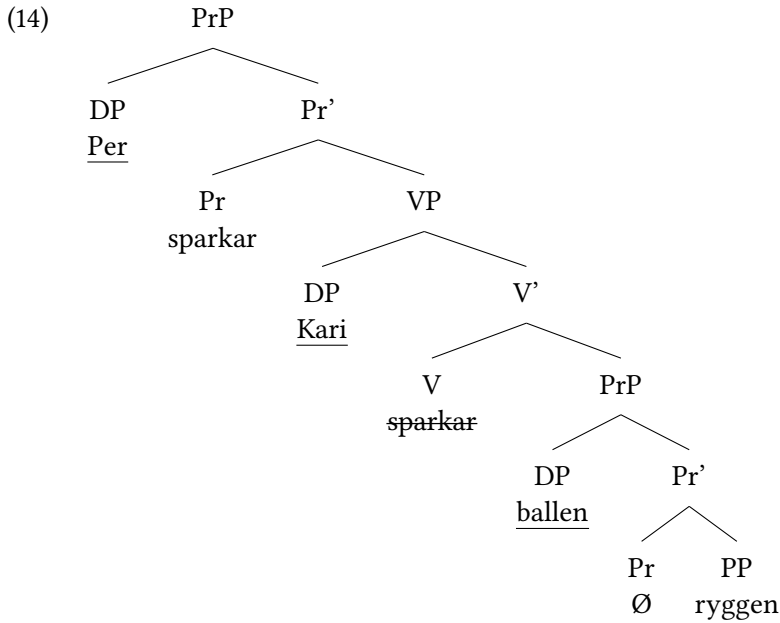
to the complement position of V. Assuming that to be the case, this fact contributes to reducing the number of exoskeletal frames allowed in the language.

Now, (11) with a PrP in the complement position of V would be a structure with a small clause complement like (5d), meaning ‘Per kicks the ball flat’. Its structure is shown in (13).



Given our earlier assumptions, there is exactly one possibility left, namely that the specifier position of VP in (13) is augmented by an indirect object DP. The expectation that such structures exist is confirmed by the existence of the ditransitive-resultative type (5e), which can be given the English translation ‘Per kicks the ball into Kari’s back’, and which has the structure shown in (14) in Norwegian.

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The remarkable thing about the reasoning above is that by assuming that PrP is a recursive category, we predict the existence of exactly the five argument structure frames that are attested in Norwegian.¹⁰ Thus, it seems that Norwegian exploits the full verb frame potential, whereas English, as we saw in Section 2, does not exploit the ditransitive-resultative type. Other languages may refuse to exploit other frames, e.g. the ditransitive type and/or the resultative type (languages that do not have double object constructions and/or resultative small clause complements). The intransitive and transitive frames seem to be exploited universally. Languages that use only a subset of the available frames, use other means (e.g. different types of adjuncts) to maintain the expressive power of the language.

¹⁰Notice, however, that this prediction requires that recursion of PrP may take place only once. Otherwise, recursion of the PrP could presumably happen infinitely, generating infinitely many frames. Thanks to Terje Lohndal for pointing this out, and also for suggesting that substituting my upper PrP for a (non-recursive) VoiceP and retaining my lower small clause PrP as the only (non-recursive) PrP available, would predict the existence of maximally five frames without any stipulation regarding recursion other than the assumption that PrP and VoiceP are non-recursive categories. Sorting out the ramifications of the two proposals will be left for future consideration.

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5 Conclusion

In this paper, I have tried to show how a version of an exoskeletal model – dubbed the Exoskeletal Frame Model – is able to account for two types of periphery type data from the empirical domains of argument structure flexibility and lexical creativity. It has become clear along the way that the terms core data and periphery data are in fact misleading. There is hardly anything peripheral about the two types of «periphery data» that I have called attention to. To the contrary, linguistic phenomena like argument structure flexibility and lexical creativity are exploited all the time by every language user, see also [Goldberg \(2003: 222\)](#) for a similar point. For that reason, it is particularly important to have access to a general grammar model that can account naturally for both the traditional core data and the so-called periphery data using a simple and coherent analysis. I have tried to show that the EFM can be such a model. I have also tried to show how and why traditional lexicalist or endoskeletal models struggle when they encounter these types of flexibility data.

My analysis has been a bird’s eye view with a somewhat programmatic flavor. I have focussed on making clear the principled differences between endoskeletalism on the one hand and exoskeletalism on the other, trying to point out the essential motivation for adopting the EFM and the exoskeletal approach, and trying to present a simple and robust model that hopefully is suitable both for bringing some linguistic insights and for didactic purposes.

The data that I have discussed come mainly from Norwegian and English. It would be interesting to see the EFM (or other exoskeletal models) applied to lexical creativity and argument structure flexibility data in other languages as well, especially since some languages appear to be quite restrictive in both areas, whereas others are more permissive, possibly having to do with varying degrees of normative pressure.

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References

- Aa, Leiv Inge. 2020. *Norwegian verb particles*. 1st edn., vol. 4. Amsterdam: John Benjamins.
- Åfarli, Tor A. 2007. Do verbs have argument structure? In Eric Reuland, Tanmoy Bhattacharya & Giorgos Spathas (eds.), *Argument structure*, 1–16. Amsterdam: John Benjamins.
- Åfarli, Tor A. 2017. Predication in syntax: Toward a semantic explanation of the subject requirement. In Piotr Stalmaszczyk (ed.), *Understanding predication*, 73–96. Frankfurt am Main: Peter Lang.
- Åfarli, Tor A. & Kristin M. Eide. 2000. Subject requirement and predication. *Nordic Journal of Linguistics* 23. 27–48.
- Åfarli, Tor A., Mari Nygård & Brita Ramsevik Riksem. 2022. Gender ‘translation’ and distributed gender: Evidence from the Norwegian DP and language mixing. *Studia Linguistica* 76(2). 626–660. DOI: [10.1111/stul.12190](https://doi.org/10.1111/stul.12190).
- Åfarli, Tor A. & Karumuri V. Subbarao. 2019. Models of grammar and the outcomes of long-term language contact: Language mixing in Dakkhini. In Albert Álvarez González, Zarina Estrada-Fernández & Claudine Chamoreau (eds.), *Diverse scenarios of syntactic complexity*, 27–52. Amsterdam: John Benjamins.
- Alexiadou, Artemis, Terje Lohndal, Tor A. Åfarli & Maren Berg Grimstad. 2015. Language mixing: A distributed morphology approach. In Thuy Bui & Deniz Özyıldız (eds.), *NELS 45: Proceedings of the Forty-Fifth Annual Meeting of the North East Linguistic Society*, vol. 1, 25–38. North Charleston: CreateSpace Independent Publishing Platform.
- Bauer, Laurie & Salvador Valera. 2005. *Approaches to conversion/zero-derivation*. Münster: Waxmann.
- Borer, Hagit. 2003. Exo-skeletal vs. endo-skeletal explanations: Syntactic projections and the lexicon. In John Moore & Maria Polinsky (eds.), *The nature of explanation in linguistic theory*, 31–67. Stanford: CSLI Publications.
- Borer, Hagit. 2005a. *Structuring sense: In name only*, vol. 1. Oxford: Oxford University Press.
- Borer, Hagit. 2005b. *Structuring sense: The normal course of events*, vol. 2. Oxford: Oxford University Press.
- Borer, Hagit. 2013. *Structuring sense: Taking form*, vol. 3. Oxford: Oxford University Press.
- Bowers, John. 1993. The syntax of predication. *Linguistic Inquiry* 24. 591–656.
- Brøseth, Heidi. 1997. *A neo-constructional approach to computer-oriented talk*. NTNU. (Doctoral dissertation).
- Chomsky, Noam. 1965. *Aspects of the theory of syntax*. Cambridge: The MIT Press.

Tor A. Åfarli

- Chomsky, Noam. 1981. *Lectures on Government and Binding*. Dordrecht: Foris.
- Chomsky, Noam. 1991. Linguistics and cognitive science: Problems and mysteries. In Asa Kasher (ed.), *The Chomskyan turn*, 26–53. Oxford: Blackwell.
- Clark, Eve & Herbert Clark. 1979. When nouns surface as verbs. *Language* 55. 767–811.
- Eide, Kristin M. & Tor A. Åfarli. 1999. The syntactic disguises of the predication operator. *Studia Linguistica* 53. 155–181.
- Eik, Ragnhild. 2014. *Setningar med possessivt objekt i norsk. Ein nykonstruksjonistisk analyse av setningar som 'ho trakka han på foten'*. NTNU, Trondheim. (MA thesis).
- Eik, Ragnhild. 2024. Ho trakka han på foten – Ein eksoskeletal analyse av possessivt objekt. *Maal og Minne* 116. 1–44.
- Embick, David. 2015. *The morpheme: A theoretical introduction*. Boston: De Gruyter Mouton.
- Frengstad, Lars Tsujimoto. 2016. *Å verbe: En eksoskeletal analyse av norske kreative verbkonstruksjoner*. NTNU, Trondheim. (MA thesis).
- Goldberg, Adele. 1995. *Constructions: A constructional grammar approach to argument structure*. Chicago: Chicago University Press.
- Goldberg, Adele E. 2003. Constructions: A new theoretical approach to language. *Trends in Cognitive Sciences* 7(5). 219–224.
- Grimstad, Maren B. 2018. *English-origin verbs in American Norwegian*. NTNU. (Doctoral dissertation).
- Grimstad, Maren Berg, Brita Ramsevik Riksem, Terje Lohndal & Tor Anders Åfarli. 2018. Lexicalist vs. exoskeletal approaches to language mixing. *The Linguistic Review* 35. 187–218.
- Jordet, Eirin. 2016. *En nykonstruksjonistisk analyse av norske presenteringssetninger*. NTNU, Trondheim. (MA thesis).
- Karlsen, Ida Kristin Holm. 2016. *Kan man spise mamma og pappa ut av huset? En ny-konstruksjonistisk rammeanalyse av barns språkforståelse*. NTNU, Trondheim. (MA thesis).
- Lohndal, Terje. 2014. *Phrase structure and argument structure: A case study of the syntax – Semantics interface*. Oxford: Oxford University Press.
- Marantz, Alec. 2013. Verbal argument structure: Events and participants. *Lingua* 130. 152–168.
- Næss, Malin Andrea. 2018. *Ikke-verbale småsetninger i norsk: En nykonstruksjonistisk analyse av ikke-verbale småsetninger med predikasjonspartiklene til og for*. NTNU, Trondheim. (MA thesis).
- Nygård, Mari. 2018. *Norwegian discourse ellipsis: Clausal architecture and licensing conditions*. Amsterdam: John Benjamins.

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- Riksem, Brita Ramsevik. 2016. *Grammatikk og rettskriving: Nye rammer for språknormering*. Oslo: Novus.
- Riksem, Brita Ramsevik. 2018. *Language mixing in American Norwegian noun phrases: An exoskeletal analysis of synchronic and diachronic patterns*. NTNU. (Doctoral dissertation).
- Riksem, Brita Ramsevik, Maren Berg Grimstad, Terje Lohndal & Tor Anders Åfarli. 2019. Language mixing within verbs and nouns in American Norwegian. *Journal of Comparative Germanic Linguistics* 22(2). 189–209.
- van Hout, Angeliek. 1996. *Event semantics and verb frame alternations*. Tilburg University. (Doctoral dissertation).

Chapter 3

English verb placement in Norwegian/English bilinguals: A Micro-cue approach

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The Micro-cue Model (Westergaard 2009) has been used to explain monolingual first and third language acquisition (Westergaard 2021) but has never been applied to simultaneous bilingual first language acquisition. We fill this gap by considering data on verb placement in English from four bilingual Norwegian/English girls, showing that they display four different patterns of crosslinguistic influence. We argue that the Micro-cue Model is perfect for capturing this variation, as it assumes that the different instantiations of verb placement and residual V2 in English are acquired separately.

1 Introduction

The Micro-cue Model argues that language acquisition takes place incrementally, property by property, through learning by parsing, irrespective of whether it is the first, the second, or the third language that is being acquired (see e.g. Westergaard 2009, 2021). Acquisition consists of building abstract structures, or micro-cues, based on the input and subsequently using these structures productively to generate new sentences.

The model has been applied to L1 and L3 acquisition and can straightforwardly capture L2 acquisition as well (see Westergaard 2021). However, it has never been used to explain simultaneous bilingual acquisition. In this study, we fill this gap by considering corpus data from four Norwegian/English bilingual children

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acquiring English verb placement in a heritage language context. For three of the children, the acquisition of verb placement in English has been studied before in [Anderssen & Bentzen \(2018\)](#). In the current study, we include a fourth child and consider the data in view of the Micro-cue Model, proposing how 2L1 can be captured within this framework based on its explanation of other acquisition contexts, particularly L1.

The acquisition of verb placement is affected differently by the simultaneous acquisition of Norwegian V2 in the four children: One child permits Norwegian V2 across the board but to varying degrees; one expands residual V2 to include main verbs in English; another expands auxiliary movement to non-subject-initial declaratives, while the final child does not seem to be affected by cross-linguistic influence (CLI) from Norwegian verb placement at all. We argue that the Micro-cue Model is the perfect model for capturing this variation, as it assumes that the different instantiations of verb placement and residual V2 in English are acquired separately.

The paper is structured as follows. In Section 2, we provide some background information on the Micro-cue Model and present the basic word order patterns in Norwegian and English. We also outline the cues for verb placement in Norwegian and English as described by [Westergaard \(2009\)](#). Section 3 describes the empirical material used in the study, how we extracted the relevant data from the corpora, and how they have been analysed. In Section 4, we present our results and discuss what they mean for the Micro-cue approach. Concluding remarks are given in Section 5.

2 Background

2.1 Extension of the Micro-cue Model to 2L1

The Micro-cue Model was first proposed for first language acquisition ([Westergaard 2009](#)). The most important intuition behind the model is the observation that children are conservative learners ([Snyder 2007](#)), but they are nevertheless sensitive to fine distinctions in the grammar and typically exhibit appropriate micro-variation from early on, such as target-like use of V2 and V3 *wh*-questions in North Norwegian ([Westergaard 2009](#)). In the acquisition process children incrementally build abstract structures, micro-cues (referred to as M in Figures 1-2), by parsing the input with the help of UG. Importantly, the structures are built based on the input rather than specified in UG. This is why children are so successful at acquiring micro-variation; it is how they build their grammars in general, property by property. In cases when children fail to (consistently) display

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the variation of the ambient language, this is usually for reasons of complexity or economy (see e.g. Anderssen & Westergaard 2010; Anderssen et al. 2012; Anderssen et al. 2014). As grammar grows when more micro-cues are added, generalizations are made. Figure 1 below illustrates first language acquisition as described by the Micro-cue Model.

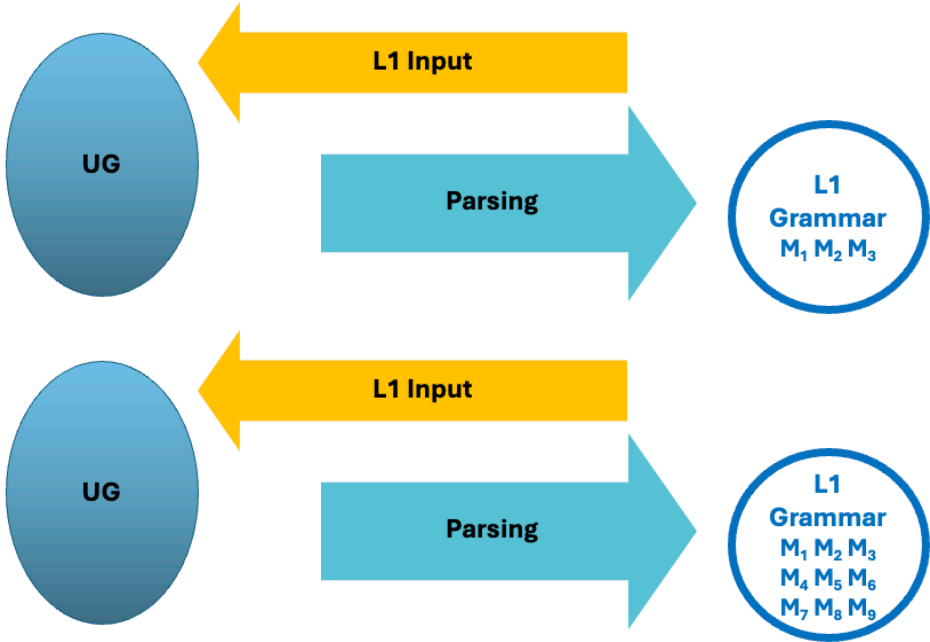


Figure 1: Illustration of L1 acquisition at two different stages according to the Micro-cue Model (Westergaard 2021: 385)

Central to the approach is also the view that learners make use of their entire linguistic repertoires in the acquisition process when more than one language is involved. For example, L3 acquisition also takes place property by property, involving a combination of the formation of new micro-cues from parsing the input and the transfer of structures from either of the previously acquired languages (Westergaard et al. 2017, Westergaard 2021). As a result, micro-cues can come from the L1, the L2 or the L3 (for illustration, see Westergaard 2021: 393).

To our knowledge, simultaneous bilingual acquisition (2L1) has never been studied within this framework. As an acquisitional context, it is different from first language acquisition because more than one language is acquired, but also different from L2 and L3 acquisition as there are no previously acquired languages to draw from. Nevertheless, it is well known that the two languages may

influence one another in 2L1 (e.g. [Serratrice 2013](#), [Van Dijk et al. 2022](#)), as they do in L2 and L3. The question is how the differences between L1 and 2L1 should be captured within the Micro-cue Model. As in L1 acquisition, 2L1 acquisition involves parsing the input guided by UG. UG is universal and the same in any acquisitional context. What is different is the fact that there is input from two languages. This is likely to affect the parser, since in a bilingual context, its job is not just to abstract micro-cues from the input for structure-building, but also to assign them to the correct one of the two grammars that are under development. Thus, it is possible for micro-cues to be misassigned due to a misanalysis of the input. This can occur because the ‘choices’ made by the parser will also be affected by its previous linguistic experience, as unlike UG, the parser is likely to be sensitive to how the input from the two languages is distributed as well as any overlap between them. Again, a possible outcome of these may be the erroneous assignment of a micro-cue to both languages rather than just one. On the assumption that parsing within the Micro-cue Model refers to the ability to comprehend and break linguistic sequences down into smaller units (syntactic, morphological, phonological, etc.), there is also reason to believe that the parsing abilities of children may differ. Thus, the building of micro-cues may be affected both by differences in the linguistic environment of the child and by individual differences between children. As a result, some children may parse the input in a target-like way and build the correct micro-cues in their two languages, while other children might mistakenly parse the input to assign the relevant micro-cue to both languages. Figure 2 illustrates what these two situations might look like.

In the next section we will briefly outline the verb placement patterns in Norwegian and English, before we consider which micro-cues are relevant for acquiring these patterns in 2.3.

2.2 Verb placement in Norwegian and English

Norwegian and English word order patterns partially overlap but also differ in important ways. In this section we outline the basic word order patterns in the two languages, focussing on four constructions that we have investigated in the child language corpora: Negated clauses, *yes/no*-questions, *wh*-questions, and non-subject initial clauses.

Both Norwegian and English are SVO languages, and the basic word order in main clauses is subject-verb-object, as illustrated in (1)-(2):

- (1) Jeg så bilen. (Norwegian)
 I saw car.DEF
 ‘I saw the car.’

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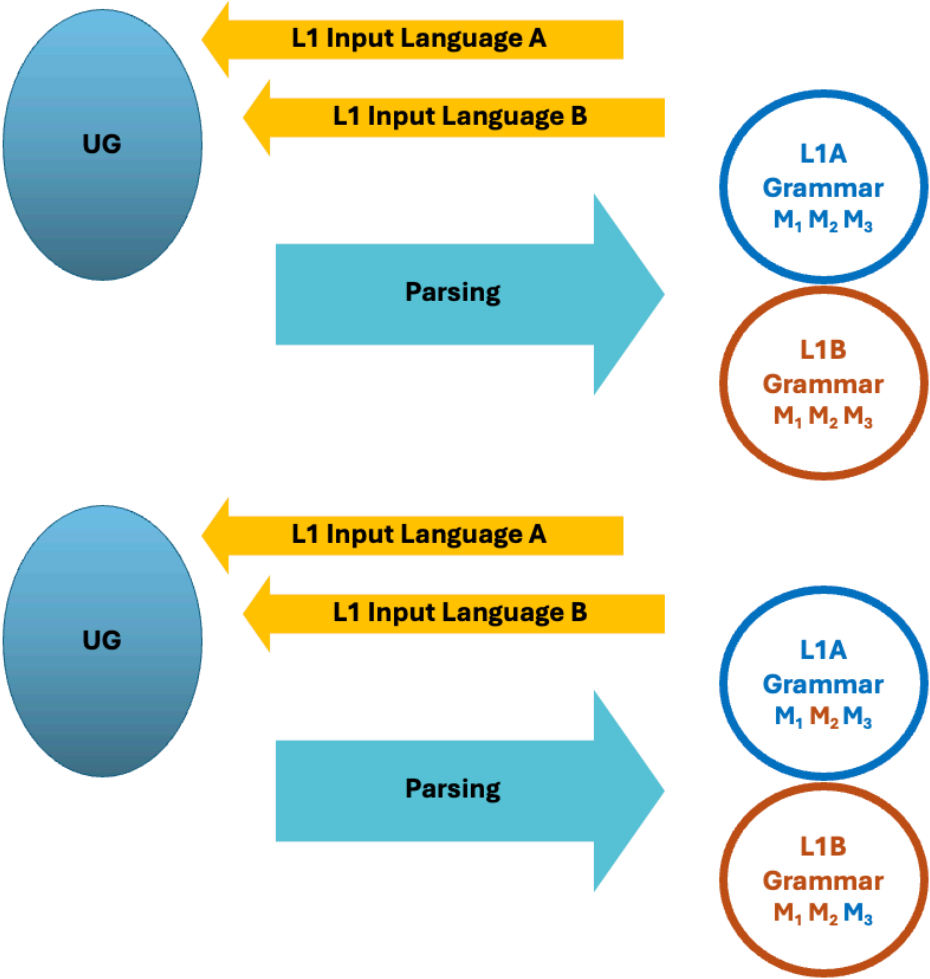


Figure 2: Illustration of 2L1 acquisition within the Micro-cue Model when the micro-cues are assigned to the target languages (top) and when they are (mistakenly) assigned to both languages (bottom).

- (2) I saw the car. (English)

Norwegian is also a V2 language, and the finite verb, be it an auxiliary or a main verb, moves to the second position in main clauses. English, on the other hand, displays only residual V2 effects, and V2 is restricted with respect to verb types (only affecting auxiliaries and the copula), and clause types (verb movement in negative and interrogative clauses, but not in non-subject-initial clauses). Thus, Norwegian and English display partially overlapping word order patterns in some constructions, but in other constructions, underlying differences surface.

First, in negated clauses, any finite verb moves across negation in Norwegian, (3). In English, on the other hand, only auxiliaries move across negation. In the absence of an auxiliary, *do*-support is required in the position preceding negation, (4):

- (3) a. Jeg **har** ikke sett bilen. (Norwegian)
 I have not seen car.DEF
 ‘I haven’t seen the car.’
 b. Jeg **så** ikke bilen.
 I saw not car.DEF
 ‘I didn’t see the car.’
- (4) a. I haven’t seen the car. (English)
 b. I **didn’t** see the car.

Turning to interrogative clauses, in Norwegian, all finite verbs undergo inversion. In English, again, verb movement only applies to auxiliaries/copula, which undergo subject-auxiliary inversion (SAI). In the absence of an auxiliary/copula, *do*-support is required for inversion. This is illustrated with *yes/no*-questions in (5)-(6), and non-subject *wh*-questions in (7)-(8)¹:

- (5) a. **Har** du sett bilen? (Norwegian)
 have you seen car.DEF
 ‘Have you seen the car?’
 b. **Så** du bilen?
 saw you car.DEF
 ‘Did you see the car?’

¹Notably, Northern Norwegian, the variety of Norwegian that the children in this study are acquiring, allows V3 in certain *wh*-questions (Westergaard 2009). However, as this will not be relevant for the discussion of the current data, we will not go into that here.

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- (6) a. **Have** you seen the car? (English)
 b. **Did** you see the car?
- (7) a. Hva **har** du kjøpt? (Norwegian)
 what have you bought
 ‘What have you bought?’
 b. Hva **kjøpte** du? what bought you
 ‘What did you buy?’
- (8) a. What **have** you bought? (English)
 b. What **did** you buy?

Finally, also in non-subject-initial clauses, inversion of the finite verb is obligatory in Norwegian, (9). In contrast, in English neither finite main verbs nor finite auxiliaries/copula undergo inversion, and there is no *do*-support, (10):

- (9) a. I dag **har** jeg kjøpt en bil. (Norwegian)
 today have I bought a car
 ‘Today I have bought a car.’
 b. I dag **kjøpte** jeg en bil.
 today bought I a car
 ‘Today I bought a car.’
- (10) a. Today I **have** bought a car. (English)
 b. Today I **bought** a car.

Note that in clauses where the topicalized element is *here* or *there*, English superficially displays a variable V2 pattern. With pronominal subjects, the pattern is as in (10) above, without SAI. However, with full DP subjects, these constructions require SAI. This is illustrated in (11):

- (11) a. Here she **comes**. (English)
 b. Here **comes** the bride.

We will not be considering the latter constructions in the current paper. We now turn to look at how the micro-cues for the verb placement patterns outlined for Norwegian and English should be formulated.

2.3 Micro-cues for verb placement in Norwegian and English

As we have seen there is extensive overlap in verb placement between English and Norwegian, which could result in misassignments of micro-cues when the two languages are acquired simultaneously. Westergaard's (2009) Micro-cue Model was developed to describe the acquisition of verb placement in monolingual Norwegian children. It was inspired by Lightfoot's (1999) cue-based model, which assumes that there is one cue for V2 ($_{CP}[XP \text{ } _CV\dots]$), but Westergaard argues that verb placement is acquired through the acquisition of several micro-cues. That is, the different realisations of both V2 and non-V2 structures are acquired property by property. An important argument for this is the observation that children are extremely sensitive to micro-variation. This is illustrated by their ability to pick up on the V2/V3 variation in *wh*-questions in Norwegian, and in the way they quickly acquire idiosyncratic non-V2 word orders in exclamative structures and with clause-initial *kanskje* (maybe) (Westergaard 2009). Consequently, while the micro-cue for V2 in declarative clauses specifies that an initial XP is followed by a (finite) verb (12), there are separate micro-cues for non-subject *wh*-questions of the type that require V2 (13) and for the type that does not (14), as well as for non-V2 with exclamatives (15) and clause-initial *kanskje* (16) (Westergaard 2009). For a complete picture the micro-cues for *yes/no*-questions in Norwegian are also provided below (17).

- (12) Micro-cue for V2 in declaratives: $_{DeclP}[XP \text{ } _{Decl}V\dots]$
- (13) Micro-cue for V2 in *wh*-questions with long *wh*-elements: $_{IntP}[XP_{[+wh]} \text{ } _{Int}V\dots]$
- (14) Micro-cue for non-V2 in *wh*-questions with short *wh*-elements: $_{IntP}[wh \text{ } _{TopP}[XP_{[-FOC]} \dots \text{ } _{IntTopP}[_{IntTop}[V]]]]$
- (15) Micro-cue for non-V2 in exclamatives: $_{ExclP}[wh\dots \text{ } _{VP}[V]]$
- (16) Micro-cue for word order in declaratives with clause-initial *kanskje* (maybe): $_{DeclP}[kanskje \text{ } XP\dots \text{ } _{VP}V]$
- (17) Micro-cue for V2 in *yes/no*-questions: $_{PolP}[_{Pol}V\dots]$

The default assumption for negative clauses (and clauses with sentence medial adverbs) in Norwegian is that this word order will be taken care of by the micro-cue for declaratives (12). However, Westergaard (2009) also discusses the possibility that children interpret subject-initial structures like (3a-b) as involving movement to I rather than to C. On the assumption that children do not move any more than they have evidence for in the input, this is not an unlikely scenario,

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and there are some indications that this might be the case. One such indication is the (erroneous) overgeneralisation of verb movement past negation and other adverbs in embedded clauses in Norwegian child language (Westergaard 2009: 191–196; Westergaard & Bentzen 2007; Ringstad & Kush 2021). The micro-cue for verb movement to I is provided in (18) (Westergaard 2009: 196).

(18) Micro-cue for V-to-I: $_{IP}[XP\ V]$

It is difficult to tease these two analyses apart and indeed also difficult to determine what input would provide the evidence necessary to revise an initial IP analysis. Thus, a more likely conclusion might be that V2 in languages like Norwegian is asymmetric, with non-subject-initial clauses involving verb movement to the C-domain and subject-initial clauses involving movement to I. Westergaard et al. (2019) argue for this view based on data from L1, L2/n, heritage languages and urban ethnolects showing different behaviours in subject-initial and non-subject-initial declaratives. It might even be that once a V-to-I analysis has been established in the child grammar, this grammar may or may not be revised as the child gets older because of the limited evidence for movement to C, making both (12) and (18) possible adult grammars.

Turning to the implementation of the Micro-cue Model to the acquisition of English, the only structures that Westergaard (2009) explicitly discusses are *wh*-questions, arguing that the micro-cues for English need to specify that only verbal elements that can occur in I may appear in the second position in *wh*-questions (ibid. 57). This is done by substituting V with I in the micro-cue in (13). Following this rationale, the same has to be assumed for *yes/no*-questions (17). There should also be a micro-cue equivalent to (18) in English, but again, only for verbal elements that can occur in I. Finally, the micro-cue for non-subject-initial clauses in English must specify that lexical verbs always remain in VP in these structures, while auxiliaries will appear in IP when they are present.

What is interesting about the description of the micro-cues for *wh*-questions, *yes/no*-questions and subject-initial declaratives in English is that the same cues could also be extracted from Norwegian. In fact, given that languages often are sensitive to the distinction between auxiliaries and lexical verbs,² it would not be surprising if separate micro-cues were built for structures with auxiliaries and structures with lexical verbs also in Norwegian. Such an approach would be in line with the basic assumption of the Micro-cue Model that children are sensitive

²For example, both children and adults have been found to be more likely to move auxiliaries than lexical verbs past negation and adverbs in embedded clauses (Håkansson & Collberg 1994; Ringstad & Kush 2021).

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to fine grained distinctions, sometimes even more fine grained than is required by the target grammar.

On this assumption, English and Norwegian have the same micro-cues for negative and interrogative structures with auxiliaries, and the different micro-cues for Norwegian and English can be summarised as follows:

- (19) Micro-cues for negated clauses and subject-initial declaratives:

Norw: Decl/IP [XP_{Decl} V...]
Eng/Norw: IP [XP I VP [... V]]

- (20) Micro-cues for yes/no-questions:

Norw: PolP [Pol V...]
Eng/Norw: PolP [Pol I...]

- (21) Micro-cues for wh-questions:

Norw: IntP [XP_[+wh] Int V...]
Eng/Norw: IntP [XP_[+wh] Int I...]

- (22) Micro-cues for non-subject-initial clauses:

Norw: DeclP [XP_{Decl} V...] AND DeclP [XP_{Decl} I...]
Eng: DeclP [XP IP [...(I) VP [... V]]]

3 Methodology

The current paper is based on corpus data from four Norwegian/English bilingual girls: Sunniva, Emily, Emma, and Hedda. The first three girls were investigated with respect to English word order and CLI in [Anderssen & Bentzen \(2018\)](#). In this paper we revisit these data in light of the addition of data from a fourth child, Hedda. All the girls grew up in Tromsø with one native English-speaking parent³ and one native Norwegian-speaking parent. In the families of Sunniva, Emily, and Emma, English was the primary home language, and both parents used English with their child and with each other. In Hedda's family, on the other hand, the mother primarily used Norwegian with Hedda, and the parents used Norwegian with each other. Thus, the language input situation in the family is quite different for Hedda compared to the other three girls. All four children attended a Norwegian language daycare from around the age of one.

³An English speaking father in the case of Sunniva/Emily and Hedda and an English speaking mother in the case of Emma.

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For Sunniva, Emma, and Hedda we have data from both Norwegian and English, while for Emily, we only have English data. For all the children, however, we have focused on the English recordings for this paper. Moreover, for Hedda, data from both languages were collected for an extensive period (from 1;1–6;1). We have investigated all existing English recordings for Sunniva, Emily and Emma, while we have selected English data collected between ages 3-5 for Hedda. The reason we have not included recordings of Hedda in English language mode prior to the age of three is that while she clearly comprehended English from an early age, no relevant structures are attested in the English data until that age.

The relevant information about the corpora is provided in Table 1 below.

Table 1: Overview of the data used in this study

Child	Age	Number of files	Number of utterances	MLU (range)
Sunniva	1;6–2;8	9	2512	1.992-3.667
Emily	2;3–3;9	4	1495	2.833-4.961
Emma	2;7–2;10	6	1831	3.074-3.998
Hedda-I	3;0–3;11	27	7435	2.000-4.277
Hedda-II	4;0–4;11	14	2548	2.317-4.712

Given that the data from Hedda spans over a two-year period and there is (obviously) development over the two years, we present and discuss the data from Hedda for the two years separately.

The data were searched manually for the four relevant constructions (negated clauses, *yes/no*-questions, *wh*-questions, and non-subject initial clauses). We excluded clauses where the topicalized element was *here/there* and V2 is variable, and rather focused on topicalized constructions where English consistently does not display V2.

4 Results and discussion

4.1 Results

Finite verb placement in negated clauses is illustrated in Table 2 for the four girls. As the table shows, all four girls fairly consistently place auxiliaries and the copula in front of negation in a target-like way. Moreover, both Sunniva and Emily also generally make use of *do*-support in clauses without other auxiliaries, and

they thus seem to have acquired this construction in English. Emma and Hedda (at least at age 3), on the other hand, have not yet fully acquired *do*-support. Emma, in particular, leaves out *do* quite frequently (73%), typically producing bare infinitives instead, but also occasionally moves the main verb across negation in a non-target-like fashion (15%). This latter strategy is even more present in Hedda's production at age 3; she moves the finite main verb across negation 45% of the time. By age 4, *do*-support seems to have fallen into place in Hedda's production, but she still employs non-target-like movement of the main verb across negation 12% of the time.⁴

Table 2: Finite verb placement in negated declaratives

Child	Aux/cop				Main V & <i>do</i> -support						Total	
	Aux-neg		*No aux		<i>do</i>	-neg	*V-neg	*No <i>do</i>			Target	
Sunniva	9/9	100%	0/9	0%	14/14	100%	0/14	0%	0/14	0%	23/23	100%
Emily	53/57	93%	4/57	7%	44/45	98%	1/45	2%	0/45	0%	97/102	95%
Emma	37/47	79%	10/47	21%	5/41	12%	6/41	15%	30/41	73%	42/88	48%
Hedda-I	83/86	97%	3/86	3%	32/69	46%	31/69	45%	6/69	9%	115/155	74%
Hedda-II	45/45	100%	0/45	0%	57/65	88%	8/65	12%	0/65	0%	102/110	93%

Table 3 shows finite verb placement in *yes/no*-questions. Again, all four girls are target-like when it comes to placement of finite auxiliaries/copula, which consistently undergo SAI in our data. Moreover, Sunniva is also quite target-like in *yes/no*-questions requiring *do*-support and only leaves out *do* 9% of the time. Emily struggles more with *do*-support and leaves it out in 20% of the cases. *Do*-support is also challenging for Emma (in particular) and Hedda at age 3. However, these two girls resort to a different strategy, and they both produce inversion of the finite main verb, Emma as much as 78% of the time, and Hedda 27% of the time. By age 4, consistent *do*-support in *yes/no*-questions has fallen into place for Hedda.

In Table 4, we see that Sunniva, Emily and Hedda generally produce target-like non-subject *wh*-questions with SAI, although both Sunniva and Emily occasionally leave out the auxiliary (7% and 9% respectively). Emma, on the other hand, quite frequently leaves the auxiliary in a non-inverted position (40% of the time).⁵

⁴In Tables 2-5 we use the following abbreviations: Auxiliary (Aux), Copula (Cop), Negation (neg), V (Verb), Inversion (Inv). Columns with non-target responses are shaded, and non-target production that likely results from cross-linguistic influence from Norwegian occurs in bold face.

⁵Note, however, that we have very few non-subject *wh*-questions from Emma, only a total of 11.

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Table 3: Finite verb placement in yes/no-questions

Child	Aux/cop						Main V & <i>do</i> -support						Total	
	Aux Inv		*No Inv		*No aux		<i>do</i> Inv		*V Inv		*No <i>do</i>		Target	
Sunniva	54/56	96%	1/56	2%	1/56	2%	10/11	91%	0/11	0%	1/11	9%	64/67	96%
Emily	53/53	100%	0/53	0%	0/53	0%	16/20	80%	0/20	0%	4/20	20%	69/73	95%
Emma	47/47	100%	0/47	0%	0/47	0%	4/18	22%	14/18	78%	0/18	0%	51/65	79%
Hedda-I	54/54	100%	0/54	0%	0/54	0%	16/22	72%	6/22	27%	0/22	0%	70/76	92%
Hedda-II	20/20	100%	0/20	0%	0/20	0%	30/30	100%	0/30	0%	0/30	0%	50/50	100%

In *wh*-questions requiring *do*-support, the numbers for all the girls are very low, but the examples we have suggest that Sunniva, Emily and Emma are more or less target-like and include *do* undergoing SAI. This is the general tendency for Hedda as well, but she also produces a few cases where the finite main verb has undergone inversion in a non-target-like way (25%).

Table 4: Finite verb placement in non-subject *wh*-questions

Child	Aux/cop						Main V - <i>do</i> -support						Total	
	Aux Inv		*No Inv		*No aux		<i>do</i> Inv		*V Inv		*No <i>do</i>		Target	
Sunniva	101/111	91%	2/111	2%	8/111	7%	7/9	78%	1/9	11%	1/9	11%	108/120	90%
Emily	29/32	91%	0/32	0%	3/32	9%	3/3	100%	0/3	0%	0/3	0%	32/35	91%
Emma	6/10	60%	4/10	40%	0/10	0%	1/1	100%	0/1	0%	0/1	0%	7/11	64%
Hedda-I	35/36	97%	1/36	3%	0/36	0%	3/4	75%	1/4	25%	0/4	0%	38/40	98%
Hedda-II	54/54	100%	0/54	0%	0/54	0%	13/20	65%	5/20	25%	2/20	10%	67/74	91%

Finally, Table 5 shows the placement of finite verbs in non-subject-initial clauses. Recall that such clauses do not display any kind of V2 effects in English. Sunniva only produces two relevant constructions, both of which are target-like without verb movement. Hedda is also more or less target-like with respect to this construction. Emily and Emma, on the other hand, produce various non-target-like word orders in these clauses. Both girls produce some non-target-like inversion of auxiliaries, although Emily does this much more than Emma (79% and 36% respectively). In clauses without auxiliaries, Emma also produces some non-target-like inversion of the finite main verb (21%). Emily, however, uses a very different strategy in these clauses. As much as 56% of the time, she employs unwarranted *do*-support which also undergoes inversion.

4.2 Discussion

In [Anderssen & Bentzen \(2018\)](#) we discussed the data from Sunniva, Emma and Emily, and showed that simultaneous acquisition of Norwegian and English can

Table 5: Finite verb placement in non-subject-initial declaratives

Child	Aux/cop				Main V - <i>do</i> -support						Total	
	No Inv		*Aux Inv		No Inv		*V Inv		* <i>do</i> Inv		Target	
Sunniva	1/1	100%	0/1	0%	1/1	100%	0/1	0%	0/1	0%	2/2	100%
Emily	7/33	21%	26/33	79%	19/43	44%	0/43	0%	24/43	56%	26/76	34%
Emma	25/39	64%	14/39	36%	23/29	79%	6/29	21%	0/29	0%	48/68	71%
Hedda-I	96/100	96%	4/100	4%	67/67	100%	0/67	0%	0/67	0%	163/167	98%
Hedda-II	43/44	98%	1/44	2%	73/75	97%	2/75	1%	1/75	1%	118/122	97%

lead to CLI with quite different outcomes, even when the children’s language situation is fairly similar. We then identified three different patterns (patterns 1, 2 and 4 below). In this paper, the addition of the fourth child, Hedda, adds a fourth CLI pattern (pattern 3 below), which completes the picture:

(23) Four patterns of CLI from Norwegian into English

Pattern 1 – Sunniva

No cross-linguistic influence

Pattern 2 – Emma

Generalization of V2 across both verb types and clause types

Pattern 3 – Hedda

Generalization of residual V2 across verb types

Pattern 4 – Emily

Generalization of residual V2 across clause types

The girls thus display different types and amounts of non-target-like behaviours. This is illustrated in Table 6.

Table 6: Non-target-like placement of main verbs and of auxiliaries

Child	Negative	Interrogatives		Non-subject-initial		
		<i>yes/no</i>	<i>wh</i>			
	MainV	MainV	MainV	Aux	MainV	<i>do</i>
Sunniva						
Emma	15%	78%		36%	21%	
Hedda I	49%	27%	25%			
Hedda II	12%		25%			
Emily				79%		56%

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These findings clearly support the idea that V2 is not acquired as *one single phenomenon*, but rather incrementally, through building various micro-cues for different types of constructions.

As we see from Table 6, Emma overgeneralizes residual V2 both across verb types and clause types. She employs movement of the finite main verb in both negative clauses (15%) and *yes/no*-questions (75%), as in (24)-(25), and she also displays inversion of both main verbs (21%) and auxiliaries (36%) in non-subject-initial clauses, as shown in (26-27).

- (24) I **hurt** not this knee now (Emma 2;8)
- (25) **Drive** daddy me to barnehage (=daycare)? (Emma 2;7)
- (26) Now **throw** I it (Emma 2;8)
- (27) Now **have** I ringed Angus (Emma 2;8)

The Micro-cue Model would explain this by assuming that Emma has erroneously assigned the Norwegian micro-cues for all these constructions to her English grammar.

Hedda also overgeneralizes residual V2, but only across verb types. Especially at age 3, Hedda moves finite main verbs across negation in negative clauses (45%), and across the subject in interrogative clauses (25%), as illustrated in (28-30). However, she rarely overgeneralizes verb movement into non-subject-initial clauses.

- (28) She **gets** not out (Hedda 3;5)
- (29) **Hear** you that song that I made? (Hedda 3;10)
- (30) What **looks** b like? (Hedda 4;4)

Like Emma, Hedda thus erroneously assigns the micro-cues for Norwegian negative and interrogative constructions to English. However, she does not mis-assign the Norwegian micro-cues for non-subject-initial clauses, which demonstrates a sensitivity to different micro-cues for different construction types.

Finally, Emily over-generalizes residual V2 for auxiliaries and copula to apply to *all* clause types (but not to lexical verbs), and consequently frequently (79%) uses SAI also in non-subject-initial clauses, (31). Moreover, a further indication that she has a residual V2 grammar for this clause type comes from the fact that she quite often (56%) employs *do*-support in such clauses, and *do* also undergoes inversion, (32).

- (31) And then **was** madame Gazelle's telephone ringing (Emily 3;8)

(32) And then **did** Candy Cat come

(Emily 3;8)

The pattern observed in Emily's data may initially seem less straightforwardly explained as CLI from Norwegian, as it aligns with neither English nor Norwegian grammar. However, we argue that the Micro-cue Model can explain this, as it assumes that children are sensitive to micro-variation. Emily is clearly aware of the cues for negative and interrogative clauses in English, specifying that only auxiliaries can move across negation or the subject. However, simultaneous input from Norwegian provides cues for verb movement also in non-subject-initial clauses. While Emma misassigns this micro-cue to also apply in English, Emily apparently combines it with internal cues from English, specifying that verb movement only applies to auxiliaries. The result is clearly a potential micro-variant, but it does not correspond to the pattern found in either English or Norwegian.

We argue that an approach assuming that V2 is acquired as one phenomenon cannot account for the various patterns attested in these four girls. The Micro-cue Model, on the other hand, predicts that precisely such variation may occur, both across and within individuals. Figure 3 illustrates this variation, showing that Emma is affected by CLI across the board (except in *wh*-questions, with only one relevant example), Hedda expands residual V2 to lexical verbs, while Emily expands it to non-subject-initial clauses.

Next, let us turn to how this can be accounted for by the Micro-cue Model. Recall that for negative and interrogative clauses, we argued that English and Norwegian share the micro-cue concerning placement of auxiliaries (examples (19)-(21), here repeated as (33)-(35)). In addition, Norwegian has a similar cue for placement of main verbs. Thus, within a Micro-cue Model, it is not surprising that CLI may lead to misassignment of the latter cue from Norwegian into English. We see this in the data from Emma and Hedda.

(33) Micro-cues relevant for negated clauses:

Norw: Decl/IP [XP_{Decl} V...]
Eng/Norw: IP [XP I_{VP} [... V]]

(34) Micro-cues for yes/no-questions:

Norw: PolP [Pol V...]
Eng/Norw: PolP [Pol I...]

(35) Micro-cues for *wh*-questions:

Norw: IntP [XP_[+wh] Int V...]
Eng/Norw: IntP [XP_[+wh] Int I...]

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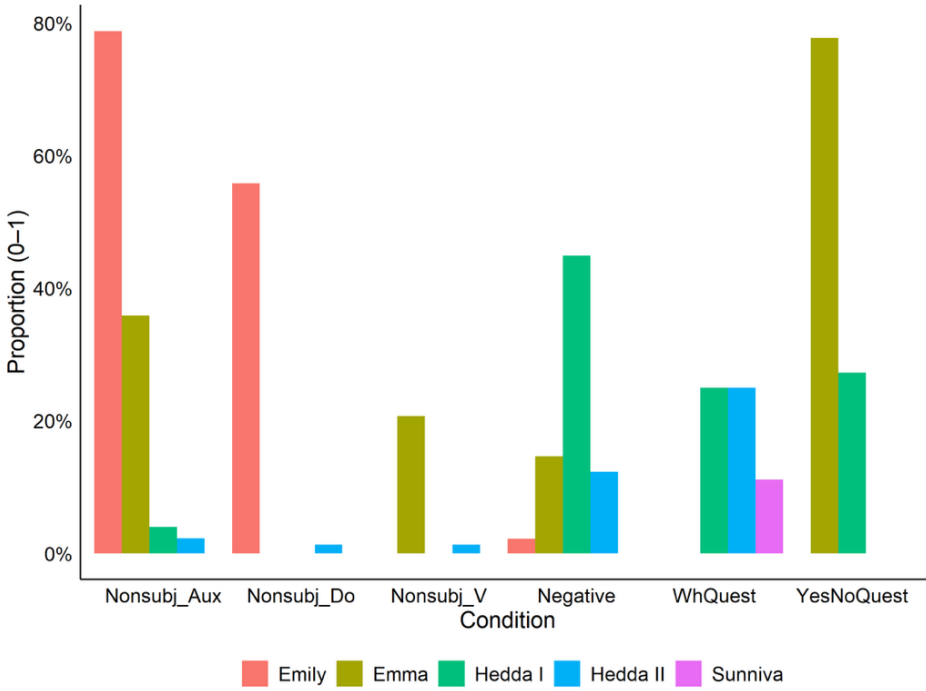


Figure 3: : Bar plot of the proportion of non-target-like structures caused by CLI from Norwegian in non-subject-initial declaratives (Nonsubj), negative structures, wh-questions and yes-/no-questions

For non-subject-initial clauses, Norwegian has micro-cues specifying inversion of both auxiliaries and main verbs, while the cue in English is that there should be no verb movement at all ((22) here repeated as (36)):

(36) Micro-cues for non-subject-initial clauses:

Norw: DeclP[XP_{Decl} V...] AND DeclP[XP_{Decl} I...]
Eng: DeclP[XP_{IP} [...(I) VP[... V]]]

As we recall from Table 6, Emma inverts both main verbs and auxiliaries in English, suggesting that she has misassigned both Norwegian micro-cues for this construction in English. Emily, on the other hand, only employs inversion with auxiliaries in such clauses in English. She thus seems to only have misassigned the Norwegian micro-cue concerning auxiliaries in English. Again, this is a possibility within the Micro-cue Model, as these micro-cues are independent of each other. Moreover, the pattern we see in Emily is probably also a result of input

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from English, as several other constructions in English have micro-cues specifying movement of auxiliaries (but not main verbs), thus supporting this as a potential hypothesis also for non-subject-initial clauses.

Even though the Micro-cue Model nicely captures the variation observed in the children of this study, several questions remain. As indicated in Section 3, the children differ from one another in other ways, and the question is whether these differences may cause the variable grammars. For example, they vary considerably in age. However, even though children clearly acquire more language with age, age is not a very good point of comparison in acquisition, as there is considerable variation among monolingual children, and this is exacerbated in bilingual contexts where language exposure is more variable. Also, while Hedda becomes more target-like between Hedda I (aged 3–4) and Hedda II (aged 4–5), the youngest child, Sunniva, behaves the most target-like overall. Sunniva produces very advanced structures early on, for example making extensive use of *do*-support already before the age of 2. Her precociousness is also confirmed by her comparatively high MLU range (cf. Table 1). Thus, there is no reason to believe that Sunniva would get less target-like or that Emily would start using V2 structures across the board (like Emma) with increasing age.

The other obvious difference between the children is Hedda's input situation compared to the other three. However, the only clear consequence of this is that Hedda is considerably older before any relevant structures are attested. It is not obvious that less English input can explain her CLI pattern. If anything, we would expect more CLI and V2 across the board, as in Emma's data, if Hedda's behaviour was caused by her input situation. Furthermore, even though the other children have English as their home language, this does not necessarily mean that their experiences are the same. In fact, the diverging behaviour with verb placement suggests that this is not the case.

Another factor that may play a role is parsing ability and its interaction with frequency. If children vary in their parsing abilities, differences between individual children are expected. Such discrepancies may also be exacerbated or counterbalanced by experiential differences, as these may bias the parser in a specific direction. However, this will have to be left for future investigations.

5 Concluding remarks

This paper examines the acquisition of verb placement in English by four bilingual children, showing that acquiring English together with the V2 language Norwegian can yield different acquisition paths. While Sunniva seems unaffected by

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CLI, Emma exhibits it in all structures. Hedda overgeneralizes residual V2 to lexical verbs, affecting negative clauses and questions, and Emily expands residual V2 to apply to non-subject-initial clauses. The Micro-cue Model can account for these variable patterns in an elegant way because it assumes that different instantiations of verb placement and (residual) V2 are acquired separately. This happens as the parser through misanalysis assigns micro-cues compatible with Norwegian to English in different combinations in the children. Simultaneous bilingual acquisition thus represents a field that would benefit from exploring the Micro-cue Model further.

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References

- Anderssen, Merete & Kristine Bentzen. 2018. Different outcomes in the acquisition of residual V2 and do-support in three Norwegian-English bilinguals: Cross-linguistic influence, dominance and structural ambiguity. *Frontiers in Psychology* 9. 1–18. DOI: [10.3389/fpsyg.2018.02130](https://doi.org/10.3389/fpsyg.2018.02130).
- Anderssen, Merete, Kristine Bentzen & Yulia Rodina. 2012. Topicality and complexity in the acquisition of Norwegian object shift. *Journal of Language Acquisition* 19(1). 39–72.
- Anderssen, Merete, Yulia Rodina, Roksolana Mykhaylyk & Paula Fikkert. 2014. The acquisition of the dative alternation in Norwegian. *Language Acquisition: A Journal of Developmental Linguistics* 21(1). 72–102.
- Anderssen, Merete & Marit Westergaard. 2010. Frequency and economy in the acquisition of variable word order. *Lingua* 120(11). 2569–2588. DOI: [10.1016/j.lingua.2010.06.006](https://doi.org/10.1016/j.lingua.2010.06.006).
- Håkansson, Gisela & Sheila Dooley Collberg. 1994. The preference for Modal + Neg: An l2 perspective applied to Swedish l1 children. *Second Language Research* 10(2). 95–124.
- Lightfoot, David. 1999. *The development of language: Acquisition, change and evolution*. Malden: Blackwell.

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- Ringstad, Tina Louise & Dave Kush. 2021. Learning embedded verb placement in Norwegian: Evidence for early overgeneralization. *Language Acquisition* 28(4). 411–432.
- Serratrice, Ludovica. 2013. Cross-linguistic influence in bilingual development: Determinants and mechanisms. *Linguistic Approaches to Bilingualism* 3. 3–25. DOI: [10.1017/S1366728904001610](https://doi.org/10.1017/S1366728904001610).
- Snyder, William. 2007. *Child language: The parametric approach*. Oxford: Oxford University Press.
- Van Dijk, Chantal, Elise van Wonderen, Elly Koutamanis, Gerrit Jan Kootstra, Ton Dijkstra & Sharon Unsworth. 2022. Cross-linguistic influence in simultaneous and early sequential bilingual children: A meta-analysis. *Journal of Child Language* 49(5). 897–929.
- Westergaard, Marit. 2009. *The acquisition of word order: Micro-cues, information structure, and economy* (Linguistik Aktuell/Linguistics Today 145). Amsterdam: John Benjamins.
- Westergaard, Marit. 2021. Microvariation in multilingual situations: The importance of property-by-property acquisition. *Second Language Research* 37(3). 379–407.
- Westergaard, Marit & Kristine Bentzen. 2007. The (non-)effect of input frequency on the acquisition of word order in Norwegian embedded clauses. In Insa Gülzow & Natalia Gagarina (eds.), *Frequency effects in language acquisition: Defining the limits of frequency as an explanatory concept*, 271–306. Berlin: Mouton de Gruyter.
- Westergaard, Marit, Terje Lohndal & Artemis Alexiadou. 2019. The asymmetric nature of V2: Evidence from learner languages. In Ken Ramshøj Christensen, Henrik Jørgensen & Johanna L. Wood (eds.), *The sign of the V – Papers in honour of Sten Vikner*, 709–733. Aarhus: Department of English, School of Communication & Culture, Aarhus University.
- Westergaard, Marit, Natalia Mitrofanova, Roksolana Mykhaylyk & Yulia Rodina. 2017. Crosslinguistic influence in the acquisition of a third language: The linguistic proximity model. *International Journal of Bilingualism* 21(6). 666–682.

Chapter 4

Mian verbs trouble linguists not learners

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Mian, a Papuan language of Papua New Guinea, shows an unusual system of verb agreement. All finite verbs obligatorily mark subject agreement, yet only a minority of transitive verbs show object agreement. This creates a striking split within the verbal lexicon. We investigate the morphological and syntactic implications of this restricted system, distinguishing it from more familiar types of inflectional irregularity (such as irregular past tense forms in English). While the absence of object agreement in Mian might appear to challenge learnability, we show that learners have robust cues: pervasive subject agreement, the higher token frequency of agreeing verbs relative to their type frequency, and the fact that a single verb form typically suffices to identify whether it shows object agreement. Comparative data from languages including Ingush, Tsez and Burmeso, situate Mian within its typological context, highlighting parallels with restricted and non canonical agreement systems elsewhere. Our analysis casts doubt on functional explanations of agreement as reference tracking, since Mian shows no compensatory strategies when agreement is absent. The Mian case thus sharpens the theoretical understanding of restricted morphological systems and their acquisition, and it broadens the typology of agreement phenomena.

1 The puzzle

Mian verbs show agreement with their subject and their object. This fact, as well as the forms realizing agreement, is an essential part of the grammar. And yet, while all verbs show subject agreement, only a minority of the transitive verbs show object agreement. We consider why linguists might find this is troubling,

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and draw comparisons with other related phenomena. The interest is not how Mian verbs agree with their object but whether. This sort of irregularity is different in nature from inflectional irregularity illustrated by English irregular past tense forms of the type *think/thought*. A learner of English might have to wait for considerable stretches of time to come across the irregular form, which often cannot be predicted from the rest of the paradigm. In Mian, in almost all instances, the learner will immediately know whether a verb shows object agreement the moment it is first produced. Mian has an unusual system, which might prompt some linguists to worry about learnability, but we show that there are good reasons for assuming that the system is fully learnable. We present the essential data on Mian (Section 2), then treat morphological issues (Section 3) and the relevance to syntax (Section 4). We next set Mian within the context of systems which are not complete, by looking at phenomena which are interestingly comparable (Section 5). This leads us to the logic of learnability (Section 6), before summing up (Section 7).

2 Mian

Mian is a Papuan language of Papua New Guinea (Fedden 2011). It belongs to the Ok family of languages, itself part of the larger Trans New Guinea family (Ross 2005; Pawley & Hammarström 2018). Our data are from the eastern dialect of Mian, which has around 1,400 speakers. The language is morphologically synthetic. It is mainly suffixing and head-marking. There is no morphological nor adpositional case marking for core arguments; grammatical relations are indicated by verbal affixes. No overt arguments are syntactically required and both pronouns and noun phrases are freely elided.

Our focus is the verbal paradigm. Example (1) illustrates subject and object agreement.

- (1) Mian (Fedden, field notes)
 naka=e gwán=o wa-têm'-Ø-e=be
 man=SG.M spider=SG.F 3SG.F.OBJ-see.PFV-REAL-3SG.M.SBJ=DECL
 'The man sees the spider.'

In (1), subject agreement (3SG.M, agreeing with *naka* 'man') is marked by the suffix *-e*, while object agreement (3SG.F, agreeing with *gwán* 'spider') is indicated by the prefix *wa-*. Tables 1 and 2 are a sketch of the paradigm of the verb 'see'. Since Mian verb roots produce hundreds of word forms, we confine ourselves to illustrating a part of the verbal paradigm using the perfective stem *têm'* in realis

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mood (which is here zero-marked, since the stem ends in a consonant). Table 1 shows subject agreement (which is suffixal) and keeps object agreement constant at 3SG.M.

Table 1: Subject agreement; object agreement held constant at 3SG.M

		SINGULAR	PLURAL
1		a-têm'-i	a-têm'-ob
2		a-têm'-eb	a-têm'-ib
3	MASCULINE	a-têm'-e	a-têm'-ib
	FEMININE	a-têm'-o	a-têm'-ib
	NEUTER 1	n/a	n/a
	NEUTER 2	n/a	n/a

As genders NEUTER 1 and NEUTER 2 contain exclusively inanimate noun, these do not appear as subjects of the verb 'see'. Table 2 shows object agreement (prefixal) and keeps subject agreement constant at 3SG.M. Historically, the forms of the object prefixes go back to proto Trans New Guinea *na- '1SG.OBJ', *ga- '2SG.OBJ', *wa- '3SG.OBJ', *ya- '3PL.OBJ' (Suter 2012: 52).

Table 2: Object agreement; subject agreement held constant at 3SG.M

		SINGULAR	PLURAL
1		na-têm'-e	ya-têm'-e
2		ka-têm'-e	ya-têm'-e
3	MASCULINE	a-têm'-e	ya-têm'-e
	FEMININE	wa-têm'-e	ya-têm'-e
	NEUTER 1	a-têm'-e	a-têm'-e
	NEUTER 2	wa-têm'-e	wa-têm'-e

But there are two other types of transitive verb. First we consider transitive verbs that take a verbal classifier, as in (2); these classifiers express the person and number of the object, as well as classify them according to six distinct classes (see Table 3 below):

- (2) Mian (Fedden, field notes)

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gwán=o om-ò-n-e=be
spider=SG.F 3SG.F_CL.OBJ-take-REAL-3SG.M.SBJ=DECL
'He picks up the spider.'

The prefix *om-* is the verbal classifier here; the classifiers distinguish not only the type (here feminine, as appropriate for 'spider'), but unusually also person and number.

Table 3 sets out a partial paradigm of a verb with an object classifier, subject agreement is kept constant at 3SG.M. Such a verb is 'take'. We use the perfective stem *ò* for illustration, finessing the allomorphy in the classifier forms; for the full set of forms, see Corbett et al. 2017: 14). Realis mood is marked with the suffix *-n*.

Table 3: Object classifiers (subject agreement held constant at 3SG.M, finessing in the classifier forms, NON-SINGULAR forms can be DUAL or PLURAL)

PERSON	CLASSIFIER	Forms	
		SINGULAR	NON-SINGULAR
1	1-CLASSIFIER ^a	nem-ò-n-e	dl-ò-n-e
2	2-CLASSIFIER	kem-ò-n-e	
3	M-CLASSIFIER	dob-ò-n-e	
	F-CLASSIFIER	om-ò-n-e	
	LONG-CLASSIFIER	tob-ò-n-e	tebel-ò-n-e
	BUNDLE-CLASSIFIER	gol-ò-n-e	gulel-ò-n-e
	COVERING-CLASSIFIER	gam-ò-n-e	gemel-ò-n-e
	RESIDUE-CLASSIFIER	ob-ò-n-e	ol-ò-n-e

^aWe call these 1_CLASSIFIER and 2_CLASSIFIER to distinguish these from the markers that are part of the ordinary person and number system.

We have now seen transitive verbs of two types, agreeing with their object in different ways. It is time to introduce two important clarifications.

First, while the object agreement shown in Table 2 is straightforward object agreement, the classifiers of Table 3 work on an absolutive basis. These mark the intransitive subject or the transitive object. We labelled them object classifiers

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because this is correct for almost every instance. There is just one intransitive verb, *mêin* ‘fall’, which has classifiers (it therefore agrees with its subject both with normal agreement (Table 1) and with classifiers (Table 3). Every other verb which takes classifiers is transitive.

The second point of clarification is that although transitive verbs may show object agreement potentially in one of two different ways, not all transitive verbs show object agreement. Here is a transitive verb which never agrees with its object:

- (3) Mian (Fedden, field notes)
- gwán=o bou-n-e=be
- spider=SG.F swat-REAL-3SG.M.SBJ=DECL
- ‘He swatted the spider.’

Verbs like *bou* ‘swat’ have the paradigm shown in Table 4. The stem of this verb is biaspectual; as above we give the realis forms (marked with *-n*).

Table 4: Verb showing no object agreement

		SINGULAR	PLURAL
1		bou-n-i	bou-n-ob
2		bou-n-eb	bou-n-ib
3	MASCULINE	bou-n-e	bou-n-ib
	FEMININE	bou-n-o	bou-n-ib
	NEUTER 1	n/a	n/a
	NEUTER 2	n/a	n/a

We now have the main facts:

- all finite Mian verbs agree with their subject;
- some transitive verbs agree with their object (two different ways for this);
- some transitive verbs do not agree with their object (we shall see that these form the majority).

Thus the syntax must include the mechanism for object agreement, but in most instances object agreement is not realized morphologically.

3 Troubling morphology

The morphological side of the issue is especially interesting in two ways. First, we ask which verbs are involved in object agreement; second, we investigate the relation between agreeing and non-agreeing verbs. The inventory of verbs which agree with their object is given in Table 5, based on a list of 245 transitive verbs.¹

Table 5 lists the verbs that have so far been identified during fieldwork.

We see from Table 5 that the overwhelming majority of transitive verbs (82.5%) do not agree with their object. Agreeing or failing to agree is a lexical matter. There is no syntactic environment in which a non-agreeing verb can be coerced into agreeing with its object; equally, there is no way of making an agreeing verb fail to agree. And a verb that agrees always does so either in person, number and gender or by taking a verbal classifier, never both. The question then is how we specify which verbs agree and which do not. We approach this question by considering two dimensions (following Bye 2015: 107). We first ask the extent to which the division is motivated (or natural) vs functionally arbitrary; in other words “why these verbs?” The second dimension is the extent to which the generalizations suggested are indeed general (applying to all items which meet the description) as opposed to lexically specified; in other words “how far do the rules apply?” We shall see that the answer to both questions is somewhat different for the two types of agreement.

Transitive verbs that agree in person, number and gender with their object—with the exception of ‘see, look’—are high on the transitivity scale (Hopper & Thompson 1980), in that they implicate or entail a change of state in the object, which makes the object rank high in affectedness (Tsunoda 1985; Beavers 2011). This makes the division motivated, but it is far from general, since there are numerous other transitive verbs, e.g. *klutaka* ‘smash’, which have a highly affected object, but which do not agree.

For the transitive verbs that agree by classifier the division is also motivated. Verbs that take a classifier—with the notable exception of *halin* ‘worry’ and *suan* ‘hate’—have lexical verb semantics of entity handling or movement. Again, the division is not fully general due to the existence of handling verbs which do not take a classifier, e.g. *mengge* ‘pull’, but there are considerably fewer handling verbs without a classifier than there are high-transitivity verbs which fail to agree in person, number and gender with their object.

¹An aspect distinction (perfective / imperfective) is at the core of the Mian verb system. As so often with aspect systems it is arguable whether we are dealing with one verb or two. We here treat aspectual stems as belonging to a single verb lexeme in Mian, but the main point is that the counting is done consistently.

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Table 5: Exhaustive list of Mian verbs showing object agreement

type of agreement	
person, number and gender	“classifier”
<i>lò, nâ</i> ‘hit, kill’; <i>fû</i> ‘grab’; <i>ntamâ</i> ‘bite’; <i>têm</i> ‘see, look at’	$\emptyset^{\wedge}$ ‘take s.o./sth.’ ^a ; <i>â</i> ‘leave sth., lose sth.’; <i>aa</i> ‘rouse (prey)’; <i>ab</i> ‘put’; <i>atdi</i> ‘throw sth. into the fire’; <i>atou</i> ‘put sth. into the fire’; <i>ba/bu</i> ‘put s.o./sth. into a bag, cover’; <i>biâ</i> ‘push s.o./sth., throw s.o./sth.’; <i>blangkê</i> ‘push s.o. out of the way’; <i>bù</i> ‘bury s.o., plant sth.’; <i>dî</i> ‘fetch (water)’; <i>êb</i> ‘take s.o./sth. (in order to carry)’; <i>fâ/ka</i> ‘put s.o./sth., care for s.o.’; <i>fâa</i> ‘lift s.o./sth.’; <i>fu</i> ‘send sth.’; <i>fuba</i> ‘put’; <i>halila/halin</i> ‘feel sorry for s.o./sth., be concerned about s.o./sth.’; <i>hâa</i> ‘chase s.o.’; <i>kimâ/kimsan</i> ‘put sth. into the fire’; <i>klafâ</i> ‘put s.o. on back (piggy-back)’; <i>ma/san</i> ‘plant sth.’; <i>meki</i> ‘hang sth. up’; <i>miki</i> ‘take (child) into arms to lull to sleep’; <i>môu</i> ‘put (pig or child) on shoulder’; <i>ò</i> ‘take s.o./sth.’; <i>silêb</i> ‘follow s.o.’; <i>ski</i> ‘turn s.o./sth.’; <i>suana/suan</i> ‘hate s.o.’; <i>tamâ</i> ‘pen in, imprison s.o.’; <i>tamâa</i> ‘step on s.o./sth.’; <i>tanâ</i> ‘light sth., set sth. on fire’; <i>tangâa</i> ‘hang up (item of clothing) to dry’; <i>tena</i> ‘elude’; <i>tlâa</i> ‘remove s.o./sth.’; <i>tôu</i> ‘put sth. above fireplace’; <i>toulêb</i> ‘take s.o./sth. into arms’; <i>ûb/ka</i> ‘give s.o./sth. to s.o.’; <i>usâ/uka</i> ‘put on (item of clothing)’; <i>textitwaa</i> ‘hide sth.’
Total 4 of 245	Total 39 of 245
Percentage of transitive verbs 1.6%	Percentage of transitive verbs 15.9%

^aThis verb is segmentally zero, yet all word forms based on this root have a LHL tonal melody. This is the reason for putting the diacritic ($\wedge$) into the representation of this verb. It seems that there used to be a non-zero verb root ‘take’, which was elided, while the tone associated with it remained.

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We now turn to the relation between the object-agreeing verbs and those which fail to agree; this is important for understanding the significance of the Mian system, and its relevance to learnability. At the most abstract level we are dealing with irregularity. But this is not just serious inflectional irregularity (whose degrees are interestingly discussed in [Marzi & Pirrelli 2023](#)). This is not the irregularity of, for instance, English *go*. This English verb is straightforwardly a verb: its feature signature matches that of other verbs. It makes exactly the same distinctions as other verbs, and all that is different about it is that some of its forms are unpredictable. Its irregularity is restricted to unexpected forms. In contrast, the Mian verbs which fail to agree with their object have a part of their paradigm missing; their feature signature does not match that of other verbs. Now the missing forms are not arbitrary gaps in the paradigm; the non-agreeing part can be specified in featural terms. Moreover, the non-agreeing verbs are not uninflectable, since they all inflect for subject agreement, and for various tense, aspect and mood values. Hence we have an interesting instance of partial uninflectability ([Palancar & Fedden 2026: xx](#)). We must be careful to distinguish this less familiar situation from run-of-the-mill irregularity.

4 Troubling syntax

The examples so far show that Mian syntax needs a rule of object agreement, a mechanism to ensure compatibility between the syntactic object and the agreeing morphology on the verb. However, in many real examples, this syntactic rule has no morphological effect, since most verbs do not show object agreement. How will this affect the syntax? We take an inferential-realisation approach to morphology, following the reasoning in [Stump \(2001: 1–30\)](#). Whether one is of the Network Morphology persuasion ([Corbett & Fraser 1993](#), [Brown & Hippisley 2012](#)) or follows Paradigm Function Morphology ([Stump 2001](#), [Stump 2023](#)), the answer will be that there is no impact on the syntax. The syntax does not ‘see’ what goes on in the morphology. And this is indeed the case: there is no syntactic effect when a transitive verb shows no object agreement. This shows that non-agreeing verbs are not defective, since they fit into all the same slots as agreeing verbs.

From a different perspective, however, there is the suggestion (going back at least to [Heath 1983](#) and often repeated) that agreement functions to facilitate reference tracking. How would we measure that? Ideally, we would like to “switch off” the agreement, and see whether this affects reference tracking. Mian provides the conditions which allow us to do this. We can compare corpus examples

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where the verb agrees with those where it fails to agree. If the failure to agree has an impact on reference tracking, we might expect to see a repair strategy, to compensate for the lack of agreement. The obvious strategy would be the inclusion of overt arguments (such as pronouns) to facilitate reference tracking. Fedden (2022) investigates this possibility, using a fully annotated Mian corpus of ~4,000 words, consisting of 14 texts (between 34 and 800 clauses for a total of ~1,700 clauses). The texts are mainly third-person narratives by fluent speakers aged 50–75 when the recordings were made in 2004.

Table 6: Overt object arguments and agreement in Mian discourse (Fedden 2022: 300)

		gender	classifier	combined
verb agrees:	n	61	218	279
	% overt object argument	33	39	38
verb does not agree:	n			397
	% overt object argument			43

Table 6 shows that there is no significant difference between transitive verb types (with and without agreement) when the proportion of overt arguments is compared. That is, the lack of agreement is not compensated for by the inclusion of additional overt arguments. This can be shown by a series of chi-square tests, for gender, classifiers, and for both combined, each vs. non-agreeing verbs (Fedden 2022: 300): The difference between the values is not significant at the .05 level (gender vs. non-agreeing: $\chi^2 = 2.09$, $p = .15$; classifiers vs. non-agreeing: $\chi^2 = 0.56$, $p = .45$; gender and classifiers combined vs. non-agreeing: $\chi^2 = 1.42$, $p = .23$).

This is a surprising result, given the common belief that agreement functions to support reference-tracking. It is therefore worth looking for comparable evidence. There is relevant data for Ingush (inh), a Nakh language, with around 300,000 speakers living primarily in Ingushetia in the North Caucasus. We find ‘an arbitrary and strictly lexical bifurcation of verbs into those that do and those that do not have gender agreement’ (Nichols 2018: 845). Again we can ask: if agreement serves to facilitate reference tracking, shall we find any sort of compensation to make up for the lack of agreement?

When the verb agrees, we find an overt argument expressed in 82% of the cases. If agreement aids reference tracking, we might expect to find more overt

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Table 7: Overt arguments and agreement in Ingush discourse (Nichols 2018: 858)

		S	A	O/T	TOTAL
Verb root agrees:	n	100	41	37	178
	% overt argument	88	59	92	82
Verb root does not agree:	n	41	38	24	103
	% overt argument	85	76	83	82

arguments when there is no agreement. Table 7 shows that this is not the case. Moreover, Forker (2018), looking at the distantly related Lak, finds similar results.

In Mian we find verbs lacking object agreement. In Ingush, with an absolutive agreement system, verbs agree or fail to agree with the subject if intransitive and with the object if transitive. Both languages allow us to evaluate the claim that agreement facilitates reference tracking, and the evidence (the lack of a repair strategy) suggests that it does not. Intuitively, the object agreement system places considerable demands on speakers, for no apparent benefit to hearers. Given this, we might anticipate rapid change in such systems, and yet, as Seržant (2021) shows, person-number agreement systems on verbs are highly stable.

5 The Mian system in its typological context

The Mian system is indeed striking. However, when we compare it with other systems, we find that some of its characteristics can be found elsewhere, if less obviously. Here we give three comparisons, to situate Mian typologically. As a baseline for calibration, we may characterize a canonical morphosyntactic system as one in which: (i) all features and their values are always distinguished; (ii) on all relevant lexical items; (iii) in all syntactic contexts. This idealization allows us to measure the various non-canonical phenomena which we encounter in real languages (see Corbett 2012: 152–199 for fuller specification). We take the three criteria in turn.

Mian shows strikingly non-canonical behaviour against the criterion of always distinguishing all features and their values (i). Rather it has a ‘partial’ system; many verbs lack person, number and gender, and *all* their values, for object agreement. As a distant comparison, consider pluralia tantum nouns, which lack a singular, even when it might be functionally valuable (see Corbett 2019 for a

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typology). Hence *these scissors are blunt* can be used of one pair or more than one pair. Similarly in Russian *sani* ‘sledge’ has only plural forms, whether used of one sledge or more than one, but it maintains case distinctions. This is similar to Mian in that the plural value is missing, not random cells of the paradigm. It differs in that one feature *value* is missing, while in Mian three complete features are missing.² Going further, instances of minor non-canonical behaviour are common, as when certain values are not distinguished owing to syncretism.

Turning now to the canonical baseline according to which all relevant items are involved (ii), Mian is a striking example of a ‘restricted system’. In Mian a minority of verbs show object agreement. We are used to more familiar instances, as in English where just the modal verbs fail to show agreement in the present tense. Or Russian where a sizable minority of nouns fail to inflect (many are borrowings, e.g. *pal'tó* ‘coat’, from French). But there are also languages more similar to Mian. In the Dagestanian language Tsez, most verbs fail to agree (details follow in Section 6.2). And in Archi, also Dagestanian, we find that just 32% of verbs agree (399/1248), and 3.4% of adverbs (13/383), see [Chumakina & Corbett \(2015\)](#).³

Consider now the point that canonical morphosyntax is consistent across syntactic contexts (iii). According to this baseline, distinctions that can be drawn must be drawn. A helpful type of comparison is reduced agreement. German attributive adjectives have more than one set of inflectional forms, depending on the syntactic context: we find the ‘strong’ form in (4), which is a dedicated neuter form, *neues*, whereas the ‘weak’ form in (5), *neue*, required after the definite article, fails to distinguish neuter gender:

- (4) German (personal knowledge)
 ein **neu-es** Buch
 INDF.NOM.N.SG new-NOM.N.SG book(N)[NOM.SG]
 ‘a new book’
- (5) das **neu-e** Buch
 DEF.NOM.N.SG new-NOM.SG book(N)[NOM.SG]
 ‘the new book’

²English / Russian type of pluralia tantum nouns are defective, leading to a syntactic gap (unlike non-agreeing Mian verbs). But this is not the only type of pluralia tantum: Tsez has animate pluralia tantum nouns, which take singular and plural agreement and so are not defective ([Comrie 2001](#), [Corbett 2019](#): 55–58).

³In Archi we have no evidence for any repair strategy when a verb does not agree, but we have no hard data comparable to that for Lak and Ingush.

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Tables 8 and 9 (Corbett 2006: 95-96) illustrate the ‘strong’ and the ‘weak’ paradigm, respectively (shading indicates syncretism). There is a third, a ‘mixed’ paradigm, which we leave aside for sake of clarity.

Table 8: ‘Strong’ forms of *gut* ‘good’ (full agreement)

	SINGULAR			PLURAL
	MASCULINE	NEUTER	FEMININE	
NOMINATIVE	gut-er	gut-es	gut-e	gut-e
ACCUSATIVE	gut-en	gut-es	gut-e	gut-e
GENITIVE	gut-en	gut-en	gut-er	gut-er
DATIVE	gut-em	gut-em	gut-er	gut-er

Table 9: ‘Weak’ forms of *gut* ‘good’ (reduced agreement)

	SINGULAR			PLURAL
	MASCULINE	NEUTER	FEMININE	
NOMINATIVE	gut-e	gut-e	gut-e	gut-en
ACCUSATIVE	gut-en	gut-e	gut-e	gut-en
GENITIVE	gut-en	gut-en	gut-en	gut-en
DATIVE	gut-en	gut-en	gut-en	gut-en

In the weak set, the forms have fewer distinctions than in the strong set, two instead of five. Reduced agreement is rather different from what we find in Mian, in two significant ways. The key one is that the Mian system is consistent across syntactic contexts: there is no way of manipulating the syntax to make a non-agreeing verb agree, or an agreeing verb fail to agree. In contrast, German reduced agreement is sensitive to the syntax: it is triggered only in attributive position. The other difference relates back to our first criterion (showing that non-canonicity can be measured according to more than one criterion). Reduced agreement in German involves a partial system, where forms exist and are indeed used for full agreement, but are not found in a given syntactic environment. In contrast, in Mian the forms simply do not exist for non-agreeing verbs.

We see that the Mian system is indeed striking, but that it involves a combination of non-canonical behaviours, which can be found individually elsewhere.

6 The logic of learnability

6.1 Learning the morphosyntactic system

A key point is that the learner needs evidence for the system as a whole, and this varies cross-linguistically in quality and—importantly—in quantity. Thus work on Bantu languages shows that their apparently complex gender systems are learned surprisingly early. The reason is that there is extensive evidence, in copious agreement:

“Thus, although Bantu languages have many more nominal gender classes than most other languages, the phonological transparency, regularity and pervasiveness of the system appears to facilitate the early acquisition of nominal agreement in these languages.” (Demuth & Weschler 2012: 86)

In comparison, Mian offers the learner pervasive evidence for person, number and gender in subject agreement: every finite verb agrees obligatorily in these features. This information carries over, to some degree, to object agreement (more obviously for agreement in those features than for object classifiers). But the learner has to establish that object agreement is not general.⁴

6.2 Learning a restricted system

The important challenge of Mian is grasping a system in which most items do not participate. A helpful study here is Gagliardi & Lidz (2014), who investigated the Dagestanian language Tsez. Here only 27% of verbs and 4% of adjectives in the dictionary agree, which is the key point of comparison. They analysed a corpus of child-directed speech (Table 10).

In Table 10 we see a striking difference between the infrequency of agreeing verbs and adjectives in the dictionary, and their relative frequency in the corpus (compare Bybee 1985: 119–121). The learner has good evidence for the agreement system from the input. The puzzle of how children learn a system—in which most of the possible participating items do not in fact participate—appears to go away.

For comparison, we offer figures for agreeing verbs in Mian in Table 11. The Mian corpus is the same that was used in Fedden (2022), a fully annotated corpus of ~4,000 words, consisting of 14 mainly third-person narrative texts. It is not a corpus of child-directed speech, but this is not necessarily a problem, since

⁴The task is harder for certain Alor-Pantar languages (such as Teiwa), where object agreement poses similar challenges to Mian, but subject agreement is generally absent (Fedden et al. 2013, Fedden et al. 2014).

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Table 10: Proportions of verbs and adjectives that show overt singular agreement in Tsez (Gagliardi & Lidz 2014: 69)

	Agreeing verbs	Agreeing adjectives
Dictionary types	27%	4%
Corpus types	60%	35%
Corpus tokens	84%	77%

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Dictionary types are calculated based on dictionary entries, corpus types refer to the frequency of agreeing (verb and adjective) lexemes in Gagliardi’s (2012) corpus, and corpus tokens refer to the frequency of all agreeing (verb and adjective) word forms in this corpus.

Kodner (2019) shows that the type frequency of grammatical phenomena is not significantly different when comparing child-directed corpora and adult corpora, whose lexicons have been trimmed to include only the most frequent items. However, we do not have a large corpus to prune, our corpus is small, and—given those caveats—this is what we can glean from it.

Table 11: Proportions of verbs that show object agreement in Mian (absolute figures in brackets); rounded to full percentages (hence percentages do not sum)

Verbs agreeing with their object						
	Agreeing in PNG		Classifier		Total	
Dictionary types	2%	(4 of 245)	16%	(39 of 245)	18%	(43 of 245)
Corpus types	3%	(3 of 118)	23%	(27 of 118)	25%	(30 of 118)
Corpus tokens	10%	(70 of 690)	32%	(220 of 690)	42%	(290 of 690)

We observe from Table 11 that in Mian, as in Tsez, agreeing verbs are more frequent in the corpus than in the dictionary. It is precisely verbs that are more frequently used in discourse which agree in Mian. For example, the verbs ‘see’ and ‘hit’ agree with their object in person, number and gender (PNG), and the verbs ‘give’, ‘take’, ‘get’, and ‘put’ take object classifiers. There is indeed more evidence for the system provided by tokens than by types. However, Mian shows an important difference from Tsez: even in the corpus, Mian agreeing transitive verbs still constitute a minority. The persistence of object agreement in Mian remains something of a puzzle.

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There is a systematic difference between Tsez (and Ingush) and Mian, which may be significant. In Tsez and Ingush verbs agree, or they do not; for those which do agree, agreement is with the absolutive argument. In Mian, finite verbs without exception show subject agreement. Thus the learner receives pervasive evidence for verb agreement. The question is the relevance of that information for learning object agreement. Some verbs show object agreement using the same features (person, number and gender) and their values, as is found in subject agreement. But these are a small minority (2% of dictionary types). More verbs show object agreement through verbal classifiers: while these do not match the subject agreeing system, they overlap to a considerable extent (recall that the classifiers work according to an absolutive system, but that verbs which take classifiers are transitive, apart from *mêin* ‘fall’, Section 2). The verbal classifiers distinguish person (an unusual trait of Mian; see Corbett et al. 2017: 240–242). They also distinguish number, though not exactly as in subject agreement (Corbett et al. 2017: 242–249). They even overlap with the gender system to some extent (Table 3). And the similarities between person, number and gender object agreement and classifiers are strengthened by the fact that they share the same slot in the paradigm: both types of object agreement occur prefixally (see examples (1) and (2)), and they cannot co-occur. All this means that the learner of Mian has evidence relevant to the system of object agreement, in addition to that provided by the minority of verbs which actually realize it.

6.3 Identifying the items which participate in the system

From learning the system, we now turn to the issue of learning exactly which verbs show object agreement. We discussed earlier the generalizations which hold over the agreeing verbs (Section 3), which may well aid the learner. Now we tackle the more basic question of how the learner can recognize the agreeing verbs. Here it is helpful to consider briefly Burmeso, a language of the Mamberamo River area of Western New Guinea. We use Burmeso because it elucidates the issue more clearly than Mian, but we will show below that the situation in Mian is similar in important respects.

Burmeso has many interesting characteristics (Donohue 2001). It has, for instance, concurrent systems of nominal classification (Corbett 2012: 176–180, Corbett et al. 2017: 33–35). For us, the important part is the classification which involves six gender values.⁶ Evidence comes from an absolutive agreement system on the verb, here illustrated by object agreement:

⁶Donohue calls these ‘noun classes’; adjectives mark the concurrent system, which Donohue terms ‘gender’.

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- (6) Burmeso (Donohue 2001: 99)
da timar g-ih-i-maru
1SG bat(II).SG II.SG-see-TODAY'S.PAST
'I saw a bat.'
- (7) da timnarid s-ih-i-maru
1SG bat(II).PL II.PL-see-TODAY'S.PAST
'I saw some bats.'
- (8) da kwehia g-ih-i-maru
1SG frog(III).SG III.SG -see-TODAY'S.PAST
'I saw a frog.'
- (9) da kwehorudo j-ih-i-maru
1SG frog(III).PL III.PL-see-TODAY'S.PAST
'I saw some frogs.'

These examples are sufficient to distinguish gender II from III; Donohue demonstrates the existence of six gender values.⁷ In Table 12, the assignment criteria for nouns are given, together with the agreements; these show interesting patterns of syncretism, indicated by the shading.

Table 12: Verbal inflection classes in Burmeso (Donohue 2001: 100, 102)

gender		inflection class			
		1: -ih-i- 'see'		2: -akwa- 'bite'	
value	assignment	SG	PL	SG	PL
I	male	<i>j-</i>	<i>s-</i>	<i>b-</i>	<i>t-</i>
II	female, animate	<i>g-</i>	<i>s-</i>	<i>n-</i>	<i>t-</i>
III	miscellaneous	<i>g-</i>	<i>j-</i>	<i>n-</i>	<i>b-</i>
IV	mass nouns	<i>j-</i>	<i>j-</i>	<i>b-</i>	<i>b-</i>
V	banana, sago tree	<i>j-</i>	<i>g-</i>	<i>b-</i>	<i>n-</i>
VI	arrows, coconuts	<i>g-</i>	<i>g-</i>	<i>n-</i>	<i>n-</i>

⁷At least gender v is 'inquorate' in that it has only two members (Donohue 2001: 102) and its agreements are a combination of agreement from other genders; it would be reasonable to treat these two nouns as exceptions.

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The forms of *-ihi-* ‘see’, in examples (6)–(9), distinguish gender and number. The key point is that besides verbs like *-ihi-* ‘see’, there is a second inflection class, illustrated by verbs like *-akwa-* ‘bite’. These have the same paradigm shape, but the phonological realizations are different (the syncretisms are identical across the classes). Burmeso verbs thus fall into two inflection classes, and these are remarkably close to being canonical (Corbett 2009). In terms of membership in the classes, Donohue (2001: 101) states explicitly: ‘... there are no obvious semantic correlations for verbs which take the different sets of prefixes, and both sets of verbs are of approximately equal size.’ Moreover, there is no obvious phonological criterion for determining which verbs belong in which inflection class (Mark Donohue, personal communication, 2 May 2013). How then is this system learnable? There are no semantic or formal criteria, and no default class.

Here we should look at the nature of the evidence. The learner’s task is to acquire an arbitrary division of the verb inventory: there are two disjoint sets of (mutually predicting) inflections for realizing the same functions. But the evidence is crystalline. Given a single form of a verb, even without knowing its morphosyntactic specification, the learner can establish unambiguously its inflection class and hence its remaining agreement forms. Any verb prefixed *j-*, *g-* or *s-*, in whatever context, must be a member of inflection class 1.

It is worth contrasting the Burmeso situation, where there is an apparently arbitrary choice but with ample evidence available, with more familiar frequency effects. It is well known that a part of the paradigm may be less frequently used (say the past subjunctive, or the dual number), which may give learning problems across the board. The issue may also be restricted, as when particular items display ‘dominance’ (Corbett 2019: 67–70 and references there). For example, the English noun *dice* is plural dominant, which has largely led to the replacement of the singular *die* (preserved in the expression *the die is cast*), by *dice*. Whether across the lexical board, or restricted to particular items, these familiar effects target the less frequent extremes of the system. Burmeso is quite different. The learner’s task is to establish, of every item, whether it is of Type A or Type B, and the choice is largely arbitrary. However, the evidence is fully available, in the most frequent core of the system just as much as in the less frequent nooks of the system. One form gives all the information required.

How does Mian compare? Surprisingly closely. To establish whether or not a verb takes object agreement or an object classifier, one form is usually enough. There are just a few instances where one form is not sufficient. A small number of Mian transitive verbs have a stem beginning in /na/, /wa/, etc., (e.g. *na* ‘do’, *walâa* ‘open’), that is in segments which are homophonous to the prefixes, but are actually part of the stem (similarly with /dob/, which is the form of one of

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the classifiers, but belongs to the stem in *dobô* ‘feel, taste, affect’). In these cases a second form of the verb might be needed, but even with such ‘difficult’ verbs this rarely happens. If the object is not of the person, number and gender combination that we would assume when analysing the segmental beginning of such a verb as a prefix, no second form is needed. For example, since the object prefix for 1SG is *na-*, the verb *na* ‘do’ would not pose a problem if its object was anything but 1SG. So in almost all instances one form is sufficient to deduce the entire pattern, whether of agreement (of either type) or the failure to agree.

To sum up, the learner of Mian encounters massive evidence for subject agreement, though it is not clear how much of this carries over to object agreement (Section 6.1). The agreeing verbs are more frequent in texts than in the dictionary, but they still constitute a minority (Section 6.2). Almost any form of any transitive verb identifies it as agreeing or non-agreeing, but there are a few exceptions (Section 6.3). Taken together, these three types of evidence show that it is not surprising that the system can be learned.

7 Conclusion

Our initial reaction to the Mian system is well founded. If object agreement is “worth it”, then it is surprising that it is found only on a minority of transitive verbs. Lexemes in the same part of speech (determined on syntactic criteria) have dramatically different behaviour. However Mian, and Ingush, show that a naïve sense of “worth it” is misplaced here, since the evidence does not support a clear functional role (reference tracking) for agreement.

This puzzle (the substantial morphological demands of the system contrasted with its lack of an obvious function) sharpens the issues of how learners acquire the restricted system of Mian. We found three types of evidence available to them, namely evidence from subject agreement, corpus frequency and straightforward identifiability. The task differs from the common issue of irregularity of form: in some respects the evidence for a Mian verb which fails to agree is more accessible than, say, the form of an English past tense (where instances such as *think*, *thinks*, *thinking* do not point unambiguously to an irregular past tense, even less to its form). Moreover, while Mian presents the issue of restricted systems starkly, forcing us to confront the logic of learnability, such systems—once recognized—are to be found in lesser form even in familiar languages.

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References

- Beavers, John. 2011. On affectedness. *Natural Language and Linguistic Theory* 29(2). 335–370.
- Brown, Dunstan & Andrew Hippisley. 2012. *Network morphology: A defaults-based theory of word structure*. Cambridge: Cambridge University Press.
- Bybee, Joan L. 1985. *Morphology: A study of the relation between meaning and form*. Amsterdam: John Benjamins.
- Bye, Patrik. 2015. The nature of allomorphy and exceptionality: Evidence from Burushaski plurals. In Eulàlia Bonet, Maria-Rosa Lloret & Joan Mascaró (eds.), *Understanding allomorphy: Perspectives from optimality theory*, 107–176. London: Equinox.
- Chumakina, Marina & Greville G. Corbett. 2015. Gender-number marking in Archi: Small is complex. In Matthew Baerman, Dunstan Brown & Greville G. Corbett (eds.), *Understanding and measuring morphological complexity*, 93–116. Oxford: Oxford University Press.
- Comrie, Bernard. 2001. How independent is Tsezic morphology? In Mary Andronis, Christopher Ball, Heidi Elston & Sylvain Neuvel (eds.), *CLS 37: The panels: 2001: Proceedings from the parasessions of the 37th Meeting of the Chicago Linguistic Society Vol. 37-2*, 367–383. Chicago: CLS.
- Corbett, Greville G. 2006. *Agreement*. Cambridge: Cambridge University Press.
- Corbett, Greville G. 2009. Canonical inflectional classes. In Fabio Montermini, Gilles Boyé & Jesse Tseng (eds.), *Selected proceedings of the 6th décembrettes: Morphology in Bordeaux*, 1–11. Somerville: Cascadilla Proceedings Project. <http://www.lingref.com/cpp/decemb/6/abstract2231.html>.
- Corbett, Greville G. 2012. *Features*. Cambridge: Cambridge University Press.
- Corbett, Greville G. 2019. Pluralia tantum nouns and the theory of features: A typology of nouns with non-canonical number properties. *Morphology* 29(1). 51–108. DOI: [10.1007/s11525-018-9336-0](https://doi.org/10.1007/s11525-018-9336-0).
- Corbett, Greville G., Sebastian Fedden & Raphael A. Finkel. 2017. Single versus concurrent feature systems: Nominal classification in Mian. *Linguistic Typology* 21(2). 209–260.
- Corbett, Greville G. & Norman M. Fraser. 1993. Network morphology: A DATR account of Russian nominal inflection. *Journal of Linguistics* 29(1). 113–142.

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- Demuth, Katherine & Sara Weschler. 2012. The acquisition of Sesotho nominal agreement. *Morphology* 22(1). 67–88. DOI: [10.1007/s11525-011-9192-7](https://doi.org/10.1007/s11525-011-9192-7).
- Donohue, Mark. 2001. Animacy, class and gender in Burmeso. In Andrew Pawley, Malcolm Ross & Darrell Tryon (eds.), *The boy from Bundaberg: Studies in Melanesian linguistics in honour of Tom Dutton* (Pacific Linguistics 514), 97–115. Canberra: Pacific Linguistics.
- Fedden, Sebastian. 2011. *A grammar of Mian* (Mouton Grammar Library 55). Berlin: De Gruyter Mouton.
- Fedden, Sebastian. 2022. Agreement and argument realization in Mian discourse. *Word Structure (Special Issue: The Many Facets of Agreement, Tania Paciaroni, Alice Idone & Michele Loporcaro eds.)* 15(3). 283–304. DOI: [10.3366/world.2022.0211](https://doi.org/10.3366/world.2022.0211).
- Fedden, Sebastian, Dunstan Brown, Greville G. Corbett, Marian Klammer, Gary Holton, Laura C. Robinson & Antoinette Schapper. 2013. Conditions on pronominal marking in the Alor-Pantar languages. *Linguistics* 51(1). 33–74.
- Fedden, Sebastian, Dunstan Brown, František Kratochvíl, Laura C. Robinson & Antoinette Schapper. 2014. Variation in pronominal indexing: Lexical stipulation vs. referential properties in the Alor-Pantar languages. *Studies in Language* 38(1). 44–79.
- Forker, Diana. 2018. Gender agreement is different. *Linguistics* 56(4). 865–894.
- Gagliardi, Annie & Jeffrey Lidz. 2014. Statistical insensitivity in the acquisition of Tsez noun classes. *Language* 90(1). 58–89.
- Gagliardi, Annie C. 2012. *Input and intake in language acquisition*. University of Maryland. (Doctoral dissertation).
- Heath, Jeffrey. 1983. Referential tracking in Nunggubuyu (Australia). In John Haiman & Pamela Munro (eds.), *Switch reference and universal grammar*, 129–149. Amsterdam: John Benjamins.
- Hopper, Paul & Sandra Thompson. 1980. Transitivity in grammar and discourse. *Language* 56(2). 251–299.
- Kodner, Jordan. 2019. Estimating child linguistic experience from historical corpora. *Glossa: A Journal of General Linguistics* 4(1). 122: 1–14. DOI: [10.5334/gjgl.92](https://doi.org/10.5334/gjgl.92).
- Marzi, Claudia & Vito Pirrelli. 2023. A discriminative information-theoretical analysis of the regularity gradient in inflectional morphology. *Morphology* 33(3). 1–51. DOI: [10.1007/s11525-023-09415-6](https://doi.org/10.1007/s11525-023-09415-6).
- Nichols, Johanna. 2018. Agreement with overt and null arguments in Ingush. *Linguistics* 56(4). 845–863.

4 *Mian verbs trouble linguists not learners*

- Palancar, Enrique L. & Sebastian Fedden. 2026. Introduction. In Enrique L. Palancar & Sebastian Fedden (eds.), *Voids in morphology: Exploring “uninflectedness”*, iii–xi. Berlin: Language Science Press.
- Pawley, Andrew K. & Harald Hammarström. 2018. The trans New Guinea family. In Bill Palmer (ed.), *The languages and linguistics of the New Guinea area: A comprehensive guide*, 21–195. Berlin: Mouton de Gruyter.
- Ross, Malcolm D. 2005. Pronouns as a preliminary diagnostic for grouping Papuan languages. In Andrew K. Pawley, Robert Attenborough, Jack Golson & Robin Hide (eds.), *Papuan pasts: Cultural, linguistic and biological histories of Papuan-speaking peoples (PL-572)*, 15–65. Canberra: Pacific Linguistics.
- Seržant, Ilja A. 2021. Cyclic changes in verbal person-number indexes are unlikely. *Folia Linguistica Historica* 42(1). 49–86.
- Stump, Gregory. 2023. *Morphotactics: A rule-combining approach*. Cambridge: Cambridge University Press.
- Stump, Gregory T. 2001. *Inflectional morphology: A theory of paradigm structure*. Cambridge: Cambridge University Press.
- Suter, Edgar. 2012. Verbs with pronominal object prefixes in Finisterre-Huon languages. In Harald Hammarström & Wilco van den Heuvel (eds.), *History, contact and classification of Papuan languages (language and linguistics in Melanesia special issue 2012)*, 23–58. Port Moresby: Linguistic Society of Papua New Guinea. <https://www.langlxmlanesia.com/Special%203%20suter23-58.pdf>.
- Tsunoda, Tasaku. 1985. Remarks on transitivity. *Journal of Linguistics* 21(2). 385–396.

Chapter 5

A corpus study of verb placement in L3 French and L3 German: Insights from learner language

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This exploratory study investigates verb placement in L3 French and German, respectively, in the written production of L1 Norwegian L2 English speakers in upper secondary schools. Verb placement in L3 in similar populations has been widely investigated using various experimental methods, generally concluding that L2 English plays a strong role in the cross-linguistic influence (CLI) observed in the L3. However, there is no consensus about whether this role is due to its L2 status or other reasons, nor about the impact that different methodologies may have on the findings. The present study seeks to complement existing knowledge focusing on the understudied area of written production. Overall findings suggest that the relevant structures are part of the learners' linguistic repertoire, and that patterns of CLI are similar to those found using other methodologies, indicating that these findings accurately reflect the underlying linguistic system. However, there is indication that CLI from either prior language is not the only explanation for learners' behavior in written production. Our data supports an analysis where general mechanisms such as principles of economy play a role in the acquisition of verb movement in L3.

1 Introduction

The present study is an exploratory study of verb placement in third language (L3) German and L3 French learner texts written by upper secondary first language (L1) Norwegian second language (L2) English students, available in the

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TRAWL corpus (Dirdal et al. 2022). We look at verb placement in subject-initial main clauses with a sentence-medial adverb (SU-I) and non-subject-initial main clauses (Non-SU-I) and investigate how cross-linguistic influence (CLI)¹ from previously learned languages (L1 and L2) affects verb placement in L3 written production. While verb placement has been extensively studied in experimental designs, this is, to our knowledge, the first comprehensive corpus study of verb placement in (free) written production in these particular language combinations. A written corpus study allows for investigating whether the target structures are indeed produced and how verb placement is realized under conditions that allow for access to explicit knowledge and metalinguistic monitoring. As such, the present study aims to contribute to the diversity in the L3 field both in terms of methodology and types of data investigated.

Section 2 provides an overview of verb placement in English, Norwegian, French and German, relevant previous research and our research questions. Data material and procedure are described in Section 3. Results are provided in Section 4 and discussed in Section 5.

2 Background

2.1 Verb placement in the four languages of the study

Placement of finite verbs varies between the languages in this study. The differences are often described in terms of verb movement: We assume that lexical verbs are generated in the Verb Phrase (VP), where they remain in English (no movement), while they subsequently move to the Inflection Phrase (IP) in French (short movement) and to the Complementizer Phrase (CP) in German and Norwegian (long movement), the latter resulting in so-called V2 word order. The resulting surface word order for the two sentence types investigated in the current study, namely non-subject initial clauses (Non-SU-I) and subject-initial clauses with certain adverbs such as *always* or *often* (SU-I), is illustrated in (1) and (2) (examples from Busterud et al. 2023).

- (1) a. [_{CP} Every morning [_{IP} Marianne [_{VP} takes the bus]]]. (X-SU-V)
 b. [_{CP} Tous les matins, [_{IP} Marianne prend [_{VP} prend le bus]]].
 Every DEF.PL mornings Marianne takes the bus
 (X-SU-V)

¹We use the terms cross-linguistic influence and transfer interchangeably to refer to systematic influence from one language to another language.

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- c. [_{CP} Jeden Morgen *nimmt* [_{IP} Marianne ~~nimmt~~ [_{VP} den Bus
Every morning takes Marianne the bus
~~nimmt~~]]]. (X-V-SU)
- d. [_{CP} Hver morgen *tar* [_{IP} Marianne ~~tar~~ [_{VP} ~~tar~~ bussen]]].
Every morning takes Marianne bus.DEF
(X-V-SU)
- (2) a. [_{CP} Peter [_{IP} [_{VP} often *eats* sushi]]]. (SU-Adv-V)
b. [_{CP} Peter [_{IP} *mange* [_{VP} souvent ~~mange~~ des sushis]]].
Peter eats often INDF.PL sushi
(SU-V-Adv)
c. [_{CP} Peter *isst* [_{IP} ~~isst~~ [_{VP} oft Sushi ~~isst~~]]]. (SU-V-Adv)
Peter eats often sushi
d. [_{CP} Peter *spiser* [_{IP} ~~spiser~~ [_{VP} ofte ~~spiser~~ sushi.]]]. (SU-V-Adv)
Peter eats often sushi

In Norwegian, German and French, auxiliaries and copular verbs appear in the same site as lexical verbs. In English they appear in the IP.² Consequently, in English, surface word order for Non-SU-I (1) is the same whether the verb is lexical or auxiliary/copular, but in the latter case, the underlying structure is the same as for French. For SU-I (2), surface word order changes, and matches that of the other three languages if the verb is auxiliary/copular in English; here too, the underlying structure is the same as that of French. This is illustrated in (3).

- (3) a. [_{CP} Tomorrow [_{IP} Marianne *will* [_{VP} stay at home]]]. (X-SU-aux)
b. [_{CP} Peter [_{IP} *is* [_{VP} often ~~is~~ sick]]]. (SU-cop-Adv)

2.2 Previous studies of L3 acquisition of verb placement

For L3 learners with L1 Norwegian, one might expect transfer of V2 word order to L3 German - which would be target-like - and to L3 French - which would result in target-like surface word order for SU-I (2) but not for Non-SU-I (1). However, previous research with speakers of an L1 with V2 learning an L3 with V2 have consistently found that these specific expectations for facilitative V2-to-V2 transfer do not necessarily play out (Bardel & Falk 2007; Dahl et al. 2022;

²A discussion of where in the IP domain auxiliaries are generated, and whether they are involved in movement operations within that domain, is beyond the scope of this paper.

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Håkansson et al. 2002). Similarly, previous studies show that L1 speakers of a V2 language do not necessarily transfer their L1 word order to a non-V2 L3 (Listhaug et al. 2021; Stadt 2019). A common explanation is that the L3 learners investigated also have English as L2 and may thus transfer the English verb in-situ word order from L2 to L3.

Such findings have been explained in different ways. According to *the L2 status factor* (Bardel & Falk 2007), transfer of non-V2 from L2 English is due to cognitive similarities between L2 and L3, giving the L2 a privileged role for transfer in L3 acquisition. Busterud et al. (2023), in contrast, suggest that principles of economy affect CLI in L3 acquisition. They argue that verb movement is a costly operation, and that CLI of non-V2 from English can be explained by its verb *in-situ* properties. Specifically, they propose that the lower accuracy found in their study on Non-SU-I in L3 German, where V2 is required, compared to Non-SU-I in French, where the target is non-V2, reflects a preference for learners not to move the verb to C. Similarly, they argue that performance on SU-I in both languages, which is more accurate than on Non-SU-I in German but less accurate than on Non-SU-I in French, is a result of short movement to I being preferred over long movement to C, but unpreferred compared to leaving the verb in situ. Importantly, Busterud et al. (2023) propose that the presence of a prior language with no verb movement (or with less verb movement compared to the target L3) is a prerequisite for this non-movement preference to interact with CLI.

Studies of L3 verb placement have used different experimental methods, including more or less controlled oral production (Bardel & Falk 2007; Bohnacker 2006; Hancock & Sanell 2009; Håkansson et al. 2002), gap filling (Stadt 2019), and acceptability judgments (Busterud et al. 2023; Dahl et al. 2022; Listhaug et al. 2021; Stadt 2019). However, to our knowledge, there is scarce data on written learner production, with the exception of Zakariassen (2021), who investigated Norwegian university students of L3 German and found them to be largely target-like on the structures in question here. Written production differs from oral data in that it may allow monitoring and accessing explicit knowledge but is also less vulnerable to pure performance slips and is therefore valuable in order to gain a complete picture of the underlying language system.

We ask the following research questions:

1. Do the participants produce the relevant syntactic structures, i.e., Non-SU-I and SU-I with a sentence-medial adverb?
2. To what extent do they produce target and non-target word order in each of the sentence types?

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3. Do we see production patterns similar to those observed in previous AJT studies with similar participants (Busterud et al. 2023; Dahl et al. 2022; Listhaug et al. 2021)?
4. What can these production data tell us about verb placement in the underlying syntactic structure for the two relevant sentence types?

We expect target and non-target production of both sentence types. Moreover, we expect production data to mirror the asymmetry for the two languages found in judgment data in Busterud et al. (2023).

3 The present study

3.1 Participants and data material

Data investigated comprises texts written by students in the first and second year of upper secondary education (Years 11 and 12 of school, age 16–18) collected in the TRAWL corpus (Dirdal et al. 2022).³ Students all had Norwegian as L1 and English as L2. They had been learning English formally in school since Year 1 (age 6), and can be assumed, like most Norwegians, to have been extensively exposed to English also outside the classroom (Council 2017, 2021). They had learned their L3, German or French, since Year 8 (age 13); at the end of their current school year, this constitutes 335 (Year 11) and 447 (Year 12) hours of L3 instruction (Kunnskapsdepartementet 2019). This participant group is very similar to that of Busterud et al. (2023), Dahl et al. (2022) and Listhaug et al. (2021) in terms of age and L2/L3 learning experience, as well as the years in which data were collected. We do not have information about their proficiency in L2 English. However, participants in the abovementioned experimental studies all had relatively high L2 English proficiency and were overall target-like with both SU-I and Non-SU-I sentences in English. There is thus reason to believe that the L2 English structures are also in place for the current learner group.

The French sub-corpus consists of 90 texts written by 31 students with a total of 37,609 words (tokens). 7 students reporting other L1s than Norwegian and/or an extended stay in a French-speaking country were excluded. After exclusion, our data material contained 8 texts (2,834 tokens) from 4 Year 11 students and 62 texts (27,509 tokens) from 24 Year 12 students (30,343 tokens in total). All students who participated in Year 11 also participated in Year 12.

³<https://tekstlab.uio.no/trawl/>

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The German sub-corpus consists of 130 texts (31,468 tokens) by 13 students. One student was excluded for reporting an extended stay in a German-speaking country. After exclusion, the data material comprised 57 texts (10,390 tokens) by 11 Year 11 students and 66 texts (17,882 tokens) by 12 Year 12 students (28,272 tokens in total); all students except one participated in both years.

Both sub-corpora consist of texts produced both in test and homework conditions. While this may impact the data, it is not clear exactly how and in which direction. A homework situation may have lower stakes but no time limitations or limitations on aids allowed. Test situations typically have higher stakes and constitute a more controlled context. However, students typically have access to online and offline aids during tests. We saw no indication of relevant differences in texts produced in the two conditions.

3.2 Procedure

As the TRAWL corpus is not tagged for syntactic function, Non-SU-I sentences were identified manually by reading each text. To identify SU-I sentences, we first searched for three selected adverbs which typically occur in the relevant position: *oft/souvent* (often), *immer/toujours* (always), and *auch/aussi* (also), and subsequently searched for the word class *adverb* to check for any misspelled occurrences of each adverb. The latter search also demonstrated that, with the exception of negation, the number of other adverbs in the relevant position was very low. Thus, the sentences included cover the vast majority of relevant cases for SU-I. Although sentences with negation were frequent, these are not analyzed here, as the specifics of French negation would complicate the comparison across L3s.

Sentences had to meet certain criteria to be included for analysis. Non-SU-I sentences had to include an element clearly identifiable as a subject and one clearly identifiable as a verb. SU-I sentences had to include an element clearly identifiable as a verb. The adverb had to be placed clause-medially and clearly meant to modify the clause rather than, for example, a following noun phrase. Subject-less coordinated SU-I clauses were included if there was a clearly identifiable verb. Any sentence in student texts that had been given in the assignment or could otherwise be identified as a quote, for example from the Internet, was excluded.

4 Results

For L3 French, we identified a total of 351 relevant sentences, of which 314 were Non-SU-I, and 37 were SU-I. A total of 46.4% of the sentences contained a lexical verb. For L3 German, the total number of relevant sentences was 632, 51.3% with lexical verbs, distributed over 419 Non-SU-I and 113 SU-I sentences. Overall, the verb appears in non-target position in 4.3% and 19.6% of the sentences in L3 French and L3 German, respectively. Distribution of sentence types, verb types, and verb placement is shown in Table 1.

Table 1: Number of verb types and verb placement per sentence type per L3 group. * indicates non-target verb placement.

French	Non-SU-I				SU-I		
	Verb type	*Verb-2	Verb-3	Total	Verb-2	*Verb-3	Total
	Lexical	2	141	143	14	6	20
	Auxiliary	2	107	109	4	1	5
	Modal	0	16	16	1	0	1
	Copula	2	44	46	9	2	11
	Total	6	308	314	28	9	37
German	Non-SU-I				SU-I		
	Verb type	Verb-2	*Verb-3	Total	Verb-2	*Verb-3	Total
	Lexical	221	59	280	40	4	44
	Auxiliary	61	21	82	16	0	16
	Modal	80	21	101	34	0	34
	Copula	39	17	56	17	2	19
	Total	401	118	519	107	6	113

Table 2 shows the distribution of verb placement in each of the sentence types as well as the percentage of non-target word order produced by students in Years 11 and 12. Since the number of texts from Year 11 is very unequal for French and German (see Section 3.1), comparisons across the learner groups by year is not felicitous, nor is a discussion of development in verb placement. We do however note that in L3 German, where there is a relatively equal distribution of texts between years and the same students participated in both years, the percentage of non-target verb placement actually increases from Year 11 to Year 12. Overall percentage of non-target verb placement is visualized in Figure 1.

In L3 French, Non-SU-I sentences were produced by all participants. Only six out of these 314 sentences had non-target word order (V2); these were produced

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Table 2: Distribution of verb placement per sentence type and percentage of non-target verb placement per year in each L3 group

Non-SU-I					SU-I				
French	Year	*Verb-2	Verb-3	Total	* %	Verb-2	*Verb-3	Total	* %
	11	0	67	67	0	1	3	4	75.0
	12	6	241	247	2.4	27	6	33	18.2
	Total	6	308	314	1.9	28	9	37	24.3
German	Year	Verb-2	*Verb-3	Total	* %	Verb-2	*Verb-3	Total	* %
	11	218	56	274	20.4	40	2	42	4.8
	12	183	62	245	25.3	67	4	71	5.6
	Total	401	118	519	22.7	107	6	113	5.3

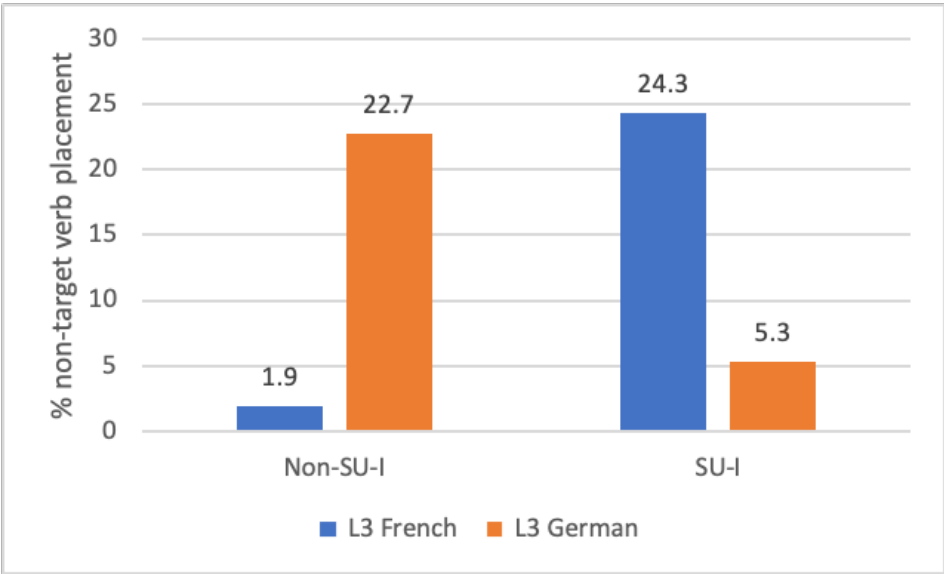


Figure 1: Percentage of non-target verb placement in Non-SU-I and SU-I sentences by L3 group

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by five different participants, all in Year 12. All five also produced Non-SU-I sentences with target verb placement. SU-I sentences were produced by 15 of the 24 participants, in both Years 11 and 12. There were nine cases (out of 37) of non-target (V3) word order, which were produced by seven participants; one produced three V3 sentences in Year 11, while the remaining six cases were produced by six different participants in Year 12. Four of the seven participants also produced target SU-I sentences, while three only produced the one non-target sentence. Overall, 11 of 24 participants produced non-target verb placement; only one of these produced non-target placement with both sentence types.

In L3 German, Non-SU-I sentences were produced by all participants. A total of 118 cases (out of 519) had non-target word order (V3), and these were produced by 11 out of the 12 participants. 56 were produced in Year 11, and 62 in Year 12. However, one participant was responsible for 45 (38.1%) of all non-target cases.⁴ This one participant also produced target word order, as did all the other. SU-I sentences were also produced by all L3 German learners. Only six out of these sentences (113) appeared with non-target word order, and they were all produced by two learners; one learner (the learner mentioned above) produced one non-target form in Year 11, while another produced one in Year 11 and four in Year 12. Both learners also produce target SU-I sentences. In sum, 11 of 12 participants produced non-target verb placement; however, all of them also produced target verb placement for each sentence type.

As a summary, in French, we see non-target verb placement in C (i.e., non-target Non-SU-I structures) in 1.9% of cases, while we see non-target verb placement in V (or, in one case, what appears to be placement of an auxiliary too low in the structure) (i.e., non-target SU-I structures) in 24.3% of cases. In German, we see lack of verb movement to C in 22.1% of cases (i.e., non-target Non-SU-I structures), and non-target verb placement in V (i.e., non-target SU-I structures) in 5.3% of cases.

5 Discussion

While we do not know the frequencies of Non-SU-I and SU-I in the learners' input, data from the present study show, as a response to our first research question, that the two structures are indeed produced by the learner group. First, the two structure types are produced by all learners in the study, with the exception of SU-I in French, which were produced by only half of the participants.

⁴Without this participant, the verb appears in non-target position in 16,5 % of Non-SU-I sentences (cf. Table 2).

This means that learners produce sentences where verb placement in L3 differs from verb placement in L1 and/or L2. Second, it is clear that Non-SU-I is highly frequent, at least in the task types given for the corpus, implying that it is a structure which the learners have experience processing. SU-I with the selected sentence medial adverbs is less frequent.⁵ However, sentences with negation are frequent in the corpus, with more than 200 examples in L3 German and more than 300 examples in L3 French. Analyzing these are beyond the scope of the present paper, but their frequencies further indicate that learners have experience with producing sentences where the cross-linguistic differences germane to our research questions are relevant.

With respect to our second research question, we find that in relatively controlled written production, surface word order is target-like in most cases, but more so in L3 French (95.7% target surface order) than in L3 German (81.4% target surface order). Regarding our third research question, we also see that the overall pattern of non-target word order is as found in previous studies of the same populations using acceptability judgment tasks (Busterud et al. 2023; Dahl et al. 2022; Listhaug et al. 2021): In L3 French, we see more frequent non-target word order in SU-I (24.3%) compared to Non-SU-I structures (1.9%). In L3 German, we see the opposite pattern, with more frequent non-target word order in Non-SU-I (22.7%) than in SU-I structures (5.3%). Our data indicate that previous findings are not an artefact of experimental task types, as non-target verb placement is produced, i.e., it is not only receptively accepted.

We now turn to our fourth research question concerning verb placement in the underlying syntactic structure. As is clear from these results, L3 French learners rarely show evidence of non-target verb placement in C (i.e., long movement), as V2 word order in Non-SU-I sentences is almost absent. This means there is little evidence of transfer of V2 from L1 Norwegian. Target-like surface word order in Non-SU-I sentences is compatible both with the target French verb placement in I, and with a placement in V; the latter could have been transferred from L2 English. In SU-I sentences in French, on the other hand, surface structure will not appear target-like with the verb in V. The fact that we see a higher ratio of non-target word order in SU-I sentences (9 out of 37 instances) compared to Non-SU-I sentences (6 out of 314) is an indication that at least some learners leave the verb in V. Therefore, one could speculate that also target-like surface word order in Non-SU-I actually reflects verb placement in V, at least in some cases. In L3 German, we saw non-target placement of verbs in Non-SU-I in almost one

⁵Additionally, participants produce sentences where the adverb is sentence-initial or sentence-final. Both options are available in both L3s.

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in four cases. This surface word order is compatible with an underlying verb placement in V, possibly transferred from English, but also with verb placement in I. A question therefore remains as to whether learners place their verbs in V or in I.

Evidence which helps disentangling underlying verb placement in either V or I comes from a sub-set of Non-SU-I sentences, namely those which also contain a sentence medial adverb. In the French sub-corpus, there are six examples of such sentences, none of which display inverted subject-verb order following the fronted element, i.e., there is no indication of verb placement in C. In four of the sentences (three with a lexical verb and one with copula *être*), the verb precedes the adverb, indicating target placement in I (which, for copula, is target-like also in English). In two sentences, the verb follows the adverb indicating non-target lexical verb placement in V, see ((4) and ((5) below. These were both produced by the same learner in Year 12.

- (4) Simultané qui nous regardons le film nous aussi mange le
 simultaneous that we watch the film we also eat the
 popcorn et bu du coca.
 popcorn and drank INDEF coke.
 ‘As we watch the film, we also eat popcorn and drink coke.’
- (5) tout de même les jeunes souvent en avoir de bons rapports avec à
 all the same the young often of have INDEF good relations with in
 semaine
 week
 ‘still, the young often have a good relationship with [the police] during
 [the] week.’

In the German sub-corpus, there are 18 examples of Non-SU-I sentences with relevant adverbs, five of which appear with non-inverted non-target verb placement. Only one of these indicates verb placement in V with the adverb preceding the (lexical) verb, see ((6) below. Four of these sentences appear with clear verb placement in I, as the verb precedes the adverb, as in the example in ((7). These five sentences are produced by two different learners: The same student who produced the example in ((6) also produced three sentences of the type in ((7), all in Year 11 and with lexical verbs, while a second student produced one such example, containing an auxiliary, in Year 12.

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- (6) Wenn ich bei ihnen bin, wir oft ansehen ein Film oder spiele
 when I with them am, we often view a film or play
 Brettspiel.
 board_games
 ‘When I am with them, we often watch a film or play board games.’
- (7) Wenn ich fertig mit meinem Schultag ich spiele sehr oft
 when I finished with my school_day I play very often
 Videospiel.
 video_games
 ‘When I [have] finished my school day, I very often play video games.’

This inspection shows that in these, admittedly few, examples, the verb is placed in I rather than in V in the majority of the cases. Thus, they do raise the question of whether target word order in Non-SU-I in L3 French and non-target word order in Non-SU-I in L3 German can simply be attributed to transfer from L2 English: In both cases, these examples constitute evidence that the verb is in fact placed in I, which is the target in French but not in German, and which, crucially, is not the underlying placement of lexical verbs in English. These examples suggest that such Non-SU-I structures with sentence-medial adverbs should be systematically investigated in future controlled experiments to disentangle the different underlying verb positions that may cause the same surface word order.

Another example of verb placement that cannot simply be the result of transfer from English comes from non-target verb placement of copula *sein/être* in SU-I. As shown in Table 1, there are two such cases in each L3 with the adverb preceding the copula, indicating verb placement in V. While we cannot know what these individual participants’ English grammar looks like, potential placement of *be* in V in their L2 English is unexpected, given that the copula moves to I in English. Moreover, placement of copula in V cannot be transfer from L1 Norwegian. These examples constitute too few tokens to draw any conclusions; however, combined with the few examples of Non-SU-I sentences with a sentence-medial adverb discussed above, we suggest that the acquisition of verb movement in L3 cannot simply be reduced to a question of transfer from L1 or L2.

In L2 acquisition, it has long been acknowledged that performance cannot be straightforwardly predicted by a simple comparison of the L1 and the L2 where any differences will lead to a non-target L2 system (Corder 1967; Odlin 1989: 24). In other words, L1 transfer is not the sole explanation for the non-target structure of the L2 grammar at any given point but is rather assumed to interact with language acquisition mechanisms, possibly guided by Universal Grammar,

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in the developing L2 grammar (Dulay & Burt 1974; Flynn & O'Neil 1988; Towell & Hawkins 1994). Similarly, there is no reason to assume that all non-target (or target) features of an L3 grammar always reflect transfer from L1 or L2.

We argue that transfer interacts with other learning mechanisms in L3 acquisition, and that the outcome is conditioned by the specific properties available in the grammars of the previous languages. In line with the proposal in Busterud et al. (2023), we suggest that learners may have an inherent preference for avoiding (unnecessary) verb movement, a preference previously documented for L1 acquisition (Roeper 1999; Waldmann 2012; Westergaard 2009). While L1 transfer may override this preference in L2 acquisition (Westergaard 2003), in a situation where two prior grammars, one with and one without verb movement, are available, learners may initially show a preference for avoiding verb movement. Crucially, during the L3 acquisition process, they may begin employing verb movement incrementally: Learners may go through an intermediate stage where they prefer short verb movement to I rather than the target long movement to C. This would explain the asymmetry in the L3 French and the L3 German data in our study and predicts that it may take the L3 German learners longer to acquire target verb placement (in Non-SU-I) compared to the L3 French learners. In the present study, learners who produce non-target word order invariably also produce target word order for the same structure. This may indicate instability in the system and reflect incremental employment of verb movement.

6 Conclusions

The present corpus study shows that learners of L3 French and L3 German produce both Non-SU-I and SU-I sentences. The findings replicate patterns of non-target word order found in previous experimental studies; L3 French learners produce more non-target word order with SU-I than with Non-SU-I sentences, whereas the pattern is the opposite for L3 German learners. In L3 French, non-target word order thus appears more prevalent as a result of verb placement in V compared to placement in C, while in L3 German, non-target word order which may reflect verb placement in either I or V is more prevalent than unambiguous placement in V. *In situ* verb placement could be CLI from English, but our overall findings indicate that the picture is more complex and that principles such as economy of movement may also play a role (Busterud et al. 2023). The advantage of a corpus study is that the data may include (unexpected) sentence types not previously tested in experimental studies. Analysis of a small number of Non-SU-I sentences where verb placement in I versus in V can be disambiguated show

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more instances of the former. Additionally, examples of copula placement in V further indicate that CLI from English may not explain all non-target production. Albeit few, these examples indicate a need for future experimental research to further our understanding of the acquisition of L3 verb placement. Additionally, while beyond the scope of the present study, an analysis of the high number of sentences with negation in the corpus would provide further insights.

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References

- Bardel, Camilla & Ylva Falk. 2007. The role of the second language in third language acquisition: The case of Germanic syntax. *Second Language Research* 23(4). 459–484.
- Bohnacker, Ute. 2006. When Swedes begin to learn German: From V2 to V2. *Second Language Research* 22(4). 443–486.
- Busterud, Guro, Anne Dahl, Dave Kush & Kjersti Faldet Listhaug. 2023. Verb placement in L3 French and L3 German: The role of language-internal factors in determining cross-linguistic influence from prior languages. *Linguistic Approaches to Bilingualism* 13(5). 693–716.
- Corder, S. P. 1967. The significance of learner's errors. *International Review of Applied Linguistics in Language Teaching* 5(4). 161–170.
- Council, Language. 2017. *Språkstatus 2017 - Språkpolitisk tilstandsrapport frå språkrådet* [Language status 2017 - Language policy report].
- Council, Language. 2021. *Språkstatus 2021 - Media and culture* [Language status 2021- Media and culture].
- Dahl, Anne, Kjersti Faldet Listhaug & Guro Busterud. 2022. The role of L1 Norwegian and L2 English in the acquisition of verb placement in L3 German. In Tania L. Leal, Elena Shimanskaya & Casilde Isabelli (eds.), *Generative SLA in the age of Minimalism: Features, interfaces, and beyond. Selected proceedings of the 15th Generative Approaches to Second Language Acquisition conference*, vol. 67 (Language Acquisition and Language Disorders), 191–212. Amsterdam: John Benjamins.

5 A corpus study of verb placement in L3 French and L3 German

- Dirdal, Hildegunn, Ingrid Kristine Hasund, Eli-Marie Danbolt Drange, Eva Thue Vold & Elin Maria Berg. 2022. Design and construction of the Tracking Written Learner Language (TRAWL) corpus: A longitudinal and multilingual young learner corpus. *Nordic Journal of Language Teaching and Learning* 10(2). 115–134.
- Dulay, Heidi C. & Marina K. Burt. 1974. Natural sequences in child second language acquisition. *Language Learning* 24(1). 37–53.
- Flynn, Suzanne & Wayne O’Neil. 1988. Introduction. In Suzanne Flynn & Wayne O’Neil (eds.), *Linguistic theory in second language acquisition*, 1–24. Dordrecht: Springer Netherlands.
- Håkansson, Gisela, Manfred Pienemann & Susan Sayehli. 2002. Transfer and typological proximity in the context of second language processing. *Second Language Research* 18(3). 250–273.
- Hancock, Victorine & Anna Sanell. 2009. The acquisition of four adverbs in a learner corpus of L2 French. *Discours* 5. 1–35.
- Kunnskapsdepartementet. 2019. *Læreplan i fremmedspråk (FSP01-02). Fastsett som forskrift. [curriculum in foreign languages (FSP01-02). Regulation]*.
- Listhaug, Kjersti Faldet, Guro Busterud & Anne Dahl. 2021. French as a third language in Norway: The influence of the prior languages in the acquisition of word order. *Oslo Studies in Language* 12(1). 125–144.
- Odlin, Terence. 1989. *Language transfer*. Cambridge: Cambridge University Press.
- Roeper, Thomas. 1999. Universal bilingualism. *Bilingualism: Language and Cognition* 2(3). 169–186. DOI: [10.1017/S1366728999000310](https://doi.org/10.1017/S1366728999000310).
- Stadt, Rosalinde. 2019. *The influence of Dutch (L1) and English (L2) on third language learning. The effects of education, development and language combinations*. Amsterdam: LOT.
- Towell, Richard & Roger Hawkins. 1994. *Approaches to second language acquisition*. Bristol: Multilingual Matters.
- Waldmann, Christian. 2012. Moving in small steps towards verb second: A case study. *Nordic Journal of Linguistics* 34(3). 331–359.
- Westergaard, Marit. 2003. Unlearning V2: Transfer, markedness, and the importance of input cues in the acquisition of word order in English by Norwegian children. *EUROSLA Yearbook* 3(1). 77–101.
- Westergaard, Marit. 2009. Usage-based vs. rule-based learning: The acquisition of word order in wh-questions in English and Norwegian. *Journal of Child Language* 36(5). 1023–1051.
- Zakariassen, Ellen. 2021. *Verbstellung im Deutschen als Drittsprache: Eine Korpusuntersuchung des Erwerbs deutscher Verbstellung von norwegischen Lernen*. NTNU Open: NTNU Master’s thesis. NTNU. (MA thesis).

Chapter 6

Directionality of cross-linguistic influence: Better insights from more comparisons

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This review focuses on syntactic and morphosyntactic processing in bilinguals when they are compared to each other (bilingual-bilingual comparison) or when the two languages of bilinguals are compared (between-language comparison). The first comparison type is especially helpful in pinpointing whether there is cross-linguistic influence, as opposed to a single comparison between, for example, a mono- and bilingual child group. We show that, thus far, there is ample evidence for cross-linguistic influence (CLI) from an L1 to an L2, but it is less clear whether it may go in the opposite direction, not the least due to a lack of proper comparison groups. We thus examine bidirectional CLI in late bilingual adults (from the L2 to the L1), with a focus on regressive CLI (rCLI). We then explore bidirectional CLI in early bilingual children. Finally, we discuss how research could go forward in this burgeoning area.

1 Introduction

While research with bilingual speakers has for a long time indicated that languages are co-activated and interact (e.g., [Grosjean 1989](#)), it remains an open question how this happens and, critically, when cross-linguistic influence (CLI) occurs and in which direction it can go. The aim of the present paper is to examine the nature and extent of bidirectional cross-linguistic influence.

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Cross-linguistic influence in terms of facilitation or interference from one language during the processing of the other language in the bilingual mind has typically been reported in the direction of a first language (L1) to a second language (L2); for a review, see e.g. Foucart (2021) and Lago et al. (2021). As highlighted by Lago et al. (2021), one challenge in experimental studies is to correctly identify CLI and separate it from more general L2 processing mechanisms. One way to do so is by including at least two bilingual groups whose L1s differ in overlap with the target language. Similar suggestions for bilingual child research have been made by van Dijk et al. (2021: 27). Comparing between bilingual groups comes with yet another advantage: it avoids the comparison with a ‘monolingual’ or ‘home country’ group, which has been shown to be problematic for various reasons (Rothman et al. 2022). Such reasons include monolingual normativity bias, lack of recognition that significant experience-based variation exists within both monolingual and multilingual groups, and indeed, the reality that multilingualism is actually the norm on a global scale.

In the following, we survey the literature on bidirectional CLI in late bilinguals, with a focus on the less well-understood influence from a later-learned language to the L1, or regressive CLI (henceforth rCLI; Brown-Bousfield & Chang 2023). We then examine bidirectional influence in early bilinguals. The focus for this review is on syntactic and morphosyntactic processing. Given the often-observed production-comprehension asymmetry (e.g., van Dijk et al. 2024), we exclusively concentrate on comprehension in our survey. Our focus is on studies that meet at least one of the following criteria: (i) comprehension was tested in both languages of the bilingual; (ii) at least two groups of bilinguals that differ in their language combination but are otherwise matched (e.g., in age, proficiency) were compared; (iii) bilinguals with the same first or heritage language (HL) that reside in different countries were compared. Following Schmid & Köpke (2017), especially in cases of partial overlap between L1 and L2, one may observe bidirectional influence, but less so for completely shared or unique structures/features. Recent research has pointed out the importance of structural similarity when it comes to CLI from previously acquired languages in the acquisition of a third language (e.g., Castle et al. 2025; Mitrofanova et al. 2023). Throughout, special attention is paid to the level of structural similarity between languages in terms of particular properties investigated as a factor that may account for (r)CLI to be observed.

In sum, we review the literature available in relation to the following questions:

1. Does rCLI occur from the L2 into the L1 in late bilinguals in syntactic and

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morphosyntactic processing?

2. Does bidirectional CLI occur in early bilingual children in syntactic and morphosyntactic processing?
3. What factors may play a role in this (r)CLI?

The paper is thus structured as follows: First, we review the literature on (r)CLI in late bilingual adults and early bilingual children. Second, we discuss what the types of comparisons our review focused on add to the field of bilingualism and provide an outlook for future research.

2 Cross-linguistic influence in late bilingual adults

In this section, we focus on bilinguals who started to learn an additional language later in life, that is, after their L1. While many researchers have tested whether the L1 influences an L2 at early stages of acquisition and beyond, only a few have investigated whether an L2 can also influence the processing of the L1. We think that research on this directionality is necessary for a better understanding of the bilingual mind, not the least because “L2-to-L1” influence or rCLI may be taken as indication for attrition of the L1 (e.g., Schmid & Köpke 2017). Attrition has been defined as the decrease of L1 performance as a result of increased L2 usage and/or L1 disuse (see Gallo et al. 2021 for an overview).

We are aware of only one study that fulfilled the criteria we set above, a visual-world eye-tracking study by Grüter & Hopp (2021). Grüter and Hopp investigated (r)CLI by means of structural preferences for globally ambiguous *wh*-questions in German, comparing four groups of late bilinguals: German L1-English L2 speakers in Germany and the US, as well as English L1-German L2 speakers in Germany and the US. The structures under investigation are frequently used in German and have word order equivalents in English, but they differ in their interpretation, as illustrated in (1). While in English a *wh*-question in the present tense (1a) can only be interpreted as a subject question and a *wh*-question in the perfect tense (1b) as an object question, it can be a subject or object question in German when there is no overt case marking information such as in (1).

- (1) a. Was leckt das Känguru?
 what lick.3sg the kangaroo
 ‘What licks the kangaroo?’ or ‘What does the kangaroo lick?’
- b. Was hat das Känguru gelect?
 what have.3sg the kangaroo licked
 ‘What has licked the kangaroo?’ or ‘What has the kangaroo licked?’

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Analyses of response and eye-movement data showed that L2 speakers in both the US and Germany were more likely to interpret *wh*-questions as subject questions when they were in the present as compared to the perfect tense, indicating influence from the L1. In contrast, there was no indication for an influence from the L2. German expats in the US displayed no difference in their interpretative biases between present-tense and perfect-tense questions. The findings thus point to an important role of order of acquisition, but to a negligible role of the immersion context when it comes to CLI. That is, although the bilinguals' languages overlapped in word order and differed in meaning, providing a case of partial overlap, CLI was only observed in one direction. Previous studies that found differences to an L1 control group indicative of L2-to-L1 influence (in one group of expats) tested less frequent and syntactically more complex structures (Dussias & Sagarra 2007; Kasparian & Steinhauer 2017). According to Grüter and Hopp, it could be for that reason that L1 routines were influenced by the L2, as these structures could have been less entrenched.

Does this mean that there is no such thing as rCLI for regularly encountered structures? Recent findings from Grüter & Roos (2024) and Schlenter and Westergaard (to appear) show that there might be. The two eye-tracking studies investigated the exploitation of morphological case marking for argument interpretation in the L1 of bi-/multilingual speakers residing abroad. Grüter & Roos (2024) tested the use of case in Japanese expats in the US in a design with four single images. Participants listened to sentences such as in (2), while presented with the images of the location (park), the agent of the event (boy), the referent that, depending on the structure, functioned as the patient/theme (2a) or recipient (2b) (pigeon), as well as the image of a potential theme (bread).

- (2) a. kooen-de otokonoko-ga hatō-o yasashiku sawatta
 park-LOC boy-NOM pigeon-ACC gently touched
 'At the park, the boy gently touched the pigeon.'
- b. kooen-de otokonoko-ga hatō-ni yasashiku pankuzu-o
 park-LOC boy-NOM pigeon-DAT gently breadcrumb-ACC
 ageta
 gave
 'At the park, the boy gently gave breadcrumbs to the pigeon.'

During the adverb segment, the expats were more likely to look at the potential theme following a dative (DAT) cue that signals a ditransitive structure (2b) than following an accusative (ACC) cue that does not (2a), indicating the predictive use of case. However, the results were influenced by the length of residence abroad,

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with a reduced use of case for long-term expats. There was no such influence for another set of sentences testing the predictive use of the semantics of classifiers. In Schlenter and Westergaard (to appear), participants were presented with two event scenes, one that depicted the action as described by a sentence and one that depicted the same action but the roles of agent and patient (transitives) or of recipient and theme (ditransitives) were reversed. An example for an item from the ditransitive sentence set is provided in (3).

- (3) a. Der Krankenpfleger bringt der Ärztin morgens
 The-NOM nurse brings the-DAT doctor in.the.morning
 den Patienten.
 the-ACC patient
 ‘In the morning, the nurse brings the patient to the doctor.’
 b. Der Krankenpfleger bringt die Ärztin morgens
 The-NOM nurse brings the-ACC doctor in.the.morning
 dem Patienten.
 the.DAT patient
 ‘In the morning, the nurse brings the doctor to the patient.’

The experiment was administered to a group of German L1 speakers in Germany and a group of German-speaking expats in Norway. While the home country group showed a higher likelihood of looks to the target scene after a DAT-marked article than after an ACC-marked article prior to the second object, the reverse held for the expat group, at least temporarily. Direct comparisons revealed a between-group difference for the ACC > DAT order, when the theme precedes the recipient, which corresponds to a less prototypical order in German. Altogether, the result indicated subtle changes in the processing of alternating argument orders in ditransitive sentences but not transitive sentences in L1 German.

Since the two latter described studies did not include a proper comparison group, it remains to be seen whether L1 processing was influenced by the societal language(s) (SL). The SLs English and Norwegian lack dative case as a marker for recipients in ditransitive sentences and themes often precede recipients in prepositional object constructions, the latter of which may explain the group-difference for ACC > DAT in Schlenter and Westergaard (to appear). Altogether, the results from the reviewed studies seem to raise more questions than they answer, including the question under which circumstances there is L2-to-L1 influence (if at all). At the same time, they show the huge potential for future research.

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Next, we turn to CLI in early bilinguals. One may assume more permeability between languages at a younger age as, following the above-mentioned reasoning by [Grüter & Hopp \(2021\)](#), languages could be less entrenched.

3 Cross-linguistic influence in early bilingual children

A meta-analysis by [van Dijk et al. \(2021\)](#) shows indication for CLI across child studies with a small to moderate effect size. Since almost all available studies at that time included the bilingual-monolingual comparison, this was the design the authors focused on. Moreover, the majority of studies employed elicited production tasks or grammaticality judgments, but few had tested (real-time) comprehension. Within the past few years, it has become evident in the field that the number of studies on comprehension in bilingual children is steadily increasing and so is the number of group comparisons. In this section, we review and discuss the results available to date.

In a self-paced listening task, [van Dijk et al. \(2022\)](#) studied CLI during the processing of Dutch sentences, such as in examples (4) and (5), in English-Dutch and German-Dutch bilingual children (5 to 9 years). The age of acquisition (AoA) for Dutch ranged from 0 to 4 years.

- (4) a. De jongen wordt geduwd door het meisje.
the boy is pushed by the girl
- b. De jongen wordt door het meisje geduwd.
the boy is by the girl pushed
'The boy is (being) pushed by the girl.'
- (5) a. Gisteren at het meisje een appel.
yesterday ate the girl an apple
'Yesterday the girl ate an apple.'
- b. * Gisteren het meisje at een appel.
yesterday the girl ate an apple

In English, passives as in (4b) that follow the order PP-V are ungrammatical and only V-PP is a viable option. In German, PP-V (4b) is preferred and clearly grammatical while V-PP (4a) is less so. Due to its V2 constraint, main clauses with an adverb in sentence-initial position as in (5) must have the finite verb in second position in Dutch. In English, the strict SVO order typically only allows the ordering as in (5b). German is like Dutch in this regard, so there is

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a complete overlap between languages for sentences such as (5a). For the passives, the German-Dutch bilinguals showed a slow-down when listening to the partially overlapping PP-V structure relative to the English-Dutch and a monolingual Dutch group. This effect was enhanced the more dominant in German the children were. The German-Dutch bilinguals also slowed down for the completely overlapping V2 structures (5a), yet only as a function of language dominance. The authors interpret this finding with co-activation of corresponding structures and, in turn, also lemmas in German, which requires inhibition. Unexpectedly, no slowdown was observed for the English-Dutch bilinguals for the partially overlapping V-PP structure (4a). One possible explanation put forward by van Dijk and colleagues is the larger lexical overlap between Dutch and German than Dutch and English that may enlarge co-activation. Overall, the results of this study thus seem to support the claim that CLI depends on the level of similarity between languages.

Several studies have tested the use of case marking in bilingual children's comprehension and production. Our primary focus is on the comprehension results. The following examples from Meir et al. (2024) and Mitrofanova et al. (Under review) illustrate how accusative case is realised in Russian (6a), Hebrew (6b) and German (6c) object-initial sentences with definite nouns.

- (6) a. ser-ogo zajčika sejčas s'est lis-a
 grey-ACC bunny-ACC now will-eat fox-NOM
 'The fox will now eat the grey bunny.'
- b. et ha-arnav ha-afor yoxal axšav ha-šual
 ACC DEF-bunny DEF-grey will-eat now DEF-fox
 'The fox will now eat the grey bunny.'
- c. den Hase-n wird im nächsten Moment der Fuchs
 the-ACC hare-ACC will in the.next moment the-NOM fox
 aufspüren
 find
 'In the next moment, the fox will find the hare.'

In Russian, ACC is marked by a suffix on the noun as well as on pre-nominal adjectives. In German, ACC is marked pre-nominally on articles (unlike in Russian) and on adjectives (like in Russian), and for weak masculine nouns such as *Hase* ('hare') post-nominally on the noun. In Hebrew, on the other hand, ACC is not marked morphologically, but by the particle *et* that precedes a definite noun. For indefinite nouns, the word order is rigidly SVO. In all three languages, SVO

corresponds to the most frequent and prototypical order. Given the restricted use in Hebrew, ACC marking seems less available in this language than in the other two. However, even though available, ACC marking in German is often not reliable. Feminine and neuter nouns do not differ between nominative (NOM) and ACC, resulting in ambiguity. In contrast, in Russian case marking is not only available, but also highly reliable. Only infrequently are arguments in transitive sentences ambiguous between NOM and ACC (Kempe 1999).

In a study that included a production and a comprehension task, Meir et al. (2024) compared the use of ACC by Russian-Hebrew bilingual children to Russian-speaking and Hebrew-speaking monolingual control groups. The children's ages ranged from 4 to 8 years and were matched, so there were no significant differences between groups. The bilingual children were all born in Israel to Russian-speaking families, so Russian was the HL and Hebrew was the SL. While all bilinguals had learned Russian from birth, the AoA for SL-Hebrew ranged from 0 to 3 years. The monolingual Russian control group, like the other groups, was tested in Israel, but had arrived there less than three months ago and had no command of Hebrew. Proficiency scores, as indicated by the parents, did not differ between HL-Russian and SL-Hebrew. The bilinguals completed all tasks in both languages.

The production elicitation task evaluated the accuracy of nouns in marked and unmarked conditions. For Russian, the unmarked condition was one wherein case inflection for ACC and NOM forms are homophonous, e.g., *mjod* 'honey-NOM = ACC', whilst in the marked condition, case inflection for ACC is different, e.g., *slon* 'elephant-NOM' – *slon-a* 'elephant-ACC'. For Hebrew, the use of an indefinite NP in object position is the unmarked condition, e.g., *koxav* 'star.INDEF', whilst the use of a definite NP in object position with the accusative marker *et* is marked, e.g., *et ha-kova* 'ACC DEF-hat'.

The results of the production task showed high performance for both the marked and unmarked conditions in the monolingual groups. The bilingual group, in contrast, was less accurate for the marked relative to the unmarked condition, that is, when ACC had to be overtly marked in both HL-Russian and SL-Hebrew. In the comprehension task, the children were introduced to three referents (e.g., carrot, bunny, fox) and then heard a transitive sentence that described an action between two of these referents, either following the order SVO or OVS as in (6). In a so-called 3-pic task, the children saw three single images of the referents and their eye movements to the referents during audio presentation was recorded, similar to previous eye-tracking studies testing case marking (e.g., Özge et al. 2022). In the 2-pic task, they were shown two events between referents (bunny eating carrot, fox eating bunny). In line with a predictive use

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of ACC case, monolingual Russian-speaking children showed a preference for the plausible agent (fox; fox eating bunny) in OVS sentences that emerged prior to the second noun. The monolingual Hebrew-speaking group only did so in the 2-pic but not the 3-pic task. Critically, the bilingual children were indistinguishable from the monolingual children in their HL-Russian and for the 2-pic task in SL-Hebrew. However, for the 3-pic task in SL-Hebrew, they showed a stronger use of ACC case than the monolinguals. The authors interpreted this latter finding as influence from HL-Russian, where case is highly reliable and thus a strong predictive cue.

Mitrofanova et al. (Under review) conducted a study that closely followed the design by Meir et al. (2024) for the comprehension task and thus allows for a good comparison. The authors contrasted the processing of case morphology in Russian- (in Russia) and German-speaking monolingual pre-schoolers as well as Russian-German bilingual pre-schoolers (in Germany). Here, the age ranged from 3 to 6 years. Replicating the findings by Meir et al. (2024), bilinguals performed similarly to the monolingual groups with one exception: in Russian, the bilingual children showed a predictive use of ACC only for the 2-pic task, unlike the monolingual group, who did so for both tasks. In German, both monolingual and bilingual children showed no difference between SVO and OVS prior to the noun. That is, there was no indication for a predictive use of morphological case in German.

To summarise the results, monolingual Russian-speaking children in Russia (Mitrofanova et al. Under review) and Israel (Meir et al. 2024) were found to predict the upcoming argument after ACC marking irrespective of the task, 2-pic or 3-pic. Similarly, bilingual children with Russian as HL used ACC predictively as well, but their use was more restricted in the study by Mitrofanova et al. One possible explanation for this difference between studies could be the younger age of the children in Mitrofanova et al.'s () study. Alternatively, it could be the result of non-facilitative influence from the SL-German. However, for such a conclusion, as stated by the authors, another bilingual group is needed where case is more reliable in the SL than it is in German. The results for the real-time use of ACC in the two SLs Hebrew and German reveal different outcomes in the two studies: while in Meir et al. (2024), it seems that HL-Russian may have an influence on ACC processing in SL-Hebrew, no such influence is seen for SL-German. This leads us to one caveat: in German, NOM and ACC are marked on the article (*der* vs. *den*) and sometimes the noun, whereas in Hebrew ACC is marked by the particle *et*. It is thus possible that the exploitation of ACC during processing was negatively influenced by the perceptual salience of the case cue, due to

the subtle difference on the article in German;¹ see also [Kim et al. \(2018\)](#), who found an influence of perceptual salience of case in HL-Korean. Moreover, note that in the comprehension experiments above, ACC was always marked twice in Russian, on the adjective and noun, but not in Hebrew (not possible) or German (possible). Hence, while the strong reliability of case marking in Russian is a likely explanation for the predictive use of ACC in both monolingual and HL-speakers, its perceptual salience is another likely explanation and so is the absence of a predictive use of ACC in the SL-German. Methodological factors can play an important role, as also shown by the differences between the 2-pic and 3-pic task.² Further note that in the above studies, semantics (e.g., animacy) and world knowledge could also have helped identify semantic roles. In a previous study by [Janssen et al. \(2015\)](#) that employed a forced-choice picture selection task similar to the one in [Schlenter & Westergaard \(2024\)](#) with reversible actions between two animate referents, Russian-Hebrew bilingual children only performed at chance level for OVS sentences, unlike an age-matched (5 years) and younger (4 years) monolingual control group and not significantly different from a group of Russian-Dutch bilinguals (Dutch has no case or object marker). While design and material choices are less critical for within-language between-group comparisons, they may complicate comparisons between languages and, for obvious reasons, comparisons across studies.

4 Discussion

Comparisons between mono- and bilingual groups often yield between-group differences (e.g., [van Dijk et al. 2022](#)). In terms of CLI, such differences are not particularly informative, as a range of other factors may be responsible here. Like others before (e.g., [Lago et al. 2021](#); [van Dijk et al. 2021](#)), we thus argue that comparisons between bilingual groups better show whether differences can be attributed to (r)CLI. Moreover, such comparisons can provide further information about the level of similarity required for CLI to occur. Note that the inclusion of an additional monolingual group might only be feasible when children are tested as it can be difficult to find and recruit truly monolingual adults with no foreign

¹For a discussion and experimental investigation how differences in perceptual salience of case cues may influence real-time processing in German, see [Schlenter & Westergaard \(2024\)](#).

²[Minor et al. \(2024\)](#) compared the use of case marking in Russian by Russian-German bilingual children (in Germany) aged 8 to 13 years between the following designs: (i) 3 individual referents, (ii) pairs of referents (NP1 + potential agent/patient) and (iii) events. This systematic investigation demonstrates that children are more likely to use case marking on the NP1 predictively when presented with event scenes than when presented with referent displays.

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language experience at all. If adults are investigated, researchers might want to include a parallel group of bilinguals (where either Language A (L1 A controls, immersed L2 A speakers) or Language B (potential A attriters, non-immersed L2 A speakers) is the societal language), as done by Grüter & Hopp (2021), rather than an L1 group for the same reason that such comparisons are more informative in terms of CLI, especially bidirectional CLI (Figure 1).

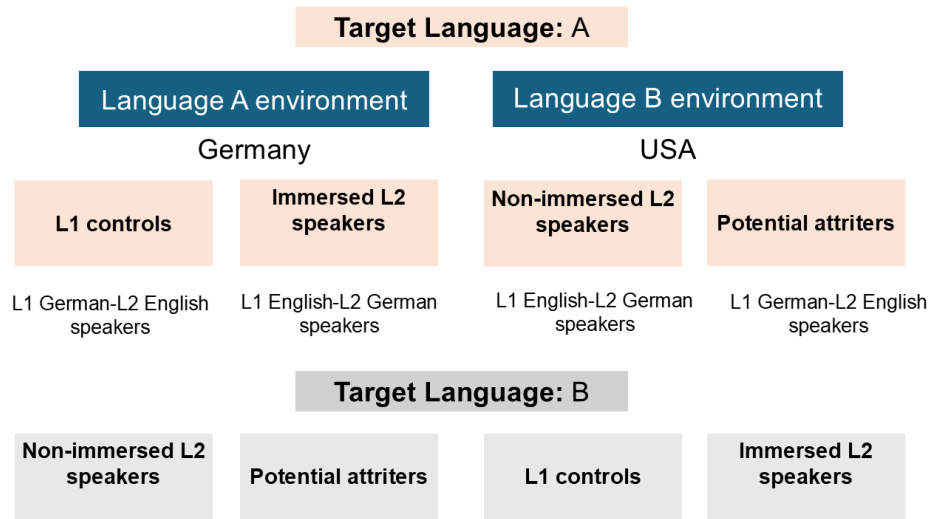


Figure 1: Proposed groups for examining bidirectional CLI, with examples from Grüter & Hopp (2021)

In addition, comparisons between bilingual groups with the same HL but different SLs can provide information about the role of the SL and, in case of L1 speakers in an L2 environment, comparisons between groups with the same L1 but different L2s can provide information about rCLI and attrition. Within-group comparisons between languages provide insight into similarities and differences between, for example, processing in a HL and a SL (Meir et al. 2017; Mitrofanova et al. Under review).

CLI has often been reported as being part and parcel of bilingual processing (e.g., van Dijk et al. 2021). However, while CLI has been shown in the direction from an L1 to an L2 (e.g., Grüter & Hopp 2021), the evidence for bidirectional influence in general is sparse when looking at syntactic/morphosyntactic processing. In fact, there was very limited evidence for it in our review. Grüter & Hopp (2021) found that, in a case of partial overlap between L1 and L2, CLI was evident in the direction from the L1 to the L2 but not the other way around. Interpretive biases in L2 German were influenced by tense (like in the L1 English, but

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see below), but there was no effect of tense in German L1 speakers who resided in an English-speaking environment. Ideally, as stated by the authors, the groups should also have been tested in English for a complete parallel design. Due to the lack of proper comparison groups, it remains to be tested whether findings for predictive processing in long-term expats were the result of rCLI (Grüter & Roos 2024; Schlenter & Westergaard To appear). Moreover, the above review revealed limited evidence for an influence from the HL to the SL for the comprehension of object-initial sentences (Meir et al. 2024). Current research needs more examples of CLI from the HL to the SL in comprehension and similar investigations where CLI from the SL to the HL would be a possibility (e.g., bilinguals with German as HL in Poland or Greece).

As regards language similarity, we saw that in cases of partial structural overlap (Grüter & Hopp 2021; van Dijk et al. 2022) processing in the L2 can be influenced by the L1. Cases of partial overlap thus seem to provide a good testing ground for other directions of CLI. What about feature overlap? The finding that the use of ACC in SL-Hebrew by Russian-Hebrew bilinguals was less restricted to the task than for a monolingual Hebrew-speaking group provides some indication for CLI, which, in turn, may indicate that case does not need to be marked the same way (i.e. as a bound morpheme) in both languages for CLI in comprehension to occur. One interesting question for future research is whether the reliability of a cue (e.g., object marker), as the results by Meir et al. (2024) may suggest, might be more decisive than feature overlap (see also Meir et al. 2020). Clearly, more research including carefully controlled designs and materials is needed to shed light on the question of whether and how language similarity and the weighting of cues in comprehension modulate CLI.

5 Conclusion

The studies we referred to here allow researchers to pinpoint CLI and its direction thanks to the inclusion of suitable bilingual-bilingual comparison groups and/or the assessment of both languages of a bilingual. Although we see the practical hindrances, we suggest that taking this extra step of including multiple comparison groups will help us to better understand how the languages in a bi/multilingual mind interact and under which circumstances CLI occurs.

References

- Brown-Bousfield, Megan M. & Charles B. Chang. 2023. Regressive cross-linguistic influence in multilingual speech rhythm: The role of language similarity. In Megan M. Brown-Bousfield, Suzanne Flynn & Éva Fernández-Berkes (eds.), *Studies in bilingualism*, vol. 65, 49–71. Amsterdam: John Benjamins. DOI: [10.1075/sibil.65.03bro](https://doi.org/10.1075/sibil.65.03bro).
- Castle, Chloe, Isabel N. Jensen, Natalia Mitrofanova & Marit Westergaard. 2025. Investigating CLI in L3 morphosyntax through artificial languages. *Second Language Research* 42(2). Advance online publication, 339–364. DOI: [10.1177/02676583251332128](https://doi.org/10.1177/02676583251332128).
- Dussias, Paola E. & Nuria Sagarra. 2007. The effect of exposure on syntactic parsing in Spanish–English bilinguals. *Bilingualism: Language and Cognition* 10. 101–116. DOI: [10.1017/S1366728906002847](https://doi.org/10.1017/S1366728906002847).
- Foucart, Alice. 2021. Language prediction in second language: Does language similarity matter? In Edith Kaan & Theres Grüter (eds.), *Bilingual processing and acquisition: Prediction in second language processing and learning*, vol. 12, 92–114. Amsterdam: John Benjamins. DOI: [10.1075/bpa.12.05fou](https://doi.org/10.1075/bpa.12.05fou).
- Gallo, Federico, Beatriz Bermudez-Margaretto, Yury Shtyrov, Jubin Abutalebi, Hamutal Kreiner, Tamara Chitaya, Anna Petrova & Andriy Myachykov. 2021. First language attrition: What it is, what it isn't, and what it can be. *Frontiers in Human Neuroscience* 15. 686388. DOI: [10.3389/fnhum.2021.686388](https://doi.org/10.3389/fnhum.2021.686388).
- Grosjean, François. 1989. Neurolinguists, beware! The bilingual is not two monolinguals in one person. *Brain and Language* 36(1). 3–15. DOI: [10.1016/0093-934X\(89\)90048-5](https://doi.org/10.1016/0093-934X(89)90048-5).
- Grüter, Theres & Holger Hopp. 2021. How permeable are native and non-native syntactic processing to crosslinguistic influence? *Journal of Memory and Language* 121. 104281. DOI: [10.1016/j.jml.2021.104281](https://doi.org/10.1016/j.jml.2021.104281).
- Grüter, Theres & Sachiko Roos. 2024. Does L1 attrition affect predictive processing? Evidence from Japanese expats in the U.S. In *5th International Symposium on Bilingual Processing in Adults and Children (ISBPAC)*. Poster presentation, 23–24 May, Swansea, UK.
- Janssen, Bibi, Natalia Meir, Anne Baker & Sharon Armon-Lotem. 2015. On-line comprehension of Russian case cues in monolingual Russian and bilingual Russian-Dutch and Russian-Hebrew children. In Elizabeth Grillo & Kyle Jepson (eds.), *Proceedings of the 39th Annual Boston University Conference on Language Development*, vol. 2, 266–278. Somerville: Cascadilla Press.

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- Kasparian, Kristina & Karsten Steinhauer. 2017. When the second language takes the lead: Neurocognitive processing changes in the first language of adult attriters. *Frontiers in Psychology* 8. 389. DOI: [10.3389/fpsyg.2017.00389](https://doi.org/10.3389/fpsyg.2017.00389).
- Kempe, Vera. 1999. Processing of morphological and semantic cues in Russian and German. *Language and Cognitive Processes* 14. 129–171. DOI: [10.1080/016909699386329](https://doi.org/10.1080/016909699386329).
- Kim, Kitaek, William O’Grady & Bonnie D. Schwartz. 2018. Case in heritage Korean. *Linguistic Approaches to Bilingualism* 8. 252–282. DOI: [10.1075/lab.16001.kim](https://doi.org/10.1075/lab.16001.kim).
- Lago, Sol, Michela Mosca & Anna Stutter Garcia. 2021. The role of crosslinguistic influence in multilingual processing: Lexicon versus syntax. *Language Learning* 71(S1). 163–192. DOI: [10.1111/lang.12412](https://doi.org/10.1111/lang.12412).
- Meir, Natalia, Olga Parshina & Irina A. Sekerina. 2020. The interaction of morphological cues in bilingual sentence processing: An eye-tracking study. In Megan M. Brown & Alexandra Kohut (eds.), *Proceedings of the 44th Boston University Conference on Language Development*, 376–389. Somerville: Cascadia Press.
- Meir, Natalia, Olga Parshina & Irina A. Sekerina. 2024. Prediction in bilingual sentence processing: Is it linked to production? *Linguistic Approaches to Bilingualism* 14. 544–576. DOI: [10.1075/lab.22102.mei](https://doi.org/10.1075/lab.22102.mei).
- Meir, Natalia, Joel Walters & Sharon Armon-Lotem. 2017. Bi-directional crosslinguistic influence in bilingual Russian-Hebrew children. *Linguistic Approaches to Bilingualism* 7(5). 514–553.
- Minor, Serge, Natalia Mitrofanova & Marit Westergaard. 2024. The interaction of linguistic and visual cues for the processing of case in Russian by Russian-German bilinguals: An eye tracking study. *Topics in Cognitive Science* 17(4). Advance online publication, 855–867. DOI: [10.1111/tops.12724](https://doi.org/10.1111/tops.12724).
- Mitrofanova, Natalia, Evelina Leivada & Marit Westergaard. 2023. Crosslinguistic influence in L3 acquisition: Evidence from artificial language learning. *Linguistic Approaches to Bilingualism* 13. 717–742. DOI: [10.1075/lab.22063.mit](https://doi.org/10.1075/lab.22063.mit).
- Mitrofanova, Natalia, Serge Minor, Marit Westergaard, Antje Sauerma, Natalia Gagarina, Duygu Özge & Irina Sekerina. Under review. *Processing of grammatical case in Russian and German by monolingual and bilingual preschoolers: Evidence from eye-tracking*.
- Özge, Duygu, Jaklin Kornfilt, Katja Maquate, Aylin C. Küntay & Jesse Snedeker. 2022. German-speaking children use sentence-initial case marking for predictive language processing at age four. *Cognition* 221. 104988. DOI: [10.1016/j.cognition.2021.104988](https://doi.org/10.1016/j.cognition.2021.104988).

6 Directionality of cross-linguistic influence

- Rothman, Jason, Fatih Bayram, Vincent DeLuca, Grazia Di Pisa, Jon Andoni Duñabeitia, Khadij Gharibi, Jiuzhou Hao, Nadine Kolb, Maki Kubota, Tanja Kupisch, Tim Laméris, Alicia Luque, Brechje van Osch, Sergio Miguel Pereira Soares, Yanina Prystauka, Deniz Tat, Aleksandra Tomić, Toms Voits & Stefanie Wulff. 2022. Monolingual comparative normativity in bilingualism research is out of ‘control’: Arguments and alternatives. *Applied Psycholinguistics* 44(3). 316–329. DOI: [10.1017/S0142716422000315](https://doi.org/10.1017/S0142716422000315).
- Schlenter, Judith & Marit Westergaard. 2024. What eye and hand movements tell us about expectations towards argument order: An eye- and mouse-tracking study in German. *Acta Psychologica* 246. 104241. DOI: [10.1016/j.actpsy.2024.104241](https://doi.org/10.1016/j.actpsy.2024.104241).
- Schlenter, Judith & Marit Westergaard. To appear. The real-time processing of morphological case by German L1 speakers in Norway: A case of attrition? *Applied Psycholinguistics* 46. DOI: [10.1017/S0142716425100295](https://doi.org/10.1017/S0142716425100295).
- Schmid, Monika S. & Barbara Köpke. 2017. The relevance of first language attrition to theories of bilingual development. *Linguistic Approaches to Bilingualism* 7. 637–667. DOI: [10.1075/lab.17058.sch](https://doi.org/10.1075/lab.17058.sch).
- van Dijk, Chantal, Jasmijn Bosch & Sharon Unsworth. 2024. Language processing in bilingual children. *Linguistic Approaches to Bilingualism* 14. 435–445. DOI: [10.1075/lab.24018.van](https://doi.org/10.1075/lab.24018.van).
- van Dijk, Chantal, Ton Dijkstra & Sharon Unsworth. 2022. Cross-linguistic influence during online sentence processing in bilingual children. *Bilingualism: Language and Cognition* 25. 691–704. DOI: [10.1017/S1366728922000050](https://doi.org/10.1017/S1366728922000050).
- van Dijk, Chantal, Elise van Wonderen, Elly Koutamanis, Gerrit Jan Kootstra, Ton Dijkstra & Sharon Unsworth. 2021. Cross-linguistic influence in simultaneous and early bilingual children: A meta-analysis. *Journal of Child Language* 49. 897–929. DOI: [10.1017/S0305000921000337](https://doi.org/10.1017/S0305000921000337).

Chapter 7

Dynamics of case in monolingual and bilingual first language acquisition of Russian

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The acquisition of case in Russian, with its apparent variability in paths and timing, poses challenges for linguistic theories as the empirical evidence found in spontaneously elicited data is rather complex.

This study investigates case acquisition by simultaneous bilinguals and compares it to monolingual acquisition using spontaneous longitudinal data from two bilingual and three monolingual participants. We track the phases in case marking acquisition and establish the developmental path of case in Russian. The findings show comparable timing and path of case acquisition across both groups, though bilingual children show dynamics in accusative case marking of a type resembling a U-shaped development with certain forms. Both bilingual and monolingual children also display overgeneralizations in the case inflections.

These results suggest that while bilingual children may experience temporary setbacks, their overall developmental trajectory remains comparable to that of monolingual peers, provided they receive consistent input in Russian.

1 Introduction

The acquisition of case in Russian is characterized by aptitudinal and temporal variation in paths and timing; it provides inconsistent empirical evidence and poses challenges for linguistic theories. On the one hand, empirical data show some general developmental patterns, such as an initial acquisition of salient

non-syncretic and simpler or more frequent forms (Ladinskaya et al. 2019), for example, accusative is usually acquired as one of the first cases (Stephany & Voeikova 2009). On the other hand, in languages with rich case system and nominal inflection, language-specific characteristics and individual variation can be said to significantly shape the developmental path, so that cases are acquired with various order and trajectories.

Also theoretically, the acquisition of case poses challenges. On the one hand, according to the Co-development Hypothesis (Rispoli 1999, Radford 1990) case shows interdependence on the syntax and lexicon domains in the acquisition process. For example, non-nominative case marking has been shown to occur when the critical number of lexical items has been acquired, and Mean Length of Utterance (MLU) exceeded two words per utterance (Gvozdev 1949/1981/1981, Voeikova & Gagarina 2002, cf. Babyonyshev 1993). On the other hand, other researchers have shown that bursts of category acquisition across different domains occur sequentially rather than simultaneously, and case development does not progress in parallel with syntax. The observation (formulated as the Hypothesis of the Switch of Acquisitional Attention) that a case spurt precedes spurts in syntactic acquisition (Gagarina & Eliseeva 2009, Gagarina & Voeikova 2009) challenges the Co-development Hypothesis, which postulates the simultaneous emergence of case marking and subject-verb agreement. Examples (1) to (3) below, taken from the corpus of spontaneous recordings of monolingual Russian-speaking children (Gagarina 2008, Gagarina & Reichel 2013) show the productive use of case marking in two-word utterances without a verb.

- (1) [Kot] s sobakoj
cat.NOM with dog.INST
'(A/the cat) with (a/the) dog' (V. 1;09.21¹)
- (2) Po lesenke [The child wants to climb down the ladder]
on ladder.DIM.INST
'(Climb down the) the ladder' (S. 2;00.20)
- (3) Na mamu [Answer to the question 'What are you doing']
on mother.ACC
'(Towards the) mother' (S. 1;10.25)

¹The numbers in the brackets 1;09.21 mean one year, nine months, twenty-one days. Age is given in this format throughout the paper.

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In the course of language development, some children utilize case marking with divergent semantic meanings in prepositional phrases prior to the emergence of predicates. This aligns with the complexity of the Russian case system, which exhibits a high degree of syncretism and interacts with other linguistic domains, such as animacy: animate masculine nouns take genitive-like inflection in the accusative, while inanimate ones take nominative-like inflection. Due to specific phonological processes in spoken Russian, such as vowel reduction and the phenomena known as *akanie* and *ikanie*, the degree of syncretism in spoken varieties is higher. The syncretism and the high amount of nominative or base forms in spoken Russian hinders the acquisition of non-nominative case in Russian, even in monolingual children, leading to a protracted learning process as compared to verb morphology (Ceytlin 2009, Gagarina & Voeikova 2009, Voeikova & Gagarina 2002). Additionally, the higher overall proportion of noun tokens in the nominative in adult speech (Krasil'nikova 1990) further challenges early case acquisition. The acquisition process is further complicated in bilingual children. The reduction of L1 input—often coinciding with kindergarten attendance—limits exposure to case-marked forms in natural communication, potentially delaying successful case acquisition.

This paper aims to examine the trajectories of case in monolingual and bilingual first language acquisition, tracking its development from emergence around the second year of life over a 12-month period until the age of three.

1.1 Theoretical and methodological background: Acquisition of case

Different theoretical approaches to the acquisition of grammar highlight various mechanisms—biological maturation, usage-based, input frequency, or innate grammar—contributing to the development of case in child language. For example, the Co-development Hypothesis posits that subject case marking and subject-verb agreement develop simultaneously in language acquisition. Supporting evidence for this hypothesis comes from the observation that errors in case marking on pronouns and agreement consistently co-occur or are absent in the speech of monolingual English-speaking children (Schütze 1997, Rispoli 1999). This point of view is also represented in Babyonyshev (1993) and Clahsen, Eisenbeiss, and Clahsen et al. (1994) and explains the order of emergence of case forms by the growing complexity of sentence structure. However, for morphologically rich languages like Russian, this hypothesis cannot be maintained, as case marking and agreement do not necessarily develop in parallel. At least some children already exhibit case marking oppositions before they produce two-word utterances with a predicate. Agreement marking is acquired later, with its speed influenced

by morphological richness, particularly in nominal and verbal inflection (Xanthos et al. 2011).

The Co-development Hypothesis is particularly appealing within certain theoretical frameworks, as it aligns with a purely linguistic explanation of acquisition patterns. Specifically, the simultaneous emergence of linguistic elements belonging to the same hierarchical level, such as finiteness and agreement, could reinforce arguments for the modular organization of the mind (Fodor 1983, van van Tiel et al. 2021). In contrast, the Hypothesis of the Switch of the Acquisitional Attention postulates that developmental spurts within different linguistic domains follow or replace one another (Gagarina & Voeikova 2009).

The usage-based theory of language development (Tomasello 2003) emphasizes that first language acquisition is rooted in the frequency and function of linguistic input rather than innate grammatical structures. Children extract patterns from repeated exposure to language in meaningful contexts. They recognize recurring constructions, specifically ‘verb islands’, generalize them, and gradually build knowledge of grammatical categories. Social interaction plays a key role in this process, as children learn by interpreting speakers’ intentions via joint attention. Early grammar is initially item-based, relying on concrete expressions before evolving into abstract structures.

For the acquisition of case, this means that the most frequent, or basic, case forms, which are likely to be nominative, appear as the first forms (see Gagarina & Voeikova 2009 for a comprehensive discussion). Oblique cases emerge later, and their acquisition is triggered by meaning and communicative intent rather than frequency and innateness. In morphologically rich languages, empirical evidence suggests that the accusative case form may serve as the default within the oblique cases (Gvozdev 1949/1981/1981). As a result, the accusative, as an oblique case, appears early and is used for an extended period due to its high frequency in the input a child receives. The question then is whether the accusative case is both the linguistic and the learner default, in the sense of Tsimpli & Hulk (2013), which would in turn lead to its early acquisition.

Before children acquire the rules for inflecting cases and construct a grammar, they use the so-called ‘frozen forms’, i.e., rote learned (memorized without analyzing the components) and contextually bound usage. This early correctness doesn’t imply productivity in the sense that a rule is acquires. However, as input increases, a wider range of case meanings, usage rules, and exceptions emerge. Throughout this process, numerous case errors occur, especially in environments with limited input, such as bilingual settings, where a child may not encounter enough diverse contexts to acquire less frequent and semantically complex cases. Rarely used case forms, which were already acquired, may be temporarily lost

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or used incorrectly because they occur infrequently in the input, are not actively used by the child, and therefore are not firmly established in the child's grammar system. The dynamic of this process is best observed through longitudinal observations of case use. Such a method of analyzing spontaneous longitudinal data is particularly valuable for the precise tracking of the developmental trajectories. This approach has high ecological validity as it relies on natural language use rather than experimental manipulation, ensuring that acquisition is observed in an authentic setting. Additionally, it allows for empirical observations over an extended period, capturing all changes in the use of case morphology across different contexts.

1.2 First language acquisition of case in Russian: Monolingual and bilingual settings

Case in Russian has been extensively studied since the 1940s using both naturalistic spontaneous observations and experimental settings. Earlier studies employed the monolingual diary Gvozdev's (1949/1981) and naturalistic spontaneous data (Gvozdev 1949/1981/1981, Ceytlin & Eliseeva 1996, Ceytlin 2000, 2009, Gagarina & Voeikova 2009, Voeikova 2011) to describe the emergence of cases in monolingual L1 acquisition. Results show that singular NOM forms are predominant in monolingual child production (Gagarina & Voeikova 2009). The first case opposition NOM-ACC occurs before age two and ACC is first marked on feminine nouns (Gvozdev 1961, Voeikova & Gagarina 2002, Gagarina & Voeikova 2009, Voeikova 2011). The spurt takes place roughly 3–4 months after the onset of oblique case production (Gagarina & Voeikova 2009), so that by age 3, all cases can be said to be acquired and productively used, although errors are still registered at age 6 (Ceytlin 2009).

More recent studies have transitioned to experimental techniques focusing on bilinguals' elicited production (Meir et al. 2021, Chrabaszcz et al. 2023) or case processing through eye-tracking methods (Meir et al. 2020) providing more controlled insights into morphological development and real-time language comprehension. One study, Schwartz & Minkov (2014), performed a short-term seven-month longitudinal recording of nine bilinguals and two monolinguals to compare the errors in the acquisition of case in these populations. The results show quantitative differences between monolinguals and bilinguals and within the latter group between simultaneous and sequential bilinguals.

Existing studies present mixed findings regarding Russian case development in bilinguals. Some suggest notable differences from monolinguals, while others indicate similarities. For instance, Anstatt (2006) observed that bilinguals ex-

hibit greater insecurity in assigning nouns to declension classes, and Meng (2001) reported persistent case errors at ages when monolinguals had already mastered them. Additionally, Klassert & Gagarina (2009) found that 25% of Russian-German bilinguals aged 4–6 failed to consistently produce correct case forms, significantly lagging behind monolingual peers, who typically master case by age three. However, data from our two simultaneous bilinguals indicate a similar developmental timeline to monolinguals in acquiring accusative and other oblique cases by the age of three.

The nominative, accusative, and dative cases emerge first in bilinguals, much like in monolinguals, as they have distinct inflectional markings, while genitive and prepositional cases tend to appear somewhat later. These findings contribute to a broader discussion on the acquisition and maintenance of Russian in bilingual settings, including heritage language contexts.

Generally, research on L1 Russian in bilingual contexts across Europe and America has been extensive since the 1980s (Karaulov 1992, Sussex 1993, Zemskaja 1995, Zemskaya 1998, 2008, Polinsky 2000, Steinke 2000, Pfandl 2002, Mustajoki & Protassova 2006, Anstatt 2008, Minor et al. 2024). Much of this research has focused on incomplete acquisition or language attrition, particularly in cases of non-balanced bilingualism, where L1 Russian becomes the weaker language (Polinsky 1997, Anstatt 2008, Armon-Lotem et al. 2011). However, relatively few studies have utilized longitudinal data to examine the earliest stages of L1 grammar acquisition in both monolingual and bilingual contexts. Despite the time-intensive nature of collecting, transcribing, tagging, and analyzing spontaneous speech, such studies are essential for understanding the developmental trajectories of bilingual and heritage speakers.

1.3 The present study

This study addresses gaps in the understanding of early L1 Russian acquisition of case by examining its developmental milestones in both monolingual and bilingual contexts. It compares longitudinal spontaneous speech data from simultaneous bilinguals with data from three age-matched monolingual Russian-speaking children, ranging from their first words to age three. The data were collected under comparable conditions to ensure consistency in methodology. By analyzing naturalistic longitudinal data, this study tracks the path and timing of case acquisition in both settings, allowing for a comparison of case trajectories. Through detailed longitudinal analyses of emerging patterns in case use and clause elements, we explore how young learners acquire both morphological and syntactic systems, focusing on the development of case distinctions.

2 Method

2.1 Participants and data

For this study, longitudinal data from two corpora of spontaneous recordings were analyzed: one bilingual corpus, including two Russian-German bilingual children from high socio-economic status (SES) backgrounds, and one monolingual one, including three children from middle SES families, (for the detailed description of this bilingual corpus, see [Gagarina & Reichel 2013](#); for the detailed description of the monolingual corpus, see [Gagarina 2008](#)). From both corpora, all recordings from the onset of case production (or the first available recordings) up to age three were extracted.

The bilingual corpus consists of speech recordings from two girls, Sonja and Anja (names have been altered). The recordings, made from the onset of the children's first intelligible words until age three, were collected in everyday settings under a one-person, one-language condition. This condition means that each month approximately one 'Russian' hour of communication with the Russian-speaking interlocutor and one 'German' hour of communication with the German-speaking interlocutor were recorded (See [Table 1](#) for overview).

Given that both children have been exposed to regular input in both languages from birth, following the 'one parent-one language' principle, this study focuses on simultaneous bilingualism. Anja started attending a German-speaking kindergarten at 10 months of age; her exposure to German became more intensive as she spent 5 to 7 hours there each day. Her mother continued to speak to her in Russian, while the communication with the father was in German. Frequent visits from German-speaking relatives and the lack of additional sources for Russian-language interactions strengthened the dominance of German. At the age of 2.6 years, Anja's input at home shifted exclusively to Russian, but exposure to German in the kindergarten was still intensive.

Sonja's exposure to Russian differed from Anja's, particularly during the first three years. Prior to attending daycare at age three, Russian input from her mother and grandmother slightly predominated over German. Following enrollment in a German-speaking daycare (max. six hours daily), German exposure began to prevail.

Thus, in Anja's case, we are dealing more with early asymmetrical simultaneous bilingualism, whereas in Sonja's case, it is more likely symmetrical or balanced bilingualism.

The numbers in [Table 1](#) show a difference in development. By the age of three, Sonja already constructs longer sentences (as indicated by MLU in the final three

Table 1: The overview of the bilingual data (adapted from [Gagarina & Reichel 2013](#)). The columns display the following information: "Age of recordings" refers to the age of the child during the recorded session; "1;06" means one year and six months; "Min." indicates the duration of the session in minutes; "N" represents the number of the analyzed utterances; and "MLU" stands for the Mean Length of Utterances measured in words.

	Anja					Sonja				
	Record.	Child	Speech	Input		Record.	Child	Speech	Input	
Age	Min	N	MLU	N	MLU	Min	N	MLU	N	MLU
1;06	0	0	0	0	0	62	333	0,937	993	3,229
1;07	94	1087	1,113	1624	3,065	62	386	0,811	963	3,459
1;08	31	303	1,099	514	2,924	33	271	0,852	640	3,200
1;09	31	252	1,179	444	3,376	17	106	1,066	233	3,060
1;10	47	262	1,221	454	3,482	21	208	1,149	356	3,441
1;11	32	276	1,243	429	3,408	33	292	1,315	457	3,352
2;00	94	619	1,347	842	3,384	62	482	2,006	811	3,619
2;01	94	570	1,401	929	3,551	9	87	3,483	900	2,778
2;02	0	0	0	0	0	59	487	2,090	558	4,470
2;03	0	0	0	0	0	73	608	2,576	655	5,330
2;04	0	0	0	0	0	80	703	2,809	785	4,839
2;05	0	0	0	0	0	62	452	2,856	542	4,913
2;06	0	0	0	0	0	73	552	3,237	613	4,622
2;07	32	258	2,605	260	3,638	81	546	2,652	617	4,173
2;08	22	188	2,489	212	4,231	122	921	2,619	1110	4,328
2;09	27	552	1,656	774	3,707	62	470	3,787	570	4,519
2;10	30	210	1,595	257	3,778	62	410	3,549	567	4,928
2;11	31	267	1,712	359	3,643	21	136	3,919	185	4,578
3;00	45	388	1,784	529	3,621	62	421	2,841	646	4,238
Total/ Mean	610	5232		7627		1056	7871		11310	

months, which averages around three words), while Anja is still in a transitional stage from one-word to two-word utterances with MLU 1.5. In Anja’s corpus, the adult MLU is 3.5, while in Sonja’s corpus, the adult input includes longer utterances, averaging 4.0 words. The dynamics of input MLU is also different; in Sonja it consistently increases, in Anja it stays more or less stable.

The bilingual data were compared to the longitudinal data from the three age-matched monolingual children: Lara (a girl) and two boys, Valja and Vasja (See Table 2). Anja, Sonja and Lara are early talkers, while the boys show slightly slower development in verb production and the development of noun and verb morphology.

The recordings for all children, except Valja, were made between 30 and 90 minutes per month in natural settings. Valja’s recordings, however, exceeded five hours per month between ages 2;2 and 2;4.

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Table 2: The overview of the monolingual data (adapted from [Gagarina 2008](#)). The columns display the following information: "Age of recordings" refers to the age of the child during the recorded session; "1;06" means one year and six months; "N" represents the number of the analyzed utterances; and "MLU" stands for the Mean Length of Utterances measured in words.

Age	Lara				Valja				Vasja			
	Child Speech		Input		Child Speech		Input		Child Speech		Input	
	N	MLU	N	MLU	N	MLU	N	MLU	N	MLU	N	MLU
1;06	81	1,123	531	2,989	89	1,742	1912	2,700	0	0	0	0
1;07	123	1.008	1807	2,821	116	1,966	1499	2,801	0	0	0	0
1;08	117	1.017	359	2,710	430	1,693	3470	2,794	0	0	0	0
1;09	887	1,072	3247	3,192	358	2,008	2247	3,694	0	0	0	0
1;10	377	1,202	902	2,560	415	1,533	1746	5,295	0	0	0	0
1;11	306	1,199	559	3,215	376	1,590	2226	3,767	0	0	0	0
2;00	268	1,679	535	3,175	412	1,842	2010	2,196	40	1,650	731	4,058
2;01	281	2,249	363	3,437	1137	2,159	2489	4,472	327	1,746	1189	6,834
2;02	310	2,590	386	3,461	1253	3,072	1926	5,028	272	1,669	814	4,957
2;03	120	2,833	186	3,301	1400	3,601	2138	4,077	387	2,052	1128	5,505
2;04	304	2,638	476	4,435	1183	3,546	1728	4,702	296	1,885	880	6,275
2;05	248	2,306	605	3,301	1227	3,275	2076	3,630	359	2,111	773	4,989
2;06	257	2,311	692	2,825	953	3,723	1812	3,658	315	2,873	389	6,671
2;07	95	2,179	165	2,970	1027	3,786	1820	2,844	261	3,222	362	6,323
2;08	154	2,565	421	3,152	1338	4,767	2307	3,548	640	3,255	1310	6,669
2;09	675	2,773	1747	3,390	533	4,640	1068	3,236	373	3,284	785	6,788
2;10	588	3,054	902	3,255	785	4,555	1628	4,241	397	3,486	775	6,748
2;11	411	3,951	654	7,946	569	2,860	811	3,109	402	4,077	572	6,259
3;00	451	3,810	805	6,192	454	3,650	851	3,593	386	3,285	584	4,758
Total/ Mean	6053		15342		14055		35764		4455		10292	

All speech data from both the children and the individuals involved in the recordings were transcribed verbatim, double-checked, and stored in CHAT format according to the CHILDES system ([MacWhinney 2000](#)). The monolingual data were collected by colleagues from the Chair of Language Acquisition (Herzen State University, St. Peterburg) and the Russian Academy of Sciences over several decades in the latter half of the last century. The transcription, tagging, and overall curation (financed by the Leibniz-Centre General Linguistics) underwent multiple phases and involved the participation of several researchers. To date, no concise, citable description or publication of the corpus exists, and the corpus itself is not publicly accessible and therefore cannot be cited. The first author of this chapter is currently working in collaboration with colleagues to prepare its publication.

3 Results

The first forms that occur in all children are the base forms; they cannot yet be called the nominative forms because no case oppositions are observed. The first non-nominative case forms in our bilingual data occur at 1;6 with nouns of the 2nd declensional class: *mame* ‘mother-dat’, *babe* ‘grandmother-dat’ with the plateau period of about half a year afterwards. The emerging non-nominative case forms include predominantly accusatives of the 2nd declensional class: *jamu* ‘pit’, *lopatku* ‘trowel-dim’, *palku* ‘stick’, *polku* ‘shelf’, *vodu* ‘water’. The contexts of several accusative case forms of the 1st declensional class are registered, but these forms show the same zero inflection as the nominative: *banan* ‘banana’, *dom* ‘house’, *sok* ‘juice’, *xleb* ‘bread’. Apart from the already mentioned dative forms, new dative nouns occur in the plural by age 2;0: *kozam* ‘goat’ and *det-jam* ‘child’. The locative is used during the six months after the onset of verb production exclusively with the noun ‘floor’: *polu*. The use of genitive inflection in the genitive singular contexts is registered with four nouns only: *Soni* ‘Sonja (the name of the girl)’, *kubika* ‘cube’, *tjoti* ‘woman’, *zajca* ‘hare’ (all are in the singular). And, finally, three nouns of the second declensional class are marked for the instrumental case: *lopatkoj* ‘trowel-dim’, *nogoj* ‘leg’, *rukoj* ‘hand’.

In Sonja’s data at 1;6, one lexeme shows a nominative/dative singular opposition: *baba/babe* ‘grandmother nom/dat’, and one lexeme shows a nominative/dative/accusative singular opposition: *mama/mame/mamu*, ‘mother.nom/dat/acc’. Within the next five months the repertoire of oppositions expands slowly: the forms, appearing in the genitive singular contexts and marked for genitive *tjoti* ‘woman’, *zajca* ‘hare’, *Soni* ‘Sonja (the name)’, *kubika* ‘cube’, and the instrumental form *nogoj* ‘leg’ occur parallel to the nominative forms of these lexemes. Two case and number oppositions are documented: nominative plural/accusative singular *utki/utku* ‘ducks/duck’ and instrumental plural/nominative singular *rukami/ruka* ‘hands/hand’.

In Anja’s data, the majority of initial accusative forms are masculine inanimate, making their marking ambiguous; at 2;11, the accusative of *slona* ‘elephant’ and *ljudej* ‘people’ is registered. Locative emerges as one of the first cases at 1;11, *na traktore* ‘on tractor’ and is used during the entire investigated period; however, instead of the Russian preposition *na* ‘on’, the German preposition *in* ‘on/in’ is used by age three, e.g. 2;11 **in zemle* ‘on earth’, **in dereve* ‘on tree’.

Interestingly, in Sonja’s data, the most common morphological errors involve noun morphology: There are 65 instances in the whole corpora; 33 of them — which constitute ca. 51% — are case errors. These errors include the incorrect use of case function and the incorrect use of case inflection. Remarkably, within the

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first six months of the acquisition of case marking, between 1;6 and 2;0, only a few errors are registered (examples 4, 5), and case inflections are used productively.

- (4) *CHI: Mama, ja posmotrju *Sonja (NOM)
Target: Mama, ja posmotrju Sonju.
mother, I.NOM look.PERF.1SG Sonja.ACC
'Mother, I will watch Sonja (in the videocamera).'
- (5) *CHI: Vot pro *zlodej (NOM)? (S. is 'reading' a book)
Target: Vot pro zlodeja?
Here about.PREP villian.ACC
'Here, it is about the villain?'

The errors number stays roughly the same over the investigated period (See Table 3).

Table 3: Overview of case errors in Sonja's data between age two and three

	Number of errors in tokens	Number of NPs per month	Proportion in percentages
2;00	4	342	1.2
2;02	1	425	0.2
2;03	1	562	0.2
2;04	5	607	0.8
2;05	3	414	0.7
2;06	0	592	0
2;07	5	590	0.8
2;08	8	870	0.9
2;09	1	644	0.2
2;10	4	595	0.7
2;11	1	275	0.4
3;00	0	489	0

Numerous case errors can also be found on adjectives, demonstrative pronouns and other NP dependent components, suggesting issues with the tracking of NP internal agreement. These errors increase with age and cover both animate and inanimate nouns in both bilingual corpora. This might be due to the increase of the more complex NPs with increasing age. Example (6) shows inappropriate

case marking of the animate noun in the accusative: this child uses nominative inflection instead of accusative:

- (6) *CHI: A ja išču *majskij (NOM) *žuk (NOM). (2;5)
 Target: A ja išču majskogo žuka.
 and I look-for.IMPF.1SG ladybug.ACC
 ‘And I’m looking for a ladybug.’

Generally, errors and loss of case marking increase with age on low frequency words or forms. For example, before age 2;3 the instrumental of *plastilina* ‘playdough, modelling material’ was frequently used in constructions like *igrat’ s plastilinom* ‘play with plasticine, modelling material’; from 2;3 onwards the child starts using another preposition, which requires the genitive, but the form stays in the instrumental: *lepit’ iz *plastilina = plastilina* (2;3) ‘model-INF from plasticine-INST’.

We also observe that nouns that were marked correctly at the onset of case production are produced with a non-target nominative inflection. The target marking, however, stabilizes later. The majority of children exhibit such patterns of initial correct inflection and later sporadic use a non-target nominative inflection. This pattern (similar to the optional infinitive in Russian, the phenomenon addressed in Gagarina 2007) occurs across different forms but might have various underlying bases (rote-learned, frequency, etc.).

In L1 acquisition of case in monolingual and bilingual contexts, the first case forms are nominative. These nominative forms can be called the Base Forms and they are used at about 30% in adult-directed Russian. In child-directed speech, parents use base forms in up to 50% of noun tokens (Gagarina & Voeikova 2009). Accusative and dative follow nominative in acquisition in monolingual and bilingual data.

On the whole, the data suggests the three developmental milestones in the longitudinal trajectories:

- Age 1;6 to 2;0 – emergence and accumulation of lexical items and noun inflections.
- Age 2;0 to 2;3/2;5 – emergence and development of productive use of inflectional noun morphology (three-word utterances emerge).
- Age 2;6 onwards, can be characterized as a protracted course of parallel development and higher dynamic (or restructuring) of case in heritage child Russian.

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The sequence in which case oppositions occur and the timing of their acquisition, while generally following developmental milestones, exhibit learner-specific variation and appear to correspond to the inflectional classes of nouns.

4 Discussion and conclusion

The study investigated the early first language acquisition of case in simultaneous bilinguals and compared it to monolingual acquisition of the age-matched children. For this purpose, longitudinal spontaneous data of two simultaneous bilingual children were analyzed and compared to the data of three monolingual children.

The results show a comparable timing and path in the acquisition of case in monolingual and bilingual contexts. The peculiarity of bilingual L1 was the attrition of the already acquired accusative case marking with its later ‘reacquisition’. Data from both monolinguals and bilinguals showed overgeneralizations, a phenomenon that Slobin called ‘inflectional imperialism’ (Slobin 1966). Occurrence of oblique cases follows nominative and accusative. This can be accounted for by positing that children build grammar incrementally by attending to language-specific cues in the input, which also show less syncretism and are more easily identifiable in the input as they have a one-form-one-meaning feature. One example of such case markings is the accusative on feminine nouns with the transparent inflection *-u*. From the perspective of successive spurts, children may initially focus on specific case marking while temporarily plateauing in other areas. For example, a “case spurt” could be a stage where morphological case takes precedence over features like word order. Early reliance on contextual cues supports the idea that case marking develops in phases, with children accumulating a critical mass of building blocks that enable further linguistic progress.

The fact that children first use the correct form and then use a non-target nominative form suggests to us that accusative is not treated as the default, rather nominative is (cf. Chvany 1982, Ljutikova 2017).

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References

- Anstatt, Tanja. 2006. Lexical and grammatical features of Russian speech in bilingual children in Germany (Based on picture-based narratives) (in Russian). In Stella N. Ceytlin (ed.), *Ontolingvistika: Nekotorye itogi i perspektivy*, 19–25. St. Petersburg: IRAY.
- Anstatt, Tanja. 2008. Aspect and tense in storytelling by Russian, German and bilingual children. *Russian Linguistics* 1. 1–26.
- Armon-Lotem, Sharon, Joel Walters & Natalia Gagarina. 2011. The impact of internal and external factors on linguistic performance in the home language and in L2 among Russian-Hebrew and Russian-German preschool children. *Linguistic Approaches to Bilingualism* 1(3). 291–317.
- Babyonyshev, Maria. 1993. Acquisition of Russian case system. *MIT Working Papers in Linguistics* 19. 1–44.
- Ceytlin, Stella N. 2000. *Language and child: Linguistics of child speech*. (in Russian). Moscow: VLADOS.
- Ceytlin, Stella N. 2009. *Essays on word formation and morphology in child speech*. (in Russian). Moskva: Znak.
- Ceytlin, Stella N. & Marina B. Eliseeva. 1996. *Children's speech: Materials for a bibliographic index*. (in Russian). St. Petersburg: Niva.
- Chrabaszcz, Anna, Natalia Ladinskaya & Anna Lopukhina. 2023. Acquisition of Russian noun case by bilingual children: Lexical cues to case assignment in real and novel words. *Language Acquisition* 32(1). 23–51.
- Chvany, Catherine V. 1982. Hierarchies in the Russian case system: For N-A-G-P-D-I, against N-G-D-A-I-P. *Russian Language Journal* 36(125). 23–39.
- Clahsen, Harald, Sonja Eisenbeiss & Anne Vainikka. 1994. The seeds of structure: A syntactic analysis of the acquisition of case marking. In T. Hoekstra & B. D. Schwartz (eds.), *Language acquisition studies in generative grammar*, vol. 8, 85–118. Amsterdam: John Benjamins.

7 Dynamics of case in mono-/bilingual first language acquisition of Russian

- Fodor, Jerry A. 1983. *The modularity of mind: An essay on faculty psychology*. Cambridge: MIT Press.
- Gagarina, Natalia. 2007. What happens when adults often use infinitives? In Insa Gülzow & Natalia Gagarina (eds.), *Frequency effects in language acquisition*, 206–237. Berlin: de Gruyter.
- Gagarina, Natalia. 2008. *First language acquisition of verb categories in Russian*. Saint Petersburg: Nauka.
- Gagarina, Natalia & Marina Eliseeva. 2009. Emergence of case meanings in Russian—Empirical evidence from one month, from one child. *Ad verba libero-rum: Linguistics, Pedagogy and Psychology* 1. 40–47.
- Gagarina, Natalia & Alexandra Reichel. 2013. The hare will melt claws and do you live zdes’? The innovations of the preschool Russian-German bilinguals. In Tatjana A. Krugljakova & T.V. Kuz’mína (eds.), *St. Petersburg school of on-tolinguistics. Festschrift SN Ceytlin*, 195–212. St. Petersburg: Zlatoust.
- Gagarina, Natalia & Maria Voeikova. 2009. Acquisition of case and number in Russian. In Ursula Stephany & Maria Voeikova (eds.), *Development of nominal inflection in first language acquisition: A cross-linguistic perspective*, 179–215. Berlin: Mouton de Gruyter.
- Gvozdev, Alexandr N. 1961. *Voprosy izucheniya detskoy rechi*. Moscow: APN RS-FSR.
- Gvozdev, Alexandr N. 1949/1981. *Ot pervyx slov do pervogo klassa: Dnevnik naučnyx nabljudenij [From the first words to the first grade: Scientific diary]*. Saratov: Izdatel’stvo Saratovskogo universiteta.
- Karaulov, Yuriy N. 1992. Über die russische Sprache des Auslands. *Fragen der Sprachwissenschaft* 6. 14.
- Klassert, Annegret & Natalia Gagarina. 2009. Sprachstandstest bei bilingualen Kindern: „Sprachstand Russisch“. *Patholink* 14(8). 7–9.
- Krasil’nikova, Elena V. 1990. *Imja suščestvitel’noe v russkoj razgovornoj reči [The noun in spoken Russian]*. Moscow: Nauka.
- Ladinskaya, Natalia, Anna Chrabaszcz & Anna Lopukhina. 2019. Acquisition of Russian nominal case inflections by monolingual children: A psycholinguistic approach. *Regional Languages eJournal*.
- Ljutikova, Ekaterina A. 2017. Configurational case in the parametric model. *Vo-prosy Jazykoznanija* 5. (in Russian), 92–122.
- MacWhinney, Brian. 2000. *The CHILDES project: Tools for analyzing talk. Vol. 1: Transcription, format and programs*. 3rd. Mahwah: Lawrence Erlbaum.
- Meir, Natalia, Marina Avramenko & Tatiana Verkhovtceva. 2021. Israeli Russian: Case morphology in a bilingual context. *Russian Journal of Linguistics* 25(4). 886–907.

- Meir, Natalia, Olga Parshina & Irina A. Sekerina. 2020. The interaction of morphological cues in bilingual sentence processing: An eye-tracking study. In Megan M. Brown & Alexandra Kohut (eds.), *Proceedings of the 44th Boston University Conference on Language Development*, 376–389. Somerville: Cascadia Press.
- Meng, Katharina. 2001. *Russlanddeutsche Sprachbiografien: Untersuchungen zur sprachlichen Integration von Aussiedlerfamilien*, vol. 21 (Studien zur deutschen Sprache). Tübingen: Gunter Narr Verlag.
- Minor, Serge, Natalia Mitrofanova & Marit Westergaard. 2024. The interaction of linguistic and visual cues for the processing of case in Russian by Russian-German bilinguals: An eye tracking study. *Topics in Cognitive Science* 17(4). Advance online publication, 855–867. DOI: [10.1111/tops.12724](https://doi.org/10.1111/tops.12724).
- Mustajoki, Arto & Ekaterina Protassova. 2006. *Russkojazyčnyj čelovek v ino-jazyčnom okruženii*, vol. 24. Helsinki: Slavica Helsingiensia.
- Pfandl, Heinrich. 2002. Četyre volny russkoj emigracii XX veka i kul’turno-jazykovoje povedenie emigrantov. In G. Hruslov (ed.), *Russkij jazyk v diaspore: Problemy sohraneniija i prepodavanija*, 9–32. Moskva: Nauka; Flint.
- Polinsky, Maria. 1997. American Russian: Language loss meets language acquisition. *Formal approaches to Slavic linguistics* 4. 370–406.
- Polinsky, Maria. 2000. A composite linguistic profile of a speaker of Russian in the US. In Olga Kogan & Benjamin Rifkin (eds.), *The learning and teaching of Slavic languages and cultures*, 437–465. Bloomington: Slavica.
- Radford, Andrew. 1990. The syntax of nominal arguments in early child English. *Language Acquisition* 1(3). 195–223.
- Rispoli, Matthew. 1999. Case and agreement in English language development. *Journal of Child Language* 26(2). 357–372.
- Schütze, Carson T. 1997. *INFL in child and adult language: Agreement, case and licensing*. Massachusetts Institute of Technology. (Doctoral dissertation).
- Schwartz, Mila & Miriam Minkov. 2014. Russian case system acquisition among Russian-Hebrew speaking children. *Journal of Slavic Linguistics* 22. 51–92.
- Slobin, Dan I. 1966. Grammatical transformations and sentence comprehension in childhood and adulthood. *Journal of Verbal Learning and Verbal Behavior* 5(3). 219–227.
- Steinke, Klaus. 2000. Russisch in der diaspora. In Lew Zybatow (ed.), *Sprachwandel in der Slavia: Die slavischen Sprachen an der Schwelle zum 21. Jahrhundert: Ein internationales Handbuch* (Linguistik International 4), 753–766. Frankfurt a. M.: Lang.

7 Dynamics of case in mono-/bilingual first language acquisition of Russian

- Stephany, Ursula & Maria D. Voeikova (eds.). 2009. *Development of nominal inflection in first language acquisition: A cross-linguistic perspective*. Berlin: Mouton de Gruyter.
- Sussex, Ronald. 1993. Slavonic languages in emigration. In Bernard Comrie & Greville G. Corbett (eds.), *The Slavonic languages*, 999–1035. London: Routledge.
- Tomasello, Michael. 2003. *Constructing a language: A Usage-Based Theory of language acquisition*. Cambridge: Harvard University Press.
- Tsimpli, Ianthi & Aafke Hulk. 2013. Grammatical gender and the notion of default: Insights from language acquisition. *Lingua* 137. 128–144.
- van Tiel, Bob, Max Blokpoel, Iris van Rooij & Andrea E. Martin. 2021. Compositionality, modularity, and the architecture of the language faculty. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 43.
- Voeikova, Maria. 2011. *Early stages of child acquisition of nominal forms of the Russian language*. Moscow: Znack.
- Voeikova, Maria D. & Natalia Gagarina. 2002. Early syntax, first lexicon and the acquisition of case forms by two Russian children. In Maria D. Voeikova & Wolfgang U. Dressler (eds.), *Pre- and protomorphology: Early phases of morphological development in nouns and verbs*, vol. 29 (LINCOM Studies in Theoretical Linguistics), 115–133. München: LINCOM Europa.
- Xanthos, Aris, Sabine Laaha, Steven Gillis, Ursula Stephany, Ayhan Aksu-Koç, Anastasia Christofidou, Natalia Gagarina, Gordana Hrzica, F. Nihan Ketrez, Marianne Kilani-Schoch, Katharina Korecky-Kröll, Melita Kovačević, Klaus Laalo, Marijan Palmović, Barbara Pfeiler, Maria D. Voeikova & Wolfgang U. Dressler. 2011. On the role of morphological richness in the early development of noun and verb inflection. *First Language* 31. 461–479.
- Zemskaja, Elena A. 1995. Once again on the language of the Russian diaspora. In J. N. Kuraulov (ed.), *Language as a system. Language as a text. Language as an ability*, 233–241. (in Russian). Moscow: Vinogradov Russian Language Institute.
- Zemskaya, Elena A. 1998. On the typical features of the Russian language of the first wave of emigrants and their descendants. *Proceedings of the Academy of Sciences, Series of Literature and Language* 57. (in Russian), 39–47.
- Zemskaya, Elena A. 2008. Active processes in the language of the Russian diaspora. In L. P. Krysin (ed.), *Modern Russian language: Active processes at the turn of the 20th–21st centuries*, 615–669. in Russian. Moscow: Languages of Slavic Cultures.

Chapter 8

Movement, morphology and Middle English multilingualism: A perfect storm of grammatical events

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The great restructuring of English grammar emerged via many separate morphosyntactic developments all related to the morphosyntactic finiteness feature, constituting a linguistic “perfect storm” occurring in a period of intense language contact. The finiteness distinction is notoriously hard to acquire for contemporary second language learners of Germanic languages; favoring the prototypical grammatical template where modals are finite. They prefer modals and (dummy) auxiliaries as exponents of finiteness and show a prolonged reluctance to move main verbs. As morphological cues of finiteness eroded in English, infinitives and participial forms of main verbs were lost (first with regular verbs), and many main verbs no longer encoded finiteness. This banned them from the V2 position, where finiteness is an absolute requirement. Do-support emerged to provide finiteness analytically, first to regular verbs. Irregular verbs maintained the finiteness distinction and kept their access to the V2 position for longer.

1 Introduction

1.1 A linguistic landslide

English modals are held to represent a higher stage in a development towards auxiliary status than their Germanic cognates, implying that modals in other Germanic languages will eventually migrate along the same grammaticalization

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path. This paper links the transition of English modals to the larger grammar restructuring of Middle English, where several morphosyntactic developments created a linguistic “perfect storm”. Perfect storms occur very rarely, almost never, and employing this notion suggests that it is rather unlikely to occur again.

Countless theoretical proposals aim to explain why Modern English modals differ from other Germanic modals and modals in earlier stages of English, debating whether the apparent landslide of linguistic changes in the Middle English period constitute one huge reanalysis of (e.g. [Lightfoot 1979](#); [Roberts 1985, 1993](#), [Roberts & Roussou 2003](#)), or many “small catastrophes” (term due to [Westergaard 2009](#); cf. also Warner, 1993). I focus here the interconnection of three linguistic developments in the relevant period: Verb movement becoming restricted to modals and auxiliaries, reduction of verb morphology, and the rise of *do*-support, all solidifying the modal-auxiliary-main verb entrenchment of Modern English.

- (1) Verb movement: Modals and auxiliaries move; main verbs do not
 - a. The monopoly of modals and auxiliaries as finiteness markers
 - b. Only modals and auxiliaries are +finite and may move to C
 - c. A temporary surge in verb movement for AUX and irregular verbs
- (2) Reduction in main verb morphology
 - a. The loss of non-finite forms eradicates the finiteness distinction
 - b. The loss of the finiteness feature in main verbs entails that
 - c. English main verbs lose access to C (regular verbs first)
- (3) The rise of *do*-support
 - a. *Do*-support emerges to provide finiteness analytically
 - b. *Do*-support appears with weak and strong verbs at different rates

I argue that these trends are all related to morphosyntactic finiteness. Moreover, the developments in 1–3 are plausibly triggered by the language contact situation where many adult speakers acquire Middle English as a second language (L2). I outline this contact situation in Section 2. Sections 3, 4 and 5 discuss the phenomena listed under 1, 2, and 3 above. Section 6 sums up and concludes the paper.

2 Modern English as an L2 grammar

2.1 Language contact induces loss of forms

According to Schendl (2012), the extensive restructuring of Middle English (ME) cannot be explained without invoking the massive language contact situation. Although the degree of general bi- and multilingualism on an individual level especially as early as the 13th century is a topic of controversy, many scholars describe England as a highly multilingual country throughout the Middle English period (cf. Schendl 2012: 506ff. and references there). Anglo-Saxons had contact with Celts, traded with speakers of Low Dutch and Low German, and hired Flemish mercenaries. Written influence came from Latin and French – French imprints left in the English vocabulary are unequivocal, and linguistic traces of Old Norse contact are clearly present in Middle English texts.

Thomason (2001: 10) claims that this “extreme language mixing led to a type of new language”, but even taking a more conservative stance, some level of bilingualism or mutual intelligibility is unavoidable under prolonged linguistic contact. Equally inevitable is one widespread impact of extended language contact, i.e. simplification.

Language contact involves multilingualism, and any such language learning situation has language users split the time for available input between two or more languages. Thus, marginal grammatical constructions may fall below a critical acquisition threshold, especially in contexts of extensive adult second language acquisition. Adult L2 learners acquire the prototypical grammatical templates first and foremost, due to “the lousy language-learning abilities of the human adult” (Trudgill 2001: 371–372).

Another hallmark of language contact-induced change is the reduction of formal distinctions across domains. Already Vogt (1948: 39) stated that whenever a language loses formal distinctions, circumstances favor foreign influence as a natural explanation. Language contact tends to reduce morphology, often co-occurring with increased analyticity, favoring the use of auxiliaries at the expense of inflection.

It is reasonable to assume that these mechanisms were operational even in the multilingual society of Medieval England. Without access to L2 speakers of that time, we seek insights from how L2 learners of modern Germanic languages acquire relevant features.

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2.2 L2 learners favor prototypical structures

After ME, English modals had shed their non-finite forms, rendering clausemate embedding impossible, thus inevitably appearing first in a verbal sequence, unlike their Germanic counterparts. In Medieval Germanic modals did appear in non-finite form, but rarely. Coupé & Kemenade (2009), for instance, find infinitival modals in Middle English and Dutch sources from this period, but always very few.

Medieval North Germanic non-finite modals too are grammatical, but infrequent. In the historic Swedish texts spanning from 1220 to 1600 studied by Lagervall (2014), many Old Swedish modals are unattested in non-finite forms, and e.g. *skola* ‘shall’ occurs as an infinitive in 0,2 % of cases, with no past participles. In the modern material *skola* ‘shall’ is always finite, with 91,6 % present tense.

Across modern Germanic, modals rarely appear in non-finite form, hence adult L2 learners often acquire the prototypical template with finite modals, and fossilize at this stage. If ME was similar, adult L2 speakers of ME would produce few or no non-finite modals. The production by high numbers of L2 learners naturally affects the ratios in the input to the next generation of L1 language learners, a reason why language contact induces substantial, rapid changes.¹

2.3 The loss of formal distinctions

Language contact erodes formal distinctions, such as the finiteness distinction in Germanic languages. Many operations in Germanic clause structure draw on the finiteness feature for their execution, and few clause types can be mastered without grasping finiteness (Hagen 2001). L2 Learners however acquire main verb finiteness morphology mostly in the very latest stages of acquisition, leading to prolonged struggles with word order as well as morphology.

In contrast, Germanic children are highly attentive to the finiteness opposition, e.g. infinitival endings (cf. Blom 2003, De Haan 1987), and in their acquisition evidence for finite forms quickly triggers target-like verb raising (Meisel 1997). Mainland Scandinavian languages systematically maintain the finiteness distinction without agreement, displaying that agreement is principally independent from finiteness. Eide (2002, 2005), Eide (2009), Eide (2016) proposes that finiteness is a primitive, akin to Comrie’s (1985) distinction of relative and absolute tense, cf. Table 1.

The integrity of this four-cell paradigm depends on input to productive morphological rules generating all four forms, although an opposition between ei-

¹Gregersen (2020: 115) finds more non-finite modals ME than previously assumed.

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Table 1: Tensed forms in Germanic languages. Norwegian weak verb *like* ‘like’, strong verb *skrive* ‘write’

	+ finite	- finite
+ past	Preterit <i>likte, skrev</i>	Past participle <i>liker, skriver</i>
- past	Present <i>likt, skrevet</i>	Infinitive <i>like, skrive</i>

ther the two past or the two non-past forms, crucially, in the productive paradigm, suffices to uphold the distinction between finite (absolute) and non-finite (relative) tenses. In a Germanic L2 grammar where the finiteness distinction is not acquired, the productive paradigm still has separate forms for non-past and past tenses (*like-liked*) but neither the present-infinitive (*I like- to like*) nor the preterit-past participle (*I liked- I have liked*) are distinct, cf. Table 2.²

Table 2: Productive paradigm of tensed forms in Modern English. Agreement -s in 3rd p. sg. present tense omitted.

+ past	Preterit / Past participle <i>liked</i>
- past	Present / Infinitive <i>like</i>

Blom (2003) discusses how the lack of cues across the finite/non-finite axis delays English children’s acquisition of finiteness-related phenomena. English children acquire finite verbs later than L1 learners of other languages (Paradis & Genesee 1996) and bilingual English children acquire finite forms earlier in their non-English language (Genesee & Nicoladis 2006). That is, like Germanic L2 grammars, the productive paradigm for English main verbs has no finiteness distinction. We discuss the development of this loss below.

2.4 The preference for analyticity

Bilingual language situations induce the replacement of inflectional morphology with analytic forms, favoring the use of auxiliaries while reducing morphologi-

²I take 3rd person sg. Present -s to express agreement, not finiteness. Cf. for instance negative imperatives; they are finite but still disallow agreement (*Don’t/*Doesn’t anyone open the door!*).

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cal markings (cf. Putnam et al. 2021: 618 and references there). This preference for analytic forms over morphology is attested also in L2 acquisition of Germanic finiteness and verb movement, and the correlation between verb raising and finiteness seen in L1 acquisition in is absent from the typical L2 acquisition path of the same languages. L2 learners prefer modals, copulas and auxiliaries as exponents of finiteness occurring to the left of sentential negation, while preferring to leave lexical verbs in situ (cf. Parodi 2000). The typical L2 acquisition path for Germanic verb movement seems little influenced by the morphosyntactic complexity and nature of the finiteness paradigms of the target language and likewise of the first language. Table 3, adapted from Dimroth (2008), sums up these claims, confirmed by a host of studies before and after.

Table 3: Phases of acquisition of verb movement and finiteness in untutored adult learners of L2 German (Dimroth 2008: 121)

Phase	Characteristics
I	Lexical verbs (tentatively marked for morphological finiteness) appear to the right of the negator. Early modals and the copula show up in finite form and position (i.e., to the left of the negator).
II	The perfect is acquired. Auxiliaries (and other nonthematic verbs) appear in finite form and position (i.e., to the left of the negator).
III	Lexical verbs occur with postverbal negation. The acquisition of target adequate morphological marking of finiteness continues.

L2 learners’ preference for analyticity further manifest in their acquiring the periphrastic present perfect before the simple past, and a tendency to replace simple preterits with present perfects. In short, periphrastic constructions are overused in L2 production as compared to native speakers, extending to the innovation of dummy auxiliaries. Early L2 learners of Germanic feature “analytic verb forms consisting of an auxiliary and a main verb in contexts where the target language requires a synthetic verb form” (Blom & de Korte 2011: 906).

Phase II of Table 3 thus shows an often-attested Germanic interlanguage grammar template where finite modals, (dummy) auxiliaries and copulas monopolize the task as finiteness markers occurring before sentential negation, leaving main verbs in situ. It is also the grammar of Standard Modern English.

3 Verb second in Germanic

3.1 Different types of V2

Germanic languages feature a finite verb as the second constituent in declarative main clauses, accounted for by [den Besten \(1983\)](#) as a leftward movement of the verb to the second (V2) position. Although English was never a categorically a V2 language, as pointed out e.g. by [van Kemenade & Westergaard \(2012\)](#) and many others, the proportions of V2 in main clause declaratives dropped radically in a couple of centuries, cf. Table 4 from [Bækken \(1998: 60\)](#). The dating of this decline of V2 in Middle English coincides with the dating for the grammaticalization of modals and other auxiliaries, a fact emerging e.g. from [Warner’s \(1993\)](#) detailed account of the history of English auxiliaries (cf. also [van Kemenade et al. 2023: 16](#)).

Table 4: Overall percentage results for inversion, [Bækken \(1998: 60\)](#)

Year	1000	1200	1500	1600	1700
%	80%	60%	20%	18,5%	8,5%

[Los \(2009: 108\)](#) holds that [Bækken \(1998\)](#), like many similar studies, fails to identify different types of Germanic V2.

- (4)
- a. No verb movement, Norwegian:
... ettersom Jon vanligvis spiser burger.
... since Jon usually eat.PRES. burger
‘... since Jon usually has a burger.’

b. Vfin 1: Subj-V, Norwegian:
Jon spiser vanligvis burger.
Jon eat.PRES. usually burger
‘Jon usually has a burger.’

c. Vfin 1: Subj-V, English:
Jon will usually have a burger.

d. Vfin2: XP-Vfin-Subj, Norwegian
Vanligvis spiser Jon burger.
Usually eat.PRES. Jon burger
‘Usually Jon will have a burger.’

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- e. V_{fin}2: XP-V_{fin}-Subj, English:
Why does John always have a burger?
- f. V_{fin}2: XP-pron-V_{fin}, HomU19 (BlHom 8) 26
Forðon we sceolan mid ealle mod [...] to Gode gecyrran.
Therefore we must with all mind to God turn
'Therefore we must turn to God with all our mind.'
- g. V_{fin}3: XP-V_{fin}-pron, ÆCHom_I_11:271
thonne sceole we huru widhstandan.
then should we at least withstand
'then we should at least withstand.'

The formalist literature on the other hand, describes three different kinds of Germanic V2. Subject-initial V2 moves the finite verb across sentence adverbs, (e.g. *usually*) but not across the visible subject (*Jon*), see (1bc) as compared to the embedded clause in (1a) without movement. A second type of V2 moves the verb across the subject (1de). In Old English certain types of constructions require moving the finite verb across phrasal subjects, but not pronoun subjects (1fg). A third type of V2 is verb movement across even pronoun subjects, e.g. *van Kemenade et al. (2023)*. I refer to this type as *high V2*.

The corpus study of *van Kemenade et al. (2023)* documents a temporary surge of high V2 (V_{fin}3) in ME. High V2 in Old English occurred in clauses with the initial adverbs *þa*, *þonne*, but restrictions on initial adverbs were relaxed in this specific period, allowing other short deictic adverbs in the first position (*here*, *there*, *etc.*). Stricter restrictions on the raised verbs however limit raising to auxiliaries and monosyllabic lexical verbs. I return to this study in Section 5.2 below.

Assuming projections FinP and ForceP on top of IP (*Rizzi (1997)*) different V2 types may be seen as the verb occupying I₀, Fin₀ and Force₀ respectively. Topics, pronominal and phrasal subjects occupy the specifier positions of these heads.

3.2 Properties of FinP and ForceP

Holmberg (2020) sees the V2 rule as a finiteness feature [+F] attracting finite verbs into Fin₀ and a raised phrase to its specifier. I assume that a deictic anchoring feature resides in Fin₀, fixating the temporal relation between the predicate event and the speech event. This is the “invisible function that finiteness serves”, distinguished from “the overt expression that finiteness takes”; *Lasser (1997: 77)*. In Germanic languages (including OE) the anchoring feature in Fin₀ is satisfied via agreement with a +finite verb. In subordinate non-V2 finite clauses this happens at a distance (cf. 1a), thus Fin₀ itself is not responsible for V2. *Truckenbrodt*

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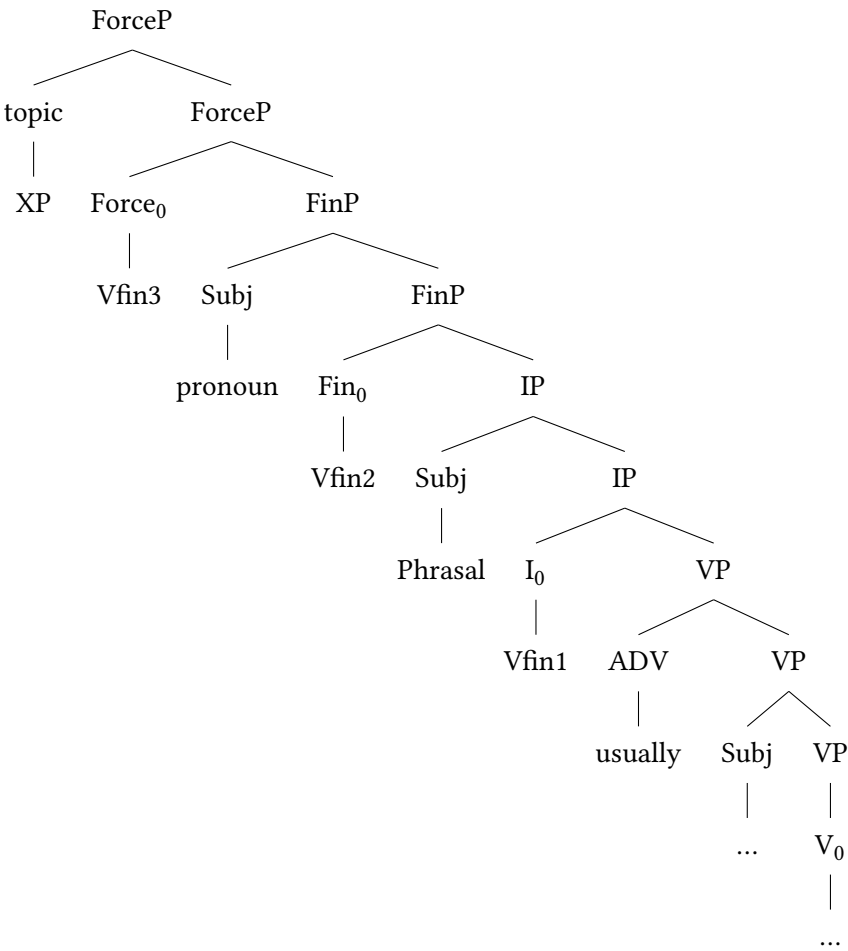


Figure 1: Three types of Germanic V2

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(2006) claims that the function of V2 is to check illocutionary force. I assume that ForceP hosts these operators (declarative, imperative, and interrogative) checking their features by agreeing with the valued features in Fin_0 under a locality restriction, attracting the finite verb to Fin_0 .

Force₀ has different requirements depending on the operator. I stipulate that Fin_0 too has two different instantiations in Modern English, positive and negative. English has negation *not*, appearing with most constituent types including non-finite verbs, and contracted negation *n't*, sentential (Σ) negation, which requires a +finite host. Σ negation is often taken to head its own clausal projection; instead, I analyze *n't* as a negative instantiation of Fin_0 . A close relation exists between Σ negation and finiteness. Dahl's (1979) survey of 240 languages in 40 language families reveals that negators are usually placed in direct relation to the finite verb, and specific finite negations are typologically widespread (Miestamo 2017: 412). In Finnish negative finite clauses, the agreement features of the lexical verb transfer to negator *ei*, parallel to English finite negation *n't*.

- (5) a. Finnish
Koirat haukkuvat ulkona.
dog-PL bark-3PL outside
'The dogs are barking outside.'
- b. Finnish
Koirat ei-vät hauku ulkona.
dog-PL neg-3PL bark. [connegative form] outside
'The dogs are not barking outside.'
- c. English
The dog barks outside.
- d. English
The dog doesn't bark outside.

4 Changes in morphology

4.1 Verb second, finiteness and agreement

OE had a rich system of person forms (up to 11 different finite forms serving 26 different functions (moods, tenses, persons, and numbers; Lass 2006: 75ff.)). Number marking is lost in the early sixteenth century; (ibid. 79) and only 3rd person sg. present remains.

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In Germanic “the second position of the clause is only available for finite verbs” (Weermann 1988: 14), i.e., the [+F] feature on [Fin + Force_{DECL}] requires matching with a verb positively encoding morphological finiteness.³ A natural idea is relating the loss of generalized V2 in English main clause declaratives to eroded finiteness markings, and many attempts to this effect exist. E.g. the RAH tradition (*Rich Agreement Hypothesis*; Rohrbacher 1999: 116), holds that verb movement needs certain “threshold features”, e.g. 1st/2nd person to be preserved.⁴

As pointed out e.g. by McWhorter (2005: 287ff.), any explanation relating verb movement solely to agreement runs counter to MSc, where both V2 and the finiteness distinction are upheld without agreement. Although the the Generative framework traditionally sees finiteness as an amalgam of tense and agreement, neither agreement nor tense is in fact necessary or sufficient for finiteness. MSc finite tenses show no agreement (+F, -AGR), while Portuguese employs agreeing infinitives (-F, +AGR). Latin used tensed infinitives (-F, +T), and many languages use mood, not tense, for finiteness (cf. Eide 2016 for details).

4.2 The loss of morphological finiteness

The loss of person markers in English clearly induces finite/non-finite syncretism. However, the crucial change is not the erosion of agreement in itself, but the loss of the finiteness distinction, entailing that non-finite parts of the tense paradigm are also relevant. Throughout Middle English finite and non-finite forms increasingly become non-distinct. The plural marking of verbs clearly declines, however, the infinitival ending *-en* also drops, from 100 percent to zero, cf. tables 5 and 6.^{5,6}

³I propose that around 1550 an abrupt change in the feature specifications of the illocution operator DECL relaxes the +F feature requirement. Before 1550 checking of [+F] attracted a +finite verb to Fin₀. After 1550 the anchoring feature in Fin₀ is satisfied via non-local AGREE with a non-conflicting verb form with only secondary features, i.e. T and AGR features, like *likes* (“affix hopping”), while the negative instantiation of Fin₀ lexicalized as contracted negation still requires +F (i.e. a modal or a finite auxiliary).

⁴Here I have crudely glossed over the fact that Rohrbacher’s proposal, like many works building on this idea, concerns movement of verbs to the I-domain (not to the C-domain). Subsequent movement to the C-domain however requires the verb to first move to IP, which is why this is relevant to the V2 rule.

⁵Roberts (1993: 261) and Roberts & Roussou (2003: 41) note that “shortly after 1500, the former infinitive ending *-e(n)* had disappeared”, a loss they relate to the restructuring of modal structures (cf. i). The loss of distinct past participles however is hardly ever discussed in this context, though Lass (2006: 57) notes that infinitives and participles constitute “the principal parts” alongside finite forms.

(i) ${}_{TP}$ Sone [${}_{TP}$ hit mæi [${}_{VP}$ t_{mæi} [${}_{TP}$ T [${}_{VP}$ limpen]
]]] → [${}_{TP}$ Soon [${}_{TP}$ it may [${}_{VP}$ happen]]]

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Table 5: Infinitival endings in English texts (adapted from Lass 2006: 80)

Text	year	% -en	% zero
PC	1154	100	0
Astrolabe	1381	44	56
Grocers'	1418	25	75
Caxton Prol	1473	2	98

Table 6: Plural endings in English texts (adapted from Lass 2006: 78)

Text	year	% -en	% zero
PC	1154	95	5
Astrolabe	1381	84	16
Grocers'	1418	52	48
Caxton Prol	1473	28	72

The prefixes and suffixes of past participles disappear at different times and paces across different texts and dialects; cf. Wojtyś (2016), for a detailed account, showing for instance that the prefixing of past participles (mainly *ge-*) go from 100 % in *The Proclamation of Henry III* (1258) to 1.3 % prefixing in *Astrolabe* (1381). Since Wojtyś has very many different tables, detailing for instance sources from the various regions of the British Isles, and since the format of the present paper is limited, I decided not to include her tables here, but note that her sources are only partially overlapping with the sources referenced by e.g. Lass (2006).

The short and blunt story is that as finite verbs lost person inflections and non-finite verbs lost their affixes, the two set of forms become non-distinct. The loss of person features and the loss of specific markings of non-finite forms taken together in effect eradicated finiteness in productive i.e. regular English main verb paradigms.

In the paradigm of irregular verbs the distinction is upheld even in modern times. Although non-productive already in Middle English as new verbs were inflected as regular (cf. table b), irregular verb paradigms were also notoriously

⁶Samples from *Petersborough Chronicle* (PC), *Chaucer's treatise on the astrolabe*, *The grocer's ordinances* and a selection of *Caxton's prologues*; cf. Lass (2006) for details.

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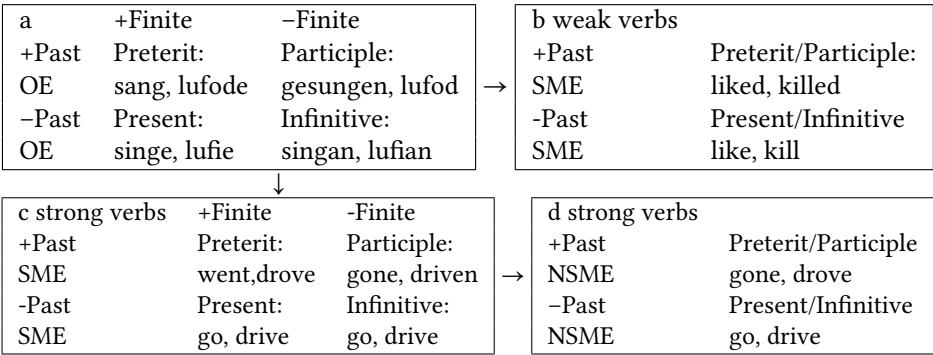


Figure 2: Loss of main verb finiteness. OE (Old English), SME (Standard Modern English), NSME (Non-standard Modern English)

unpredictable. Grammarians in the 16th to 18th centuries note the vast variability for irregular past-participle forms, cf. Lass (2006: 99ff.). Even where the distinction between finite and non-finite past exists, the two forms appear as if at random, the past participle affix *-en* is often missing, *writ(en)*, *bit(ten)*, *broke(n)*, and many verbs feature one past form: *began*, *took*, *drove*, *got*, *sang*, *won*, *flung*, *found*. The preterit/past participle distinction hence was hence unreliable already in late Middle English.

Leveling of irregular preterits and past participles (*You coulda went*, *We haven't spoke*, *Here's what I would have wrote* etc), is observed across modern English varieties, by authors across frameworks; e.g. Eide (2016), Tortora (2018), Chatten et al. (2024). Invoking Figure 2, leveling represents two +past forms competing for one cell in the paradigm, completing a development at work since the 14th century (on finiteness paradigms in Germanic languages, cf. Eide 2009).

5 The emergence of do-support

5.1 Do-support as analytical finiteness

In this section, I advocate the idea that *do*-support arises as a means to provide finiteness analytically in a clausal structure where main verbs are longer finite, but certain constructions still require a finite verb.

Assume that Force and Fin continue to require morphological finiteness to check [+F] in a set of syntactic operations for a significant number of speakers. Then a repair strategy must emerge to provide [+F] where main verbs, specifically regular ones, no longer provide it. This paves the way for *do*-support, cf. Figure 3.

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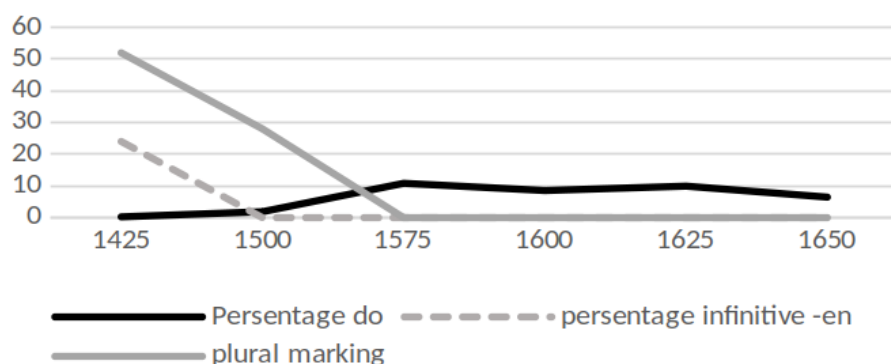


Figure 3: Spread of *do*-support, loss of infinitive *-en* and plural marking. Numbers from Ellegård (1953: 160–161) and Lass (2006: 78, 80)

Periphrastic *do*-constructions exist in other Germanic languages, but the kind of *do*-support observed in present-day English is quite unique (e.g. McWhorter 2005). Ellegård's (1953) study on the emergence of *do*-support in the history of English sparked speculations into how this grammatical feature arose. Many authors suspect that *do*-support is somehow related to the loss of general main verb movement, but a causal chain has been hard to state. Also, Warner (1993: 221) notes:

The coincidence of date [between the grammaticalization of modals as auxiliaries and the rise of *do*-support] strongly suggest that modals and *DO* are interconnected. Any linguistic history must give some account of this interconnection if it is to be convincing.

In the present context, the uniqueness of English *do*-support stems from the fact that English is alone among the Germanic languages in not maintaining the finiteness distinction for the productive paradigms of main verbs. The interrelation between English modals and *do*-support is that both are intrinsically finite, with no non-finite forms.

Many proposals link the common behavior of modals and *do* to their status as auxiliaries, but this is insufficient to describe their behavior in other contexts. For instance, modals and *do* are all banned from e.g. subjunctives (3); (data from Schütze 2003; Eide 2016). Crucially, non-finite auxiliaries *have* and *be* (and main verbs) are fine in these constructions. Hence the lack of non-finite forms, not auxiliary status, prevents modals and *do* from appearing here. The fact that contracted (i.e. finite) negation is banned from these structures as well (?), corroborates that the +finite feature itself is the offensive trait.

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(6) Subjunctives

- a. It is vital that John be here on time.
- b. It is vital that John be smiling on the photograph.
- c. It is vital that Rover have/*haven't eaten before we arrive.
- d. * It is vital that John do not be late.
- e. # It is vital that John will not come unprepared (*subjunctive).
- f. It is vital that John do the dishes.
- g. It is vital that he show up.

This is not to say that auxiliary status is unrelated to morphological finiteness. Rather, the verbs ending up as auxiliaries (modals, *have* and *be*) were all irregular when they started to behave syntactically different from regular main verbs. Irregular verbs maintained their finiteness distinction (cf. Section 4.2.), and their access to the I and C domains of the clause; in short, were able to undergo verb movement.

5.2 Regular v. irregular, raising and *do*

According to Figure 2 above, ME regular verbs lost, while irregular verb paradigms maintained, a morphological finiteness distinction. If finiteness grants access to Fin_0 and Force_0 , regular and irregular verbs should behave differently regarding *do*-support and raising. This is what we find – for a certain period of time.

Do-support occurs more often, and earlier with regular (weak) verbs than with irregular (mostly strong) strong verbs, and the historic trajectories of these two verb types with *do*-support are quite different.

Moretti (2024) investigates *do*-support from 1350–1499, arriving at “the top 20 most attracted infinitives of auxiliary *do* that are statistically significant” (ibid. 68). These are verbs of low frequency, often of French origin. Verbs imported into a language typically adhere to the regular paradigm, and almost all the 20 verbs are regular (*apperen*, *understanden*, *spescien*, *crien*, *bigilen*, *seisen*, *deklaren*, *shouen*, *comen*, *appropren*, *dedicaten*, *overmaccen*, *recchen*, *racen*, *wecchen*, *asken*, *braggen*, *maagnifien*, *reneuen*, *satisfien*). *Understanden* appears as formulas of one writer and is less relevant.

Do-support occurred in at most 10% of all affirmative declaratives, although the frequency of such clauses made affirmative declarative *do* outnumber the other construction patterns by a large margin; cf. Budts (2021: 13). The “low” percentage of *do*, relatively speaking, is likely to be due to *do*-support occurring

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mainly with the low-frequent, regular verbs. Even in Present-Day English the most common verbs are irregular verbs.⁷

Warner (2011) investigates *do*-support with regular and irregular verbs across in ME negative declaratives. His numbers are shown in Figure 4, demonstrating that regular and irregular verbs behave differently as regards *do*-support at least in this construction type (Ellegård, the source of these data, fails to give the corresponding percentages for affirmative declaratives).

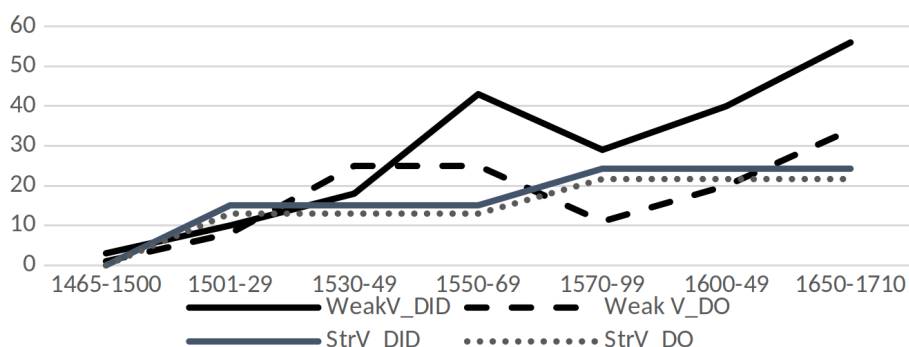


Figure 4: Percentages of the past and present of the third person singular *do* in negative declaratives. Based on the numbers in Warner (2011: tables 18-1 & 18-3)

Figure 5 illustrates the time window where plural markings and infinitival markings are lost for Middle English main verbs. The figure also shows how *do*-supports emerges in the same period.

Section 3.1 mentioned a temporary surge of high V2 (Vfin3) over a period in late Middle English observed by van Kemenade et al. (2023). This surge is clearly visible in one of the graphs in Figure 5. According to these authors, the OE restriction on specific initial adverbs was relaxed, leading to more raising, while the class of verbs raised was restricted to “auxiliaries and monosyllabic lexical verbs”. That is, more raising, but only for the few.⁸

⁷Lieberman et al. (2007: 714): “Although less than 3% of modern verbs are irregular, the ten most common verbs are all irregular (*be, have, do, go, say, can, will, see, take, get*).” Of the 50 most frequent English today around 35 are irregular.

⁸The split between auxiliaries (modals, *have* and *be*) and lexical verbs is presumably quite easy to understand since these two verbal types have very different distribution in English, emerging in this period. The reasons to distinguish between unaccusative verbs and others may seem less self-evident, but unaccusative verbs often feature postverbal subjects (even in Present-

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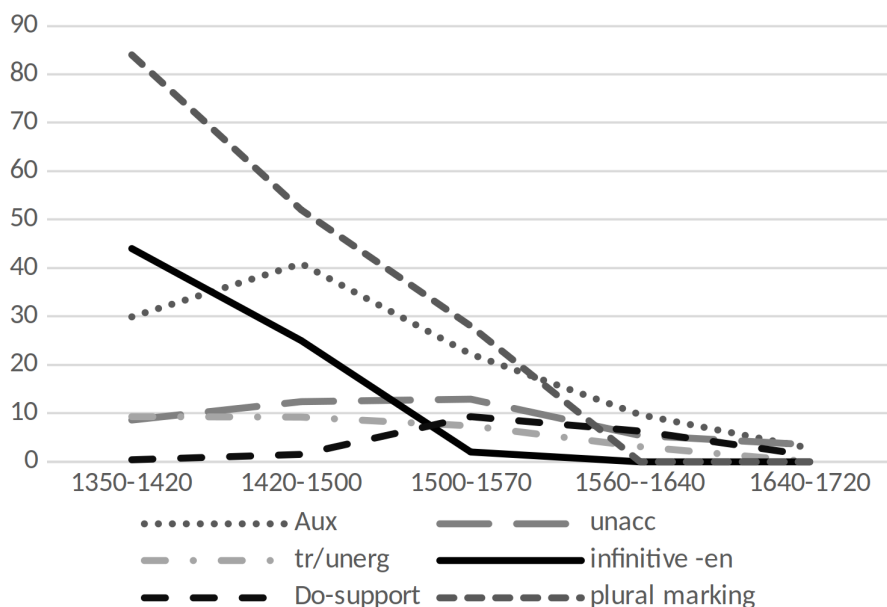


Figure 5: High V2, infinitive -en, plural marking and do-support. Numbers from [van Kemenade et al. \(2023: 13\)](#), [Ellegård \(1953: 159\)](#) and [Lass \(2006: 78,80\)](#)

[van Kemenade et al.](#) explain the grouping of verbs allowed to raise (auxiliaries and certain lexical verbs) via prosody and stress patterns. However, observing that monosyllabic verbs are also overwhelmingly irregular may provide another explanation of why these lexical verbs specifically continued to raise for an extended period in Middle English. Irregular verbs that raise are short, but they also show finiteness, unlike regular verbs.

6 Conclusion

In this paper I discuss how several features of Modern English syntax are shared with Germanic second language interlanguage grammars, e.g. the preference for analytic constructions over morphological inflections, the reluctance to move

Day English), resembling a verb-raising structure while often, *de facto*, featuring a different construction altogether. Also, a great percentage of the relevant unaccusatives are copulas ([van Kemenade 2012: 826](#), [van Kemenade et al. 2023: note 1](#)), very often BE; also irregular, overtly finite, and monosyllabic.

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main verbs, the loss of formal distinctions, and the absence of the main verb finiteness distinction.

Do-support and verb raising have been approached from many different angles. The hypotheses put forth in the present paper advocate a direct relation between morphology and syntax which have been seen in the Generative Enterprise as something of a Holy Grail, cf. Holmberg & Roberts (2013: 112):

Morphological differences [...] are extremely easy to observe [...] for children acquiring their first language[...] Indeed, there is clear evidence from studies of children's language acquisition that they are highly sensitive to morphology, particularly inflectional morphology [...], [and given] a systematic relation between morphological structure and syntactic structure, morphological differences may reflect more abstract syntactic differences among languages.[...]. [If so], morphological properties may be an important cue employed in the acquisition of syntax.

I have described certain developments of the extensive restructuring of English grammar during Middle English from a morphosyntactic angle. Particularly the new perspective on the finiteness distinction as a primitive distinction in principle independent from agreement offers potential for new insights. The study predicted and attested how regular and irregular verbs behave differently as regards *do*-support and verb movement, because of their morphological finiteness.

The patterns observed here are temporary and only account for parts of the emergence of the great restructuring of English grammar. Later these patterns become undone; irregular main verbs cease to count as “finite enough” to maintain their access to the C-domain, and occur with *do*-support at the same rate as regular verbs. Modals and *do* lose their non-finite forms, hence in a sense also lose their finiteness distinction, *have* ends up having no more of a morphological finiteness distinction than regular verbs, *be* maintains its many forms.

However, it does seem significant that the verbal elements that finally end up as auxiliaries share with monosyllabic lexical verbs partaking in verb raising in ME, irregular paradigms that maintain the finiteness distinction, and the developments I have addressed may be turn out to have played a more substantial part in the greater picture.

References

- Bækken, Bjørg. 1998. *Word order patterns in Early Modern English*. Oslo: Novus.
 Blom, Elma. 2003. *From root infinitive to finite sentences: The acquisition of verbal inflections and auxiliaries*. University of Utrecht. (Doctoral dissertation).
 Blom, Elma & Siebe de Korte. 2011. Dummy auxiliaries in child and adult second language acquisition of Dutch. *Lingua* 121(5). 906–919.

8 *Movement, morphology and Middle English multilingualism*

- Budts, Sara. 2021. *On periphrastic do and the modal auxiliaries: A connectionist approach to language change*. Universiteit Antwerpen. (Doctoral dissertation).
- Chatten, Alicia, Kimberly Baxter, Erwanne Mas, Jailyn Pena, Guy Tabachnik, Daniel Duncan & Laurel MacKenzie. 2024. “i’ve always spoke like this, you see.” Preterite-to-participle levelling in American and British Englishes. *American Speech* 99(1). 3–46.
- Comrie, Bernard. 1985. *Tense*. Cambridge: Cambridge University Press.
- Coupé, Griet & Ans Van Kemenade. 2009. Grammaticalization of modals in English and Dutch: Uncontingent change. In Paola Crisma & Giuseppe Longobardi (eds.), *Historical syntax and linguistic theory*, 250–270. Oxford: Oxford University Press.
- Dahl, Östen. 1979. Typology of sentence negation. *Linguistics* 17. 79–106.
- De Haan, George. 1987. A theory-bound approach to the acquisition of verb placement in Dutch. In Ger de Haan & Wim Zonneveld (eds.), *Formal parameters of generative grammar: OTS yearbook 1987*, 15–30. Dordrecht: ICG.
- den Besten, Hans. 1983. On the interaction of root transformations and lexical deletive rules. In Werner Abraham (ed.), *On the formal syntax of the Westgermania*, 47–131. Amsterdam: Benjamins.
- Dimroth, Christine. 2008. Age effects on the process of L2 acquisition? Evidence from the acquisition of negation and finiteness in L2 German. *Language Learning* 58(1). 117–150.
- Eide, Kristin M. 2009. Finiteness: The haves and the have-nots. In Artemis Alexiadou, Jorge Hankamer, Thomas McFadden, Justin Nuger & Florian Schäfer (eds.), *Advances in comparative Germanic syntax* (Linguistik Aktuell 141), 357–390. Philadelphia: John Benjamins.
- Eide, Kristin Melum. 2002. *Norwegian modals*. Norwegian University of Technology & Science. (Doctoral dissertation).
- Eide, Kristin Melum. 2005. *Norwegian modals*. Berlin: Mouton De Gruyter.
- Eide, Kristin Melum. 2016. Finiteness, inflection, and the syntax your morphology can afford. In Kristin Melum Eide (ed.), *Finiteness matters: On finiteness-related phenomena in natural languages*, 121–168. Amsterdam: John Benjamins.
- Ellegård, Alvar. 1953. *The auxiliary do: The establishment and regulation of its use in English*. Stockholm: Almqvist & Wiksell.
- Genesee, Fred & Elena Nicoladis. 2006. Bilingual acquisition. In E. Hoff & M. Shatz (eds.), *Handbook of language development*, 324–342. Oxford: Blackwell.
- Gregersen, Sune. 2020. *Early English modals: Form, function, and analogy* (LOT Dissertation Series 579). Amsterdam: LOT.

Kristin Melum Eide

- Hagen, Jon Erik. 2001. Finittkategoriens kritiske karakter i norsk som andrespråk. In Helene Uri & Anne Golden (eds.), *Andrespråk, tospråklighet, norsk: Festskrift til Anne Hvenekilde*, 45–57. Oslo: Unipub Forlag.
- Holmberg, Anders. 2020. On the bottleneck hypothesis of verb second in Swedish. In Rebecca Woods & Sam Wolfe (eds.), *Rethinking verb second*, 40–60. Oxford: Oxford University Press.
- Holmberg, Anders & Ian Roberts. 2013. The syntax-morphology relation. *Lingua* 130. 111–131.
- Lagervall, Marika. 2014. *Modala hjälpverb i språkhistorisk belysning*. Göteborgs universitet. (Doctoral dissertation).
- Lass, Roger. 2006. Phonology and morphology. In Richard Hogg & David Denison (eds.), *A history of the English language*, 43–108. Cambridge: Cambridge University Press.
- Lasser, Ingeborg. 1997. *Finiteness in adult and child German* (MPI Series in Psycholinguistics 8). Wageningen: Ponsen & Looijen.
- Lieberman, Erez, Jean-Baptiste Michel, Joe Jackson, Tina Tang & Martin Nowak. 2007. Quantifying the evolutionary dynamics of language. *Nature* 449. 713–716.
- Lightfoot, David. 1979. *Principles of diachronic syntax*. Cambridge: Cambridge University Press.
- Los, Bettelou. 2009. The consequences of the loss of verb-second in English: Information structure and syntax in interaction. *English Language and Linguistics* 13(1). 97–125.
- McWhorter, John. 2005. *Defining creole*. Oxford: Oxford University Press.
- Meisel, Jürgen. 1997. The acquisition of the syntax of negation in French and German: Contrasting first and second language development. *Second Language Research* 13(3). 227–263.
- Miestamo, Matti. 2017. Negation. In Alexandra Y. Aikhenvald & R. M. W. Dixon (eds.), *The Cambridge handbook of linguistic typology*, 405–439. Cambridge: Cambridge University Press.
- Moretti, Lorenzo. 2024. A quantitative exploration of the functions of auxiliary do in Middle English. *English Language and Linguistics* 29(1). 1–21.
- Paradis, Johanne & Fred Genesee. 1996. Syntactic acquisition in bilingual children. *Studies in Second Language Acquisition* 18(1). 1–25.
- Parodi, Teresa. 2000. Finiteness and verb placement in second language acquisition. *Second Language Research* 16(4). 355–381.
- Putnam, Michael T., Lara Schwarz & Andrew D. Hoffman. 2021. Morphology of heritage languages. In Silvana Montrul & Maria Polinsky (eds.), *The Cambridge handbook of heritage languages and linguistics*, 613–643. Cambridge: Cambridge University Press.

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- Rizzi, Luigi. 1997. The fine structure of the left periphery. In Liliane Haegeman (ed.), *Elements of grammar: Handbook of generative syntax* (Kluwer International Handbooks of Linguistics), 281–337. Dordrecht: Kluwer. DOI: [10.1007/978-94-011-5420-8_7](https://doi.org/10.1007/978-94-011-5420-8_7).
- Roberts, Ian. 1985. Agreement parameters and the development of English modal auxiliaries. *Natural Language and Linguistic Theory* 3. 21–58.
- Roberts, Ian. 1993. *Verbs and diachronic syntax*. Dordrecht: Kluwer.
- Roberts, Ian & Anna Roussou. 2003. *Syntactic change: A minimalist approach to grammaticalization*. Cambridge: Cambridge University Press.
- Rohrbacher, Brenhard W. 1999. *Morphology-driven syntax: A theory of V-to-I raising and pro-drop*. Amsterdam: John Benjamins.
- Schendl, Herbert. 2012. Middle English: Language contact. In Alex Bergs & Laurel Brinton (eds.), *Historical linguistics of English* (HSK 34.1), 505–551. Berlin: de Gruyter.
- Schütze, Carson T. 2003. When is a verb not a verb? *Nordlyd* 31(2). 400–415.
- Thomason, Sarah G. 2001. *Language contact. An introduction*. Edinburgh: Edinburgh University Press.
- Tortora, Christina. 2018. Evidence for generalized verbal periphrasis in English. In Huba Bartos, Marcel Den Dikken, Zoltán Bánréti & Tamás Váradí (eds.), *Boundaries crossed, at the interfaces of morphosyntax, phonology, pragmatics and semantics* (Studies in Natural Language and Linguistic Theory), 163–180. Cham: Springer.
- Truckenbrodt, Hubert. 2006. On the semantic motivation of syntactic verb movement to C in German. *Theoretical Linguistics* 32(3). 257–306.
- Trudgill, Peter. 2001. Contact and simplification: Historical baggage and directionality in linguistic change. *Linguistic Typology* 5(2/3). 371–374.
- van Kemenade, Ans. 2012. Rethinking the loss of verb second. In Terttu Nevalainen & Elizabeth Closs Traugott (eds.), *The Oxford handbook of the history of English*, 822–834. Oxford: Oxford University Press. DOI: [10.1093/oxfordhb/9780199922765.013.0067](https://doi.org/10.1093/oxfordhb/9780199922765.013.0067).
- van Kemenade, Ans, Roland Hinterhölzl & Tara Struik. 2023. Word order change, architecture and interfaces: Evidence from the development of V to C movement in the history of English. *Journal of Historical Syntax* 7(25). 1–55. DOI: [10.18148/hs/2023.v7i25.189](https://doi.org/10.18148/hs/2023.v7i25.189).
- van Kemenade, Ans & Marit Westergaard. 2012. Syntax and information structure. In Anneli Meurman-Solin, Maria Jose Lopez-Couso & Bettelou Los (eds.), *Information structure and syntactic change in the history of English*, 87–118. Oxford: Oxford University Press.

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- Vogt, Hans. 1948. Dans quelles conditions et dans quelles limites s'exercer sur le système morphologique d'une langue l'action du système morphologique d'une autre langue? In Michel Lejeune (ed.), *Actes du VIe Congrès Internationale de Linguistes*, 458–460. Paris: Klincksieck.
- Warner, Anthony. 1993. *English auxiliaries: Structure and history* (Cambridge Studies in Linguistics 66). Cambridge: Cambridge University Press.
- Warner, Anthony. 2011. Do with weak verbs in early Modern English. In David Denison, Ricardo Bermúdez-Otero, Chris McCully & Emma Moore (eds.), *Analyzing older English*, 285–305. Cambridge: Cambridge University Press.
- Weermann, Fred. 1988. *The V2 conspiracy. A synchronic and a diachronic analysis of verbal positions in Germanic languages*. Berlin: De Gruyter Mouton.
- Westergaard, Marit. 2009. Word order in Old and Middle English: The role of information structure and first language acquisition. *Diachronica* 26(1). 65–102.
- Wojtyś, Anna. 2016. *Past participle marking in Medieval English. A corpus-based study in historical morphology*. San Diego: Æ Academic Publishing.

Chapter 9

Eliciting grammatical gender: Some methodological reflections

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A frequently used behavioral task to study grammatical gender involves a picture-based elicitation task. For Norwegian, Rodina & Westergaard (2015) adopted a paradigm originally developed by Blom et al. (2008) for grammatical gender. Since then, this paradigm has been used by many scholars (Busterud et al. 2019, Rodina & Westergaard 2021, van Baal et al. 2023) to elicit indefinite and definite determiners and pronominal and postnominal possessives. As part of the research project Grammatical Gender in Norwegian Dialects: Variation, Acquisition & Change (GenVAC), we developed this methodology further so that we also could elicit plurals and anaphoric pronouns. Elicitation data have been crucial in demonstrating that the Norwegian grammatical gender system is in the process of being reduced from three grammatical genders (masculine, feminine, and neuter) to two (common and neuter). In this chapter, we discuss methodological challenges that emerge when trying to capture a changing grammatical gender system using elicitation tasks. We will discuss the methodological choices made in the GenVAC project, including both positive and negative consequences. In particular, we see that participants produce more feminine markers as the experiment session progresses, and we discuss some factors that may explain this result, involving the availability of more than one gender system in the input, lexical effects, the role of the interviewer, and task effects more generally.

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1 Introduction

Across many Norwegian dialects, grammatical gender is changing from a three-gender system to a two-gender system (Lødrup 2011, Rodina & Westergaard 2015, 2021, Busterud et al. 2019, van Baal et al. 2023, 2024, Solbakken et al. 2024). The two-gender system involves a merger of masculine and feminine gender, which for instance can be seen in that the indefinite feminine article *ei* is replaced by the masculine indefinite article *en* in front of traditionally feminine nouns. That is, *ei bok* ‘a book’ becomes *en bok*.

Most of this research has relied on a picture-based elicitation task where it is possible to control the distribution of nouns that are included. Rodina & Westergaard (2015) and Busterud et al. (2019) did two experiments; the first one tested the presence of feminine gender in the dialect, whereas the second tested if speakers are sensitive to semantic and morphophonological gender cues with four subclasses of traditionally feminine nouns. Interestingly, Busterud et al. (2019: 156) observe the following: ‘... the rates for the indefinite *ei* are somewhat higher in Experiment 2 than in Experiment 1 across all age groups, possibly due to self-priming’. They suggest that this finding may be because only feminine nouns were included in the second experiment, with the exception of four neuter fillers, whereas the first experiment included all three genders equally distributed. This finding motivates the present contribution, which seeks to investigate whether the trend whereby the rate of feminine marking varies between experiments also holds true for the ongoing project *Grammatical Gender in Norwegian Dialects: Acquisition & Change* (GenVAC). This is a large cross-dialectal investigation of grammatical gender where several different types of experiments are conducted to probe the status of especially feminine grammatical gender. In the current chapter, we ask whether the production of feminine gender forms increases in the course of the experiments? We will investigate this based on data from two locations that have been shown to be in the middle of changing from a three-gender to a two-gender system (van Baal et al. 2024): Lyngdal in the south of Norway and Bodø in the north.

This chapter is organized as follows. Section 2 provides a brief overview of the relevant tests from which we are using data in the present contribution. Section 3 discusses elicitation as a methodology in more general terms. In Section 4, we investigate whether the production of feminine gender changes across the three experiments. Section 5 provides an in-depth discussion of three specific traditionally feminine lexical items. In Section 6, we offer a discussion of our results before Section 7 concludes the chapter.

2 An overview of the GenVAC experimental methodology

The GenVAC project uses a range of different experiments (see [van Baal et al. 2023](#) for a detailed overview). In this article, we look at the production of grammatical gender markers in three of the picture-based elicitation tasks, which elicit prenominal gender forms: a task for indefinite singular articles, a task for prenominal possessives, and a task designed to elicit anaphoric pronouns, but which also elicits indefinite articles, which are the forms that are of interest to the current study. In what follows, we briefly present each of the three tasks.

In three-gender dialects of Norwegian, indefinite singular articles co-vary with gender: feminine (*ei*), masculine (*en*), and neuter (*et*). The experiment on singular articles elicits indefinite and definite singular nouns modified by an adjective. The participant looks at a computer monitor and sees two images of the same object. However, the two images have different colors. The investigator asks what the participant sees on the screen, and the target answer consists of two phrases. Then one of the pictures disappears, and the participant is asked to tell what disappeared. An example based on the masculine noun *traktor* ‘tractor’ is provided in (1).

- (1) a. Investigator:
 Hva ser du her?
 what see you here
 ‘What do you see here?’
 Expected answer:
 Jeg ser en rød traktor og en grønn traktor.
 I see a.M red tractor and a.M green tractor
 ‘I see a red tractor and a green tractor.’
 b. Investigator:
 Hva skjedde nå?
 what happened now
 ‘What happened now?’
 Expected answer:
 Den grønne traktor-en forsvant.
 the.M green tractor-DEF.M disappeared
 ‘The green tractor disappeared.’

32 nouns are included split across the three genders: masculine (n=8), feminine (n=16), and neuter (n=8). For each gender, half of the nouns end in an unstressed

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vowel (weak declension class). The order of nouns is randomized, though adjusted manually to ensure that the same gender appears at most two times in a row, but all participants see the items in the same order. See [van Baal et al. \(2023\)](#) for more on the methodology and [van Baal et al. \(2024\)](#) for an in-depth presentation of the results.

The possessives task seeks to elicit prenominal and postnominal possessives, which are marked for grammatical gender in Norwegian. In the task, the participant and the investigator own different objects, and a creature called *Knut* steals these objects. The participant is to say what is stolen and from whom. After piloting and consulting a different protocol in [Rodina & Westergaard \(2021\)](#), it was decided to do the experiment twice: In the one round, the participants were instructed to produce prenominal possessives (2a), and in the other round, they were instructed to use the postnominal possessive (2b). The order of the two experiments varied between participants. This is because it proved impossible to create a natural way to elicit both orders within one experiment ([van Baal et al. 2023: 35](#)).

- (2) a. Investigator:
 Hva skjedde nå?
 what happened now
 ‘What happened now?’
 Participant:
 Knut stjal mitt tog.
 Knut stole my.N train
 ‘Knut stole my train.’
 b. Investigator:
 Hva skjedde nå?
 what happened now
 ‘What happened now?’
 Participant:
 Knut stjal tog-et mitt.
 Knut stole train-DEF.N my.N
 ‘Knut stole my train.’

The order of the tasks is counterbalanced, and 20 nouns are included: 8 feminine nouns, 8 masculine nouns, and 4 neuter nouns. All nouns are also included in the experiment on singular indefinite articles. Only the pattern for prenominal possessives will be discussed further in this chapter. See [van Baal et al. \(2023\)](#) for

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more on the methodology and Solbakken et al. (2024) for a detailed presentation of the results.

Lastly, the pronominal experiment investigates the production of anaphoric pronouns across sentence boundaries. The purpose is to study the use of pronouns referring to nouns of different grammatical genders and animacy. In the task, participants are first asked to say what they see, using the indefinite singular form. Then they see the object in four different pictures, and they are asked to describe the pictures without using the noun. The task includes 24 nouns (8 feminine, 8 masculine, and 8 neuter nouns). Some of the nouns are included in the previous experiments, others only appear in this experiment. Since we are only concerned with the production of the indefinite singular article and noun in this chapter, not the form of the pronoun, we are not going to explain this methodology further; see van Baal et al. (2023) for details.

The GenVAC project collected data from seven different locations and from either two or three different age groups in each location. In this chapter, we consider data from 19 high school pupils in Lyngdal and 20 in Bodø. These two locations have been chosen because our previous work demonstrates that they are in the midst of going through a change from a two-gender to a three-gender system (see van Baal et al. 2023, 2024 and Solbakken et al. 2024).

3 Elicitation experiments: Some pros and cons

Elicitation is a frequently used methodology in linguistic research. In this section we discuss some advantages and possible problems with this methodology.

One major advantage is that it offers a controlled way of collecting production data from speakers. It is possible to control the material as the investigator gets to determine what the participant is exposed to and how the participant is supposed to answer. That is, it is possible to constraint the production context in whatever way is suitable for the experiment at hand (Gillioz & Zufferey 2020: 60). This makes it possible to control for interfering confounding factors, such as rare words, weird or complex sentences, etc. Furthermore, controlled experiments allow the investigator to probe phenomena that do not occur very frequently in for example a spoken corpus, thus it is ‘possible to determine whether people are capable of producing a specific linguistic form or not’ (Gillioz & Zufferey 2020: 61). The downside of this structured and constrained approach is that it becomes less natural. The data has not been collected in a natural environment for the speakers, which in turn may affect the quality and nature of the data.

Another advantage of structured elicitation also relates to the quantitative nature of experiments. They are often set up so that it is possible to carry out

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a quantitative analysis of the data. Depending on the area of interest, this is often instrumental to fully understand the phenomenon at hand. However, the flipside of this is of course that not all types of data or questions are amenable to quantitative analysis; rather, a qualitative approach may be better called for. In the latter cases, constrained production experiments are often not ideal.

Sometimes it may not be clear for the participant what the task involves and what they are asked to do. The lack of a natural production context may affect what speakers say, and they may also be slightly frustrated that they must follow a particular format. For these reasons, Gillioz & Zufferey (2020: 63) argue that ‘... elicitation tasks generally indicate a lower level of competence compared to the observation of productions in a corpus, and thus provide a conservative image of the linguistic level of the participants’. In sum, as with any method, there are pros and cons related to elicited production experiments. We have utilized these experiments in the GenVAC project because they offer the right tool to answer our questions, which first and foremost relate to how gender markers are produced. By providing participants with constrained contexts, it is possible to record their production of various gender markers as naturally as possible. Finally, structured elicitation maximizes the possibility for comparisons across locations and age groups, using the same methodology.

4 Development in the production of feminine forms over experiment time

Speakers’ production is always influenced by the context in which they find themselves, and the structured, less natural property of elicitation tasks may also influence the types of responses that researchers get from such tasks. As mentioned in Section 1, Busterud et al. (2019: 156) observed in their experiments eliciting grammatical gender that there was a development in speakers’ responses, with more feminine forms in the second experiment than in the first. This could be due to random variation, it could be because of internal differences in the two experiments, or it could simply be because speakers were more used to the experimental setting in the second experiment. In this chapter, we investigate whether similar differences between tasks are present in the GenVAC data. Figure 1 plots the rate of prenominal feminine markers (*ei* ‘a.F’ or *mi* ‘my.F’) for each feminine noun in the three relevant tasks described in Section 2.

The data are taken from highschoolers in the towns of Lyngdal and Bodø. Figure 1 shows the group level production (%) of feminine forms (*ei*, *mi*) with each feminine noun, listed from left to right in the order in which they were seen by

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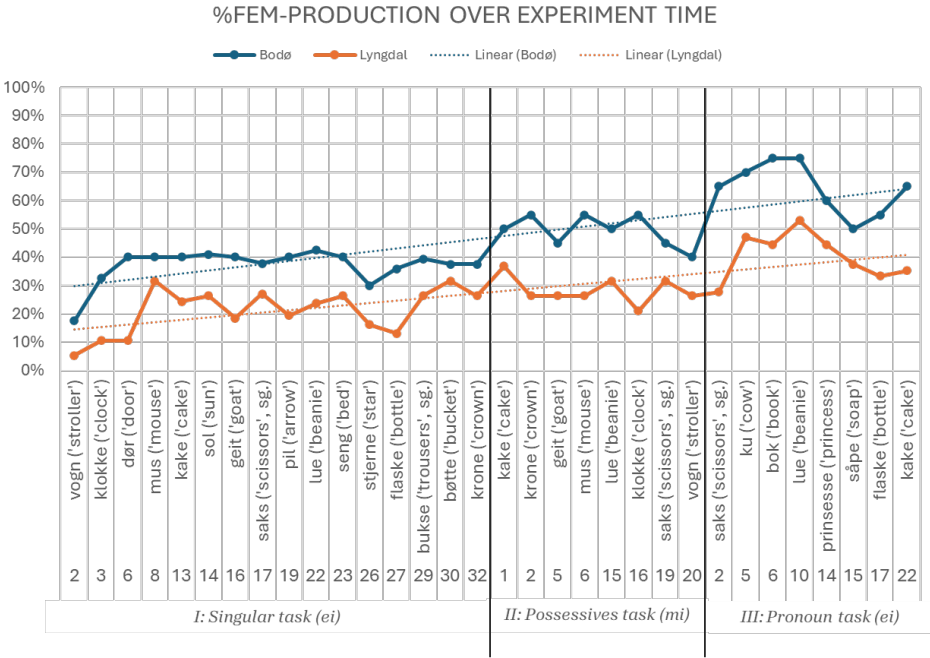


Figure 1: Development of the production of prenominal feminine forms across the three tasks

the participant (x-axis). The vertical lines mark the boundaries between the tasks, and the number under each noun signals where it occurred within each task. For example, the noun *vogn* 'stroller' was the second noun in the singular task. *Vogn* was combined with the feminine article *ei* 18% of the time in the Bodø group, and 5% of the time in the Lyngdal group. All participants did the three tasks in the same order with a break in between,¹ and all participants saw the items in the same order. As can be seen from the data points and dotted trend lines,

¹All the participants started with the singular task and did an additional plural task in-between the singular task and the possessives task. It varied between participants whether they started with the prenominal or postnominal subtask for possessives (see Section 2). Figure 1 only includes the results for prenominal possessives, since these show a clear, undisputed distinction between two- and three-gender systems (on prenominal vs. postnominal possessives, see among others [Svenonius 2017](#); [Solbakken et al. 2024](#)). Some participants from Lyngdal also did an eye-tracking experiment in-between the possessives task and the pronoun task. The stimuli in the eye-tracking experiment included the feminine article *ei*, which could in theory have primed participants to produce more *ei* after this experiment. However, the Bodø group did not do the eye-tracking experiment, so potential priming from this task cannot explain the data investigated in this chapter.

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there seems to be a clear increase in the number of feminine forms produced throughout the experiment session.

To assess the statistical reliability of this trend, we modelled Fem-production as a function of the fixed effects Condition (i.e. task) and Location in a binomial generalized mixed effects model, using the *glmer*-function of the package *lme4* (Bates et al. 2015) in R (R Core Team 2022). We included by-Subject random slopes for Condition, and random intercepts for Subject and Word. P-values were computed via likelihood ratio tests with the *afex* package (Singmann et al. 2024). In addition, pairwise comparisons of the three conditions were conducted with the *emmeans* function (Lenth 2022). We found a statistically reliable effect of Condition ($\chi^2 = 20.65$, $df(2)$, $p < .001$), and a marginal effect of Location ($\chi^2 = 3.02$, $df(1)$, $p = .082$). The pairwise comparisons reveal a significant difference in the production of feminine prenominal markers from Task 1 to Task 3 ($p < .0001$). The differences between Task 1 and Task 2 ($p = 0.0714$) and between Task 2 and Task 3 ($p = 0.0766$) do not come out as statistically significant. In other words, we see a clear increase in the use of feminine markers from the singular task to the pronoun task, with the possessives task as an intermediary step.

To further assess the impact of experiment time on Fem-production, we also tested the effect of experiment time within each Condition. We did not find a statistically significant effect of time within Conditions.² This indicates that the differences between Conditions is what is most pronounced in our data.

Since all participants responded to all tests and items in the same order, it won't be possible to fully tease apart the role of *experiment time* versus *internal properties of each task* (to be discussed below). Future studies should therefore aim to counterbalance the order of tasks and items between subjects to gain a deeper insight into these aspects and their influence on feminine marking. However, our data in combination with the observation from Busterud et al. (2019) do strongly suggest that speakers produce fewer feminine gender forms in earlier tasks than in the later tasks. In Section 6 we offer some reflection on what this means for studies of grammatical gender and language change. However, first we take a closer look at three items that were included in all three tasks.

²This was done by subsetting the data frame for Condition and modelling Fem-production as a function of the fixed effects Location + InCondTime, the latter being a specification of where the item occurred in each task (cf. the item-numbers in Figure 1). We included random intercepts for Subjects and by-Subject random slopes for InCondTime. The effect of InCondTime in each Condition was: Singular task ($p = .361$), Possessives task ($p = .215$), Pronoun task ($p = .295$).

5 The nouns *kake*, *saks* and *lue* across experiments

Three feminine nouns were used in all three tasks: *kake* ‘cake’, *saks* ‘scissors’(singular), and *lue* ‘beanie’. Comparing these nouns across the three tasks, we see a similar tendency: The participants tend to produce more feminine prenominal forms in the later experiments. We see only two exceptions to this tendency: in the Lyngdal group, the score goes slightly down from the possessives task to the pronoun task for the nouns *kake* (from 37% to 35%) and *saks* (from 32% to 28%).

Table 1: The production of the feminine nouns *kake* ‘cake’, *saks* ‘scissors’, and *lue* ‘beanie’ across the three tasks

	Bodø			Lyngdal		
	I: sing	II: poss	III: pron	I: sing	II: poss	III: pron
<i>kake</i>	40%	50%	65%	24%	37%	35%
<i>saks</i>	38%	45%	65%	27%	32%	28%
<i>lue</i>	43%	50%	75%	24%	32%	53%

This comparison indicates that there must be intraspeaker variation for the same noun, a point which can be further illustrated by the production of the speakers *Bodø-B-01k* and *Lyngdal-B-04k*:

Table 2: The production of the feminine nouns *kake* ‘cake’, *saks* ‘scissors’, and *lue* ‘beanie’ for two speakers

	Bodø-B-01k			Lyngdal-B-04k		
	I: sing	II: poss	III: pron	I: sing	II: poss	III: pron
<i>kake</i>	M	F	F	M	M	F
<i>saks</i>	M	M	M	M	F	F
<i>lue</i>	M	F	F	M	M	F

Both speakers start out with masculine (or common) marking in Task 1 but change their gender marking in Tasks 2 and 3. We note that there are also speakers who show opposite mixing patterns, and we see variation in our data related to how much individual speakers mix, although the details are left for future research. During a process of language change, intraspeaker variation is not sur-

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prising. Importantly, we only see this type of variation with traditionally feminine nouns; the gender marking for both masculine and neuter nouns is stable. Thus, at least some speakers who produce the masculine article *en* with a noun like *kake* in Task 1, nevertheless have the knowledge that this noun can also take the feminine marking. In other words, they make use of both the two-gender and the three-gender system. Had we only considered the results from Task 1, this could have been taken to show that feminine gender had been lost or not acquired for these speakers. With the inclusion of more tasks, we see that the tendency to choose masculine marking in Task 1 and feminine marking in Task 3 must be due to other factors. This does not mean that the results from Task 1 are invalid or not interesting. Rather, it indicates that the design or sequence of the tasks taps into different aspects of speakers' grammatical competence and production.

6 Discussion

The data presented in Sections 5 and 6 show that speakers in both the Bodø and Lyngdal groups produce more feminine forms later in the experiment session, in line with the observation by [Busterud et al. \(2019\)](#). Here we offer some speculation as to why that might be.

It is highly likely all three-gender speakers in our experiments also have full mastery of the two-gender system, such that the two systems actually exist in parallel in speakers' minds. This is so because all of these participants have Norwegian Bokmål as their main written language and most of their written input is likely to be in Bokmål. This is the dominant written language in Norway, which speakers will most often see in books, media, et cetera. The Bokmål written system allows speakers to choose between a three-gender grammar and a two-gender grammar, but the two-gender grammar is the grammar most commonly used in writing and it is considered more formal ([Dyvik 2012, 2018](#); see also [Sandengen 2023](#) on the prevalence of formal norms in written Norwegian). The relevance of this for elicitation procedures is that the structured and somewhat unnatural property of elicitation tasks (cf. Section 3) could lead speakers into a more formal or even written mode. For participants with both gender systems, this could activate their two-gender grammar and thus influence their responses. Furthermore, the elicitation procedure gives the impression of a test, and in the case of highschoolers, they were being all tested in school, which underlines the formal aspect. The formal, test-like nature of the elicitation procedure can also help explain why the participants produce more masculine forms in Task 1 than

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in Task 3: As the experiment session progresses, they are likely to become more relaxed and familiar with the format, which could bring them out of the formal mode and thereby lead them to produce more feminine gender forms.

Another explanation for the differences between tasks could lie in the design of the tasks themselves. Task 1 (singular indefinites) is highly formulaic with little variation and very similar types of responses throughout the tasks. Task 2 (possessives) is similarly formulaic, but has slightly more variation, as speakers do not only have to name the items that they see, but also pay attention to who owns what (thus producing 1st or 2nd person possessives). Task 3 (pronouns) elicits the most varied responses, with longer sentences where speakers have to describe the placement of one item in relation to a number of other items while also making choices about which verbs and prepositions to use to describe the pictures. This aspect of Task 3 can be said to introduce a more natural, less formulaic way of speaking, which again could lead to more feminine forms for those speakers who master both the two- and the three-gender system.

Busterud et al. (2019) speculated that the higher rate of feminine marking in their experiment 2 could be due to self-priming, as their second experiment contained mainly feminine nouns.³ Self-priming does not seem to be a likely explanation in our tasks, however, since all tasks contain nouns of all three genders, and the proportion of feminine nouns is lowest in Task 3. If anything, self-priming would suggest that the high amount of masculine marking on feminine nouns in Task 1 should lead to even more masculine marking in Task 3. The observed patterns are also not easily explained by lexical effects (cf. the comparison of nouns across tasks in Section 5), or e.g. animacy, although this is something that could be investigated further in future work.

Finally, it is worth reflecting on the role of the interviewer. The interviewers in the GenVAC project speak different dialects of Norwegian and use the two- and three-gender systems differently. It is not inconceivable that the interviewers influence the participants by the way they speak (i.e. priming), or simply via their rapport. In the Bodø group, each participant did all tests with the same interviewer, but it varied which interviewer they met. In the Lyngdal group, the participants did Task 1 with one interviewer, and Tasks 2 and 3 with a different interviewer. While we haven't seen any indication that different interviewers elicit different types of responses, we also have not investigated this thoroughly. The procedure was designed in such a way that the interviewer would avoid

³As a reviewer points out, there is additional support for a self-priming effect in the data provided in Rodina & Westergaard (2015), where the proportion of *ei* is notably higher in Experiment 2 compared to Experiment 1.

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using feminine nouns and gender marking during the tests (e.g. to exemplify the procedure), but we can't rule out that participants were influenced by the interviewer for example while chatting prior to testing.

As illustrated through this discussion, a number of different factors could be playing together to influence speakers' production of feminine forms and the variation between tasks. While this introduces complexity to the data and their interpretation, it also offers an opportunity to deepen our understanding of both gender marking and language change more generally. In our data we see that whatever the cause of the variability in feminine marking between tasks, not all speakers are equally sensitive to such factors: The GenVAC project has tested speakers from seven different locations, and overall, the results vary greatly between locations and age-groups (van Baal et al. 2024). Both the inter- and intra-speaker variation have a geographical and age-related component, as we see considerably more intra-speaker mixing of gender systems in the locations that are in the midst of the change (Solbakken et al. 2024). This type of variation comes to light through the series of experiments that we have conducted.

7 Conclusion

In this chapter, we have discussed various methodological challenges that emerge when investigating an ongoing change in grammatical gender through experimental studies. Our focus has been on methodological choices made in the GenVAC project and how these may have influenced our findings. Overall, we find that participants produce more feminine markers later in the experimental session. In the discussion, we consider various factors that may conspire to yield this result, involving the availability of more than one gender system in the input, the role of the interviewer, and task effects more generally. In ongoing work, we are also comparing elicitation against what choices and explicit evaluations the participants make when trying to construct sentences involving gender markers. Future work should also ensure that the order of tests and items is balanced to tease apart the different factors that may influence the production of feminine markers.

Our results suggest that in situations of language change, the variable that is unstable is the target of task effects or factors relating to the experimental context. From this point of view, it is not surprising that the production of feminine markers differs throughout a session consisting of multiple experiments. Rather, it would have been surprising if the production of masculine or neuter markers had been subject to the same variation.

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References

- Bates, Douglas, Martin Maechler, Ben Bolker & Steve Walker. 2015. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* 67(1). 1–48.
- Blom, Elma, Daniela Polišínská & Fred Weerman. 2008. Articles, adjectives and age of onset: The acquisition of Dutch grammatical gender. *Second Language Research* 24. 297–331.
- Busterud, Guro, Terje Lohndal, Yulia Rodina & Marit Westergaard. 2019. The loss of feminine gender in Norwegian: A dialect comparison. *Journal of Comparative Germanic Linguistics* 22. 141–167.
- Dyvik, Helge. 2012. Norm clusters in written Norwegian. In Gisle Andersen (ed.), *Exploring newspaper language: Using the web to create and investigate a large corpus of modern Norwegian*, 193–220. Amsterdam: John Benjamins.
- Dyvik, Helge. 2018. ‘jorda’ eller ‘jorden’? hunkjønn i bokmål [‘the earth.f’ or ‘the earth.c’? Feminine gender in Bokmål]. Accessed on June 17, 2024. <https://nogramtall.w.uib.no/2018/02/05/hunkjonn-i-bokmal/>.
- Gillioz, Christelle & Sandrine Zufferey. 2020. *Introduction to experimental linguistics*. London: ISTE Ltd. & John Wiley & Sons Inc.
- Lenth, Russel V. 2022. *emmeans: Estimated Marginal Means, aka Least-Squares Means*. R package version 1.8.1-1. <https://CRAN.R-project.org/package=emmeans>.
- Lødrup, Helge. 2011. Hvor mange genus er det i Oslo-dialekten? [‘How many grammatical genders does the Oslo dialect have?’] *Maal og Minne* 2. 120–136.
- R Core Team. 2022. *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna. <https://www.R-project.org/>.
- Rodina, Yulia & Marit Westergaard. 2015. Grammatical gender in Norwegian: Language acquisition and language change. *Journal of Germanic Linguistics* 27. 145–187.
- Rodina, Yulia & Marit Westergaard. 2021. Grammatical gender and declension class in language change: A study of the loss of feminine gender in Norwegian. *Journal of Germanic Linguistics* 33. 235–263.

Ragnhild Eik, Terje Lohndal, Yvonne van Baal & Hedda Solbakken

- Sandengen, Andrea. 2023. «ikke at det er feil sånn egentlig, bare at det er, sånn, det er litt annerledes.» En undersøkelse av unge skriveres forestillinger og oppfatninger om bokmålsnormen. [“Not that it’s really wrong, it’s just, really, a bit different”. An investigation of young writers’ ideas and opinions about the bokmål norm.] Norwegian University of Science & Technology. (MA thesis).
- Singmann, Henrik, Ben Bolker, Jake Westfall, Frederik Aust & Mattan S. Ben-Shachar. 2024. *afex: Analysis of Factorial Experiments*. R package version 1.3-1. <https://CRAN.R-project.org/package=afex>.
- Solbakken, Hedda, Yvonne van Baal, Ragnhild Eik & Terje Lohndal. 2024. The decline of feminine possessives in Norwegian: An empirical and theoretical investigation of gender and declension class. *Glossa: A Journal of General Linguistics* 9(1). 1–36.
- Svenonius, Peter. 2017. Declension class and the Norwegian definite suffix. In Vera Gribova & Stephanie S. Shih (eds.), *The morphosyntax-phonology connection: Locality and directionality at the interface*, 325–360. New York: Oxford University Press.
- van Baal, Yvonne, Ragnhild Eik, Hedda Solbakken & Terje Lohndal. 2024. The decline of feminine gender: A cross-dialectal study of seven Norwegian dialects. *Journal of Comparative Germanic Linguistics* 27(9). 1–35.
- van Baal, Yvonne, Hedda Solbakken, Ragnhild Eik & Terje Lohndal. 2023. Endringer i grammatisk kjønn på tvers av dialekter: Et eksperimentelt paradigme [‘Changes in grammatical gender across dialects: An experimental paradigm’]. *Norsk Lingvistisk Tidsskrift* 41. 17–58.

Chapter 10

Naming what has not been done

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Fábregas & González Rodríguez (2020) analyse inhibited events as instances where negation is merged low, over Init(iation)P, with the effect that causation is reinterpreted as inhibition (not-causation) of an event. This predicts that there should be instances of lexical verbs whose initiation layer actually means exerting force to inhibit an event that otherwise was expected to happen. This work argues that the prediction is borne out and provides examples from Spanish, showing that their behaviour is the one predicted by the original analysis.

1 Negative events in general

Fábregas & González Rodríguez (2020) argue for the existence of negative events as objects of natural languages, following Stockwell et al. (1973: 250–251) and against approaches where structures such as the one underlined in (1) represent facts, states or other types of entities (cf. Horn 1989, Asher 1993, Verkuyl 1993, de Swart & Molendijk 1999; see in particular Bernard & Champollion 2024 for a recent, semantically explicit analysis).

- (1) I saw John not lock the door.

The goal of this article is to provide evidence for an immediate prediction of this analysis: That there should be lexical verbs that already encode negative events. The rationale for this is that, in Fábregas & González Rodríguez' (2020) analysis, the negative event involves introducing negation within the syntactic area where lexical verbs are built, therefore presupposing negation can be part of

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the information encoded by a lexical verb. This article shows that the prediction is borne out.

Part of the criticism that the negative event concept receives is due to the difficulty of identifying the specific conditions under which one can talk about an eventuality that, ultimately, involves failing to perform an expected event. Not every situation where one does not lock a door can be felicitously described by (1): For instance, speakers have a clear intuition that (1) cannot be used to describe a situation where John is walking his dog in the park, and yet it is strictly true that in this situation John does not lock the door. (1) is felicitous in situations where the natural course of events triggers the expectation that John should lock the door at that particular point in time, and that point in time passes without that event being performed. For instance, (1) works fine if John is the last person to leave a lab whose door must be locked when nobody is there.

For this reason, among others, Fábregas & González Rodríguez (2020) propose to analyse negative events as instances of a low negation that directly operates over the causation component of the verbal event, which was labelled Init(iation) in their analysis (Ramchand 2008, 2018), as represented in (2).

- (2) [TP ...saw [E_{VP} [John] Evt [NegP not [InitP kiss the bride]]]]

In contrast, negated event readings, where negation denies the existence of an event, have the structure in (3).

- (3) [NegP not [TP see ...[E_{VP} [John] Evt [InitP kiss the bride]]]]

This triggers two distinct readings: (4a) represents the negative event, where the event is existentially bound, and therefore there is an event. (4b) represents a negated event, where the existence of an event is denied. Assuming that the existential operator is introduced at the TP level, the scope follows from the location of Neg.

- (4) a. $\exists e[\text{inhibit-kiss}'(e)]$
 b. $\neg \exists e[\text{kiss}'(e)]$

Assuming that Init has a causative semantics and assigns an 'initiator' theta role to the external argument, as the entity who starts the eventuality, the effect of negation over it is to create the complementary predicate for 'cause', which

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we label 'inhibit'.¹ (4b), then, means that John refrains from causing an event that he was otherwise expected to cause in that particular situation, given our common knowledge of the natural course of events and the rules and norms in our culture. Given this, eventualities whose external argument does not have initiator entailments are expected to be degraded with low negation: An external argument incapable of initiating an event is equally incapable of inhibiting it. This explains why (5) is degraded with respect to (1).

- (5) ?? I saw John not die at the attack.

Beyond this, the inhibition of the event has consequences for Aktionsart: Given that the event that otherwise could be cause is inhibited, these verbs are non-dynamic, and consequently reject manner modifiers that presuppose some event progression, like *slowly*. Unless the adverb is the scope of the negation, (6b) is ungrammatical.

- (6) a. I saw John slowly kiss the bride.
b. # I saw John not slowly kiss the bride.

Being non-dynamic, they must also be atelic, and durative. For this reason, negative events approach some types of stative predicates, specifically those that Maienborn (2003) calls Davidsonian-states: There is an event which consists essentially of keeping some state of affairs unchanged. Note that we can measure the duration during which an event is inhibited with a *for*-phrase.

- (7) I saw John not kiss the bride for half an hour.

This contrasts with negated events, where negation denies the existence of any event, where measure phrases are out because an event that does not exist cannot have any extension. Again, (8) is grammatical only if negation does not deny the existence of the event, but scopes only over the *for*-phrase (...*not for half an hour but for a whole hour*).²

¹Note that we are assuming, with Ramchand (2018), that Init does not introduce the external argument in its specifier: the external argument is introduced in spec, EvtP, and its thematic interpretation is determined by the complement of Evt. Without negation, that complement is Init and consequently the specifier gets initiator entailments; when Neg combines with Init, the constituent is interpreted as Inhibition. The specifier of an EvtP whose complement is NegP-InitP gets thematic entailments of an inhibitor. We are grateful to an anonymous reviewer for the request to clarify this.

²As an anonymous reviewer points out, the presence of the *for*-phrase in (i) forces the negative event reading, but the external argument does not control the event, which might seem to

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- (8) I did not see John kiss the bride (#for half an hour).

The negative event structure, which has been reproduced in (2), makes a prediction that has been so far unexplored: There should be lexical verbs that denote negative events. The reason is that in (2) negation is contained below Evt, which is the head that defines a Davidsonian event from the material provided by Init, Proc and Res. Consequently, negation is in these cases contained within the syntactic material that corresponds to a lexical verb. We then expect some verbal exponents to be able to lexicalise this structure, including the negation. This would result in a verb whose lexical meaning already denotes a negative event.

We will argue that lexical negative event verbs exist, and that among others the verbs in (9) belong to this class.

- (9) abstenerse 'to not vote', omitir 'to omit', prescindir 'to do away with', saltarse 'to skip', plantar 'to stand someone up'...

The rest of this article is structured as follows. In Section 2, we delimit the set of potential verbs by discarding verbs that involve cessation, reversion or restitution, which ultimately involve changes in the state of affairs. In Section 3 we discuss verbs with an overt negative component, as a prefix, that have a negative event interpretation, and in Section 4 we present the verbal classes that involve negative events without a negative prefix. The conclusions are presented in Section 5.

2 Verbs with a negative component that are not negative event verbs

Admittedly, there are not many verbs that express negative events, that is, failing to perform an event. The vast majority of verbs with a negative component denote the cessation of a previous state of affairs or the reversion to a previous

block that reading. However, the lack of control over the event does not prevent the external argument from having initiator entailments. Following Ramchand (2008: 24), we assume that the initiator refers to the argument whose properties or behaviour are responsible for making the event happen, volitionally or non-volitionally, as well as consciously or unconsciously. The external argument in (i) complies with this definition, unlike what happens with inanimate entities, which cannot be the goal of a transference event (*John received a package/ *The house received a package*).

- (i) John didn't receive any news from her for several months.

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state. None of these classes of verbs falls into the definition of a negative event because in both cases they imply changes, and are therefore dynamic.

Let us start with verbs denoting the cessation of a previous event, verbs that mean 'to stop or to interrupt some event'. These verbs are dynamic and telic. Some of the members of the class are in (10):

- (10) abandonar 'to abandon', abolir 'to abolish', abortar 'to abort', cancelar 'to cancel', cejar 'to cease', dejar 'to leave', deshabitar 'to leave empty', jubilarse 'to retire', recesar 'to recess', suspender 'to suspend'...

All these verbs involve situations where there was a previous state of affairs, and the verbs themselves denote the change that leads to the cessation of the event. For example, if someone abolishes a law, a change is performed in the external reality, removing some regulation that previously existed. Similarly, if someone suspends the celebration of a previously planned event, this also involves a change, and, correlatively, both verbs are construed as telic events.

A second frequent interpretation of a negative notion in a lexical verb is the restitution of a previous state, with or without the additional reversion of a previous process. This is one of the most frequent readings of the English prefix *un-* (Bauer et al. 2013: Chapter 17) and Spanish *des-* (Gibert-Sotelo 2017). Some verbs involving reversion or restitution are cited in (11):

- (11) despintar 'to remove paint', devolver 'to return', reparar 'to repair', reponer 'to replenish', restaurar 'to restore', restituir 'to retribute', retroceder 'to retreat', volver 'to return'...

It is, additionally, clear that the combination of a prefix with an opposition or negation meaning and a verb generally results in a change of state meaning. Consider for instance the comparison between *deshabitar* 'to leave empty' as a verb and the version without a prefix, *habitar* 'to inhabit'. The first can only mean to leave a house where one was living (12), and lacks the non-dynamic meaning expected from the negative event, while the second (13) allows a non-dynamic meaning equivalent to 'to live in a place'.

- (12) Spanish (personal knowledge)

- a. La familia des-habit-ó la casa.
the family un-inhabit-PFV.PAST the house
'The family abandoned the house.'

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- b. * La familia des-habit-a esta casa desde 2018.
 The family un-inhabit-T_{HV} this house since 2018
 Intended: ‘The family does not live in this house since 2018.’

- (13) Spanish (personal knowledge)
 La familia habit-a esta casa desde 2018.
 the family inhabit-T_{HV} this house since 2018
 ‘The family inhabits this house since 2018.’

3 Negative event verbs with a negative prefix

While it is empirically correct to say that negative components generally trigger change of state readings, this is not an absolute principle, and it is possible to identify verbs where negation triggers a negative event meaning on the verb.

The negative prefix *in-* (Gibert-Sotelo 2017, Fábregas 2023) barely combines with verbal bases. In the few instances where it does, its behaviour is mixed. In two of the few documented cases, it triggers a change of state reading that blocks a stative, non-dynamic construal for the verb: *in-comunicar* ‘to incommunicate, to isolate’ and *in-subordinar* ‘to mutiny, to stop following orders’. However, the negative reading is documented in a few cases: *pagar* ‘to pay’ acts as the base of the verb *im-pagar* ‘to fail to pay, to default’ (14a), *in-cumplir* ‘to not comply with’ is the negative event version of *cumplir* ‘to comply with’ (14b) and *admitir* ‘to admit’ has a negative event version *in-admitir* ‘to not admit’ in legal speech (14c).

- (14) Spanish (personal knowledge)
- El afectado comenzó a im-pagar su hipoteca en 2013.
 The affected start.3SG.PFV to not-pay his mortgage in 2013
 ‘The affected person began to default on his mortgage in 2013.’
 - El soldado in-cumplió las órdenes del general.
 The soldier not-comply.with.3SG.PFV the orders of the general
 ‘The soldier failed to comply with the general’s orders.’
 - El tribunal in-admitió la demanda.
 The court not-admit.3SG.PFV the lawsuit
 ‘The court rejected the lawsuit.’

While there are few verbs derived with *in-* that denote negative events, *des-* is much more frequent with verbal bases. Correlatively, it is easier to find examples

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of negative events with this prefix. In Spanish we have (among many others) *atender* 'to tend (to someone)' > *des-atender* 'to neglect', *cuidar* 'to take care' > *des-cuidar* 'to neglect', *cansar* 'to get tired' > *des-cansar* 'to rest', *entenderse* 'to get on well' > *des-entenderse* 'to disregard', *obedecer* 'to obey' > *des-obedecer* 'to disobey', all cases that conform to the semantic interpretation of a negative event, 'failing to {tend / take care / obey / follow / pay attention...}', where one has the expectation that the event should be carried. For instance, (15a) does not make sense unless there was some expectation that María should take care of Juan; in (15b) we presuppose that Jan was supposed to follow the laws of Spain, and the sentence would be infelicitous otherwise: For instance, if Jan is a Norwegian citizen that lives in Norway and has no interaction with Spain or Spanish laws.

(15) Spanish (personal knowledge)

- a. María ha des-cuidado a Juan.
 María have.3SG un-taken.care.of to Juan
 'María has neglected Juan.'
- b. Jan ha des-obedecido las leyes de España.
 Jan have.3SG dis-obeyed the laws of Spain
 'Jan has disobeyed Spanish laws.'

The verbs identified in this section display all the properties of negative events as presented in (1) above. To begin with, in all cases the subject must be an entity able to control the eventuality. Many of these verbs require animate subjects; some non-animates may be allowed in some instances, as in (16), but note that they still comply with the definition of initiator as the entity whose internal properties initiate a state or event (cf. Folli & Harley 2008 and their notion of 'teleological agent'). A set of rules, for instance, can control whether some regulation is followed or not, through its internal properties, and an analysis can neglect some aspects to the extent that its internal conditions satisfy or do not satisfy those requisites.

(16) Spanish (personal knowledge)

- a. Esta normativa in-cumple la legislación europea.
 This rule un-follows the legislation European
 'This rule fails to comply with the European legislation.'
- b. Tu análisis des-cuida el papel de la frecuencia.
 Your analysis un-tends the role of the frequency
 'Your analysis fails to take frequency into account.'

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Secondly, all these verbs denote non-dynamic eventualities that are close to states, as shown by the compatibility with *for*-phrases and the incompatibility with adverbials that presuppose some internal event progression.

(17) Spanish (personal knowledge)

- a. La nación im-pagó su deuda durante seis meses.
The nation un-pay.3SG.PFV its debt for six months
'The nation defaulted for six months.'
- b. El soldado des-obedeció la orden durante una semana.
The soldier dis-obey.3SG.PFV the order for one week
'The soldier failed to comply with the general's orders.'
- c. El tribunal in-admitió la demanda durante seis meses.
The court not-admit.3SG.PFV the lawsuit for six months
'The court rejected the lawsuit.'

(18) Spanish (personal knowledge)

- a. El ministro des-atendió (*lentamente) sus asuntos.
The minister un-tend.3SG.PFV slowly his affairs
Intended: "The minister (*slowly) neglected his affairs."
- b. Luis in-cumplió (*lentamente) la ley.
Luis not-comply.with.3SG.PFV (slowly) the law
Intended: "Luis (*slowly) failed to comply with the law."
- c. Marta des-cuidó (*lentamente) a su marido.
Marta un-take.care.3SG.PFV DOM her husband
'Marta (*slowly) neglected her husband.'

4 Lexical negative event verbs without negative prefixes

4.1 Lexical verbs that inhibit an event expressed by another verb

Let us move now to a number of lexical verbs that have the required properties of negative events, and do not contain any negative prefix. We will divide them into two main classes. The first one includes a number of verbs which prefer construals where they introduce another non-finite verb, and whose meaning is, generically 'to inhibit, to fail to do', without specifying the concrete action that fails to be performed. At least in one of their meanings, the following verbs fall into the class.

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- (19) abstenerse de <infinitive> 'to abstain from'
 dejar de <+infinitive> 'to stop'
 evitar <infinitive> 'to avoid'
 obviar <infinitive> 'to omit'
 olvidar <infinitive> 'to forget'
 omitir <infinitive> 'to omit'

The negative event use of these verbs, combined with another lexical verb, is illustrated in (20):

- (20) Spanish (personal knowledge)
- a. Juan se abstuvo de hablar en la reunión.
 Juan REFL abstain.3SG.PFV from speak at the meeting
 'Juan abstained from speaking at the meeting.'
 - b. Luis evitó saludar a María en la fiesta.
 Luis avoid. 3SG.PFV to.greet María at the party
 'Luis avoided greeting María at the party.'
 - c. El presidente obvió comentar la reunión.
 The president omit.3SG.PFV to.comment the meeting
 'The president omitted commenting the meeting.'
 - d. María olvidó apagar el horno.
 María forget.3SG.PFV to.switch.off the stove
 'María forgot to switch the stove off.'
 - e. Pedro omitió limpiar el salón.
 Pedro omit.3SG.PFV to.clean the living.room
 'Pedro skipped cleaning the living room.'

Even though these predicates do not specify the type of event that fails to be performed, the combination of these verbs is not completely unrestricted; *obviar* 'to omit' combines best with lexical verbs that denote speech events, such as *comentar* 'to comment', *decir* 'to say', *hablar* 'to speak', *mencionar* 'to mention', etc. Note that they cannot be considered auxiliaries, because they impose their own restrictions to the external argument. As expected from negative event predicates, their subjects must be animate, initiator-like entities. Unless the subject is personified, the following sentences are ungrammatical.

- (21) Spanish (personal knowledge)

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- a. # El tren se abstuvo de parar en la estación.
the train REFL abstain.3SG.PFV from stop at the station
Intended: ‘The train abstained from stopping at the station.’
- b. # El coche evitó arrancar.
The car avoid.3SG.PFV to.start
Intended: ‘The car avoided starting.’
- c. # La aplicación olvidó guardar los datos del usuario.
the app forget.3SG.PFV to.store the data of.the user
Intended: ‘The app forgot to store the user’s data.’

All these verbs share the properties of being non-dynamic and denoting situations whose temporal extension can be measured:

(22) Spanish (personal knowledge)

- a. Juan se abstuvo de hablar durante diez minutos.
Juan REFL abstain.3SG.PFV from speak for ten minutes
‘Juan abstained from speaking for ten minutes.’
- b. Luis evitó saludar a María durante una hora.
Luis avoid.3SG.PFV to.greet to María for one hour
‘Luis avoided greeting María for one hour.’
- c. El presidente obvió (#lentamente) comentar la reunión.
The president omit.3SG.PFV (slowly) to.comment the meeting
‘The president (#slowly) omitted commenting the meeting.’

This group of verbs may be expanded: While the verb *dejar de* <infinitive> is generally used as a dynamic change of state verb equivalent to *to stop* <gerund> (*Pedro dejó de leer* ‘Pedro stopped reading’), it is also possible to identify a non-dynamic use equivalent to *fail* <infinitive>, specially below negation.

(23) Spanish (personal knowledge)

- Un buen ciudadano no deja de cumplir sus obligaciones.
a good citizen not leaves of meet his obligations
‘A good citizen does not fail to satisfy his obligations.’

4.2 Lexical verbs without verbal complements

There are also lexical verbs that include in their meaning which type of event fails to be performed. In fact, two of the verbs from the previous section can also

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be restrictively used without additional non-finite verbs, in which case they get a very specific interpretation: *abstenerse* 'to abstain', without a second verb, means 'to not vote' in some election or decision, while *obviar* and *omitir* 'to omit', under the same circumstances, mean 'not to mention'.

(24) Spanish (personal knowledge)

- a. Pedro se abstuvo en la votación.
Pedro REFL abstain.3SG.PFV in the voting
'Juan abstained from voting in the election.'
- b. Juan obvió en el informe su encuentro con el embajador.
Juan omit.3SG.PFV in the report his encounter with the ambassador
'Juan failed to mention in the report his encounter with the ambassador.'

A class of predicates involves those meaning 'to hold back, not to intervene in a situation', among them *contenerse* 'to refrain', *inhibirse* 'to inhibit oneself', *refrenarse* 'to refrain', *reprimirse* 'to repress oneself'.

(25) Spanish (personal knowledge)

- a. Juan se inhibió durante la negociación.
Juan REFL inhibit.3SG.PFV during the negotiation
'Juan held back during the negotiations.'
- b. Marta se contuvo durante la pelea.
Marta REFL contain.3SG.PFV during the fight
'Marta held back during the fight.'
- c. Pedro se reprimió en la fiesta.
Pedro REFL repress.3SG.PFV at the party
'Pedro held back at the party.'

Some verbs are highly specific with respect to the type of event that fails to be performed: *ayunar* 'to fast' denotes 'to not eat'; *callar* is equivalent to 'to not speak', *plantar* means 'not to meet someone that one was supposed to meet', and *prescindir* means 'to not use'.

(26) Spanish (personal knowledge)

- a. El monje ayunó durante tres días.
The monk fast.3SG.PFV for three days
'The monk fasted for three days.'

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- b. Juan calló durante tres minutos antes de responder.
 Juan shut.up.3SG.PFV for three minutes before of answer
 ‘Juan was silent for three minutes before answering.’
- c. Pedro plantó a María.
 Pedro stand.up.3SG.PFV to María
 ‘Pedro stood María up.’
- d. Juan prescindió del teléfono móvil durante dos semanas.
 Juan do.away.3SG.PFV of.the phone mobile for two weeks
 ‘Juan did away with the mobile phone for two weeks.’

Additionally, the verb *ausentarse*, in addition to a change of state meaning ‘to leave a meeting’, allows for a non-dynamic reading, ‘to not attend a meeting’, as in (27).

(27) Spanish (personal knowledge)

El presidente se ausentó de la reunión programada
 the president REFL not.attend.3SG.PFV from the meeting scheduled
 con los sindicatos.
 with the unions
 ‘The president failed to attend the meeting scheduled with the unions.’

Finally, there is a number of verbs that mean ‘to skip’, that is, to fail to appropriately interact with some entity in some context. Among these verbs, we find *comerse*, which in (25a) is specifically interpreted as ‘to fail to produce some linguistic utterance’, *ignorar* (‘not to pay attention to someone’ in 25b), *saltarse* (‘to fail to stop at some intended location’ in 25c), and the idioms *pasar por alto* ‘to overlook, to fail to notice’ and *dejar de lado* ‘to ignore’.

(28) Spanish (personal knowledge)

- a. Juan se comió varias palabras en su discurso.
 Juan REFL eat.3SG.PFV several words in his speech
 ‘Juan omitted several words in his speech.’
- b. Luis ignoró a María durante toda la fiesta.
 Luis ignore.3SG.PFV to María during all the party
 ‘Luis ignored María at the party.’
- c. El conductor se saltó la parada.
 The driver REFL jump.3SG.PFV the stop
 ‘The driver skipped the stop.’

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- d. María pasó por alto la carta que había en su buzón.
 María pass.3SG.PFV by high the letter that was in her mailbox
 ‘María failed to notice the letter in her mailbox.’

With respect to the behaviour of these verbs, all of them require external arguments that have enough agency to trigger the expectation that they were supposed to perform that action, and to inhibit it. Additionally, they are all non-dynamic in the sense that, during their running time, there are no changes along the relevant dimensions.

(29) Spanish (personal knowledge)

- a. Juan plantó (*lentamente) a María.
 Juan stand.up.3SG.PFV to María
 ‘Juan stood María up.’
- b. Luis ignoró (*lentamente) a María.
 Luis ignore.3SG.PFV (slowly) to María
 ‘Luis (*slowly) ignored María.’
- c. El monje ayunó (*lentamente).
 the monk fast.3SG.PFV (*slowly)
 ‘The monk (*slowly) fasted.’
- d. El presidente se contuvo (*lentamente) en el debate.
 the president REFL contain.3SG.PFV (slowly) in the debate
 ‘The president (*slowly) held back during the debate.’

When it comes to its telicity or atelicity, while it is true that most of these verbs are clearly atelic as expected from their non-dynamicity, those negative event verbs that convey the inhibition of ‘attending a meeting’ (*plantar* ‘to stand someone up’, *ausentarse* ‘to not attend’) are difficult to measure. The *for*-phrase with *ausentarse* is interpreted in the change-of-state version of the verb, ‘to leave a meeting’, and then the result state of being outside the meeting is measured. Note that *in*-phrases are also not allowed.

(30) Spanish (personal knowledge)

- a. Juan plantó a María (??durante dos horas).
 Juan stand.up.3SG.PFV to María (for two hours)
 ‘Juan stood María up (??for two hours).’

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- b. El presidente se ausentó de la reunión (#durante
the president REFL not.attend.3SG.PFV from the meeting for
dos horas).
two hours
‘The president left the meeting for two hours.’ (not ‘The president
failed to attend the meeting for two hours’)

(31) Spanish (personal knowledge)

- a. Juan plantó a María (*en dos horas).
Juan stand.up.3SG.PFV to María (in two hours)
‘Juan stood María up (*in two hours).’
- b. El presidente se ausentó de la reunión (*en una
the president REFL not.attend.3SG.PFV from the meeting (in one
hora).
hour)
‘The president left the meeting (*in one hour).’

Given that *in*-phrases are also out with these verbs, and given that they are non-dynamic verbs, we believe that the difficulty of measuring its running time comes from an independent factor: These verbs denote the inhibition of the event of attending a meeting, or meeting someone at an agreed point in time. The running time of the event, then, has been pre-established when the meeting was scheduled, and its inhibition involves precisely not to meet at those designated times. Pragmatically, then, establishing an independent running time for the negative event is impossible, because the inhibition fully depends on whatever time coordinates had been agreed for the meeting that fails to happen.

Before ending this section, we would like to point out an important asymmetry with respect to syntactic negative events: Unlike them, lexical negative events do not license Negative Polarity Items, and as far as we can tell this is a general property cross-linguistically (cf. *not know anything* ~ **ignore anything*). In our current account, there are two possible explanations for this: One is that in lexical negative events Neg is directly compounded with Init, not its own projection, so that it fails to c-command NPIs. The second option is that NPIs always have to syntactically move to a position immediately dominated by Neg, therefore breaking the syntactic constituent Neg-Init that is lexicalised together by the verbal exponent in a lexical negative event, while not interfering in cases where Neg corresponds to *no* and is not lexicalised as part of the verbal exponent. We leave discussion of the second option, which requires a detailed analysis of NPIs, for further research.

4.3 Lexical predicates that are close to this class

Before ending this article, we want to mention one potential verb class that may be sufficiently close to the lexical negative event class, but which we hesitate to classify in this domain.

In Section 2 we mentioned the class of verbs involving *cancelling* (as in *John cancelled the meeting*) and noted that they cannot be considered negative event predicates because they involve a change on the state of affairs. However, there is a close class of verbs that seem to involve the inhibition of an effect that may otherwise have happened: Thwarting verbs. Consider the following sentence:

- (32) Spanish (personal knowledge)
 La policía frustró el robo.
 the police thwart.3SG.PFV the robbery
 ‘The police thwarted the robbery.’

It could be argued that the police inhibited the performance of a robbery; in contrast with the cancelling class, the thwarting the robbery does not result in a change in the state of affairs, but rather on the absence of a change that may have otherwise happened. The verbs in the following list have a meaning close to this class:

- (33) bloquear ‘to block’, desbaratar ‘to thwart’, evitar ‘to avoid’, impedir ‘to prevent’, malograr ‘to ruin’, vetar ‘to veto’
- (34) Spanish (personal knowledge)
- El decano {bloqueó / vetó} la contratación del profesor.
 the dean {block.3SG.PFV / veto.3SG.PFV} the hiring of the professor.
 ‘The dean blocked / vetoed the hiring of the professor.’
 - Sherlock desbarató la conspiración de Moriarty.
 Luis thwart.3SG.PFV the conspiracy by Moriarty
 ‘Sherlock thwarted Moriarty’s conspiracy.’
 - Juan malogró su oportunidad de alcanzar la fama.
 Juan ruin.3SG.PFV his opportunity of reach the fame
 ‘Juan ruined his chance to achieve fame.’
 - Supermán impidió un terremoto.
 Superman prevent.3SG.PFV an earthquake
 ‘Superman prevented an earthquake.’

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While all these verbs ultimately inhibit a possible result, we believe that they should not be classified as negative event verbs. Here is why: First of all, in contrast to the previous classes of verbs, these verbs allow too easily for indirect causer subjects (Pesetsky 1995, Sichel 2010) that cannot control the inhibition of the event, but may lead to states that prevent it from happening: *The bad weather thwarted John's plans*, *The sudden worsening of the economy prevented a strike*, etc. Secondly, in contrast to the verbs discussed so far, when these verbs take animate subject, they imply the active participation of the external argument in blocking that effect: In order to prevent, block or veto something, the animate subject must take some measures that lead to that blocking. Third, these verbs easily allow construals where the subject blocks the effect of some event that was being performed by another participant (for instance, a policeman that thwarts a robbery is different from whoever was intending to rob), while in negative event verbs the subject inhibits an event that he or she could have performed (for instance, a monk that fasts is not preventing others from eating).

5 Conclusions

In conclusion, the prediction that some lexical verbs should denote negative events, understood as events where an agentive subject inhibits some eventuality that he or she was expected to perform, is borne out. While the negative component of lexical verbs is more frequent in cessation, reversion and restitution readings, negative events are also produced by negative prefixes, as expected from Fábregas & González Rodríguez (2020) proposal that negative events involve negation in the lexical area. Also, negative events can be lexically named without negative components, as expected from the claim that 'inhibition' is a possible ingredient of the low lexical area in the verbal domain. The aspectual and argumental properties of these lexical negative events are, as far as we can tell, the same ones identified for derived lexical verbs involving syntactic negation.

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References

- Asher, Nicholas. 1993. *Reference to abstract objects in discourse*. Dordrecht: Kluwer.
- Bauer, Laurie, Rochelle Lieber & Ingo Plag. 2013. *The Oxford reference guide to English morphology*. Oxford: Oxford University Press.
- Bernard, Timothée & Lucas Champollion. 2024. Negative events and compositional semantics. *Journal of Semantics* 40(4). 585–620.
- de Swart, Henriette & Arie Molendijk. 1999. Negation and the temporal structure of narrative discourse. *Journal of Semantics* 16. 1–42.
- Fábregas, Antonio. 2023. *Diccionario de afijos del español contemporáneo*. Oxford: Routledge.
- Fábregas, Antonio & Raquel González Rodríguez. 2020. On inhibited eventualities. *Natural Language and Linguistic Theory* 38. 729–773.
- Folli, Raffaella & Heidi Harley. 2008. Teleology and animacy in external arguments. *Lingua* 118(2). 190–202.
- Gibert-Sotelo, Elisabeth. 2017. *Source and negative prefixes: On the syntax-lexicon interface and the encoding of spatial relations*. Universitat de Girona. (Doctoral dissertation).
- Horn, Laurence R. 1989. *A natural history of negation*. Chicago: University of Chicago Press.
- Maienborn, Claudia. 2003. *Die logische Form von Kopula-Sätzen*. Berlin: Akademie Verlag.
- Pesetsky, David. 1995. *Zero syntax: Experiencers and cascades*. Cambridge: MIT Press.
- Ramchand, Gillian. 2008. *First phase syntax. Verb meaning and the lexicon*. Cambridge: Cambridge University Press.
- Ramchand, Gillian. 2018. *Situations and syntactic structures*. Cambridge: MIT Press.
- Sichel, Ivy. 2010. Event-structure constraints on nominalizations. In Artemis Alexiadou & Monika Rathert (eds.), *The syntax of nominalizations across languages and frameworks*, 159–198. Berlin: Mouton de Gruyter.
- Stockwell, Robert, Paul Schachter & Barbara H. Partee. 1973. *The major syntactic structures of English*. New York: Holt, Rinehart & Winston Inc.
- Verkuyl, Henk J. 1993. *A theory of aspectuality: The interaction between temporal and atemporal structure*. Cambridge: Cambridge University Press.

Chapter 11

Using production data to study pragmatic development

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While production data, in particular those including spontaneous speech, have been considered a useful source of evidence in studies of the development of grammar, their adequacy for studies of pragmatic development has been questioned. Production data for pragmatic phenomena are often hard to interpret and typically thought to underestimate children's performance compared to comprehension data. In this paper we survey a set of studies using production data, both spontaneous and elicited, to investigate children's language development, including pragmatic aspects. We argue that they provide a valuable source of evidence, especially for studies of early pragmatic development and phenomena where controlled comprehension experiments may not be feasible to run with a given age group, for instance when studying subtle intonational differences or non-literal uses of language. Finally, we discuss to what extent production data can be seen as providing a window into children's developing pragmatic competence.

1 Introduction

Ever since its beginnings, research on linguistic development – including vocabulary, phonology, morphology and syntax – has relied on analyses of samples of spoken language from child learners. In the early days, naturalistic studies, which typically consisted of diary notes (e.g., [Leopold 1949](#); [Stern & Stern 1907](#)) or recordings (e.g., [Isaacs 1930, 1933](#)) of the speech of individual children – often the researchers' own – provided the main empirical source for theories of language acquisition ([Johnson & Zamuner 2010](#)). One notable exception is [Piaget's](#)

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(1923) early systematic observations of the spontaneous speech of 4–7-year-old children in the classroom, which inspired a host of later observational studies of language development.¹ The historical focus on spontaneous speech, typically gleaned from anecdotal samples, as the main source of production data is today shifted to experimental and quantitative methods, for instance, in the form of elicitation tasks, sentence completion tasks and corpus studies, often including large samples of spontaneous speech, enabling the use of quantitative methods.

Spontaneous production data are considered a useful source of evidence for studies of many aspects of children's language development (with grammar being a case in point, see [Westergaard 2009](#)). However, there is one domain of language development in which the general usefulness and adequacy of production data – and in particular spontaneous productions – have been largely questioned, namely pragmatics. This branch of developmental linguistics investigates how children learn to go beyond encoded word and sentence meanings to get at the speaker-intended, communicated meaning ([Falkum 2019b](#); [Matthews 2014](#); [Zufferey 2015](#)). As in other domains of developmental linguistics, the use of spontaneous productions as a source of evidence of children's pragmatic competence requires the data to contain a representative number of occurrences of the phenomenon in question ([Tomasello & Stahl 2004](#), [Zufferey 2015](#): 32f). However, drawing conclusions about the acquisition of pragmatic phenomena provides additional concerns given that the focus of interest in this domain is how children (learn to) express their own intentions and make inferences about others' intentions based on their utterances and the contextual situation. For instance, investigating the production of early metaphors might be problematic because it may be debatable whether the observed use counts as a metaphor in the adult sense or is evidence of a more basic process ([Pouscoulous 2014](#)). Thus, a pertinent criticism of using spontaneous productions as evidence of pragmatic competence is that it involves an interpretation of the speaker's communicative intention which is inevitably speculative ([Zufferey 2015](#)). In addition, production data is typically thought to underestimate children's performance compared to comprehension data across domains of developmental linguistics ([Clark 2016](#)).

In this paper we discuss a set of production data, both spontaneous and elicited, used to investigate children's language development, including pragmatic aspects. We argue that it provides a valuable source of evidence, especially for studies of early pragmatic development and phenomena where controlled comprehension experiments may not be feasible, including non-literal uses of language (e.g., metonymy, metaphor). Finally, we discuss to what extent production

¹See [Levelt \(2012\)](#) for a comprehensive overview of early language acquisition research.

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data can be seen as providing a window into children's developing pragmatic competence.

The paper is organized as follows. In Section 2, we look into examples of production studies used to glean information about children's acquisition of grammar, taking word order and negation as examples. We point out how production tasks focused on this aspect of children's language development also involve pragmatic considerations on the part of the child, which the researcher has to make inferences about. In Section 3, we discuss selected studies investigating children's language productions, focusing on aspects of pragmatic development, including intonation, additive focus particles, and creative uses of language in the form of lexical innovation and 'figurative' uses. We show how this data provide a valuable source of evidence, especially for studies of early pragmatic development and certain pragmatic phenomena for which there might be some initial competence present in children but the experimental designs required to test their comprehension may be too complex for the age groups in question. Finally, in Section 4, we discuss to what extent production data can be seen as providing a window into children's developing pragmatic competence, and discuss methodological measures for improving the soundness of production studies for pragmatic development research.

2 Production data and the acquisition of syntax

In studies of children's acquisition of syntax, various types of production data have been used. For instance, Westergaard (2009) used a longitudinal corpus of spontaneous production data from three Norwegian-speaking children to investigate the acquisition of word order, and Ringstad & Westergaard (2023) used children's productions of embedded *that*-clauses to investigate whether Norwegian-speaking children rely on frequency or syntactic complexity (in terms of syntactic movement) in their acquisition of this construction. In Norwegian, when embedded clauses contain a negation, the subject can either precede (more complex structure) or follow (less complex structure) the negation: *Petra er glad for at (barnevakta_{SUBJ}) ikke_{NEG} (barnevakta_{SUBJ}) lagar kjøttkake* ('Petra is happy that the babysitter isn't making meatloaf'). By investigating adults' production of such embedded clauses in three speech corpora, they found differences with regard to frequency: while preceding subjects were highly frequent and could be realized either as a personal pronoun or a lexical nominal phrase, following subjects had a low frequency in adult speech (and, therefore, likely to be infrequent in children's language input) and were normally realized as a nominal phrase.

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In an elicitation task, Ringstad & Westergaard (2023) manipulated the subject type (pronoun vs. nominal phrase) to see whether 3–6-year-old Norwegian-speaking children use both subject positions and whether they distinguish the two types of subjects in their productions of embedded clauses. Participants heard short context stories, for instance: “Petra is being watched by her babysitter. The babysitter often cooks meatloaf for dinner.”, followed by a statement containing a negation, “Today she doesn’t make meatloaf” and information about the main character’s feelings: “Petra is happy about that”. Participants were then asked a question eliciting an embedded *that*-clause: “What is Petra happy about?”. Results showed that children used both subject positions, but with a slight overweight of the less complex structure with a following subject, and that they produced both types of subjects in both positions. However, the results also showed a high variability: While some children produced both pronouns and lexical nominal phrases in both positions, other children produced only preceding subjects or only following subjects, but in either case they used both lexical nominal phrases and pronouns. Overall, it seems that around the age of five, children have acquired subject placement in embedded *that*-clauses. Ringstad and Westergaard take this to suggest that frequency has less influence than complexity on children’s early syntactic acquisition: Despite the preceding subject structure clearly being the most frequent in children’s language input, children often produced the syntactically less complex structure.

The study of Ringstad and Westergaard focuses on syntactic complexity, suggesting that referring expressions might be harder for children to process when they are realized lexically as nominal phrases than as pronouns, and that children might produce pronominal subjects in the complex position earlier than lexical subjects, since they have a lower processing load than lexical subjects. It is relevant to point out that what the authors see as syntactic complexity in this context is not necessarily symmetrical with pragmatic complexity. From a pragmatic point of view, lexical subjects, which typically refer unambiguously in a context, are less complex than pronouns which are inherently underspecified and require more by way of pragmatic inference to access the contextually appropriate referent. Children’s ability to use pronouns in a subject position to refer to discourse-given referents has been found to be adult like from the age of 3 to 4 years (Matthews et al. 2006). Although the pragmatic complexity of pronouns is outside the scope of Ringstad & Westergaard (2023), they point out that “[m]ore detailed studies of the subjects’ pragmatic and information structural properties could yield further insights of what triggers the two subject positions” (ibid: 9). We take it to illustrate that analyses of production data for language acquisition, even when the goal is to investigate children’s syntactic competence, inevitably

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leads the researcher into the pragmatic domain, considering how syntax interacts with pragmatics.

3 Production data in studies of pragmatic development

In this section we highlight examples of production studies that focus on aspects of pragmatic development, including additive focus particles, intonation, and creative uses of language in the form of lexical innovation and ‘figurative’ uses. Our aim is to show how this data provides a valuable source of evidence for studies of early pragmatic development.

3.1 The production of additive focus particles

Additive focus particles (e.g., *also*) contribute to utterance meaning by including a set of alternatives to an utterance’s focused syntactic constituent, for instance the subject or object (König 1991). Consider the following use of the Norwegian additive focus particle *også*: (i) *JONAS spiser OGSÅ is* (‘Jonas, too, is eating ice cream’),² and (ii) *JONAS spiser også IS* (‘Jonas is eating ice cream too’). While the utterance in (i) conveys that in addition to Jonas, someone else is eating ice cream, the utterance in (ii) communicates that in addition to eating ice cream, Jonas is eating something else. The alternatives have to be inferred on the basis of lexical, syntactic and intonational cues and the contextual situation (Fretheim 1987, König 1991), introducing a level of linguistic complexity which makes them interesting to study from a developmental perspective.

There are only a few developmental studies of the production of the additive focus particle. Nderstigt (2006) analyzed the productions of a German-speaking girl (0;10–2;0) and found that she started to produce the accented *auch* at around 18 months, six months before the unaccented *auch*. Her findings were later confirmed by Höhle et al. (2009) who showed that *auch* is frequently produced by children in their early multi-word utterances, and Höhle et al. (2009), who found that young German-speaking children produce the accented *auch* before the unaccented version. Importantly, children were never found to produce accented *auch* in contexts requiring an unaccented *auch*, suggesting that they know when and how to produce the focus particle appropriately. Similar results have been found for Japanese-speaking children’s early productions of *mo* (‘also’) (Matsuoka et al. 2006). While studies of Norwegian-speaking children’s early productions of *også* are currently limited to anecdotal observations in the form of

²Capital letters indicate a tonal rise to an extra high tone.

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diary notes, we might expect Norwegian children to produce the additive focus particle earlier than comprehension can be demonstrated, that is, after the age of 3 years (Helganger et al. To appear). Taken together, the production studies show that children are able to appropriately use the additive focus particles by the age of 2 years, suggesting that they have also acquired the pragmatic abilities required to use this lexical item in relevant contexts. Thus, while no comprehension studies to date have been able to capture this early pragmatic competence, production studies seem to provide valuable insights into this developing ability.

3.2 Intonational production

Helganger & Falkum (2023) is another study we believe shows that production data can provide valuable insights into young children's developing pragmatic competence. We investigated Norwegian 2–5-year-olds' productions of so-called polarity focus in a semi-structured elicitation task. Polarity focus is the case where a speaker signals her attitude about the truth or falsity of a description of some state of affairs by intonational means (e.g., *Jonas SPISER is*, conveying that 'Jonas is eating ice cream, despite what you seem to think.'). Pragmatically, the use of polarity focus requires the ability to represent and attribute a thought, to assess the reliability of its content, and to express her attitude (of endorsement or denial) toward its truth-conditional content. This suggests that the use of polarity focus requires a higher-level metarepresentational ability (Wilson 2012) and so-called 'epistemic vigilance' towards utterance content, that is, the ability to evaluate the reliability of communicated information (Sperber et al. 2010).

We designed a task that consisted of four test conditions with increasing complexity. Participants heard a handpuppet express his prior positive or negative belief about what was to be depicted in a series of upcoming pictures. For instance, in the negative belief condition, the handpuppet would say "I believe that the boy is not reading a book" and then hide, before the relevant picture (which in this condition did not match the handpuppet's expressed belief) was shown to the child. The child was then expected to produce a target utterance in response to the handpuppet's utterance when seeing the picture (e.g., *Gutten LESER bok*, 'The boy DOES read a book, despite what you just said'). We also included a control condition in which the handpuppet said "I don't know what the girl does" before the child was shown a picture of a girl hugging a teddy bear. The use of polarity focus would not be relevant in this condition since there is no thought to express an endorsement or denial towards. Results showed that from the age of 2 years children produced polarity focus only in the test conditions and never in the control condition, suggesting that they are proficient users of this

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intonation pattern from this early age. Thus, this study shows that very young children demonstrate quite complex pragmatic abilities in production when provided with an appropriate task. Given the data from Ringstad & Westergaard (2023), in which children seemed to prefer syntactically simpler constructions in production, our study shows that intonational production may be a fruitful way to investigate early pragmatic competence because it enables children to communicate rather complex content without having to verbalise it.

3.3 Early creativity: lexical innovation and ‘figurative’ uses

In this section we show how the study of children’s early lexical productions has yielded important insights for theories of pragmatic development and the mechanisms underlying figurative uses of language.

In her pioneering work, Clark (1982) observed that children learn early that the lexicon can be used creatively, and that this plays an important role in acquisition. When children have not acquired the conventional word form to express a particular meaning, they have several options available to them (Clark 1991: 47): One strategy is to overextend the meanings of already acquired words, for instance, to use the word *dog* to refer to all four-legged animals (Clark 1973; Rescorla 1980). Another strategy is to use general-purpose words, for instance, deictic terms like *this* or *that* to refer to objects and events (Clark & Sengul 1978; Garton 1985), or highly underspecified verbs like *do*, *go*, *get* or *put* to refer to actions (Clark 1996). A further strategy young children typically rely on is word-formation. In a series of foundational studies, Clark and her colleagues established the important communicative function of word coinage in language development, as a means to fill chronic vocabulary gaps, both in English and cross-linguistically (see Clark 1993, 2016, and references therein). Most of this work focuses on production, discussing data from diary studies and a range of elicitation experiments. They all attest to an early ability to use and understand neologisms of various kinds, including denominal verbs, compounds, as well as novel nouns, verbs and adjectives derived by affixation. Coining new words is a strategy that allows young children with a developing lexicon to express meanings that they otherwise have no words for.

Some authors have observed that young children’s lexical innovations have an affinity with metonymy (Bonnett & Tamine-Gardes 1982, 1984; Gibbs 1994; Konieczna & Kleparski 2006). Metonymy is a type of non-literal use where one refers to something via a salient property, as in “The moustache took your orange juice”, referring to the person with the moustache. In a discussion of lexical innovation in French children, Bonnett & Tamine-Gardes (1984) note that many early neologisms appear to be metonymically motivated, reflecting properties,

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characteristics, parts or places that children associate with the objects referred to. One example is the child Guénaële (4;1), who used the expression *noisette* ('hazel nut') to refer to her stuffed squirrel. Another productive word coinage strategy that young English-speaking children resort to for denoting subcategories is compounding (e.g., Clark 1993; Clark et al. 1985; Clark et al. 1986).³ As Konieczna & Kleparski (2006) observe, many of the novel noun-noun compounds that children produce (e.g., *clown-boy* for 'the boy who is a clown') appear to be 'metonymically motivated', highlighting a salient characteristic of the entity in question.

A small number of experimental studies have explored children's production and comprehension of metonymy (Falkum et al. 2017; Köder & Falkum 2020; Rundblad & Annaz 2010; Van Herwegen et al. 2013). They all point to an early developing capacity to produce and understand metonymic uses, in line with other forms of word coinage processes. From a theoretical point of view, production data – both spontaneous and elicited – have proven particularly useful, both in developing and supporting the claim that metaphor and metonymy rely of different pragmatic mechanisms (Falkum 2019a)⁴ and in showing how children's metonymic uses have an affinity with lexical innovation, both being processes where the child exploits salient associative relations to compensate for gaps in vocabulary. We illustrate the last point below.

In a study exploring pre-schoolers' production of referential metonyms, Falkum et al. (2017) investigated two communicative functions of metonymy. One task aimed at testing whether metonymy offers a useful referential strategy for children when they lack a conventional label, in place of fuller referring expressions that might be more demanding syntactically and/or conceptually (Task 1). Another task looked at whether young children are able to take a salient property of an individual and use it to create a new, metonymic name (Task 2).

In Task 1, children aged 3–5 and a group of adults were shown pairs of simple games using familiar objects, one after the other. The games were new to the children and were not named by the experimenter, who simply gave each

³This is also a productive option children learning other Germanic languages, such as Swedish (Mellenius 1996, 1997; Söderbergh 1979) and German (Schipke & Kauschke 2011), and likely also Norwegian, though there are, to the best of our knowledge, no existing studies.

⁴The claim is that although metaphor and metonymy have certain characteristics in common (they both target individual words or phrases, they both contribute content to the proposition explicitly expressed, and they both lie on a continuum of literal and figurative uses), early metonymic uses may be the result of a more basic, associative process which the child exploits communicatively from the earliest stages of language acquisition.

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child the instructions for playing them. Then she put the two games out of sight and asked “Which of the two games would you like to play?”, thus requiring the child to come up with a referring expression for one of the games. After letting the child play the chosen game, the experimenter asked “So, do you want to play the other game too?”, to which the child always answered affirmatively. The experimenter then asked the child to tell her which one it was (“Which one did you want to play?”), eliciting a referring expression for the second game too. The results showed three prevalent referential strategies: (i) Metonymic uses (e.g., *the stickers*), (ii) compound nouns (e.g., *the sticker game*), (iii) modified noun phrases (e.g., *the game with the stickers*). All the children produced significantly more metonyms than compounds or modified noun phrases. Also, 3-year-olds and 4-year-olds were more likely to use a metonym or a compound noun than a modified noun phrase. However, 5-year-olds and adults produced significantly more modified noun phrases than the younger children). The adults, however, were equally likely to go for any of (i), (ii) or (iii). We took these results as supporting the view that metonymy offers a useful communicative strategy for young children when they lack ready-made category label, similar to the functions of other word coinage strategies. The results also suggest that metonymy might be *particularly* useful as an early referential strategy, since this option is often simpler than a fuller descriptive phrase morphologically and/or syntactically, and may be less costly in terms of short-term memory or planning for speaking (cf. Ringstad & Westergaard 2023). Also, that younger children showed a preference for both metonyms and compound nouns over modified noun phrases suggests that metonymy and compounding may serve similar communicative functions for this group, thus supporting the view that they might be different outcomes of the same underlying phenomenon.

In Task 2, the same participants were presented with a set of pictures of people and animals, each with a salient property (a large nose, a strange hat, etc.), and children were asked to help the experimenter come up with a name for the person or animal on display (“Can you help me find names for my friends? What would you call him/her/it?”). The names produced by the participants were classified into (i) metonymic uses (e.g., *Black Lips*, *The Hat*), (ii) compounds (e.g., *Scarf Cat*, *The Blanket Girl*), (iii) modified noun phrases (e.g. *Duck with Shoes on*, *The Flying Magic Horse*), and (iv) proper names (e.g., *Laura*, *Arie*). Results showed that overall, participants produced metonyms about one-third of the time in this task, suggesting that this is a productive naming strategy for both pre-schoolers and adults. Taken together, the results of Falkum et al.’s (2017) production studies show that for both children and adults, metonymic uses may serve a function

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similar to nominal compounding, as a productive option for referring to entities that lack a conventional label or proper name.

4 Production data as a window into pragmatic development?

What can production data reveal about children's developing pragmatic competence? As we have mentioned, the common view is that one should be careful reading too much into children's productions since this involves a speculative interpretation of their communicative intentions (Zufferey 2015). However, we hope to have shown in this short paper that production data can be particularly useful for studies of early pragmatic development, and phenomena where controlled comprehension experiments may not be feasible (Falkum 2019a; Helganger & Falkum 2023).

From a methodological viewpoint, we need to consider how we can improve our production studies. The methods used in production studies span from observations of spontaneous speech in one or several children (Höhle et al. 2009; Nederstigt 2006; Westergaard 2009) to sentence-completion tasks (Höhle et al. 2009; Müller et al. 2009) and elicitation tasks (Falkum et al. 2017; Helganger & Falkum 2023; Ringstad & Westergaard 2023), which are controlled in the sense of providing participants with contextual cues but also invite them to produce spontaneous responses. An advantage of spontaneous speech in a naturalistic communicative setting is its high ecological validity (Eisenbeiss 2010). However, this method may be problematic when investigating child language phenomena that are infrequent (Tomasello & Stahl 2004) and leaves the researcher with little control and potentially a large amount of irrelevant data (Eisenbeiss 2010). An advantage of the sentence-completion task is that it is highly controllable, ensuring data of the phenomenon in question. However, this method is associated with low ecological validity and does not provide children with a communicative setting where they can put their full pragmatic competence to use. An advantage of elicitation tasks is that they invite spontaneous responses in a semi-controlled communicative setting. A possible improvement of such designs could be to combine the quantitative measures from elicitation with qualitative analyses of the unstructured parts of the conversation.

From a theoretical point of view, one question to consider is what is cognitively required for the production of utterances. Recently, Park & Clark (2022) have taken some initial steps to describe how productions of communicative acts

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can be characterized theoretically. Referring to relevance theory (Sperber & Wilson 1986/1995) and this framework's account of the procedure that hearers follow when they comprehend utterances in context ('the relevance-theoretic comprehension heuristic', Wilson & Sperber 2004), Park & Clark present what they call a 'relevance-focused *production* heuristic' (Park & Clark 2022: 180; our italics), which we reproduce below:

Relevance-focused production heuristic:

1. Follow a path of least effort in formulating and producing communicative acts (verbal and nonverbal) which will give rise to positive cognitive effects for members of your audience
2. Produce an utterance which permits members of your audience to find an interpretation which meets their expectations of relevance (one which is optimally relevant)
3. Look for evidence of how addressees are understanding your communicative acts
4. Reformulate or produce new communicative behaviour when there is evidence that expectations of relevance have not been met or that you have been misunderstood

This production heuristic tells us that in addition to being responsible for producing relevant communicative signals, the speaker is expected to monitor her interlocutor(s) in order to detect whether her communicative intentions have been understood. So if Park and Clark are correct, contrary to the common view that production data underestimate children's performance compared to comprehension data, the process of production may be just as – or even more – complex than comprehension, by requiring more by way of perspective-taking abilities. However, it is possible that children produce utterances without following such a production heuristics, and that they rely on some lower-level competence. For instance, going back to Helganger & Falkum's (2023) study of the production of polarity focus, it could be that children's productions were based on their imitation of intonation patterns available in their input. However, if that were the case, we should expect to see some productions of polarity focus also in the control condition, which we did not. Children in our study only produced polarity focus in appropriate contexts, suggesting that from early on, they can use intonation as a linguistic-pragmatic cue in utterance production. This shows that children's productions can provide valuable insights into early pragmatic development, especially in cases where controlled comprehension experiments may

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be too demanding for young participants. It should be mentioned, of course, that there are comprehension methods, including eye-tracking-while-listening, that can be conducted with very young children (even before they start producing language). These methods are also suitable for studying the acquisition of many pragmatic phenomena, such as reference assignment, perspective taking, and many others. Our point, however, is that for certain pragmatic phenomena these methods might simply not be feasible, either because it is difficult to set up ecologically valid experiments (for instance, this has proven particularly difficult for the study of subtle intonational differences with consequences for pragmatic interpretation) as well as non-literal uses of language, where it may be difficult for young children to sit through the experiments where the presentation of large chunks of language are often required to set up a context that would make a non-literal use appropriate.

5 Conclusion

In this paper we have reviewed studies investigating children's productions in several linguistic domains, focusing specifically on syntax, intonation, lexical innovation, and figurative uses of language. We have argued that all of these studies involve consideration of children's pragmatic abilities, and that production studies enable children to demonstrate their early pragmatic competence. Contrary to the current view within the field of developmental pragmatics, where production studies – in particular those involving spontaneous speech – are seen as speculative, we argue that we run the risk of losing important insights if we undermine the value of production data. We have suggested some ways forward, both methodological and theoretical, and to start acknowledging production studies as valuable sources of evidence in the same way as within other areas of developmental linguistics.

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References

- Bonnett, Clairelise & Joëlle Tamine-Gardes. 1982. Les noms construits par les enfants: Description d'un corpus. *Langages* 66. 67–101.

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- Bonnett, Clairelise & Joëlle Tamine-Gardes. 1984. *Quand l'enfant parle du langage. Connaissance et conscience du langage chez l'enfant*. Bruxelles: Pierre Mardaga.
- Clark, Eve V. 1973. What's in a word? On the child's acquisition of semantics in his first language. In Timothy E. Moore (ed.), *Cognitive development and the acquisition of language*, 65–110. New York: Academic Press.
- Clark, Eve V. 1982. The young word maker: A case study of innovation in the child's lexicon. In Eric Wanner & Lila R. Gleitman (eds.), *Language acquisition: The state of the art*, 390–425. Cambridge: Cambridge University Press.
- Clark, Eve V. 1991. Acquisitional principles in lexical development. In Susan A. Gelman & James P. Byrnes (eds.), *Perspectives on language and thought. Interrelations in development*, 32–71. New York: Cambridge University Press.
- Clark, Eve V. 1993. *The lexicon in acquisition*. Cambridge: Cambridge University Press.
- Clark, Eve V. 1996. Early verbs, event types, and inflections. In Carolyn E. Johnson & John H.V. Gilbert (eds.), *Children's language*, vol. 9, 61–73. Mahwah: Lawrence Erlbaum.
- Clark, Eve V. 2016. *First language acquisition*. 3rd edn. Cambridge: Cambridge University Press.
- Clark, Eve V., Susan A. Gelman & Nancy M. Lane. 1985. Compound nouns and category structure in young children. *Child Development* 56(1). 84–94.
- Clark, Eve V., Barbara F. Hecht & Randa C. Mulford. 1986. Coining complex compounds in English: Affixes and word order in acquisition. *Linguistics* 24. 7–29.
- Clark, Eve V. & C.J. Sengul. 1978. Strategies in the acquisition of deixis. *Journal of Child Language* 5. 457–447.
- Eisenbeiss, S. 2010. Production methods in language acquisition research. In E. Blom & S. Unsworth (eds.), *Experimental methods in language acquisition research*, 1st edn., vol. 27, 11–34. Netherlands: John Benjamins.
- Falkum, Ingrid L. 2019a. Metaphor and metonymy in acquisition: A relevance-theoretic perspective. In Kate Scott, Billy Clark & Robyn Carston (eds.), *Relevance, pragmatics and interpretation*, 205–217. Cambridge: Cambridge University Press.
- Falkum, Ingrid L. 2019b. Pragmatic development: Learning to use language to communicate. In Jessica S. Horst & Janne Von Koss Torkildsen (eds.), *International handbook of language acquisition*. London: Routledge.
- Falkum, Ingrid L., Marta Recasens & Eve V. Clark. 2017. 'the moustache sits down first': On the acquisition of metonymy. *Journal of Child Language* 44(1). 87–119. DOI: [10.1017/S0305000915000720](https://doi.org/10.1017/S0305000915000720).

Ingrid Lossius Falkum & Line Sjøtun Helganger

- Fretheim, Thorstein. 1987. Pragmatics and intonation. In J. Verschueren & M. Bertuccelli-Papi (eds.), *The pragmatic perspective: Selected papers from the 1985 International Pragmatics Conference*, 395–420. Amsterdam: John Benjamins.
- Garton, Alison. 1985. The production of 'this' and 'that' by young children. *First Language* 6(16). 29–39.
- Gibbs, Raymond W. 1994. *The poetics of mind*. Cambridge: Cambridge University Press.
- Helganger, Line S. & Ingrid L. Falkum. 2023. Intonational production as a window into children's early pragmatic competence: The case of the Norwegian polarity focus and two jo particles. *Frontiers in Psychology* 14. DOI: [10.3389/fpsyg.2023.1116842](https://doi.org/10.3389/fpsyg.2023.1116842).
- Helganger, Line S., Camilo R. Ronderos, Franziska Köder & Ingrid L. Falkum. To appear. Children's sensitivity to intonational cues in ambiguity resolution: The Norwegian additive focus particle også ('also'). *Journal of Child Language*.
- Höhle, Barbara, Frauke Berger, Anja Müller, Michaela Schmitz & Jürgen Weisenborn. 2009. Focus particles in children's language: Production and comprehension of auch 'also' in German learners from 1 year to 4 years of age. *Language Acquisition* 16(1). 36–66. DOI: [10.1080/10489220802584550](https://doi.org/10.1080/10489220802584550).
- Isaacs, Susan. 1930. *Intellectual growth in young children*. London: Routledge & Kegan Paul Ltd.
- Isaacs, Susan. 1933. *Social development in young children*. New York: Harcourt Brace.
- Johnson, Elizabeth & Tania Zamuner. 2010. Using infant and toddler testing methods in language acquisition research. In Elma Blom & Sharon Unsworth (eds.), *Experimental methods in language acquisition research*, 73–93. Amsterdam: John Benjamins.
- Köder, Franziska & Ingrid L. Falkum. 2020. Children's metonymy comprehension: Evidence from eye-tracking and picture selection. *Journal of Pragmatics* 156. 191–205. DOI: [10.1016/j.pragma.2019.07.007](https://doi.org/10.1016/j.pragma.2019.07.007).
- Konieczna, Ewa & Grzegorz A. Kleparski. 2006. Motivation in language: The case of metonymically motivated innovative nouns in children's speech. *Odissea* 7. 103–117.
- König, Ekkehard. 1991. *The meaning of focus particles: A comparative perspective*. London: Routledge.
- Leopold, Werner F. 1949. *Speech development of a bilingual child*. Evanston: Northwestern University Press.
- Levelt, Willem J. M. 2012. *A history of psycholinguistics: The pre-Chomskyan era*. Oxford: Oxford Academic.

11 Using production data to study pragmatic development

- Matsuoka, Kazumi, Nobuhiro Miyoshi, Koji Hoshi, Masanobu Ueda, Izumi Yabu & Miki Hirata. 2006. The acquisition of Japanese focus particles: Dake (only) and mo (also). In *Proceedings of the 30th Boston University Conference on Language Development*.
- Matthews, Danielle. 2014. *Pragmatic development in first language acquisition*. Amsterdam: John Benjamins.
- Matthews, Danielle, Elena Lieven, Anna Theakston & Michael Tomasello. 2006. The effect of perceptual availability and prior discourse on young children's use of referring expressions. *Applied Psycholinguistics* 27. 403–422. DOI: [10.1017/S0142716406060334](https://doi.org/10.1017/S0142716406060334).
- Mellenius, Ingmarie. 1996. Children's comprehension of Swedish nominal compounds. In Carolyn E. Johnson & John H. V. Gilbert (eds.), *Children's language*, vol. 9, 167–182. Mahwah: Lawrence Erlbaum.
- Mellenius, Ingmarie. 1997. *The acquisition of nominal compounding in Swedish* (Travaux de l'institut de linguistique de Lund 31). Lund: Lund University Press.
- Müller, A., B. Höhle, M. Schmitz & J. Weissenborn. 2009. Information structural constraints on children's early language production: The acquisition of the focus particle auch ('also') in German-learning 12- to 36-month-olds. *First Language* 29(4). 373–399. DOI: [10.1177/0142723709105314](https://doi.org/10.1177/0142723709105314).
- Nederstigt, Ulrike. 2006. Additive particles and scope marking in child German. In Veerle van Geenhoven (ed.), *Semantics in acquisition*, 303–328. Dordrecht: Springer.
- Park, Kyu Hyun & Billy Clark. 2022. A relevance-focused production heuristic. *Journal of Pragmatics* 187. 176–185. DOI: [10.1016/j.pragma.2021.11.007](https://doi.org/10.1016/j.pragma.2021.11.007).
- Piaget, Jean. 1923. *The language and thought of the child*. London: Routledge.
- Pouscoulous, N. 2014. 'the elevator's buttocks': Metaphorical abilities in children. In D. Matthews (ed.), *Pragmatic development in first language acquisition*, 239–259. Amsterdam: John Benjamins.
- Rescorla, Leslie A. 1980. Overextension in early language development. *Journal of Child Language* 7(2). 321–335.
- Ringstad, Tina & Marit Westergaard. 2023. Children's acquisition of word order variation: A study of subject placement in embedded clauses in Norwegian. *Journal of Child Language* 5. 1–25. DOI: [10.1017/S0305000923000259](https://doi.org/10.1017/S0305000923000259).
- Rundblad, Gabriella & Dagmara Annaz. 2010. Development of metaphor and metonymy comprehension: Receptive vocabulary and conceptual knowledge. *British Journal of Developmental Psychology* 28. 547–563. DOI: [10.1348/026151009X454373](https://doi.org/10.1348/026151009X454373).

Ingrid Lossius Falkum & Line Sjøtun Helganger

- Schipke, Christine S. & Christina Kauschke. 2011. Early word formation in German language acquisition: A study on word formation growth during the second and third years. *First Language* 31(1). 67–82.
- Söderbergh, Ragnhild. 1979. Barnets språkliga kreativitet belyst av exempel från ordböjning och ordbildning [The child's linguistic creativity, as shown in examples of inflection and word formation]. *Språk och språknorm. Skrifter utgivna av Svenska språknämnden* 67. 236–245.
- Sperber, Dan, Fabrice Clément, Christophe Heintz, Olivier Mascaró, Hugo Mercier, Gloria Origgi & Deirdre Wilson. 2010. Epistemic vigilance. *Mind & Language* 25(4). 359–393.
- Sperber, Dan & Deirdre Wilson. 1986/1995. *Relevance: Communication and cognition*. Oxford: Blackwell.
- Stern, Clara & William Stern. 1907. *Die Kindersprache: Eine psychologische und sprachtheoretische Untersuchung*. Leipzig: Barth.
- Tomasello, M. & D. Stahl. 2004. Sampling children's spontaneous speech: How much is enough? *J. Child Lang* 31(1). 101–121. DOI: [10.1017/S0305000903005944](https://doi.org/10.1017/S0305000903005944).
- Van Herwegen, Jo, Dagmara Dimitriou & Gabriella Rundblad. 2013. Development of novel metaphor and metonymy comprehension in typically developing children and Williams syndrome. *Research in Developmental Disabilities* 34(4). 1300–1311. DOI: [10.1016/j.ridd.2013.01.017](https://doi.org/10.1016/j.ridd.2013.01.017).
- Westergaard, Marit. 2009. *The acquisition of word order: Micro-cues, information structure, and economy* (Linguistik Aktuell/Linguistics Today 145). Amsterdam: John Benjamins.
- Wilson, Deirdre. 2012. Metarepresentation in linguistic communication. In Deirdre Wilson & Dan Sperber (eds.), *Meaning and relevance*, 230–258. Cambridge: Cambridge University Press.
- Wilson, Deirdre & Dan Sperber. 2004. Relevance theory. In Laurence R. Horn & Gregory Ward (eds.), *The handbook of pragmatics*, 607–632. Oxford: Blackwell.
- Zufferey, Sandrine. 2015. *Acquiring pragmatics: Social and cognitive perspectives*. London: Routledge.

Chapter 12

Cross-linguistic influence in L3 Norwegian: An empirical study of Ukrainian-English speakers in Norway

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In this empirical study, we build on previous L3 work by examining the L3 acquisition of Norwegian among L1 Ukrainian-L2 English multilinguals using an adapted method presented in [van Osch et al. \(2023\)](#). Specifically, we conduct a gap filling task on adverb placement based on similarities and differences in the languages, which allows us to identify whether cross-linguistic influence (CLI), if present, comes from the L1, the L2 or both previously acquired languages. Based on related results and analyses ([Kolb et al. 2022, 2023](#), [van Osch et al. 2023](#)), we anticipate CLI from both previously acquired languages. Our findings show that L3 learners have acquired surface-level Norwegian word order in main clauses and overgeneralize the same pattern to subordinate clauses. We did not find evidence for surface-level CLI from L1 Ukrainian or L2 English. While our results do not support direct CLI from either Ukrainian or English, analyzing them through the lens of the Linguistic Proximity Model (LPM, [Mykhaylyk et al. 2015](#); [Westergaard et al. 2017](#); [Westergaard et al. 2019](#)), requires us to consider more abstract patterns of CLI based on individual structural properties. This perspective leads us to new questions to pursue in future studies.

1 Introduction

This empirical study investigates cross-linguistic influence (CLI) in third language (L3) acquisition to determine whether CLI occurs from the first language (L1), the second language (L2) or both, and whether CLI in L3 acquisition is

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modulated by extra-linguistic variables such as L2/L3 age of onset, proficiency, and use. We examine the acquisition of adult L3/L4¹ Norwegian among recent Ukrainian migrants to Norway, predominantly refugees, whose first language is either Ukrainian, Russian or both, and who have learned English in a school setting. Since all speakers know both Ukrainian and Russian and because the languages do not differ on the properties under investigation (see e.g., Kolb et al. 2022 for a study on CLI involving Russian adverbs), we will view English as the L2 and Norwegian as the L3 for the sake of exposition.

This study builds on previous L3 theoretical advancements, particularly the Linguistic Proximity Model (LPM: Mykhaylyk et al. 2015, Westergaard et al. 2017; Westergaard 2021), to examine the source of CLI in L3 acquisition and whether CLI occurs property by property based on structural similarity. The LPM suggests that structural similarity is a key factor leading to CLI on a property-by-property basis. Different models attribute CLI in L3 acquisition to a variety of factors, such as order of acquisition (Bardel & Falk 2007, Jin 2009), language use (Fallah et al. 2016, Fallah & Jabbari 2018), overall typological similarity (Rothman 2015), structural similarity (Westergaard et al. 2017; Westergaard 2021, Slabakova 2017), and cumulative input (Cabrelli & Iverson 2023).

Using a gap-filling task designed to examine adverb placement across main and subordinate clauses with lexical and modal verbs, we investigate whether CLI in Norwegian emerges from the L1, the L2 or both (see Section 2.2 for details). In line with Westergaard's micro-cue approach, we focus on fine-grained grammatical distinctions, here between different clause and verb types, rather than global word order patterns, e.g., SVO versus SOV. We do this in order to contribute to the understanding of how L3 learners use these specific grammatical cues when acquiring further languages.

2 Theoretical preliminaries and empirical predictions

2.1 Introduction

This section outlines this study's theoretical underpinnings and motivations for investigating CLI in adverb placement among Ukrainian-English-Norwegian multilinguals. We present the Linguistic Proximity Model (LPM; Westergaard et al. 2017; Westergaard et al. 2019), contextualized with other perspectives on L3 acquisition. We then discuss the relevant syntactic differences in Ukrainian, English, and Norwegian that are the domains for our investigation.

¹As most participants speak both Ukrainian and Russian, Norwegian functions technically as a fourth language (L4).

2.2 The Linguistic Proximity Model

The LPM is anchored in generative theories on language acquisition. An important facet of this model – and Westergaard’s research in language acquisition more broadly – is that it contributes to debates on understanding the role of universal grammar (UG) in L2, L3, and beyond (Ln) acquisition. A specific conceptual issue is whether differences in outcomes in L1 versus Ln acquisition are due to lack of, or reduced, access to UG in the latter. Westergaard’s research on the acquisition of syntactic structures in the first language (Westergaard 2009), second language (Westergaard 2003), and third language (Mykhaylyk et al. 2015, Westergaard et al. 2017; Westergaard 2021), suggests that there is no reduced or restricted role in Ln acquisition: “first language (L1), second language (L2) and third language (L3) acquisition are fundamentally the same process” (Westergaard 2021: 379). The core differences in these developments and trajectories, then, stem from (1) whether or not there is preexisting acquired structure, and if so, (2) the specific make-up and organization of those existing representations that allows for the parsing of new and potentially different linguistic input. While the LPM is primarily a model that considers syntactic acquisition, its influence stretches beyond. For instance, recent research on Ln phonological acquisition (Archibald 2022, 2023; Wu 2024) and general theorizing on multilingual morphophonological representations (e.g., Natvig et al. 2025) adopts this perspective and line of argumentation.

Westergaard (2021) takes the position that the acquisition of any language occurs via the parsing of input for grammatical comprehension (e.g., Lightfoot 2006, 2020; Westergaard 2009, 2014). Parsing requires some form of abstract knowledge, here in the form of micro-cues, which are constructed from innate categories and small-scale structural relationships, rather than large-scale rules or constraints on word order (see Westergaard 2009 for a full discussion of these representations and their relationships to Norwegian V2 acquisition). In terms of Ln acquisition, then, micro-cues from the acquisition of previous languages are available to parse Ln input; instances where the existing grammar(s) cannot parse the relevant input necessitate the acquisition of novel representations. The benefit of this approach is that it provides an explicit framework for assessing both facilitative and non-facilitative CLI based on fine-grained structural similarities and differences in all the languages involved.

Because the LPM predicts, in an L3 acquisition context, that both the L1 and the L2 may influence the L3, the model contrasts with competing theories of L3 acquisition. Specifically, theories that predict primary influence from the L1 (Jin 2009), the L2 (Bardel & Falk 2007), or either based on use and communication

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(Fallah et al. 2016, Fallah & Jabbari 2018) or typological proximity (Rothman 2015, Rothman et al. 2019). Although the language of communication may potentially be relevant for the present study and is worthy of future investigation, most structural models focus primarily on initial stages of L3 acquisition. Because the participants in this study are at various acquisitional stages (see Section 3) and we cannot examine the initial stages of their Norwegian acquisition, the LPM is most appropriate for examining their syntactic judgments in Norwegian and English. Furthermore, we are interested in variation within distinct syntactic contexts – main clauses and subordinate clauses – and the potential for asymmetric patterns of influence from one or multiple languages on another. We will now take on a more detailed discussion of the relevant structures in Ukrainian, English, and Norwegian.

2.3 Syntactic structures under investigation

In this section, we compare and contrast the general preferences in Ukrainian, English, and Norwegian for the properties under investigation. We focus on the relative positions of verbs and adverbs in the three languages since these surface patterns allow us to investigate CLI from L1 or L2, or both, on a property-by-property basis. While the preferred ordering in Ukrainian is adverb-verb (see baseline data presented in Section 4.1), both English and Norwegian make structural distinctions that affect adverb position relative to the verb. These considerations are likewise distinct in each language. While Norwegian finite verbs precede adverbs in main clauses and follow them in subordinate clauses, English contrasts modal and lexical verbs, with modal verbs occurring before adverbs and lexical verbs after them. In the Norwegian examples in (1), both finite modals (*vil* ‘will, want’) and lexical verbs (*jobber* ‘work’) occur before the adverb in main clauses (1a, 1c). In subordinate clauses, however, both follow the adverb (1b, 1d).

(1) Norwegian

- a. Main clause with a modal verb
 Jon vil alltid jobbe.
 John will always work
 ‘John will always work.’
- b. Subordinate clause with a modal verb
 Jon som alltid vil jobbe.
 John REL always will work
 ‘John, who will always work.’

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- c. Main clause with a lexical verb
Jon jobber alltid.
John works always
'John always works.'
- d. Subordinate clause with a lexical verb
Jon som alltid jobber.
John REL always works
'John, who always works.'

For English, on the other hand, clause type is irrelevant for the surface order of adverbs and verbs. Modal verbs precede the verb in both main and subordinate clauses (2a, 2b), whereas lexical verbs occur after the adverb in the same environments (2c, 2d).

(2) English

- a. Main clause with a modal verb
John will always work
- b. Subordinate clause with a modal verb
John, who will always work
- c. Main clause with a lexical verb
John always works
- d. Subordinate clause with a lexical verb
John, who always works

According to Mykhaylyk & Ko (2010) and Mykhaylyk (2013), neither clause type nor verb type is a deciding factor for adverb placement in Ukrainian, as both modal and lexical verbs typically follow the adverb in both clause types, as in (3a–3d). However, as will be discussed in Section 4.1, Ukrainian may also exhibit subtle distinctions between modal and lexical verbs, at least for some speakers.

(3) Ukrainian

- a. Main clause with a modal verb
Джон завжди буде працювати
Dzhon zavzhdy bude pratsiuvaty
John always will work
'John will always work.'

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- b. Subordinate clause with a modal verb
 Джон який завжди буде працювати
 Dzhon jakyj zavzhdy bude pratsiuvaty
 John REL always will work
 ‘John, who will always work.’
- c. Main clause with a lexical verb
 Джон завжди працює
 Dzhon zavzhdy pratsiuje
 John always works
 ‘John always works.’
- d. Subordinate clause with a lexical verb
 Джон який завжди працює
 Dzhon jakyj zavzhdy pratsiuje
 John REL always works
 ‘John, who always works.’

Space does not allow us to go into a detailed analysis of the structures at play, but we draw on broad insights from the generative syntactic literature. First, Norwegian is considered a ‘V2’ language, so the verb moves to the left periphery in main clauses (e.g., [Westergaard 2009](#)), capturing the higher position of the verb relative to the adverb. However, this movement is either not available for subordinate clauses, or the structures are different (see [Westergaard et al. 2019](#) for a discussion of different approaches), so finite verbs do not raise over the adverb. For English, we follow [Lightfoot & Westergaard \(2007\)](#) by assuming that modals are assigned a different category as lexical verbs and are generated above adverbs, whereas lexical verbs are in a lower position. This explains distinctions based on verb type, but not clause type since English does not employ verb movement into the left periphery. Finally, in contrast to Norwegian, there is no required verb movement into the C-domain in Ukrainian. Ukrainian also differs from English in that it does not demonstrate a categorical distinction between modal and lexical verbs. We assume that all verbs are generated lower than adverbs since both sentential and manner adverbs typically precede the verb in Ukrainian ([Mykhaylyk & Ko 2010](#), [Mykhaylyk 2013](#)). Accordingly, the preference for adverb-verb order in Ukrainian can be understood as a structure that does not undergo verb movement (see, however, [Kallestinova & Slabakova 2008](#) for Russian). These cross-linguistic differences in surface patterns are summarized in [Table 1](#).

In an Ln-acquisition context, Ukrainian provides the structure for parsing Norwegian subordinate clauses, i.e., the domains where verb movement to the

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Table 1: Cross-linguistic Variation in Norwegian, English, and Ukrainian

Clause type	Verb type	Norwegian (L3)	English (L2)	Ukrainian (L1)
Main	Lexical	V-Adv	Adv-V	Adv-V
Main	Modal	V-Adv	V-Adv	Adv-V
Sub	Lexical	Adv-V	Adv-V	Adv-V
Sub	Modal	Adv-V	V-Adv	Adv-V

CP layer is not available. Likewise, because, on the view presented here, all Ukrainian verbs are generated lower than the adverb, English structures with lexical verbs can be interpreted based on the existing grammar. Furthermore, splitting English verbs into two types necessitates the acquisition of a new category, or at least the assignment of a subtype of verbs into a different category, to capture their distinct behaviors. The acquisition of postverbal adverbs in Norwegian main clauses, however, implies internalizing verb movement to a higher position above the adverb.

2.4 Research questions and predictions

We developed the following research questions and predictions:

Research Questions

RQ1. Does cross-linguistic influence – if present – occur from the L1, the L2 or both previously acquired languages?

RQ2. Do experiential variables, such as age of onset, proficiency, and current language use modulate the L3 acquisition process?

Predictions

Following the LPM and in line with recent research findings, we have formulated the following predictions:

P1. We predict CLI to occur from both L1 and L2 with facilitation and non-facilitation from both previously acquired languages (e.g., [Listhaug et al. 2021](#), [Stadt et al. 2018, 2020](#), [van Osch et al. 2024](#)).

- If L3 learners are sensitive to verb type by distinguishing between lexical and modal verbs, this indicates CLI from L2 English.

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- If L3 learners show an overall preference for adverb-verb word order with no significant differences between conditions, this indicates CLI from L1 Ukrainian.
- If L3 learners are sensitive to clause type by distinguishing between subordinate and main clauses, this indicates learning of L3 Norwegian.

P2. We predict that one or several of the following factors influence task performance in line with (partially contradictory) previous findings:

- L2 and/or L3 proficiency (e.g., [Sánchez & Bardel 2017](#), [Listhaug et al. 2021](#), [Stadt et al. 2018, 2020](#))
- L2 exposure ([Tremblay 2006](#), [Stadt et al. 2018, 2020](#))

3 The study

3.1 Overview

The following four measures were used: (1) gap filling task (GFT) on adverb placement in Norwegian, (2) GFT on adverb placement in English, (3) an adapted version of the Peabody Picture Vocabulary Task (PPVT) in English and (4) a language background questionnaire. The methodological design has been adapted from [van Osch et al. \(2023\)](#). The GFT in Norwegian is the main experiment, while the GFT has also been conducted in English to investigate the participants' preferences on adverb placement in L2 English. An adapted version of the PPVT with 20 items was used as a proficiency measure in English, and variables on age of onset, length of exposure, self-rated proficiency and language use were collected via the language background questionnaire. The L3 data was collected in Norwegian classrooms at the university.

3.2 Participants

The participants in this study are adult L3 Norwegian learners from Ukraine living in Norway (n=34). Many participants indicated having acquired both Russian and Ukrainian from birth. This was not considered a problem since adverbs-verb is the canonical order in Russian (see [Kolb et al. 2022](#)). Two participants were excluded due to having other additional first languages, leading to a total of 32 participants (28 female, 4 male). Furthermore, we analyzed data from the English GFT to verify that participants demonstrated the expected patterns in their L2

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English grammars. This step was essential to investigate cross-linguistic influence; to draw meaningful conclusions about potential cross-linguistic influence from English into Norwegian, we first needed to ensure that the participants’ English grammars were not themselves shaped by their L1. The results revealed considerable individual variation, particularly in subordinate clauses involving modal verbs. To maintain the integrity of the analysis, we excluded participants whose accuracy of adverb placement fell below 66%.

After exclusion, 18 participants remained, of which the main characteristics are summarized in Table 2 and Table 3.

Table 2: Description of included participants

N	18
Age at testing, Range (Mean)	18–49 (28,11)
Norwegian level	Beginner (n=7) Intermediate (n=11)
L1	Ukrainian (n=4) Ukrainian/Russian (n=12) Russian (n=2)
L2/3	English
L3/4	Norwegian
Further languages	Polish (n=5) French (n=2) Spanish (n=1) Chinese (n=1) German (n=1)
Age of Onset in English, Range (Mean)	6–45 (14.08)
Length of Exposure to English, in years, Range (Mean)	4–33 (13.25)
Length of Exposure to Norwegian, in years, Range (Mean)	0–3 (1.38)

Both beginner and intermediate groups received 11 hours of Norwegian instruction, two hours of tutoring, and two hours of listening practice per week. At the time of testing, the beginner group had a total of 5 weeks of instruction while the intermediate group was in their third learning year.

Table 3: Proficiency and language use of included participants

N	18
Proficiency in English, PPVT score 0–20, Range (Mean)	6–19 (14.21)
Self-reported proficiency in English, 1–4, Range (Mean)	1–4 (2.47)
Self-reported proficiency in Norwegian, 1–4, Range (Mean)	1–2 (1.58)
Current use of Ukrainian and Russian, in percent, Range (Mean)	20–100 (63.81)
Current use of English, in percent, Range (Mean)	0–61.5 (18.31)
Current use of Norwegian, in percent, Range (Mean)	0–50 (16.6)

3.3 Methodology

The gap filling task in Norwegian included six items per condition², leading to a total of 24 items. Our four experimental conditions focused on placement of the adverb always in main and subordinate clauses with lexical (play, dance, work, sing, walk, swim) and modal verbs (will); (4) shows an example of a main clause with a lexical verb, (5) a main clause with a modal, (6) a subordinate clause with a lexical verb, and (7) a subordinate clause with a modal. Two filler item conditions targeted topicalization as in (8) and embedded questions as in (9). Norwegian differs from both English and Ukrainian in condition (4), and patterns with both English and Ukrainian in condition (6), while it patterns with English in condition (5), and with Ukrainian in condition (7). The conditions differ based on verb type and clause type (see Table 1 for an overview). The adverb always was chosen as it led to single word translations in all three languages and plausible sentences.

- (4) Main clause with a lexical verb

Dette er Vidar. Han jobber alltid hjemme.

This is Vidar. He work-PRES always at.home

‘This is Vidar. He always works at home.’
- (5) Main clause with a modal verb

Dette er Karl. Han vil alltid gå i parken.

This is Karl. He will.PRES always walk in park-DEF

‘This is Karl. He will always walk in the park.’
- (6) Subordinate clause with a lexical verb

²One item in the main clause modal condition contained an error and was excluded from the analysis.

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Min bestefar Jørgen, som alltid synger karaoke, er lykkelig.
 My grandfather Jørgen REL always sings-PRES karaoke is happy
 ‘My grandfather Jørgen, who always sings karaoke, is happy.’

- (7) Subordinate clause with a modal verb

Min bestefar Martin, som alltid vil jobbe på skolen er
 My grandfather Martin REL always will.PRES work on school-DEF is
 positiv.
 positive
 ‘My grandfather Martin, who will always work at the school, is positive.’

- (8) Topicalization

Når sola skinner, går Suzanne i parken.
 When sun-DEF shine-PRES walk-PRES Suzanne in park-DEF
 ‘When the sun shines, Suzanne walks in the park.’

- (9) Embedded questions

Jeg vet ikke når Ronja leker i parken.
 I know.PRES not when Ronja play-PRES in park-DEF
 ‘I don’t know when Ronja plays in the park.’

This experimental design allows us to investigate whether there is CLI from L1 Ukrainian, L2 English or both and whether structural similarity is a determining factor since Norwegian differs from and patterns with both previously acquired languages for one condition each, and patterns with only one of the previously acquired languages for one condition each.

The GFT in English is an adapted version of the Norwegian GFT with six items per condition and thus included the same four experimental and two filler conditions. It was designed to investigate the participants’ word order preferences in their L2.

The L3 and L2 data were collected in a classroom setting. The participants filled out a pencil and paper version of the Norwegian GFT individually by writing the verb in the gap before or after the adverb, e.g., Dette er Vidar. Han ____ - alltid ____ hjemme. (jobber). Afterwards, the participants filled out the English GFT and the language background questionnaire. For the PPVT in English, the participants were shown four pictures on a screen and were asked to match a word that was presented visually and orally to one of the four pictures by circling the respective number on their answer sheet.

To establish baseline data, we administered the Norwegian version of the GFT to native speakers of Norwegian and a Ukrainian version – featuring direct

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translations of all stimuli – to a separate group of native Ukrainian speakers. Both sets of baseline data were collected online using the Gorilla experiment builder (www.gorilla.sc), with participants recruited via the online platform Prolific (www.prolific.com) and through personal networks. For English, we draw on baseline data from a previous study ([van Osch et al. 2023](#)), which employed comparable sentence structures but focused on copular verbs rather than modals.

4 Results

4.1 Baseline data from native speakers of Ukrainian, Norwegian, and English

The Norwegian native speakers (n=18) behaved categorically in line with the expected pattern. They overwhelmingly chose postverbal adverbs in main clauses and preverbal adverbs in subordinate clauses (Figure 1).

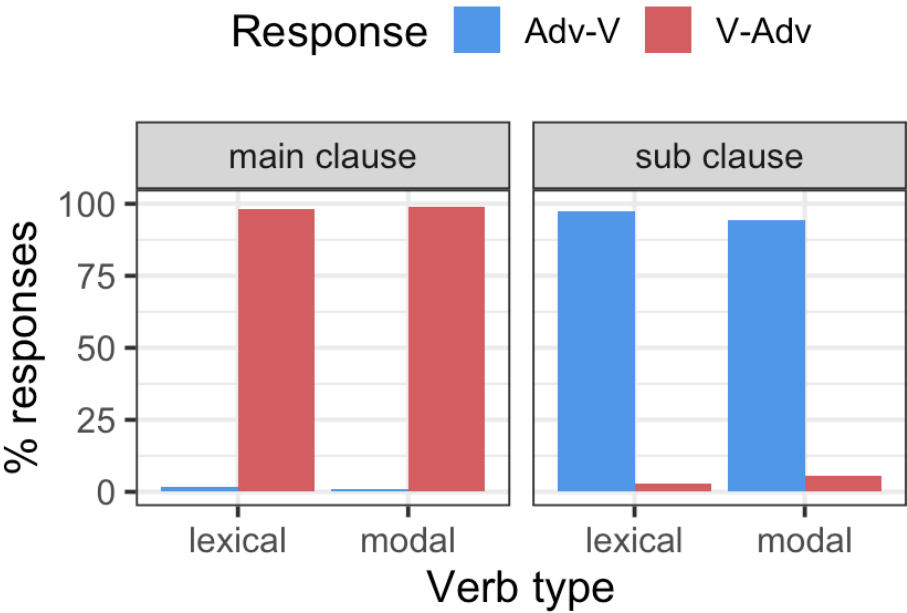


Figure 1: Responses on the Norwegian gap filling task by native speakers

In the Ukrainian baseline task, the 15 native Ukrainian speakers showed a general preference for placing the verb after the adverb across all conditions, align-

ing with patterns described in the literature (Mykhaylyk & Ko 2010, Mykhaylyk 2013). However, modal verbs were also occasionally positioned before the adjective (Figure 2). According to personal communication with a native speaker of Ukrainian who is a linguist, the use of the auxiliary *буде* (‘will’) in combination with the adverb *завжди* (‘always’) may sound somewhat unnatural, as habitual actions are typically expressed in the present tense in Ukrainian. This aspect of the language may have contributed to greater variability in responses within this condition.

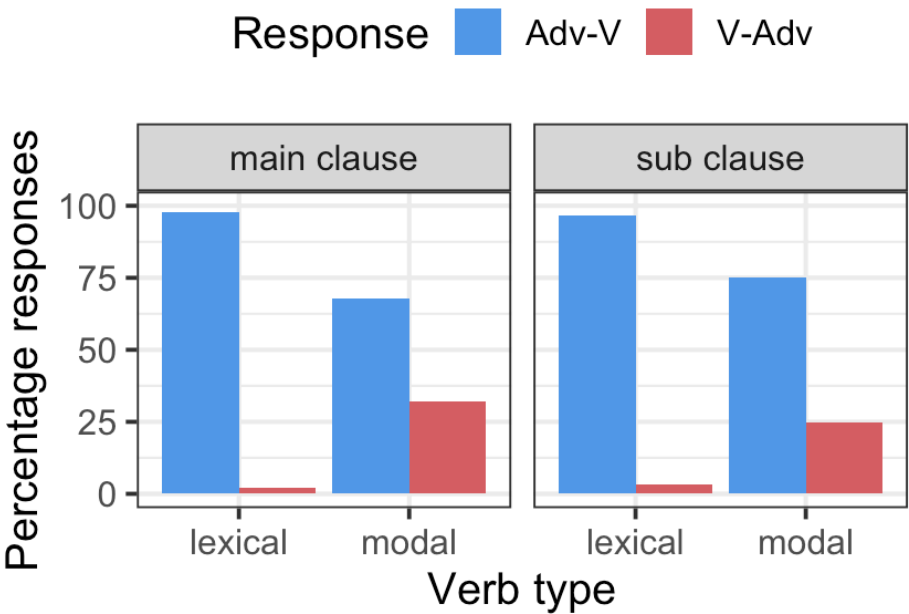


Figure 2: Responses on the Ukrainian gap filling task by native speakers

For English, we draw on data collected in a previous study conducted by two of the present authors. That study involved native English speakers and featured sentences closely resembling those used in the current experiment, with the key differences being the use of the copular verb *to be* instead of modal verbs, and the inclusion of both *often* and *always* as adverbs (e.g., *My neighbor Joseph, who is often active, walks every evening*). Data were collected via the Gorilla experiment platform, with participants recruited through Prolific. A total of 54 native English speakers participated. As illustrated in Figure 3, they responded categorically in line with the expected pattern, preferring postverbal adverbs

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for copular verbs and preverbal adverbs with lexical verbs, irrespective of clause type.

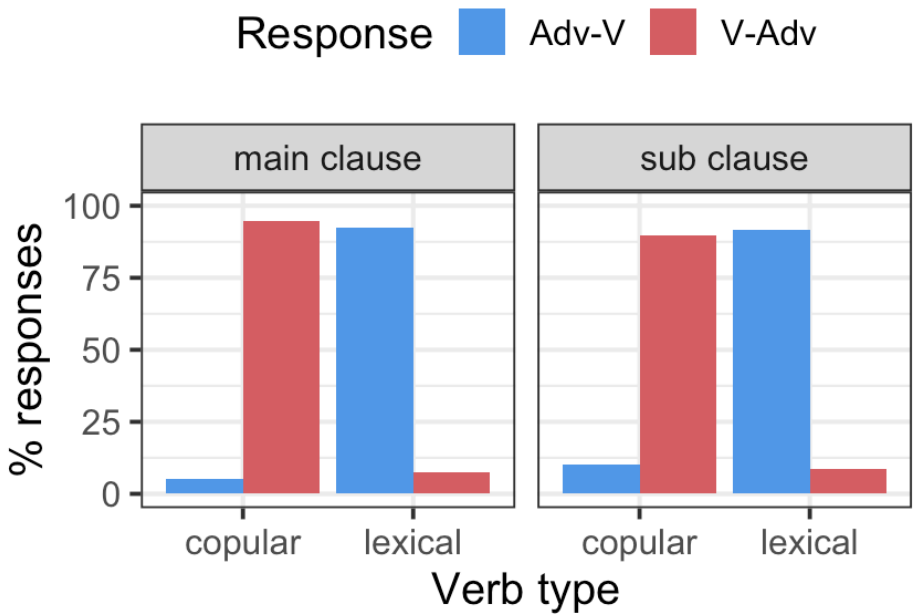


Figure 3: Responses on the English gap filling task by native speakers

4.2 Norwegian gap filling task

The responses on the Norwegian task (n=18) are depicted in Figure 4. The L3 learners show a preference for V-Adv in all conditions. However, this preference is stronger in the main clause than the subclause condition.

For the statistical analysis, all binary variables were sum-coded using orthogonal contrasts, and all numerical variables were centered and scaled. A generalized linear mixed effects model was run on the data, with clause type and verb type as linguistic predictors. In addition, various extra-linguistic variables related to proficiency in, and use of, Norwegian, English and Ukrainian/Russian, and their interaction with the linguistic variables, were added to the model in a stepwise manner, only if they significantly improved model fit. The optimal model included a significant intercept ($\beta = 2.99$, $SE = 0.67$, $z = 4.44$, $p < 0.001$), indicating

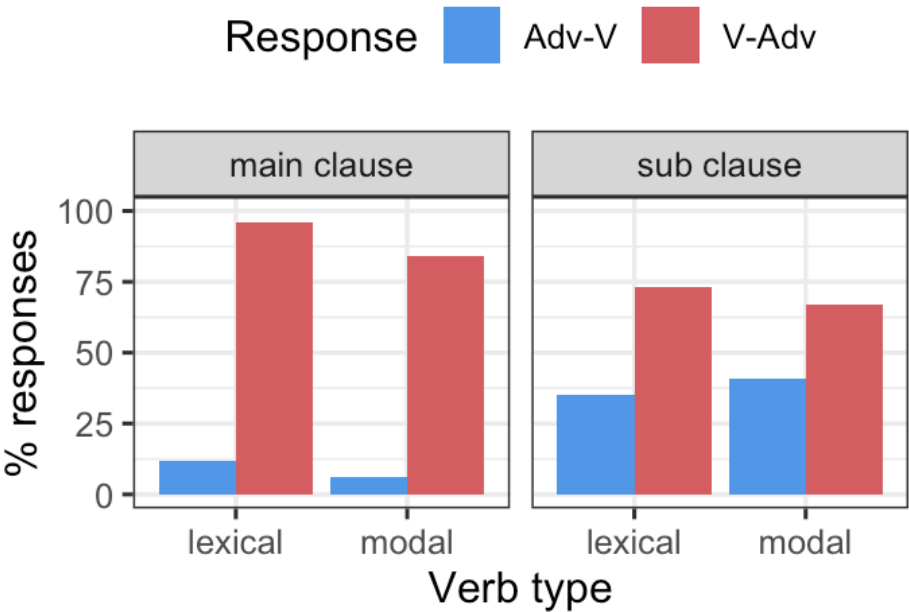


Figure 4: Percentage responses on the Norwegian gap filling task

an overall preference for verb-adverb order. There was also a strong and significant interaction between clause type and instructional level ($\beta = -2.47$, $SE = 0.78$, $z = -3.16$, $p = 0.002$), indicating that intermediate students, more than beginner students made a difference between main clauses and subordinate clauses, choosing preverbal adverbs more frequently in the later type. Post hoc testing revealed the effect of clause type was only significant for intermediate learners ($z = 6.95$, $p < 0.001$), not for beginner learners ($z = 0.46$, $p = 0.98$) (Figure 5).

While the model did not yield a significant main effect of verb type, the interaction between verb type and the proportion of use of English was marginally significant ($\beta = -0.80$, $SE = 0.41$, $z = -1.94$, $p = 0.05$), which suggested that an increased use of English was associated with a higher preference for verb-adverb with lexical verbs relative to modal verbs in L3 Norwegian (Figure 6).

5 Discussion

We now return to our research questions, namely whether CLI in Norwegian comes from the L1 (Ukrainian or Russian), L2 (English), or both (RQ1) and

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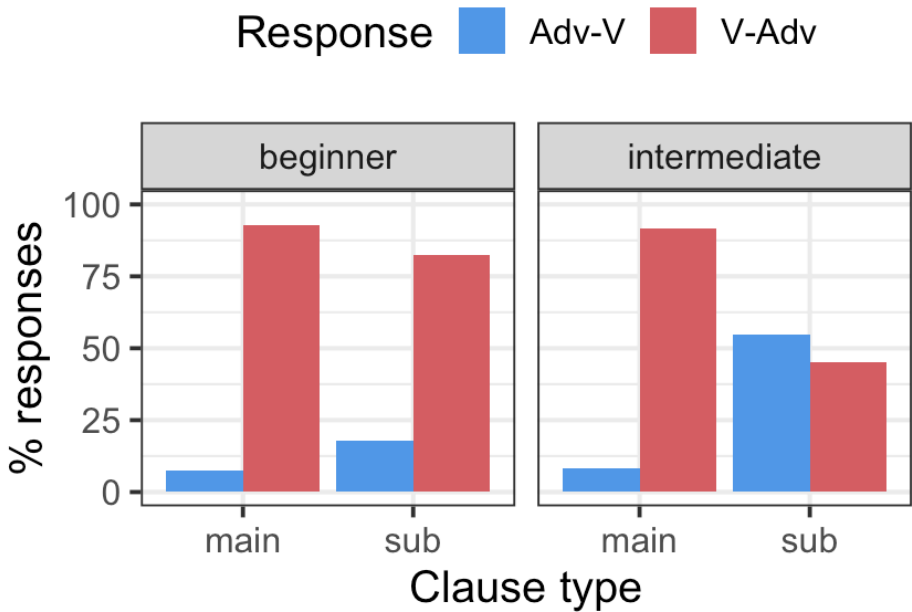


Figure 5: Frequency of responses (in percentages) on the Norwegian gap filling task, divided by instructional level and clause type

whether English age of acquisition, proficiency, and use modulate L3 acquisition (RQ2). The low number of participants included in the analysis ($n=18$) means that our results may not be representative, but they indicate patterns in L3 acquisition that serve as a basis for follow up research. Our findings in L3 Norwegian show an overall preference for V-Adv word order (e.g., *han jobber alltid*) in all four conditions. However, we found stronger preferences for V-Adv in main clauses compared to subordinate clauses. This effect was modulated by instructional level and use of English. Thus, we find, on the one hand, that the linear strings of verbs and adverbs in Norwegian do not provide evidence for direct CLI from either language, contrary to e.g., [Listhaug et al. \(2021\)](#); [Stadt et al. \(2018, 2020\)](#); [van Osch et al. \(2024\)](#) and our first prediction (P1). Rather, the results raise the question of whether English V-Adv for modal verbs may facilitate learning of surface-level Norwegian word order in main clauses, and overgeneralization of the same word order in subordinate clauses, in line with the target structure in Norwegian in main clauses but not in subordinate clauses. This will have to be investigated further. On the other hand, it is not clear from the GFT results what the underlying structures deriving this word order are. That is, do the

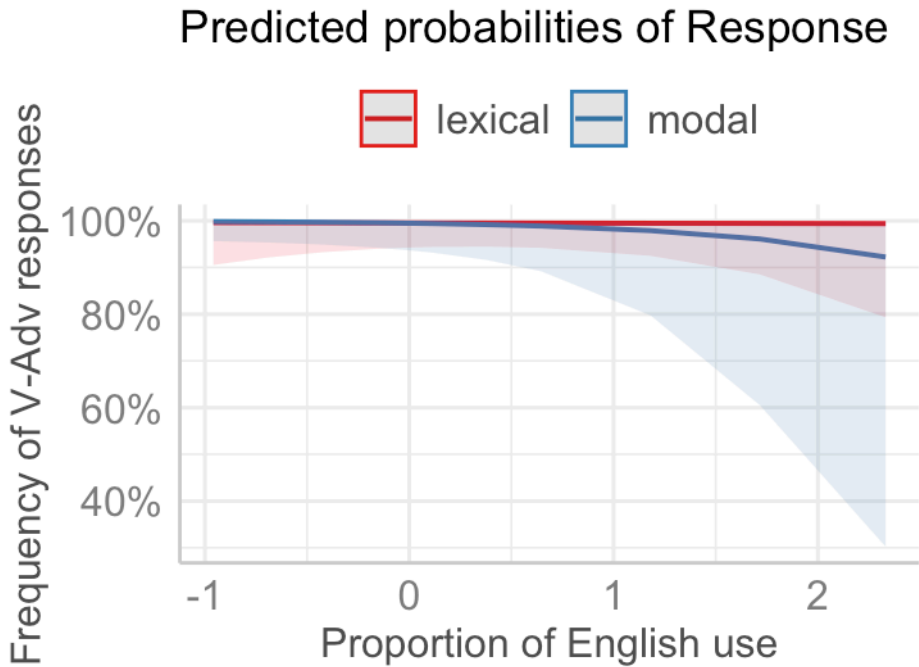


Figure 6: Predicted probabilities of V-Adv responses as a factor or proportion of use of English and verb type

participants acquire verb movement into the left periphery, or into a shallower structure? We elaborate on the need to investigate this question further.

Because we have only included participants who perform at ceiling for the English GFT, one indication of English CLI would have been a distinction in Norwegian modal and lexical verbs. This does not occur; verb-adverb order is preferred regardless of verb type (Figure 4). Moreover, the Ukrainian preference for adverb-verb is also dispreferred in Norwegian, so CLI from the L1 is likewise not supported by the results. Evidence from the beginner group (Figure 5) clearly shows a verb-adverb preference in both clause types; we suspect that the greater frequency of main clauses compared to subordinate clauses is the primary factor for target-like performance in main clauses and overgeneralization to subordinate clauses. Intermediate learners, on the other hand, demonstrate learning of the Norwegian pattern in subordinate clauses by showing a slight preference for adverb-verb in this environment.

As for RQ2, whether experiential variables influence L3 acquisition, we found a marginal effect of the proportion of English use, yet in the opposite direction

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from our expectations. The more the participants use English, the more they prefer verb-adverb for lexical verbs, rather than modal verbs, in Norwegian. We tentatively interpret these results to indicate that the more the participants use English, the more likely they are to create a distinction between lexical and modal verbs and adopt that distinction into Norwegian, even if its realization is reversed from that in English. However, further empirical evidence is required to gain a better understanding of the underlying mechanisms responsible for this pattern, especially since the baseline data suggest that there may be more freedom in adverb position in Ukrainian for certain models than for lexical verbs to begin with (see Figure 2).

The verb preceding the adverb indicates verb movement, but it is still unclear from our data whether or not the verb in main clauses moves into an extended CP layer like in Norwegian, or to a lower position, more consistent with English. Teasing this apart involves examining V2 in main clauses; the syntax of Ukrainian subordinate clauses is similar enough to those required for Norwegian, so there may not be any need to acquire a new structure, in accordance with the LPM (Westergaard 2021).

In sum, V2 in main clauses is consistent with Norwegian-like structures, where an SVO preference is aligned with English. A further prediction from the latter is that if an abstract categorization of Norwegian verbs is based on English modals, i.e., positioned higher than the adverb, then these structures cannot result in adverb-verb order in subordinate clauses. Here we find a relatively high average probability of verb-adverb responses in subordinate clauses, although it is clear that some participants have learned that adverb-verb structures are also possible. Because the structures that are responsible for these patterns are unclear, examining the specific kinds of syntactic representations involved in creating the surface-level order leads us to additional questions to pursue.

6 Conclusion and outlook

This empirical study provides insight into L3 acquisition, specifically focusing on the acquisition of Norwegian among L1 Ukrainian-L2 English multilinguals through the lens of the Linguistic Proximity Model. Our findings suggest that L3 learners have acquired surface-level Norwegian word order in main clauses and overgeneralize by using the same word order in subordinate clauses. According to the LPM, these results suggest that English may facilitate the processing and parsing of V-Adv word order in Norwegian since it exists in English, although only with modal verbs. An additional step in the acquisition of Norwegian word order is learning the distributions of the V-Adv and Adv-V across

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clause types. This appears to be what intermediate learners demonstrate, although with variation between the two options. No evidence for surface-level CLI from L1 Ukrainian or L2 English was found as there is neither a preference for the adverb-verb word order in line with Ukrainian nor a categorical distinction between modal and lexical verbs in Norwegian in line with English. More empirical evidence is necessary in order to investigate whether abstract CLI occurred. That is, we are unable to determine what the structures involved in L3 Norwegian are without further examining the relationships between verbs, adverbs, and other syntactic representations. An investigation of the range of grammatical options in Ukrainian is also required. As such, surface similarity was not found to be a determining factor. However, the current use of English was found to be a predictor: participants with increased use of English were more likely to distinguish between lexical and modal verbs, albeit in an unexpected way. This suggests an avenue for future study into the extent that L2 English may influence verb categorization in L3 Norwegian. As a whole, this study and our interpretation of its results highlight the dynamic nature of L3 acquisition that involves several (extra-)linguistic factors, as well as the importance of viewing language acquisition and CLI on a property-by-property basis and of understanding the fine-grained representations of those properties (Westergaard 2009, 2021).

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References

- Archibald, John. 2022. Segmental and prosodic evidence for property-by-property transfer in L3 English in Northern Africa. *Languages* 7(1). 28.
- Archibald, John. 2023. Phonological parsing via an integrated I-language: The emergence of property-by-property transfer effects in L3 phonology. *Linguistic Approaches to Bilingualism* 13(6). 743–766.
- Bardel, Camilla & Ylva Falk. 2007. The role of the second language in third language acquisition: The case of Germanic syntax. *Second Language Research* 23(4). 459–484.
- Cabrelli, Jennifer & Michael Iverson. 2023. Why do learners overcome non-facilitative transfer faster from an L2 than an L1? The cumulative input threshold hypothesis. *International Journal of Multilingualism* 21. 1594–1620.

Nadine Kolb, David Natvig & Brechje van Osch

- Fallah, Nader & Ali Akbar Jabbari. 2018. L3 acquisition of English attributive adjectives: Dominant language of communication matters for syntactic cross-linguistic influence. *Linguistic Approaches to Bilingualism* 8(2). 193–216.
- Fallah, Nader, Ali Akbar Jabbari & Ali Mohammad Fazilatfar. 2016. Source(s) of syntactic cross-linguistic influence (CLI): The case of L3 acquisition of English possessives by Mazandarani–Persian bilinguals. *Second Language Research* 32(2). 225–245.
- Jin, Fufen. 2009. Third language acquisition of Norwegian objects: Interlanguage transfer or L1 influence? In Yan-kit Ingrid Leung (ed.), *Third language acquisition and universal grammar*, 144–161. Bristol: Multilingual Matters.
- Kalvestinova, E. & R. Slabakova. 2008. Does the verb move in Russian? In *Proceedings of FASL 16*, 199–214. Michigan Slavic Publications.
- Kolb, Nadine, Natalia Mitrofanova & Marit Westergaard. 2022. Crosslinguistic influence in child L3 English: An empirical study on Russian-German heritage bilinguals. *International Journal of Bilingualism* 26(4). 476–501.
- Kolb, Nadine, Natalia Mitrofanova & Marit Westergaard. 2023. Structural similarity in third language acquisition. *Linguistic Approaches to Bilingualism* 13(5). 607–613.
- Lightfoot, David. 2006. *How new languages emerge*. Cambridge: Cambridge University Press.
- Lightfoot, David. 2020. *Born to parse: How children select their languages*. Cambridge: Cambridge University Press.
- Lightfoot, David & Marit Westergaard. 2007. Language acquisition and language change: Inter-relationships. *Language and Linguistics Compass* 1(5). 396–415.
- Listhaug, Kjersti Faldet, Guro Busterud & Anne Dahl. 2021. French as a third language in Norway: The influence of the prior languages in the acquisition of word order. *Oslo Studies in Language* 12(1). 125–144.
- Mykhaylyk, Roksolana. 2013. Bilinguals vs. monolinguals: Where is the difference? *Studia Linguistica* 67. 101–122. DOI: [10.1111/stul.12007](https://doi.org/10.1111/stul.12007).
- Mykhaylyk, Roksolana & Heejeong Ko. 2010. Optional scrambling is not random: Evidence from English-Ukrainian acquisition. In Merete Anderssen, Kristine Bentzen & Marit Westergaard (eds.), *Variation in the input: Studies in the acquisition of word order*, 207–241. Dordrecht: Springer.
- Mykhaylyk, Roksolana, Natalia Mitrofanova, Yulia Rodina & Marit Westergaard. 2015. The Linguistic Proximity Model: The case of verb-second revisited. In Elizabeth Grillo & Kyle Jepson (eds.), *Proceedings of the 39th Boston University Conference on Language Development*, vol. 2, 337–349. Somerville: Cascadia Press.

12 Cross-linguistic influence in L3 Norwegian

- Natvig, David, Michael T. Putnam & Alexander K. Lykke. 2025. Stability in the integrated bilingual grammar: Tense exponency in North American Norwegian. *Nordic Journal of Linguistics* 48(1). 57–90.
- Rothman, Jason. 2015. Linguistic and cognitive motivations for the typological primacy model (TPM) of third language (L3) transfer: Timing of acquisition and proficiency considered. *Bilingualism: Language and Cognition* 18(2). 179–190.
- Rothman, Jason, Jorge González Alonso & Eloí Puig-Mayenco. 2019. *Third language acquisition and linguistic transfer*. Cambridge: Cambridge University Press.
- Sánchez, Laura & Camilla Bardel. 2017. Transfer from an L2 in third language learning: A study on L2 proficiency. In Tanja Angelovska & Angela Hahn (eds.), *L3 syntactic transfer: Models, new developments and implications*, 223–250. Amsterdam: John Benjamins.
- Slabakova, Roumyana. 2017. The scalpel model of third language acquisition. *International Journal of Bilingualism* 21(6). 651–665.
- Stadt, Rosalinde, Aafke Hulk & Petra Sleeman. 2018. The influence of L1 Dutch and L2 English on L3 French: A longitudinal study. *Journal of the European Second Language Association* 2(1). 63–71.
- Stadt, Rosalinde, Aafke Hulk & Petra Sleeman. 2020. L1 Dutch vs L2 English and the initial stages of L3 French acquisition. In Camilla Bardel & Laura Sánchez (eds.), *Third language acquisition: Age, proficiency and multilingualism*, 237–261. Berlin: Language Science Press. DOI: [10.5281/zenodo.4138749](https://doi.org/10.5281/zenodo.4138749).
- Tremblay, Marie-Claude. 2006. Cross-linguistic influence in third language acquisition: The role of L2 proficiency and L2 exposure. *Cahiers Linguistiques d'Ottawa/Ottawa Papers in Linguistics* 34. 109–119.
- van Osch, Brechje, Nadine Kolb, Merete Anderssen, Alicia Luque & Marit Westergaard. 2024. Cross-linguistic influence in adverb placement in L3 Spanish: Comparing offline and online measures. In *Bilingualism in the Hispanic and Lusophone world (BHL 2024)*. Unpublished conference presentation.
- van Osch, Brechje, Nadine Kolb, Merete Anderssen, Rosalinde Stadt, Alicia Luque & Marit Westergaard. 2023. Cross-linguistic influence from English as a heritage language in L3 acquisition. In *HL@Cross*. Unpublished conference presentation.
- Westergaard, Marit. 2003. Unlearning V2: Transfer, markedness, and the importance of input cues in the acquisition of word order in English by Norwegian children. *EUROSLA Yearbook* 3(1). 77–101.

Nadine Kolb, David Natvig & Brechje van Osch

- Westergaard, Marit. 2009. *The acquisition of word order: Micro-cues, information structure, and economy* (Linguistik Aktuell/Linguistics Today 145). Amsterdam: John Benjamins.
- Westergaard, Marit. 2014. Linguistic variation and micro-cues in first language acquisition. *Linguistic Variation* 14(1). 26–45. DOI: [10.1075/lv.14.1.02wes](https://doi.org/10.1075/lv.14.1.02wes).
- Westergaard, Marit. 2021. Microvariation in multilingual situations: The importance of property-by-property acquisition. *Second Language Research* 37(3). 379–407.
- Westergaard, Marit, Terje Lohndal & Artemis Alexiadou. 2019. The asymmetric nature of V2: Evidence from learner languages. In Ken Ramshøj Christensen, Henrik Jørgensen & Johanna L. Wood (eds.), *The sign of the V – Papers in honour of Sten Vikner*, 709–733. Aarhus: Department of English, School of Communication & Culture, Aarhus University.
- Westergaard, Marit, Natalia Mitrofanova, Roksolana Mykhaylyk & Yulia Rodina. 2017. Crosslinguistic influence in the acquisition of a third language: The linguistic proximity model. *International Journal of Bilingualism* 21(6). 666–682.
- Wu, Junyu. 2024. *Perception of L3 French vowel contrasts by L1 Mandarin-L2 English learners: A contrastive hierarchy perspective*. Vancouver: University of Victoria. (Doctoral dissertation).

Chapter 13

Bilectalism, dominance, and micro-cues: Vowel insertion and deletion in Brazilian and European Portuguese bilectals

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European Portuguese speaking Brazilians who acquired European Portuguese late in life (late bilectals) were rated as more Brazilian-sounding than speakers who acquired EP early in life (early bilectals). Both groups show traces of the other variety in each speaking mode (European and Brazilian Portuguese, respectively), which distinguishes them from monolectals. We propose that this is best accounted for in a model of indexed violable constraints, where a basic ranking and indexed constraints are activated for the different varieties if rankings diverge. Cross-linguistic influence (CLI) emerges from two sources: First, micro-cues are not detected and the corresponding differences in Elementary Ranking Conditions are not implemented as reranking/indexation. Moreover, constraint indexes may occasionally be switched on/off for the wrong variety or lost entirely, resulting in a permanent ranking deviance from the respective monolectal ranking.

Keywords: language acquisition, bilectalism, phonology, Optimality Theory, micro-cues

1 Introduction

Based on [Kupisch et al.](#)'s (2023) study of Brazilian and European Portuguese bilectals¹, we propose a model of bilectal phonologies in Optimality Theory, taking

¹Speakers with proficiency in more than one language variety have been called “bidialectals” (e.g., [Siegel 2012](#), [Cornips 2014](#), [Lundquist & Vangsnes 2018](#)). Since the dialect-language dis-

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a micro-cue approach. [Kupisch et al.](#) examined how monolectal users of Brazilian and European Portuguese rated the speech of bilectals who had acquired the Brazilian variety (BP) from birth and the European variety (EP) either entirely in Portugal (henceforth EO – early-onset bilectals), or after relocation from Brazil to Europe (henceforth LO – late-onset bilectals). In BP-mode, late bilectals were perceived as slightly divergent from homeland speakers in Brazil, suggesting that their phonologies have drifted due to contact with EP; early bilectals were substantially divergent from homeland speakers in Brazil, suggesting that they acquired a different phonological system from the start. At the same time, the early bilectals were perceived as unambiguously European in EP-mode. In this experiment of Lect identification, native judges relied exclusively on phonetic and phonological cues, as all syntactic, morphological and lexical differences were avoided when creating the stimuli.

Similar to the judges in the aforementioned experiment, learners can be assumed to rely on (phonetic or phonological) micro-cues. We explore the theory of micro-cues ([Lightfoot 1979, 1999](#); [Westergaard 2009, 2014, 2021](#)) to explain the perceived deviation of Lect 2 acquisition as well as the potential deviation of Lect 1 (where Lect1 and Lect2 can be acquired simultaneously or sequentially). We formalize this in Optimality Theory ([Prince & Smolensky 1994/2004](#))², in which micro-cues are extensional traits of properties; what is understood as parameter setting in syntax corresponds to setting or shifting of (intensional) property values (for Property Theory, see [Alber et al. 2016](#); [Alber & Prince 2021](#)). The bidirectionality of cross-linguistic influence (CLI) in the population of BP/EP bilectals is seen as evidence of intertwined grammars in the form of indexed constraints within one core hierarchy rather than co-phonologies. We illustrate the theory with an analysis sketch of vowel/zero differences between the two varieties. [Holliday et al. \(2024\)](#) run low-pass filtered as well as unfiltered stimuli of several varieties of American English by American English listeners, and conclude that “[o]verall, listeners appear to be more sensitive to segmental phonological than prosodic information in Lect identification” (p. 1). [Clopper & Pisoni \(2004\)](#) find a negative correlation between Lect identification and the similarity of the vowel systems of the Lects of American English to be identified. These results on Lect identification in English do not necessarily apply to Portuguese listeners, who might have different perceptual priorities. [Sancier & Fowler \(1997\)](#) report that a Brazilian Portuguese-American English bilingual is identified by raters as more

tion is controversial, we use “Lect”.

²Readers unfamiliar with OT are referred to [Gouskova \(2015\)](#); for more in-depth reading, see [Iosad \(2018\)](#), [Krämer \(2018\)](#) and [Zsiga \(2024\)](#).

foreign-sounding right after a period of immersion in English than after a period of immersion in Portuguese. The only cue for these judgements is that he has different VOT's in Portuguese depending on whether he was immersed in English or Portuguese before the recordings. (English has short lag VOT contrasting with a long lag VOT, while both varieties of Portuguese contrast negative VOT with short lag VOT). Anecdotal evidence (p.c., João Veloso, among others) indicates that Portuguese listeners also rely predominantly on segmental information, in particular the striking differences between BP and EP in unstressed vowel reduction, deletion and epenthesis (described in detail by Mateus & d'Andrade 2000 and Barbosa & Albano 2004). We discuss these patterns in our formal analysis.

2 Phonological differences between EP and BP

BP and EP differ considerably at the phonetic and phonological level. According to Mateus & d'Andrade (2000: 129, henceforth M& d'A), “foreign listeners first encountering the two varieties of Portuguese get the strange impression that they are facing two different languages.” Vigário (2022) provides an overview of the differences between BP and EP, including also prosodic differences (see also Kupisch et al. 2023). For reasons of space, we will focus here on the most salient differences, i.e., phonotactically driven vowel reduction, deletion and insertion, that will also play a central role in the grammatical sketches that illustrate the theoretical discussion below.

EP is characterized by more extensive vowel reduction which, in the extreme, results in vowel deletion, while a key characteristic of BP is vowel epenthesis in obstruent clusters (Barbosa & Albano 2004). The respective forms of the verb *desprestigiar* ‘depreciate’ in (1) differ as follows: EP either deletes all unstressed mid vowels or raises them to high central, neither of which is an option in BP. Both varieties display vowel reduction in unstressed syllables, but differ in the details. One such detail is that BP tolerates tense mid vowels in unstressed syllables, while EP raises them to high central (a vowel non-existent in BP) or deletes them. Another difference in (1) is the realization of the rhotic in the coda, which is produced as a tap in EP, as opposed to its realization as a tap, velar fricative or its deletion in BP.

- (1) Some differences between EP and BP

desprestigiar ‘to depreciate’

- a. EP: [dʃprɨʔtʃɨʒjaɾ]/[dʃprɨʔtʃɨʒjaɾ] (M& d'A: 44)
 b. BP: [desprestʃiʒjaɾ]/[desprestʃiʒjax]/[desprestʃiʒja]

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The consonant clusters created by mid vowel deletion in EP in (1a) are impossible in BP. Even lexical strings of obstruents are frequently broken up by an epenthetic vowel, as in (2).

(2) Vowel epenthesis in BP (adapted from M& d'A: 45)

[pinew]	<i>pneu</i>	‘tire’
[pisikoloziɐ]	<i>psicologia</i>	‘psychology’
[abisuxdu]	<i>absurdo</i>	‘absurd’

The realization of /e/ in an unstressed syllable or /r/ in a coda might be used by learners to deduce a phonological process or high ranking of an active markedness constraint, i.e., vowel reduction or coda rhotic avoidance, or it might be used by native listeners as a cue to identify a speaker of Portuguese as Brazilian or European. We can also assume that Brazilian learners of EP (and vice versa) differ in their level of phonological awareness. Even if they have high phonological awareness, they might not notice all micro-cues and acquire all ranking relations of the other Lect. The use of micro-cues and its dependence on specific phonological awareness thus explain why learners retain a foreign accent in their L2 (in addition to psycho-affective motivational and identity-related factors). However, it does not explain the development a foreign accent in their native variety.

3 Language Acquisition in OT

An OT grammar is a constraint ranking or a set of constraint rankings that converge on all attested and acceptable outputs in a language. OT assumes that observed forms of speech emerge from the satisfaction of conflicting constraints. Constraints are arranged in language-specific dominance hierarchies. Only those candidate forms which satisfy higher ranked constraints in comparison to competing forms are selected as grammatical outputs. The acquisition of a grammar consists of learning a language-specific hierarchy of constraints (Prince & Smolensky 1994/2004).

3.1 L1 Acquisition in OT

Infants start with an unranked set of constraints or the general order of all markedness constraints above all faithfulness constraints, the state before any ambient linguistic data has been considered as evidence for grammar construction. The most widely accepted algorithm for learning constraint hierarchies is that of Error Driven Constraint Demotion, originally defined for L1 acquisition

(e.g., [Tesar & Smolensky 2000](#)). Reranking can consist of moving a constraint up past other constraints, or constraints can be moved down in the hierarchy, i.e., their importance diminished with respect to other constraints. Restrictiveness considerations ensure that constraint movement is always downward in the hierarchy, i.e., demotion means that selected constraints become hierarchically less important. Given the initial state (all markedness constraints dominate all faithfulness constraints), learning consists of demoting individual constraints. Markedness constraints lacking evidence for demotion remain at the top of the hierarchy, keeping the grammar maximally restrictive.

3.2 Bilingual language acquisition in OT

A bilingual/bilectal speaker has at least two grammars, that is, two (sets of) constraint rankings that converge on different outputs, one for each Lect, and must choose the appropriate one in the right context. (A bilingual/bilectal with three languages has three grammars, etc.) The learner could be assumed to create an exact copy of H1 (Hierarchy 1, the grammar acquired first), which would correspond to “full transfer”—terms used in the domain of syntax and then create H2 for L2 from this by successive demotion of those constraints that are ranked less highly in the L2 target system. Since H0 (the initial state before acquisition has started: no ranking or all Markedness above all Faithfulness; [Gnanadesikan 1995/2004](#); [Tesar & Smolensky 2000](#)) has been modified by learning L1, it is no longer available and the only possible starting point for L2 learning is H1. This explains why late bilinguals/bilectals often speak their L2 with an L1 accent beyond effects of fine motor control; i.e., if learning is incomplete not all necessary re-ranking operations have been carried out. Since the algorithm is error driven, a speaker first has to detect a difference between their rendition of an L2 form and the target realization, i.e., a micro-cue, to initiate the relevant constraint demotion. Phonological differences that remain undetected by the learner thus stay in the L2 system.

The problem with the assumption of two independent constraint hierarchies is that it does not easily explain effects of CLI in bilingual speakers. For example, it has been shown that in early bilingual language acquisition complexity encountered in one target language can speed up learning in the other ([Lleó et al. 2003](#)).

Bilingualism can also be conceived in OT as co-phonologies. Languages often display subregularities or outright irregularities. In English, for example, different affixes induce different stress assignment, as in *parent* – *parental* versus *parenting* where *-al* causes stress shift and *-ing* does not. At first blush, one could

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argue that the hypothesis of languages having just one phonology is wrong and English simply has several, i.e., several co-phonologies. In the co-phonology approach (henceforth CPA; Anttila 2002; Inkelas & Zoll 2003), each morpheme selects a ranking.

- (3) Co-phonology 1: A >> B >> C >> D >> E >> F
Co-phonology 2: A >> B >> C >> E >> D >> F

Alternatively, co-phonology effects can be modelled by constraint indexation. For example, loanwords frequently display subregularities. Ito & Mester (1999) show for loanwords in Japanese that in each layer of the lexicon (i.e., native Yamato vocabulary, Chinese loans, English borrowings) markedness restrictions are less tight. They propose an analysis with a core hierarchy of markedness constraints and a set of indexed faithfulness constraints, with indexes for each lexical class (see Pater 2000, 2006 for constraint indexation). The faithfulness constraints for the native lexical items are ranked lowest, while the peripheral morphemes are linked to faithfulness constraint higher up in the hierarchy:

- (4) Stratification of the lexicon by faithfulness (M = markedness constraints)
M1 >> Faith^{Foreign} >> M2 >> Faith^{Loan} >> M3 >> Faith^{Sino} >> M4 >> Faith
(native) >> M5

In such a model of the lexicon, L2 could be yet another, more marginal, layer of the lexicon and of associated faithfulness. At the beginning of L2 learning, this set of Faithfulness constraints starts at the same level as the faithfulness constraints for the native morphemes. At the end of L2 acquisition, all the necessary layers of markedness should have been demoted below this set of faithfulness constraints. The model plausibly captures that typologically close languages are easier to learn, as they involve fewer reranking operations.

- (5) Late L2 acquisition in the Stratal Faithfulness model
 - a. Monolingual state
M1 >> Faith^{Foreign} >> M2 >> Faith^{Loan} >> M3 >> Faith^{Sino} >> M4 >> Faith,
Faith^{L2} >> M5
 - b. Target state (L2 completely acquired)
Faith^{L2} >> M1 >> Faith^{Foreign} >> M2 >> Faith^{Loan} >> M3 >> Faith^{Sino} >> M4
>> Faith, >> M5

Faith^{L2} has to move from one end of the hierarchy to the other, and, unsurprisingly, hardly ever gets that far (incidence of native-like pronunciation is very

low). However, the situation is not that simple, neither for loanwords (e.g., [Jurgec 2010](#)), nor for exceptional morphemes and exceptional phonology ([Pater 2006](#)), nor for bilingualism/bilectalism. If we restrict the theory to indexed Faithfulness constraints, as Ito & Mester do for loanwords, phonological processes in L2 cannot be acquired, since they require high ranking of an M constraint that is ranked low in L1. [Pater's](#) (2006) approach allows for both indexed markedness constraints (for exceptional processes) and indexed faithfulness constraints (for exceptions to otherwise regular processes/patterns).

Subregularity or exceptionality is analyzed in the constraint indexation approach (henceforth CIA) as schematically depicted in (6). E is the relevant constraint that is cloned and coindexed with selected morphemes. If E is a faithfulness constraint, we are dealing with the exceptional blocking of regular phonology, and if E is a markedness constraint, we are dealing with an exceptional phonological process.

(6) Sub-regularities as lexical indexation

A >> B >> C >> E^X >> D >> E >> F

In Section 4, we connect these models, the CPA and the CIA, more concretely with questions concerning the analysis of bilingualism/bilectalism.

4 Modeling coexisting phonologies in OT

We discuss the expectations raised by each approach, then present an example case how EP vowel deletion and BP vowel epenthesis are reconciled in bilingual OT grammars in the indexation approach.

4.1 Predictions of the CPA and the CIA

In the CPA, the two grammars of a bilectal are two separate hierarchies. Late bilectals use their L1 constraint hierarchy as the basis, make a copy and rearrange constraints to accommodate L2 regularities. Given that the acquisition algorithm is *error driven* demotion, it can be expected that they do not detect all their errors (perception is biased by the L1) and accordingly fail to use them as ranking arguments, resulting in divergent acquisition of L2 EP with surface effects from BP. CLI from L2 EP on L1 BP is not expected to happen since both languages are strictly separated. The BP ranking is fixed and strictly separated from the L2 ranking. The learner could occasionally detect a divergent ranking condition and

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accidentally rerank the respective constraints in the L1 hierarchy. This would result in the application of a regularity of L2 in L1 but not in L2, a CLI that, to our knowledge, has not been reported.

Early bilectals can be expected to have a H1, originally created on the basis of BP input, which does not deviate very much from H^0 yet. With relocation into an EP-speaking environment/or being born in Portugal, they split this incompletely arranged H1 into two hierarchies and continue learning. Based on input, they start ranking the constraints in each hierarchy. Presumably, they observe more errors in their own productions for the language from which they receive more input and consequently rearrange more constraints. Since the former L1 BP input is probably dwarfed in comparison to the EP input after relocation (especially for children who attend school or daycare and establish a tight social network with local peers), one expects the learner to detect fewer ranking arguments and thus retain a more rudimentary grammar. CLI from EP to BP is not expected, however, since the BP and EP hierarchies are strictly separated, unless, as sketched above, the learner accidentally erroneously reranks constraints in the BP hierarchy in response to EP input. The only CLI of BP on EP is expected to stem from the earliest rankings, since the EP hierarchy is built on the foundations of an early and still unfinished BP hierarchy.

In the CIA, late bilinguals enrich H1 by adding indexed clones of L2 constraints. There are three sources of error/CLI. First, a learner does not notice a micro-cue and does not create an indexed constraint to account for the regularity. This would correspond to divergent L2 acquisition and automatic use of L1 rankings in L2 mode. Second, constraints are correctly indexed and placed in the hierarchy, but in actual use some indexes are overlooked. This has consequences in both L1 mode and L2 mode. If an index is overlooked in L1 mode, a constraint that should be low ranked has a higher position in the hierarchy and determines candidate choice, when it should not. If an index is overlooked in L2 mode, it is not activated in its higher position and does not influence candidate choice in the way it should in L2, and accordingly a L1 pattern prevails. It is crucial for this approach that indexes can accidentally be ignored. The most likely source of CLI results from a constraint that is not activated when needed, i.e., the BP ranking prevails in EP mode, while the activation of an EP specific constraint in BP mode is less likely (but possible). The third source of CLI is permanent: A constraint originally established as a clone with an index to account for a regularity of the L2 loses its index but remains in the position necessary for the L2.

For early bilectals the situation is more complex in the indexing approach. An early bilectal could introduce indexes for both languages, which results in indexation and consequently cloning of almost every constraint, creating de facto two

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separate hierarchies. Since the EO bilectals were not rated as 100% BP, we can conclude that early bilectals cannot be balanced—a fairly common assumption in bilingualism research. Since EO bilectals received higher ratings for their EP than for their BP, they might use the EP ranking as their basic spine and index constraints for BP. Since the EO bilectals receive more EP than BP input from early on, an early established BP ranking that is already in place might be modified on the basis of EP data, and henceforth the constraints for BP would be indexed. Regarding quantity of input in the different Lects, the same situation prevails as in the CPA: less input provides fewer instances of micro-cues and this triggers fewer ranking arguments; more input provides data for more ranking arguments and a more exhaustively and more correctly ranked constraint set. As shown below, the learning process is more complex and co-indexation for both Lects is unavoidable.

4.2 An illustration

As discussed in Section 3, in colloquial EP many unstressed vowels are deleted, yielding more consonant clusters, while in BP additional vowels surface because there are more severe restrictions on consonant clusters. While we have not done any detailed quantitative analysis of our speech samples, we noted that the late bilectals delete fewer vowels in EP mode than EP monolinguals (e.g., producing *seguida* instead of expected *s'guida* ‘after’), epenthesize less in BP mode than BP monolinguals, and even delete vowels in this mode (e.g., *o m'nino* instead of expected *o menino* ‘the boy/lad’). The early bilectals, on the other hand, delete vowels as expected from a colloquial EP speaker, producing forms such as *m'nino*, but also *f'gir* for *fugir* ‘escape’, *c'mer* for *comer* ‘to eat’.

To keep it simple, we assume that EP vowel deletion is caused by one markedness constraint against unstressed vowels, *Unstressed.³ BP vowel epenthesis is caused by various phonotactic restrictions represented here by a constraint against consonant clusters: *CC. These constraints interact with faithfulness constraints against deletion, i.e., IO-Max, and against epenthesis, i.e., IO-Dep (McCarthy & Prince 1999).

(7) CON

a. Faithfulness

Max: ‘Assign one violation mark for each segment in the input that is not present in the output.’

³We are aware that deletion only affects underlying tense mid vowels. A more thorough analysis requires an exact definition of the involved constraints.

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Dep: ‘Assign one violation mark for each segment in the output that is not present in the input.’

b. Markedness

*U(nstressed): ‘Assign a constraint violation for each unstressed tense mid vowel.’

*CC. ‘Assign a violation for each pair of adjacent obstruents.’

In accordance with the simplifications just introduced, the candidate set consists of schematic CV sequences rather than actual examples. In this minitypology we distinguish between Standard EP (neither vowel deletion nor epenthesis), and Colloquial EP (CEP) as well as BP.

(8) GEN

a. Input: /CCuCuCV/⁴ (e.g., hypothetical *pselegó*)⁵

b. Candidate set

	<i>Representation</i>	<i>Comment</i>	<i>Variety</i>
i.	CCuCuCV	Faithful	EP
ii.	CCCCV	Deletes all unstressed (mid) Vs	CEP
iii.	CvCuCuCV	Avoids CC	BP
iv.	CCuCCV	Deletes some unstressed Vs	Universally bounded ⁶
v.	CvCCCCV	Deletes all unstressed Vs, avoids initial CC	Universally bounded
vi.	CV	Deletes all unstressed Vs and Cs in clusters	not Portuguese
vii.	CuCuCV	Deletes Cs in clusters	not Portuguese
viii.	CuCV	Deletes most unstressed Vs and Cs	Universally bounded

⁴(C=consonant, V=vowel, u=unstressed vowel, v=unstressed epenthetic vowel)

⁵We are not aware of any Portuguese words that combine both phonological conditions, containing an initial consonant cluster that violates the phonotactics of the core lexicon and several unstressed mid front vowels. To streamline the discussion of the analysis we therefore use this schematic skeleton instead of illustrating ranking arguments with more tableaux.

⁶Bounded candidates (iv and viii) can become potential winners if additional constraints are considered.

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We consider the three candidates only, representing EP, CEP and BP, respectively, and the Elementary Ranking Conditions that have to be met for each.

The tableau in (9) shows the constraint violations of the candidates. Universally bounded candidates are shaded. Candidate (d) bounds candidate (e), because it has a subset of the violations of (e). Candidate (h) beats (g) on *U, but if *U is more important than Max, which is necessary for (h) to beat (g), both lose against (f), regardless of the ranking of the other constraints. We will not consider the bounded candidates in the following ranking arguments.

(9) Master tableau

	/CCuCuCV/	*Unstressed	*CC	Max	Dep
(EP) a.	CCuCuCV	**	*		
(EPC) b.	CCCCV		***	**	
(BP) c.	CvCuCuCV	**			*
d.	CCuCCV	*	**	*	
e.	CvCCCV	*	**	**	*
f.	CV			*****	
g.	CuCuCV	**		*	
h.	CuCV	*		***	

(10) shows the elementary ranking conditions for the faithful candidate to win against all the other candidates. W-marked constraints favour the designated winner (first column) over the loser to which it is compared in the respective row, and L-marked constraints favour the loser. For the faithful EP candidate to win, every L-marked constraint has to be dominated by at least one W-marked constraint.

(10) Faithful map (Standard EP)

	/CCuCuCV/	*Unstressed	*CC	Max	Dep
a.	CCuCuCV	CCCCV	L	W	
b.	CCuCuCV	CvCuCuCV	W		L
c.	CCuCuCV	CV	L	W	
d.	CCuCuCV	CuCuCV		L	W

The first comparison shows us that *U should be dominated by either *CC or Max. Comparison (b) shows that it can't be *CC that dominates *U, unless Dep is ranked above both. Candidates (c and d) show that Max must dominate *CC and *U.

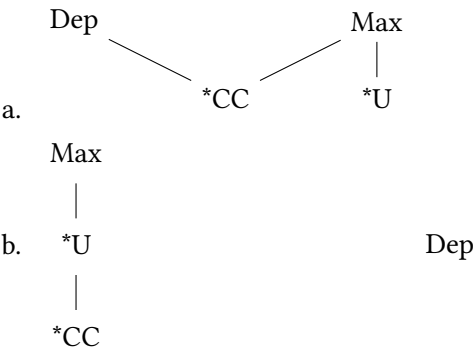
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This boils down to the following ranking conditions that have to prevail for the faithful candidate to win.

- (11) Ranking conditions for the faithful map / SEP:
- a. Dep or *U » *CC;
 - b. Max » *CC, *U

The two resulting rankings are displayed in the diagrams in (12).

- (12) Hasse diagrams Faithful map (Standard EP)



The Comparative Tableau for selecting the CEP candidate as the winner is given in (13). It shows that *U must dominate both *CC and Max. Comparison (c) shows that additionally Max dominates *CC. The ranking of Dep is again irrelevant.

- (13) CCCC CV (CEP)

		/CCuCuCV/	*Unstressed	*CC	Max	Dep
a.	CCCCV	CCuCuCV	W	L	L	
b.	CCCCV	CvCuCuCV	W	L	L	W
c.	CCCCV	CV		L	W	
d.	CCCCV	CuCuCV	W	L	L	

ERCs: *U » *CC, Max; Max » *CC

- (14) Hasse diagram CEP map

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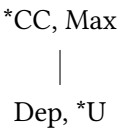


The Comparative Tableau with the BP candidate as the designated winner reveals that *U and Dep should be dominated by Max and *CC. Comparison (a) shows that *CC dominates both *U and Dep. Comparisons (c and d) show that Max crucially dominates *U and Dep.

(15) CvCuCuCV (BP)

/CCuCuCV/*Unstressed *CC				Max	Dep
a.	CvCuCuCV	CCuCuCV	L W		L
b.	CvCuCuCV	CCCCV	L W	W	L
c.	CvCuCuCV	CV	L	W	L
d.	CvCuCuCV	CuCuCV	L	W	L

(16) Hasse diagram BP



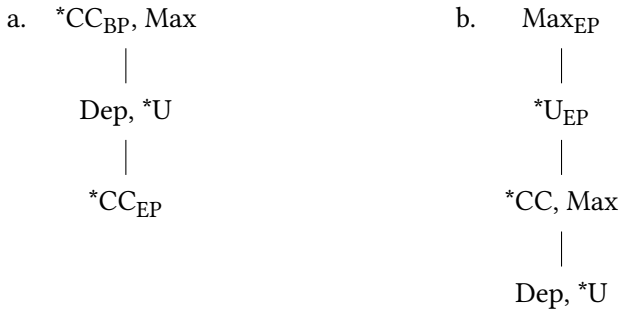
A comparison of the Hasse diagrams of the three Lects reveals the reranking operations a BP speaker needs to go through to arrive at an EP-like grammar. To get to SEP (12), they either demote *CC below Max and below Dep or *CC below Max and then below *U. Either way, two reranking operations are needed.

To acquire CEP (14), a BP speaker demotes both *CC and Max below *U and additionally *CC below Max.

Analysing the bilectal grammar with constraint indexation we immediately encounter the problem that our BP learner of EP cannot just insert an indexed constraint *CC_{EP} for the new Lect in the hierarchy, because it will be ranked below the unindexed original. The copy of *CC does not only have to be demoted, the original also needs to be tagged as BP such that it can be temporarily ignored when switching to EP (17a). Alternatively, the learner indexes Max and *U for EP and demotes the complete BP hierarchy below these two constraints (17b).

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If, in (17a), tagging of *CC in its original position is omitted and the constraint instead just demoted, we deal with the kind of CLI that makes the LO speaker sound a little bit less authentic in their native BP. We conclude that (17a), the mixed index system, is the more accurate analysis.

The EO learner does not have that problem to the same degree, since they will most likely demote many such M constraints and create a superset grammar that can code-switch with minimal effort.

A conclusion we draw from this simple example is that claiming that asymmetric bilectals dispose over a grammar consisting of the dominant hierarchy enriched with constraint copies indexed for the non-dominant Lect is oversimplifying. Bilectals need tagged constraints for both lects. We are thus dealing with a system in which a shared core hierarchy consists of un-indexed constraints, which are interleaved with constraints tagged for each of the Lects. The difference between L1 and L2, then, is one of quantity of overlap with the core spine and number of indexed constraints. We hypothesize that speakers have fewer indexed constraints and more agreement with the core ranking for their L1 than for L2.

That the constraints for both modes in both groups are in the same hierarchy makes the grammars more likely to change. Additionally, for the LO speakers, chances are high that some micro-cues for L2 ranking conditions are overlooked and the corresponding rerankings and indexing never happened.

5 Conclusion

The formal analysis of the differences between EP and BP in unstressed vowels shows which conflicting Elementary Ranking Conditions bilectals have to accommodate when constructing their grammars. The observation that bilectal learners have to index constraints for both Lects helps to explain why there is CLI

also on the primary Lect, or Lect acquired first. The lower degree of authenticity ascribed to LO bilectals by the raters in Kupisch et al.'s (2023) experiment can be attributed to two sources, unstable use of constraint indexes and overlooked micro-cues in the acquisition of Lect 2. If a micro-cue is overlooked by a learner, necessary Elementary Ranking Conditions are not detected, and the corresponding ranking operations are not carried out. The grammar remains fragmentary.

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References

- Alber, Birgit, Natalie Delbusso & Alan Prince. 2016. From intensional properties to universal support. *Language* 92. e88–e116.
- Alber, Birgit & Alan Prince. 2021. *The structure of OT typologies. Chapter 1: Introduction to property theory*. Unpublished manuscript, ROA 1381. <http://roa.rutgers.edu>.
- Anttila, Arto. 2002. Morphologically conditioned phonological alternations. *Natural Language and Linguistic Theory* 20(1). 1–42.
- Barbosa, Plínio A. & Eleonora C. Albano. 2004. Illustrations of the IPA: Brazilian Portuguese. *Journal of the International Phonetic Association* 34(2). 227–232. DOI: [10.1017/S0025100304001756](https://doi.org/10.1017/S0025100304001756).
- Clopper, Cynthia & David Pisoni. 2004. Some acoustic cues for the perceptual categorization of American English regional dialects. *Journal of Phonetics* 32(1). 111–140.
- Cornips, Leonie. 2014. Socio-syntax and variation in acquisition: Problematising monolingual and bidialectal acquisition. *Linguistic Variation* 14(1). 1–25.
- Gnanadesikan, Amalia. 1995/2004. Markedness and faithfulness constraints in child phonology. In René Kager, Joe Pater & Wim Zonneveld (eds.), *Constraints in phonological acquisition*, 73–108. Originally available on Rutgers Optimality Archive, ROA-67 (1995). Cambridge: Cambridge University Press.
- Gouskova, Maria. 2015. Optimality theory in phonology. In Bernd Heine & Heiko Narrog (eds.), *The Oxford handbook of linguistic analysis*, 2nd, 545–566. Oxford: Oxford University Press.

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- Holliday, Nicole, Rachel Burdin & Paul Reed. 2024. Can listeners differentiate varieties using prosodic information? Lect identification in a persona-matching task. *Glossa: A Journal of General Linguistics* 9(1). DOI: [10.16995/glossa.10124](https://doi.org/10.16995/glossa.10124).
- Inkelas, Sharon & Cheryl Zoll. 2003. *Is grammar dependence real?* Manuscript, UC Berkeley and MIT, ROA-587.
- Iosad, Pavel. 2018. Optimality theory: Motivations and perspectives. In S.J. Hannahs & Anna Bosch (eds.), *The Routledge handbook of phonological theory*, 13–36. London: Routledge.
- Ito, Junko & Armin Mester. 1999. The phonological lexicon. In N. Tsujimura (ed.), *The handbook of Japanese linguistics*, 62–100. Oxford: Blackwell.
- Jurcic, Peter. 2010. Disjunctive lexical stratification. *Linguistic Inquiry* 41(1). 149–161.
- Krämer, Martin. 2018. Current issues and directions in Optimality Theory – Constraints and their interaction. In S.J. Hannahs & Anna Bosch (eds.), *The Routledge handbook of phonological theory*, 37–67. London: Routledge.
- Kupisch, Tanja, Tammer Castro, Martin Krämer & Marit Westergaard. 2023. Phonological influence in bilingual speakers of Brazilian and European Portuguese. *International Journal of Bilingualism* 8(3). 406–423.
- Lightfoot, David. 1979. *Principles of diachronic syntax*. Cambridge: Cambridge University Press.
- Lightfoot, David. 1999. *The development of language: Acquisition, change and evolution*. Malden: Blackwell.
- Lleó, Conxita, Imme Kuchenbrandt, Margaret Kehoe & Cristina Trujillo. 2003. Syllable final consonants in Spanish and German monolingual and bilingual acquisition. In Natascha Müller (ed.), *(in)vulnerable domains in multilingualism* (Hamburg Studies on Multilingualism), 191–220. Amsterdam: John Benjamins.
- Lundquist, Björn & Øystein Vangsnes. 2018. Language separation in bidialectal speakers: Evidence from eye tracking. *Frontiers in Psychology*. DOI: [10.3389/fpsyg.2018.01394](https://doi.org/10.3389/fpsyg.2018.01394).
- Mateus, Maria Helena & Ernesto d’Andrade. 2000. *The phonology of Portuguese*. Oxford: Oxford University Press.
- McCarthy, John & Alan Prince. 1999. Faithfulness and identity in prosodic morphology. In René Kager, Harry van der Hulst & Wim Zonneveld (eds.), *The prosody–morphology interface*, 218–309. Cambridge: Cambridge University Press.
- Pater, Joe. 2000. Nonuniformity in English stress: The role of ranked and lexically specific constraints. *Phonology* 17. 237–274.

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- Pater, Joe. 2006. The locus of exceptionality: Morpheme-specific phonology as constraint indexation. In Ehren Reilly, Leah Bateman & Michael O’Keefe (eds.), *UMOPL 32: Papers in Optimality Theory iii*, 259–296. Amherst: GLSA.
- Prince, Alan & Paul Smolensky. 1994/2004. *Optimality Theory: Constraint interaction in generative grammar*. Oxford: Blackwell.
- Sancier, Michele L. & Carol A. Fowler. 1997. Gestural drift in a bilingual speaker of Brazilian Portuguese and English. *Journal of Phonetics* 25(4). 421–436.
- Siegel, Jeff. 2012. *Second dialect acquisition*. Cambridge: Cambridge University Press.
- Tesar, Bruce & Paul Smolensky. 2000. *Learnability in Optimality Theory*. Cambridge: MIT Press.
- Vigário, Marina. 2022. Portuguese. In Christoph Gabriell, Randall Gess & Trudel Meisenburg (eds.), *Manual of Romance phonetics and phonology*, 839–881. Berlin: Mouton de Gruyter.
- Westergaard, Marit. 2009. *The acquisition of word order: Micro-cues, information structure, and economy* (Linguistik Aktuell/Linguistics Today 145). Amsterdam: John Benjamins.
- Westergaard, Marit. 2014. Linguistic variation and micro-cues in first language acquisition. *Linguistic Variation* 14(1). 26–45. DOI: [10.1075/lv.14.1.02wes](https://doi.org/10.1075/lv.14.1.02wes).
- Westergaard, Marit. 2021. Language acquisition, microcues, parameters, and morphosyntactic change. In Richard D. Janda, Brian D. Joseph & Barbara S. Vance (eds.), *Handbook of historical linguistics*, 357–374. Hoboken: Wiley Publishers.
- Zsiga, Elizabeth C. 2024. *The sounds of language. An introduction to phonetics and phonology*. Oxford: Wiley-Blackwell.

Chapter 14

Unshifted subjects or shifting particles? On the acquisition of particle- and subject placement

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Young speakers in certain areas of Norway display a word order deviation regarding placement of the subject with respect to verb particles: Both children and teenagers occasionally place verb particles in front of subjects in non-subject initial clauses, in contrast to the adult pattern. In this article we investigate whether this word order deviation is a result of a particle shifting across the subject, or a subject surfacing in situ, i.e., inside the verb phrase. Based on spontaneous and elicited examples of particle placement from children and teenagers, we can observe a development in which the particle placement becomes increasingly target-like. Still, the pattern observed goes against trends of economy and conservativity observed in other studies of word-order development: The children appear to start with a relatively free word order, which slowly becomes more restricted. As we will argue, the development is difficult to describe in terms of preference either for or against syntactic movement.

1 Introduction

An interesting finding in previous research on the development of constituent order in Norwegian is a systematic pattern where Norwegian children tend to place subjects in an unshifted position, i.e., in a position to the right of sentence adverbials, most notably negation. The early overuse of unshifted subjects has been observed for both pronouns and full phrases, in both main and embedded

clauses (Anderssen & Westergaard 2010, Anderssen et al. 2010, Westergaard 2011, Ringstad & Westergaard 2023). Westergaard (2014) argues that this falls under a more general pattern where children avoid syntactic movement due to a syntactic economy principle. A learner guided by this syntactic economy principle would not postulate 1) more structure, or 2) more syntactic operations than there is evidence for in the input. If the pre-adverbial subject placement is the result of a syntactic movement operation (1a), the post-adverbial position (1b) would thus be preferred as it is derived with fewer syntactic operations:

(1) Norwegian

- a. I går leste Jon_i ikke t_i boka.
 yesterday read Jon not t book.DEF
 ‘Yesterday Jon didn’t read the book.’
- b. I går leste ikke Jon boka.
 yesterday read not Jon book.DEF
 ‘Yesterday Jon didn’t read the book.’

In addition to subject placement, Norwegian children display small deviations in their production in two other constructions that can both be accounted for in terms of economy: object shift and choice of possessive structure (Anderssen et al. 2010, Anderssen & Westergaard 2010).

In this short article we will discuss another word order deviation in young speakers’ production that on the surface looks very similar to the unshifted subjects mentioned above, namely the placement of subjects to the right of verb particles. This word order has been labeled *long particle shift* (Lundquist et al. 2019), in parallel to *long object shift*, in which an object is placed to the left of a subject (see section 4). As we will show, young speakers of the Trøndersk dialect (and children in other dialects) sporadically produce verb particles preceding subjects; an order that is not licit in the adult language. We will argue that in most of these cases, the order is not a result of a subject staying in a low position, but rather of a particle shifting leftwards across the subject. These speakers thus produce a word order that is derived from more operations than there is evidence from in the input, i.e., a shifting operation of the particle. This goes against the economy principle suggested in Westergaard (2014). We will discuss this pattern together with other shifting operations in Mainland North Germanic and propose that shifts should not be treated as syntactic movements, but rather as outcomes of direct linearization statements. In this view, the language learner slowly acquires more linearization statements stated over syntactic functions during the acquisition of target-like constituency order.

2 Long particle shift

As we will see in Section 3, no Germanic language allows shifting or scrambling of a verb particle out of the verb phrase. However, this very pattern has been attested in young speakers in some dialects of Norwegian. Author 1 of this paper was first made aware of shifting particles in Norwegian by Jan Terje Faarlund (p.c., 2017), who mentioned that Hanne Siri Sund had a collection of particle-subject orders, collected from teenagers speaking Trøndersk (2014–2016), a collection to which we later got access, and which included examples like (2):

- (2) Trøndersk, age 12 (From Hanne Siri Sund, p.c)
 Og da kjørt inn han i en sånn strømboks.
 and then drove in he in a kind.of electricity.box
 ‘And then he drove into a kind of electricity box.’

Sentences of this type were later included in the test battery for the North Germanic elicitation project that resulted in the Nordic Word order Database (NWD, see [Lundquist et al. 2019](#), [Lundquist & Tengesdal 2022](#)). In the first round, over 2500 sentences with particles and subjects were elicited in Norway, Sweden and Denmark, mainly from teenagers; among them, only four elicited sentences displayed the particle-subject order, and these showed signs of hesitation indicating that they were better analyzed as speech errors than naturally occurring variation. However, in a later data collection for the NWD in Fosen (Trøndelag), the particle-subject order was found quite frequently: about 15% of the sentences with noun phrase subjects displayed this order, and more than half of the participants (10/19) produced at least one instance of long particle shift (see (3) for an example, see the NWD online database for all sentences and results):

- (3) Teenager of Fosen dialect, (From Nordic Word order Database)
 I går rydda opp den nye studenten på kjøkkenet.
 yesterday cleaned up the new student.DEF on kitchen.DEF
 ‘Yesterday, the new student cleaned up in the kitchen.’

At the same time as author 1 was made aware of shifting particles, author 2 made observations about deviant particle placement in Trøndersk-speaking children’s production, for example (4–5):

- (4) Trøndersk, age 4:6
 Æ villa ha fire vafla og så smurt på pappa tre.
 I wanted to.have four waffles and then buttered on dad three
 ‘I wanted four waffles and then dad made three.’

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- (5) Trøndersk, age 4:5
 No sto opp vi.
 now got up we
 ‘Now we got up.’

In most of the observed examples, the particle directly follows the main verb, but in some cases the particle shifts to a position between a subject and negation, as in (6).

- (6) Trøndersk, 3:6 years old
 så rekk æ opp ikke dit.
 so reached I up not there
 ‘So I didn’t reach up there.’

Interestingly, examples of particle-subject are also found in children from other dialect areas: two examples are even given in Westergaard & Anderssen (2015) from children in Tromsø (discussed in the context of possessive placement):

- (7) Tromsø, age 2:1.29 (from Westergaard & Anderssen 2015)
 nei no døtt ned mannen på foten min. (Ina, age 2;1.29)
 no now fell down man.DEF on foot.mine
 ‘No, now the man fell down on my foot.’
- (8) Tromsø, age 2:2 (from Westergaard & Anderssen 2015)
 så døtt av hjulan.
 so fall off wheels.DEF
 ‘Then the wheels fall off.’

It is likely that this “deviating” word order appears in child language in all Norwegian dialects. However, the results from the NWD suggest that the order is more long-lasting and frequent in the Trøndersk dialect than other dialects. Note that long particle shift cannot be found in the Norwegian part of the Nordic Dialect Corpus (NDC, Lindstad et al. 2009) which mainly contains speakers above 20 years of age (see Tengesdal et al. 2024). Furthermore, all older speakers from Trøndelag that we have asked, judge sentences with long particle shift as fully ungrammatical. This suggests that “target-like” subject-particle placement is indeed acquired, only very late (see K. et al. 2012 for evidence that ability for grammar learning is intact until late adolescence).

14 *Unshifted subjects or shifting particles?*

Although the unshifted subjects discussed in e.g. [Anderssen & Westergaard \(2010\)](#) support an economy principle or a general dispreference for syntactic movement in language learners, the particle-subject order cannot easily be explained by the same mechanism. The examples of long particle shift above could be the result of the particle shifting across the subject, but they could also result from the subject remaining in a low, post-verbal position. The latter analysis would be in line with a general economy principle, while the first would not; the derivation would consist of an extra operation: a shift of the particle across the subject (and/or incorporation of the particle into the verb). The two options are illustrated in (9)–(10) below. For simplicity, we assume that the verb has moved to the V2-position in both cases, but it is also possible that it surfaces *in situ* (10), or in T (in 9) with the particle incorporated or cliticized to the verb.¹

- (9) Norwegian: shifted particle analysis

CP[No datt TP[ned_i mannen_j vP[t_j t_i på foten min]]].
 now fell down man.DEF on foot.DEF my

‘The man fell down on my foot now.’

- (10) Norwegian: low subject analysis

CP[No datt vP [ned mannen på foten min]].
 now fell down man.DEF on foot.DEF my

‘The man fell down on my foot now.’

In adult Norwegian, a structure like (9) would be illicit as verb particles must surface inside the verb phrase (with some marginal exceptions of topicalized particles). (10) would be illicit since definite subjects must surface externally to the verb phrase. The question is thus what kind of error the child who produced (7) has committed: Is it an extra movement of the particle (9), or an *in-situ* placement of the subject (10)? The second option is not entirely implausible, and would align with the economy principle, but it would entail that the child has not acquired one of the most salient features of Norwegian clausal syntax: The finite clause requires an overt, vP-external subject. However, null subjects are common in child language, and (7) could be analyzed as an existential clause with a dropped expletive subject (11). Note that the child in such case has not acquired the definiteness restriction of existential clauses: Only indefinite subjects can surface vP-internally.

¹It is in principle possible that the learner has stored the particle verbs as one word. This analysis is however not possible in cases where particles only precede NP subjects and not pronominal subjects. Given the often transparent semantic contribution of the verb and the particle, and relatively frequent input of separated verbs and particles, it is unlikely that a learner could maintain a single word hypothesis for a long time.

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- (11) Norwegian: existential construction

CP[No datt TP[det _{VP}[ned en mann på foten min]]].
 now fell it down a man on foot.DEF mine
 ‘Now there fell a man down on my foot.’

Support for the low subject hypothesis can be found in examples such as (12–13) below from [Larsen & Louise \(2014\)](#), which have subjects after a non-finite verb and therefore must be analyzed as vP-internal subjects.

- (12) Trøndersk, age 2:7

så ikke kan _{VP}[komme rotta og mus inni der].
 so not can come rats and mice in there
 ‘So rats and mice cannot come in there.’

- (13) Trøndersk, age 2:0

har itj _{VP}[kommi Magnus bilde].
 has not come Magnus picture
 ‘Has not come picture of Magnus.’

Example (12) has an indefinite subject and would be acceptable in adult language with an expletive subject, while (13) deviates from adult language in the same way as (7): It contains a vP-internal definite subject (and no expletive). In principle, we could simply state that both (12) and (13) follow the simple economy principle: The subjects are left *in situ*. There are however cases where a low position is difficult to defend. In the analysis in Section 5, we will use the following guidelines for evaluating the position of the particle and the subject:

1. Particles preceding negation must be “shifted” particles.
2. Particles preceding subject pronouns are most likely shifted, at least if produced by a child aged 3 or older (at which age subject pronoun placement is more or less target-like, [Westergaard 2008](#); [Anderssen et al. 2010](#))
3. Particles preceding agentive subjects of (non-reflexive) transitive verbs must be shifted, as these are always external, i.e., transitive agentive verbs never occur in existential constructions (**Det har spist en mann en pølse*, lit. ‘there has eaten a man a hot dog’).
4. Subjects following a non-finite verb must be in a low position.

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As we will argue below, there are good reasons to assume that all definite subjects are vP external, at least in the older children's and teenagers' production. Before analyzing the attested long particle shifts, we will discuss the syntax of particles and other "shifts" in Norwegian grammar.

3 Particles and regular particle shift

Verb particles have been described as intransitive prepositions (Emonds 1976, Svenonius 1996, Faarlund 2019: 137): They realize only the figure argument, while leaving the ground argument implicit or expressed as a PP, see e.g., the semantically transparent locative use of the particle in (14a). However, the meaning provided by the particle when combined with a verb is not always fully transparent, and Norwegian and other Germanic languages are full of lexicalized particle verbs (14b-c):

(14) Norwegian

- a. Han kastet ut hunden_{FIG} (fra huset_{GROUND}).
 he threw out dog._{DEF} from house._{DEF}
 'He threw the dog out of the house.'
- b. Hun slo opp med kjæresten.
 she hit up with boyfriend._{DEF}
 'She broke up with the boyfriend.'
- c. Han skjelte ut eleven.
 he scolded out pupil._{DEF}
 'He scolded the pupil.'

There is still debate about the correct syntactic analysis of particle verbs, and some questions remain unsettled: Should there be a unified analysis for locative and more abstract/non-transparent particles, and if so, should particles be analyzed as heads or phrases (Wurmbrand 2000, Ramchand & Svenonius 2002, Aa 2020, Larsson & Lundquist 2022)? Despite the semantic variability, the syntactic and prosodic behavior of verb particles is surprisingly uniform within Norwegian. Particles typically form a prosodic unit with the verb, which is best described as a maximal prosodic word. The prosodic word can either be realized with main stress on the verb, often accompanied with accent 2 (similar to a compound), (15a), or with main stress on the particle (??), similar to phrasal stress (see Kristoffersen 2000), but realizations with stress on both verb and particle are also attested (Tengesdal et al. 2024).

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- (15) a. Ole ²[TOK ut] bilen.
 Ole took out car.DEF
 ‘Ole took the car out.’
 b. Ole ¹[tok UT] bilen.
 Ole took out car.DEF
 ‘Ole took the car out.’

The accent 2 realization (15a) is most common in Trøndersk dialect area but is also frequent in Eastern Norwegian. In Norwegian, verbs form maximal prosodic words not only with particles, but also with following phonologically light elements like pronouns and articles. However, the compound-like realization of particle verbs is not found in any other multiword context, except for some names, and verb-negation sequences in some dialects (see [Tengesdal et al. 2024](#) and references therein). Particle verbs thus have both semantic and prosodic properties associated with simple words. Despite this, verb particles are independent syntactic words ([Åfarli 1985](#)), as seen in the fact that the verb strands the particle under V2 movement. This leads to word orders where inverted subjects and sentence adverbs intervene between the verb and the particle:

- (16) I går tok ikke Ole ut bilen.
 yesterday took not Ole out car.DEF
 ‘Yesterday Ole didn’t take the car out.’

The Germanic languages exhibit substantial variation in word order shifts and scrambling: German allows scrambling of both NP’s and PP’s, Icelandic only allows shifts of NP’s and pronouns, and Mainland North Germanic only allows for pronominal object shift ([Thráinsson 2001](#)). Yet, no Germanic language allows shifting or scrambling of a verb particle to a vP external position (see [Zeller 2003](#) and references therein). Thus, sentences with the structures of (17–18) are ungrammatical in all Germanic languages:

- (17) *Jeg gikk ut_i ikke vP[t_i i går].
 I went out not yesterday
 ‘I didn’t go out yesterday.’
 (18) *Han skjelte ut_i ikke vP[t_i eleven i går].
 he scolded out not pupil.DEF yesterday
 ‘He didn’t scold the pupil yesterday.’

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The distribution of verb particles is thus more similar to non-finite verbs in auxiliary constructions than to selected arguments. Like non-finite verbs, verb particles always surface inside the verb phrase. However, in some of the VO-Germanic languages, most notably Norwegian and English, particles display a shifting-like behavior with respect to direct objects inside the vP, so-called particle shift (19). The patterns in Norwegian and English are very similar; pronouns more or less obligatorily surface before particles, while the particle-noun phrase order is freer. Norwegian however has a preference for the Particle-NP order (Lundquist & Tengesdal 2022), and this preference is especially strong in the Trøndersk dialects where particles may occasionally also precede pronominal objects (Larsson & Lundquist 2014). This leads to a high input of verbs directly followed by particles in Trøndersk, but as far as we can tell, children acquire the order Verb-Pronoun-Particle early (e.g., *slipp oss ut*, ‘let us out’), which suggests that particles are treated as independent syntactic words (although this needs to be studied in greater detail). For English, several corpus studies of verb particle constructions in children’s spontaneous speech have found that the word order V-NP-Particle, (19a), typically appears in children’s speech earlier than, or at the same time as, V-Particle-NP, (19b), (Brown & Hanlon 1970, Snyder & Stromswold 1997, Snyder 2007). No children are found to produce the V-Particle-NP order before the V-NP-Particle order. For pronouns, neither children nor adults produce the order V-Particle-Pronoun (19c).

- (19) a. Sue finished her dinner up.
 b. Sue finished up her dinner.
 c. Sue finished (*up) it up.

It is not obvious that the developmental pattern would be the same in Norwegian, given tendencies of “low” arguments (as in low subjects), and also the strong preference for the V-Particle-NP order in adult Norwegian.

4 Subject shift, object shift and long object shift

Subject shift (see examples 1-2) shows similarities with particle shift in the sense that both shifts involve word order alternation within one syntactic domain: the vP for particle shift and the TP/IP for subject shift. Furthermore, these two shifts are not dependent on verb movement, and they show similar patterns with respect to full noun phrases and pronouns: Subject pronouns shift more or less obligatorily around negation or a sentence adverb, while there is more variation

for NP's. Norwegian differs from Danish and insular Scandinavian in its general preference for non-shifted subjects; even non-contrastive, context-given short subjects such as names and definite simple noun phrases are preferably placed after negation in Norwegian (appr. 70–80%, see [Lundquist & Tengesdal 2022](#)), while low placement of subjects in Danish and Icelandic is rare and has an effect on interpretation (see e.g. [Svenonius 2002](#)).

The variable placement of subjects also leads to variable placement of object pronouns in the midfield in Norwegian. When the main verb surfaces in the V2-position, pronominal objects tend to surface before negation and other adverbs (i.e., object shift, 20a, [Holmberg 1986](#)). However, objects never shift across a subject (20b), i.e., so-called long object shift is not allowed in Norwegian. Objects can however precede a “true” subject in existential constructions (20c), i.e., when there is a formal subject in the subject position. Note that NP subjects may optionally shift, which leads to shifting of the object (20d).

- (20) a. Studenten registrerte {seg} ikke {*seg} i tide.
 student.DEF registered REFL not REFL in time
 ‘The student didn’t register in time.’
- b. I fjor registrerte {*seg} ikke studenten {seg} i tide.
 last year registered REFL not student.DEF REFL in time
 ‘Last year the student didn’t register in time.’
- c. Det registrerte {seg} ikke {*seg} mange studenter {*seg} til
 there registered REFL not REFL many students REFL to
 kurset i fjor.
 course.DEF last year
 ‘Not many students registered for the course last year.’
- d. I fjor registrerte studenten {seg} ikke {*seg} i tide.
 last year registered student.DEF REFL not REFL in time
 ‘Last year the student didn’t register in time.’

The four examples above indicate that object placement cannot be explained solely in terms of available syntactic positions or information structure. The reflexive object has the same information structural status in (20a–20d), and the available landing sites for a shifted object should be the same in all four examples. Yet, shifting the object across the negation is only licit if the object does not cross the syntactic subject. Although there are various formal analyses of the restrictions on object shift, none of them directly captures the simple fact that object pronouns cannot precede the syntactic subject (see e.g. [Fox & Pesetsky](#)

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2005 and Erteschik-Shir 2005). For addressing both the issue of (long) object shift and (long) particle shift, we will propose that the grammar contains simple linearization statements, which the child slowly acquires through the input. The adult grammar must contain the following statements to account for object shift: In the TP, subjects precede objects, verbs precede objects, and unstressed/light object pronouns precede adverbs. We will approach the long particle shift in the same way in Section 6.

The examples in (20) illustrate four subject positions in Norwegian. The vP internal position is only available in existential constructions (20c). Note that you can also find extraposed subjects in spoken Norwegian, with a doubled pronoun in one of the canonical subject positions (21).

- (21) Den var veldig fin, den nye dukka.
 it was very nice, the new doll
 ‘The new doll was very nice.’

Given that subject pronouns are sometimes dropped in children’s speech, it is not always clear if the subject is in an “unshifted” position, or VP-internal (as in existential constructions) or extraposed. This uncertainty carries over to the cases of long particle shift; subjects that follow particles may be in a vP-internal or extraposed position. This will be taken into account when diagnosing the position of the particles and subjects.

5 Overview of the preliminary material

So far, no larger study on the acquisition of subject-particle placement has been carried out on Norwegian. Below we will analyze 31 examples that have been found in child language corpora, or have been observed by ourselves or colleagues. In addition, we will discuss the attested long particle shifts from Fosen (Trøndelag) in the Nordic Word order Database.

The Fosen results reveal some interesting patterns: (1) the particle-subject order is not found with pronominal subjects, only with NP subjects (in approximately 15% of the elicited sentences), (2) particle-subject order is found with both typical unaccusative verbs (*fall down*) and unergative verbs (*clean up*, transitive verbs were not tested), and (3) the particle-subject order is found with both definite and indefinite subjects. The participants sometimes inserted an expletive subject (*det*) in sentences with indefinite subjects in order to avoid particle-subject orders, e.g. *i går brant det ned tre hus* (lit. “yesterday burnt it down three

houses”, not counted as long particle shift). This never occurred with definite subjects, which suggests that the participants knew the definiteness restrictions of the existential constructions. Hence, the particle-subject order found with these speakers should not be analyzed as existential constructions with dropped formal subjects. Furthermore, the fact that particles never precede pronominal subjects rules out a potential clitic or compound analysis of the particle in particle-subject-orders. Note also that the subjects in the material are never sentence final, which rules out extraposition.

Of the 31 examples of spontaneous speech that we have gathered, two come from the Tromsø corpus (age: 2;1, 2;2.), two from the Ringstad corpus (age: 2;5, 2;8) and 27 examples from a wider age range (3;5–21), all from Trøndelag, collected by Ringstad and Hanne Siri Sund. It is highly likely that this order is attested in child speech in other dialects too; we simply do not have the data. Of 31 utterances, only four (12.9%) contain transitive verbs, and these are all produced by children below the age of six. Out of 15 examples from children/teenagers above nine, all verbs are intransitive. Thus, it appears that the teenagers have developed a more restricted and more target-like placement of particles compared to the children.

In four sentences, the particle precedes negation, which we count as a clear diagnosis of a shifted particle, but all of these are produced by children between the age three and four, e.g. “pappa sto opp ikke” (4;6). There are seven instances of particles preceding pronominal subjects (see example (4) above), and there appear to be an age effect here as well: None of the 13 examples of long particle shift from speakers above the age of twelve were across pronominal subjects, while almost half of the examples from the younger speakers have pronominal subjects (7/16, 44%). The examples we have from speakers of age 13 and above all have intransitive verbs and noun phrase subjects (just like the examples from NWD). Most of the subjects are definite (10/13) and although subjects are sometimes sentence-final (6/13), they do not appear to be extraposed. For example, in the utterance in (22), it is unlikely that a speaker at the age of 18 would drop a non-initial subject. All the examples from children and teenagers have the main verb in V2 position, i.e., the sentences are main clauses without auxiliary verbs. In other words, there are no unambiguous cases of vP-internal subjects, cf. example (9)-(10) above. Note that corpus studies of particle placement (Tengesdal et al. 2024) have shown that as few as 25% of the instances of particle verbs have the main verb in V2 position; most examples involve complex tenses, embedded clauses and various infinitival constructions. The fact that we find particle-subject order only when the main verb appears in V2 position (22a, but not 22b),

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and that shift never takes place over a verb (22c), suggests that we are looking at a “shift” of the particle governed by verb movement, similar to object shift.

- (22) a. Trøndersk, age 18
 No gikk ut Luren.
 now went out Lure.DEF
 ‘Now the Lure went out.’
 b. Low definite subject, only attested in young children
 * No har gådd ut Luren.
 now has gone out Lure.DEF
 ‘Now the Lure has gone out.’
 c. Shift over verb, never attested
 * No har ut gådd Luren.
 now has out gone Lure.DEF
 ‘Now the Lure has gone out.’

However, it should be acknowledged that some of the very early attestations of particle-subject orders could favor a low subject analysis. This includes the two examples from the Tromsø corpus (age 2;1–2;2), which both involve typical unaccusative particle verbs (*fall off*, *fall down*). The subjects in both examples are definite, which would speak against a low subject analysis in principle, but at this very early stage of acquisition, it is not obvious that the rules of subject placement are in place. The Trøndersk examples from the younger population show clearer evidence of particles shifting out of the vP: particles often precede either a negation, pronoun or the subject of a transitive verb (in total, 7/12 examples).

Although the examples from teenagers in the Trøndersk area all involve (a) intransitive verbs and (b) NP subjects, it is unlikely that these could be analyzed as vP-internal subjects. The main arguments against this analysis are (i) absence of subjects following non-finite verbs, and (ii) evidence from the NWD data that speakers of this age know the definiteness restrictions on low subjects/existential constructions.

6 Discussion, summary and remaining questions

From the data we have available today, we feel confident in concluding that long particle shift is a real phenomenon, found in young speakers in the Trøndersk dialect area and possibly children of other dialects. We conclude this from the fact that the word order is not found in adolescent speakers from Northern and

Eastern Norway (NWD) or in older speakers from the Trøndersk area (see [Larsson & Lundquist 2014](#), [Tengesdal et al. 2024](#)). Although particle-subject orders can be found in younger children of other dialects, it is hard to prove that these involve shifted particles rather than non-target-like, low subjects.

From the small set of examples collected we can see some developmental patterns in the distribution of shifted particles: Particle placement becomes more restricted the older the speakers get, as illustrated in Table 1.

Table 1: Developmental patterns in the distribution of shifted particles

Age	Particle >Neg	Particle >Transitive subject	Particle pronominal subject	> Part subject intransitive verb	NP-of
≤ 4;6	✓	✓	✓	✓	
≤ 5;6		✓	✓	✓	
≤ 12			✓	✓	
≤ 21				✓	

Although our material is small, it indicates that (a) teenagers only produce particle-subject in a very restricted context, most likely only in clauses with NP-subjects and intransitive verbs (see the NWD results), and (b) younger children produce shifted particles in more contexts. There are several ways of capturing the developmental patterns. For example, it could be argued that the particle initially targets any landing site between the finite verb and the base position, and slowly, the learner restricts the movement, ending up with only two sites: before or after the object (if regular particle shift is treated as a shift of the particle, not the object). This developmental trajectory would be the opposite of the one predicted by Westergaard’s economy principle, which predicts no movement at the earliest stages, in the absence of evidence for movement (and in this case, there is no particle-subject or particle-negation order in the input, at least not from adults). Alternatively, the developmental trajectory could be captured as the acquisition of ordering statements. As we saw in the examples in (20) above, the restrictions on object shift cannot fully be captured in terms of possible landing sites of the object; a direct ordering statement is needed: Subjects precede objects in the TP. From this perspective, word order acquisition would just be the acquisition of explicit ordering statements, for example the following (where “>” means “precedes”):

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- I. Verb > Particle (the main verb precedes the particle)
- II. Neg/Adverbs > Particle
- III. Subject > Particle

If the grammar contains no explicit ordering statements, the word order is free. A grammar that contains ordering statements I and II above, but not III, allows both regular particle shift (*kaste {katten} ut {katten} ut*) and long particle shift (*nå gikk {ut} luren {ut}*), but no shifting of particles over negation or verbs. A grammar that contains all three ordering statements would allow only regular particle shift (target grammar). In the absence of ordering statements, the produced word order will be influenced by factors like accessibility, prosodic well-formedness and availability of stored chunks (for example, particle-verb, verb-negation or verb-pronoun sequences) – factors that could explain the more subtle pattern with respect to different types of arguments (pronouns vs. NPs, transitive vs. intransitive subjects). The ordering statements can be thought of as surface filters that rule out certain orders that have been created in the syntax. Alternatively, one could imagine syntax being completely “a-linear”, with ordering statements acting on syntactic functions created in syntax.

A problem with the current dataset is that we do not know the proportion of particle-subject orders in children or even if there is a decline in this order during language learning. There is little doubt that the children and teenagers at all stages produce both target-like and deviant particle placement. The number of contexts for particle-subject order seems to shrink, and at some stage, the construction is presumably no longer part of the grammar. However, to be able to evaluate the development of target-like particle placement, we need larger corpus studies or elicitation experiments.

There are two questions that we have not addressed yet. The first is why long particle shift is found in the Trøndersk dialect area in particular, and the second is why similar “long” shifts do not take place with other verb-complements, for example light pronouns, e.g. we find “*no gikk ut Luren*”, but not “*nå kjeder seg luren*” (lit. “Now bores himself the Lure”). It is possible that a prosodic analysis could answer both these questions: Compound accents in particle verbs are more common in Trøndersk than in other dialects, and compound accents are not found in verb-pronoun sequences. The compound realization could initially be analyzed as evidence for syntactic proximity between the verb and particle. These are questions for future studies, with a larger dataset.

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References

- Aa, Leiv Inge. 2020. *Norwegian verb particles*. 1st edn., vol. 4. Amsterdam: John Benjamins.
- Áfarli, Tor Anders. 1985. Norwegian verb particle constructions as causative constructions. *Nordic Journal of Linguistics* 8(1). 75–98. DOI: [10.1017 / S0332586500001268](https://doi.org/10.1017/S0332586500001268).
- Anderssen, Merete, Kristine Bentzen, Yulia Rodina & Marit Westergaard. 2010. The acquisition of apparent optionality: Word order in subject and object shift constructions in Norwegian. In Merete Anderssen, Kristine Bentzen & Marit Westergaard (eds.), *Variation in the input. Studies in the acquisition of word order*, vol. 3, 241–270. Dordrecht: Springer. DOI: [10.1007/978-90-481-9207-6_10](https://doi.org/10.1007/978-90-481-9207-6_10).
- Anderssen, Merete & Marit Westergaard. 2010. Frequency and economy in the acquisition of variable word order. *Lingua* 120(11). 2569–2588. DOI: [10.1016/j.lingua.2010.06.006](https://doi.org/10.1016/j.lingua.2010.06.006).
- Brown, Roger & Camille Hanlon. 1970. Derivational complexity and order of acquisition in child speech. In John R. Hayes (ed.), *Cognition and the development of language*, 155–207. New York: Wiley.
- Emonds, Joseph. 1976. *A transformational approach to English syntax*. New York: Academic Press.
- Erteschik-Shir, Nomi. 2005. Sound patterns of syntax: Object shift. *Theoretical Linguistics* 31(1-2). 47–93. DOI: [10.1515/thli.2005.31.1-2.47](https://doi.org/10.1515/thli.2005.31.1-2.47).
- Faarlund, Jan Terje. 2019. *The syntax of mainland Scandinavian*. Oxford: Oxford University Press.
- Fox, Danny & David Pesetsky. 2005. Cyclic linearization of syntactic structure. *Theoretical Linguistics* 31(1-2). 1–45. DOI: [10.1515/thli.2005.31.1-2.1](https://doi.org/10.1515/thli.2005.31.1-2.1).
- Holmberg, Anders. 1986. *Word order and syntactic features in the Scandinavian languages and English*. University of Stockholm. (Doctoral dissertation).

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- K., Hartshorne J., Tenenbaum J. B. & Pinker S. 2012. A critical period for second language acquisition: Evidence from 2/3 million English speakers. *Cognition* 177. 263–277. DOI: [10.1016/j.cognition.2018.04.007](https://doi.org/10.1016/j.cognition.2018.04.007).
- Kristoffersen, Gjert. 2000. *The phonology of Norwegian*. Oxford: Oxford University Press.
- Larsen, Ringstad & Tina Louise. 2014. *Byggeklossar i barnespråk [Building blocks in child language]*. NTNU. (MA thesis).
- Larsson, Ida & Björn Lundquist. 2014. Objektsplacering vid partikelverb i norska dialekter och äldre svenska. In Janne Bondi Johannessen & Kristin Hagen (eds.), *Språk i norge og nabolanda*, 99–131. Oslo: Novus.
- Larsson, Ida & Björn Lundquist. 2022. The development of Swedish particle placement. In Ida Larsson & Erik Petzell (eds.), *Morphosyntactic change in Late Modern Swedish*, 145–194. Berlin: Language Science Press. DOI: [10.5281/zenodo.5792957](https://doi.org/10.5281/zenodo.5792957).
- Lindstad, Arne Martinus, Anders Nøklestad, Janne Bondi Johannessen & Øystein Alexander Vangsnes. 2009. The Nordic dialect database: Mapping microsyntactic variation in the Scandinavian languages. In Kristiina Jokinen & Eckhard Bick (eds.), *NEALT proceedings series*, vol. 4.
- Lundquist, Björn, Ida Larsson, Maud Westendorp, Eirik Tengesdal & Anders Nøklestad. 2019. Nordic word order database: Motivations, methods, material and infrastructure. *Nordic Atlas of Language Structures (NALS) Journal* 4(1). 1–33. DOI: [10.5617/nals.7529](https://doi.org/10.5617/nals.7529).
- Lundquist, Björn & Eirik Tengesdal. 2022. Argument placement in Norwegian. *Nordic Atlas of Language Structures (NALS) Journal* 7(1). 31–72. DOI: [10.5617/nals.10099](https://doi.org/10.5617/nals.10099).
- Ramchand, Gillian & Peter Svenonius. 2002. The lexical syntax and lexical semantics of the verb-particle construction. In Line Mikkelsen & Christopher Potts (eds.), *WCCFL 21 proceedings*, 387–400. Somerville: Cascadilla Press.
- Ringstad, Tina & Marit Westergaard. 2023. Children’s acquisition of word order variation: A study of subject placement in embedded clauses in Norwegian. *Journal of Child Language* 5. 1–25. DOI: [10.1017/S0305000923000259](https://doi.org/10.1017/S0305000923000259).
- Snyder, William. 2007. *Child language: The parametric approach*. Oxford: Oxford University Press.
- Snyder, William & Karin Stromswold. 1997. The structure and acquisition of English dative constructions. *Linguistic Inquiry* 28. 281–317.
- Svenonius, Peter. 1996. The optionality of particle shift. *Working Papers in Scandinavian Syntax* 57. 47–75.

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- Svenonius, Peter. 2002. Subject positions and the placement of adverbials. In Peter Svenonius (ed.), *Subjects, expletives and the EPP*, 201–242. Oxford: Oxford University Press.
- Tengesdal, Eirik, Ida Larsson & Björn Lundquist. 2024. Prosodic variation in particle constructions in three Norwegian dialect areas. *Maal Og Minne* 116(2). 295–344. DOI: [10.52145/mom.v116i2.2316](https://doi.org/10.52145/mom.v116i2.2316).
- Thráinsson, Höskuldur. 2001. Object shift and scrambling. In Mark Balten & Chris Collins (eds.), *The handbook of contemporary syntactic theory*, 148–202. Oxford: Blackwell.
- Westergaard, Marit. 2008. Verb movement and subject placement in the acquisition of word order. Pragmatics or structural economy? In Pedro Guijarro-Fuentes, María Pilar Larranaga & John Clibbens (eds.), *First language acquisition of morphology and syntax. Perspectives across languages and learners*, 61–86. Amsterdam: John Benjamins.
- Westergaard, Marit. 2011. Subject positions and information structure: The effect of frequency on acquisition and change. *Studia Linguistica* 65(3). 299–332. DOI: [10.1111/j.1467-9582.2011.01183.x](https://doi.org/10.1111/j.1467-9582.2011.01183.x).
- Westergaard, Marit. 2014. Linguistic variation and micro-cues in first language acquisition. *Linguistic Variation* 14(1). 26–45. DOI: [10.1075/lv.14.1.02wes](https://doi.org/10.1075/lv.14.1.02wes).
- Westergaard, Marit & Merete Anderssen. 2015. Word order variation in Norwegian possessive constructions: Bilingual acquisition and attrition. In Janne Bondi Johannessen & Joseph C. Salmons (eds.), *Germanic heritage language in North America*, 20–45. Amsterdam: John Benjamins.
- Wurmbrand, Susi. 2000. The structure(s) of particle verbs. Unpublished manuscript.
- Zeller, Jochen. 2003. Moved preverbs in German: Displaced or misplaced? In Geert Booij & Jaap van Marle (eds.), *Yearbook of morphology 2003*, 179–212. Dordrecht: Springer Netherlands.

Chapter 15

Crosslinguistic influence and the role of micro-cues in gender assignment: Ukrainian refugee children acquiring Polish

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Over 6 million Ukrainians were forced to leave their homes and sought refuge in Europe as the result of the Russian invasion of Ukraine in 2022, with over 2 million of them being children. These children faced the need to rapidly learn the languages of their new host countries and adapt to schooling in the new language. We present an elicited production study aimed at investigating the acquisition of grammatical gender in the new language, Polish, with a special focus on morpho-phonological gender cues. Our experimental group are Ukrainian children aged 6–12 years ($n = 64$), who have been living in Poland for 4–77 months and acquiring Polish in an immersive context. We compare them to monolingual speakers of the same age ($n = 30$). Our findings indicate that bilingual children tended to use feminine agreement with palatalized consonants, which serve as ambiguous gender cues, more often than their monolingual peers, particularly for the ‘-ć’ cue. This tendency increased with age, suggesting a stronger cross-linguistic influence from the L1 in older children.

1 Introduction

1.1 Acquisition of gender and the role of micro-cues

An important finding that has been replicated in research spanning over a decade is children's early sensitivity to the so-called phonological gender cues in both mono- and bilingual language acquisition (see Rodina & Westergaard 2015, Mitrofanova et al. 2018, 2021, Tanja et al. 2022). Phonological cues are internal regularities in the form of words that help deduce the nouns' grammatical gender. For example, in many languages, nouns ending in certain vowels or consonants are more likely to belong to a particular gender class (e.g., in Spanish, nouns ending in -a are typically feminine, while those ending in -o are likely to be masculine). Phonological cues tend to be one of the earliest and most reliable indicators for children learning grammatical gender, especially in systems where these cues are prominent and regular (Rodina & Westergaard 2012, Westergaard 2014, Mitrofanova et al. 2021, Kupisch et al. 2022).

Furthermore, children appear to acquire language incrementally, paying attention to small regularities in the input, the so-called micro-cues, generalizing from the linguistic environment without needing massive exposure to particular structures. They make incremental generalizations over limited input, refining their grammatical knowledge as they encounter more examples (see the micro-cue model of L1 acquisition in Westergaard 2009, 2014).

In the current study, we will build on this work on grammatical gender with a special focus on the sensitivity to micro-cues, and use the pseudo-words paradigm to the multilingual context involving two closely related Slavic languages, Polish and Ukrainian.

1.2 Gender assignment in Ukrainian and Polish

In the domain of grammatical gender, most Slavic languages distinguish between masculine, feminine and neuter, with neuter being the least frequent gender, and masculine being arguably the morphologically unmarked/default gender. Masculine is considered to be the morphological default, as it is associated with the least marked agreement form, is also typically assigned to borrowings (Corbett 1991: 78), and is most often used as a fall-back strategy in various acquisition contexts (Janssen 2016, Rodina & Westergaard 2012; Mitrofanova et al. 2018, 2021, Vykhovalnets' & Horodens'ka 2004). Both Ukrainian and Polish have a relatively transparent system of morphophonological noun-internal gender cues where the shape of the noun can serve as a reliable albeit probabilistic cue to its grammatical

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gender. For instance, nouns ending in *-a* are likely to be feminine, those ending in *-o* or *-e* are likely to be neuter, while nouns ending in a non-palatalized consonant are very likely to be masculine. However, Polish cues are more consistent than Ukrainian ones and lead to fewer ambiguities. For instance, in Ukrainian, neuter nouns, in addition to the majority ending in *-o* or *-e*, can also end in *-a*. These typically refer either to young beings (e.g., /dʲiɤtʃa/ (girl), /dʲɪtɪnʲtʃa/ (infant), /tɛlʲa/ (calf), /katʃɛnʲa/ (duckling) or to abstract nouns, usually with the suffixes *-annia*, *-ennia*, *-innia*, where the final consonant is palatalized, as seen in /malʲɔʋanʲa/ (drawing), /zvʲilʲnʲɛnʲa/ (dismissal), /stvɔrʲijɛnʲa/ (creature). Such neuter nouns ending in *-a* are absent in Polish. Nevertheless, the *-a* ending is predominantly and strongly associated with the feminine gender in Ukrainian, as it is in Polish.

More problematic are nouns ending in palatalized consonants, which can indicate both masculine and feminine gender. The distribution of nouns ending in palatalized consonants varies across the Slavic languages, with cognates often having different gender features across the languages (see examples in Table 1 below, where these nouns are masculine in Polish but feminine in Ukrainian and Russian).

Table 1: Gender of cognate nouns ending in a palatalized consonant

Polish	Ukrainian	Russian	English
/tʃɛp/ M	/tʲinʲ/ F	/tʲɛnʲ/ F	shadow
/rɨɕ/ M	/rɨsʲ/ F	/rɨsʲ/ F	lynx
/tʃɛl/ M	/tʃɛlʲ/ F	/tʃɛlʲ/ F	goal

In Ukrainian, determining the gender for nouns ending in a palatalized consonant is traditionally considered challenging (cf. /biʎ/ (*pain*) M vs. /tʲinʲ/ (*shadow*) F; Bezpoiasko et al. 1993). The only unambiguous indicator is the derivational suffix *-istʹ*, which reliably signals feminine gender, e.g., /vʲɪdpovʲɪdɔtʲnʲistʲ/ (*responsibility*). Certain nouns exhibit instability in their gender classification, with both masculine and feminine forms being considered standard, or one form regarded as normative and the other as colloquial (e.g., /tʃɔrʲnɔbɪlʲ/ (*tarragon*) MF). In other cases, only *-tsʹ* and *-dzʹ* are strongly associated with masculine gender (e.g., /ɦudʒʲ/ (*bump*) M, /mʲɨsʲɪtsʲ/ (*moon*) M), whereas other consonants, such as *-lʹ*, *-sʹ*, *-nʹ*, *-tʹ*, *-zʹ*, *-dʹ*, are ambiguous in terms of gender assignment (see further details in Zahnitko 1996). However, the gender assignment of individual ambiguous gender cues in Ukrainian has not been systematically studied before.

Similar to Ukrainian, Polish inanimate nouns ending in a palatalized consonant can be either masculine or feminine gender, with a higher proportion of these nouns belonging to masculine gender compared to Ukrainian. Previous corpus studies have found that Polish nouns ending in *-ń* and *-ś* tend to be masculine, while nouns ending in *-ć* are comparably likely to be masculine or feminine (Stefańczyk 2007). Language acquisition data show that both monolingual and bilingual children learning Polish have difficulty acquiring the declensional patterns of feminine nouns ending in a palatalized consonant (Smoczyńska 1985, Brehmer & Rothweiler 2012). In addition, adult native speakers of Polish retrieve the feminine gender of nouns ending in a palatalized consonant more slowly than for transparent feminine nouns ending in ‘-a’, even when controlling for noun frequency (Długosz 2023).

Since Russian is one of the commonly spoken first languages among children in Ukraine, it is important to note that previous studies (Nesset 2003, Mitrofanova et al. 2018, 2021, Urek n.d.) focusing on Russian have identified several phonological factors that may serve as cues for either masculine or feminine gender in nouns ending in a palatalized consonant. These factors include the place and manner of articulation of the final palatalized consonant, stress patterns, and the characteristics of the nucleus (see Nesset 2003, Urek n.d.). Simplifying the complex picture a little bit, we can say that for mono- and disyllabic nouns, most final palatalized consonants (with the exception of rhotics) tend to be associated with feminine in Russian, with both adults and children showing sensitivity to these fine-grained tendencies (as evident from their gender assignment to nonce words, see Mitrofanova et al. 2018, 2021, Urek n.d.).

To sum up, in contrast to generally transparent feminine (i.e., final *-a*) and masculine (final non-palatalized consonant) nouns, there are cases of variation between Polish versus Ukrainian and Russian. Specifically, nouns ending in a palatalized consonant tend to be masculine in Polish (Krajewski 2005). Further research on Ukrainian is currently ongoing, with preliminary data showing nouns ending in palatalized consonants may be comparably likely to be masculine or feminine. Finally, in Russian, such nouns are predominantly feminine (Urek n.d.).

1.3 The present study

With the intriguing cross-linguistic differences in mind, this study investigates bilingual Ukrainian-Polish children’s sensitivity to gender cues in Polish. We designed and conducted an elicited production task, in which we asked the children to produce color adjectives with a set of nonce nouns ending in palatalized *-ń*, *-ś*, *-ć* and compared them to nonce nouns with transparent masculine or feminine

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gender cues (final nonpalatalized consonant and final *-a*, respectively). Two sets of nonce nouns were created: one for Ukrainian and one for Polish, conforming with the language-specific phonotactics and avoiding phonological neighbors.

We selected the Polish nonce words to (i) conform to Polish phonotactic rules, (ii) avoid resemblance to any real words, and (iii) lack phonological neighbors. Similar criteria were used for creating Ukrainian stimuli, where we modified the stimuli set from [Mitrofanova et al. \(2018\)](#), adjusting it to Ukrainian phonotactics and phonological neighborhood. For Polish, to meet criterion (iii), we used the YourPond Tool by [Marian et al. \(2012\)](#) to process Polish nouns from the KELLY Project word list, which includes the 9,000 most useful words for learners of Polish. The naturalness of the items was assessed by one of the authors, who is a native speaker of Polish. Bilingual Ukrainian-Polish children completed the task in two versions: in Polish and Ukrainian, while monolingual speakers participated only once.

2 Method

2.1 Participants

The experimental group consisted of 75 Ukrainian children (36 boys), aged 5 to 13 years (mean age = 9), who had been living in Poland for a period ranging from 4 to 77 months. Seven participants from Western Ukraine were migrants who settled in Poland between 2017 and 2021. In contrast, the vast majority of participants, comprising 68 children, were refugees who arrived in Poland following the onset of the full-scale invasion, between February 2022 and August 2023. These children were acquiring Polish in an immersive context by attending Polish schools (66 children) or kindergartens (10 children). Data collection was conducted in two Polish cities, Poznań and Rawicz (Greater Poland Voivodeship).

Out of the 75 participants, 11 were excluded from the analysis as they encountered difficulties completing the task in Polish. Consequently, the final analysis is based on data from 64 bilingual children. The majority of parents (51.6%) reported speaking exclusively in Ukrainian with their child at home, while 11.3% indicated using exclusively Russian. Additionally, 17.7% of parents reported using both Ukrainian and Russian, and a comparable proportion (16.1%) spoke Ukrainian and Polish. A smaller group (4.8%) reported using all three languages – Ukrainian, Russian, and Polish – in communication with their child.

According to parental reports, children were more frequently exposed to Ukrainian both through hearing and speaking the language at home, during interactions with Ukrainian-speaking peers, through media consumption, and

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via online Ukrainian schooling. The 11.3% of children who were exposed only to Russian at home still heard and used Ukrainian in other settings, such as school, media, or online activities. As anticipated, the primary environments for acquiring Polish were Polish-medium schools or kindergartens, along with public spaces. While school was the most frequent setting for both hearing and speaking Polish, parents reported that the children used Polish outside of school as well: in stores, with friends, and partly at home.

A group of Polish monolingual controls ($n = 30$; 14 boys) aged between 7 and 12 years (mean age = 10.4, $SD = 1.7$) was recruited and tested in person in Poznań and Rawicz. Another group of Ukrainian monolingual controls ($n = 15$), consisting of adult Ukrainians (2 males) aged between 21 and 39 years (mean age = 32.8), primarily from the eastern and southern regions of Ukraine, was tested online¹. We were unable to test monolingual Ukrainian-speaking children in Ukraine due to the ongoing war, and we opted not to conduct the testing online, as this would have significantly reduced the experimenter's control over the procedure. Importantly, this limitation applied only to child participants due to the need for closer supervision; online testing was not considered problematic for adult controls.

Ukrainian and Russian remain the most widely spoken languages among Ukrainians. Despite the significant linguistic shifts following Russia's full-scale invasion of Ukraine – such as a sharp decline in Russian use and an increased preference for Ukrainian – both languages are still familiar to most participants. The character of the combination of languages is fluid and varies across regions, both in its distribution and characteristics. While it may be difficult to generalize participants' linguistic preferences based solely on their place of residence, recognizing broader regional trends is meaningful. Sociological, linguistic, and ethnographic studies (see Kulyk 2017, 2023, Sokolova 2022, Soroka et al. 2023) highlight certain patterns: the eastern and southern regions are often characterized as predominantly Russian-speaking, the northern region tends to exhibit greater bilingualism, and the central and western regions are generally considered more Ukrainian-speaking.

¹As a reviewer pointed out, the adult baseline may not be ideal for studies involving children. We are planning a baseline data collection with Ukrainian children living in Ukraine, using a larger set of items that will hopefully provide further empirical data on gender assignment for nouns ending in palatalized consonants in Ukrainian. In the current study, we do not make statistical comparisons between adults and children; our main focus is gender assignment in Polish, for which we include a control group of age-matched Polish monolingual children. Data from Ukrainian adults is used only as a reference point and is not included in direct statistical comparisons.

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The participants in the current study came from all geographic regions of Ukraine. In a questionnaire completed by parents before the experiment, we asked them to indicate their place of residence prior to the full-scale invasion. The largest proportion of respondents (22.4%) identified their place of residence as Ukraine without providing further details, while the remaining participants specified their location as being in Western Ukraine (19.7%), Southern Ukraine (18.4%), Central Ukraine (17.2%), Eastern Ukraine (11.8%), and Northern Ukraine (10.5%).

Thus, our focus is not on dialectal differences within the Ukrainian language – although such differences certainly exist across regions – but rather on regional differences in language exposure and usage patterns.

However, while we acknowledge that regional variation in exposure to Ukrainian and Russian exists, the inclusion of region as a variable in our analysis did not improve the models and was therefore not included in the final models.

This complexity of language exposure is further deepened by another sociolinguistic factor – the widespread phenomenon of *Surzhyk*, a Ukrainian-Russian mixed speech. The characteristics and features of this “mixture” vary depending on geography, and Hentshel and Taranenko emphasize that this is not a “schematic dichotomy”, but rather “gradual differences in a non-bilingual country” (Hentshel’ & Taranenko 2015). However, until recently, *Surzhyk* was not commonly included in the response options provided to participants. Even if it had been included, its use may have been underreported by the participants themselves due to factors such as stigma or the lack of social prestige associated with it. The variability of speech across many regions of Ukraine makes it difficult to precisely classify the linguistic situation, as it is inherently fluid and not strictly defined. *Surzhyk* was occasionally mentioned in parental reports of our participants, and the participants themselves sometimes spoke either about *Surzhyk* or in *Surzhyk*.

It is also important to note that even Russian-speaking individuals often have at least passive competence in Ukrainian (Lakhtikova 2017). The geographical distribution may play a role in the treatment of gender due to the variation in gender assignment between Ukrainian and Russian. There appears to be a difference in which gender is assigned to nonce nouns ending in a palatalized consonant across the three Slavic languages. Monolingual Russian-speaking children (tested in Mitrofanova et al. 2018) tend to assign feminine to nonce nouns in the palatalized condition (78% feminine). Adult Ukrainian speakers tested online assigned feminine at a substantially lower rate to these nonce nouns (50%). Finally, the probability of assigning feminine to nouns ending in a palatalized consonant was even lower in Polish monolingual children (30%), as will be evident in the

results section. Therefore, although is difficult to disentangle the influence of Ukrainian and Russian as background languages, the overall prediction is that both would lead to a higher percentage of feminine responses compared to the Polish monolinguals.

2.2 Design and materials

To elicit gender-marked adjectives in Ukrainian and Polish, we replicated the experiment using nonce nouns ending in a palatalized consonant (see Mitrofanova et al. 2018, 2021, Urek n.d.). We employed nonce words to investigate children’s sensitivity to more abstract morphophonological cues, rather than to test their knowledge of individual lexical items. Children were asked to produce [color adjective + noun] combinations using a set of nonce nouns. Among the 17 nonce words, 4 ended in a hard consonant and were associated with masculine gender (e.g., /pʊnʲip/ in Ukrainian or /mɔ.ban/ in Polish); 4 ended in -a and were associated with feminine gender (e.g., /tranʰa/ in Ukrainian or /fuf.ga/ in Polish); and 9 words ended in a palatalized consonant, which were the target of our experiment (e.g., /drʊnʲ/ in Ukrainian or /mseɲ/ in Polish). The items were matched for final segment phonology in both languages, e.g., Polish -ś, -ń, -ć and the corresponding Ukrainian -сь, -нъ, -ть (-s’, -n’, -t’). In the present paper, we report only the Polish data.

Table 2: Experimental items in Polish

Condition	Micro-cue	Nouns
	M	/mɔ.ban/, /tu.rip/, /ra.pɔl/, /vɔ.mut/
	F	/fuf.ga/, /tɔp.va/, /bam.ga/, /fum.ta/
MF	-ń	/tlɔɲ/, /naɲɲ/, /mseɲ/
	-ś	/ɲɔɕ/, /frɔɕ/, /bliɕ/
	-ć	/dlitɕ/, /glɔtɕ/, /vdɛtɕ/

2.3 Procedure

The experiment was preceded by a practice block, in which participants were shown familiar objects and asked to describe their color (e.g., *To jest zielony kubek*, ‘This is a green cup’ or *To jest czarny dom*, ‘This is a black house’). The

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experimenter then introduced participants to the nonce objects depicted in the images and asked them to describe their color. The lead-in sentences were structured to avoid providing any cues regarding the grammatical gender of the target noun.

The elicitation procedure was as follows:

Experimenter: “This is what we call a /pʊnʲip/. What color are they?” Child: “A blue-MASC /pʊnʲip/ and a golden-MASC /pʊnʲip/”.

Experimenter: “And now, what disappeared?” Child: “The blue-MASC /pʊnʲip/”.

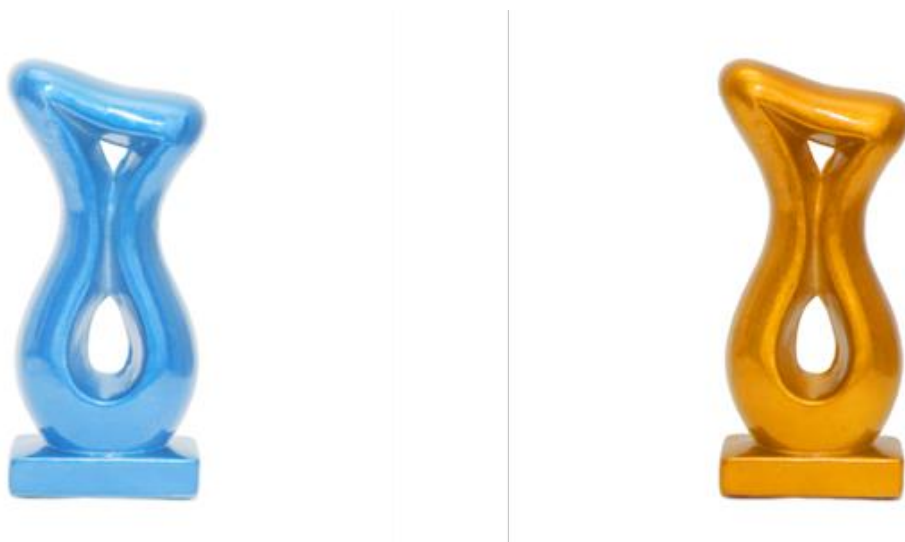


Figure 1: Example of an experimental stimulus

Each participant was tested individually in Ukrainian (or Russian, depending on the child’s preference) and Polish by two experimenters: one was a bilingual Ukrainian-Russian speaker, and the other a native speaker of Polish. The participants’ responses were audio-recorded, transcribed, and coded by the authors of this study.

3 Results

Recall that there seems to be variation within Slavic languages regarding palatalized micro-cues:

- 78% feminine assignment to nonce nouns ending in a palatalized consonant in Russian (Mitrofanova et al. 2018; monolingual children aged between 3;0 and 7;0 years, $M = 5;2$),

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- 50% feminine assignment in the palatalized condition in Ukrainian (note that this is based on data collected with Ukrainian adults, and will need to be adjusted in future research involving children),
- 30% feminine assignment in the palatalized condition in Polish (6–12 y.o. monolingual children, see below).

For the data analysis, we counted the number of feminine responses for each item. Participants were instructed to produce three instances of adjectival gender agreement, resulting in a range of feminine responses from 0 to 3.

Figure 2 illustrates the distribution of feminine responses in Polish across the three conditions: F, M, and MF.

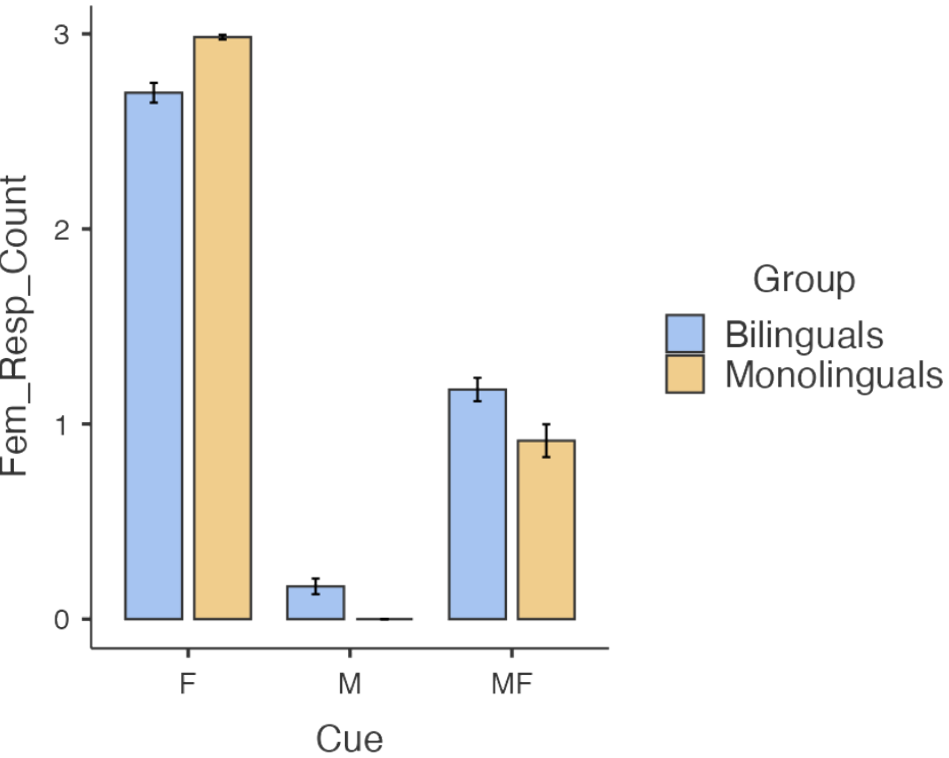


Figure 2: Feminine assigned to nonce nouns across three conditions: M, F and MF

Both groups consistently used feminine agreement in the F condition and masculine agreement in M condition. In the MF condition, masculine agreement was predominant ($M = 1.09$, $SD = 1.49$), although bilingual children appeared to use

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feminine agreement more frequently than monolingual children. Given this observation, we conducted a more detailed analysis of the results in the MF condition.

We fitted a generalized linear mixed-effects model with a Poisson distribution and log link function using the GAMLj module (Gallucci 2019) in Jamovi (version 2.4). The model included group, age (centered), and their interaction as fixed effects, with participant and item as random intercepts.

The effect of group was marginally significant, estimate = -0.51, SE = 0.31, OR = 0.60, $z = -1.69$, $p = .092$, with bilinguals using feminine agreement more frequently ($M = 0.74$, SE = 0.14) than monolinguals ($M = 0.44$, SE = 0.13). The effect of age was not significant, estimate = 0.05, SE = 0.08, OR = 1.06, $z = 0.66$, $p = .510$. Although not significant, estimate = -0.06, SE = 0.16, OR = 0.94, $z = -0.37$, $p = .711$, there was a trend towards more feminine responses in bilinguals than monolinguals with increasing age (see Figure 3).

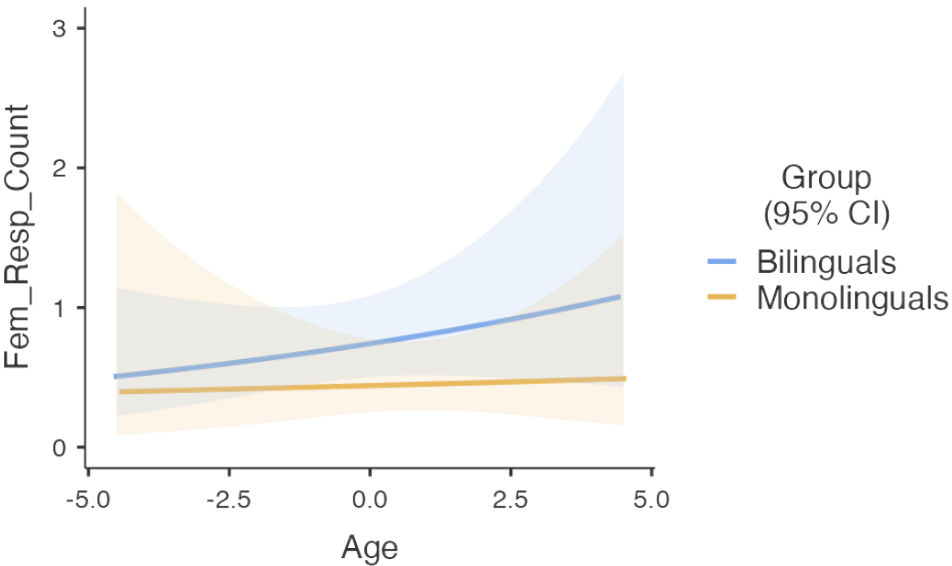


Figure 3: Effect of age (centered) on the probability of assigning Feminine gender to nonce nouns ending in a palatalized consonant

For each micro-cue in the MF condition ($-\acute{c}$, $-\acute{s}$, $-\acute{n}$), we ran additional models with group, age (centered), and their interaction as fixed effects, and participant and item as random intercepts.

For the $-\acute{c}$ cue, we found a significant effect of group, estimate = -0.87, SE = 0.44, OR = 0.42, $z = -1.97$, $p = .049$. Bilinguals were more likely to use feminine

agreement ($M = 0.68$, $SE = 0.17$) compared to monolinguals ($M = 0.29$, $SE = 0.12$). The effect of age was not significant, estimate = -0.01 , $SE = 0.12$, $OR = 0.99$, $z = -0.05$, $p = .960$, nor was the interaction between group and age, estimate = -0.18 , $SE = 0.23$, $OR = 0.84$, $z = -0.76$, $p = .449$.

For the *-ś* cue, we found no significant effects of group, estimate = -0.67 , $SE = 0.44$, $OR = 0.51$, $z = -1.51$, $p = .131$, age, estimate = 0.07 , $SE = 0.12$, $OR = 1.07$, $z = 0.58$, $p = .560$, or their interaction, estimate = -0.12 , $SE = 0.23$, $OR = 0.89$, $z = -0.52$, $p = .602$.

Finally, for the *-ń* cue, we found no significant effects of group, estimate = -0.19 , $SE = 0.24$, $OR = 0.83$, $z = -0.78$, $p = .437$, age, estimate = 0.05 , $SE = 0.07$, $OR = 1.05$, $z = 0.77$, $p = .442$, or their interaction, estimate = -0.03 , $SE = 0.13$, $OR = 0.97$, $z = -0.25$, $p = .803$.

To sum up, the analysis revealed a marginally significant effect of group, indicating that bilingual children tended to use feminine agreement in the MF condition more frequently than monolingual children. A significant difference between the two groups was observed only for the *-ć* cue, with bilingual children producing more feminine agreement than their monolingual peers. Finally, while not statistically significant, there was a descriptive trend indicating an increase in feminine agreement in the MF condition with age, pointing to stronger CLI from L2 to L1 in older children.²

4 Discussion

In the current paper, we focus on the results of the Polish task comparing Ukrainian-Polish bilingual children to age-matched Polish monolingual children. Recall that masculine is the default gender for both Ukrainian and Polish, hence assigning feminine would go against the default strategy. If we observe a prevalence of feminine assignment to nouns with palatalized micro-cues in the Ukrainian group compared to Polish monolinguals, a reasonable interpretation would be crosslinguistic influence from Ukrainian and/or Russian, where such nouns are more likely to be feminine than in Polish³.

The second prediction was regarding the effect of age at the immersion onset: we expected larger effects of CLI from Ukrainian and/or Russian to Polish in

²We included home exposure to Ukrainian/Russian – quantified from parental questionnaires – as a covariate, but it did not improve the model and was not a significant predictor of feminine assignment in either L1 or L2.

³However, there appears to be substantial variation in Ukrainian regarding these cues and further research (involving more items and a larger participant sample) is needed to uncover the underlying patterns.

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older children compared to younger ones. The literature has proposed two potential justifications for this prediction. First, younger children tend to be faster at becoming dominant in the new language and showing signs of attrition in their L1 than older children. Second, older children tend to rely more on their L1 when acquiring the L2 and have more consolidated representations in their L1 than younger children (see Flores 2010, 2020).

First of all, for palatalized consonants, Polish monolinguals preferred masculine agreement over feminine agreement, which aligns with the distributional patterns of these cues in Polish. Ukrainian children showed at-ceiling performance with transparent M and F micro-cues in Polish (final non-palatalized consonant for M and final *-a* for F). Note that the same cues also exist in the children's native language, Ukrainian and/or Russian, although in Ukrainian, final *-a* may also be associated with neuter gender in specific cases, as mentioned earlier (these cases occur even more rarely in Russian). It is reasonable to suggest that they transferred the micro-cues from their L1 into the L2 (cf. Kupisch et al. 2022), resulting in facilitative cross-linguistic influence.

At the same time, participants showed a different pattern in assigning gender to nouns ending in a palatalized consonant: bilingual children tended to assign feminine gender to these nouns more frequently than age-matched Polish monolinguals, particularly when the noun ended in *-ć*. This tendency goes against the default M strategy and is likely an artefact of cross-linguistic influence from their first language, Ukrainian and/or Russian, where these nouns have a higher probability of being feminine than in Polish. Specifically, we found that bilingual children used more feminine agreement with nonce nouns ending in *-ć* compared to Polish monolingual children. The between-group differences did not reach statistical significance for the other micro-cues. This result may be explained by the differing distribution of individual micro-cues in Polish. While the ending *-ć* is more likely to indicate feminine gender, the endings *-ś* and *-ń* are slightly more strongly associated with masculine gender. Although fine-grained patterns are evident for individual palatalized endings in Polish, similar distinctions are not well documented in Ukrainian. Existing descriptions (e.g., Zahnitko 1996) suggest that gender assignment for such nouns is influenced by multiple factors and lacks consistency, with only a few endings (e.g., [ts], [dz]) reliably linked to masculine gender. The increased use of feminine agreement with the *-ć* cue is compatible with two explanations: a) CLI from the L1(s) and b) that the children have picked up this cue from input in Polish.

We suggest CLI as the primary explanation, since assigning feminine gender to nonce nouns ending in a palatalized consonant contradicts the default masculine assignment strategy in Polish. Even if we assume that bilingual children are, for

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some reason, more sensitive to feminine exemplars such as *wieś* ('village') in their new L2 and base their gender assignment to nonce words on this, CLI likely remains the underlying cause, especially given that monolingual Polish children assign fewer feminine responses to the same nonce nouns.

Importantly, we observed considerable variation in using feminine versus masculine agreement with palatalized endings. Even though masculine agreement was predominant, again reflecting children's sensitivity to the distributional patterns in Polish, both bilingual and monolingual children also used feminine agreement.

The results of our study provide support for the micro-cue model (Westergaard 2009, 2014), in that they show that children are sensitive to fine distinctions in the input.

In line with our expectations, we also observed an age trend, albeit not significant: younger bilingual children were more similar to their monolingual peers than older children, who in turn were more likely to assign feminine gender to nouns with palatalized micro-cues compared to younger bilinguals and Polish monolinguals. This tendency is in line with previous studies, which have shown that the ability to acquire new languages declines with age. Consequently, L1 to L2 crosslinguistic influence is expected to be more pronounced in older learners and conversely, L2 to L1 influence is predicted to be stronger in younger learners (Schmid & Köpke 2007, Flores 2010, 2020).

It is important to highlight that our study participants were children who may still have been in the process of acquiring their L1 when the immersion in the new L2 started. This is why L2 acquisition and L1 attrition may go hand in hand. It may be the case that the observed age trend, i.e., younger children being more likely to use masculine, may be due to two reasons: that they have already internalized Polish palatalized cues, or that they have not fully acquired these cues in Ukrainian and are relying on the masculine default strategy more than the older children. Similarly, in older learners, potential CLI effects may be difficult to disentangle from those driven by slower acquisition rates of Polish compared to younger children. Furthermore, there is great variability between speakers in whether feminine or masculine gender is assigned to nouns ending in palatalized consonants, as evident from Figure 3. This variability was observed in both the bilingual and monolingual groups, highlighting the absence of clear gender assignment preferences in this context. Moreover, due to this variability, group-level differences may not consistently reflect patterns at the individual level across all children. Future research with a larger set of items and participant sample may help clarify these issues. The first step would be collecting data with

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monolingual Ukrainian children of the same age without exposure to Polish, and such research is already underway.

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References

- Bezpoiasko, Olena, Kateryna Horodens'ka & Vitali Rusanivskyi. 1993. *Hramatyka ukraïns'koï movy. Morfolohiia* [Grammar of the Ukrainian language: Morphology]. Kyiv: Lybid.
- Brehmer, Bernhard & Monika Rothweiler. 2012. The acquisition of gender agreement marking in Polish: A study of bilingual Polish-German-speaking children. In Kurt Braunmüller & Christoph Gabriel (eds.), *Multilingual individuals and multilingual societies*, 81–100. Amsterdam: John Benjamins.
- Corbett, Greville G. 1991. *Gender*. Cambridge: Cambridge University Press.
- Długosz, Kamil. 2023. The role of noun termination in grammatical gender assignment in Polish: A psycholinguistic study. *Die Welt der Slaven* 68(2). 291–303.
- Flores, Cristina. 2010. The effect of age on language attrition: Evidence from bilingual returnees. *Bilingualism: Language and Cognition* 13(4). 533–546.
- Flores, Cristina. 2020. Attrition and reactivation of a childhood language: The case of returnee heritage speakers. *Language Learning* 70(1). 85–121.
- Gallucci, Marcello. 2019. *GAMLj: General analyses for linear models*. Jamovi module. <https://gamlj.github.io/>.
- Hentshel', Gerd & Oleksandr Taranenko. 2015. Movnyi landshaft tsentral'noi Ukraïny: Ukraïns'ka mova, rosiys'ka mova, “Surzhyk” (Uzhyvannia — movna kompetentsiia — natsional'ne pozytsionuvannia) [Linguistic landscape of Central Ukraine: Ukrainian language, Russian language, “Surzhyk” (usage — linguistic competence — national positioning)]. *Movoznavstvo* 4. 3–25.
- Janssen, Bibi E. 2016. *The acquisition of gender and case in Polish and Russian: A study of monolingual and bilingual children*. Universiteit van Amsterdam. (Doctoral dissertation).

Oleksandra Hrebenshchykova, Kamil Dhugosz & Natalia Mitrofanova

- Krajewski, Grzegorz. 2005. The role of grammatical gender in the acquisition of noun inflection in Polish. *Psychology of Language and Communication* 9(1). 75–84.
- Kulyk, Volodymyr. 2017. Language attitudes in independent Ukraine: Differentiation and evolution. *Harvard Ukrainian Studies* 35(1–4). 265–292.
- Kulyk, Volodymyr. 2023. Ukrainians now (say that they) speak predominantly Ukrainian. *Ukrainian Analytical Digest* 1. 6–11.
- Kupisch, Tanja, Miriam Geiss, Natalia Mitrofanova & Marit Westergaard. 2022. Structural and phonological cues for gender assignment in monolingual and bilingual children acquiring German: Experiments with real and nonce words. *Glossa: A Journal of General Linguistics* 7(1). DOI: [10.16995/glossa.5696](https://doi.org/10.16995/glossa.5696).
- Lakhtikova, Anastasia. 2017. Understanding passive bilingualism in Eastern Ukraine. *Critical Multilingualism Studies* 5(1). 144–173.
- Marian, Viorica, James Bartolotti, Sarah Chabal & Anthony Shook. 2012. CLEAR-POND: Cross-linguistic easy-access resource for phonological and orthographic neighborhood densities. *PLoS ONE* 7(8). e43230.
- Mitrofanova, Natalia, Yulia Rodina, Olga Urek & Marit Westergaard. 2018. Bilinguals' sensitivity to grammatical gender cues in Russian: The role of cumulative input, proficiency, and dominance. *Frontiers in Psychology* 9. 1894. DOI: [10.3389/fpsyg.2018.01894](https://doi.org/10.3389/fpsyg.2018.01894).
- Mitrofanova, Natalia, Yulia Rodina, Olga Urek & Marit Westergaard. 2021. Sensitivity to microvariation in bilingual acquisition: Morphophonological gender cues in Russian heritage language. *Applied Psycholinguistics* 43(1). 41–79.
- Neset, Tore. 2003. The assignment of gender and declension to Russian nouns in soft consonants: Predictability and rule interaction. *Journal of Slavic Linguistics* 11(2). 287–322.
- Rodina, Yulia & Marit Westergaard. 2012. A cue-based approach to the acquisition of grammatical gender in Russian. *Journal of Child Language* 39(5). 1077–1106.
- Rodina, Yulia & Marit Westergaard. 2015. Grammatical gender in Norwegian: Language acquisition and language change. *Journal of Germanic Linguistics* 27. 145–187.
- Schmid, Monika S. & Barbara Köpke. 2007. Bilingualism and attrition. In Barbara Köpke, Monika S. Schmid, Merel Keijzer & Susan Dostert (eds.), *Language attrition: Theoretical perspectives*, 1–8. Amsterdam: John Benjamins.
- Smoczyńska, Magdalena. 1985. The acquisition of Polish. In Dan I. Slobin (ed.), *The crosslinguistic study of language acquisition, Vol. I: The data*, 595–686. Hillsdale: Lawrence Erlbaum Associates.
- Sokolova, Svitlana. 2022. The problem of choosing the language of communication: Ukrainian realities. *Cognitive Studies | Études cognitives* 22. Article 2649.

15 Crosslinguistic influence and the role of micro-cues in gender assignment

- Soroka, Yuliia, Natalia Kudriavtseva & Igor Danylenko. 2023. Language and social inequalities in Ukraine: Monolingual and bilingual practices. *Ideology and Politics Journal* 24(2). 58–85.
- Stefańczyk, Wiesław T. 2007. *Kategoria rodzaju i przypadku polskiego rzeczownika: Próba synchronicznej analizy morfologicznej [The category of gender and case of the Polish noun: An attempt at a synchronous morphological analysis]*. Kraków: Wydawnictwo Uniwersytetu Jagiellońskiego.
- Tanja, Kupisch, Natalia Mitrofanova & Marit Westergaard. 2022. Phonological vs. natural gender cues in the acquisition of German by simultaneous and sequential bilinguals (German–Russian). *Journal of Child Language* 49(4). 661–683. DOI: [10.1017/S0305000921000039](https://doi.org/10.1017/S0305000921000039).
- Urek, Olga. N.d. Categorization of ambiguously cued novel nouns: Evidence from grammatical gender assignment in Russian. Submitted.
- Vykhovanets', Ivan & Kateryna Horodens'ka. 2004. *Teoretychna morfolohiia ukraïns'koï movy [Theoretical morphology of the Ukrainian language]*. Kyiv: Pulsary.
- Westergaard, Marit. 2009. *The acquisition of word order: Micro-cues, information structure, and economy* (Linguistik Aktuell/Linguistics Today 145). Amsterdam: John Benjamins.
- Westergaard, Marit. 2014. Linguistic variation and micro-cues in first language acquisition. *Linguistic Variation* 14(1). 26–45. DOI: [10.1075/lv.14.1.02wes](https://doi.org/10.1075/lv.14.1.02wes).
- Zahnitko, Anastasiia. 1996. *Teoretychna hramatyka ukraïns'koï movy. Morfolohiia [Theoretical grammar of the Ukrainian language: Morphology]*. Donetsk.

Chapter 16

Processing of locative prepositional phrases by young children: An eye-tracking study

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In this study, we use Visual World eye-tracking to investigate processing of locative PPs by Russian-speaking 2- and 3-year-old children (n=63). Experimental trials involved pictures representing contrasting locative configurations: “in” vs “under” and “on” vs “behind”. The relevant linguistic cues involved locative preposition and case marking on the Ground NP: “in” and “on” assign locative case, while “under” and “behind” assign instrumental case. In the control condition pictures differed in the choice of objects involved (figure or ground). The analysis revealed a significant positive effect of age group in the experimental conditions which went beyond the differences observed in the control conditions. This suggests that the 3-year-olds’ better performance on locative PP comprehension cannot be fully explained by attention to the task or differences in the processing of lexical nouns. These results align with previously reported production patterns, with 2- but not 3-year-olds omitting a substantial portion of prepositions from their early utterances and support the Underspecified P account of preposition omission at the telegraphic speech stage (Mitrofanova 2017).

1 Introduction

Spatial cognition and spatial language have been the focus of extensive research in language typology and developmental psychology (Piaget & Inhelder 1956, Johnston & Slobin 1979, Landau & Jackendoff 1993, Bowerman & Choi 2001, a.o.).

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There is considerable variation in how spatial relations are expressed across languages. This variation includes differences in the range and specificity of spatial meanings encoded grammatically, the way spatial features are combined within individual grammatical elements, and more (Pederson et al. 1998, Levinson 2003, Levinson & Wilkins 2006). Early acquisition has been reported to be influenced by the perceptual and cognitive salience of spatial contrasts, with children first acquiring spatial relations that are more visually and physically apparent, progressing from simpler, more concrete relations (e.g., containment and support) to more abstract and complex ones (e.g., left/right or in front of/behind). In a crosslinguistic elicited production study, Johnston & Slobin (1979) propose the following order of acquisition of locative expressions: *in*, *on*, *under*, *beside* being learned prior to *between*, *front*, and *back* (p. 537). Leikin (1998) reports a slightly modified sequence (*in*, *on*, *behind*, *under*, *above*, *in front of*, and *to*), with minimal variation across languages, including Russian (p. 92). The intriguing fact is that prepositions appear in production relatively late, around 2;6–3 years of age. But can we confidently assert that children lack an understanding of individual locative prepositions before they begin producing them? Our study seeks to answer this question by examining how 2- and 3-year-old Russian-speaking children process locative prepositions.

The omission of unstressed grammatical elements is a hallmark of early child language production, often referred to as the TELEGRAPHIC SPEECH STAGE (TSS, Bloom 1970; Brown & Fraser 1964). The elements typically omitted include grammatical morphemes (Matthews & Theakston 2006, Phillips 2010), determiners (Eisenbeiss 2000, Guasti et al. 2008), auxiliaries (Theakston & Lieven 2005; Westergaard 2016), and prepositions (Leikin 1998; Nicholas 2011; Tomasello 1987). Two types of theories have been proposed to account for such omissions. FULL-COMPETENCE ACCOUNTS (Boser et al. 1992; Lust 2006; Poeppel & Wexler 1993) assume that children's underlying representations are target-like from early on and attribute omissions to factors outside of grammar, e.g., limited attention and working memory span, production limitations, non-target like pragmatic skills, etc. DIMINISHED COMPETENCE APPROACHES, on the other hand, (Clahsen et al. 1996; Duffield 2008; Eisenbeiss 2000; Pinker 1984) take omissions to reflect non-target-like aspects of the children's underlying grammar.

The two types of theories make contrasting predictions with respect to comprehension of functional elements by young children. FULL-COMPETENCE approaches predict full comprehension of functional elements at the omission stage in production. Initial support for this prediction comes from studies which have revealed sensitivity to functional elements by young children (e.g., children show longer reaction times when listening to sentences lacking obligatory

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functional elements, etc.; see Bernal et al. 2007, Höhle et al. 2006, Santelmann & Jusczyk 1998; Shady 1996; Shi 2014). Importantly, however, early sensitivity to the presence (and even the type) of functional elements does not entail target-like comprehension of these elements. It is conceivable that children at early stages may be aware of the presence and type of certain functional elements in the structure, without having acquired their target-like semantics and distributional properties. It is thus possible that the observed sensitivity to the presence and type of functional elements precedes target-like comprehension of these elements (cf. e.g., Soderstrom 2008).

Only a few studies couched within generative linguistics have directly addressed the relation of production and comprehension of functional elements (Beyer & Hudson Kam 2009; Johnson et al. 2005; Miller & Schmitt 2012) and locative prepositions (P) and their relationship to realization of case in particular. At the same time, locative prepositions possess the most concrete and easily visualizable semantics, which makes them ideal for testing children's comprehension at the early stages of grammatical development.

To investigate the relation between production and comprehension of locative prepositions at the telegraphic speech stage, Mitrofanova (2017) conducted a series of experiments with a group of monolingual typically developing Russian-speaking children aged 2–4. As in Nicholas (2011), the children were first tested on an elicited production task. The results showed that the 2-year-olds omitted prepositions at a significantly higher rate than the 3-year-olds (32% vs. 6%). In utterances involving P omission, the children generally used a case-marked form of the noun, but overgeneralized one specific case, locative, and did not use instrumental case, which is required by a subset of prepositions in adult Russian (e.g., produced *stole* 'table.LOC' instead of *pod stolom* 'under table.INSTR'). In cases where the children produced an overt P, the case marking was target-like: instrumental case with prepositions 'under' and 'behind' and locative case with prepositions 'in' and 'on'.

Subsequently, the same children performed a sentence-picture matching comprehension task where they were asked to point to a picture (1 out of 4) described by an audio stimulus. The target picture always had one competitor that differed only with respect to the locative configuration involved (e.g., a teddy bear *in* a boat vs. a teddy bear *under* a boat), while the other two competitors involved a different figure animal (e.g., mouse) or a different ground object (e.g., box). The results revealed a significant negative correlation between the accuracy of comprehension of locative Ps on the sentence-picture matching task and the rate of preposition omission in the elicited production task, with comparable omission

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rates for the four prepositions. In other words, children who omitted prepositions at higher rates performed significantly worse on the comprehension task than children with lower rates of P omission.

The analysis proposed in [Mitrofanova \(2017\)](#) for P omission in Russian argues against the FULL-COMPETENCE hypothesis ([Nicholas 2011](#)) and is in line with DIMINISHED COMPETENCE approaches to early grammatical development, which do not presuppose target-like grammatical competence at the early stages of acquisition ([Clahsen 1990](#), [Westergaard 2009](#)). According to [Mitrofanova \(2017\)](#), children's grammar at early stages includes an underspecified Place category with a generalized locative semantics, while lacking finer-grained locative contrasts. This approach accounts for the diminished comprehension of locative Ps at the stage of their omission in production as well as the fact that Russian children between 2 and 3 years of age overwhelmingly use the same case marking on ground nominals in utterances involving P omission irrespective of the implied locative relation ([Mitrofanova 2017](#)).

The current study contributes online processing data to the pool of evidence. First, a new eye-tracking experiment is set up to test the predictions of the two competing accounts of preposition omission at age 2: the FULL-COMPETENCE and the DIMINISHED COMPETENCE accounts. If the DIMINISHED COMPETENCE approaches are on the right track, we expect to find a significant difference between 2- and 3-year-olds in terms of comprehension of locative Ps, which would mirror the previously reported difference between these two groups in terms of production of overt locative Ps. On the other hand, if the FULL-COMPETENCE accounts that attribute P omission to performance limitations (e.g., [Nicholas 2011](#)) are correct, it is predicted that there will be no significant difference between 2- and 3-year-olds in terms of comprehension of locative Ps. Second, we aim to avoid a potential methodological limitation in the previous Russian study ([Mitrofanova 2017](#)), namely, the fact that the off-line method that was employed could have masked children's implicit comprehension of locative prepositions ([Bergmann et al. 2012](#), [Sekerina 2015](#); [Sekerina et al. 2004](#)). Specifically, the children's inability to select the correct picture in the explicit sentence-picture matching task could have been influenced by limited attention span, difficulty following the instructions, and action bias.

2 The present study

2.1 Participants

The participants were monolingual Russian-speaking children: 28 2-year-olds (six girls, $M_{\text{age}} = 2;6$ years, range: 2;2–2;11) and 35 3-year-olds (five girls, $M_{\text{age}} = 3;5$ years, range: 3;0–3;11) recruited and tested at a preschool in Moscow, Russia in 2015 and 2019. The study was registered and approved by Sikt (sikt.no). The parents of participating children signed an informed consent. The children received a small toy as a thank-you gift after the experiment.

2.2 Design and materials

The experiment contained 16 test trials (8 control and 8 experimental), each consisting of a preamble and a locative utterance presented together with two pictures, a target and a competitor. Four spatial configurations -IN, ON, UNDER, or BEHIND -were selected based on their case marking properties, high frequency, visualizability, and early acquisition in Russian (Leikin 1998). Across all the experimental trials, IN-configurations were contrasted with UNDER-configurations, while ON-configurations were contrasted with BEHIND-configurations. All ground objects in the task were inanimate (*table, sofa, boat, cupboard, etc.*). Figure objects were 50% animate (*mouse, teddy, bee, doll* – which are animate in Russian) and 50% inanimate (*napkin, cup, spade and book*). In the figure contrast condition, animate figures were contrasted with other animates, and inanimate figure with inanimates, e.g., *a mouse in a box* vs *a kitten in a box*; *a book under a toy truck* vs *a ball under a toy truck*). See Appendices A and B for the full list of stimuli. Locative utterances included one of the four spatial Ps, i.e., *na* ‘on’, *v* ‘in’, *pod* ‘under’, or *za* ‘behind’. We created two experimental lists for the experiment. Both lists had the experimental PP contrast condition, and differed in the Control condition: List A involved a Figure contrast condition and List B included a Ground contrast condition (see Appendices A and B of the supplementary materials). In the Figure Control condition (??), the two pictures differed with respect to the figure object (e.g., *a baby mouse* and *a kitten*) while the spatial configuration was the same (e.g., *in the box*). In the Ground control condition (??), the two pictures differed with respect to the ground object (e.g., *a box* and *a cupboard*), while the figure object and the spatial configuration were the same (*a baby mouse* + IN). In the Experimental condition (??), the figure and ground objects were the same for both pictures (e.g., *a baby mouse* and *a box*), but the spatial configuration differed (e.g., *a baby mouse under a box* versus *a baby mouse in a box*.) The experimental condition was included in both lists, and all

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prepositions were tested in the course of the task. All of the pictures represented objects familiar to 2-year-olds and were specially drawn for the study. After listening to the sentence, the children were asked to point to the picture that best matched the utterance they just heard. Their offline choices (pointing) and eye-movements during listening were recorded.

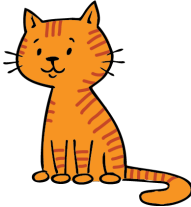
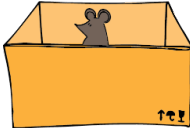
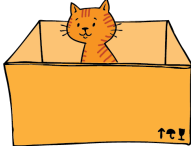
Figure Control Condition		
Preamble		
	<i>Myshonok.</i> baby.mouse Target	<i>Kotenok.</i> kitten Competitor
Locative utterance		
	Posmotri! Myshonok Look! baby.mouse.NOM in big.INST boxINST 'Look! The baby mouse is in a big box.'	v bol'shoj korobke.

Figure 1: Examples of experimental stimuli

In the experimental condition, the spoken sentence matched both pictures during the presentation of the subject NP, e.g. *myshonok* ‘baby mouse_{NOM}’ in (??), and was only disambiguated at the preposition (and the associated case marking on the Ground noun). In order to select the target picture, the child needed to know the meaning of the preposition, e.g., *under* vs *in* in (??) or the case marking associated with this P (with the prepositions ‘in’ and ‘on’ assigning prepositional case, and the prepositions ‘under’ and ‘behind’ assigning instrumental case).

The stimuli in the control conditions were disambiguated either at the first NP (Figure condition) or at the second NP (Ground condition). The control condition did not require knowledge of locative prepositions as they were disambiguated based on the knowledge of the lexical nouns involved in the study.

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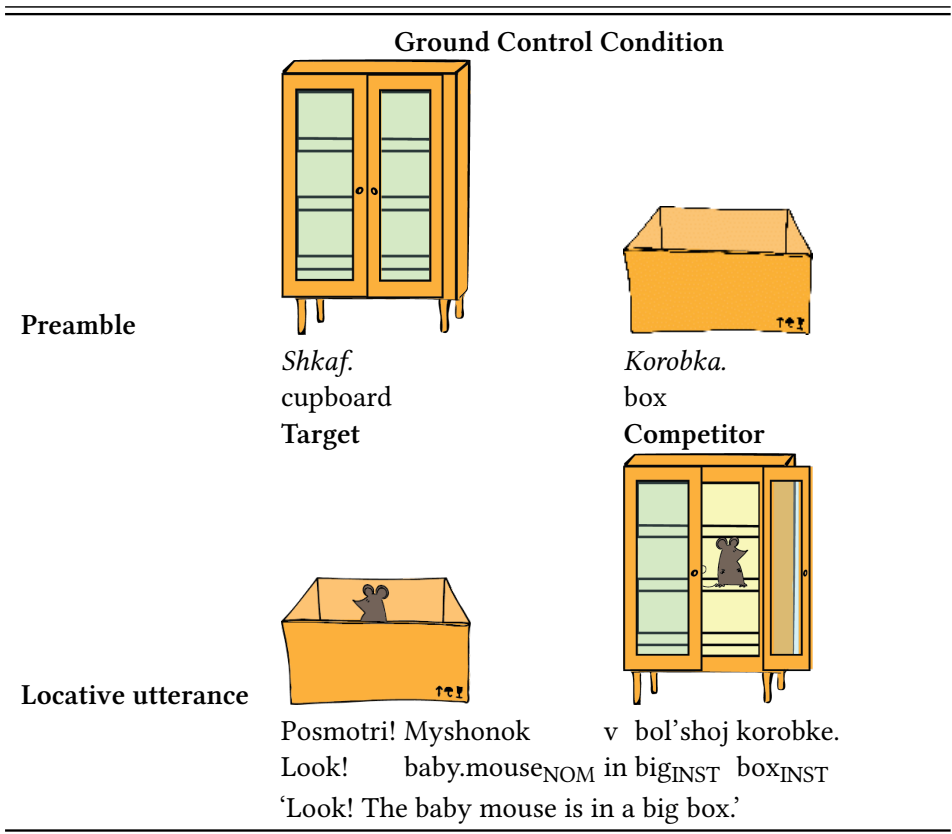


Figure 2: Examples of experimental stimuli

The spoken stimuli were recorded by a female professional radio announcer in a soundproof booth, sampling at 22,050 Hz, and were pronounced with a child-friendly prosody. There were short pauses after each named object in the preamble. Another short pause was inserted between the instruction to look and the experimental sentence with the preposition. In all the experimental stimuli the P was followed by an adjective to increase the distance between the P and its complement N.

The 16 trials were split into two blocks of 8. This made it possible for each target picture to appear once in the control and once in the experimental conditions. The two occurrences of the same target picture were always separated by 7 trials. There was a short break containing an attention-getter (a picture of ice cream) in between the two blocks. Latin-square design was used to determine

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Experimental Condition	
Preamble	
	<i>Myshonok.</i> <i>Korobka.</i> baby.mouse box Target Competitor
Locative utterance	
	Posmotri! Myshonok v bol'shoj korobke. Look! baby.mouse _{NOM} in big _{INST} box _{INST} 'Look! The baby mouse is in a big box.'

Figure 3: Examples of experimental stimuli

the order of stimuli presentation within each block. The order of the target and competitor pictures was counterbalanced so that the target picture appeared on the left and on the right side an equal number of times and did not appear on the same side more than twice in a row.

2.3 Procedure

Children were tested individually in a quiet room of their preschool. Prior to the experiment, each child was familiarized with the equipment and the task requirements and played with the stick used for pointing. After obtaining the children's assent, they were seated comfortably in front of the stimuli monitor fitted with a remote eye-tracking camera, and an eye-tracking calibration was performed.

We used a remote free-viewing eye-tracker SMI Red-500 to record the children's eye movements during the experiment. Eye movements were sampled at a rate of 60Hz. Visual stimuli were presented on a 22-inch monitor while spoken utterances were simultaneously played through two speakers located on either side of the monitor. The materials were presented via the SMI Experiment Center

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software. Each trial started with the presentation of a preamble. After approximately 500 ms, two pictures of spatial configurations appeared, and the locative utterance was played simultaneously.

The children used a long stick to point to one of the two pictures by tapping on the monitor. No feedback was provided. The study was conducted by two experimenters. One interacted with the child, and the other monitored the eye-tracker, checked calibration, and advanced to the next trial as soon as the child tapped one of the pictures. The experiment lasted about 10 minutes.

3 Results

3.1 Accuracy

In the experiment, the children were asked to point to the picture that matched the locative utterance. Three 2-year-olds and three 3-year-olds refused to point to the pictures due to shyness and were excluded from the analysis of offline responses. Figure 4 illustrates mean response accuracy for the Control and Experimental conditions in the two age groups of children.¹

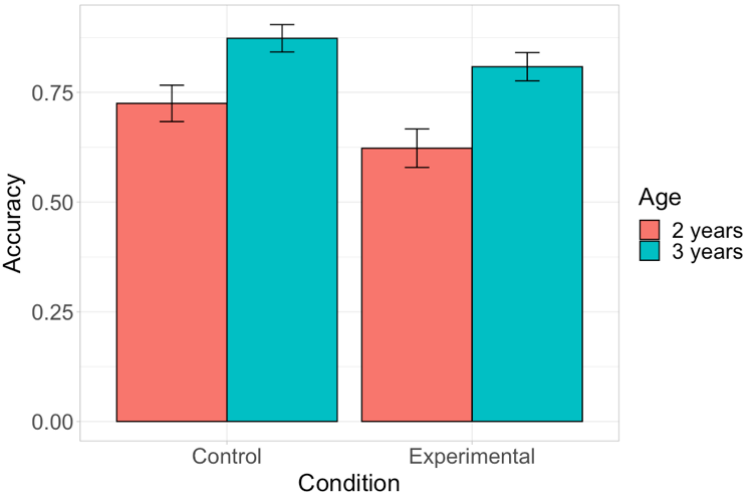


Figure 4: Accuracy scores on the sentence-picture-matching task

To analyze the participants’ comprehension of locative prepositions we fit a mixed effects logistic regression with response accuracy in the Experimental con-

¹In case of self-correction (5% of the responses) the last response was counted, while non-available responses (6% of all answers) were excluded from the dataset.

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dition (0 or 1) as the dependent variable (Jaeger 2008, Quené & Bergh 2008). Age (2 versus 3 years) was included as a predictor. Furthermore, to control for general effects of attention and lexical processing, we included the participants' mean accuracy on the Control condition as a covariate. Participants and items were included as random intercepts. The model revealed a significant effect of mean Control condition accuracy, with higher accuracy on the Control condition associated with a higher probability of correct response in the Experimental condition ($B = 2.73$, $z = 3.95$, $p < 0.001$). We also found a significant effect of Age, with 3-year-olds more likely to respond correctly on the Experimental condition compared to the 2-year-olds ($B = 0.63$, $z = 2.26$, $p = 0.02$). Thus, the 3-year-olds were better at matching the correct spatial configuration picture with the locative P than the 2-year-olds.² Moreover, the effect of Age was significant while controlling for performance on the Control task, where the children needed to identify spatial configurations based on the names of the Figure and Ground objects.

3.2 Eye movements

In order to investigate the children's online comprehension of locative utterances, we conducted an analysis of their eye-movement patterns. Figure 5 illustrates the proportions of looks to the Target and Competitor pictures in the Experimental condition by Age, calculated in 50 ms time bins after the onset of the critical word (the locative preposition).

To analyze Target preference in the Experimental condition in each age group, we conducted cluster-based permutation analyses separately for each group (Maris & Oostenveld 2007; see also Hahn et al. 2015; Minor et al. 2022, 2023, Oakes et al. 2013, Yang et al. 2020). Given that children need at least 300 ms to plan and execute a saccade (Trueswell 2008; Yang et al. 2002), we analyzed gaze patterns in 3,000-ms time windows starting from 300 ms after the onset of the preposition.

²We thank one of the reviewers for asking about accuracy rates for individual prepositions. Recall that we tested two prepositional contrasts: 1) 'in' vs 'under' and 2) 'on' vs 'behind'. Offline accuracy results show that 3-year-olds had comparably high accuracy on all Ps (over 75%), while 2-year-olds were more accurate with respect to 'in' and 'on' (82% and 69%, respectively) vs 'under' and 'behind' (50% and 48%, respectively). These results are in accord with previous studies suggesting earlier acquisition of 'in' and 'on' vs 'under' and 'behind'. However, it is possible that some children may have had a bias towards the more prototypical 'in' and 'on' spatial configurations in cases when they were unsure about the meaning of the prepositions. Unfortunately, we cannot disentangle these two hypotheses in this experiment, as our contrasts always involved a more frequent/prototypical vs a less frequent P, instead of *all* possible contrasts between the prepositions. We welcome future research on this topic.

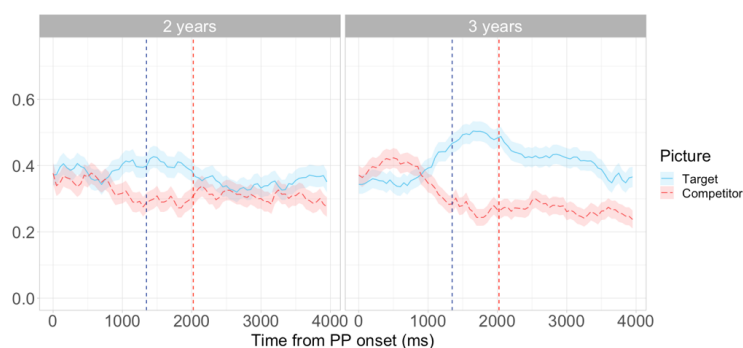
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Figure 5: Proportions of looks to Target and Competitor pictures in the Experimental condition by Age. Plots start at PP onset; vertical blue lines represent average onsets of the Ground noun; vertical red lines represent average sentence offsets.

For each age group, the analysis consisted of the following steps (following [Minor et al. 2023](#)). First, all looks outside of the pictures were removed, and proportions of looks to Target were calculated in seventy 50-ms time bins within the window of interest. Almost all (99.85%) of the calculated proportions were either 1 or 0; the remaining proportions were binarized (all proportions ≥ 0.5 were replaced with 1, all others with 0). Then, for each time bin we fit an intercept-only mixed effects logistic regression modelling the probability of looks to Target, with Participants and Items included as random intercepts. Clusters were identified based on adjacent time bins where the probability of looks to Target was above chance with an estimated p -value below the inclusion threshold. In order to increase the power of the test to detect weak but long-lasting effects and avoid artificial break-up of clusters, the inclusion threshold was set at 0.1 prior to the analysis (this choice does not affect the overall validity of the cluster-based permutation test, see [Maris & Oostenveld 2007](#); [Hahn et al. 2015](#); [Minor et al. 2023](#)). The z -values for all the time bins in each cluster were summed up to produce the sum statistic for the cluster.

The null hypothesis was that the probability of looks to the Target and Competitor pictures in the Experimental condition was equal. Hence, to obtain the distribution of the sum statistic under the null hypothesis, we randomly permuted the picture labels (Target vs. Competitor). Permutation was performed by participant, i.e., for each participant the picture labels were permuted in either all or none of the trials with 0.5 probability ([Minor et al. 2023](#)). We then identified clusters in the permuted dataset in same way as in the original dataset. The largest cluster sum statistic was recorded. We performed 1000 permutations, generating

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a distribution of the sum statistic under the null hypothesis. We then obtained p -values for the cluster statistics in the original dataset as the proportion of the null hypothesis distribution with equal or more extreme values. Clusters were considered significant if their sum statistic had a p -value below 0.05.

For the 3-year-olds, the analysis revealed one significant cluster stretching from 1,100 to 3,000 ms after preposition onset (sum $z = 137.5$, $p < 0.001$), indicating that the participants exhibited above-chance preference for the Target picture in the Experimental condition. For the 2-year-olds, the largest cluster stretched from 1,100 to 1,700 ms and was only marginally significant (sum $z = 23.72$, $p = 0.093$).

Next, we conducted a cluster-permutation analysis to compare the probability of target looks between the age groups. The analysis followed the same steps as above with the following modifications. First, data from both age groups were included in the analysis. Next, Age was included as a predictor in the mixed effects logistics regressions fit to data in each time bin, and time bins were included in clusters based on the p -value of the Age effect. Similarly, the cluster sum statistic was taken as the sum of the z -statistics of the Age effect in each time bin within the cluster. Finally, the null hypothesis was that the probability of Target looks was the same for both Age groups. Hence, to obtain the null hypothesis distribution, we permuted the Age labels by participant. Several clusters were detected where the probability of looks to the Target picture was higher in 3-year-old group compared to the 2-year-olds, but none of them reached statistical significance.

4 Discussion and conclusion

The results of the study indicate a marked difference between 2- and 3-year-olds with respect to the comprehension of locative prepositions. The analysis of the participants' offline accuracy revealed a significant effect of age, in addition to attention and lexical knowledge of the nouns involved in the study. This suggests that the difference between the age groups is specifically related to locative prepositions and cannot be fully explained by differences in the processing of lexical nouns or attention to the task. The analysis of participants' eye-movements revealed a similar pattern, with only 3- but not 2-year-olds demonstrating a significant preference for the Target picture in the condition involving contrasts between individual locative prepositions.

The observed discrepancy between 2- and 3-year-old children mirrors a previously established contrast between these age groups with respect to the produc-

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tion of prepositions. The elicited production study with a large sample of 2–4-year-old children (Mitrofanova 2017) reports that a substantial portion (32%) of locative utterances produced by 2-year-old children lacked an obligatory preposition, while the rate of omission in the production of 3-year-olds was negligible (6%). Furthermore, the same study reported a significant negative correlation between offline comprehension accuracy and the rate of P omission in production in individual children, suggesting that at the telegraphic speech stage, the children who do not fully understand the meaning of the prepositions omit them in their own production.

Taken together, these findings are consistent with the so-called DIMINISHED COMPETENCE approaches (see e.g., Clahsen et al. 1996; Mitrofanova 2017; Westergaard 2009). Specifically, with respect to locative Ps, the reported data fit well with the Underspecified P Hypothesis (UPH), proposed in Mitrofanova (2017), which assumes that children’s grammar at early stages includes an underspecified Place category with a generalized locative semantics, while lacking finer-grained locative contrasts. The UPH predicts that 2-year-olds’ ability to distinguish between individual locative prepositions in comprehension should be diminished as compared to older children, which is consistent with the results of the current study.

In a broader context, the underspecification approach can be generalized from Ps to other types of functional categories and can resolve the apparent contradiction between children’s early sensitivity to the presence and type of functional categories (Kedar 2007; Shi 2014) and lack of target-like comprehension of functional elements at early stages (Beyer & Hudson Kam 2009; Johnson et al. 2005). This combination of findings is unexpected both under approaches that assume the complete lack of functional categories in child grammar at early stages (Felix 1987; Radford 1990; Radford & Ramos 2001) and those that postulate target-like competence with respect to functional categories at the stage of their omission in production (Boser et al. 1992; Nicholas 2011; Poeppel & Wexler 1993). According to the underspecification approach (Mitrofanova 2017; cf. also Clahsen 1990; Clahsen et al. 1994), children’s acquisition of functional elements proceeds in two stages. Specifically, children are initially able to associate individual functional items with the underspecified functional categories already present in their grammar, without establishing the full range of semantic distinctions between individual instantiations of the same category. The semantic and syntactic properties of individual functional elements are acquired at a later stage (Mitrofanova 2016).

In conclusion, eye-tracking offers a powerful tool for exploring real-time language processing and comprehension, providing valuable insights into children’s

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linguistic competence during the early stages of development. Future research using this paradigm could expand to examine a broader range of locative prepositions across diverse languages and integrate production and processing methodologies within the same groups of children. We look forward to seeing this line of investigation advance further.

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References

- Bergmann, C., M. Paulus & P. Fikkert. 2012. Preschoolers' comprehension of pronouns and reflexives: The impact of the task. *Journal of child language* 39(4). 777–803. DOI: [10.1017/S0305000911000298](https://doi.org/10.1017/S0305000911000298).
- Bernal, S., J. Lidz, S. Millotte & A. Christophe. 2007. Syntax constrains the acquisition of verb meaning. *Language Learning and Development* 3(4). 325–341.
- Beyer, T. & C. L. Hudson Kam. 2009. Some cues are stronger than others: The (non)interpretation of 3rd person present -s as a tense marker by 6- and 7-year-olds. *First Language* 29(2). 208–227.
- Bloom, L. 1970. *Language development: Form and function in emerging grammars*. Cambridge: MIT Press.
- Boser, Katharina, Barbara Lust, Lynn Santelmann & John Whitman. 1992. The syntax of V2 in Early German grammar: The strong continuity hypothesis. 22. 51–66.
- Bowerman, Melissa & Soonja Choi. 2001. Shaping meanings for language: Universal and language-specific in the acquisition of spatial semantic categories. In Melissa Bowerman & Stephen C. Levinson (eds.), *Language acquisition and conceptual development*, 475–511. Cambridge: Cambridge University Press.
- Brown, Roger & Colin Fraser. 1964. The acquisition of syntax. *Monographs of the Society for Research in Child Development* 29. 43–79.
- Clahsen, H. 1990. Constraints on parameter setting: A grammatical analysis of some acquisition stages in German child language. *Language Acquisition* 1(4). 361–391.

16 *Processing of locative prepositional phrases by young children*

- Clahsen, H., S. Eisenbeiss & M. Penke. 1996. Lexical learning in early syntactic development. In H. Clahsen (ed.), *Generative perspectives on language acquisition: Empirical findings, theoretical considerations and crosslinguistic comparisons*, vol. 14, 129–159. Amsterdam: John Benjamins.
- Clahsen, Harald, Sonja Eisenbeiss & Anne Vainikka. 1994. The seeds of structure: A syntactic analysis of the acquisition of case marking. In T. Hoekstra & B. D. Schwartz (eds.), *Language acquisition studies in generative grammar*, vol. 8, 85–118. Amsterdam: John Benjamins.
- Duffield, N. 2008. Roots and rogues in German child language. *Language Acquisition* 15(4). 225–269.
- Eisenbeiss, S. 2000. The acquisition of the determiner phrase in German child language. In M.-A. Friedemann & L. Rizzi (eds.), *The acquisition of syntax: Studies in comparative developmental linguistics*, 26–62. London: Longman.
- Felix, Sascha. 1987. *Cognition and language growth*. Dordrecht: Foris.
- Guasti, M. T., A. Gavarró, J. De Lange & C. Caprin. 2008. Article omission across child languages. *Language Acquisition* 15(2). 89–119.
- Hahn, N., J. Snedeker & H. Rabagliati. 2015. Rapid linguistic ambiguity resolution in young children with autism spectrum disorder: Eye tracking evidence for the limits of weak central coherence. *Autism Research* 8(6). 717–726.
- Höhle, B., M. Schmitz, L. M. Santelmann & J. Weissenborn. 2006. The recognition of discontinuous verbal dependencies by German 19-month-olds: Evidence for lexical and structural influences on children's early processing capacities. *Language Learning and Development* 2(4). 277–300.
- Jaeger, Florian T. 2008. Categorical data analysis: Away from ANOVAs (transformation or not) and towards logit mixed models. *Journal of Memory and Language* 59(4). 434–446.
- Johnson, V. E., J. G. de Villiers & H. N. Seymour. 2005. Agreement without understanding? The case of third person singular /s/. *First Language* 25(3). 317–330.
- Johnston, R. J. & D. I. Slobin. 1979. The development of locative expressions in English, Italian, Serbo-Croatian and Turkish. *Journal of Child Language* 6. 529–545.
- Kedar, Y. 2007. *Functional categories in early language acquisition: The case of determiners*. Cornell University. (Unpublished Ph.D. Dissertation).
- Landau, B. & R. Jackendoff. 1993. What and where in spatial language and spatial cognition. *Behavioral and Brain Sciences* 16(2). 217–265.
- Leikin, M. 1998. Acquisition of locative prepositions in Russian. *Journal of Psycholinguistic Research* 27(1). 91–108.

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- Levinson, S. C. & D. P. Wilkins. 2006. *Grammars of space: Explorations in cognitive diversity*. Cambridge: Cambridge University Press.
- Levinson, S. R. 2003. *Space in language and cognition: Explorations in cognitive diversity*. Cambridge: Cambridge University Press.
- Lust, B. 2006. *Child language: Acquisition and growth*. Cambridge: Cambridge University Press.
- Maris, E. & R. Oostenveld. 2007. Nonparametric statistical testing of EEG-and MEG-data. *Journal of Neuroscience Methods* 164. 177–190.
- Matthews, D. E. & A. L. Theakston. 2006. Errors of omission in English-speaking children’s production of plurals and the past tense: The effects of frequency, phonology, and competition. *Cognitive Science* 30(6). 1027–1052.
- Miller, K. & C. Schmitt. 2012. Variable input and the acquisition of plural morphology. *Language Acquisition* 19(3). 223–261.
- Minor, S., N. Mitrofanova, G. Guajardo, M. Vos & G. Ramchand. 2023. Aspect processing across languages: A visual world eye-tracking study. *Frontiers in Language Sciences*.
- Minor, S., N. Mitrofanova & G. Ramchand. 2022. Fine-grained time course of verb aspect processing. *PLOS One* 17(2).
- Mitrofanova, Natalia. 2016. *Paths and places: Aspects of grammar and acquisition*. Tromsø: UiT Norges Arktiske Universitet.
- Mitrofanova, Natalia. 2017. Early underspecification of functional categories: Evidence from the acquisition of locative PPs in Russian. *Language Acquisition* 25(4). 341–365.
- Nicholas, K. E. 2011. *Children’s omission of prepositions in English and Icelandic*. University of Arizona. (Unpublished Ph.D. Dissertation).
- Oakes, L., H. Baumgartner, F. Barrett, I. Messenger & S. Luck. 2013. Developmental changes in visual short-term memory in infancy: Evidence from eye-tracking. *Frontiers in Psychology* 4. 697.
- Pederson, E., E. Danziger, D. Wilkins, S. Levinson, S. Kita & G. Senft. 1998. Semantic typology and spatial conceptualization. *Language* 74(3). 557–589. DOI: [10.2307/417793](https://doi.org/10.2307/417793).
- Phillips, C. 2010. Syntax at age two: Cross-linguistic differences. *Language Acquisition* 17(1-2). 70–120.
- Piaget, J. & B. Inhelder. 1956. *The child’s conception of space*. London: Routledge & Kegan Paul.
- Pinker, S. 1984. *Language learnability and language development*, vol. 7. Cambridge: Harvard University Press.
- Poepfel, D. & K. Wexler. 1993. The full competence hypothesis of clause structure in early German. *Language* 69(1). 1–33.

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- Quené, H. & H. v. d. Bergh. 2008. Examples of mixed-effects modeling with crossed random effects and with binomial data. *Journal of Memory and Language* 59(4). 413–425.
- Radford, A. 1990. *Syntactic theory and the acquisition of English syntax: The nature of early child grammars of English*. Oxford: Blackwell.
- Radford, Andrew & Eliane Ramos. 2001. Case, agreement and EPP: Evidence from an English-speaking child with SLI. *Essex Research Reports in Linguistics*.
- Santelmann, L. M. & P. W. Jusczyk. 1998. Sensitivity to discontinuous dependencies in language learners: Evidence for limitations in processing space. *Cognition* 69(2). 105–134.
- Sekerina, I. A. 2015. Online evidence for children's interpretation of personal pronouns. In L. Serratrice & S. Allen (eds.), *The acquisition of reference*, 213–239. Amsterdam: John Benjamins.
- Sekerina, I. A., K. Stromswold & A. Hestvik. 2004. How do adults and children process referentially ambiguous pronouns? *Journal of Child Language* 31(1). 123–152.
- Shady, M. E. 1996. *Infants' sensitivity to function morphemes*. The State University of New York: University at Buffalo. (Doctoral dissertation).
- Shi, R. 2014. Functional morphemes and early language acquisition. *Child Development Perspectives* 8(1). 6–11.
- Soderstrom, M. 2008. Early perception–late comprehension of grammar? The case of verbal -s: A response to de Villiers & Johnson (2007). *Journal of Child Language* 35(3). 671–676.
- Theakston, A. L. & E. V. M. Lieven. 2005. The acquisition of auxiliaries BE and HAVE: An elicitation study. *Journal of Child Language* 32(3). 587–616.
- Tomasello, M. 1987. Learning to use prepositions: A case study. *Journal of Child Language* 14(1). 79–98.
- Trueswell, J. 2008. Using eye movements as a developmental measure within psycholinguistics. In I. A. Sekerina, E. M. Fernández & H. Clahsen (eds.), *Language processing in children*, 73–96. Amsterdam: John Benjamins.
- Westergaard, Marit. 2009. *The acquisition of word order: Micro-cues, information structure, and economy* (Linguistik Aktuell/Linguistics Today 145). Amsterdam: John Benjamins.
- Westergaard, Marit. 2016. Word order and finiteness in acquisition: A study of English and Norwegian wh-questions. In K. M. Eide (ed.), *Finiteness matters. On finiteness-related phenomena in natural language*, 257–286. Amsterdam: John Benjamins.

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Yang, O., M. P. Bucci & Z. Kapoula. 2002. The latency of saccades, vergence, and combined eye movements in children and in adults. *Investigative Ophthalmology and Visual Science* 43. 2939–2949.

Yang, Wenchun, Angel Chan, Franklin Chang & Evan Kidd. 2020. Four-year-old Mandarin-speaking children’s online comprehension of relative clauses. *Cognition* 196. DOI: <https://doi.org/10.1016/j.cognition.2019.104103>.

Appendix

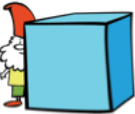
Table 1: Experimental and control conditions

Spatial contrasts	con-	Target	PP contrast	Figure contrast	con-	Ground contrast
in vs under		a mouse in a box	a mouse under a box	a kitten in a box		a mouse in a cupboard
		a napkin in a glass	a napkin under a glass	a spoon in a glass		a napkin in a plate
behind vs on		a teddy behind a stool	a teddy on a stool	a dog behind a stool		a teddy behind a case
		a doll behind a cube	a doll on a cube	a dwarf behind a cube		a doll behind a house
under vs in		a book under a toy car	a book in a toy car	a ball under a toy car		a book under a basket
		a bee under a flower	a bee in a flower	a mouse under a flower		a bee under a leaf
on vs behind		a spade on a sofa	a spade behind a sofa	a shoe on a sofa		a spade on a stool
		a cup on a table	a cup behind a table	an apple on a table		a cup on a plate

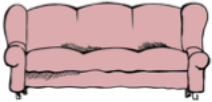
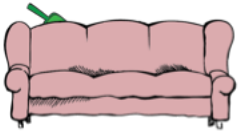
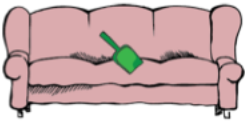
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TABLE B: EXPERIMENTAL PROCEDURE.

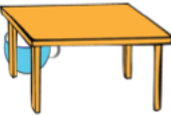
List A: Figure Control

1P			Myshonok. Korobka. Mouse. Box.
1E			Posmotri, gde zdes': Myshonok v bolshoj korobke. Look, where is here: The mouse is in a big box.
4P			Gnom. Kukla. Dwarf. Doll.
4C			Posmotri, kukla za bolshim kubikom. Kakaja eto kartinka? Look, the doll is behind a big cube. Which picture is that?
5P			Mashina. Knizhka. Car. Book.
5E			Posmotri, gde zdes': Knizhka pod igrushechnoj mashinoj Look, where is here: The book is under a toy truck.

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8P			Jabloko. Chashka. Apple. Cup.
8C			Posmotri, chashka na kukolnym stolike. Kakaja eto kartinka? Look, the cup is on a doll table. Which picture is that?
3P			Taburetko. Mishka. Stool. Teddy bear.
3E			Posmotri, gde zdes': Mishka za malenkej taburetkoj Look, where is here: The teddy bear is behind a small stool.
2P			Salfetka. Lozhka. Napkin. Spoon.
2C			Posmotri, salfetka v stekljannom stakane. Kakaja eto kartinka? Look, the napkin is in a glass cup. Which picture is that?
7P			Divan. Lopatka. Sofa. Spade.
7E			Posmotri, gde zdes': Lopatka na novom divane Look, where is here: The spade is on a new sofa.

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5P			Mashina. Korzina. Truck. Basket.
5C			Posmotri, gde zdes': Knizhka pod igrushechnoj mashinoj Look, where is here: The book is under a toy truck.
8P			Chashka. Stol. Cup. Table.
8E			Posmotri, chashka na kukolnym stolike. Kakaja eto kartinka? Look, the cup is on a doll table. Which picture is that?
3P			Taburetka. Jashchik. Stool. Case.
3C			Posmotri, gde zdes': Mishka za malenkej taburetkoj Look, where is here: The bear is behind a small stool.

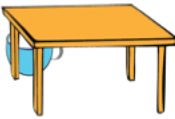
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6P			Pchela. Myshka. Bee. Mouse.
6C			Posmotri, pchela pod krasivym tsvetkom. Kakaja eto kartinka? Look, the bee is under a beautiful flower. Which picture is that?

Part2

1P			Myshonok. Kotenok. Mouse. Kitten.
1C			Posmotri, gde zdes': Myshonok v bolshoj korobke. Look, where is here: The mouse is in a big box.
4P			Kubik. Kukla. Cube. Doll.
4E			Posmotri, kukla za bolshim kubikom. Kakaja eto kartinka? Look, the doll is behind a big cube. Which picture is that?

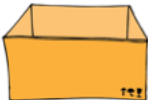
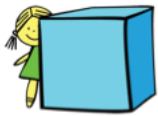
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5P			Mjachik. Knizhka. Ball. Book.
5C			Posmotri, gde zdes': Knizhka pod igrushečnoj mashinoj Look, where is here: The book is under a toy truck.
8P			Chashka. Stol. Cup. Table.
8E			Posmotri, chashka na kukolnym stolike. Kakaja eto kartinka? Look, the cup is on a doll table. Which picture is that?
3P			Sobachka. Mishka. Dog. Teddy bear.
3C			Posmotri, gde zdes': Mishka za malenkej taburetkoj Look, where is here: The bear is behind a small stool.

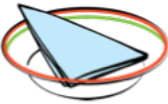
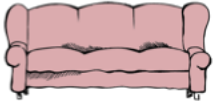
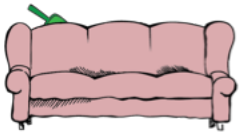
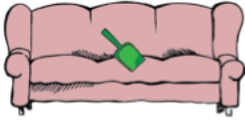
Natalia Mitrofanova, Serge Minor & Irina Sekerina

List B: Ground Control

PART 1

1P			Myshonok. Korobka. Mouse. Box.
1E			Posmotri, gde zdes': Myshonok v bolshoj korobke. Look, where is here: The mouse is in a big box.
4P			Kubik. Domik. Block. House.
4C			Posmotri, kukla za bolshim kubikom. Kakaja eto kartinka? Look, the doll is behind a big cube. Which picture is that?
5P			Mashina. Knizhka. Car. Book.
5E			Posmotri, gde zdes': Knizhka pod igrushechnoj mashinoj Look, where is here: The book is under a toy truck.

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8P			Stolik. Tarelka. Table. Plate.
8C			Posmotri, chashka na kukolnym stolike. Kakaja eto kartinka? Look, the cup is on a doll table. Which picture is that?
3P			Taburetko. Mishka. Stool. Teddy bear.
3E			Posmotri, gde zdes': Mishka za malenkej taburetkoj Look, where is here: The teddy bear is behind a small stool.
2P			Stakan. Tarelka. Glass. Plate.
2C			Posmotri, salfetka v stekljannom stakane. Kakaja eto kartinka? Look, the napkin is in a glass cup. Which picture is that?
7P			Divan. Lopatka. Sofa. Spade.
7E			Posmotri, gde zdes': Lopatka na novom divane Look, where is here: The spade is on a new sofa.

Chapter 17

Subordination in Swedish narratives of monolingual and bilingual children

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This paper investigates clausal subordination in oral narratives elicited from 72 monolingual Swedish and 102 bilingual Turkish/Swedish children age 4–7, keeping task and context constant. The cross-sectional analysis shows a clear increase in the frequency and variety of subordinate clauses with age for both groups, which suggests that children’s expressive Swedish language competence in the domain of subordination develops strongly beyond age 4;0, unlike earlier claims in the literature. Subordinators are very rarely omitted. The monolinguals produce higher proportions of subordinate clauses than their bilingual age-peers, and the difference between the groups is particularly evident for relative clauses. Implications for language assessment in clinical practice are discussed. The study adds knowledge in a domain where empirical work has so far concentrated on case studies of monolingual Swedish children age 2–3.

1 Introduction

Subordination, the use of clausal embedding, is considered a sign of syntactic complexity and advanced language development in children across languages. Subordinate clauses embedded in another clause (the matrix clause) either fill one of the grammatical roles of that clause (such as subject or object) or serve as modifiers of matrix-clause constituents (such as an adverbial clause modifying a verb phrase, or a relative clause modifying a noun phrase). The use of subordinate clauses thus allows a speaker to elaborate on referents and events within the same utterance and to linguistically express more complex events and relations.

Ute Bohnacker. 2026. Subordination in Swedish narratives of monolingual and bilingual children. In Merete Brendeford Anderssen, Anne Dahl, Terje Lohndal, Natalia Mitrofanova & Yulia Rodina (eds.), *Variation, acquisition and change*, 293–310. Berlin: Language Science Press. DOI: ??

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Subordinate clauses (SCls) are generally said to emerge between age 2 and 3 in the spontaneous speech of children (reports about the exact age vary somewhat across studies and languages), and typically-developing children are expected to be able to produce subordinate clauses by age 4. These expectations also hold for Swedish child language. Textbooks (e.g. [Nettelbladt & Salameh 2007](#):157, 182–185) and language assessment tools widely used in Swedish clinical practice (e.g. CELF, Gramba, GrUS) build on the assumption that if a child aged 4 to 5 is unable to produce well-formed SCls, in particular relative clauses, something is likely to be wrong, and the child may be language-impaired. The empirical foundation for this assumption is surprisingly weak though, as there are no large-scale reference or baseline data for the acquisition of subordination in Swedish. Save for a few longitudinal case studies of monolingual Swedish children age 1;6–3;11 ([Lundin 1987](#); [Waldmann 2008](#)), only one study has been published ([Håkansson & Hansson 2000](#)). In this small-scale elicitation study, 10 typically developing (TD) children (age 3;1–3;7) and 10 language-impaired (LI) children age (4;0–6;3) were administered a sentence repetition task and a picture-based sentence completion task with negated relative clauses (e.g. experimenter says: *Titta, här är flickan som sover och här är flickan...* ('Look, here is the girl that's sleeping and here is the girl ...'), expected completion by child: *...som inte sover* (...that's not sleeping')). As a group, the 10 TD children performed significantly better than the LI children, and omitted the relativiser/subordinator *som* ('that') in only 17% of all instances, whilst the LI children did so 77%.¹ This finding has given rise to the view that subordinator omissions and/or the lack of clausal subordination in Swedish beyond age 4;0 may be considered a clinical marker of language impairment.²

Yet we do not know how Swedish-speaking children beyond age 4;0 perform as regards subordination on more ecologically valid tasks, e.g. during narration or spontaneous speech (i.e. less contrived tasks than sentence repetition or completion of negated relative clauses). How frequent are SCls in the oral productions of children older than 4;0, what are the types and functions of these SCls, and what kind of development with age may we expect? There is simply a lack of research on the development of subordination in Swedish-speaking children in the later preschool and school-entry years (age 4–7).³ From studies of other lan-

¹When tested again 6 months later, both groups had improved (TD: 8% vs LI: 37% omissions).

²For other languages as well, it has been suggested that subordinator omissions, fragmentary relative clauses and/or the lack of clausal subordination altogether are a hallmark of language impairment, signalling a problem with marking or constructing hierarchical/dependency relations (e.g. [Novogrodsky & Friedmann 2006](#)).

³In Sweden, grade 1 of primary school starts at age 7. Most children attend preschool prior to

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guages, we know that children's development in this domain is protracted and not complete by 4;0 (e.g. [Berman & Slobin 1994](#); [Justice et al. 2006](#)).

Children growing up with two languages (simultaneous bilinguals or early L2 learners) have also been reported (anecdotally) to struggle with Swedish subordination constructions. The extent to which subordination is a challenge for such learners is unknown, though it certainly need not be because of any language impairment, but may rather be due to insufficient L2-input and/or cross-linguistic influence from the L1, where subordination may be linguistically encoded in other ways than in Swedish (e.g. not necessarily with overt subordinators).

This lack of research on the development of subordination motivated the current study of clausal subordination in the narrative production data of 174 children, 72 Swedish monolinguals (age 4–6) and 102 Turkish/Swedish bilinguals (age 4–7). With task and context kept constant, all children told two comparable picture-elicited narratives to a naïve listener in a Swedish-language setting. The study investigates the frequency of SCIs produced by these children, their morphological forms and syntactic and semantic functions, and the use (or omission) of lexical subordinators. How do age-matched monolingual and bilingual children compare to each other in controlled, relatively short narrative samples, and is there any development with age? The results will advance our knowledge of child language development in a way that is also useful to practitioners (speech-language pathologists, teachers), as the data may serve as a reference point or baseline for clausal subordination in Swedish-speaking children at late preschool and school-entry age.

2 Subordinate clauses in Swedish from an acquisition perspective

Swedish has a variety of subordinate constructions embedded in matrix (main) clauses, just like many other languages. In Swedish, these subordinating constructions prototypically take the form of a finite clause, introduced by a subordinator (complementiser, i.e. a subordinating conjunction or relativiser), and contain an overt subject and a tense-marked finite verb, as shown in (1).

- (1) a. Sen var det en fjäril [SCI som katten ville ta].
 then was there a butterfly that cat-the wanted catch
 (attributive, relative SCI)

‘Then there was a butterfly that the cat wanted to catch.’

that, often from age 1 or 2, and a prep year for school (grade 0).

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- b. Fjärilen flög iväg [_{SCI} när katten sprang efter den].
butterfly-the flew away when cat-the jumped after it
(adverbial, temporal SCI)
'The butterfly flew away when the cat jumped for it.'
- c. Katten tyckte [_{SCI} att fisken såg mumsig ut]. (complement,
cat-the thought that fish-the saw yummy out
object SCI)
'The cat thought that the fish looked yummy.'

Swedish SCIs are commonly classified according to syntactic function, semantic function, and form (Teleman et al. 1999:462–472). Syntactically, the subordinate clause can function as an attribute, adverbial, or complement (subject/object) in the matrix clause. Semantically, the subordinate clause relates to the matrix clause in a certain way (e.g. temporal, causal, conditional). Formally, Swedish subordinate clauses are introduced by different subordinators (e.g. *som* 'as/that', *att* 'that', *när* 'when', *medan(s)* 'while', *eftersom* 'because', *för att* 'because/in order to', *så att* 'so that'),⁴ and/or exhibit a particular word order.⁵ Finite SCIs as illustrated in (1) are very common in Swedish, and this also holds for informal speech (e.g. Jørgensen 1978; Waldmann 2008). Whilst nonfinite participial and infinitival subordinating constructions also exist, these are traditionally not considered clauses by Swedish grammarians and are comparatively rare.⁶

Swedish is also known for relative clauses being produced from a comparatively early age, already at around age 2–2;6 by some of the 5 children in the Stockholm corpus from the early 1970s (Lundin 1987), and by some of the 4 children in the Gothenburg corpus from the mid-1980s (Waldmann 2008). These studies document considerable variation in the frequency and types of SCIs in the

⁴In some contexts, the subordinator can be zero: *som* is optional in certain object relative clauses, and so is *att* in certain complement clauses.

⁵This particular SCI word order only shows up in a small fraction of SCIs, namely those containing a negation or certain adverbials (e.g. *inte* 'not', *alltid* 'always', *aldrig* 'never'), as in *Sen var det en fjäril [_{SCI} som katten inte ville ta]* (then was there a butterfly that cat-the not wanted catch). Here the finite verb follows both subject and negation/adverbial in the SCI (which is different from the verb-second word order of Swedish main clauses, where the verb precedes the negation). There exists a large body of theoretical and empirical research on this topic. However, as the current paper is not about verb placement in negated SCIs (which are comparatively rare anyway) but about the general development of SCIs in Swedish child language, word order will not be explored further here (cf. Waldmann 2008).

⁶In Turkish, the L1 of the bilingual participants of this study, the situation is different. Turkish is particularly rich in *nonfinite* subordinating constructions with a subordinating suffix on the verb (for details, see Bohnacker & Karakoç 2020). The repertoire of SCIs in Turkish is the same as in Swedish though, including relative, adverbial and complement clauses.

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spontaneous speech samples of individual children before age 3, variation that may signal individual differences in development, but may also be attributable to the disparate contextual settings the children were recorded in. Interestingly, by age 3;0, all these children produced full-fledged relative clauses (Lundin 1987:176; Waldmann 2008:189–192). It has been suggested that Swedish relative clause acquisition is facilitated by their simple and regular form, with an invariant subordinator/complementiser *som* ('that/as') initiating the finite relative clause. Another facilitating factor may be their high frequency in the input. Swedish relative clauses are often used in presentational contexts, for instance to introduce new characters when telling a story, as in (1a).⁷ Concerning SCLs other than relative clauses, the children in Lundin's and Waldmann's case studies were able to spontaneously produce a variety of SCLs by age 3;6 and hardly ever omitted subordinators any more.

The present paper investigates clausal subordination in a controlled study of narratives elicited from 174 Swedish-speaking children age 4–7. The following research questions are asked: Which SCLs do the children produce, how many, and for what function? Is there any development with age, and how do age-matched monolingual and bilingual children differ in this domain?

3 Method

3.1 Participants

174 children growing up in urban areas of eastern central Sweden participated in the study; 72 were monolingual Swedish (age 4;0–6;10) and 102 were Turkish/Swedish bilinguals (age 4;0–8;1). Informed parental consent was obtained in writing and families and children could terminate their participation at any time. Background information was collected via parental questionnaires. The data were collected as part of a larger child bilingualism research project at Uppsala University led by the author (BiLI-TAS),⁸ including an affiliated doctoral project (Lindgren 2018). None of the participating children had received a diagnosis of

⁷By contrast, Turkish relative clauses have no such freestanding relativiser but are nonfinite, morphologically complex participial constructions with oblique subject case-marking. They are much less frequent in conversation and storytelling than relative clauses in languages like Swedish. This could be because Turkish relatives are placed *before* the matrix noun, which makes them unlikely candidates for introducing new story characters or building narrative progression.

⁸BiLI-TAS stands for 'Bilingualism and Language Impairment, Turkish, Arabic, Swedish' (research funded by *Vetenskapsrådet* (421-2013-1309), and *Riksbankens Jubileumsfond* (P19-0644:1)).

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language impairment or any neuropsychiatric disorder. All children were able to speak Swedish well enough to follow the experimenter’s Swedish instructions and verbally respond in Swedish. An overview of the participants is given in Table 1, broken down by age group.

Table 1: Participants

	4 years	5 years	6 years	7 years	Total
Monolinguals	24	24	24	--	72
Mean age	4;5	5;6	6;6	--	5;5
Age range	4;0–4;10	5;0–5;11	6;0–6;10	--	4;0–6;10
Bilinguals	25	23	26	28	102
Mean age	4;5	5;5	6;5	7;5	6;0
Age range	4;0–4;11	5;1–5;11	6;1–6;11	7;0–8;1	4;0–8;1

The monolinguals grew up with only Swedish spoken in the home and they attended Swedish preschools. The bilinguals were mainly exposed to Turkish in their homes and to Swedish at (pre)school. 92% of the bilingual children had lived in Sweden from birth, whereas most of their parents were born in and had grown up in Turkey and moved to Sweden as adults. The majority of the bilinguals (67%) grew up in families with two Turkish-L1 parents who both spoke only or mostly Turkish to the child and to each other; the remaining children had parents who spoke Turkish to varying degrees to the child; only 8% had a parent who reported Swedish to be (one of) their L1(s). Thus, the bilinguals received less exposure to Swedish than the monolinguals. However, most bilingual children (85%) had started to hear Swedish regularly before age 3;0. The children attended Swedish (pre)school for a major part of the day, as is typical in Sweden.⁹ We lack information on (pre)school activities for the individual children, but book reading and storytelling activities are commonplace in Swedish (pre)schools, so all children should be familiar with such practices at least to some extent. The bilingual children came from a wide variety of socio-economic backgrounds regarding parental occupation and education (ranging from a few years of primary education to doctorates). The monolingual children came from somewhat more homogeneous, higher socio-economic backgrounds, as many parents had at least some tertiary education.¹⁰

⁹On average, children had started preschool at 23 months, with a few late starters (60–72 months) in the bilingual group due to recent relocation to Sweden.

¹⁰Space limitations preclude a detailed description of the participants’ backgrounds and recruit-

3.2 Materials, procedure and data treatment

Oral narratives were elicited using the Multilingual Assessment Instrument for Narratives (MAIN; Gagarina et al. 2019). MAIN consists of 4 carefully designed coloured fold-out picture sequences (Cat, Dog, Baby Birds, Baby Goats) and standardised protocols. The picture-based stories are parallel in terms of length and story structure. MAIN was administered in the ‘telling mode’ (story generation), where the child tells the story from the pictures to a naïve listener who cannot see the pictures, without any prior story model provided by the experimenter (Gagarina et al. 2019). Each child told two MAIN stories in Swedish: Cat or Dog, and Baby Birds or Baby Goats, as part of a larger test battery also involving vocabulary tasks.¹¹ The narratives are relatively short; it usually took a child 1–2 minutes to tell a story. The stories were video and audio recorded, transcribed orthographically from the audio using the CHAT-format by experienced native-speaker transcribers, carefully checked against the video in their entirety, and scored for narrative macrostructure. The present study focuses on the children’s use of clausal subordination in the Swedish MAIN stories. For other aspects of the children’s performance, see e.g. Bohnacker et al. (2022). Table 2 summarises the Swedish data, and Table 3 gives an overview of the amount of data per child (total number of words) in the two groups.

Table 2: Number of children, narratives and word tokens

	Children	Narratives	Word tokens (excl. repetitions/fillers)
	N	N	N
Monolingual Swedish	72	144	10,553
Bilingual Turkish/Swedish	102	203	16,705
Total	174	437	27,258

The transcribed narratives were manually searched for SCLs, and each item was classified according to syntactic function, semantic function, morphological form and the use (or omission) of lexical subordinators. SCLs were extracted together with their immediately preceding context and entered into Excel sheets by a linguistically trained Swedish-speaking assistant. The author, an experienced

ment here, but see Bohnacker et al. (2022).

¹¹On a separate occasion, the bilinguals told two parallel MAIN stories in Turkish. Subordination in these Turkish narratives is investigated in Bohnacker & Karakoç (2020).

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Table 3: Narrative length (total number of words per child for both narratives), means and ranges, by age group

	Monolinguals				Bilinguals				
Age group	4y	5y	6y	Total	4y	5y	6y	7y	Total
(N)	(24)	(24)	(24)	(72)	(25)	(23)	(26)	(28)	(102)
Mean	132.6	138.5	168.8	146.6	141.6	166.6	161.7	183.3	164.6
Range	74–	91–	114–	74–	60–	40–	98–	95–	40–
	240	234	238	240	375	319	293	333	375

linguist, checked all entries for accuracy and consistency of classification. Differences and unclear cases were resolved through discussion.

Prototypically, Swedish SCLs are introduced by a subordinator and contain a subject and a finite verb. However, as some of the children’s SCLs were fragmentary, unclear or nontarget in form, lenient inclusion criteria were applied to also capture clauses with omitted or incomplete subordinators, nonstandard morphological verb forms or word order. The data also contained some clauses for which SCL status could not be clearly determined, as their form and function was identical to that of a coordinated main clause; such instances were counted separately as ‘ambiguous’. Ambiguous clauses are introduced by a connective that is a borderline coordinating/subordinating conjunction (*så* ‘so (that)’, *för* ‘for/because’, *fast* ‘but/(even) though’) and also occur in adult Swedish (e.g. [Teleman et al. 1999:467](#); [Bohnacker 2006:452–453](#)). Ambiguous clauses were kept separate from the SCL counts to stay on the side of caution. Nonfinite subordinate infinitival constructions (without subject) were counted separately as ‘nonfinite’. Obvious instances of repetitions, breakoffs and restarts were only counted once.

4 Results

4.1 Overall frequencies of clausal subordination

An overview of the different types of finite, unclear and nonfinite subordinate constructions in the narratives is provided in Table 4.

The monolinguals produced 306 instances of subordinations in their 144 narratives, a large majority of which were finite SCLs (N=232). The bilinguals produced 285 subordinations in their 203 narratives; again the majority were finite SCLs (N=215). The monolingual data also contained 62 finite clauses that were

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Table 4: Different types of subordinating constructions, raw numbers

	Monolinguals				Bilinguals				
	4y	5y	6y	Total	4y	5y	6y	7y	Total
Age group (N children)	(24)	(24)	(24)	(72)	(25)	(23)	(26)	(28)	(102)
Finite SCLs	48	70	114	232	33	37	56	89	215
Finite ambiguous clauses (subordinate or coordinated)	11	15	36	62	5	7	3	12	27
Unclear subordinate fragments	0	0	0	0	8	5	2	0	15
Nonfinite subordinate constructions (without subject)	1	5	6	12	3	7	6	12	28
Total	60	90	156	306	49	54	67	113	285

ambiguous concerning their subordinate or coordinate clause status (see preceding section);¹² the bilingual data contained 27 such instances. In addition, the monolinguals produced 12 nonfinite subordinate constructions (infinitival, without a subject) and the bilinguals 28 such instances, as well as 15 unclear subordinate fragments. Both groups used predominantly finite SCLs for subordination in Swedish, and in what follows, only these finite SCLs will be studied further.

Table 5: Finite subordinate clauses, by age group, raw numbers, range, mean

	Monolinguals				Bilinguals				
	4y	5y	6y	Total	4y	5y	6y	7y	Total
Age group (N children)	(24)	(24)	(24)	(72)	(25)	(23)	(26)	(28)	(102)
Raw number	48	70	114	232	33	37	56	89	215
Range	0–5	0–11	0–14	0–14	0–9	0–7	0–6	0–11	0–11
Mean	2.0	2.92	4.75	3.22	1.32	1.61	2.15	3.18	2.11

As shown in Table 5, in the monolingual group, the 4-year-olds already produce substantial numbers of SCLs in their Swedish narratives, on average 2 per child. The increasing means for each age group suggest that SCLs become more frequent with age, having more than doubled from age 4 to 6. As indicated by the large ranges however, there is a lot of within-group variation, see Fig. 1. 21% of the monolingual 4-year-olds (5/24) do not yet produce any SCL in their narratives, but at age 5 and 6, only one and two children, respectively, do not yet produce any, and most children produce several SCLs.

¹²An ambiguous example would be *Musen va(r) så snäll, så den hjälpte den där killen* (mouse-the was so kind so it helped this there boy-the), produced by 6-year-old MoSwe615. This could either be a consecutive SCL introduced by *så* ('The mouse was so kind (so) that it helped that boy'), or two main clauses, where the second is introduced by connective *så* ('The mouse was so kind. So it helped that boy:').

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The bilinguals produce proportionally fewer SCLs in their narratives than the monolinguals their age, and even at age 7 when the bilinguals are one year older than the monolinguals. SCLs are not frequent in the narratives of the 4-year old bilinguals, but the increase in means suggests that SCLs become more frequent with age, especially so by age 7 (Table 5). Again, there is a lot of within-group variation, as shown by the large ranges and Fig. 1. 52% of the 4-year-old bilinguals (13/25) and 36% of the 5-year-olds (8/22) do not yet produce any SCLs in their Swedish narratives (the diamonds on the x-axis (zero)). Such children become fewer as age goes up.

In the scatterplots, some individuals appear to stick out, for instance, a few bilinguals who already at age 4 produce several SCLs in their narratives, in contrast to their age peers. It would be interesting to statistically investigate the relationships between SCL frequency, child age, vocabulary knowledge, and background factors (such as length/amount of exposure to Swedish, preschool attendance, SES, literacy activities), though this cannot be done within the confines of this short chapter.

4.2 Overt subordinators or not?

Omissions of subordinators are unattested in the monolingual group and extremely rare (below 3%) in the bilinguals, as shown in Table 6. Both monolinguals and bilinguals occasionally produce incorrect forms (such as *att* ‘that’ instead of *som* ‘as’, *vad* ‘what’ instead of phonologically related *var* ‘where’, *varför* ‘why’ instead of *(där)för att* ‘because’, or incomplete *att* instead of the two-word subordinator *(där)för att* ‘because’). Such erroneous forms are mainly found in the 4- and 5-year-olds, and are reminiscent of what has been reported for the spontaneous speech of younger monolinguals (Lundin 1987; Waldmann 2008).

Table 7 shows the types of subordinators used by the children to introduce SCLs. For the monolinguals, the variety of subordinators appears to increase with age, but three forms, the relativiser *som*, and the subordinating conjunctions *att* ‘that’ and *när* ‘when’, greatly predominate (72.8%). Some subordinators only occur once or twice in the entire data. By far the largest proportion of SCLs (48.3%) is introduced by the relativiser *som*. The Turkish/Swedish bilinguals also introduce a large majority of their SCLs (75.8%) with just three subordinators (*som*, *att*, *när*), but for them, the predominance of *som* (33.0%) is not as pronounced as for the monolinguals. The range of subordinators used by the bilinguals does not appear to get broader with age, unlike for the monolinguals, even though the sheer number of SCLs increases, having nearly tripled by age 7.

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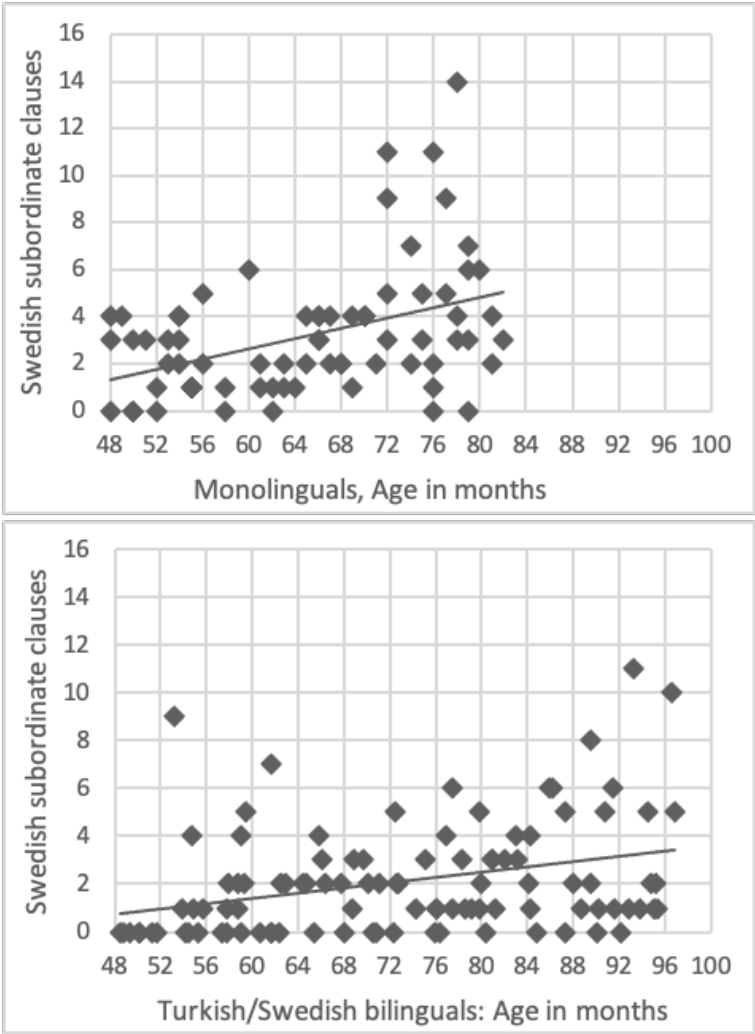


Figure 1: Subordinate clauses produced by individual children in their Swedish MAIN stories, raw figures (y-axis) by age in months (x-axis). A dot may represent more than one child.

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Table 6: Subordinator provision, by age group, raw numbers and proportions

	Monolinguals					Bilinguals					
	4y	5y	6y	Total	%	4y	5y	6y	7y	Total	%
Targetlike	45	68	111	224	96.5	25	28	53	85	191	88.8
<i>Overt/complete</i>	44	65	108	217	(93.5)	25	26	53	84	188	(87.4)
<i>Shortened</i>	1	2	0	3	(1.3)	0	0	0	0	0	(0.0)
<i>Zero</i>	0	1	3	4	(1.7)	0	2	0	1	3	(1.4)
Non-targetlike	3	2	3	8	3.5	8	9	3	4	24	11.2
<i>Incorrect/incomplete</i>	3	2	3	8	(3.5)	6	7	2	3	18	(8.4)
<i>Omitted</i>	0	0	0	0	(0.0)	2	2	1	1	6	(2.8)
Total	48	70	114	232	100	33	37	56	89	215	100

Table 7: Types of subordinators, by age group, raw figures and proportions

	Monolinguals					Bilinguals					
	4y	5y	6y	Total	%	4y	5y	6y	7y	Total	%
<i>som</i> (relativiser)	19	32	61	112	48.3	12	6	28	25	71	33.0
<i>när</i> ‘when’	0	13	15	28	21.5	2	7	9	38	56	26.0
<i>att</i> ‘that’	9	9	11	29	12.5	6	8	7	15	36	16.7
<i>medans</i> ‘while’	1	1	9	11	4.7	--	--	--	--	--	--
<i>hur</i> ‘how’	4	4	2	10	4.3	0	0	2	0	2	0.9
<i>så att</i> ‘so that’	7	2	0	9	3.9	0	0	1	2	3	1.4
<i>vad</i> ‘what’	3	5	0	8	3.4	1	0	4	2	7	3.3
<i>för att</i> ‘in order to’	1	1	4	6	2.6	1	5	0	5	11	5.1
<i>därför att</i> ‘because’	0	1	2	3	1.3	5	5	0	0	10	4.7
zero form	0	1	2	3	1.3	2	4	1	2	9	4.2
<i>var</i> ‘where’	0	1	2	3	0.9	0	0	1	0	1	0.5
<i>ända tills</i> ‘until’	0	0	2	2	0.9	--	--	--	--	--	--
<i>där</i> ‘where’ (relativiser)	1	0	1	2	0.9	1	0	0	0	1	0.5
<i>förutom att</i> ‘except that’	0	0	2	2	0.9	--	--	--	--	--	--
<i>fast</i> ‘although’	2	0	0	2	0.9	--	--	--	--	--	--
<i>utan att</i> ‘without’	0	0	1	1	0.4	0	0	1	0	1	0.5
<i>om</i> ‘whether/if’	1	0	0	1	0.4	0	1	1	0	2	0.9
<i>varför</i> ‘why’ (nontarget)	--	--	--	--	--	2	1	1	0	4	1.9
<i>efter (att)</i>	--	--	--	--	--	1	0	0	0	1	0.5
Total	48	70	114	232	100	33	37	56	89	215	100

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4.3 Syntactic and semantic functions

Table 8 breaks down the children’s SCLs by syntactic and semantic function. For the monolinguals, the majority are relative clauses (48.3%), followed by adverbial clauses (28.9%), which holds true for all age groups. Interestingly, this distribution is different in the Turkish/Swedish bilinguals though: They produce proportionally fewer relative clauses (34.0%) than adverbial clauses (39.5%).

Table 8: Functions of finite subordinate clauses, by age group, raw numbers and proportions

	Monolinguals					Bilinguals					
	4y	5y	6y	Total	%	4y	5y	6y	7y	Total	%
Relative clauses	20	30	62	112	48.3	12	6	28	27	73	34.0
Subject-subject	13	20	47	80	(34.5)	11	3	23	19	56	(26.0)
Object-subject	2	4	4	10	(4.3)	0	2	3	5	10	(4.7)
Other	5	6	11	22	(9.5)	1	1	2	2	7	(3.3)
Adverbial clauses	11	20	36	67	28.9	10	19	13	43	85	39.5
Temporal	1	14	26	41	(17.7)	3	7	9	38	57	(26.5)
Causal	1	2	6	9	(3.9)	4	11	1	5	21	(9.8)
Other Advl	9	4	10	17	(7.3)	3	1	3	0	7	(3.3)
Complement clauses	14	17	13	44	19.0	8	8	12	12	40	18.6
Other*	3	3	3	9	3.9	3	4	3	7	17	7.9
Total	48	70	114	232	100	33	37	56	89	215	100

The large majority of relative clauses are subject relatives, where the relativiser *som* functions as the subject. Most subject relatives serve as modifiers of the subject in the matrix (main) clause, as in (2a), as indicated in Table 8 by the numbers for ‘subject-subject’ (monolinguals: 71.4% (80/112); bilinguals: 76.7% (56/73)). A few subject relatives modify the object of the matrix clause (‘object-subject’), as illustrated in (2b). Object relatives, where the relativiser functions as the object of the relative clause, as in (2c), are infrequent; they are included in the numbers for ‘Other’ relatives. This distributional pattern concerning relative clauses holds for monolinguals and bilinguals alike, and aligns with reports in the child acquisition literature.

- (2) a. sen kom en pojke [SCL som ville fiska]. (subject-subject)
 then came a boy that wanted fish.INF
 ‘Then came a boy who wanted to fish.’ (BiTur6-09)
- b. och sen mamma nu kom och hjälpte den där lilla [SCL som var
 and then mum now came and helped the there little that was

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på vattnet]. (object-subject)
on water-the

‘And then the mother came and helped the little one that was in the water.’ (BiTur6-21)

- c. hunden såg korvarna [SCl **som han villde ha**]. (object-object)
dog-the saw sausages-the that he wanted have
‘The dog saw the sausages that he wanted to have.’ (BiTur7-17)

Complement clauses in both groups are mostly *att*-('that') clauses, as illustrated in (1c); the remainder are indirect interrogatives, mostly introduced by *hur* 'how' or *vad* 'what' (e.g. *jag vet inte vad den är* 'I don't know what this is').¹³

For both groups, the adverbial clauses are predominantly temporal (61% (41/57) for the monolinguals, 67% (57/85) for the bilinguals), which is not particularly surprising in the context of telling a story of events taking place simultaneously and sequentially. For examples see (3a–3b). Adverbial clauses also express causality (e.g. (3c)), though much less often than temporality. Other adverbial functions (e.g. consecutive, spatial, purpose) are infrequent and have been combined in Table 8 into the category of 'Other Advl'.¹⁴

- (3) a. sen [SCl **när han skulle gå**], han tappa(de) sin boll.
then when he should go he lost his ball
‘Then when he was about to go, he lost his ball.’ (BiTur6-07)
- b. och [SCl **medans han försökte ta loss den**, så tog hunden hans
and while he tried take off it so took dog-the his
korvar].
sausages
‘And while he was trying to get it off, the dog took his sausages.’
(MoSwe6-11)
- c. räven vill äta lammen [SCl **för att han kan inte hitta mat**].
fox-the wants eat lambs-the for that he can not find food
‘The fox want to eat the lambs because he can't find any food.’
(BiTur7-18)

¹³The category 'Other*' in Table 8 includes 13 attributive *att*-clauses (monolinguals: 3; bilinguals: 10), and 11 clauses that were clearly subordinate in form but whose syntactic function could not be determined, as they occurred stand-alone without a matrix (monolinguals: 4; bilinguals: 7).

¹⁴The 'Other Advl' category also includes a few mixed/unclear cases, where the precise semantic function of the adverbial clause could not be determined (e.g. consecutive/final; concessive/adversative).

5 Discussion and conclusion

This paper has investigated clausal subordination in a controlled sample of narratives elicited from 174 Swedish-speaking children age 4–7. I have reported how frequently the children produce SCLs, which type, for what function, and explored similarities and differences between age-matched monolinguals and Turkish/Swedish bilinguals. As the existing acquisition literature on subordination has focused on case studies of monolingual Swedish children age 2–3, the large-scale data presented here make an important empirical contribution for the preschool and school-entry years.

Keeping task and context constant, the cross-sectional data show a clear increase in the frequency and variety of SCLs with age, indicating that mastery of this domain is not complete by age 4;0, as suggested previously (e.g. [Håkansson & Hansson 2000](#): 330; [Nettelbladt & Salameh 2007](#): 157). Rather, clausal subordination continues to develop after age 4. Even though a development with age is seen for both monolinguals and bilinguals, the monolinguals as a group produce far higher proportions of SCLs than the bilingual children in their (L2) Swedish.

The majority of SCLs in the narrative data are relative clauses (introduced by *som*), followed by temporal adverbial clauses (introduced by *när*). This is similar to what has been reported for the spontaneous speech of younger Swedish-speaking children ([Lundin 1987](#); [Waldmann 2008](#)), even though such data age (2–3) tend to have higher proportions of causal adverbial clauses and complement clauses (introduced by *att* ‘that’) than the narrative data of the present study. This could be a task effect, as narratives favour temporal structuring.

When the monolinguals and bilinguals use SCLs, omissions of subordinators are extremely rare. This finding is line with claims in the literature that typically-developing Swedish-speaking children do not omit subordinators beyond age 3;7 ([Håkansson & Hansson 2000](#), recall Section 1). Our data show this for the first time on a large scale.

Narratives are generally regarded as an ecologically valid and rich source of linguistic data when studying the expressive language competence of children at the discourse level ([Justice et al. 2006](#)), and therefore recommended to be used for assessment in clinical practice. So how do the narratives elicited with the MAIN materials fare with regard to subordination? Nearly all monolingual 5- and 6-year-old children spontaneously produced at least one subordinate clause, and often several. Yet 21% of the monolingual 4-year-olds and 52% of the bilingual 4-year-olds did *not* produce any SCLs, and nor did substantial portions of the 5- and 6-year-old bilinguals. These findings have direct consequences for practitioners: Firstly, we cannot expect a typically-developing Swedish-speaking 4-year-old

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to produce SCIs in a short narrative sample, though we might expect this for monolinguals by age 5. Secondly, baseline data on SCI production in monolingual children should not be used as a yardstick for bilingual children.

The Turkish/Swedish children age 4–5 produced remarkably few relative clauses when compared to their monolingual Swedish peers. Relative clauses did increase in frequency in the bilinguals' narratives by age 6–7, but even at that age they were still much less common than in the monolinguals. Again, this finding has repercussions for language assessment practice. Widely used assessment tests and textbooks build on the assumption that typically-developing children age 4 can be expected to produce well-formed SCIs clauses, especially relative clauses introduced by the relativiser *som*. We suggest that this assumption needs to be revised. Most of the 4- and 5-year-olds bilinguals, did not produce any relative clauses in their narrative speech samples. This does not mean that they develop atypically though. It is likely that the Turkish-Swedish bilinguals lag behind since they have had less exposure to Swedish. But it might also be the case that they underuse Swedish relative clauses for some time under the influence of Turkish, where relative clauses are very infrequent (see [Bohnacker & Karakoç 2020](#)).

Owing to space constraints, the findings are reported in relatively condensed form here, with descriptive statistics only. In future work, they will be complemented by more detailed analyses and inferential statistics, to shed light on what the contributing factors might be behind the distributional and developmental patterns observed in the present study.

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References

- Berman, Ruth A. & Dan Isaac Slobin. 1994. *Relating events in narrative: A cross-linguistic developmental study*. New York: Psychology Press.

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- Bohnacker, Ute. 2006. When Swedes begin to learn German: From V2 to V2. *Second Language Research* 22(4). 443–486.
- Bohnacker, Ute & Birsal Karakoç. 2020. Subordination in children acquiring Turkish as a heritage language in Sweden. In Fatih Bayram (ed.), *Studies in Turkish as a heritage language*, 155–204. Amsterdam: John Benjamins.
- Bohnacker, Ute, Josefin Lindgren & Buket Öztekin. 2022. Storytelling in bilingual Turkish-Swedish children: Effects of language, age and exposure on narrative macrostructure. *Linguistic Approaches to Bilingualism* 12(4). 413–445.
- Gagarina, Natalia, Daleen Klop, Sari Kunnari, Koula Tantele, Taina Välimaa, Ute Bohnacker & Joel Walters. 2019. *MAIN: Multilingual assessment instrument for narratives. Revised version* (ZAS Papers in Linguistics 63).
- Håkansson, Gisela & Kristina Hansson. 2000. Comprehension and production of relative clauses: A comparison between Swedish impaired and unimpaired children. *Journal of Child Language* 27. 313–333.
- Jørgensen, Nils. 1978. *Underordnade satser och fraser i talad svenska: Funktion och byggnad*. [Subordinate clauses and phrases in spoken Swedish: Function and structure]. Lund: Ekstrands.
- Justice, Laura, Ryan Bowles, Joan Kaderavek, Teresa Ukrainetz, Sarita Eisenberg & Ronald Gillam. 2006. The index of narrative microstructure: A clinical tool for analyzing school-age children's narrative performances. *American Journal of Speech-Language Pathology* 15. 177–191.
- Lindgren, Josefin. 2018. *Developing narrative competence: Swedish, Swedish-German and Swedish-Turkish children aged 4–6*. Uppsala University. (Doctoral dissertation).
- Lundin, Barbro. 1987. *Bisatser i små barns språk. En analys av fem barns första bisatser*. [Subordinate clauses in early child language. An analysis of the first sub clauses of five children]. Lund University. (Doctoral dissertation).
- Nettelbladt, Ulrika & Eva-Kristina Salameh. 2007. *Språkutveckling och språkstörning. Del I. Fonologi, grammatik och lexikon*. [Language development and language disorder. Part 1: Phonology, grammar and lexicon]. Lund: Studentlitteratur.
- Novogrodsky, Rama & Naama Friedmann. 2006. The production of relative clauses in syntactic SLI: A window to the nature of the impairment. *Advances in Speech-Language Pathology* 8(4). 364–375.
- Teleman, Ulf, Staffan Hellberg & Erik Andersson. 1999. *Svenska akademien grammatik. Vol. 4: Satser och meningar*. [The Swedish academy grammar: Clauses and sentences]. Stockholm: NorstedtsOrdbok.

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Waldmann, Christian. 2008. *Input och output: Ordföljd i svenska barns huvudsatser och bisatser*. [*Input and output: Word order in Swedish children's main clauses and subordinate clauses*]. Lund University. (Doctoral dissertation).

Chapter 18

Is a center a unit or a group of individuals? Agreement of Norwegian predicate adjectives

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In this article, we explore the polysemous Norwegian word *senter* ‘center’ and similar collective nouns, focusing on their agreement properties. Based on corpus data, we show that the relevant collective nouns can combine with predicate adjectives in both the singular and the plural. We relate the distribution to the “agency hypothesis” (Nessel et al. 2022), whereby plural agreement is favored when predicate adjectives imply carrying out an action, and the “individuation hypothesis” (Nessel & Janda 2023), according to which plural agreement is favored when a collective noun is conceptualized as a group of individuals. Our findings support Westergaard’s (2014) idea about “micro-cues”, whereby children are sensitive to fine syntactic distinctions from early on.

1 Introduction

Agreement has received considerable attention in theoretical linguistics in recent years (e.g., Mitrofanova et al. 2021, Klassen et al. 2022, and Kupisch et al. 2022). In this article, we explore a well-known but still somewhat understudied agreement construction. In Norwegian, if the subject is a collective noun, a predicate adjective may occur in the singular or the plural. We explore three collective nouns: *senter* ‘center’, *styre* ‘board (of directors)’, and *ledelse* ‘leadership’. Consider the following examples, where the predicate adjective is in the singular in the (a)

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examples, while the plural is used in the (b) examples:¹

- (1) a. Hamsunsenterets Bodil Børset pekte på at **senteret** er helt **avhengig**_{SG} av infrastruktur og av samarbeid med lokalt næringsliv for å kunne lykkes.
‘Bodil Børset of the Hamsun Center pointed out that the center is completely dependent on infrastructure and collaboration with local businesses.’
b. **Senteret** var helt **avhengige**_{PL} av å få igjennom dette.
‘The center was completely dependent on getting this proposal passed.’
- (2) a. **Styret** er **enig**_{SG} i dette og vil bestemt understreke behovet for fortsatt omstilling og effektivisering ved sykehuset.
‘The board of directors agrees with this and emphasizes the need for continued restructuring and effectivization at the hospital.’
b. **Styret** er **enige**_{PL} i dette, og oppdaterer de to foregående referatene etter dette prinsippet.
‘The board of directors agrees with this and will update the two preceding minutes according to this principle.’
- (3) a. **Ledelsen** er **aktiv**_{SG} i pedagogiske spørsmål.
‘The leadership is active in pedagogical questions.’
b. Det oppfattes positivt at **ledelsen** er **aktive**_{PL} i profilering av selskapet.
‘It is considered positive that the leadership is active in profiling the company.’

While the rivalry between singular and plural predicate adjectives is well known (Western 1921, Knudsen 1973, Åfarli & Vangsnes 2021), recent empirical studies of corpus data are few (but see Nessel et al. 2022 and Nessel & Janda 2023). The conditions that motivate the choice between singular and plural agreement are not fully understood. The aim of the present article is to contribute to filling this knowledge gap through a corpus study of the three collective nouns in (1)–(3).

¹Throughout the article, numbered examples are from the NoWaC corpus, described in Section 2. For the convenience of the reader, in each numbered example we boldface the subject and the predicate adjective. Adjectives in the singular are marked with subscript _{SG}, while subscript _{PL} indicates that the adjective is in the plural.

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Our contribution can be summarized as follows. First, we show that for the collective nouns under scrutiny plural agreement occurs in about 10% of the examples. This is a lower proportion of plural agreement than in comparable collective nouns analyzed in earlier studies. Second, while factors such as copula verb lemma and noun definiteness do not seem to affect the choice between singular and plural agreement, there is weak evidence that participles are less likely than adjectives to combine with predicate adjectives in the plural, and that high frequency promotes plural agreement. Third, we show that there is considerable variation among predicate adjectives with regard to their preferences for singular or plural agreement, a fact that concurs with Westergaard's (2014) observation that research should focus on micro-cues rather than assuming that language acquisition proceeds exclusively from general to specific. Finally, our findings lend support to the agency and individuation hypotheses, according to which plural agreement is more likely if the subject carries out an action and is conceptualized as a group of individuals rather than a unified whole (Nesset et al. 2022).

In Section 2 we present our data, before we carry out a comparison with earlier studies in Section 3. An analysis of factors influencing the choice between singular and plural agreement ("micro-cues") is carried out in Section 4, while the agency and individuation hypotheses are discussed in Section 5. Section 6 summarizes our findings.

2 Data

We extracted all relevant examples of the three collective nouns, *styre*, *senter*, and *ledelse* from the NoWaC corpus, an internet corpus containing more than 700 million Norwegian words (bokmål).² Our corpus searches were carried out as follows. We searched for the relevant nouns immediately followed by a copula verb (*være* 'be' or *bli* 'become'), which in turn was immediately followed by an adjective in the singular or plural. Searches were performed so as to return both indefinite and definite forms of the relevant nouns (e.g., indefinite *senter* and definite *senteret*) and different inflected forms of the copula verb (e.g., present tense *er* and past tense *var* of *være*). Searches were carried out in December 2023.

The database was manually checked, and irrelevant examples were removed. In some examples, an adverb was incorrectly classified as an adjective in the corpus (e.g., *svært aktiv* 'very active', where the corpus incorrectly identified the

²For information about the NoWaC corpus, see <https://www.hf.uio.no/iln/om/organisasjon/tekstlab/prosjekter/nowac/>.

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adverb *svært* as a neuter singular form of the adjective *svær* ‘big’). Such examples were manually annotated for the correct adjective and included in the database. Some Norwegian adjectives have homophonous singular and plural forms (e.g., *bra* ‘good’). Since we are interested in the factors conditioning the choice between singular and plural adjectives, adjectives like *bra* were excluded from the analysis. After the removal of irrelevant examples, the database consisted of 810 examples that were subjected to further analysis.³

The dataset was manually annotated for the following features:

- (4) a. Noun lemma: *senter* ‘center’, *styre* ‘board (of directors)’, *ledelse* ‘leadership’
- b. Copula verb lemma: *bli* ‘become’, *være* ‘be’
- c. Noun definiteness (morphological): indefinite vs. definite forms of the subject noun
- d. Adjective frequency: the frequency of the adjective lemma in the corpus
- e. Adjective/participle: the status of the agreement target as an adjective or a participle
- f. Adjective/participle lemma: 214 different adjectives/participles
- g. Type of indefinite: four subtypes of subjects with morphologically indefinite nouns

The distinction between adjectives and participles is notoriously difficult to draw in Norwegian (Dyvik 2019). Participles undergo grammaticalization, whereby their connection to a verb becomes less obvious for the language users. Testimony to this grammaticalization process is the negative derivational prefix *u-* ‘un-’, which frequently attaches to adjectives, as shown in examples like *usikker* ‘unsure’ from *sikker* ‘sure’. However, we also find this prefix in items like *ubetalt* ‘unpaid’ from *betalt* ‘paid’. Historically, at least, *betalt* is a participle of the verb *betale* ‘pay’, but the fact that there is no verb “*ubetale*” ‘unpay’ suggests that language users treat *betalt* not as a participle, but as an adjective, to which the adjectival prefix can be added. For the purposes of the present study, we applied a simple and well-known criterion (Lie 2003: 77): If it is possible to place the adverb *svært* ‘very’ immediately before the word in question, the word was classified as an adjective. The same criterion was used by Nessel et al. (2022: 131) and Nessel & Janda (2023). This facilitates comparison of the results from these studies to the results reported in the present study.

³Our database and statistical code are available at <https://doi.org/10.18710/1JQ1QJ>.

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A feature that has been shown to influence the rivalry between singular and plural in Norwegian predicate adjectives is the distance between the agreement controller (the subject noun) and the agreement target (the predicate adjective/participle). [Nesset et al. \(2022\)](#) found that the larger the distance, the more likely is a plural form of the agreement target. However, distance was not included as a factor in the present study, because the NoWaC corpus is not syntactically annotated. In corpora without syntactic annotation, searches for examples with a long distance between controller and target are not feasible, since they return more irrelevant than relevant examples.

3 Overall distribution – Comparison with earlier studies

Table 1 and Figure 1 compare the distribution of singular and plural for the nouns under scrutiny in the present study to data from the same corpus discussed by [Nesset et al. \(2022: 129\)](#), who analyzed a different set of nouns.⁴ The table provides raw numbers (in the columns with #) and the proportion of plural agreement (in the rightmost column with %). The nouns are given in ranked order, with the nouns with the highest percentage of plural on the top. In the figure, the nouns are ranked from left to right.

Two conclusions can be drawn based on Table 1 and Figure 1. First, we see that there is considerable variation among nouns. The most highly ranked noun, the soccer team Rosenborg, has 85% of examples in the plural, while the noun at the other end of the scale, *regjering* ‘government’, only has 9% of examples with plural agreement. Second, the three nouns under scrutiny in the present study cluster at the low end of the spectrum, with 10–12% of the examples in the plural.

4 Micro-cues: What factors influence the rivalry between singular and plural?

In order to shed light on the factors that influence the choice between singular and plural of adjectives, we carried out a statistical analysis. As shown in the previous section, plural agreement is relatively rare for the collective nouns under scrutiny in the present study. Of the 810 examples in our dataset, 717 have a singular form of the predicate adjective, which represents 89% of our data.

We commenced by checking the distribution of singular vs. plural agreement across the values of each of the possible factors listed in (4) above. Table 2 shows

⁴[Nesset et al. \(2022\)](#) also discuss data from other sources, but they are not included here, since they are not directly comparable to the data gathered for the purposes of the present study.

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Table 1: The sg-pl rivalry for predicate adjectives with collective nouns in the present study (in italics) and Nessel et al. 2022 (all other nouns)

	#SG	#PL	%PL
Rosenborg (name of soccer team)	5	28	85%
RBK (name of soccer team)	8	36	82%
Telenor (name of company)	13	25	66%
flertall ‘majority’	54	56	51%
politi ‘police’	136	126	48%
publikum ‘audience’	105	89	46%
Arbeiderpartiet (name of political party)	35	25	42%
ungdom ‘youth’	103	69	40%
høyreside ‘right wing’	12	7	37%
parti ‘party’	22	11	33%
EU ‘European Union’	94	25	21%
familie ‘family’	209	41	16%
forsvar ‘defense’	95	17	15%
kommune ‘municipality’	306	48	14%
<i>styre</i> ‘board of directors’	433	60	12%
<i>ledelse</i> ‘leadership’	168	20	11%
<i>senter</i> ‘center’	116	13	10%
regjering ‘government’	391	37	9%

the data pertaining to the three nouns included in this study. A chi-square test shows that Noun lemma is not a significant factor.

Table 2: Distribution of singular vs. plural of predicate adjectives with three collective nouns

Noun lemma	#SG	#PL	%PL
<i>styre</i> ‘board of directors’	433	60	12%
<i>ledelse</i> ‘leadership’	168	20	11%
<i>senter</i> ‘center’	116	13	10%

In Table 3, it is shown that the proportion of plural agreement is slightly higher for *bli* than for *være*. However, a chi-square test indicates that Copula verb lemma is not a significant factor.

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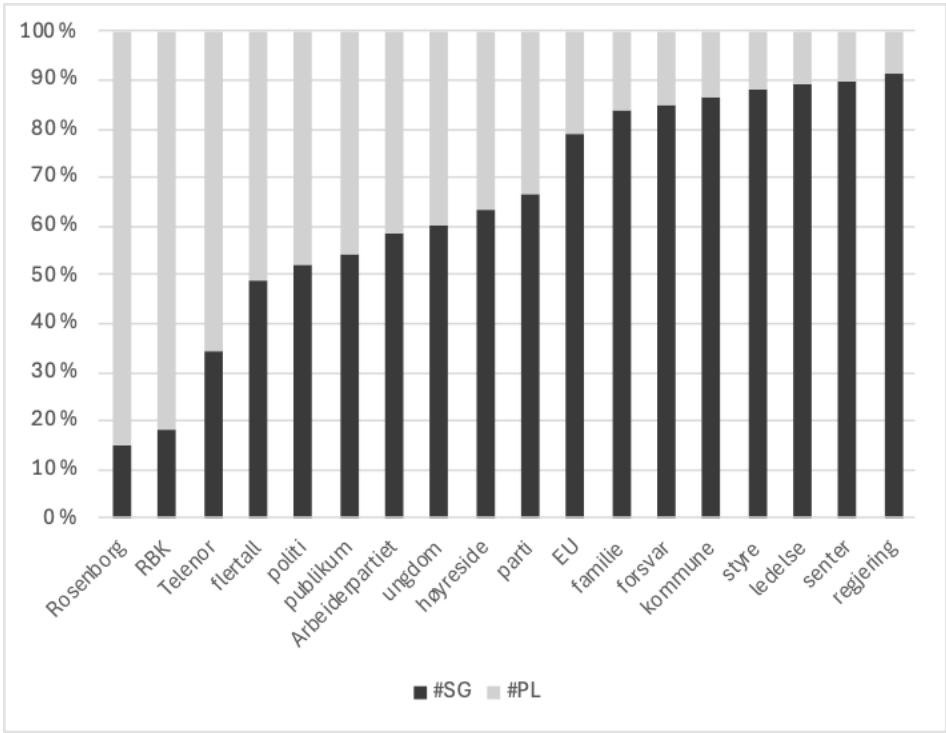


Figure 1: The sg-pl rivalry for predicate adjectives with collective nouns in the present study and [Nesset et al. 2022](#)

Table 3: Distribution of singular vs. plural of predicate adjectives across the values of copula verb lemma

Copula verb lemma	#SG	#PL	%PL
<i>være</i> ‘be’	651	80	11%
<i>bli</i> ‘become’	66	13	16%

Table 4 shows the data pertaining to morphological definiteness of the target nouns. Nouns with the definite markers *-en* (*ledelsen* ‘the leadership’) or *-et* (*senteret* ‘the center’ and *styret* ‘the board of directors’) were annotated as definite, while nouns without these markers were classified as indefinite. As shown, the percentage of plural agreement is slightly higher for indefinite nouns.

Table 4: Distribution of singular vs. plural of predicate adjectives across the values of morphological noun definiteness

Morphological definiteness	#SG	#PL	%PL
definite	606	70	10%
indefinite	111	23	17%

The examples with morphologically indefinite nouns are of four different types. First, we have examples where the indefinite noun is part of a name:

- (5) **Antirasistisk Senter** er **glade**_{PL} for at man har et høyt ambisjonsnivå.
‘The Antiracist Center is glad that they are highly ambitious.’

We also have examples where a possessive phrase precedes the indefinite noun:

- (6) Real Madrids sportslige **ledelse** var **enig**_{SG} om å vrake David Beckham.
‘Real Madrid’s athletic managers agreed to get rid of David Beckham.’

In a few cases, the morphologically indefinite noun is preceded by the definiteness marker *den* ‘that’:

- (7) Både den administrative og politiske **ledelse** er **fullpakket**_{SG} av tidligere sulisværing.
‘Both the administrative and the political leadership are packed with people from Sulithjelma.’

The remainder of the examples are what we may call “true indefinites”:

- (8) Et slikt **senter** er veldig **aktuelt**_{SG} både for helsevesenet og politikere.
‘Such a center is very relevant for both the healthcare system and for politicians.’

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Table 5 shows the distribution of singular and plural agreement for the four categories of morphologically indefinite nouns. For the possessive type, the percentage of plural is close to the general percentage of plural for definite nouns in Table 4, suggesting that this type of examples behave like definite nouns. For names and *den* + indefinite nouns, our numbers are too small to draw strong conclusions. However, it is possible that names may increase the likelihood of plural agreement. We notice that there are several proper names among the nouns that are most likely to take plural agreement in Table 1 in Section 1. In our data, we have no examples of “true indefinites” with plural agreement, so it is possible that this type disfavors plural agreement. This is an interesting topic for future research.

Table 5: Distribution of singular and plural of predicate adjectives with four types of indefinite nouns

	#SG	#PL	%PL
Name	6	12	67%
Possessive + indefinite noun	75	11	13%
<i>Den</i> + indefinite noun	4	0	0%
True indefinite	26	0	0%

The corpus frequency of the predicates varied from 1 (for hapaxes) to 1,939,662 for *god* ‘good’ with a mean of 82,521. We logarithmically transformed this variable, yielding a minimum value of 0, mean of 9.14, and maximum of 14.5. A logistic regression model of frequency as a fixed predictor and lemma as a random predictor shows that frequency has a significant but weak effect, with higher frequency associated with more use of plural.

Table 6 presents the data pertaining to the status of the predicate as an adjective as opposed to a participle. A chi-square test shows that this relationship is significant at the $p < 0.05$ level, but that the effect size (Cramer’s $V = 0.08$) is very small. There is therefore weak evidence that participles are even less likely to appear with plurals than adjectives. This is in harmony with the findings of [Nesset et al. \(2022\)](#). However, our result must be approached with caution given the uncertainty in drawing a distinction between adjectives and participles mentioned above.

Table 7 displays all the predicate lemmas that appear 5 or more times in our database, along with the percentage of plural forms. As shown, there is considerable variation, since the proportion of plural agreement ranges from 60% for *kritisk* ‘critical’ to 0% for a number of adjectives.

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Table 6: Distribution of singular vs. plural of predicate adjectives across the values of adjective/participle

Adjective/participle	#SG	#PL	%PL
adjective	640	90	12%
participle	77	3	4%

Table 7: All predicate lemmas attested ≥ 5 times in our dataset, arranged according to the percentage of plural forms

Predicate lemma	number of examples	%PL
<i>kritisk</i> ‘critical’	5	60%
<i>glad</i> ‘happy’	9	56%
<i>positiv</i> ‘positive’	19	42%
<i>enig</i> ‘in agreement’	63	41%
<i>aktiv</i> ‘active’	5	20%
<i>inkompetent</i> ‘incompetent’	5	20%
<i>svak</i> ‘weak’	5	20%
<i>tydelig</i> ‘clear’	5	20%
<i>bevisst</i> ‘conscious’	7	14%
<i>ferdig</i> ‘finished’	9	11%
<i>fornøyd</i> ‘satisfied’	19	11%
<i>åpen</i> ‘open’	27	11%
<i>klar</i> ‘ready’	40	10%
<i>avhengig</i> ‘dependent’	12	8%
<i>ansvarlig</i> ‘responsible’	66	6%
<i>opptatt</i> ‘busy’	38	3%
<i>vedtaksfør</i> ‘in quorum’	32	3%
<i>beslutningsdyktig</i> ‘in quorum’	99	1%
<i>god</i> ‘good’	5	0%
<i>sammensatt</i> ‘composed’	8	0%
<i>takknemlig</i> ‘grateful’	5	0%
<i>tilfreds</i> ‘satisfied’	19	0%
<i>valgt</i> ‘elected’	11	0%
<i>viktig</i> ‘important’	7	0%
<i>villig</i> ‘willing’	9	0%
<i>åpnet</i> ‘opened’	5	0%

18 Is a center a unit or a group of individuals?

The picture that emerges from this and the previous section is a system with considerable variation. The choice of singular or plural agreement depends on the preferences of individual nouns and adjectives, as well as factors such as the distinction between adjectives and participles and frequency. It stands to reason that in order for children to acquire such a linguistic system, they must be sensitive to a number of factors that influence the choice between singular and plural agreement – what Westergaard (2014) refers to as “micro-cues”.

5 The agency and individuation hypotheses

Nesset et al. (2022: 139) proposed the agency hypothesis, whereby sentences where the subject represents an agent that may carry out an action are more likely to display plural agreement. They found support for this hypothesis in an experiment with the adjectives *klar* ‘ready’ (which implies the readiness to undertake an action) and *stor* ‘big’ (which does not carry such an implication). Plural was significantly more common with *klar*, as predicted by the agency hypothesis.

Is there any evidence for the agency hypothesis in the data under scrutiny in the present article? Admittedly, it is difficult to operationalize the notion of agency in the wide sense it is used by Nesset et al. (2022). However, consider the following examples with three adjectives that have a relatively high proportion of plural agreement, as shown in Table 7 above. Arguably, these examples involve some level of agency that may promote plural agreement.

- (9) Styret ble **enig**_{PL} om å lete etter et navn som er lettere å kommunisere.
‘The board of directors agreed that they should look for a name that is easier to communicate.’
- (10) Dette er en grov forskjellsbehandling som **Antirasistisk Senter** er sterkt **kritiske**_{PL} til.
‘This is an example of gross discrimination, to which the Antiracist Center is highly critical.’
- (11) Styret er **positive**_{PL} til Boreas inntreden som en betydelig ny aksjonær i selskapet.
‘The board of directors is positive to Boreas entering the company as an important shareholder.’

In (9), the board of directors has come to an agreement (which involves negotiations, i.e., actions) that in turn will give rise to the renaming of the company

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(another action). Example (10) involves issuing a critical statement (an action), and in (11) the board of directors issues a positive statement about the new owners of the company. Although our dataset is too small to facilitate a systematic investigation, sentences like (9) – (11) are at least compatible with the agency hypothesis.

In a study of Norwegian soccer teams as subjects of predicate adjectives, Nessel & Janda (2023: 296–298) argue that individuation has an impact on agreement. According to the individuation hypothesis, plural is more likely when the collective noun is conceptualized as a group of individuals, while singular is the expected option when the collective noun is construed as a single whole. Consider the following examples featuring the soccer clubs Viking and Brann (from Nessel & Janda 2023: 297):

- (12) Viking er **tafatte**_{PL} og gir stadig ballen fra seg til langt mer aggressive Rosenborg-spillere.
‘Viking are helpless and are always letting the ball go to the much more aggressive Rosenborg players.’
- (13) Brann er **avhengig**_{SG} av sponsorer for å ha sjansen til å levere sportslige resultater.
‘Brann is dependent on sponsors to have a chance to deliver competitive results.’

Example (12) portrays the players of the soccer team Viking as individuals and hence uses an adjective in the plural, while singular agreement occurs in (8), where the soccer club Brann is described as a unified organization, not a group of individuals.

Examples from the present study lend some support to the individuation hypothesis. For instance, sentence (9) above describes a process whereby the individual members of the board of directors reach an agreement. However, examples like (10) and (11) show that plural agreement is also possible when a center or a board of directors acts as a unified whole.

It is worth pointing out that agency and individuation do not always pull in the same direction; while in (10) and (11) the collective nouns show a considerable degree of agency, they are low on individuation. Examples of this type represent an interesting topic for future research, as does the interplay of agency, individuation and other factors, such as the distance between subject and predicate adjective, which was not explored in the present study.

6 Concluding remarks

In this empirical study of corpus data, we have shown that the Norwegian collective nouns *senter* ‘center’, *styre* ‘board (of directors)’, and *ledelse* ‘leadership’ are compatible with singular and plural agreement, although singular is by far the most frequent option. This sets the nouns under scrutiny apart from collective nouns analyzed in previous studies, which show a higher proportion of plural agreement. We have demonstrated that factors such as frequency and the distinction between adjectives and participles may affect the choice between singular and plural agreement, and that individual adjectives display different preferences for agreement. Our data furthermore indicate that agency and individuation may be relevant for agreement.

The picture that emerges is a complex system with considerable variation where a number of factors show a complicated interplay. For such a system to be acquired, children need to be sensitive to “micro-cues”, as proposed by Westergaard (2014).

References

- Åfarli, Tor A. & Øystein A. Vangsnes. 2021. Formell og semantisk adjektivkongruens i norsk. In K. Hagen, A. Hjelde, K. Stjernholm & Ø.A. Vangsnes (eds.), *Bauta: Janne bondi johannessen in memoriam*, vol. 11, 527–540. Oslo: University of Oslo.
- Dyvik, Helge J.J. 2019. *Er vi fokusert eller fokuserte? Partisippenes indre kamp i bokmål*. NORGRAMTALL: Data om norsk fra NorGramBank. <https://nogramtall.w.uib.no/2019/02/19/er-vi-fokusert-eller-fokuserte-partisippenes-indre-kamp-i-bokmal/>.
- Klassen, Rachel, Nadine Kolb, Holger Hopp & Marit Westergaard. 2022. The interaction between syntactic and lexical gender congruency in L1 Spanish-L2 German speakers. *Applied Psycholinguistics* 43(6). 1221–1256.
- Knudsen, Trygve. 1973. Bøyningen av adjektiver som predikatsord. *Maal og Minne* 1–2. 29–41.
- Kupisch, Tanja, Miriam Geiss, Natalia Mitrofanova & Marit Westergaard. 2022. Structural and phonological cues for gender assignment in monolingual and bilingual children acquiring German: Experiments with real and nonce words. *Glossa: A Journal of General Linguistics* 7(1). DOI: [10.16995/glossa.5696](https://doi.org/10.16995/glossa.5696).
- Lie, Svein. 2003. *Innføring i norsk syntaks*. 5th. Oslo: Universitetsforlaget.

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- Mitrofanova, Natalia, Yulia Rodina, Olga Urek & Marit Westergaard. 2021. Sensitivity to microvariation in bilingual acquisition: Morphophonological gender cues in Russian heritage language. *Applied Psycholinguistics* 43(1). 41–79.
- Nessel, Tore, Hans-Olav Enger & Laura A. Janda. 2022. Er politiet sikker eller sikre? Adjektivkongruens ved kollektiver i norsk. *Arkiv för nordisk filologi* 137. 123–144.
- Nessel, Tore & Laura A. Janda. 2023. Norwegian agreement clashes on the football field. In T.I. Davidjuk, I.I. Isaev, Ju.B. Mazurova, S.G. Tatevosov & O.V. Fedorova (eds.), *Jazyk kak on est': Sbornik statej k 60-letiju Andreja Aleksandroviča Kibrika*, 293–298. Moscow: Buki Vedi.
- Westergaard, Marit. 2014. Linguistic variation and micro-cues in first language acquisition. *Linguistic Variation* 14(1). 26–45. DOI: [10.1075/lv.14.1.02wes](https://doi.org/10.1075/lv.14.1.02wes).
- Western, August. 1921. *Norsk riksmåls-grammatikk*. Kristiania: Aschehoug.

Chapter 19

Plural generic statements in Norwegian as L3

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This study investigates the effect of previously acquired languages on the acquisition of article use in generic and specific NPs in Norwegian as a third language (L3); in particular, how multilinguals acquire the form-to-meaning mapping (FMM) between the universal meaning of genericity and its expressions by plural noun phrases. We tested three learner groups: English native speakers residing in Norway and two groups of Polish speakers with English as an L2, one group residing in Norway, and the other group enrolled in Norwegian classes at university in Poland. We hypothesized that all learners would be highly accurate with the bare plural, as it is a shared FMM. In contrast, the Polish speakers were expected to be less accurate in accepting the definite plural form, given that Polish lacks definite marking with articles. We tested our hypotheses using a truth value judgment task providing generic and non-generic contexts, with test sentence subjects expressed as either bare or definite plurals. Our findings indicate that all learner groups had acquired the Norwegian FMMs, although the English group demonstrated higher accuracy compared to the Polish groups. Unexpectedly, the Polish speakers showed higher accuracy with the definite than with the bare form, and the foreign language learners were more accurate compared to the immersed learners. We argue that this is a consequence of the learner setting, i.e., the foreign language learners receiving more explicit instruction on article use. Additionally, we argue that dialectal variation in the expression of the definite plural form may have influenced the responses of the immersed learners.

Keywords: Genericity, L3 acquisition, plural NP, Form to meaning mapping

1 Characterizing generics and their expressions

Generic expressions are used to categorize and depict regularities in the environment through generalizing over classes of entities or events (Carlson 1995; Krifka et al. 1995; Mari et al. 2012). This linguistic phenomenon is pervasive across languages, often leveraging the determiner system rather than dedicated grammatical structures (Carlson 2011). While genericity as a semantic category contains different types of meanings, we focus here on so-called *characterizing generics*, henceforth referred to simply as *generics*. They draw on the prototypical traits of a class of objects, allowing for generalizations that do not necessarily apply to every instance, yet hold true for the whole class.

The generic reference in (1) denotes a property of a prototypical referent, providing the typical property of the banana. The generic meaning does not fall apart if there are exceptions: bananas may be green when they are not ripe, or brown when they are overripe; this does not contradict the truth value of bananas, in general, being yellow. With the definite form as in (2), the sentence refers to the bananas in the relevant context. This is referred to as Maximality (Ionin & Montrul 2010). In sentence (2), the definite form signals that reference is made to the entire set of contextually relevant bananas, rather than to bananas in general. This means that the property ascribed (e.g., being yellow) applies to the bananas under discussion. The principle of maximality ensures that the definite description does not permit partial reference—i.e., it refers to the entire relevant group. This is what distinguishes the episodic reading, where the property is specific to a particular situation, from the generic reading in (1), where the statement applies to the category as a whole. Thus, sentences as in (2) exemplify the episodic reading.

(1) Bananas are yellow.

(2) The bananas are yellow.

Languages may differ in form–meaning mapping (FMM) regarding which NP form denotes genericity and which denotes maximality. In this section, we summarize FMMs for the languages relevant to this study. While in English definite plural NPs express maximality as in (2), we follow Carlson (1977: a.o.) in treating English bare plurals as ambiguous. The generic reading of (1), which according to Carlson works similarly to a universal quantifier, contrasts with the reading in (3), which has the flavor of an existential quantifier (there exist some doctors who...). We did not test this interpretation but consider it in the discussion.

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(3) Doctors tried to save the dying boy. (Carlson 1977: 414)

In Norwegian, the bare plural can denote characterizing genericity as in English, see (4), whereas the definite plural denotes maximality (5), just as in English (2).¹ Thus English and Norwegian work in a similar fashion with respect to these FMMs, although they differ in the complete generic paradigm (Halmøy 2008, 2010).

- (4) Bananer er gule.
bananas are yellow
'Bananas are yellow.'
- (5) Bananene er gule.
bananas-textscdef are yellow
'The bananas are yellow.'

Polish does not have articles, resulting in a bare plural form that can indicate either generic or episodic meanings. Therefore, while the generic interpretation is represented by the bare plural in all three languages, the episodic meaning is indicated by the definite form in Norwegian and English, and by the bare form in Polish. These comparisons are summarized in Table 1.

Table 1: Overview of the FMM of the plural forms across the three languages

Language	Form	Generic	Episodic
Polish	Definite plural		
	Bare plural	√	√
English	Definite plural	#	√
	Bare plural	√	#
Norwegian	Definite plural	#	√
	Indefinite plural	√	#

¹In addition, according to Halmøy (2010: 57) the Norwegian bare plural cannot have a kind generic reading (e.g. #*Elger er truet av utryddelse* 'Moose are threatened by extinction.') We did not test kind generic meanings in the task we report on in this chapter. Halmøy states that this form is indefinite, rather than ambiguous between specific and indefinite. However, from a morphological standpoint, the form is bare, i.e., it is a plural without any other functional morpheme, and we thus treat it here as a bare form.

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2 Previous findings and theoretical underpinnings

The distinction between definite and bare plural subjects in generic and episodic conditions has been studied for the English–Spanish contrast in adult multilinguals (Ionin & Montrul 2010, Ionin et al. 2013). Spanish only allows definite plural subjects, which are ambiguous between generic and episodic interpretations, while bare plurals are ungrammatical. Ionin & Montrul (2010), Ionin et al. (2013) found that English learners of L2 Spanish were more successful in detecting the ungrammaticality of the bare plural than Spanish learners of L2 English were in acquiring the distinction between the generic (indefinite) and episodic (definite) readings.

We have a similar learning situation in this experiment: some forms are ambiguous and others unattested. As different generic meanings are expressed by different functional morphemes or context across languages, these mismatches augment acquisition difficulty. In addition, some functional morphemes are polysemous within a language. Thus, we can expect not only native language (L1) influence but also additional challenges due to the diversity of FMMs. Therefore, concrete predictions for difficulty might consider not just the expressions of isolated generic meanings (characterizing generics in our experimental study) but also the whole genericity-expressing paradigm of a language.

According to the Bottleneck Hypothesis (BH), functional morphology is the bottleneck in additional language acquisition (Slabakova 2014, 2019). Slabakova (2019) states that if there are no *syntax–semantics mismatches*, the L2/L3 semantics is easily acquired. As Table 1 shows, English speakers face no mismatches in L2 Norwegian, whereas Polish speakers face the acquisition of the definite plural article in Norwegian and its mapping onto the episodic meaning as a new FMM.

Based on the FMM description in Table 1 and the BH, we predict that, since the bare form is used to convey the generic meaning in all three languages, it should be advantageous for the L1 Polish groups to assign that form to generic meaning in L3 Norwegian. Discrepancies are attested with the FMM in the episodic reading, however, as the definite form is unavailable in Polish, creating a mismatch in the FMM compared to Norwegian. This may entail that acquiring the mapping for this condition will be more challenging. Nevertheless, the L1 Polish speakers have the possibility of transferring their L2 English system, which is identical to the Norwegian system in the relevant respects.

3 Current study

We tested the acceptance of bare vs. definite plural forms in generic and episodic contexts by L2 speakers of Norwegian with L1 English, and by two groups of L3 Norwegian speakers with L1 Polish and L2 English. The two L3 groups differed in how they acquired the L3: we included both immersive learners residing in Norway and classroom-instructed learners residing in Poland. Since it is established that genericity is acquired early in the L1, we are confident that all the participants in our study, being adult learners, have generic meanings pinned down. What they need to acquire is the FMM of the Norwegian plural NPs onto the different contexts. This is exactly what the current study is testing.

We address the following research questions:

1. Is L1 influence attested in third language acquisition (L3A)?
2. Is learning a new morphological form easier in L3A, compared to readjusting the meaning of a native form?
3. Is the immersive or classroom instructional setting more beneficial for acquiring the FMM under discussion?

In order to gauge the language of influence, we use a subtractive language group design (Westergaard et al. 2023). The target language is kept constant and the performance of an L2 group (English–Norwegian) is compared to that of L3 groups (Polish–English–Norwegian).² Arguably, if the L2 and L3 groups perform *differently*, the subtracted language (Polish) exerts (some) influence in the acquisition process. If the bilingual and multilingual groups perform *similarly*, then the subtracted language is not influential.

It is possible that multilinguals are at a general disadvantage compared to bilinguals: if so, we expect lower accuracy in the two Polish groups compared to the English group, on all experimental conditions. On the other hand, if we establish that the Polish groups are not equally accurate across the generic and episodic conditions, we can attribute that finding to the linguistic contrast described in Table 1. In other words, if the Polish natives over-accept the bare form in the episodic condition and under-accept the definite in the generic condition, this would be evidence of Polish influence, because they have the possibility of transferring the bare form, but not the definite form from their L1.

²It is practically impossible to find participants for a second L2 group, L1-Polish L2 Norwegian, without knowledge of English.

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While our group design allows us to establish L1 influence (RQ1), we are also interested in whether the native language influence is possible to overcome through acquisition. We can answer this question positively if we find learners who show they distinguish between acceptable and unacceptable forms. However, a negative answer is not possible since learning may occur later than testing. Furthermore, compared to the L1, there is a new form for the Polish groups, the definite, to add to the L3 FMMs and a known native form, the bare plural, whose meaning has to be adjusted (Table 1). If we see an influence from L1 Polish in the data, RQ2 asks which process is easier in L3A.

RQ3 relates to the difference in accuracy between what we label “immersed” and “foreign language (FL)” learners. Depending on which learner group is more accurate, we may discern the benefits of the L3 setting. We formulate our prediction in light of the BH: the challenge of acquiring the Norwegian FMM for the Polish learners lies in acquiring the definite plural form, manifested through morphology. Whereas genericity is rarely a part of foreign language curricula, articles certainly are part of the instruction, and knowledge of articles constitutes a necessary steppingstone for acquiring genericity.

4 Methodology

4.1 Participants

A total of 125 participants were tested, divided into four groups. A summary of their demographics is outlined in Table 2. The Norwegian native speakers (Nor group, $n=33$) and the English native speakers (Eng group, $n=39$) were tested in Norway. Our immersive Polish learners residing in Norway ($n=27$) are henceforth labelled Pol-N. The foreign language group ($n=26$) studied Norwegian at the universities of Poznan and Szczecin, and are designated as Pol-P. The participants were recruited through social media and through channels established by the research project this study was a part of. They received a 200 NOK gift card as compensation.

Within the Pol-P group, 6 participants had started to learn Norwegian less than a year before testing, 12 declared to have started learning it 1–3 years ago, 2 participants started 3–5 years ago, and 3 participants had been learning Norwegian for over 5 years. Background information was missing for the 3 remaining Pol-P participants. Fifteen participants declared to have made short trips to Norway during their education. Thus, the Pol-N and the Pol-P groups differed categorically on exposure.

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Twelve immersive learners (Pol-N) declared to have taken Norwegian courses spanning from beginner to upper-intermediate. However, only 3 declared to be enrolled in a Norwegian class at the time of testing. Most of the immersive participants had been residing in Norway for a long time (5 longer than 10 years, 8 between 5–10 years, and 5 from 3–5 years) and were thus likely to have had extensive and varied exposure to the dialectal variation of Norwegian.

Table 2: Summary of the demographics of the participants

Group	n.	Age range	Mean age	Average education level ³
Nor group (NS)	33	18–77	29.1	2.81
Eng group (L2)	39	24–75	42.9	3.51
Pol-N (L3)	27	24–68	37.2	3.95
Pol-P (L3)	26	19–38	23	2.61

Values are averages based on a scale where 1 represents elementary school, 2 represents high school or vocational school, 3 represents a bachelor’s degree, 4 represents a master’s degree, and 5 represents a doctoral degree

4.2 Materials

The experimental design was inspired by [Ionin & Montrul \(2010\)](#). It is a Truth Value Judgment Task (TVJT) where a description of a situation is followed by a *summary sentence* containing either a definite or a bare plural NP. This sentence then had to be judged as True or False within the given context.

The context introduced a regular property of an NP (e.g., *yellow bananas*) and an exception (e.g., *blue bananas*). An image on the same screen depicted the exceptional situation. The difference between the generic and episodic conditions was in the adjective in the target sentence. Example (6) shows the context and the target sentences, in English for convenience.

- (6) There is a supermarket in town that sells unusual fruit and vegetables.
While the banana is usually a yellow fruit, their bananas are blue, and
they taste like vanilla ice cream.
- Summary sentences (test items)
- Generic meaning

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- | | | |
|-----|-------------------------|-------|
| (a) | Bananas are yellow. | True |
| (b) | The bananas are yellow. | False |

Episodic meaning

- | | | |
|-----|-----------------------|-------|
| (a) | Bananas are blue. | False |
| (b) | The bananas are blue. | True |

The task was a 2x2 design testing the acceptance of the two forms (definite plural vs. bare plural) in two conditions (generic vs. episodic) (see example 6). There was a total of 9 contexts, yielding 36 test items. The context–test item pairs were divided across three lists, and the trials were separated in two blocks, allowing the participants to take planned breaks.

The fillers (n=24) contained items with the same format as the target sentences and included a True or False statement in relation to the context. These items were used to assess the participants’ understanding of Norwegian. A full list of test items is included in the appendix.

4.3 Procedure

The task was designed in OpenSesame Web (Mathôt et al. 2012) and distributed online through the JATOS platform (Lange et al. 2015). An image depicting the episodic situation (exceptional situation) appeared on the screen for 1000ms, and then the description of the context appeared, together with a smaller version of the image placed on the left-hand side of the screen. The participants were instructed to read the context carefully and to press the spacebar when they were done reading. Then, the test sentence appeared, together with True and False answers indicated at the bottom; the letter keys E and O were used respectively to provide the judgment.

5 Results

5.1 Data coding

We used R Studio to conduct the analysis. The O and E keypresses were coded as 0 and 1 respectively, to represent the True and False judgments.

5.2 Comprehension measure

Our fillers consisted of a description of a picture and a short summary; the participants were asked to say whether the summary was True or False (24 items,

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equally divided between true and false). Based on these, we calculated a comprehension measure based on the accuracy of each participant; in Figure 1 each dot represents an individual’s accuracy on the fillers.

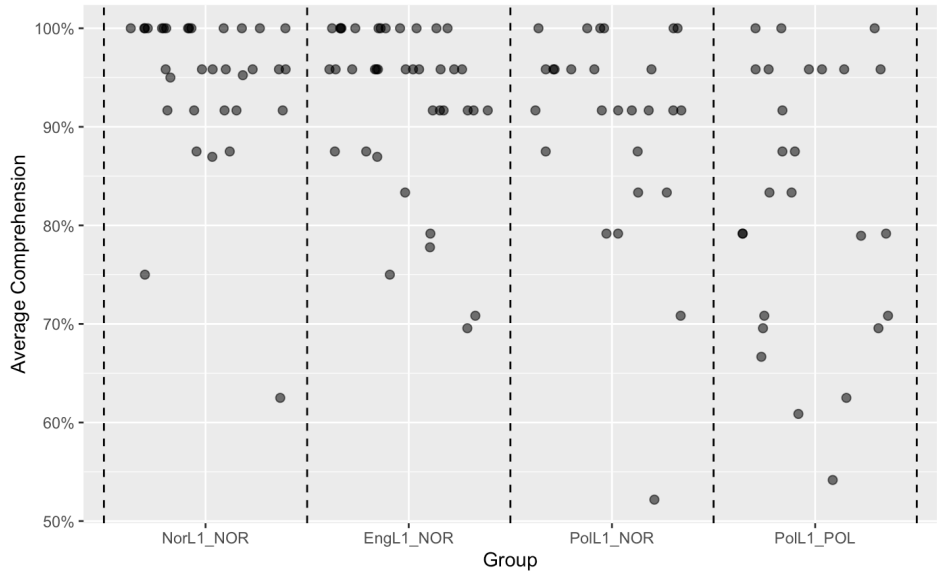


Figure 1: Average accuracy on filler items across groups

We plotted an analysis of variance: comprehension was the dependent variable, and group was the independent variable. The result was significant at $p < .001$, and we thus conducted a Tukey post-hoc test. Significant differences were found between every group, with the Nor group having unsurprisingly the best comprehension score, followed by the Eng group, the immersive learners (Pol-N), and with the FL learners (Pol-P) scoring the lowest.

5.3 Overview of the target responses

Figure 2 displays an overview of the responses across participant groups, divided per condition and item form. All responses delivered faster than 1000ms were excluded from the analysis. The task thus yielded 1469 observations; all further analysis is conducted on these.

The statistical analysis was conducted separately for each condition, resulting in two distinct models. The first model pertains to the generic condition where acceptance of bare plurals is anticipated, and the second model corresponds to the episodic condition which requires the definite form. For both conditions, we

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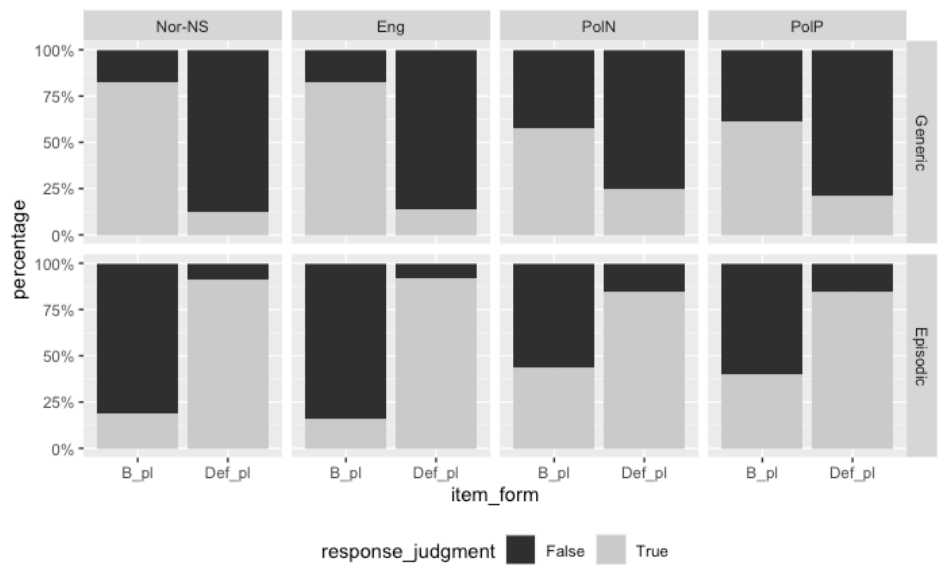


Figure 2: Overview of plural NP choices mapping to generic and episodic meanings

plotted general linear mixed effects regressions from the lme4 package (Bates et al. 2015) in R. In both models, the dependent variable was the response (0 vs. 1), with group, NP form, and the comprehension measure serving as the independent variables; participant was set as a random effect.⁴ The comprehension measure was centred so that the intercept represents the predicted response for a participant with average comprehension. Centring also reduces correlation between predictors, leading to more stable model estimates. The Nor group was always set as the reference category. We adjusted the intercept for the NP form to the most representative form for the condition (bare for generic, definite for episodic). The model predictions are displayed in Figures 3 and 4.

In the generic model (Figure 3) we found that the Nor group accepted the definite form to a significantly lower degree ($p < .001$) compared to the bare form as expected. The fixed effect of the comprehension measure showed a positive trend (Estimate = 9.227, $SE = 5.359$, $p = .085$), suggesting that higher proficiency was associated with higher log odds of a correct response. Although this effect is marginal ($p \approx .085$), it indicates that participant proficiency contributes to accuracy. At a group level in the fixed effects, there was only one group that

⁴`glmer(response_value ~ item_form * group * comprehension_c + (1 | id),`

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differed from the control intercept, Pol-N, which accepted the bare form to a lesser degree ($p < .05$). Furthermore, the interaction of form*group was also significant only for Pol-N, as they accepted the definite to a higher degree ($p < .01$). There was also an interaction for form and comprehension for Pol-N: As comprehension of Norwegian increases, the acceptance of the definite decreases ($p < .05$). The interactions between individual groups and comprehension were not found to be significant, and neither was the three-way interactions between form*group*comprehension.

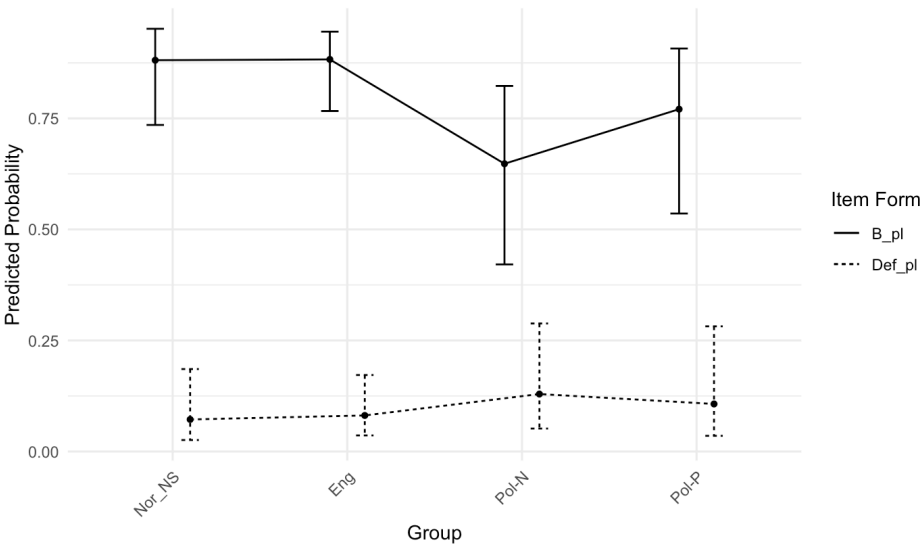


Figure 3: Predicted probability of acceptance by group and item form in the Generic condition

The episodic model (Figure 4) had the definite form as the intercept form. As expected, the native group showed a strong rejection of the bare form ($p < .001$). No differences were found at the group level, and higher comprehension did not contribute to enhanced accuracy. In the form*group interaction, the Pol-N group differed once again: they accepted the bare form significantly more in the episodic condition ($p < .05$) compared to the native controls, and the other learner groups. We found a positive interaction for comprehension*form: a higher comprehension contributed to a higher rejection for the bare form ($p < .05$). There were no significant interactions of group*comprehension. In the three-way interactions, a marginal effect was found for the Pol-N group ($p = .0755$) as they were found to reject the bare form more with increasing comprehension.

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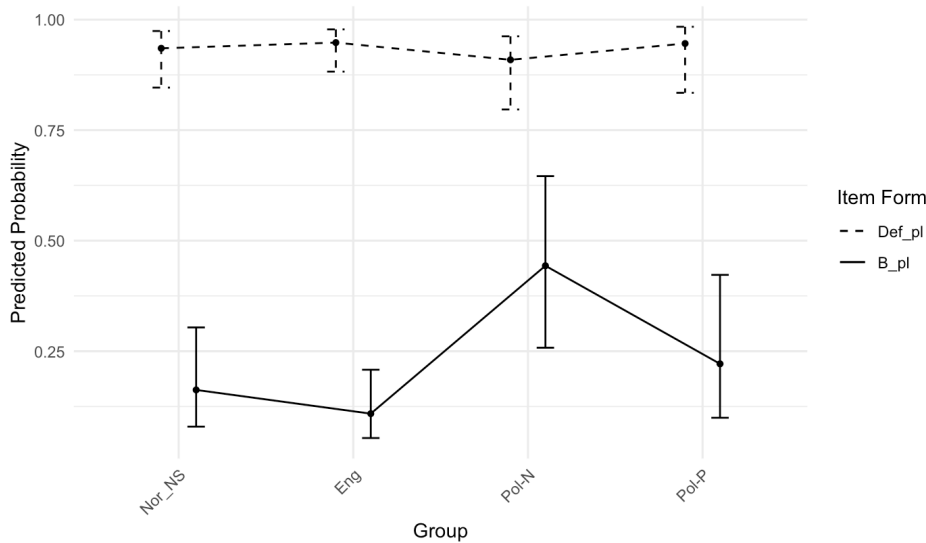


Figure 4: Predicted probability of acceptance by group and item form in the Episodic condition

5.4 The acceptance of the variants in L2 English

Before interpreting the results, we must consider whether it was a possibility for the Polish groups to be influenced by their L2 English. We administered the same TVJT in English. Here, the Eng group served as native controls, while the Nor L2 group, with its overlap in FMM to English, represented an interesting comparison to the Polish one. The analysis consisted of two separate models per condition. The models were simplified compared to the ones presented in Section 5.3, as we were not interested in their proficiency/comprehension levels in English. Given the space constraints and the chapter’s primary focus, we provide only a brief overview of the L2 English results.

The Polish groups’ acceptance patterns were qualitatively similar to native controls: they accepted the bare form more in the generic condition, and the definite form more in the episodic condition, though this contrast was weaker than for the native group. This is corroborated by individual models on each Polish group: both groups make the relevant distinctions in their use of the bare and the definite plural.

The generic model on all groups showed that both Polish groups accepted the bare form less consistently than the controls, with Pol-P ($p < .001$) deviating

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more than Pol-N ($p < .05$). Both groups were found to accept the definite more ($p < .001$) when compared to the controls. In the Episodic model, the Pol-N group showed a marginal deviation from the controls ($p = .063$) in the fixed effects, as they accepted the definite form less consistently. Both Polish groups significantly over-accepted the bare form ($p < .001$), a pattern also observed in the Norwegian group ($p < .01$).

These findings suggest both Polish groups had acquired the relevant L2 English contrasts, albeit with varying precision, making facilitative L2 influence plausible. However, since Pol-N's deviation from controls in the episodic condition was only marginal, they appear more attuned to the L2 English FMM than Pol-P.

6 Discussion

The current study aimed to explore the acceptance of plural forms in generic and episodic contexts by L3 speakers of Norwegian. The tested groups were Polish L1 speakers with English as their L2 who differed in the setting of their L3 acquisition (immersive vs. instructed).

It is important to note that all participant groups (two target and two control groups) showed sensitivity to the tested distinctions and *qualitatively* patterned with the native controls. In other words, all groups distinguished between True and False test sentences in generic and episodic contexts. Still, there were quantitative differences, in that the contrasts were less categorical in the L3 groups. These *quantitative* differences help us answer the research questions.

Starting with the Eng group, they patterned with the Nor group in both conditions. This was expected given that the generic FMMs are similar in English and Norwegian for plural NPs. However, both Polish groups showed significant differences from the control groups. In principle, these differences can be attributed either to the general disadvantage of multilingual learners in having to manipulate three languages in the brain, or to the non-facilitative influence of Polish, given that it lacks articles. However, it should also be noted that the Eng group performed better than both Polish groups on the fillers, which implies higher proficiency in Norwegian.

Turning to our concrete predictions, we postulated in Section 3 that if the three learner groups had behaved in the same way, this would suggest that the subtracted language, Polish, had no influence on the acquisition process. This expectation was not met; hence, we can safely conclude that the L1 did impact the acquisition process in some way. Both Polish groups differed quantitatively

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from the controls and the English L2 speakers of Norwegian when looking at the raw data (Figure 2). However, when the level of Norwegian comprehension was taken into account, we found that only Pol-N differed from the other groups. The differences found within the *generic condition* show a lower acceptance of the bare form in this group compared to all other groups, as well as an increased acceptance of the definite form. In the *episodic condition*, in which the definite plural is expected, the Pol-N group deviated from the controls to a lesser degree, as they accepted the definite form in a way comparable to the controls. This was unexpected as we anticipated that both Polish groups would show a heightened precision in the generic condition, as there is surface overlap between the three languages in this condition (i.e., the use of the bare form). To master the episodic condition, however, acquisition and mapping of the definite form is needed.

We also conjecture that if multilinguals are at a general disadvantage compared to bilinguals, both Polish groups should be equally likely to consider bare NPs in episodic conditions as 'True' and definite NPs in generic conditions 'False'. Our results show that this was not the case. The Pol-N group was less accurate in the generic condition than in the episodic condition, as demonstrated by the two models in Section 5.3.

Furthermore, we identified distinct behaviour not just across conditions but also across form acceptance. Contrary to our expectations, the Pol-P multilinguals were more accurate with definite NPs than with bare NPs, suggesting that language acquisition had overcome this aspect of negative Polish L1 influence. Pol-P's acceptance of the definite form matched that of the Nor and Eng groups, suggesting that the form had been largely acquired, indicating that the CLI from L1 Polish is not insurmountable.

RQ2 is theoretically more intriguing. We have established that the Polish groups had successfully acquired the FMM of a form absent from their L1: the definite plural. The difficulty lies in acquiring the exact uses of the form and especially the restrictions on its use. Nevertheless, our results on the same task in English show that the Polish learners have a grasp of the FMM in English, which can be a source of facilitation for the FMM in the L3. Overgeneralization, or some usage adjustment, of a newly acquired form is to be expected. In the episodic condition, there can be no overgeneralization of this new form, because it is always appropriate, so the only non-target performance is a lingering over-acceptance of the bare form. In the generic condition, the bare plural is under-accepted, and the new definite form is differentially rejected, see Figure 2 and discussion below. Thus, the overall results in the Polish groups, and Pol-N in particular, may indicate that adjusting the meaning of a form absent from the L1 may be difficult to overcome even with potential facilitation from the L2.

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In addition, it is not unexpected that generic FMMs are harder than non-generic FMMs to adjust. Firstly, every-day situations as described in our episodic condition are (anecdotally) much more frequent in natural input compared to generic descriptions, hence more opportunities for acquisition present themselves. Secondly, as remarked above, bare plurals in English and Norwegian are ambiguous between generic and existential readings; compare examples (1) and (3). Thirdly, Polish bare plurals can be used in episodic, generic and existential contexts, so they are at least three-way ambiguous. In fact, one could ask whether Polish bare plurals are indeed comparable to those in the other two languages, as Polish has only one plural form while Norwegian and English have two. This linguistic configuration necessitates a lot of paradigm adjustment, in the sense that forms with multiple meanings in the L1 have to be remapped onto a different set of forms and meanings in the L3. It could be expected, therefore, that bare plurals, although available from the L1, were hesitantly employed.⁵

Finally, let us turn to RQ3. Recall that the immersive (Pol-N) group deviated more strongly from the Nor group than the foreign-language (Pol-P) group. According to our comprehension measure (accuracy on fillers), Pol-N scored higher than Pol-P, which is likely due to higher and longer exposure to Norwegian. However, the two groups acted similarly in their judgments across the generic and episodic conditions, even though the immersive group scored significantly higher on their comprehension. Thus, including the comprehension measure in our statistical models showed that Pol-P were more accurate compared to Pol-N. This leads us to the role of instruction for the acquisition of the article system.

While Pol-P contained FL classroom learners whereas Pol-N learners resided in Norway, some of the latter had also received some instruction in the past, which might have provided metalinguistic awareness of grammatical features of Norwegian, including the definite article. We may, however, speculate that the instruction that the Pol-P group was exposed to was both more advanced and more targeted to speakers of an article-less language. This could explain why Pol-P were more target-like in their use of the definite plural article, especially when their lower comprehension is taken into account, and implies that formal instruction is particularly beneficial to the acquisition of the relevant FMMs.

Another potential reason for the observed difference among the L3 groups may be the dialectal variation in Norwegian. Sandøy (1998) offers a concise overview of the variation in the expression of the two plural forms in Norwegian varieties

⁵In addition, as discussed in Footnote 1, English and Norwegian bare plurals do not share all meanings, which prompts Halmøy to designate the latter not as bare but indefinite plurals. This fact may increase the paradigm ambiguity.

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and shows that the definite plural is the form prone to more variation. This suggests that the immersive learners may have been exposed to higher variation in plural forms when compared to the instructed learners, who were exposed mostly, if not only, to the Norwegian *Bokmål* standard, in which the task was presented. This may have influenced the results of our study, as the immersive learners may not have interpreted all intended definite forms as definite. Previous findings on second language acquisition have established that learners benefit from stable FMMs (DeKeyser 2005; Slabakova 2008, 2016). In our study, the classroom setting may have offered a stronger FMM for definite plurals than immersion into Norwegian.

7 Conclusions

This study investigated the influence of previously learned languages on the L3 acquisition of FMM of plural forms to generic and episodic meanings, with special attention to the learning context. Based on the results of our study, we conclude that the FMMs can indeed be acquired in L3 Norwegian, as both Polish learner groups made a contextual distinction between the two forms and their usage in Norwegian.

However, it does not seem to be the case that the presence of a functional morpheme, in this case the bare plural, across the learners' three languages provides a straightforward advantage. The multilingual learners were less accurate in the generic condition, where the bare form is appropriate, compared to the episodic condition, which required the definite form. We attributed the hesitance in bare plural choices to a FMM adjustment as well as L1 influence.

The FL group was found to be more accurate than the immersive group and we attribute this to two factors: (i) stronger knowledge of the article system due to explicit instruction, and (ii) more stable exposure to the written standard used in the task, as varieties of Norwegian (to which the immersive learners were exposed) have different morphemes for marking definiteness. In summary, we demonstrated that multilingual linguistic representations can be adjusted to the target ones, but the shape and ease of this adjustment benefits from stable form-meaning mappings.

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References

- Bates, Douglas, Martin Maechler, Ben Bolker & Steve Walker. 2015. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* 67(1). 1–48.
- Carlson, Greg. 1977. *Reference to kinds in English*. University of Massachusetts at Amherst. (Doctoral dissertation).
- Carlson, Gregory. 2011. Genericity. In Klaus von Heusinger, Claudia Maienborn & Paul Portner (eds.), *Semantics: An international handbook*, 1153–1185. Berlin: De Gruyter Mouton.
- Carlson, Gregory N. 1995. Truth conditions of generic sentences: Two contrasting views. In Gregory N. Carlson & Francis Jeffrey Pelletier (eds.), *The generic book*, 224–237. Chicago: The University of Chicago Press.
- DeKeyser, Robert M. 2005. What makes learning second-language grammar difficult? A review of issues. *Language Learning* 55. 1–25. DOI: [10.1111/j.0023-8333.2005.00294.x](https://doi.org/10.1111/j.0023-8333.2005.00294.x).
- Halmøy, Madeleine. 2008. Number, (in)definiteness and Norwegian nouns. In *Proceedings of Sinn und Bedeutung* 12, 180–195.
- Halmøy, Madeleine. 2010. *The Norwegian nominal system: A Neo-Saussurean perspective*. Department of Language & Culture, University of Tromsø. (Doctoral Degree Monograph).
- Ionin, Tania & Silvina Montrul. 2010. The role of L1 transfer in the interpretation of articles with definite plurals in L2 English. *Language Learning* 60(4). 877–925.
- Ionin, Tania, Silvina Montrul & Mónica Crivos. 2013. A bidirectional study on the acquisition of plural noun phrase interpretation in English and Spanish. *Applied Psycholinguistics* 34(3). 483–518.
- Krifka, Manfred, Francis Jeffrey Pelletier, Gregory N. Carlson, Alice ter Meulen, Gennaro Chierchia & Godehard Link. 1995. Genericity: An introduction. In Gregory N. Carlson & Francis Jeffrey Pelletier (eds.), *The generic book*, 1–124. Chicago: University of Chicago Press.

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- Lange, Kristian, Simone Kühn & Elisa Filevich. 2015. Just another tool for online studies” (JATOS): An easy solution for setup and management of web servers supporting online studies. *PLoS One* 10(6). 1–10. DOI: [10.1371/journal.pone.0130834](https://doi.org/10.1371/journal.pone.0130834).
- Mari, Alda, Claire Beyssade & Fabio Del Prete. 2012. *Genericity*, vol. 43. Oxford: Oxford University Press.
- Mathôt, Sebastiaan, Daniel Schreij & Jan Theeuwes. 2012. Opensesame: An open-source, graphical experiment builder for the social sciences. *Behavior Research Methods* 44(2). 314–324.
- Sandøy, Helge. 1998. The diffusion of a new morphology in Norwegian dialects. *Folia Linguistica* 32(1–2). 83–100.
- Slabakova, Roumyana. 2008. *Meaning in the second language*. Berlin: Mouton de Gruyter.
- Slabakova, Roumyana. 2014. The bottleneck of second language acquisition. *Foreign Language Teaching and Research* 46(4). 543–559.
- Slabakova, Roumyana. 2016. *Second language acquisition*. Oxford: Oxford University Press.
- Slabakova, Roumyana. 2019. The bottleneck hypothesis updated. In Tania Ionin and Matthew Rispoli (ed.), *Three streams of generative language acquisition research: Selected papers from the 7th meeting of Generative Approaches to Language Acquisition–North America*, 319–346. Amsterdam: John Benjamins.
- Westergaard, Marit, Natalia Mitrofanova, Yulia Rodina & Roumyana Slabakova. 2023. Full transfer potential in L3/Ln acquisition: Crosslinguistic influence as a property-by-property process. In Jennifer Cabrelli, Adel Chaouch-Orozco, Jorge González Alonso, Sergio Miguel Pereira Soares, Eloi Puig-Mayenco & Jason Rothman (eds.), *The Cambridge handbook of third language acquisition*, chap. 9, 219–242. Cambridge: Cambridge University Press.

Chapter 20

Micro-cues are spans

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In this chapter, we explore the notion that *micro-cues* (Westergaard 2009a,b, 2014, 2021) can aid in conceptualizing the observed conservative nature of gains and changes in (multilingual) language acquisition, historical syntactic change, and dialect variation. In a related sense, segments of syntactic structures commonly known as *spans* (Blix 2021a, 2021b; Svenonius 2016, 2020) share a number of similarities with micro-cues; e.g., sub-segments of larger hierarchical structures, both consisting of contiguous heads and features, etc. In this chapter, we introduce and support the claim that micro-cues are inherently spans. We support this thesis by demonstrating how a spanning-based approach is capable of accounting for the acquisition and historical development of elements of syntax.

Keywords: L1 acquisition, historical syntax, language variation, V2-ordering, spanning

1 Introduction

Since the inception of generative grammar, theorists have sought to develop a generalizable framework that adheres to *descriptive* and *explanatory adequacy*. Assuming a core assumption of Minimalist desiderata (Chomsky 1995), the generative component of grammar, i.e., its syntax, is responsible for the creation of well-formed, licit syntactic objects that can be externalized, i.e., spelled out, in order to be deciphered and interpreted by components of the mind responsible for sound (Sensori-Motor) and meaning (Conceptual-Intentional). The advent of the Minimalist Program ushered in a new era of theorizing in its shift from top-down,

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(weakly) representational constraints on well-formedness of structure (e.g., X'-theory) and additional conditions and filters held to evaluate these structures at different levels (e.g., D- & S-structure, Theta Criterion, Case Filter, etc.) to a bottom-up derivational approach. This shift rendered the vast majority of these conditions and filters as epiphenomenal, resulting in a computational approach to syntax that views parameters (of various sizes; see e.g., Roberts 2019) as third factor entities (following Chomsky's 2005 discussion of the three core factors of language design).

The (sometimes) radical shift towards the externalization of syntactic objects (Boeckx 2014) and the elimination, or at the very least, radical reconceptualization of traditional parameters (Roberts 2019, Eguren et al. 2016, Biberauer 2019), has resulted in a current version of Minimalism that embraces an 'open' conceptualization of UG (Lightfoot 2020) in which the licensing and interpreting of features takes primacy (Slabakova 2009, Lohndal & Putnam 2024, Stroik & Putnam 2013). The challenge with this nuanced version of Minimalism is arriving at a consistent algorithm responsible for determining the well-formedness of syntactic objects to ensure their legibility 'at the interfaces.' In her extensive research on the development of syntax in historical, L1 acquisition, and bi/multilingual contexts, Marit Westergaard (2009aa,b, 2014, 2021) and colleagues propose that linguistic cues existing in ambient input are often much smaller and more fine-grained than traditional parameters suggest. Westergaard calls these units of grammar **micro-cues**.

Among a host of positive attributes of the micro-cues approach, two stand out. First, the notion of linguistic 'cues' establishes a connection with emergentist, functional, and statistical approaches to language development (e.g., Holme 2013, MacWhinney 2005, 2019) and promotes a hybrid approach to language development reliant on some version of competition. Second, the actual size of a (micro-)cue can vary substantially, and can ostensibly be realized as a single morpheme or an entire idiom. This second point functions as somewhat of a double-edged sword, because, in its current form, there exist no mechanisms responsible for reigning in the potential overgeneration of the number of actual (micro-)cues that would be legible at the interfaces of sound and meaning. In the absence of a consistent set of generative mechanisms responsible for producing licit structures, it is extremely difficult, if not impossible, to understand what counts as a viable cue/construction, how they are generated and stored, and in the case of language change and L2/3 acquisition, how they compete with each other (e.g., see Diessel 2019 & Höder 2012 for Construction Grammar-based approaches).

In this chapter, we advance a tenable solution to this issue by proposing that the generation of micro-cues can be enhanced through the theory of **spanning**.

Although there are a number of competing ideas and suggested algorithms to classify spanning (Blix 2021a, 2021b; Ramchand & Svenonius 2014; Svenonius 2016, 2020), we adopt a simplified version in this chapter that claims a well-formed span is a series of contiguous heads in a syntactic configuration that are recognized as a single unit at either/both (of the) external interface(s).¹ The remainder of this chapter will adhere to the following structure: In Section 2 we provide a more detailed overview of micro-cues and their application in language development to date. We then turn to a discussion of spanning, making the case for the augmentation of the concept of micro-cues by its incorporation into a spanning analysis. We illustrate the advantages to this approach in Section 3, by demonstrating how a spanning-based approach can be used in research on L1 acquisition (Section 3.1) and language change and variation (Section 3.2). We conclude this chapter in Section 4, where we summarize the advantages of our approach and provide a series of questions for future research.

2 Micro-cues = Spans

In this section we lay out the conditions under which Westergaard's appeal to micro-cues can be effectively supplemented by a spanning-based mechanism. In Section 2.1 we provide a brief overview of the conceptualization of micro-cues, focusing primarily on defining these units. We provide a working definition of spans and the spanning procedure we adopt throughout the remainder of this chapter in Section 2.2. Here we illustrate how spanning provides a clear and straightforward axiom regarding how micro-cues can be derived. We highlight and discuss the architectural consequences of our treatment of spans and how the implementation of spans is capable of routinizing the generation of micro-cues in Section 2.3.

2.1 Micro-cues

Lightfoot's (2006) cue-based approach conceptualizes a cue as a piece of syntactic structure provided by UG. Westergaard (2009a) extends Lightfoot's theory, developing a theory of micro-cues (also referred to as "smaller-scale" cues) which posits that micro-cues are not necessarily provided by UG. Instead, Westergaard's theory suggests that children are endowed with categories and structures

¹The adoption of a 'spanning'-like algorithm to generate licit syntactic objects is not only restricted to derivational, proof-theoretical accounts of syntax; e.g., see Osborne et al. (2012) for a treatment of *catenae* derived from a dependency-based model of syntax.

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and they possess the ability to succinctly parse and detect fine-grained, relevant features from linguistic input. A micro-cue is therefore defined as a piece of abstract syntactic structure made up of syntactic primitives, the result of parsing (Westergaard 2014: 38). Micro-cues are defined by two key properties; (i) they must be derived within the specific context of a particular word order, clause type, or information structure, and (ii) they result from the interaction between the I-language faculty and the triggering experience present in the input.

Micro-cues are fundamental in a child's acquisition process according to Westergaard's research that presents L1 acquisition data as support for the existence of micro-cues. She argues that for a theory to adequately account for variation, it must be well-supported by a set of unambiguous micro-cues. Additionally, Westergaard (2009a) suggests that in order to derive a micro-cue for verb-second (V2) word order, an L1 Norwegian learner would only concentrate on non-subject-initial sentences, disregarding subject-initial sentences, as they do not contain relevant information. For instance, the micro-cue in (1a) specifies information about the V2 word order in declaratives while the micro-cue in (2a) provides word order for V2 in exclamatives.

- (1) a. Micro-cue for V2 in declaratives:
 $DeclP[XP_{Decl}[V \dots]]$ (Westergaard 2014: 37)
- b. Peter **snakker** ikke italiensk.
 Peter speaks not Italian
 'Peter doesn't speak Italian.' (Westergaard 2009b: 17)
- (2) a. Micro-cue for V2 in exclamatives:
 $ExIP[wh_{Excl}V]$ (Westergaard 2009b: 54)
- b. Hvor **er** han sød!
 Where/how is he sweet
 'How nice he is!' (Westergaard 2009b: 54)

Westergaard (2009b) observes that while micro-cues such as those in (1a) and (2a) specify the precise context of their application, each operates independently, and they are not acquired in any specific sequence. Notably, the acquisition of one micro-cue does not depend on the acquisition of the other.

Early L1 learners are conservative in their use of a given structure and do not tend to over-generalize. While some over-generalization is present in the literature, it is restricted to lexical classes or sub-categories. This conservative nature is most prominently displayed not in rare overgeneralizations, but in the frequent underuse of complex syntactic operations like Subject-Auxiliary Inversion (SAI),

where learners default to simpler, non-inverted forms. [Roeper \(1999\)](#) provides examples of these instances of over-generalization concerning V2 inversion in the *wh*-questions of L1 English acquirers. Occasionally, early L1 English speakers will over-generalize verbs that are semantically similar to the original cue, the verb ‘be,’ such as ‘call’ or ‘mean’ (cf. [Westergaard 2014](#): 41). These instances of over-generalization are class-specific and provide evidence that its scope is restricted, accounting for the general tendency of speakers to remain conservative in their production of target forms. We find that this model appropriately addresses Constructionist findings that account for limited evidence of over-generalization in early L1 speech from a Generative perspective.

2.2 Spans

Much of syntactic theory assumes features and roots as primitives. In syntactic derivation, processes such as Merge join these elements, resulting in binary-branching structures that ultimately go on to be pronounced (‘lexicalized’) and endowed with meaning. Within this type of theory, a span represents a certain sequence of features in a binary-branching structure, or a “contiguous sequence of heads in a head-complement relation” ([Svenonius 2016](#): 6). A more technical definition comes from [Blix \(2021b\)](#): 7) in (3).

(3) Span

An n -tuple of heads $\langle X_n, \dots, X_1 \rangle$ is a span in a syntactic structure S , if and only if $X_{n-1}P$ is the complement of X_n in S .

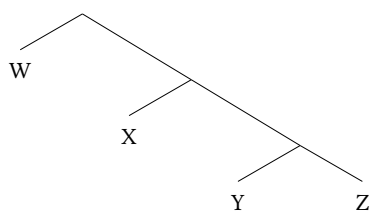
A theory with spans posits that in addition to storing features and roots in your mental lexicon, a span may also be stored and associated with a specific meaning or pronunciation. If a stored span matches a derived syntactic structure, it is inserted, meaning that the associated semantics and phonology are added to the derivation, generally at an interface with the semantics and phonology modules, respectively.

Versions of spanning have been utilized within frameworks such as Nanosyntax and Distributed Morphology, where they, depending on the framework, compete for insertion by way of either a superset- or subset-based matching protocol. For the remainder of this chapter, we will utilize a **superset** matching process in which a span will lexicalize any structure of which it is a superset. To illustrate the basic assumptions of this approach, consider the following scenario. The structure in (4) represents a sequence of functional heads, a span, in line with the definition provided in (3). The span in (4) will match and lexicalize the derived

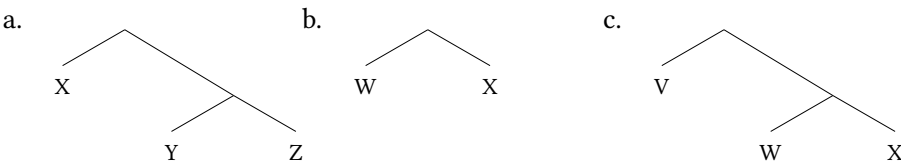
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syntactic structures (5a) and (5b) because those structures are contained within the span (the span is a superset of those structures). However, the structure in (5c) is not well-formed; i.e., does not match any lexically stored span, feature, or root, and, as a result, cannot be lexicalized by the span in (4).

(4) Lexically stored span



(5) Possible matches



The representative ‘matches’ between well-formed syntactic configurations (i.e., spans) and exponency can vary across languages, and can, in some instances, contribute to inter- and intra-speaker variation within a given community of speakers. Of prime importance here is the fact that spans enable the lexicalization of multiple functional heads as one unit of language.² The hypothesis we introduce, and subsequently defend, in the following subsection is that the tree-structures in (5) simultaneously fit the definition of ‘micro-cue’ and ‘span.’

²Appealing to spans as licit and interpretable units of language is in line with realizational, late-insertion models of morphosyntax. There are, of course, differences between these models, most notably, between *Distributed Morphology* (DM) (Embick & Noyer 2007) and *Nanosyntax* (Baunaz et al. 2018, Starke 2009, Caha et al. 2025). Whereas most spanning-based approaches loosen the restriction of terminal nodes being the only targets for lexicalization, most require a relatively strict adherence to a One Feature-One Head architecture (Putnam 2020). Contrariwise, approaches such as DM allow ‘feature bundling,’ i.e., the presence of multiple features on a single head. It remains to be seen if some version of this approach is commensurate with a spanning-based approach to spellout, which is a topic that requires much additional detailed research. For the remainder of this chapter, we do not engage in further comparison of these two frameworks, but simply wish to acknowledge this debate.

2.3 How micro-cues are spans

Spanning theory augments micro-cues with a matching mechanism that it currently lacks. In this subsection, we illustrate the importance of making a connection between spans and micro-cues. Both spans and micro-cues are learned and stored patterns of syntactic structure. They are sequences of syntactic heads that define well-formed structure in a specific language. Spans and micro-cues are equivalent units. In the rest of this chapter, we refer to micro-cues as developed by Westergaard as ‘micro-cues’ and the stored syntactic structures as ‘spans.’

By definition, a span, being a well-formed syntactic structure, is acquired by noticing, integrating, and eventually parsing the structure from the input. As in micro-cue theory, the development and acquisition of these structures progresses from specific contexts to more general ones. The more general structures that are encountered often become a ‘default’ optimal structure.³ These optimal structures are considered well-formed and will not, as a result of the structure itself, crash during derivation. Exceptions to the default, i.e., additional optimal structures, are acquired in the same fashion – specific to general, though how general the span becomes depends on the range of contexts in which it is permissible.

Spans may be developed for all structures in a language – the default unmarked structures and the marked structures – but they are most useful in marked cases where they signal a well-formed structure that otherwise breaks the norm. For example, the micro-cue in (1a) could be considered the default case – the verb appears in second position in a declarative phrase in Norwegian. This pattern occurs often and is routinized. The micro-cue displayed in (6a), on the other hand, exhibits a non-V2 pattern and has a very specific context in which it occurs. The focus-sensitive adverb *bare* ‘just’ in (6b) can be optionally placed between the clause-initial subject and the verb. The adverb is adjoined to FocP, while the verb moves to the *Foc*[°] head.

- (6) a. Micro-cue for word order in subject-initial declaratives with focus-sensitive adverbs:

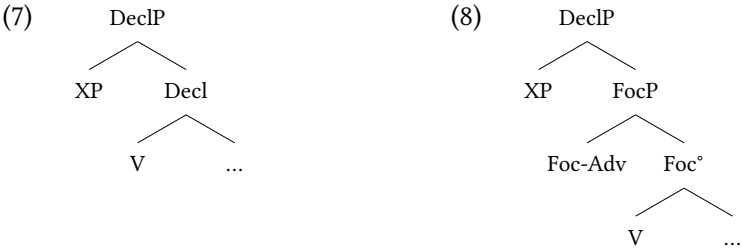
$DeclP[XP_{FocP}[Foc-Adv_{Foc^{\circ}}[V]]]$ (Westergaard 2009b: 58)

³How a given structure comes to be considered the default optimal one over the course of acquisition has been strongly linked to the notion of noticing the *frequency* and *distribution* of a structure in input. Yang’s (2016) Tolerance Principle illustrates the latter algorithmically with use of a calculated threshold for rule development based on the number of different contexts in which a specific structure is encountered during acquisition. In this chapter, we do not detail the specifics that may factor into the acquisition and differentiation of structures (e.g., number of encounters, frequency or quality of input, etc.), but are instead most concerned with the aftermath.

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- b. Han bare svarte ikke.
he just answered not
'He simply didn't answer.' (Westergaard 2009b: 58)

It is here where the span corresponding to this micro-cue acts as an important alternative to the default optimal V2 declarative pattern. The existence of this span allows such a structure to be recognized as well-formed. To further illustrate this point, we represent this state of affairs with a syntactic tree. A micro-cue such as (1a), V2 in declaratives, can be translated straightforwardly into the span in (7). In (1a) the verb has moved from its base position to the edge of the CP-clause and appears now as the head of DeclP (which can also be understood as a CP where the head C would be endowed with a clause-typing head/feature value of DECLARATIVE). This segment of syntactic structure contains a contiguous hierarchical string of syntactic elements, making it eligible to be considered a span.



Contrast the tree-let of the ‘default’ V2-ordering found in Norwegian declarative clauses, (7), with marked structure introduced as the micro-cue in (6a). In this non-default example, the focus adverb is in second position, and the verb only undergoes head-movement to Foc, resulting in a V3-ordering in declarative clauses (i.e., XP–Foc–Adv–V). Once again, this section of syntactic structure is eligible to be classified as a span and is illustrated in (8). This configuration can ‘override’ the expected V2-ordering in declarative clauses for speakers of (varieties of) Norwegian who prefer a V3-ordering whenever the second element in the sequence at the edge of a declarative CP is a focus-adverbial.

In the next section, we illustrate how the insertion processes that spanning theory introduces can provide a more economic solution for the storage of unmarked structure in the domains of L1 acquisition and historical syntactic development.

3 Empirical evidence

Equipped with a theoretical treatment of micro-cues and spans in the previous section, we now demonstrate how the former can be successfully augmented by the latter without contradiction or lack of empirical coverage. In Section 3.1, we revisit previous research by Westergaard in the domain of L1 acquisition. In particular, we take a closer look at the acquisition of the V2 ordering of finite verbs in Norwegian, while also accounting for ‘exceptions’ (i.e., non-V2-structures) in the ambient data. Finally, in Section 3.2 we turn to how we can apply a spanning approach to Westergaard’s research on the V2-parameter in historical English (i.e., Old to Middle English) and modern Norwegian data.

3.1 L1 acquisition

Westergaard’s research in the domain of L1 acquisition based on the micro-cue approach suggests that a child’s language ability must be at least partially genetically endowed. From an early age, children effortlessly distinguish between various syntactic structures by parsing the input, building micro-cues in their internal grammar, and eventually generating new sentences from these micro-cues. The spanning perspective we adopt aligns with this process of acquisition.

The specific work we feature in this section is Westergaard’s (2009a, 2014) examination of the L1 acquisition of the V2 and non-V2 variation present in the Norwegian dialect from the Tromsø region. In Norwegian, a prototypical V2 language, the finite verb is typically required to be positioned second in the main clauses of both subject-initial (9) and non-subject-initial declaratives (10).

- (9) Jeg **liker** ikke denne boka.

I like not this book.DEF

‘I don’t like this book.’

(Westergaard 2009b: 36)

- (10) Da **snakker** han engelsk./*da han **snakker** engelsk.

then speaks he English

‘Then he speaks English.’

(Westergaard 2009b: 17)

However, in certain declarative clauses, such as (i) those that contain a focus-adverb (11a), (ii) those that begin with the lexical item *kanskje* (‘maybe’) (11b), and (iii) exclamatives, non-V2 order emerges. In fact, some degree of variation in verb placement occurs with the lexical item *kanskje*, with a strong (approximately 95%) preference, by both children and adults, for the non-V2 structure (Westergaard 2008).

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- (11) a. Han bare **svarte** ikke.
 he just answered not
 ‘He simply didn’t answer.’ (Westergaard 2009b: 58)
- b. Kanskje **vinner** fotballaget i dag. / Kanskje fotballaget **vinner** i
 maybe wins soccer-team.DEF today
 dag.
 ‘Maybe the soccer team will win today.’ (Westergaard 2009a: 75)

These examples, which are available in the ambient input a child receives, are successfully acquired; however, the important question remains of how children (i) notice, (ii) incorporate, and (iii) store specific and more general patterns and structures as their grammars grow. When a child is provided with sufficient input, such as the examples previously given in (9)-(11), they begin to construct their own representations of what they perceive to be the optimal structure(s) in a given context. As mentioned in Section 2.1, they begin by forming specific micro-cues and generalizing them as they experience and encounter more (varied) input over time. According to the micro-cues approach, the cues in (12)-(14) are the final, specific, cues at which an L1 speaker of most varieties of Norwegian would ultimately arrive.

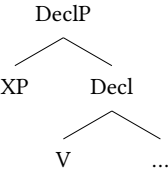
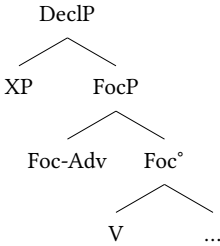
- (12) Micro-cue for V2 in declaratives:
 $DeclP[XP_{Decl^*}[V \dots]]$ (Westergaard 2014: 37)
- (13) Micro-cue for word order in subject-initial declaratives with focus-sensitive adverbs:
 $DeclP[XP_{FocP}[Foc-Adv_{Foc^*}[V]]]$ (Westergaard 2009b: 58)
- (14) Micro-cue for word order in declaratives with clause-initial *kanskje* ‘maybe’:
 $TopP[kanskje \text{ } XP \dots VP[V]]$ (Westergaard 2014: 37)

Westergaard (2009b: 186–188) notes another interesting exception to canonical V2-structures in Norwegian in which the finite verb appears after the subject in exclamatives and certain embedded structures, resulting in a V3- or later placement. These clause types do not occur in the prototypical V2-word order in certain varieties of Norwegian, making them less common in the input and minimally represented in child data. The presence of target-consistent non-V2 word order in exclamatives within child data provides evidence for treating ‘micro-cues as spans,’ suggesting that children are sensitive to minor syntactic distinctions in the input and do not overgeneralize across different clause types. The

micro-cue for non-V2 word order in exclamatives in Norwegian and the corresponding empirical data is presented in (15). However, for the rest of this section, focus will be placed on declarative structures presented in (12)-(14).⁴

- (15) a. Micro-cue for non-V2 in exclamatives:
ExIP[wh ... *VP*[V]] (Westergaard 2009b: 188)
- b. kor store mage han har
where/how big stomach he have.PRES
'What a big stomach he has!' (Westergaard 2009b: 188)

The micro-cues provided by Westergaard can be directly translated into spans. For example, the default declarative structure is represented by (16). The span in (17), however, is equally well-formed and its existence does not block the production of the default form. These trees correspond to the micro-cues in (12) and (13), respectively.

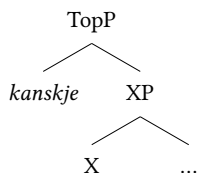
- (16) General declaratives
- 
- (17) Declaratives with a Foc-Adv in second position
- 

Our spanning-based translation of micro-cues diverges from Westergaard’s original conception of them in a way that warrants some attention. Notice that the span in (18) does not fully match the corresponding micro-cue (14). The span does not contain a V node, while the micro-cue does include a V as well as much of the declarative clause. Although this approach is descriptively adequate, it does beg the question as to whether or not it can be simplified on the basis of an appeal to locality. Westergaard (2009a) rightly points out that with respect to the placement of finite verbs in declarative clauses that begin with *kanskje*, not only is V3 a possibility, but so are other non-V2 iterations. This situation contrasts with declaratives that contain a focus-adverbial in second position (17), where the span and micro-cue share a higher degree of symmetry.

⁴It is worth noting that exclamatives in Danish, a closely related Scandinavian language, generally maintain a V2-order.

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- (18) Declaratives beginning with *kanskje* ‘maybe’



In a developing grammar, the span we propose in (18) can be understood to capture the local cue at the left-edge of declarative clauses that ban V2-ordering. Although the general trend to place the finite verb in third position will eventually be reinforced and solidified throughout the acquisition process, the span in (18) captures the local condition that triggers this environment. By restricting the ideal span size to a more local domain, the spans a learner has to store are much smaller. In this respect, assuming that our hypothesis of equating micro-cues with spans is on the right track, we have an instance of spans providing a economic and succinct representation to micro-cues.

Due to time and space considerations, we cannot develop a full and detailed account as to how these various spans are activated and compete with one another. Taking Westergaard’s claim at face value that children are generally ‘conservative learners’ and that they tend to move from specific to more general hypotheses about underlying linguistic representations, they would seem to have a (hyper-)tendency to notice and incorporate these structures into their developing grammar. It would also make sense that more specific, less frequent spans could become difficult to retrieve or activate over time as they are infrequently activated due to competition with more frequent forms. Modeling the competition and potential override of less frequent, more specific spans over more general ones could be achieved by appealing to spell-out rules found in late-insertion, realizational models of morphosyntax such as *Distributed Morphology* (Embick & Noyer 2007) and *Nanosyntax* (Starke 2009, Baunaz et al. 2018).

3.2 Language change & variation

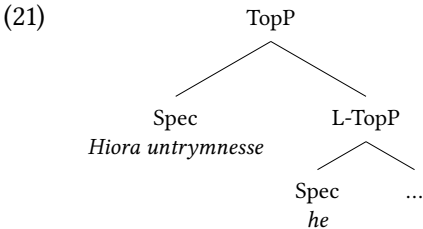
Another domain in which we apply a span-based analysis of micro-cues in a more hyper-local approach is the historical development of V2 and non-V2 word order in Old English (OE) and Middle English (ME), as analyzed by Westergaard (2009c). We claim that language change proceeds through the deactivation and eventual loss of spans because spans that are no longer consistently reinforced in the input gradually weaken and are ultimately eliminated from the grammar. This view allows us to account for the gradual shift from V2 to non-V2 word

order in the development of English. Historically, English exhibited a mixture of word order variation, with both V2 and non-V2 declaratives existing for many centuries. A widely accepted theoretical argument for this diachronic variation is grammar competition, in which two different parameter settings compete with one another. Westergaard (2009b) asserts that approaches supporting grammar competition overemphasize the role of UG by suggesting that UG not only accounts for universal principles but also language variation. Instead, she argues that diachronic language change occurs in small stages, leading to a gradual loss of micro-cues, and meaning language change is a slow process rather than a sudden shift. This can be illustrated in the transition from variable word order to primarily non-V2 in English, both of which can be seen in the OE data in (19).

- (19) a. On his dagum **sende** Gregorius us fulluht.
in his days sent Gregory us baptism
‘In his time, Gregory sent us Christianity.’ (Westergaard 2009c: 66)
- b. & fela ðinga swa gerad man **sceal don**.
and many things so wise man must do
‘And such a wise man must do many things.’ (Westergaard 2009c: 66)

As illustrated in (19), the child acquiring OE is exposed to conflicting triggers in the input. In (19a), the V2 word order is clearly active with the finite verb *sende* (‘sent’) appearing in second position. This indicates that V2 was not only one of the available options, but also an active span within the syntax. From there, non-V2 word order progressed from being restricted to pronominal subjects in the early periods of ME, as in the example in (20) and its corresponding syntactic tree in (21) where the fronting of the object (*Hiora untrymnesse*) in a sentence with a pronomial subject *he* results in V3, to occurring more frequently with both pronominal subjects and full DPs. By late ME, non-V2 was the default, and only specific structures, such as those with initial DP subject, retained a V2 order.

- (20) Hiora untrymnesse he **sceal** ðrowian on his heortan.
their weakness he shall atone in his heart
‘He shall atone in his heart for their weakness.’ (Westergaard 2009c: 73)



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(Westergaard 2009c: 89)

Over the course of time, the general decrease in examples exhibiting a V2-structure with non-DP subjects led to a reduction in the activation of the corresponding span, restricting its use solely to DP contexts. However, the presence of both V2 and non-V2 structures (such as (19) in OE) in the same period and the subsequent ME expansion of acceptable non-V2 environments suggest that speakers during this time period were incrementally reducing the conditions under which V2 applied, rather than suddenly abandoning it altogether.⁵ This aligns with the spanning analysis in (4), where dueling spans reflecting both V2 and non-V2 structures can be present in the grammar. This state of affairs allows individual speakers to match and lexicalize the different – and, in some instances, competing – syntactic structures as represented in (5a) and (5b). Furthermore, the gradual deactivation and eventual elimination of specific spans illustrate how language changes over time. Initially, multiple competing structures may coexist, but as exposure to certain structures decreases, they become increasingly confined to the edge. This selection of specific spans suggests that, rather than undergoing sudden reanalysis, syntactic change progresses through the gradual restructuring of spans/micro-cues that are no longer productive in larger syntactic structures.

We close this subsection by mentioning how a spanning approach to micro-cues can also account for variation found within a language and continuum of related languages. Consider once again the variation in the role that *kanskje* ‘maybe’ plays in Norwegian, as illustrated in (11b) and (18). Both V2 and non-V2 structures are possible when the complementizer *kanskje* is present, but speakers exhibit a strong preference for the non-V2 structure. This variation in verb placement can be accounted for within a span-based approach to micro-cues showing how micro-cues shape variation by allowing syntactic flexibility rather than strict adherence to a single syntactic structure. Westergaard (2009c) notes that information structure also plays a role in distinguishing between V2- and non-V2-ordering in clauses beginning with *kanskje*. *Kanskje* appears in both syntactic structures, but there is a preference for *kanskje* to occur with non-V2 order in spontaneous speech, as shown in (22b).

- (22) a. Micro-cue for non-V2 in declaratives with *kanskje*:
 $_{TopP}[\text{kanskje } _{TP}[\text{SU V...}]]$ (Westergaard 2009c: 84)

⁵More precisely, we observe a shift away from hardwired representational parameters found in previous instantiations of UG in favor of an ‘open’ version of UG that is feature-based (Lightfoot 2020, Slabakova 2009, Lohndal & Putnam 2024, Eguren et al. 2016).

b. kanskje **dem** krangla.

maybe they fight.PAST

‘Maybe they were fighting.’

(Westergaard 2009c: 84)

Taken together, the ‘micro-cues are spans’ hypothesis enables us, as linguists, to consider L1 acquisition, historical development, and variant/dialectal diffusion as related processes. The incremental process of creating small(er) spans, i.e., micro-cues, which are more specific in initial stages of acquisition and change, and are, potentially, later deactivated or eliminated due to lack of sufficient qualitative and quantitative input, is in line with Westergaard’s proposal of acquisition being a conservative process that progresses from specific preferences to more generalized ones. Micro-cues served as a precursor to current spanning-based approaches, which now renders them epiphenomenal, i.e., the operation responsible for generating interpretable syntactic objects is the same one responsible for L1 acquisition, historical change, and variant/dialectal diffusion.

4 Conclusion

In this chapter we have made the case for the use of micro-cues as well-formed syntactic objects in a spanning-based analysis. The spanning approach adopted here provides a more precise heuristic of locality than what is found in the existing literature on micro-cues. An additional attractive facet of adopting the ‘micro-cues are spans’ approach is that the well-formedness principles of what derives a licit syntactic object are isomorphic to those that derive spans. Finally, interpreting micro-cues as spans makes it possible to extend analyses focusing on language acquisition, historical development, and comparative studies (including dialectal distinctions) to the morphemic level (Fábregas & Putnam 2020).

Before concluding, we would like to address a query mentioned to us by an anonymous reviewer concerning how the approach sketched out here is superior to potential competitor approaches (especially with respect to generative/formal ones). Beyond establishing a more consistent algorithm and heuristic of what counts as a (micro-)cue, this nascent approach provides a way to juggle between the *granularity* of the elements of syntactic trees, i.e., spans, that are targeted by language learners and language producers that operate within the restricted confines of a Universal Grammar (Bosch & Biberauer 2024, Roberts 2019, Lohndal & Putnam 2024, Lightfoot 2020). It bears repeating that we do not view this as a weakness of previous research that made use of micro-cues, but rather, as an opportunity to align these previous findings with more recent derivational

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machinery. In closing, if the equivalency of micro-cues and spans is on the right track, we should view Westergaard's proposal as being avant garde and ahead of its time when it was originally developed.

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References

- Baunaz, Lena, Karen De Clercq, Liliane Haegeman & Eric Lander (eds.). 2018. *Exploring nanosyntax*. Oxford: Oxford University Press.
- Biberauer, Theresa. 2019. Factors 2 and 3: Towards a principled approach. *Catalan Journal of Linguistics*. 45–88. DOI: <https://doi.org/10.5565/rev/catjl.219>.
- Blix, Hagen. 2021a. Phrasal spellout and partial overwrite: On an alternative to backtracking. *Glossa: A Journal of General Linguistics* 6(1). 62. 1–17.
- Blix, Hagen. 2021b. Spans in South Caucasian agreement: Revisiting the pieces of inflection. *Natural Language & Linguistic Theory* 39. 1–55.
- Boeckx, Cedric. 2014. *Elementary syntactic structures*. Cambridge: Cambridge University Press.
- Bosch, Núria & Theresa Biberauer. 2024. Emergent syntactic categories and increasing granularity: Evidence from a multilingual corpus study. In Hayat Abdullah, Ali Al Thagafi & Jupitara Ray (eds.), *Proceedings from the 48th annual Boston University Conference on Language Development*, 101–116. Somerville: Cascadilla Press.
- Caha, Pavel, Karen De Clercq & Guido Vanden Wyngaerd (eds.). 2025. *Nanosyntax and the lexicalization algorithm*. Oxford: Oxford University Press.
- Chomsky, Noam. 1995. *The minimalist program*. Cambridge: MIT Press.
- Chomsky, Noam. 2005. Three factors in language design. *Linguistic Inquiry* 36(1). 1–22.
- Diessel, Holger. 2019. *The grammar network. How linguistic structure is shaped by language use*. Cambridge: Cambridge University Press.
- Eguren, Luis, Olga Fernández-Soriano & Amaya Mendikoetxea (eds.). 2016. *Rethinking parameters*. Oxford: Oxford University Press.

- Embick, David & Rolf Noyer. 2007. Distributed Morphology and the syntax/morphology interface. In Gillian Ramchand & Charles Reiss (eds.), *Oxford handbook of linguistic interfaces*, 289–324. Oxford: Oxford University Press.
- Fábregas, Antonio & Michael T. Putnam. 2020. *Passives and middles in Mainland Scandinavian: Microvariation through exponency*, vol. 338. Berlin: Mouton de Gruyter.
- Höder, Steffen. 2012. Multilingual constructions: A diasystematic approach to common structures. In K. Braunmüller & C. Gabriel (eds.), *Multilingual individuals and multilingual societies*, 241–257. Amsterdam: John Benjamins.
- Holme, Randall. 2013. Emergentism, connectionism, and complexity. In Julia Herschensohn & Martha Young-Scholten (eds.), *The Cambridge handbook of second language acquisition*, 605–626. Cambridge: Cambridge University Press.
- Lightfoot, David. 2006. *How new languages emerge*. Cambridge: Cambridge University Press.
- Lightfoot, David. 2020. *Born to parse: How children select their languages*. Cambridge: Cambridge University Press.
- Lohndal, Terje & Michael T. Putnam. 2024. The importance of features and exponents: Dissolving Feature Reassembly. *Linguistic Approaches to Bilingualism* 14(1). 1–36.
- MacWhinney, Brian. 2005. A unified model of language acquisition. In J.F. Kroll & A.M.B. de Groot (eds.), *Handbook of bilingualism: Psycholinguistic approaches*, 49–67. Oxford: Oxford University Press.
- MacWhinney, Brian. 2019. Language attrition and the Competition Model. In Monika S. Schmid & Barbara Köpke (eds.), *The Oxford handbook of language attrition*, 7–17. Oxford: Oxford University Press.
- Osborne, Timothy, Michael T. Putnam & Thomas Groß. 2012. Catenae: Introducing a novel unit of syntactic analysis. *Syntax* 15(4). 354–396.
- Putnam, Michael T. 2020. One Feature - One Head: Features as functional heads in language acquisition and attrition. In Pedro Guijarro Fuentes & Cristina Suárez-Gómez (eds.), *New trends in language acquisition within the generative perspective*, 1–23. Dordrecht: Springer.
- Ramchand, Gillian & Peter Svenonius. 2014. Deriving the functional hierarchy. *Language Sciences* 46. 152–174.
- Roberts, Ian. 2019. *Parameter hierarchies and Universal Grammar*. Oxford: Oxford University Press.
- Roeper, Thomas. 1999. Universal bilingualism. *Bilingualism: Language and Cognition* 2(3). 169–186. DOI: [10.1017/S1366728999000310](https://doi.org/10.1017/S1366728999000310).

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- Slabakova, Roumyana. 2009. Features or parameters: Which one makes second language acquisition easier, and more interesting to study? *Second Language Research* 25(2). 313–324.
- Starke, Michel. 2009. Nanosyntax: A short primer to a new approach to language. *Nordlyd* 36(1). 1–6.
- Stroik, Thomas S. & Michael T. Putnam. 2013. *The structural design of language*. Cambridge: Cambridge University Press.
- Svenonius, Peter. 2016. Spans and words. In Daniel Siddiqi & Heidi Harley (eds.), *Morphological metatheory*, 199–220. Amsterdam: John Benjamins.
- Svenonius, Peter. 2020. A span is a thing: A span-based theory of words. In Mariam Asatryan, Yixiao Song & Ayana Whitmal (eds.), *Proceedings of NELS 50*, vol. 3, 193–206. Cambridge: MIT Press.
- Westergaard, Marit. 2008. Acquisition and change: On the robustness of the triggering experience for word order cues. *Lingua* 118(12). 1841–1863.
- Westergaard, Marit. 2009a. Microvariation as diachrony: A view from acquisition. *The Journal of Comparative Germanic Linguistics* 12(1). 49–79.
- Westergaard, Marit. 2009b. *The acquisition of word order: Micro-cues, information structure, and economy* (Linguistik Aktuell/Linguistics Today 145). Amsterdam: John Benjamins.
- Westergaard, Marit. 2009c. Word order in Old and Middle English: The role of information structure and first language acquisition. *Diachronica* 26(1). 65–102.
- Westergaard, Marit. 2014. Linguistic variation and micro-cues in first language acquisition. *Linguistic Variation* 14(1). 26–45. DOI: [10.1075/lv.14.1.02wes](https://doi.org/10.1075/lv.14.1.02wes).
- Westergaard, Marit. 2021. Microvariation in multilingual situations: The importance of property-by-property acquisition. *Second Language Research* 37(3). 379–407.
- Yang, Charles. 2016. *The price of linguistic productivity: How children learn to break the rules of language*. Cambridge: MIT Press.

Chapter 21

Maintaining word order flexibility in contact with a V2 grammar: Evidence from semi-spontaneous narratives of Norwegian-Russian bilingual children

Yulia Rodina

This study investigates word order patterns in semi-spontaneous spoken narratives produced by Russian-speaking children living in Norway (n=27, mean age=6;7). The analysis of the distribution of subject-initial vs. non-subject-initial declarative main clauses, as well as the verb positioning in relation to overtly expressed subjects and objects, shows that Norwegian-Russian children operate two distinct grammars: the Russian word order grammar, which closely resembles that of monolinguals, and differs from their Norwegian word order grammar. In other words, there is no indication of a quantitative or qualitative reduction in word order flexibility in heritage Russian, nor is there strong evidence that the Norwegian verb second order is affected by cross-linguistic influence from Russian.

1 Introduction

Word order studies on Russian heritage language (HL) in adolescents and adults in contact with English and German have shown a reduction in syntactic variability as well as stability and an increase in variability in the HL word order system (cf., [Laleko 2024](#) for an overview). In this paper, I investigate the word order patterns produced by bilingual Norwegian-Russian children residing in Norway. Unlike previous studies, the comparison is made not only between bilingual

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and monolingual speakers, but also across the two languages of bilinguals. Norwegian and Russian have distinct word order grammars. While Russian shows variation in the ordering of constituents based on information structure requirements, Norwegian follows a verb second (V2) word order where the finite verb must appear in second position in main clauses. Overall, I show that Norwegian-Russian children have a high degree of word order variation in their HL and a robust V2 grammar in Norwegian, their societal language.

This paper is organized as follows. Section 2 outlines the main word order properties of Russian and Norwegian. Section 3 reviews previous research on word order in heritage Russian speakers. The research questions and predictions are presented in Section 4, while Section 5 describes the participants, methodology, and data analysis. Section 6 reports the results of the study, and Section 7 offers a general discussion of the findings.

2 Word order in Russian and Norwegian

Russian is an SVO language, but it exhibits significant flexibility in the ordering of sentence constituents governed by syntax-discourse factors, i.e. the information structure in terms of topic (given/old information) and comment/focus (new information). To illustrate, the examples in (1) are not equivalent in terms of their information structure (from Kallestinova 2007: 1). In contrast to the canonical SVO structure in (1a) typical in neutral contexts, the orderings in (1c and d) follow a given-before-new principle with the fronted object *Ivana* as the topic. The example in (1c) is a felicitous response to the question “Who visited Ivan?”, and the example in (1d) is a possible answer to the question “What did Boris do to Ivan?”. Non-canonical word orders with inversion illustrated in (1e and f) are used frequently to introduce new referents in storytelling in Standard Russian. It should be noted that in the non-canonical examples in (1b-f), SVO orders are possible across all contexts, with the focused constituent receiving prosodic stress *in situ* (cf. Jasinskaja 2016).

- | | | | |
|-----|----|--------------------------------------|-----|
| (1) | a. | Boris navestil Ivana. | SVO |
| | | Boris.NOM visited Ivan.ACC | |
| | | ‘Boris visited Ivan.’ | |
| | b. | Boris Ivana navestil. | SOV |
| | | Boris.NOM Ivan.ACC visited | |
| | c. | Ivana navestil Boris. | OVS |
| | | Ivan. ACC visited Boris.NOM | |

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- d. Ivana Boris navestil. OSV
Ivan.ACC Boris.NOM visited
- e. Navestil Boris Ivana. VSO
visited Boris.NOM Ivan.ACC
- f. Navestil Ivana Boris. VOS
visited Ivan.ACC Boris.NOM

Table 1 shows the proportions of different word order types in spoken and written registers in Russian, based on the overview in [Laleko \(2024\)](#). SVO is the preferred order in both modalities, while the frequency of non-canonical orders varies substantially, with SOV being the next most common order in spoken Russian.

Table 1: Distributional patterns in Russian written and spoken corpora

	SVO	SOV	OSV	OVS	VOS	VSO
Spoken (Miller & Weinert 1998)	42%	34%	11%	1%	0%	0%
Written (Miller & Weinert 1998)	79%	1%	4%	11%	2%	1%
Written (Billings 2015)	89.6%	4.4%	1.8%	2.4%	0.2%	1.6%

Like Russian, Norwegian is an SVO language but also a V2 language. This means that the finite verb occupies the second position in main clauses. V2 word order is only distinguishable from SVO syntax in declarative sentences that begin with a non-subject, such as in (2). For instance, in spontaneous production approximately 30–40% of declaratives start with a non-subject in Germanic V2 languages (cf. [Westergaard 2009](#)). For comparison, the proportions of non-subject-initial declaratives in English are less than 10%, due to a preference for subjects to appear sentence-initially ([Yang 2001](#): 242). In light of this evidence, it will be important to consider the distribution of subject- and non-subject-initial declaratives in the two languages of bilinguals in the present study.

- (2) I går besøkte Boris Ivan. XVSO
yesterday visited. Boris.NOM Ivan.ACC
‘Yesterday, Boris visited Ivan.’

In sum, Norwegian word order is less flexible than Russian, and the structures SOV, OSV, VSO, and VOS in (1b, 1d, 1e, and 1f) are ungrammatical in Norwegian. Nevertheless, object-fronted OVS structures, as in (1c), are allowed in Norwegian

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when discourse referents are familiar to the hearer and function as discourse topics:

- (3) Den bilen, den har jeg sett før. (Borthen 2003: 46)
 that car-DEFSUFF.MASC, it-MASC have I seen before
 ‘That car, I have seen it before.’

3 Previous research

Research on word order in Russian HL has primarily focused on adult HSs in the USA and Germany, examining word order patterns in both written and spoken corpora and comparing them to data from baseline speakers (Polinsky 2006, Isurin & Ivanova-Sullivan 2008, Dubinina & Polinsky 2013, Laleko & Dubinina 2018, Brehmer & Usanova 2015, Zuban et al. 2021). For Heritage Russian in contact with English, the word order patterns are characterized by increased use of SVO and a reduction in word order flexibility/variation such that non-canonical word order structures, including subject-verb inversion (VS, OVS) and object fronting (SOV, OSV) occur infrequently or are completely absent (Isurin & Ivanova-Sullivan 2008, Dubinina & Polinsky 2013, Laleko & Dubinina 2018). For example, in oral narratives adult HSs showed a preference for the canonical SVO (71.15%) word order and used non-canonical word orders with inversion (OVS) and dislocation (OSV) significantly less frequently than monolinguals (Laleko & Dubinina 2018). Similarly, Polinsky (2006), based on the data gathered through interviews, reports the predominant use of SVO with both nominal and pronominal arguments/objects, while SOV is expected with pronominal arguments in homeland/Standard Russian. The longitudinal data of a 9-year-old in Isurin (2005) revealed an increased use of SVO and no inverted orders (VS or XVS) at the end of the study, even though the latter were present in the first recording. Across different studies, overuse of SVO is attributed to cross-linguistic influence (CLI) from English. Some studies assume a connection between inflectional morphology and word order such that the decline in word order variation may correspond to reduced sensitivity to/knowledge of case morphology in HSs (Polinsky 2011, 2018).

In contrast, HSs of Russian in Germany are argued to retain a high degree of word order variation, demonstrating maintenance of the pragmatic flexibility of verb placement in Russian (Brehmer & Usanova 2015, Zuban et al. 2021). Based on the evidence from verb placement in declarative main clauses, Brehmer & Usanova (2015) conclude that the V2/SV position of finite verbs does not occur at the expense of other verb placement patterns since other positions (all of

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which are ungrammatical in German) were attested in written texts of Russian-speaking adolescents, including verbs in the initial, third, fourth or fifth positions. The frequencies of these non-canonical verb placements were low compared to the predominant canonical V2/SV structures. However, the different verb placement patterns showed similarities between HSs and Russian monolinguals. CLI from German was nevertheless attested in subordinate clauses where a statistically significant trend towards V-final was found. This trend was not confirmed for Russian-German bilinguals in Zuban et al. (2021). In the semi-spontaneous spoken and oral narratives, the word order distributions of bilinguals closely resembled those of monolinguals in both main and embedded clauses. In contrast, the study found that Russian-English bilinguals differed by overusing SVO word order compared to both monolinguals and Russian-German bilinguals, yet only in the embedded clauses. Zuban et al. (2021) explored several alternative explanations for the observed asymmetry, including the structural complexity of the embedded clauses and extra-linguistic factors related to the varying communicative contexts used for data elicitation.

Most recently, Laleko (2024) made a more nuanced account of the word order system in heritage Russian based on a series of acceptability judgment tasks testing the effects of information structure (topicality or givenness) on the placement of constituents with English-dominant heritage Russian speakers in the USA at two proficiency levels and Russian baseline speakers. The study uncovered fine-grained asymmetries in the acceptability of the non-canonical orders: with the weakening of OVS and OSV, but the retention and strengthening of SOV. The latter is argued to be an input-driven change that echoes an ongoing change in spoken homeland Russian and provides evidence against convergence on SVO in heritage Russian in an English-speaking environment.

4 Research questions and predictions

The present study examines the acquisition of word order in Heritage Russian in contact with the V2 system of Norwegian, focusing on preschool and school-aged children, and explores word order patterns in both languages. The study addresses two main research questions:

- RQ1: Do Norwegian-Russian children preserve the word orders generated in the baseline languages?
- RQ2: Is there evidence of CLI in the word order patterns of Norwegian-Russian bilingual children?

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To explore whether child bilinguals maintain the flexibility of word order in Russian, it is necessary to compare their word order patterns to those of Russian-speaking monolingual peers.¹ In line with previous research on HR in the USA, it can be predicted that language contact between Russian and Norwegian may lead to a narrowing of word order options and overuse of SV/SVO. At the same time, it may still be possible for the bilinguals to retain a high degree of word order variation, as was found for Russian in contact with a V2 language – German, or an SOV structure in heritage Russian in the USA. Thus, special attention will be given to the distribution of subject-initial and non-subject-initial declarative main clauses as well as verb and object placement in declarative main clauses in both HL Russian and SL Norwegian.

In addition, the role of CLI will be studied through an examination of word order patterns in Russian and Norwegian of bilingual children. In Norwegian, it will be crucial to examine non-subject-initial declaratives. Since V2 languages typically have a high proportion of non-subject-initial declaratives (the context for V2) in spontaneous speech, the children are expected to produce a high number of non-subject-initial declaratives in Norwegian where V2 is target-like. V2 violations and a reduced proportion of non-subject-initial declaratives may signal CLI from Russian. A combination of non-V2 word order and a severe reduction in non-subject-initial declaratives has been found in heritage Norwegian in the USA (Westergaard & Lohndal 2019). However, in German, the societal language of German-Russian bilingual adolescents in Brehmer & Usanova (2015), the target-like V2 positioning of the verb was produced 90% of the time. This study does not report on the proportion of non-subject-initial declaratives. Finally, in Russian, CLI from Norwegian can be observed in verb placement preferences: V1, V2, V3, and V4 are all possible in Russian, with V2 and V3 occurring most frequently (cf. Brehmer & Usanova 2015). Therefore, an overuse of V2 and a lack of other verb positions could indicate CLI from Norwegian.

5 Research questions and predictions

The data in the present study were collected within the contexts of two different larger research projects *Micro-variation in Multilingual Acquisition and Attrition*

¹Comparison with Russian monolinguals is necessary because of the variable word order patterns in Russian. In the absence of a similar word order flexibility in Norwegian, it was less crucial to include Norwegian controls in the present study. Furthermore, the data from Norwegian monolinguals collected in the projects presented in the Methodology section were limited to a small sample of preschool children only and, therefore, cannot serve as an adequate comparison group.

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Situations (MiMS) and *Bilingual Acquisition: Russian-Norwegian children in Norway* (BARN). Findings from these projects based on narrative data analyses are reported in [Mitrofanova et al. \(2018\)](#), [Rodina et al. \(2023\)](#), and [Rodina \(2016\)](#).

The present study is based on the language samples of 27 Norwegian-Russian children (mean age 6;7, age range 4;5–9;6) born in Norway and 20 Russian monolinguals (mean age 8;3, age range 4;0–9;6) from Ivanovo, Russia. Fifteen bilinguals were from mixed Norwegian-Russian households (i.e. families with one Russian- and one Norwegian-speaking parent) and 12 were from Russian-speaking families. Ten children attended Norwegian preschools at the time of testing, while 17 went to public schools with instruction in Norwegian. Half of the monolingual children attended preschool; the other half attended public schools. The language samples were obtained using narrative elicitation material in the *Multilingual Assessment Instrument for Narratives* (MAIN) ([Gagarina et al. 2019](#)). Two MAIN stories were used to elicit oral narratives with bilinguals: “Baby Goats” in Norwegian and “Baby Birds” in Russian. The monolingual participants were presented with “Baby Birds”. The elicitation procedure was conducted according to the MAIN guidelines. The bilingual children were tested on two separate days, with one session for each language, approximately a week apart.

The narrative data of Russian monolinguals (L1 Rus), bilinguals in Russian (2L1 Rus), and bilinguals in Norwegian (2L1 Nor) were transcribed and manually annotated for clause type (main and subordinate), position of the verb (V1, V2, V3, etc.), presence of a sentential adverbial (X), and word order pattern (S, V, O). Incomprehensible and incomplete utterances were excluded from the analysis. In both languages, only declarative main clauses with overt subjects and finite verb forms were included in the analysis. The following structures were excluded in Russian: seven idiomatic expressions like *žili-byli ptički* ‘once upon a time there were birds’ and five possessive constructions with the preposition *u* ‘at’ illustrated in (4). The latter were excluded due to their unique syntactic properties, the analysis of which falls outside the scope of the present paper. For the same reason, 38 Norwegian main clauses with the pronoun *det* ‘there/it’ in subject position – e.g., *Det kom en rev* ‘There came a fox’ – were also excluded.

- (4) u mamy byli ptenčiki. (L1 Rus, age 7;4)
 at mother.GEN were baby-birds.NOM
 ‘Mother had baby-birds.’

After these exclusions, the analysis was based on 637 declarative main clauses in total and focused on

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1. the distribution of subject-initial vs. non-subject-initial declarative main clauses with a sentential adverbial (X),
2. the positioning of the verb in relation to the subject in clauses with overtly expressed subjects (i.e., the SV – VS distinction), and
3. the positioning of the verb in relation to overtly expressed subject and direct or indirect object (i.e., the distinction between SVO, SOV, VSO, VOS, OVS, OSV patterns).

6 Results

I first present the frequency data for subject-initial and non-subject-initial declarative main clauses, followed by the analysis of verb placement in declarative main clauses and finally by the analysis of main clauses with objects. Statistical analysis of the data was performed using mixed logistic regression modeling with *glmer* from the *lme4* package in *R Studio* (Bates et al. 2015). In line with the research objectives outlined in Section 4, the analysis focuses on comparing word order patterns in the two languages of bilinguals. Additionally, it includes a comparison of bilinguals' speech patterns in Russian with those of Russian-speaking monolinguals.

6.1 The distribution of subject-initial and non-subject-initial declarative main clauses

In total, bilinguals produced 257 declarative main clauses in Norwegian and 177 in Russian. Russian-speaking monolinguals produced 203 clauses. The lower number of declarative sentences in heritage Russian compared to Norwegian can be attributed to the shorter narratives produced in Russian. Reduced narrative productivity in heritage Russian was previously observed by Rodina et al. (2023), based on a larger dataset that also included participants from the present study. Table 2 presents the frequencies of subject-initial and non-subject-initial declarative main clauses produced by bilinguals and monolinguals. Eight additional structures in Russian with VS word order are not included in Table 2 and are discussed separately below.

In Russian, bilinguals produced similar proportions of subject-initial and non-subject-initial declaratives: 49% (87/177) and 48% (85/177) respectively. Subject-initial declaratives had a higher frequency than non-subject-initial declaratives in the Russian monolingual data, 70% (143/203) vs. 28% (57/203). The opposite

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pattern occurred in the bilinguals’ Norwegian productions, where non-subject-initial declaratives had a higher frequency than subject-initial structures, 66% (170/257) vs. 34% (87/257), respectively. All bilinguals produced non-subject-initial declaratives in Norwegian except one child (aged 6;3). To analyze the proportions of non-subject initial declaratives both with and without objects, I fitted a mixed linear logistic regression for Non-subject-initial as the dependent variable with Language (2L1 Nor, 2L1 Rus, L1 Rus) as a fixed factor/main effect and Participant as a random factor. The model showed a significant effect of Language on the production of non-subject-initial clauses. Bilinguals were significantly more likely to use non-subject-initial structures in Norwegian than in Russian ($\beta = 0.89$, $SE = 0.22$, $z = 4.07$, $p < .001$). Furthermore, **Russian monolinguals** were significantly less likely to produce non-subject-initial orders ($\beta = -0.88$, $SE = 0.30$, $z = -2.96$, $p = .003$).

Table 2: Word order frequencies in subject-initial and non-subject-initial declarative main clauses

	Non-subject-initial declaratives		Total	Subject-initial declaratives		Total
	+object	–object XSV/XVS		+object	–object	
2L1 Nor	72	11/87	170	33	54	87
2L1 Rus	36	33/16	85	36	51	87
L1 Rus	22	26/9	57	79	64	143

The target-like XVS order was used 88% (87/98) of the time in Norwegian. Eight bilinguals produced 11 errors violating V2, i.e. XSV order. Both XVS and XSV were used by bilinguals and monolinguals in Russian. However, in contrast to Norwegian, the XSV order was more frequent than XVS in the Russian data of both bilinguals and monolinguals: bilinguals produced 33 XSV and 16 XVS constructions, while monolinguals produced 26 XSV and nine XVS constructions.

Additionally, verb initial (VS) structures were found in the Russian data of both bilinguals and monolinguals – five and three structures respectively. All of these structures were used to introduce new referents, as illustrated in the example in (5). VS structures were never used by bilinguals in Norwegian. Monolinguals also produced two VSO structures introducing new referents, presented in Table 3 below.

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- (5) prišla koška. (2L1 Rus, age 7;3)
 came cat
 ‘A cat came.’

6.2 Verb placement in declarative main clauses

Figure 1 shows the distribution of verb placement in the total production of bilinguals and Russian-speaking monolinguals. In Russian, both bilinguals and monolinguals allow verbs in the initial, second, third, and fourth positions, with V2 being the most frequent structure, followed by V3. V2 was the most frequent order in Norwegian (92%). For the two most frequent word orders in both languages – V2 and V3, I fitted two mixed linear logistic regression models with V2 or V3 as the dependent variable, Language (2L1 Nor, 2L1 Rus, L1 Rus) as a fixed factor, and Participant as a random factor. There was a significant effect of Language in both models. The children were significantly more likely to use V2 in Norwegian than in Russian, regardless of whether Russian was spoken by bilinguals or monolinguals ($\beta = 2.41$, $SE = 0.30$, $z = 8.14$, $p < .0001$). No significant difference emerged between bilinguals and monolinguals in Russian ($\beta = 0.18$, $SE = 0.27$, $z = 0.67$, $p = .501$). For V3, the children were significantly less likely to produce V3 word order in Norwegian compared to Russian bilingual production ($\beta = -1.97$, $SE = 0.30$, $z = -6.50$, $p < .001$). Furthermore, there was no significant difference between Russian monolinguals and bilinguals in Russian ($\beta = -0.09$, $SE = 0.31$, $z = -0.30$, $p = .766$).

Eight bilinguals produced 17 non-target-like V3 structures in Norwegian: 11 XSV, 6 XSVO. None of them used V3 exclusively. Two bilinguals (aged 5;5 and 8;6) were responsible for ten of the errors. Additionally, an exceptional but target-like V3 pattern was found in three structures with the adverb *bare* ‘just’, illustrated in (6a). The typical V2 word order is also grammatical with this adverb and one bilingual child was sensitive to the variation in word order with it and was able to use both V2 and V3 orders, as in (6).

- (6) a. hun bare *løpte. (2L1 Nor, age 8;6)
 she just ran
 ‘She just ran.’
 Target form: *Hun bare løp.*
- b. og mamman var bare i vannet. (2L1 Nor, age 8;6)
 and mother was just in water
 ‘And mother was just in the water.’

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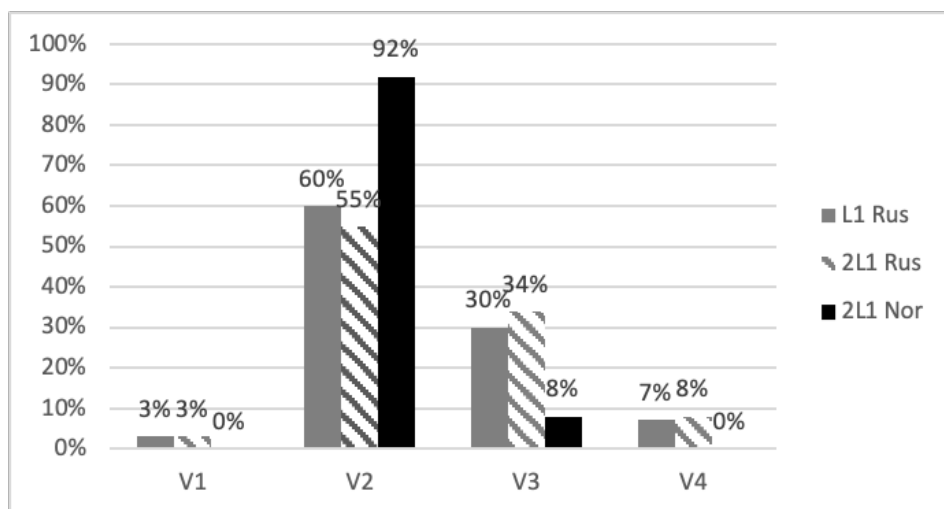


Figure 1: Verb placement in declarative main clauses

In Russian, the V2 order was used exclusively by six bilinguals. To compare, 21 out of 27 bilinguals used V3 word order in Russian, illustrated in (7). One child used V3 exclusively; others used a combination of various positions. V3 occurred even in the data of low proficiency children, who made case and gender errors, as shown in (8).

- (7) a sobaka potom ukusila xvost. (2L1 Rus, age 6;3)
and dog.NOM.SG then bit tail.ACC.SG
'And then the dog bit (the fox's) tail.'
- (8) i potom koška *begal za *sobačka. (2L1 Rus, age 4;9)
and then cat.F.NOM.SG ran.NOM.SG after dog.F.NOM.SG
'And then the cat was running after the dog.'
Target form: *I potom koška begala za sobačkoj.*

6.3 Analysis of main clauses with objects

In total, structures containing S, V, and O were 101/203 (50%) in L1 Rus, 72/177 (41%) in 2L1 Russian, and 105/257 (41%) in 2L1 Norwegian. In the subject-initial main clauses presented in Table 3, SVO was the most frequent word order: 91% (30/33) in 2L1 Nor, 86% (31/36) in 2L1 Rus, and 76% (60/79) in L1 Rus. SOV was the next most frequent word order in Russian monolinguals who also used OSV,

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OVS, and VSO structures. Non-canonical word orders (SOV, OSV, OVS) were also used by bilinguals in Russian, although very infrequently. OVS was the only non-canonical structure in 2L1 Nor. To analyze the proportion of SVO out of all subject-initial declaratives with objects, I fitted a mixed linear logistic regression model with SVO as the dependent variable, Language (2L1 Nor, 2L1 Rus, L1 Rus) as a fixed factor, and Participant as a random factor. There were no significant differences in SVO use between 2L1 Rus and either L1 Russian ($\beta = -0.79$, $SE = 0.65$, $z = -1.22$, $p = .224$) or 2L1 Norwegian ($\beta = 0.38$, $SE = 0.82$, $z = 0.47$, $p = .641$).

Table 3: The distribution of different word order types in subject-initial main clauses with objects

	SVO	SOV	OSV	OVS	VSO	Total
2L1 Nor	30	0	0	3	0	33
2L1 Rus	31	3	1	1	0	36
L1 Rus	60	15	1	1	2	79

Finally, Table 4, shows the distribution of structures containing S, V, and O in non-subject initial main clauses. In Norwegian, bilinguals used the target-like XVSO 89% of the time (64/72). Bilinguals made six errors violating V2 in Norwegian, i.e. XSVO order. Only three XVSO structures occurred in 2L1 Rus. XSVO was the most frequent order in Russian: 78% (28/36) in 2L1 Rus and 73% (16/22) in L1 Rus. Russian monolinguals used four other non-canonical orders, two of which were also produced by bilinguals in Russian (XSOV and XVSO). To analyze the proportion of XSVO out of all non-subject-initial declaratives with objects, I fitted a mixed linear logistic regression model with XSVO as the dependent variable, Language (2L1 Nor, 2L1 Rus, L1 Rus) as a fixed factor, and Participant as a random factor. The model revealed a significant effect of Language. The model showed that there was no difference in the proportion of XSVO structures in the Russian data of bilingual and monolingual children ($\beta = -0.23$, $SE = 0.44$, $z = -0.51$, $p = .608$). Bilinguals were significantly less likely to produce XSVO in Norwegian than in Russian ($\beta = -2.63$, $SE = 0.49$, $z = -5.35$, $p < .001$).

7 Discussion and conclusions

Semi-spontaneous oral narrative data from Norwegian-Russian bilingual children and a group of Russian-speaking monolinguals were analyzed to explore

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Table 4: The distribution of different word order types in non-subject-initial main clauses with objects

	XSVO	XSOV	XOVS	XVSO	XVOS	Total
2L1 Nor	6	0	0	64	2	72
2L1 Rus	28	5	0	3	0	36
L1 Rus	16	3	1	1	1	22

word order patterns in the societal and heritage languages of bilinguals, focusing on word order variation, restructuring, and CLI.

Overall, the analysis of word order patterns in main clauses with and without objects showed that bilinguals behaved similarly to monolinguals regarding the distribution of all word order patterns in Russian, while showing differences between Russian and Norwegian. First of all, this is evident in the patterns obtained for subject-initial and non-subject-initial declaratives (Table 2). A characteristic feature of Norwegian is a high proportion of non-subject-initial declaratives which triggers the activation of the V2 grammar (Putnam & Sánchez 2013). This feature seems to be preserved in the current data sample since the proportion of non-subject-initial declaratives is significantly higher in Norwegian (66%) than in heritage Russian (48%). The comparison with Russian monolinguals also reveals a significantly higher proportion of non-subject-initial declaratives in heritage Russian. However, unlike Norwegian, where the target-like XVS order was used 88% of the time, the distribution of XVS and XSV orders differed in heritage Russian. It was more similar to baseline Russian, where XSV occurred predominantly – 67% in heritage and 74% in baseline Russian speakers.

This pattern receives further support in the analysis of verb placement (Figure 1). Similarly to German-Russian bilinguals in Brehmer & Usanova (2015), Norwegian-Russian bilinguals exhibit no extension of the V2 position of finite verbs at the expense of other verb placement patterns. Not only V2 but also V1, V3, and V4 orders are found in heritage Russian in proportions similar to those of Russian monolinguals. The proportion of V2 structures is also significantly lower in heritage Russian (55%) than in Norwegian (92%) since V3 is a second-preferred option for bilinguals in Russian (34%). V3 word order was used by 21 bilinguals and only six children used V2 exclusively in Russian. It should be noted that the V1 order, ungrammatical in Norwegian but typical for introducing new referents in storytelling in Standard Russian, was found in the Russian bilingual and monolingual data only, although infrequently. Thus, there is no “narrowing of

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options” with respect to verb placement in heritage Russian in Norway. The V2 word order in Norwegian is also generally maintained, with occasional (17/257) V2 violations made by ten children.

Finally, the distribution of structures containing S, V, and O in subject-initial and non-subject-initial main clauses was conducted to draw parallels with previous research on heritage Russian adults in the USA where overuse of SVO was attributed to CLI from English, a strict SVO language. The fact that both Norwegian and Russian have SVO as the basic underlying word order is reflected in the bilinguals’ production where SVO was the most frequent word order for all participant groups (Table 3). The overuse of SVO is not attested in the Norwegian-Russian data since no differences in the proportions of SVO in Norwegian and heritage Russian as well as in heritage Russian and baseline Russian were found. Note, however, that the percentage of SVO was still lowest in Russian monolinguals – 76% vs. 86% in heritage Russian and 91% in Norwegian. With regards to the retention of word order flexibility in main clauses with objects where Russian allows SVO, SOV, OSV, VSO, and VOS, while Norwegian allows SVO and OVS, the data are less clear. Like monolinguals, bilinguals produced three non-canonical orders in Russian – SOV, OSV, and OVS. However, these orders were used sporadically (14%, 5/36) by only three children. In contrast, 14 monolinguals used non-canonical orders more frequently (24%, 19/79), with SOV being the most common non-canonical order. The distribution of canonical and non-canonical orders with objects in the present study is different from what was found for adult Russian heritage speakers in Germany and the USA by Zuban *et al.* (2021). In their corpus investigation, SVO was also the most frequent order but it occurred with lower frequencies (between 55% and 65%) in heritage and baseline Russian speakers. All other non-canonical orders were attested with the highest frequencies for OVS (between 17% and 29%) and SOV (between 8% and 11%). It may still be the case that the differences between the results of the present study and Zuban *et al.* (2021) are due to the differences in the sample size. Thus, further evidence on the word order flexibility in main clauses with objects is needed to provide more definitive conclusions.

To conclude, the Norwegian-Russian bilingual children in the present study operate with two grammars in their language learning: the Russian word order grammar of the bilingual children closely resembles that of monolinguals and differs from the Norwegian word order grammar. The distributional patterns in declarative main clauses with and without objects suggest that there is no loss of flexibility in word order in heritage Russian. The influence of Russian on Norwegian is minimal, if it exists at all. Further research addressing the effects

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of information structure on word order variation is needed to confirm these conclusions.

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References

- Bates, Douglas, Martin Maechler, Ben Bolker & Steve Walker. 2015. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* 67(1). 1–48.
- Billings, Stephanie Kay. 2015. *A corpus-based analysis of Russian word order patterns*. Brigham Young University. (MA thesis).
- Borthen, Kaja. 2003. *Norwegian bare singulars*. Norwegian University of Science & Technology. (Doctoral dissertation).
- Brehmer, Bernhard & Irina Usanova. 2015. Let's fix it? Cross-linguistic influence in word order patterns of Russian heritage speakers in Germany. In Holger Peukert (ed.), *Transfer effects in multilingual language development*, 161–188. Amsterdam: John Benjamins.
- Dubinina, Irina & Maria Polinsky. 2013. Russian in the U.S. In Michael Moser & Maria Polinsky (eds.), *Slavic languages in migration*, 161–187. Wien: LIT.
- Gagarina, Natalia, Daleen Klop, Sari Kunnari, Koula Tantele, Taina Välimaa, Ute Bohnacker & Joel Walters. 2019. *MAIN: Multilingual assessment instrument for narratives. Revised version* (ZAS Papers in Linguistics 63).
- Isurin, Ludmila. 2005. Cross-linguistic transfer in word order: Evidence from L1 forgetting and L2 acquisition. In James Cohen, Kara T. McAlister, Kellie Rolstad & Jeff MacSwan (eds.), *Proceedings of the 4th international symposium on bilingualism*, 1115–1130. Somerville: Cascadilla Press.
- Isurin, Ludmila & Tanya Ivanova-Sullivan. 2008. Lost in between: The case of Russian heritage speakers. *Heritage Language Journal* 6(1). 72–104. DOI: [10.46538/hlj.6.1.4](https://doi.org/10.46538/hlj.6.1.4).
- Jasinskaja, Katja. 2016. Information structure in Slavic. In Caroline Féry & Shinichiro Ishihara (eds.), *The Oxford handbook of information structure*, 709–732. Oxford: Oxford University Press.

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- Kallestinova, Elena D. 2007. *Aspects of word order in Russian*. University of Iowa. (Doctoral dissertation).
- Laleko, Oksana. 2024. The complexity of word order change in a flexible system: On stability and variation in heritage Russian word order. In Maria Polinsky & Michael T. Putnam (eds.), *Formal approaches to complexity in heritage language grammars*, 73–99. Berlin: Language Science Press.
- Laleko, Oksana & Irina Dubinina. 2018. Word order production in heritage Russian: Perspectives from linguistics and pedagogy. In Sabine Bauckus & Sofia Kresin (eds.), *Connecting across languages and cultures: A heritage language festschrift in honor of Olga Kagan*, 191–215. Bloomington: Slavica.
- Miller, Jim & Regina Weinert. 1998. *Spontaneous spoken language: Syntax and discourse*. Oxford: Oxford University Press.
- Mitrofanova, Natalia, Yulia Rodina, Olga Urek & Marit Westergaard. 2018. Bilinguals' sensitivity to grammatical gender cues in Russian: The role of cumulative input, proficiency, and dominance. *Frontiers in Psychology* 9. 1894. DOI: [10.3389/fpsyg.2018.01894](https://doi.org/10.3389/fpsyg.2018.01894).
- Polinsky, Maria. 2006. Incomplete acquisition: American Russian. *Journal of Slavic Linguistics* 14. 191–262.
- Polinsky, Maria. 2011. Reanalysis in adult heritage language: A case for attrition. *Studies in Second Language Acquisition* 33(2). 305–328.
- Polinsky, Maria. 2018. *Heritage languages and their speakers*. Cambridge: Cambridge University Press.
- Putnam, Michael T. & Liliana Sánchez. 2013. What's so incomplete about incomplete acquisition? A prolegomenon to modeling heritage language grammars. *Linguistic Approaches to Bilingualism* 3(4). 478–508.
- Rodina, Yulia. 2016. Narrative abilities of preschool bilingual Norwegian-Russian children. *International Journal of Bilingualism* 21(5). 617–635.
- Rodina, Yulia, Alexandra Bogoyavlenskaya, Natalia Mitrofanova & Marit Westergaard. 2023. Russian heritage language development in narrative contexts: Evidence from pre- and primary-school children in Norway, Germany, and the UK. *Frontiers in Psychology* 14. 1101995. DOI: [10.3389/fpsyg.2023.1101995](https://doi.org/10.3389/fpsyg.2023.1101995).
- Westergaard, Marit. 2009. *The acquisition of word order: Micro-cues, information structure, and economy* (Linguistik Aktuell/Linguistics Today 145). Amsterdam: John Benjamins.
- Westergaard, Marit & Terje Lohndal. 2019. Verb second word order in Norwegian heritage language: Syntax and pragmatics. In David W. Lightfoot & Jonathan Havenhill (eds.), *Variable properties in language: Their nature and acquisition*, 91–102. Washington: Georgetown University Press.

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- Yang, Charles. 2001. Internal and external forces in language change. *Language Variation and Change* 12(3). 231–250.
- Zuban, Yulia, Maria Martynova, Sabine Zerbian, Luka Szucsich & Natalia Gagarina. 2021. Word order in heritage Russian: Clause type and majority language matter. *Russian Linguistics* 45. 253–281.

Chapter 22

Exploring the Micro-Cue Model: Contextual and referential effects on sentence comprehension in Russian-German bilingual children

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Preschoolers often exhibit difficulties comprehending non-canonical OVS sentences, possibly due to factors related to processing, information structure and/or case marking. We examined how topic-first contexts, pronominal topics and language proficiency influence offline OVS comprehension in the dual languages of Russian-German bilingual. The topic-first context did not influence OVS accuracy in neither language. When the object was realized as a pronoun in OVS sentences, comprehension accuracy improved in Russian but not in German. We discuss the results by adapting the Micro-cue model (Westergaard 2009, 2014) to comprehension: a) children start from more specific structures that are later generalized, b) children are sensitive to information structure from early on, and c) comprehension is influenced by issues of economy.

1 Introduction

1.1 The Micro-cue model

Acquiring word order variation can be seen as interaction/interplay between information structure, processing and economy (e.g. Westergaard 2009, 2014). In this paper, we relate the Micro-cue model (Westergaard 2009, 2014) that aims to

account for this complex relationship to word order comprehension in bilingual children.

The Micro-cue model accounts for L1 word order acquisition from a generative perspective. Children acquire their language incrementally. They use Universal Grammar (UG) and the grammar build so far to parse the input, and revise their grammar when parsing fails. Moreover, children's application of syntactic operations is influenced by economy (their occurrence in the input) and contextual factors. Initially, word order variation may only occur in very specific contexts that are defined by micro-cues and only later generalized to other contexts. Micro-cues are language-specific cues and may take specific forms, e.g. specific lexical items, information structure characteristics, type of referring expression (Westergaard 2014).

In bilingual child language acquisition, the same basic mechanisms are involved, yet the entire language repertoire might be used during parsing (Westergaard 2021, see Mitrofanova et al. 2022 for empirical support).

In the present study, we investigate the impact of information structure and the type of referring expression on the comprehension of subject- and object-initial transitive sentence. We will apply three basic assumptions of the Micro-cue model to sentence comprehension: a) children start from more specific structures (e.g. involving pronominal topic referents) that are later generalized (e.g. to lexical NPs), b) children are sensitive to information structure from early on, and c) comprehension is influenced by issues of economy, e.g., pronominal topic referents may ease comprehension because sentence-initial given objects are realized as pronouns in child-directed speech (cf. Fleischhauer 2009).

1.2 Comprehension of word order and case in children

Many languages allow some word order flexibility concerning the ordering of the subject and object in a sentence. For instance, in languages like German, the subject usually precedes the direct object (SVO), see (1), but the object before subject word order (OVS) in (2) is also possible.

- (1) **Der** Hahn fängt gleich **den** Igel.
the.NOM rooster catches now the.ACC hedgehog
'The rooster will catch the hedgehog.'
- (2) **Den** Hahn fängt gleich **der** Igel.
the.ACC rooster catches now the.NOM hedgehog
'The hedgehog will catch the rooster.'

In these examples, case marking at the article indicates the subject vs. object status of the noun phrases (NPs). In (1), nominative case of the article *der* indicates that the first NP *der Hahn* is the subject, and accusative case of the article *den* indicates that the second NP *den Igel* is the object. In (2), case marking indicates that the first NP is the object and the second NP the subject.

Preschoolers often have difficulties comprehending sentences in the OVS order if they cannot rely on cues such as animacy or plausibility (e.g., Dittmar et al. 2008, Cristante & Schimke 2018). These difficulties might arise from several reasons: First, they may not have fully acquired the case system yet (Dittmar et al. 2008, Smolik 2015). Second, OVS sentences may impose processing difficulties because they are syntactically more complex (Frazier 1987) or because children weight morphological information different from adults (Slobin & Bever 1982). Third, OVS sentences are usually infelicitous without an appropriate context (Crain & Thornton 1998).

Children acquiring Slavic languages such as Russian usually master the comprehension of OVS sentences earlier compared to children acquiring Germanic languages (Slobin & Bever 1982, Sauermann & Gagarina 2018). Slavic languages usually have more word order flexibility and a transparent case marking system with case marking on the noun itself (Xanthos et al. 2011, Slobin & Bever 1982). Examples (3) and (4) illustrate case marking of feminine nouns in Russian: the final inflection -a indicates nominative case and the inflection -u accusative case.

- (3) Obez'ana sejčas pojmaet myšku.
 monkey. NOM now catches.PVF mouse.ACC
 'The monkey will catch the mouse.'
- (4) Obez'anu sejčas pojmaet myška.
 monkey.ACC now catches.PVF mouse.NOM
 'The mouse will catch the monkey.'

Nevertheless, 3- to 4-year-old children acquiring Slavic language like Czech may have comprehension difficulties if test sentences involve nouns with less transparent or ambiguous case marking (Smolik 2015).

Non-canonical sentence comprehension can be improved if the test sentences are presented with supportive context (e.g., Grünloh et al. 2011), although not all contexts improve comprehension accuracy (Sauermann & Höhle 2016, Smolik 2015). Particularly relevant to our study are the studies by Smolik (2015), Sauermann & Höhle (2016) and Sauermann & Gagarina (2018).

Smolik (2015) investigated the impact of given-before-new vs. new-before-given contexts and receptive vocabulary skills on SVO and OVS comprehension

in monolingual Czech learning 2;9-to-4;7-year-olds, using a picture-selection task. Children with high lexicon comprehended OVS sentences above chance level, and children with low lexicon at chance level. Context did not influence sentence comprehension, but in low lexicon children OVS comprehension was marginally higher in new-before-given context compared to given-before-new context. This is in line with findings from Grünloh et al. (2011) and Sauermann & Höhle (2016) who found that sentence-initial contrastive focus can ease OVS comprehension in the German-speaking 4–5-year-olds.

Sauermann & Höhle (2016) and Sauermann & Gagarina (2018) examined the impact of the topic-first context (i.e., given-before-new), and type of referring expression in German and Russian monolingual 4- and 5-year-olds using a variant of the truth value judgement task. Sentences such as (1)-(2) and (3)-(4) were presented in an neutral all-new context (*“Look what is happening.”*) or in a topic-first context, that marks the first NP as an aboutness topic (cf. Krifka 2008) (*“Now you hear a story about the rooster/monkey.”*). In a third condition, the sentence-initial object in the topic-first context was realized as a pronoun while the subject was realized as a lexical NP, see (5) and (6).

- (5) **Ihn** fängt gleich der Igel.
 he.ACC catches now the.NOM hedgehog
 ‘The hedgehog will catch him.’
- (6) **Eë** sejčas pojmaet myška
 she.ACC now catches.PVF mouse.NOM
 ‘The mouse will catch her.’

In Russian monolinguals, neither context nor the type of referring expression influenced comprehension accuracy because they already performed at ceiling. In German monolinguals, the topic-first context per se did not influence comprehension accuracy: when the subject and object were realized as lexical NPs, OVS comprehension was at guessing rate. However, when the object was realized as a pronoun, comprehension accuracy was above guessing rates, suggesting that non-canonical sentence comprehension may be improved when one of two referents is realized as a pronoun (cf. Friedmann et al. 2009)

Bilingual preschoolers were shown to have lower comprehension accuracy compared to their monolingual peers (e.g., Boila 2024, Meir et al. 2020, Cristante & Schimke 2018). These differences may result from different processing strategies arising from the influence of another language and/or cognitive factors (Janssen et al. 2015, Pléh et al. 1987). Alternatively, mono- and bilingual children

may use the same processing mechanism but differ in proficiency or grammar development (e.g. [Cristante & Schimke 2018](#), [Meir et al. 2020](#), [Mitrofanova et al. 2022](#)).

[Boila \(2024\)](#) and [Cristante & Schimke \(2018\)](#) examined OVS sentence comprehension in the majority language German in Italian-German bilingual preschoolers and Turkish-German 7-year-olds using eye-tracking. The bilingual children in both studies had lower comprehension accuracy compared to their monolingual German-speaking peers and, in contrast to the monolingual peers, were not sensitive to the case marking cue at the object-initial NP during processing.

[Meir et al. \(2020\)](#) showed that Hebrew-Russian preschoolers may transfer processing strategies of subject- and object-initial sentences from one language to the other. The bilingual children were less accurate in the production of case marking in Russian and Hebrew compared to monolingual preschoolers. Eye-tracking results revealed that Russian monolinguals used case marking early for predictive processing. The bilingual children were also sensitive to case marking at the sentence-initial noun in Russian, but the integration of the cue was delayed. Monolingual Hebrew children did not use case marking for predictive processing. The bilingual children, however, showed a sensitivity to case marking at the first NP and an advantage for case integration at the second NP, suggesting that they may transfer the use of case for sentence processing from Russian to Hebrew.

1.3 The present study

The present study contributes to the discussion whether bilingual and monolingual children rely similarly on information structure factors during offline sentence comprehension. Specifically, we build up on the studies by [Sauermann & Höhle \(2016\)](#) and [Sauermann & Gagarina \(2018\)](#) and use a variant of the truth-value judgement task to examine whether a topic-first context and the type of referring expression affect OVS comprehension in bilingual Russian-German preschoolers. In addition, we consider whether language proficiency may account for individual differences in comprehension accuracy (cf. [Smolik 2015](#)), i.e., in the integration of contextual vs. grammatical information during sentence comprehension.

It is hypothesized that bilingual and monolingual children are sensitive to similar information structure factors. Therefore, it is predicted that the type of referring expression influences accuracy in bilingual children. That is, it is expected that bilingual children demonstrate higher OVS comprehension accuracy if the topic object is realized as a pronoun rather than as a lexical NP. The topic-first context may improve comprehension in Russian because it is a felicitous context

in Russian (Bailyn 2012). In German, it may not influence or impair comprehension, because previous comprehension studies from the monolingual German-learning children did not find clear effects, and preschoolers place given and new objects in the first position of OVS sentences in spontaneous speech (Fleischhauer 2009).

2 Method

2.1 Participants

27 bilingual typically-developing Russian-German children (mean age: 5;01, range 4;5 - 5;9; 7 boys) were tested in two bilingual Russian-German kindergartens in Berlin. All children acquired Russian from birth, with at least one parent being a Russian L1 speaker. Onset of acquisition for German varied from age 0 to 3.

Proficiency in Russian was assessed by the Russian language proficiency test for multilingual children (SRUK, Gagarina et al. 2010) that assessed the productive and receptive lexicon for nouns and verbs, the production of morphology (verb flexion and case) and comprehension of syntax (analogous to TROG-D). Particularly relevant for the present study was the subtest on case marking which elicited the production of accusative and dative singular using questions about events depicted on pictures.

Proficiency in German was assessed using the TROG-D test (Fox-Beyer 2006), which tested the comprehension of various grammatical constructions and the cross-linguistic lexical task (CLT, Rinker & Gagarina 2017) to assess the production and comprehension of nouns and verbs.

2.2 Design and materials

The experiment had a 2 x 3 within-participant design, with Word order (SVO vs. OVS) and Context (newNP vs. topicNP vs. topicPron¹) as independent variables. Word order was modified by the case marking at the noun (in Russian, Table 1) or article (in German, Table 2) of both arguments indicating the subject or object status of the argument. Context captured the information status and the realization of the sentence-initial referent.

¹We choose the label *newNP* to indicate that the first NP has no special status in this condition because the *neutral* context is used. The first referent in the *topicNP* and *topicPron* condition is an *aboutness-topic* but also *given*. Yet, we acknowledge that information structure dimensions usually distinguish *given* vs. *new* and *topic* vs. *comment* (Krifka 2008).

In the *newNP* context, the context sentence (“*Look what is happening here?*”) did not mention any of the referents and both referents were realized as lexical NPs in the test sentence. In the *topicNP* context, the first referent was defined as aboutness-topic by the context sentence (“*And there is a story about the monkey.*”) and realized as a lexical NP. In the *topicPron* context, the first referent was defined as topic by the context sentence and realized as an unambiguous personal pronoun, with *ona* (“she”) and *er* (“he”) indicating nominative case and *eë* (“her”) and *ihn* (“him”) indicating accusative case in Russian and German respectively. The second NP was always realized as a lexical NP.

Twelve experimental items were created. Each item consisted of an introductory sentence and a subsequent test sentence. The subject and object of the test sentence were feminine animal names in Russian, and masculine animal names in German. Due to the restricted set of masculine animal names that the children may know, 3 of the 24 German animal names (*Affe*, *Löwe*, *Hase*) were weak nouns, that have the bound morpheme *-n* in accusative, dative and genitive case. One weak noun *Bär* was used without the bound morpheme as it is used in colloquial German. Different animal pairs were presented in Russian and German because of genus differences between both languages. The sentences involved one of two actions: catch (*pojmat’* in Russian, *fängt* in German) or tickle (*poščekotat’* in Russian, *kitzelt* in German).

2.3 Procedure

The procedure of the Russian and German version of the experiment was identical. The experimental movies were presented on a laptop using Microsoft PowerPoint. Participants were involved in a betting game in which a cartoon character would predict what two animals shown in each scene would do. Each animal pair was shown on the screen for 1000 ms before the onset of the introductory context sentence. The predictions were formulated using the target sentences and were followed by a movie showing the correct or reversed action. Half of the trials in each condition showed the correct action and the other half the reversed action. The participants had to say whether the cartoon character’s prediction was right or wrong.

Participants were shown 4 practice items and 24 movies, 12 movies in the *newNP* context and 12 in the two topic-first contexts (*topicNP*, *topicPron*). The movies in the *newNP* context were shown in the first half of the experiment and movies in the topic-first contexts in the second part. The animal pairs in the second part were the same pairs as in the first half, as the set of animal names

Table 1: Example of an item of the Russian experiment

	SVO	OVS
	<i>Davaj posmotrim, čto tut sejčas prozjojdēt.</i> (‘Look what will happen here.’)	
new NP	<i>Ja dumaju, čto</i> <i>obez’ana sejčas pojmaet myšku.</i> ‘I think that the monkey will	<i>Ja dumaju, čto</i> <i>obez’anu sejčas pojmaet myška.</i> ‘I think that the mouse will catch
	catch the mouse.’	the monkey.’
	<i>A teper’ davaj posmotrim istoriju pro obez’anu.</i> (‘And now let us watch a story about a/the monkey.’)	
topic NP	<i>Ja dumaju, čto</i> <i>obez’ana sejčas pojmaet myšku.</i> ‘I think that the monkey will	<i>Ja dumaju, čto</i> <i>obez’anu sejčas pojmaet myška.</i> ‘I think that the mouse will catch
	catch the mouse.’	the monkey.’
	<i>A teper’ davaj posmotrim istoriju pro obez’anu.</i> (‘And now let us watch a story about a/the monkey.’)	
topic Pron	<i>Ja dumaju, čto</i> <i>ona sejčas pojmaet myšku.</i> ‘I think that she will catch the	<i>Ja dumaju, čto</i> <i>eě sejčas pojmaet myška.</i> ‘I think that the mouse will catch
	mouse.’	her.’

was restricted. Each participant saw 6 items in each word order in the newNP context and 3 items in each word order in the topicNP and topicPron contexts.

The experimental items were arranged in blocks of 6 movies, but participants could take breaks after each trial. There was a longer break after the first two blocks.

The Russian and German experiment were conducted by two female Russian L1 and German L1 experimenters respectively. The children took part in the experiment and proficiency tests in each language on different days, with an interval of around 2 weeks between both experiments.

3 Results

3.1 Accuracy

Due to their heterogenous response pattern, the children were assigned to three response groups. The criteria for the groupings were based on binomial tests.

Table 2: Example of an item of the German experiment

	SVO	OVS
	<i>Schau mal was hier passiert.</i> (‘Look what will happen here.’)	
new NP	<i>Ich wette,</i>	<i>Ich wette,</i>
	der Hahn fängt gleich den Igel.	den Hahn fängt gleich der Igel.
	‘I bet the rooster will catch the hedgehog.’	‘I bet the hedgehog will catch the rooster.’
	<i>Nun kommt eine Geschichte über den Hahn.</i> (‘And now there is a story about a/the rooster.’)	
topic NP	<i>Ich wette,</i>	<i>Ich wette,</i>
	der Hahn fängt gleich den Igel.	den Hahn fängt gleich der Igel.
	‘I bet the rooster will catch the hedgehog.’	‘I bet the hedgehog will catch the rooster.’
	<i>Nun kommt eine Geschichte über den Hahn.</i> (‘And now there is a story about a/the rooster.’)	
topic Pron	<i>Ich wette,</i>	<i>Ich wette,</i>
	er fängt gleich den Igel.	ihn fängt gleich der Igel.
	‘I bet he will catch the hedgehog.’	‘I bet the hedgehog will catch him.’

Yes-bias children showed a response bias in that they either gave 18 or more yes-responses or 18 or more no-responses out of the 24 trials. SVO-bias children got only two or less of the OVS sentences correct, and apparently treated OVS as SVO sentences. Non-biased children were all children who did not fall into the first two groups.

Logit linear mixed effects (lme) models (Baayen 2008, Jaeger 2008) were calculated in the R environment. For each language, we calculated separate models to estimate the fixed effects of Word Order (SVO, OVS), Context (newNP, topicNP, topicPron), and their interaction on the comprehension accuracy. Participants and Items were treated as random factors in the models for the Russian data, but due to convergence errors only Participants were kept in the models for the German data. The contrast coding of Word Order was sum contrast (+/-0.5) that resembled traditional ANOVA analyses. The contrast coding of Context was the treatment contrast, with newNP condition as the baseline.

Model fitting was performed in a stepwise fashion, starting with the most complex model that included the full factorial set of random effects (random slope-adjustment for all fixed effects and interactions for both Participants and

Items). Due to convergence errors with the maximal model, the complex model was trimmed down in a stepwise fashion using log-likelihood tests for model comparisons (Baayen 2008). The procedure always resulted in the choice of the simplest model that included only intercept adjustments.

Russian

Five bilingual children showed a yes(- or no)-bias (2 of them a no-bias) and two children a subject-first-bias, the remaining 20 children are considered as non-biased children. In the following, we report only the results of the non-biased children.

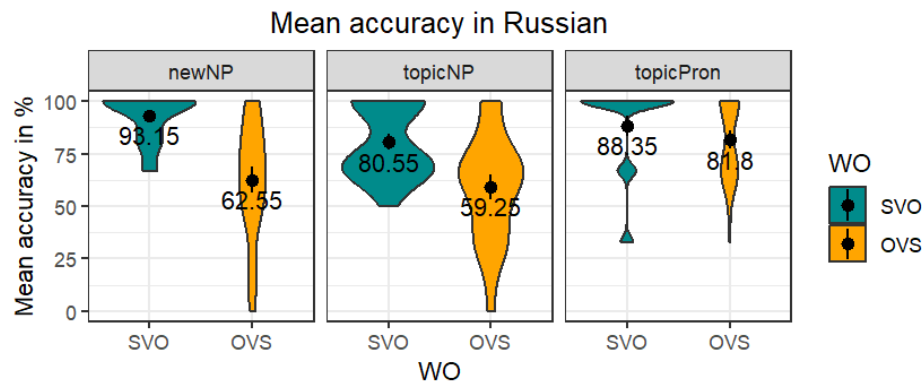


Figure 1: Accuracy of SVO and OVS sentences in the three contexts in Russian

Figure 1 shows the mean accuracy of SVO and OVS sentence comprehension in the three contexts. Logit lme models revealed a main effect of Word Order ($b = 1.14$, $SE = 0.22$, $z = 5.268$, $p < .001$) resulting from higher accuracy in SVO compared to OVS sentences, but this effect was modulated by an interaction with Context, i.e. a reduced word order effect for the topicPron condition ($b = -0.85$, $SE = 0.34$, $z = -2.479$, $p < .005$) due to higher accuracy for OVS sentences. Planned comparisons conducted by using logit lme models that test performance of OVS sentences against the intercept (50% chance level) respectively, support that OVS sentence comprehension was above 50% chance level in the topicPron condition ($b = 1.91$, $SE = 0.63$, $z = 3.024$, $p = .0025$) whereas it was at chance level in the newNP ($b = 0.66$, $SE = 0.38$, $z = 1.762$, $p = .0781$) and topicNP ($b = 0.43$, $SE = 0.41$, $z = 1.047$, $p = .295$) condition.²

²To account for multiple testing, the p-values should be multiplied by 3 (Bonferroni correction, cf. Baayen 2008).

Moreover, the model revealed that the overall accuracy in topicNP condition was lower in comparison to the newNP condition ($b = -0.71$, $SE = 0.30$, $z = -2.335$, $p < .05$), suggesting that realizing a given referent as a lexical NP caused comprehension problems.

German

Three children showed a yes-bias (no no-bias) and two children a subject-first bias in German, leaving 22 children in the non-biased group. None of the children showed a bias in both languages.

Figure 2 shows the mean accuracy of SVO and OVS sentence comprehension in the three contexts.

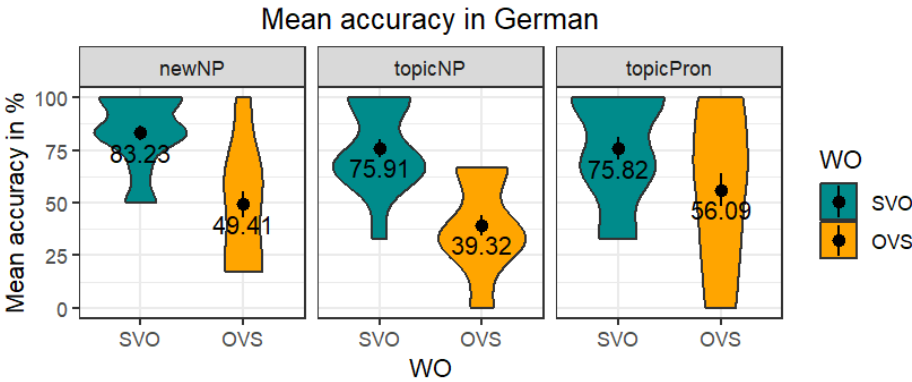


Figure 2: Accuracy of SVO and OVS sentences in the three contexts in German

Linear mixed effects models revealed a main effect of Word Order ($b = 0.83$, $SE = 0.15$, $z = 5.652$, $p < .001$) resulting from higher accuracy in SVO compared to OVS sentences, as well as a marginal effect of the topicNP condition, reflecting slightly lower overall accuracy in this condition compared to the newNP condition ($b = -0.44$, $SE = 0.24$, $z = -1.824$, $p = .0681$). There was no interaction between Word Order and Context. Planned comparisons revealed that sentence comprehension of OVS sentences was at 50% chance level in all three conditions.

3.2 Relationship between proficiency and comprehension

Figures 1 and 2 illustrate that the “non-biased” children varied in the performance especially on OVS sentences. To assess whether this variation arose from language proficiency, correlations between language proficiency and OVS comprehension were calculated.

For Russian, there were no significant correlations between SVO comprehension and language proficiency. For OVS sentences, there were moderate positive correlations with case production ($\rho = 0.389$, $p < 0.05$) and productive lexicon for nouns ($\rho = 0.388$, $p < .05$) if all children were considered, but no correlations if only the non-biased children were considered. Figure 3 shows the relationship between case production and OVS accuracy for all children. Most non-biased children (and yes-biased) performed well on case production regardless of the OVS comprehension, whereas the two SVO-biased children had lower case production scores.

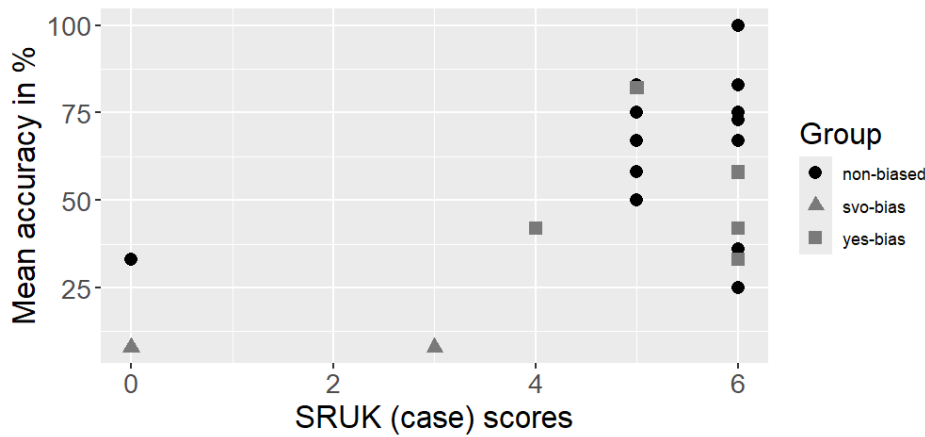


Figure 3: Relationship between case production and OVS comprehension in Russian

The centred proficiency scores of each measure were added in separated lme models predicting accuracy in the non-biased children because both measures were highly correlated. The models revealed effects of case production ($b = 0.28$, $SE = 0.13$, $z = 2.225$, $p < .05$) and productive lexicon for nouns ($b = 0.12$, $SE = 0.05$, $z = 2.428$, $p < .05$), but these effects did not interact or change the nature of the effects reported for the model without the proficiency measures. They indicate that proficiency had a rather general impact on comprehension even on the non-biased children.

For German, there were also no significant correlations between SVO accuracy and language proficiency. OVS comprehension was moderately and positively correlated with TROG-D, productive lexicon for nouns and verbs, and perceptive lexicon for verbs regardless of whether all or only the non-biased children were considered (for non-biased children: TROG: $\rho = 0.594$, $p < .01$, verb_comp: ρ

= 0.602, $p < .01$, verb_prod: $\rho = 0.753$, $p < 0.001$, noun_prod: $\rho = 0.679$, $p < 0.001$). Figure 4 exemplifies the correlation between TROG-D scores and OVS comprehension, indicating that the correlations occurred in all response groups.

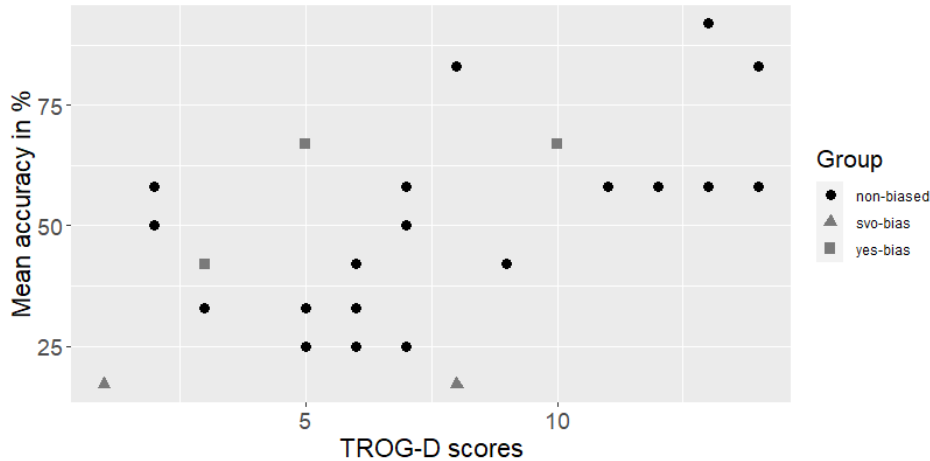


Figure 4: Relationship between TROG scores and OVS comprehension in German

The centred proficiency measures were added separately to the lme models predicting comprehension in the non-biased children. They revealed main effects of the proficiency performance for TROG-D ($b: 0.10$, $SE = 0.03$, $z = 3.555$, $p < 0.001$), verb comprehension ($b: 0.04$, $SE = 0.02$, $z = 2.792$, $p < 0.01$), verb production ($b: 0.05$, $SE = 0.02$, $z = 2.146$, $p < 0.05$) and noun production ($b: 0.04$, $SE = 0.01$, $z = 3.065$, $p < 0.01$). There were no interactions with proficiency, indicating that proficiency had an impact on comprehension, regardless of word order and context.

4 Discussion

The present study investigated the factors influencing OVS comprehension in bilingual Russian-German pre-schoolers. It revealed two main findings: 1) The type of referring expression but not context improve OVS comprehension in Russian, while none of these factors eases comprehension in German. 2) Language proficiency had a general impact on comprehension in both languages, but OVS comprehension was more strongly correlated with proficiency in German.

The results indicate that the same factors that were found to influence sentence comprehension (i.e. type of referring expression), also influence comprehension

in bilingual children. Similar to the German-speaking monolinguals (Sauermann & Höhle 2016), a topical object pronoun but not the topic-first context per se eased OVS comprehension in Russian. A topical pronoun may ease comprehension by reducing processing costs, e.g., similarity-based interference between the subject and object referent (Friedmann et al. 2009, Gordon et al. 2001), or by supporting discourse integration (Gagarina 2012, Gordon et al. 1993).

The children in our study seem to be sensitive to the link between topicality and the type of referring expression because they showed lower SVO and OVS comprehension accuracy in the topicNP condition, i.e. when the initial NP was a topic and given, compared to the newNP condition when it was new. These difficulties occurred in Russian and marginally in German and may reflect the repeated name penalty (Gordon et al. 1993), i.e., children's expectations that given topic referents are realized as a pronoun but not as a lexical NP (Gagarina 2012).

The repeated name penalty may explain why the topic-first context per se did not improve comprehension in Russian and German. Alternatively, the topic-first context may not ease comprehension because it merely eases the integration of information during sentence processing (Schumacher & Hung 2012) but does not provide a clear cue to the object status of the initial referent. Finally, monolingual and bilingual children may not profit from the topic-first or the corresponding given-before-new order in comprehension if they also produce new-before-given word orders (cf. Fleischhauer 2009, Smolik 2015).

The results show an impact of language proficiency on Russian and German. In contrast to Smolik (2015), who found that language proficiency interacted with word order and context, our study revealed that language proficiency influenced comprehension regardless of context and word order. While the low number of items may have contributed to the lack of an interaction effect, differences in the proficiency in each language may play a role. In Russian, the correlations between proficiency and OVS comprehension mainly accounted for the different response groups and suggest that the problem of the non-biased children may not be a lack of knowledge of case marking but rather using and integrating this information during processing. In German, proficiency also influenced comprehension more generally, but here the low proficiency in grammar and lexicon seemed to be an issue also for the non-biased children. Possibly, the children are at an earlier stage on the developmental path in German compared to Russian. This may explain why a topical pronoun did not ease comprehension in German.

Our results agree with three assumptions resulting from the Micro-cue model (Westergaard 2014, 2021). First, our results suggest that children may start from more context-specific structures that are later generalized given that the bilingual children had problems understanding OVS sentences in contexts when both

the subject and object were realized as lexical NPs. Second, the repeated name penalty suggests that the bilingual children were sensitive to information structure. Third, OVS comprehension seems to be influenced by economy because it was improved in contexts that might be closer to contexts that children find in their input, e.g., the sentence-initial object was a topical pronoun rather than a new lexical NP, especially in Russian. Finally, our result support the assumption that similar factors influence comprehension in bilingual and monolingual children.

While our results support the Micro-cue model, corpus research is required to identify the micro-cues bilingual preschoolers may rely on, and online studies to assess whether children consider these and additional cues during processing (cf. [Minor et al. 2024](#)).

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References

- Baayen, R. Harald. 2008. *Analysing linguistic data*. Cambridge: Cambridge University Press.
- Bailyn, John Frederick. 2012. *The syntax of Russian*. Cambridge: Cambridge University Press.
- Boila, Chiara. 2024. *The processing of OVS and passive sentences in German monolingual and German-Italian simultaneous bilingual children*. Universität Potsdam. (Doctoral dissertation).
- Crain, Stephen & Rosalind Thornton. 1998. *Investigations in universal grammar: A guide to experiments on the acquisition of syntax and semantics*. Cambridge: MIT Press.
- Cristante, Valentina & Sarah Schimke. 2018. Die Verarbeitung von Passivsätzen und OVS-Sätzen im kindlichen L2-Erwerb. In Sarah Schimke & Holger Hopp (eds.), *Sprachverarbeitung im Zweitspracherwerb*, 169–192. Berlin: de Gruyter.
- Dittmar, Miriam, Kirsten Abbot-Smith, Elena Lieven & Michael Tomasello. 2008. German children's comprehension of word order and case marking in causative sentences. *Child Development* 79(4). 1152–1167.

Antje Sauermann & Natalia Gagarina

- Fleischhauer, Elisabeth. 2009. *Word order variation in German child language: A single case corpus analysis*. University of Potsdam. (MA thesis).
- Fox-Beyer, Annette V. 2006. *TROG-D—Test zur Überprüfung des Grammatikverständnisses*. Idstein: Schulz-Kirchner Verlag.
- Frazier, Lyn. 1987. Sentence processing: A tutorial review. In Max Coltheart (ed.), *Attention and performance XII: The psychology of reading*, 549–586. East Sussex: Lawrence Erlbaum.
- Friedmann, Nama, Andrea Belletti & Luigi Rizzi. 2009. Relativized relatives: Types of intervention in the acquisition of A-bar dependencies. *Lingua* 119. 67–88.
- Gagarina, Natalia. 2012. Discourse cohesion in the elicited narratives of early Russian-German sequential bilinguals. In Kurt Braunmüller & Christoph Gabriel (eds.), *Multilingual individuals and multilingual societies*, 101–120. Amsterdam: John Benjamins.
- Gagarina, Natalia V., Annegret Klassert & Nathalie Topaj. 2010. *Sprachstandstest Russisch für mehrsprachige Kinder [Russian language proficiency test for multilingual children]* (ZAS Papers in Linguistics 54). DOI: [10.21248/zaspil.54.2010.403](https://doi.org/10.21248/zaspil.54.2010.403).
- Gordon, Peter C., Barbara J. Grosz & Laura A. Gilliom. 1993. Pronouns, names, and the centering of attention in discourse. *Cognitive Science* 17. 311–347.
- Gordon, Peter C., Randall Hendrick & Marcus Johnson. 2001. Memory interference during language processing. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 27(6). 1411–1423.
- Grünloh, Thomas, Elena Lieven & Michael Tomasello. 2011. German children use prosody to identify participant roles in transitive sentences. *Cognitive Linguistics* 22(2). 393–419.
- Jaeger, Florian T. 2008. Categorical data analysis: Away from ANOVAs (transformation or not) and towards logit mixed models. *Journal of Memory and Language* 59(4). 434–446.
- Janssen, Bibi, Natalia Meir, Anne Baker & Sharon Armon-Lotem. 2015. On-line comprehension of Russian case cues in monolingual Russian and bilingual Russian-Dutch and Russian-Hebrew children. In Elizabeth Grillo & Kyle Jepson (eds.), *Proceedings of the 39th Annual Boston University Conference on Language Development*, vol. 2, 266–278. Somerville: Cascadia Press.
- Krifka, Manfred. 2008. Basic notions of information structure. *Acta Linguistica Hungarica* 55. 243–276.
- Meir, Natalia, Olga Parshina & Irina A. Sekerina. 2020. The interaction of morphological cues in bilingual sentence processing: An eye-tracking study. In Megan M. Brown & Alexandra Kohut (eds.), *Proceedings of the 44th Boston Uni-*

- versity Conference on Language Development, 376–389. Somerville: Cascadilla Press.
- Minor, Serge, Natalia Mitrofanova & Marit Westergaard. 2024. The interaction of linguistic and visual cues for the processing of case in Russian by Russian-German bilinguals: An eye tracking study. *Topics in Cognitive Science* 17(4). Advance online publication, 855–867. DOI: [10.1111/tops.12724](https://doi.org/10.1111/tops.12724).
- Mitrofanova, Natalia, Olga Urek, Yulia Rodina & Marit Westergaard. 2022. Sensitivity to microvariation in bilingual acquisition: Morphophonological gender cues in Russian heritage language. *Applied Psycholinguistics* 43(1). 41–79.
- Pléh, Csaba, Alexandr Jarovinskij & Alexandr Balajan. 1987. Sentence comprehension in Hungarian-Russian bilingual and monolingual preschool children. *Journal of Child Language* 14. 587–603.
- Rinker, Tanja & Natalia Gagarina. 2017. *Cross-linguistic lexical tasks (CLT): German. part of COST IS0804 LITMUS battery*. Unpublished material.
- Sauermann, Antje & Natalia Gagarina. 2018. The roles of givenness and type of referring expression in the comprehension of word order in Russian-speaking children. *Linguistics Vanguard* 4(s1). 20160094.
- Sauermann, Antje & Barbara Höhle. 2016. Impact of context and type of referring expression on sentence comprehension in German-speaking children. In Laurel Perkins, Rachel Dudley, Juliana Gerard & Kasia Hitczenko (eds.), *Proceedings of the 6th Conference on Generative Approaches to Language Acquisition North America (GALANA 2015)*, 90–101. Somerville: Cascadilla Proceedings Project.
- Schumacher, Petra B. & Yu-Chen Hung. 2012. Positional influences on information packaging: Insights from topological fields in German. *Journal of Memory and Language* 67. 295–310.
- Slobin, Dan I. & Thomas G. Bever. 1982. Children use canonical sentence schema: A crosslinguistic study of word order and inflections. *Cognition* 12. 229–265.
- Smolik, Filip. 2015. Word order and information structure in Czech 3- and 4-year olds' comprehension. *First Language* 35(3). 137–153.
- Westergaard, Marit. 2009. *The acquisition of word order: Micro-cues, information structure, and economy* (Linguistik Aktuell/Linguistics Today 145). Amsterdam: John Benjamins.
- Westergaard, Marit. 2014. Linguistic variation and micro-cues in first language acquisition. *Linguistic Variation* 14(1). 26–45. DOI: [10.1075/lv.14.1.02wes](https://doi.org/10.1075/lv.14.1.02wes).
- Westergaard, Marit. 2021. Microvariation in multilingual situations: The importance of property-by-property acquisition. *Second Language Research* 37(3). 379–407.

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Xanthos, Aris, Sabine Laaha, Steven Gillis, Ursula Stephany, Ayhan Aksu-Koç, Anastasia Christofidou, Natalia Gagarina, Gordana Hrzica, F. Nihan Ketrez, Marianne Kilani-Schoch, Katharina Korecky-Kröll, Melita Kovačević, Klaus Laalo, Marijan Palmović, Barbara Pfeiler, Maria D. Voeikova & Wolfgang U. Dressler. 2011. On the role of morphological richness in the early development of noun and verb inflection. *First Language* 31. 461–479.

Chapter 23

The effectiveness of cues in gender processing by Russian-speaking children

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Eye movements of 27 Russian-speaking 5-to-6-year-old children who viewed 4-picture animated displays in a Visual World study were recorded. Participants listened to “In the sky flew.F [a] silver.F bird.F” and clicked on the Target ‘bird.F’. In the 2x2 design, the gender of the Target (M/F) was crossed with referential ambiguity in the visual display, i.e., two referents of the same gender (AMB) vs. only one (UNAMB). The accuracy in clicking on the Target was at ceiling: 97%. The cluster-based permutation analysis of eye movements (1,000 permutations) that predicted the empirical logit of proportion of fixations to the Target, identified only one late significant cluster where the looks to the Target were higher in UNAMB than AMB conditions. Thus, children used gender agreement cues only in the facilitatory manner, F cues were no more perceptually salient than M cues in the early adjective region, and did not benefit from cue coalition, i.e., the doubling of cues. These results can be best explained by complexity of the design combined with the less efficient gender agreement cues in Russian.

1 Introduction

Prediction in language comprehension has been the driving force in psycholinguistic research for the past 20 years. By now, it is firmly established that monolingual adults anticipate upcoming words at all linguistic levels, including morphosyntactic features such as grammatical gender. For example, when hearing a German phrase that contains gender-marked modifiers, such as the determiner and adjective (e.g., *ein/der kleiner* [N] ‘a/the.M small.M [N]’), German listeners

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successfully predict both lexical (i.e., activation of a general lexical item in its grammatical form–noun) and morphosyntactic (i.e., activation of grammatical gender–masculine, singular) information from the context (Luke & Christianson 2015).

A seminal theory of *Prediction-by-Production* (Pickering & Gambi 2018) argues that language comprehension critically depends on predictive mechanisms in production. Two types of prediction are distinguished in comprehension, namely, *anticipation* and *integration*. Anticipation happens in the form of preactivation of an upcoming word that has not appeared in speech yet and is a hallmark of true prediction. Integration, in contrast, takes place while the word is already being processed. It simply facilitates integration of the word into context, but it does not boost comprehension. The debate on the role of these two types of prediction has been fueling empirical research both in adult psycholinguistics and in language acquisition.

In first language acquisition, the key empirical questions that have been recently investigated with respect to whether children can predict the head noun from gender-marked modifiers are as follows:

1. How early do children become sensitive to gender cues? It can happen as early as at the age of 2 to 3 (Lew-Williams & Fernald 2007; Smolik & Bláhová 2019), but it dramatically improves between the ages of 5 to 10 (Günther et al. 2023).
2. Are there cross-linguistic differences? Yes, grammatical gender cues are more powerful in Romance than Germanic languages (Baron et al. 2022; Brouwer et al. 2017).
3. Do types of cues differ in their strength and does their effect increase when there are more than one cue? Phonologically transparent cues are stronger (Ivanova-Sullivan et al. 2024), but the effect of doubling of cues, known as *cue coalition*, is inconsistent (Günther et al. 2023).
4. How does individual variation in the form of linguistic experience (e.g., vocabulary scores) and working memory affect the effectiveness of gender cues in real-time language processing? (Kray et al. 2024; Özkan et al. 2022; Smolik & Bláhová 2019).

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1.1 Can children predict grammatical gender of nouns from their modifiers?

Development of sensitivity to gender agreement cues on modifiers plays an important role in investigating predictive abilities in children. Previous research within the Micro-Cue theory of language acquisition by Westergaard and colleagues (Westergaard 2014) investigated monolingual Russian children's use of gender agreement cues in an elicited production task (Rodina & Westergaard 2015). The experiment showed that 5-year-olds were already at ceiling (96%) in producing nouns following the adjectives in correct gender, although nouns with opaque gender resulted in lower accuracy (87%).

Rapid adoption of eye-tracking for language acquisition has made it possible to extend the predictive effects of gender agreement in production in children to real-time language comprehension. Three earlier experiments—in Spanish (Lew-Williams & Fernald 2007), French (Van Heugten & Shi 2009), and a recent one in Czech (Smolik & Bláhová 2019)—with 2- and 3-year-olds revealed that they could anticipate the noun based on the preceding gender-marked modifier. Children looked at the target noun in the Informative (UNAMB, pictures of two nouns in different gender) condition significantly earlier than in the Uninformative (AMB, pictures of two nouns in the same gender) condition. However, in Czech, the grammatical gender agreement effect was modulated by toddlers' receptive vocabulary scores that revealed the impact of individual differences in language experience for early sensitivity to gender cues.

However, the abovementioned studies relied on the preferential-looking paradigm in its modified version, i.e., looking-while-listening. This method has shortcomings. Although the experimental items tested children's spoken word recognition (e.g., *Encuentra la pelota*. "Find the.F ball.", Lew-Williams & Fernald 2007), their number was very small (e.g., eight nouns), and there was no separation between the gender-marked modifier and the noun. Moreover, while the force-choice looking task works well for infants and toddlers, it does not suit older children and adults.

Adaption of the more powerful *Visual World eye-tracking Paradigm* (VWP, Sekerina 2014) allowed for the subsequent experiments to investigate the predictive effects of gender agreement cues in children exactly in the same way as in adults. In this design, children usually view two pictures simultaneously presented on the screen and listen to a spoken sentence (e.g., Dutch: *Zie je het groene huis?* "Do you see the.N green.N house.N?", Brouwer et al. 2017). The pictures represent two colored objects of either different gender (UNAMB, or Informative condition), for example, neuter vs. common in Dutch or of the same gender (AMB, or Uninformative condition).

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mative condition). If children use gender-marked determiners and adjectives to anticipate the Target noun, they are expected to look at its picture faster in the UNAMB than in the AMB condition. If the looks happen early, i.e., before the Target noun onset in speech (anticipatory processing) or early during its duration (facilitatory processing), then children are sensitive to the gender marking on modifiers and use the gender cues predictively.

Such VWP experiments with monolingual children were recently conducted with 4-to-7-year-olds in Dutch (Brouwer et al. 2017), 5–10-year-olds in German (J et al. 2019; Günther et al. 2023; Lemmerth & Hopp 2019), and with 5-to-6-year-olds in Bulgarian and Russian (Ivanova-Sullivan et al. 2024). Brouwer and colleagues found that only 53% of Dutch-speaking children, specifically, those who consistently and correctly produced the marked neuter gender in the vocabulary test, were able to use the determiners to anticipate the Target noun. This group of children did not differ in their eye-movement patterns from Dutch adults: both groups showed significantly earlier anticipatory looks to the Target noun (i.e., starting at 300 ms after the article onset) in the UNAMB condition. In addition, they discovered an unexpected asymmetry between the two genders in Dutch: during the noun region, only the neuter determiner *het*, but not the common *de* “the”, induced the facilitatory processing.

Similar gender agreement sensitivity in the UNAMB condition was found by Lemmerth & Hopp (2019), J et al. (2019), and Günther et al. (2023) for monolingual 5-to-10-year-old German children. When the pictures were of different gender, even 5-year-olds looked at the Target noun during the adjective and before the noun encounter in speech. But each study revealed interactions of other factors with the condition. For example, Lemmerth and Hopp manipulated the type of modifiers, with half being gender-marked definite determiners *der/die/das* and the other half, a sequence of two gender-marked adjectives (e.g., *ein kleiner gelber* [N] “a small.M yellow.M [N]”). Cholewa and colleagues changed the definite articles to indefinite articles and had only one adjective, effectively doubling the grammatical gender cues (the first on the indefinite article, the second on the adjective). Günther and colleagues were interested in the ‘cue-coalition’ effect as they combined the grammatical gender with a semantic relation cue between the adjective and the noun (e.g., *ein wilder Löwe* “a.M wild.M lion.M”). The modifier type did not matter although the difference in the first fixation time between the determiners and adjectives was numerically larger for the latter. The cue coalition enhanced the gender prediction effect, but it was modulated by age, with 5-year-olds not benefiting from the doubling of gender information early enough compared to older children.

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1.2 The previous and the present studies with Russian-speaking children

Two previous eye-tracking studies based on the UNAMB/AMB manipulation in gender agreement were conducted in Russian, a language of the experiment in the present study.

Aumeistere et al. (2022) used the looking-while-listening method and compared monolingual Russian adults and 3-to-6-year-old children. Russian does not have articles, so their stimuli were questions with gender-marked adjectives and Target noun in the diminutive form (e.g., *Gde krasivyy zelenyy stol'čik?* “Where [is] pretty.M green.M chair.DIM.M?”). The eight Target nouns were phonologically transparent for gender, and only F and M were used. The second adjective was included to increase the distance between the first gender-marked adjective and the Target noun to test whether one gender agreement cue is enough for children to anticipate the noun. Predictive looks in this case should happen during the second adjective and before the noun onset. Both adults and children started looking at the Target noun by the time it appeared in speech, at about 1,200 ms. There were more anticipatory looks, and they were earlier in the UNAMB condition for adults. Surprisingly, the younger 3–4-year-olds were as capable of prediction as the older 5–6-year-olds demonstrating no age effect.

In contrast to Aumeistere et al.’s (2022) results, our previous study (Ivanova-Sullivan et al. 2024) found no evidence for the predictive effect of the gender agreement cues in the group of monolingual Russian-speaking children of the same age. Our participants in the standard VWP experiment that used 2 side-by-side pictures with the UNAMB/AMB manipulation showed only the facilitatory, and not predictive, processing as they looked at the Target noun only when they were listening to the noun.

We speculated that the difference between Aumeistere et al.’s (2022) and our results was caused by methodology: We used a full set of adult-like materials (59 nouns, 7 colors), all three genders, including the neuter, and both phonologically transparent and opaque nouns (e.g., *lopata* “shovel.F” vs. *krov'* “blood.F”). Moreover, Aumeistere and colleagues did not provide the exact moment in which children’s looks to the Target noun separated between the UNAMB and AMB trials. Visual inspection of the time course of fixations to the target noun in both conditions (see Fig. 3B: 878) shows a very slight divergence that seems to be confined to the first 200 ms of the noun.

The mixed findings from the two previous Russian studies prompted us to conduct a new VWP experiment with Russian-speaking preschool children in an attempt to shed light on which factors make prediction from gender agree-

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ment cues less efficient in Russian. We focused on two research questions, i.e., (1) whether strength and type of gender cues (on verbal predicate and modifying adjective) differ in their predictive power, and (2) whether cue coalition (doubling of gender cues, one on the verb and another on the adjective) can enhance a weak facilitatory effect and turn it into an anticipatory one.

The experiment described in this chapter is a part of the larger multigroup project on the predictive effect of gender agreement cues in three different Russian-speaking populations, i.e., monolingual and bilingual young adults (reported in [Fuchs & Sekerina 2024](#)), and monolingual 5-to-6-year-olds (reported here). The design, materials, procedure, and analysis were identical for all three groups.

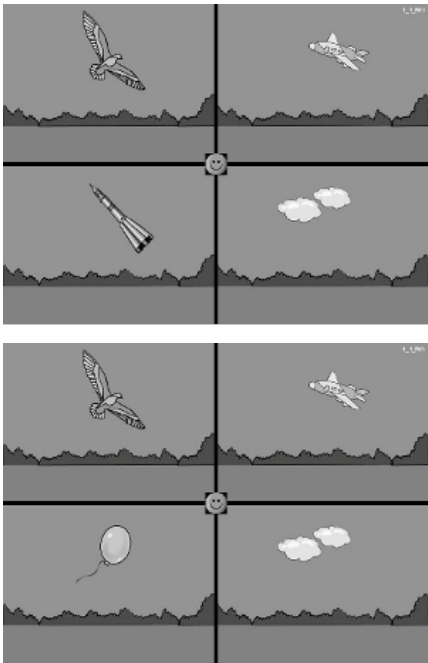
2 Method

2.1 Participants

There were 27 5-to-6-year-old children (8 boys, $M_{age} = 68$ months, range 60–83) who were tested in-person in a kindergarten in Moscow. All children were native Russian speakers without language disorders, based on parental report. No children were excluded based on their performance. They received a toy (value of \$3) for their participation.

The experiment was carried out in accordance with the ethical principles of psychologists and code of conduct of the American Psychological Association and was approved by the College of Staten Island IRB. All children gave oral assent while their parents signed an informed consent in accordance with the Declaration of Helsinki.

2.2 Design and materials



- (1) a. **M-UNAMB condition**
Po nebu letel-Ø
In sky was-flying.M.SG
serebrist-yj samolet-Ø.
silver.M.SG airplane.M.SG
‘A silver airplane was flying
in the sky.’
- b. **M-AMB condition**
Po nebu letel-Ø
In sky was-flying.M.SG
serebrist-yj samolet-Ø.
silver.M.SG airplane.M.SG
‘A silver airplane was flying
in the sky.’
- c. **F-UNAMB/AMB condition**
Po nebu letel-a
In sky was-flying.F.SG
serebrist-aja ptic-a.
silver.F.SG bird.F.SG
‘A silver bird was flying in
the sky.’
- d. **PL condition**
Po nebu letel-i
In sky were-flying.PL
serebrist-ye oblaka.
silver.PL clouds.PL
‘Silver clouds were flying in
the sky.’

Figure 1: An example of the UNAMB (a) and AMB (b) visual displays paired with the same spoken sentence (a or b). The Target noun is the airplane, F distractors are the bird and the rocket, PL distractor is the clouds. The F distractor rocket in (a) is replaced with the M distractor balloon in (b).

The experiment was conducted using the VWP, but in contrast to the standard VWP experiments on gender agreement cues with 2 side-by-side static pictures

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described in the introduction, we created 4-picture animated displays (Figure 1). Thirty experimental items consisted of a single spoken instruction in Russian (1a-1c) presented simultaneously with a picture visual display:

In Figure 1, the Target noun is *samolet* ‘airplane.M’ that is contrasted with two F distractor nouns (i.e., a rocket and a bird) and one noun in plural (e.g., clouds). In the M-AMB (Uninformative) condition, one of the two F distractor pictures was replaced with a M Competitor picture (i.e., a balloon). The same two conditions were replicated with the Target noun in F (e.g., *ptica* ‘bird.F’; (1b)). In the control condition, the Target noun was in plural (e.g., *oblaka* ‘clouds.N.PL’; (1c)). The images in the visual displays were in color and were animated using the .tiff format. People and objects in each of the four quadrants were performing the action described by the verb. Thus, in Figure 1, the airplane, rocket, bird, and clouds were slowly flying for the duration of the entire trial. Animation was added to the design because we believed that depicting moving events, such as flying, running, climbing, floating etc., using static pictures would diminish the ecological validity of the stimuli. However, at the time of conducting this study, there were no methodological studies in the Visual World paradigm that compared static pictures with dynamic pictures or video clips. [Full-detailed description of the visual displays and images in them is available in Fuchs & Sekerina (2024)].

The spoken instructions were in the form of the locative inversion construction in Russian, namely, PP.LOC-VERB-NP.NOM, in which the agreeing verb and adjective preceded the Target noun that controls the agreement in gender and number for both. The instructions were recorded with normal speed and natural prosody by the author who is a native Russian speaker.

The 30 items (visual displays paired with a spoken instruction) consisted of six items in each of five conditions (M-UNAMB, M-AMB, F-UNAMB, F-AMB, PL). There were no specially designed fillers because the experimental sentences did not contain any special constructions. Five versions of the experiment were created for the between-participants design, with experimental items rotated through the five conditions, in a Latin Square design. Child participants were randomly assigned to one of the five versions.

2.3 Procedure

The experiment was presented to the children as a Powerpoint© slide show that combined the visual displays with four animated pictures and auditory sentences played through the speakers. The task was to select one picture out of four in the visual display that matched the spoken sentence by clicking on it. Children were seated in front of the 17-inch HP laptop that ran the slideshow and was connected

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to a remote free-viewing eye-tracking system (ETL-500) from ISCAN, Inc. and recorded participants' eye movements. Eye movements were sampled at a rate of 30 times per sec and were recorded on a digital SONY DSR-30 video tape-recorder. Prior to the experiment, children underwent a short calibration procedure, and the experiment lasted 10 minutes.

3 Results

Eye movements were extracted from videotape using a SONY DSR-30 video tape-recorder with frame-by-frame control and synchronized video and audio. Data were prepared using the package *eyetrackingR* (Dink & Ferguson 2015) and analyzed using the package *clusterperm* (Barr 2022) in R version 4.2.1 (R Development Core Team 2021). To analyze the proportion of looks to the Target noun over time across conditions, we implemented a permutation-based cluster analysis to identify clusters of time bins in which there was a significant difference between conditions in the proportion of fixations to the Target noun (Ito & Knoeferle 2022). The onset of the critical region for fixations to the Target noun was the offset of the verb agreement marker (cf. Figure 2). Data were binned into time bins of 50 ms. Generalized mixed effects models were fitted to the data in each time bin, predicting the empirical logit of the proportion of fixations to the Target noun in the given bin. We used Condition (UNAMB vs. AMB) and Gender (M vs. F) as fixed effects, with an interaction term between them. The data were permuted 1,000 times to assess the significance of the clusters.

Children's accuracy was at ceiling for choosing the correct picture, 97%, on par with the control group of monolingual adults, 99.4% (reported in Fuchs & Sekerina 2024). This suggests that the sentence-picture selection task was easy for children. Figure 2 presents children's proportion of fixations to the Target noun as a function of the Condition (UNAMB vs. AMB) and Gender (M vs. F). Visual inspection of the plot suggests that the proportion of fixations to the Target overlapped a lot across the two conditions and two genders.

As far as eye movements are concerned, cluster-based permutation analysis shown in Table 1 identified one significant cluster for the Condition effect. The looks to the Target noun were higher during the period that started at 400 ms and continued until 850 ms after the end of ADJ, way well into the noun. However, this separation in the looks between the UNAMB and AMB conditions happened quite late in the sentence, namely, when the Target noun was already encountered in speech. This is compatible with the delayed interpretation of the gender agreement effect as facilitatory, instead of anticipatory, in nature. This finding is

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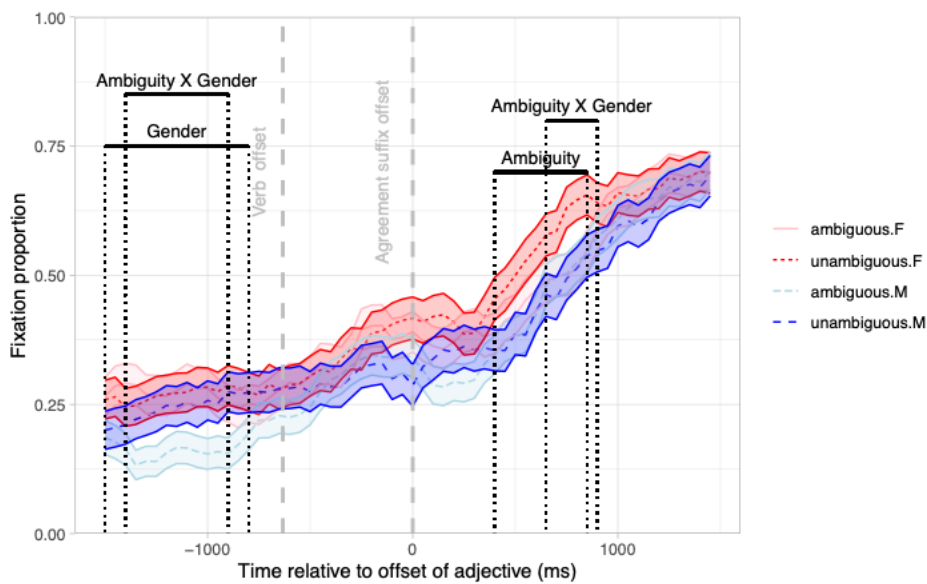


Figure 2: Children’s proportion of fixations on the Target noun as a function of Condition (UNAMB vs. AMB) and Gender (M vs. F). Significant clusters are shaded.

in line with our previous findings for Russian (Ivanova-Sullivan et al. 2024), and in contrast to cross-linguistic evidence of gender agreement anticipatory effect in children’s processing in German, Dutch, and Spanish. There was also one significant cluster found for Gender; however, it occurred very early, between -1,500 and -800 ms, i.e., before the first gender marker on the verb even became available in speech. At this point, no gender information was available to children rendering this difference uninterpretable. Two significant clusters were identified for the interaction between Condition and Gender. The first appeared earlier, overlapping with the cluster for Gender between -1400 and -900 ms. As with the Gender cluster, this interaction occurs too early to be interpretable, and may be spurious. The second cluster overlapped with the cluster for Condition, between 650 and 900 ms, and suggests that the feminine marker was more facilitatory than the masculine marker.

4 Discussion

In the present VWP experiment, that is a part of the larger multigroup project (see Fuchs & Sekerina 2024 for monolingual and bilingual adult experiments),

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Table 1: Significant clusters identified by the cluster-based permutation analysis

Variable	Cluster Mass	β	p -value	Cluster Start Time	Cluster End Time
Condition	107.30	0.24	< 0.001	400	850
Gender	292.06	-0.30	< 0.001	-1500	-800
Condition×Gender					
Cluster 1	73.70	0.27	< 0.001	-1400	-900
Cluster 2	52.85	-0.32	< 0.001	650	900

we recorded eye movements of 5-to-6-year-old Russian-speaking children who viewed 4-picture visual displays that were contrasted in informativeness of gender agreement cues (UNAMB vs. AMB). The spoken sentences paired with the visual displays were in the Locative-V-Adj-N order that allowed us to investigate cue coalition, when two gender agreement markers, one on V and another on the adjective, preceded the Target noun. In the explanation of the findings that follows, we discuss them with respect to (a) the previously published cross-linguistic results for children, including Russian, and (b) the research questions that informed the present study.

First, consider cross-linguistic results from Dutch (Brouwer et al. 2017), German (J et al. 2019; Günther et al. 2023; Lemmerth & Hopp 2019), and Russian (Aumeistere et al. 2022). In all of them, 5-to-10-year-old children were able to use the gender agreement cue on the modifiers (i.e., determiners and adjectives) before the Target noun occurred in speech and therefore were engaged in anticipatory processing. In contrast, our cluster-based permutation analysis of eye movements identified only one significant cluster in which the looks of our 5-to-6-year-old Russian-speaking children to the Target noun were higher in the UNAMB than in the AMB condition, but it occurred quite late. Following Pickering & Gambi (2018)’s theory of *Prediction-by-Production* that specifically contrasts anticipation and integration in comprehension, we have to conclude that the late facilitatory looks in our study reflect *integration*, and not anticipation, because they took place only during the Target noun.

Second, there is the case of asymmetry in the perceptual saliency of gender agreement markers between M and F in Russian. In M forms, the gender marker is phonologically null on past tense verbs and nouns and is one syllable shorter

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on adjectives (-Vj) whereas in F forms, it is fully morphologically specified (-a and -Vja). Based on stronger perceptual saliency of F cues, we expected them to be more effective than M cues in facilitating the integration of the Target noun. Although the F cues did increase the looks to the Target compared to the M cues, it happened late, only in the second interaction cluster, establishing the absence of difference between two genders.

Next, we turn to the reasonable idea that cue coalition (the doubling of gender cues, one on the verb and another on the adjective) can enhance a weak facilitatory effect and turn it into an anticipatory one. However, the true anticipatory effect of gender agreement cues in the present experiment never materialized even in the presence of cue coalition because the doubling of gender cues did not strengthen their predictive ability. Thus, our present results confirmed the delayed use of gender agreement cues found in our previous experiment with Russian-speaking children, with just a single gender marker on the adjective (Ivanova-Sullivan et al. 2024).

We consider two potential explanations for the only significant result, i.e., the difference between the looks in UNAMB and AMB conditions (albeit late), and lack of cue coalition effects. The first has to do with suboptimal processing set-up, namely, potential high cognitive load brought about by our methodology. Predictive processing in children could be diminished by high cognitive load in the experiment. The task of matching the sentence with one of the 4 pictures in the visual display requires combining in real time different sources of information, namely, complex visual display and gender cues embedded in spoken sentences with non-canonical word order. In addition, in contrast to the standard VWP design with 2 side-by-side static pictures with force-choice used in the previous Russian VWP experiment by Aumeistere et al. (2022), our visual displays were animated, with 4 objects performing an action described by the verb (e.g., the airplane, the rocket, and bird and the clouds were all simultaneously flying in the sky against the backdrop of some trees).

There is some corroborating evidence that children's developing processing abilities could be overburdened in a situation when multiple sources of information compete for their limited attention. For example, in our VWP experiment with 6-year-olds (Sekerina & Trueswell 2011), we found that Russian-speaking children did not compute contrast sets in the noncanonical word order sentences with the split Scrambling construction (e.g., *Red put star ...*) in real time. This was despite the presence of the visual display with the objects and the pitch accent on the adjective *red* that signaled contrastiveness. Instead, they had to hear the head noun to confidently pick out the Target referent, effectively resorting to a 'wait-and-see' strategy.

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However, this explanation that the methodological complexity of the present study caused the higher cognitive load that prevented the children from using gender agreement cues in the anticipatory manner cannot explain similar findings of Ivanova-Sullivan et al. (2024). In that VWP study, a simpler design with 2 side-by-side static pictures were used with a different but comparable group of Russian-speaking children (Ivanova-Sullivan et al. 2024). The sentences were short (*Where here is red blood?*) and paired with only 2 pictures (the Target noun *red blood.F* and the Competitor *red chair.M/red shovel.F*), and even this simple design did not result in increasing the children's sensitivity to gender cues on the adjectives. Their looks between the Target and Competitor still diverged late, in the silence region, indicating only facilitatory use of gender agreement information.

We need to entertain a second potential explanation. Ivanova-Sullivan and colleagues proposed that the weaker, facilitatory nature of gender agreement cues in Russian has to do with their lesser strength. The mostly opaque gender markers on adjectives (usually unstressed), phonological transparency and opacity of gender markers on nouns (the nouns for the third declension class are ambiguous between M and F), pervasive vowel reduction of unstressed vowels, and absence of gender-marked determiners make gender agreement cues in Russian weaker than in other languages. In Dutch, German, Spanish, and Bulgarian, gender agreement cues occur not only on adjectives and but also on determiners, and are more transparent and regularly distributed, allowing them to be more rapidly and efficiently processed resulting in prediction, even at young ages (e.g., Lew-Williams & Fernald 2007). With the lack of such strong cues in Russian, it is unsurprising that they are integrated later during speech processing, resulting only in a facilitatory effect. Future studies should look into the anticipatory vs. facilitatory use of gender in other languages in order to tease apart the predictive use on verbs vs. adjectives vs. determiners on verbal suffixes in languages like Hebrew and Arabic in which gender information is not marked on articles but is marked on verbs and adjectives.

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References

- Aumeistere, Anna, Sybrine Bultena & Susanne Brouwer. 2022. Wisdom comes with age? The role of grammatical gender in predictive processing in Russian children and adults. *Applied Psycholinguistics* 43. 867–887. DOI: [10.1017/S0142716422000170](https://doi.org/10.1017/S0142716422000170).
- Baron, Alisa, Katrina Connell & Zenzi M. Griffin. 2022. Grammatical gender in spoken word recognition in school-age Spanish-English bilingual children. *Frontiers in Psychology* 13. 788076. DOI: [10.3389/fpsyg.2022.788076](https://doi.org/10.3389/fpsyg.2022.788076).
- Barr, Dale J. 2022. *Cluster-permutation tests for time-series data from factorial experiments*. <https://rdrr.io/github/dalejbarr/clusterperm/>.
- Brouwer, Susanne, Simone Sprenger & Sharon Unsworth. 2017. Processing grammatical gender in Dutch: Evidence from eye movements. *Journal of Experimental Child Psychology* 159. 50–65. DOI: [10.1016/j.jecp.2017.01.007](https://doi.org/10.1016/j.jecp.2017.01.007).
- Dink, Jacob W. & Brock Ferguson. 2015. *EyetrackingR: An R library for eye-tracking data analysis*. <http://www.eyetrackingr.com>.
- Fuchs, Zuzanna & Irina A. Sekerina. 2024. New evidence for the role of morphological markedness of gender agreement cues in monolingual and heritage-bilingual facilitative processing. In Lauren Clemens, Vera Gribova & Gregory Scontras (eds.), *Syntax in uncharted territories. Essays in honor of Maria Polinsky*, 215–232. University of California.
- Günther, Thomas, Annika Kirschenkern, Axel Mayer, Frederike Steinke & Jürgen Cholewa. 2023. The development of predictive gender processing strategies during noun phrase decoding: An eye-tracking study with German 5-to-10-year-old children. *Journal of Speech-Language-Hearing Research* 66. 3907–3924. DOI: [10.1044/2023_JSLHR-23-00100](https://doi.org/10.1044/2023_JSLHR-23-00100).
- Ito, Aine & Pia Knoeferle. 2022. Analysing data from the psycholinguistic visual-world paradigm: Comparison of different analysis methods. *Behavior Research Methods* 55. 3461–3493. DOI: [10.3758/s13428-022-01969-3](https://doi.org/10.3758/s13428-022-01969-3).
- Ivanova-Sullivan, Tanya, Natalia Meir & Irina A. Sekerina. 2024. Fine-grained differences in gender-cue strength affect predictive processing in children: Cross-linguistic evidence from Russian and Bulgarian. *Journal of Experimental Child Psychology* 242. 105868. DOI: [10.1016/j.jecp.2024.105868](https://doi.org/10.1016/j.jecp.2024.105868).

23 The effectiveness of cues in gender processing by Russian children

- J, Cholewa, Neitzel I, Bürsgens A & Günther T. 2019. Online-processing of grammatical gender in noun-phrase decoding: An eye-tracking study with monolingual German 3rd and 4th graders. *Front. Psychol* 10. 2586. DOI: [10.3389/fpsyg.2019.02586](https://doi.org/10.3389/fpsyg.2019.02586).
- Kray, Jutta, Linda Sommerfeld, Arielle Borovsky & Katja Häuser. 2024. The role of prediction error in the development of language learning and memory. *Child Development Perspectives* 18. 190–203. DOI: [10.1111/cdep.12515](https://doi.org/10.1111/cdep.12515).
- Lemmerth, N. & H. Hopp. 2019. Gender processing in simultaneous and successive bilingual children: Cross-linguistic lexical and syntactic influences. *Language Acquisition* 26(1). 21–45. DOI: [10.1080/10489223.2017.1391815](https://doi.org/10.1080/10489223.2017.1391815).
- Lew-Williams, Casey & Anne Fernald. 2007. Young children learning Spanish make rapid use of grammatical gender in spoken word recognition. *Psychological Science* 18(3). 193–198. DOI: [10.1111/j.1467-9280.2007.01871.x](https://doi.org/10.1111/j.1467-9280.2007.01871.x).
- Luke, Steven G. & Kiel Christianson. 2015. Predicting inflectional morphology from context. *Language, Cognition and Neuroscience* 30. 735–748. DOI: [10.1080/23273798.2015.1009918](https://doi.org/10.1080/23273798.2015.1009918).
- Özkan, Deniz, Aylin C. Küntay & Susanne Brouwer. 2022. The role of verbal and working memory skills in Turkish-speaking children’s morphosyntactic prediction. *Applied Psycholinguistics* 43(6). 1305–1328. DOI: [10.1017 / S0142716422000388](https://doi.org/10.1017/S0142716422000388).
- Pickering, Martin & Chiara Gambi. 2018. Predicting while comprehending language: A theory and review. *Psychological Bulletin* 144(10). 1002–1044. DOI: [10.1037/bul0000158](https://doi.org/10.1037/bul0000158).
- R Development Core Team. 2021. *R: A language and environment for statistical computing*. Version 4.2.1, Vienna, Austria: R Foundation for Statistical Computing.
- Rodina, Yulia & Marit Westergaard. 2015. Grammatical gender in bilingual Norwegian–Russian acquisition: The role of input and transparency. *Bilingualism: Language & Cognition* 20(1). 197–214. DOI: [10.1017/S1366728915000668](https://doi.org/10.1017/S1366728915000668).
- Sekerina, Irina A. 2014. Visual world eye-tracking paradigm. In Patricia J. Brooks & Vera Kempe (eds.), *Encyclopedia of language development*, 657–658. Thousand Oaks: Sage Publications.
- Sekerina, Irina A. & John C. Trueswell. 2011. Interactive processing of contrastive expressions by Russian children. *First Language* 32(1-2). 63–87. DOI: [10.1177 / 0142723711403981](https://doi.org/10.1177/0142723711403981).
- Smolik, Filip & Veronika Bláhová. 2019. Czech 23-month-olds use gender agreement to anticipate upcoming nouns. *Journal of Experimental Child Psychology* 178. 251–265. DOI: [10.1016/j.jecp.2018.10.004](https://doi.org/10.1016/j.jecp.2018.10.004).

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Van Heugten, Marieke & Rushen Shi. 2009. French-learning toddlers use gender information on determiners during word recognition. *Developmental Science* 12(3). 419–425. DOI: [10.1111/j.1467-7687.2008.00788.x](https://doi.org/10.1111/j.1467-7687.2008.00788.x).

Westergaard, Marit. 2014. Linguistic variation and micro-cues in first language acquisition. *Linguistic Variation* 14(1). 26–45. DOI: [10.1075/lv.14.1.02wes](https://doi.org/10.1075/lv.14.1.02wes).

Chapter 24

A spanning approach to the categories of the English verb and their linearization

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In English, auxiliaries linearize in the T position, preceding medial adverbs, while lexical verbs linearize in the verb phrase, following medial adverbs. In a span-based approach to word order, linearization of words in the extended projection is determined by the position of a spell-out label @. The English pattern suggests that the distribution of @ is not a fixed property of a head like T, but is partly dependent on the category of the complement of that head. I propose to encode this observation in a functional lexicon which is not simply an unstructured set of formatives, but is organized into a finite state automaton, where properties of the functional lexicon are distributed over the states and transitions. The finite state automaton defines an extended projection. States encode properties which are fixed for a category, where category is defined in terms of distribution. Transitions carry information that can vary with the lower category in the extended projection, including information affecting linearization.

1 Introduction

A basic difference between English grammar and French grammar is apparent in the relative order of the verb and medial adverbs, which has been analyzed in terms of the verb moving to T in French, but remaining in the verb phrase in English (some version of a verb-raising analysis of French goes back at least to Emonds 1978).

- (1) a. Marit marche souvent.
 Marit walks often
 ‘Marit often walks.’

Peter Svenonius. 2026. A spanning approach to the categories of the English verb and their linearization. In Merete Brendeford Anderssen, Anne Dahl, Terje Lohndal, Natalia Mitrofanova & Yulia Rodina (eds.), *Variation, acquisition and change*, 413–437. Berlin: Language Science Press. DOI: ??

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- b. Marit often walks.

Formally, this can be described in terms of French T having a “strong” feature which attracts the verb, causing it to move (Pollock 1989, Chomsky 1991, Lasnik 1995).¹

In span-based approaches to word linearization stemming from Brody 2000a, heads do not move. A word spells out a span, a sequence of heads in a head-complement relation in an extended projection like V-*v*-T. When specifiers and adjuncts appear in between heads in the span, for example an adverb adjoined to *v*P, or a noun phrase in the specifier of VP, the word spelling out the V-*v*-T span must linearize to the left or right of that item. Brody proposed to regulate the linearization of heads in their extended projections by means of a spell-out label @ (“at”). On that view, French has @ in T, while English has @ lower down, in *v*. The label @ is then like a strong feature in the sense of being a locus of parameterization, but the linearization of words in their extended projections is not due to movement.² If arguments of the verb generally occupy specifiers which linearize to the left of V, then @ in V gives OV order, according to Brody’s proposal.

An interesting complication to this picture is that English finite auxiliaries and modals and the copula obligatorily precede negation, and tend to precede medial adverbs.

- (2) a. The walk isn’t arduous. (*The walk not is arduous)
 b. The leaves may not have fallen. (*The leaves not may have fallen)
- (3) a. The tourists will always remember the scenery. (more marked: *always will remember*)
 b. The leaves have probably turned. (more marked: *probably have turned*)

According to Baker 1991, the marked order in which the auxiliary follows a medial adverb in examples like the parenthesized options in (3) generally requires stress on the auxiliary, and following Wilder 1997 it can be associated with verum focus, in which the contradiction of the assertion is a salient proposition in the discourse (one which is challenged by the speaker).

¹Pollock related the strength of the feature to the projection of θ roles, whereas Lasnik argued for a morphological motivation.

²Like a strong feature, @ can either be connected to other syntactic properties, as in Pollock’s analysis of verb movement, or not, as on Lasnik’s. The question is whether “head movement” has interpretive effects, and the jury is out, as summarized in the review by Dékány 2018.

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An adverb may precede the auxiliary even when the auxiliary precedes negation; the relative order determines the semantic scope, so that the two examples in (4) have different meanings.

- (4) a. They obviously didn't leave.
b. They didn't obviously leave.

Following Laka 1990, there is a polarity head Σ in the clause, located below the tense head T in English. Σ appears when there is sentential negation or verum focus, and is absent in affirmative declaratives without marked focus (as in Wilder's implementation). In order for it to be interpreted as negative, it must obtain a negative feature from a negative operator in or below its projection. Suitable negative operators include the adverbial *not*, which appears below Σ , a negative DP like *nobody*, or *n't* which appears in Σ as a clitic specifier.³

The word orders discussed so far can be accounted for as follows. The unmarked position for an auxiliary in a finite clause in English is in T, as in the unmarked examples in (3). Middle field adverbs typically adjoin to projections below T, so they follow an auxiliary in T. Σ is present below T if there is negation or verum focus, in which case the finite auxiliary may appear in Σ and fail to raise to T. This accounts for the marked orders in (3), if the adverbs there are adjoined to Σ P. The negative clitic *n't* is introduced in Σ , appearing as an enclitic on the auxiliary to establish the obligatory order in (2a). The freestanding form *not* adjoins to a projection below Σ , following a finite auxiliary whether it remains in Σ or raises to T, as in (2b).

My objective therefore is to develop a formal grammar in which the highest auxiliary in a finite clause (including modals) obligatorily linearizes at least as high as Σ when Σ is present, and optionally higher in T. In the absence of Σ , the highest auxiliary obligatorily linearizes in T. In the absence of an auxiliary, the verb linearizes in the verb phrase, with finite inflection, unless it is the copula, in which case it behaves as though it were an auxiliary. Along the way I will also account for various other details of the English system of auxiliaries.

³The clitic status of *n't* is evident from the fact that it appears outside tense (as in *ha-d-n't*, **have-n't-ed*) despite being introduced in a projection structurally lower than T (as evidenced by the pattern of *do*-support, Chomsky 1957). It is morphologically a clitic in the sense defined in Svenonius 2016, a deficient specifier spelled out in the same word-formation cycle as its host. Its clitic status allows it to linearize as an enclitic, outside inflection, and the fact that it is introduced inside the word cycle, between the base Aux and the inflection T, allows it to condition allomorphy of the exponence of its host, as in *won't* (cf. Zwicky & Pullum 1983, who observe that that is not possible for simple clitics).

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Projections above T are not going to be treated in this paper, but I will briefly mention the first obvious extension. Stance adverbials like *frankly* and *unfortunately* can easily follow a tensed auxiliary, in English, as indicated in (5).

- (5) a. The days of summer are frankly numbered. (more marked: *frankly are numbered*)
- b. The tourists will obviously remember the scenery. (more marked: *obviously will remember*)

Given what I have said so far, these sentences would seem to require the stance adverbials to be attached below T, perhaps to ΣP . However, that is somewhat surprisingly low, given that such adverbials are interpreted outside the truth-conditional proposition which includes the semantic content of T (tense). Wilder 1997 suggests that high adverbials are parenthetical in that position. Cinque 1999 proposes that there are higher heads than T (Wilder 1997 also makes use of an Agr position above T). In that case, the unmarked examples in (5) might involve the auxiliary linearizing in a position above T such as Fin[ite].

Ernst 2002 proposes that the semantic interpretation of syntactic projections is flexible to a certain degree, reducing the number of fine-grained discriminations needed in the syntax. In that case, the examples in (5) could be accommodated without positing a higher head. I will assume that the account that I develop here can be extended to accommodate examples like those in (5) one way or the other, and will proceed to describe the English clause up to T. Further extension would cover subject-aux inversion.

To return to the main story line, the functional structure including the sequence ...-(Σ)-T-(Fin) is by assumption the same for auxiliaries and lexical verbs, yet only auxiliaries linearize there in English. In an analysis with head movement driven by feature checking, the different position of auxiliaries compared to lexical verbs can be described in terms of a featural distinction between the two, but in a theory where linearization of the words in the extended projection is determined only by the distribution of @ in the span, @ on a functional head like T should simply affect the highest available word in the projection line irrespective of its featural make-up (as in V2 languages, with @ on C).

In fact, the problem is more general, if auxiliaries in English more generally occupy the position of the category of their inflection, as argued by Harwood 2014, rather than the position immediately above their complement. If that is correct, then for example the position of a perfect-inflected auxiliary *been* is in the Perfect head which gives it its inflection, whether it is the auxiliary of a passive or the auxiliary of a progressive. Since main verbs with perfect inflection

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remain in the verb phrase (as witnessed by the fact that they are not separated from their objects by adverbs), this would seem to be another instance of an inflectional head inducing different linearization on auxiliaries as opposed to main verbs.

The analyses of Adger 2003 and Bjorkman 2011, in which inflections are copied downward, make the opposite prediction, and I will not resolve the empirical question here. However, I can briefly illustrate the issue. Suppose that there is a class of adverbs which are preferentially adjoined higher than a progressive-inflected passive auxiliary but lower than a perfect-inflected progressive auxiliary. These would be toward the lower end of the adverb scale of Cinque 1999. The examples in (6) show that adverbs can appear there; additional examples would be needed to show whether this position is the preferred one for some senses of some adverbs.⁴

- (6) a. They have been constantly being shown soothing images.
- b. They have been gradually being taken advantage of.

Neither of these adverbs requires the progressive. We can therefore ask what the preferred relative order is with respect to a perfect-inflected passive auxiliary. If auxiliaries generally linearize according to their base category, then a perfect-inflected passive auxiliary should linearize like the progressive-inflected passive auxiliaries in (6), after the adverbs. If auxiliaries generally linearize according to their inflection, then a perfect-inflected passive auxiliary should linearize like the perfect-inflected progressive auxiliaries in (6), before the adverbs. Relevant kinds of examples are illustrated in (7).

- (7) a. They have {been} constantly {been} shown soothing images.
- b. They have {been} gradually {been} taken advantage of.

Suppose for the sake of argument that there is a preference for the auxiliary-adverb order in these cases. If it can also be shown that the same adverbs are not equally good after the progressive-inflected passive auxiliary in examples like

⁴There are a few relevant examples in the Corpus of Contemporary American English (www.english-corpora.org), including: *It's been constantly being enriched...*; *...people might have been gradually being reprogrammed....* Testing with the adverbs discussed by Cinque, I found that *completely* was too low, and preverbal *again* was too high, as in *He has again been being completely taken advantage of*, where neither adverb seems to sit easily between *been* and *being*. Note that adverbs like *completely* can also modify adjectives, so in order to distinguish the passive auxiliary from the copula, it can be important to avoid passive participles which are ambiguously adjectival, hence my use of passive participles with objects.

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(6), then that would be an argument that nonfinite auxiliaries linearize in the position of their inflection, not of their base category. Since the preferences are not yet clear, I concentrate on the clearer case of T.

To summarize, the problem addressed in this paper is the problem of the linearization of verbal heads in the extended projection in English. Auxiliaries linearize in T, but main verbs don't. If linearization is controlled by the label "@" at the point of linearization, then this seems to mean that T has a property when it combines with auxiliaries that it lacks when it combines with main verbs. The question is how this information is encoded in the grammar.

An interesting hypothesis in Minimalist Grammar is that all variation is encoded in the listed items of the lexicon (Borer 1984, Chomsky 1986). This becomes especially clear when Minimalist Grammar is fully formalized as in Stabler 1997, Stabler 2011. Because English has past and present tense, there must be distinct formatives $T_{[PRES]}$ and $T_{[PAST]}$ (in some languages they have different properties, e.g., Arabic as analyzed by Benmamoun 2000). Both have the property that they can combine with a lexical verb, and when they do, they lack the property @, if my span-based premise about linearization is correct. Both heads also have the property that they can combine with an auxiliary, but when they do, it seems they must have the property @. What's more, the same may be true of one or more nonfinite inflectional heads, according to Harwood 2014 and the discussion surrounding (6–7) above.

To avoid the conclusion that there are multiple strong and weak instances of inflectional heads, one could posit a feature on auxiliaries along with a mechanism that requires items with that feature to linearize in T, as is done for example in Baker 1991, who posits a feature [+Special]. However, syntax seems to make sparing use of ad hoc features, so they should be used judiciously.

My alternative solution, in a nutshell, will be to suggest that distribution of the label @ be separated from the lexical entries for the formatives and from the category T. I will suggest instead that T in English has @ when it combines with auxiliaries and lacks it when it combines with main verbs. The distribution of @ must be learned in any case, the question is just whether the distribution of @ is learned for each formative, or for something else (as I propose here). In my analysis, @ is not a feature of formatives, and for that reason I avoid calling it a feature at all.

Effectively, I am suggesting that properties of an extended projection can be extracted from the collection of functional heads that constitute it. At this point, this provides a more succinct statement of the grammar but does not make different formal predictions compared to a Minimalist Grammar in which all parametrically varying properties are encoded as features on functional heads. However,

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since I shift the locus of variation in a concrete way to an aspect of the grammar distinct from the heads themselves, I open up for a different perspective on cross-linguistic variation, one which might make original predictions as the theory develops. The matter bears on what Rizzi 2017 has called the format of parameters, and where in the grammar children acquiring a language locate listed information.

Construction Grammar is another framework in which language-specific lexically listed information is not restricted to a list of formatives (as applied to the English auxiliary system in Sag et al. 2020), but space precludes a detailed comparison here. More immediately relevant here is Westergaard's (2014) theory of micro-cues, in which children acquiring language are sensitive not only to the properties of individual heads, but to the properties of heads in the immediate context of subtrees (which in many of the cases she discusses could simply be spans, sometimes including a feature relating a head in the span to a class of specifier).

2 The extended projection of the verb in English grammar

In a formal Minimalist Grammar following the strict interpretation of the Borer-Chomsky conjecture, the syntax of a language consists of a set of formatives. These formatives have features which interact with principles of Universal Grammar to generate all and only the grammatical sentences of the language. In a late insertion theory (Halle & Marantz 1993, see also Svenonius to appear for discussion), there is also a set of exponents which are matched to those formatives in a spell-out procedure, and in a spanning version of late insertion, exponents may spell out spans (head-complement sequences).

Formatives are traditionally organized into distributional classes, or categories (Bloomfield 1933). English verb forms can be organized into five important distributional classes, each one identifying a category: the infinitive, the finite forms of the verb, the perfect participle, the passive participle, and the progressive participle (as indicated in the leftmost column of the table in Table 1, and setting aside the imperative and subjunctive). The perfect and passive participles are systematically identical (syncretic), and except for the copula and auxiliary *be*, present tense forms other than the third singular are syncretic with the infinitive. For regular weak verbs, as well as for many strong verbs, the past is identical to the perfect/passive participle.

The present and past forms together constitute only a single distributional class because there are no syntactic contexts in which a past tense verb is possible

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Table 1: pls provide a caption

Distributional Categories	Interpretable features distinguishing formatives	Agreement features distinguishing surface forms	Examples of exponents		
			Copula	Strong	Regular
Infinitive (T _{inf})			be	break	walk
Finite (T)	pres	1sg	am		
		2sg or pl	are		
		3sg	is	breaks	walks
	past	1sg or 3sg	was	broke	walked
		2sg or pl	were		
Perfect (Perf)			been	broken	
Passive (Pass)			—		
Progressive (Prog)			being	breaking	walking

but a present tense verb is not.⁵ Therefore, there is a category finite T in English which can surface in featurally distinct forms, as illustrated in Table 1. Some of those forms express agreement with the subject, so that if we distinguish underlying features from feature values acquired in the course of the derivation, then there are at least two formatives of the category finite T in English, which we can call T_[PAST] and T_[PRES]. They can be assumed to both have the same phi-probe for subject agreement features, with person and number being filled in through an Agree operation which copies the nearest appropriate features onto the probe (Chomsky 2001).

3 Finite state generation of extended projections

The grammar contains information indicating which formatives can take which formatives as complements in the projection of extended projections, for exam-

⁵In “sequence of tense” contexts, like *I didn’t think it was/*is Friday today*, as analyzed in Zagana 2014, the conditioning context for the tense is another tense; this is similar to cases of agreement and (e.g., negative) concord, and can be distinguished from the contexts which define categories.

ple whether T can combine directly with V or whether there is some other intermediate category such as Asp or Voice which must combine with V first. Every extended projection is a string of categories, and the part of the grammar which generates them is a simple Finite State Automaton (FSA), as discussed in [Svenonius 2012, 2021, 2024](#).⁶

A Finite State Automaton consists of a set of states, including an initial state and one or more accepting states, a set of symbols, and a state-transition function, which is a set of triples of two states and a symbol. In the case of extended projections, formatives are the symbols, the states are the syntactic categories to which the formatives belong or the phrasal projections of the formatives, and the state-transition function is the specification of which categories take which complements.⁷

An example is given in Figure 1, where edges are labeled with names of formatives. Accepting states are complete phrases which can be embedded as nonheads in another extended projection (i.e., states at which the operations in the workspace can be terminated). Here, AspP and TP are accepting states because progressive AspPs and finite TPs can be embedded as nonheads in other extended projections.

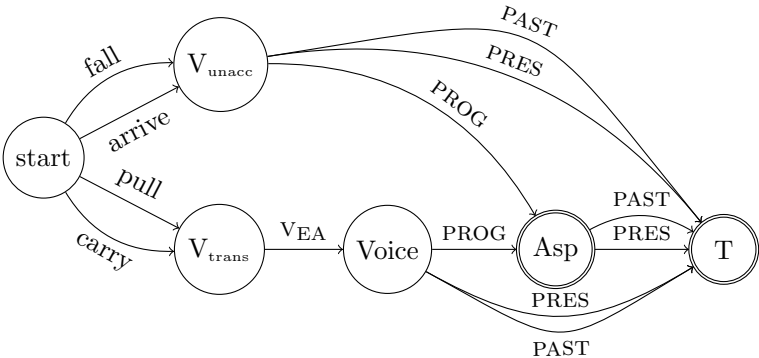


Figure 1: pls provide a caption

Unaccusative verbs, V_{unacc} , can project either to Asp or T, meaning that either Asp or T can merge with VP to create an AspP or a TP. Obligatorily transitive verb

⁶[Kobele to appear](#) uses flowcharts which are similar to my FSA as an expository device. [Hanson 2024](#) observes that extended projections and other aspects of grammar can be described by simple FSA and investigates their computational restrictiveness in terms of a notion of strict locality.

⁷See [Adger 2013](#) for an example of a grammar in which the Hierarchy of Projections is encoded as a list of pairs of categories.

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roots, on the other hand, must first merge with an external argument introducing head, Voice (Kratzer 1996). The VoiceP can then merge with Asp or T just like V_{unacc} .

To produce the morphological spell-out of this set of extended projections, more information could be added, for example with what is known in computational morphology as a finite state transducer where the present tense transitions from V_{unacc} and Voice are spelled out as *-s* (notated as PRES:-s), and the present tense transition from Asp to T is spelled out as *is* (notated PRES:is). However, my focus here is on the generation of the syntactic structure of extended projections, not their morphophonological exponence.

A lexicon of formatives that encodes the FSA in Figure 1 can be represented as in Table 2. I represent alternative complementation in the form of sets under the heading ‘functional complementation’. At this stage of the exposition, the lexicon in Table 2 should be seen as a notational variant of the FSA in Figure 1 (apart from the minor detail that I have added theta roles and ϕ -probes and EPP to Table 2 in the last column and left them out of Figure 1). The annotation ‘Complete’ in Table 2 corresponds to a double circle in Figure 1 and means that a category can be embedded in another extended projection, as mentioned above (or possibly used as a matrix utterance, in the case of TP).

Table 2: pls provide a caption

Category	Formatives	Functional complementation	Other category-wide features
V_{unacc}	fall, arrive, ...	–	θ
V_{trans}	pull, carry, ...	–	θ
Voice	v_{EA}	V_{trans}	θ , Accusative
Asp	PROG	{Voice, V_{unacc} }	Complete
T	PRES, PAST	{Asp, Voice, V_{unacc} }	ϕ -probe, EPP, Nominative, Complete

Note that every feature listed in the third and fourth columns is constant for the entire category. Formatives in a category may also have features that distinguish them from other formatives. For example, the formative PAST must have a feature [PAST] which is absent from the formative PRES and which is copied in sequence of tense (see note 5), to allow a semantically present tense in an embedded clause to spell out with a past exponent.

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Importantly, note that PRES and PAST are not distinct in state-transitions. Every state which can transition to T can do so either with PRES or with PAST, so that is a predictable property of T. This means that to represent the tenses as distinct transitions, as I did expositively in Figure 1, misses a generalization. Functional complementation is not a property which could distinguish PAST from PRES, but rather a property of T. Functional complementation differs in this way from PAST, which might be a feature of T, with PRES simply reflecting the absence from T of PAST. Note that sequence of tense copies past, but cannot copy present, which is explained if present is not a distinct feature.

The fact that there are two tenses can be more insightfully represented by understanding the state T in the FSA to include whatever formatives of category T exist, whether that is T with optional [PAST] or two distinctly listed formatives $T_{[PRES]}$ and $T_{[PAST]}$. In either case, the opposition of [PAST] to the T without it is a single fact about T, not a set of facts about each of the transitions to T. Recognizing this fact about T, we can use abbreviated FSAs which leave out the indications of how many formatives exist to allow projection to a given state, as long as nothing about their syntax impacts the structure of the FSA.

Similarly, in Figure 1 every transition to Asp is PROG; other aspects are not included in this diagram. Perfect aspect is introduced below, but it has a distinct syntax. For that reason, we can dispense with the label PROG on the transitions to Asp in the FSA and recognize that the aspect Asp_{PROG} , can be identified with the formative that derived the state as well as with the state itself.⁸

Stripped of the redundant transition labels, the category labels on the states in Figure 2 indicate the sets of formatives that can be combined at that stage of the construction of an extended projection. For example, the transition to V_{unacc} in Figure 2 stands in for all unaccusative verbs.

According to Figure 2, V_{unacc} and V_{trans} are not the same category. However, if we define a category in terms of external distribution, then V_{unacc} is the same category as Voice in Figure 2, because they combine with the same formatives.⁹

The formal analysis of the extended projection in the form of an FSA is neutral to the question of whether there are silent unmarked categories in the clause corresponding to the marked ones, as argued by Cinque 1999 (for whom the unmarked categories are necessary in order to explain adverb ordering). I briefly

⁸This was an aspect of a certain class of FSA discussed by Moore 1956, and FSA in which states rather than transitions are identified with outputs are therefore sometimes called Moore machines. However, I am not suggesting that extended projections should be formally identified with Moore machines, and in fact I will be putting the transitions to use below.

⁹The labels under the column “Category” in (Table 2) indicate a more fine-grained sense of category, for which both internal and external subcategorization relations are significant.

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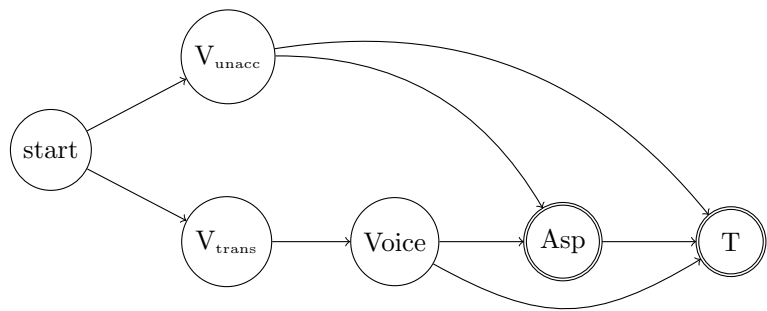


Figure 2: pls provide a caption

digress here from the main thrust of the discussion to illustrate this point because it is ultimately important for the analysis whether there is unaccusative Voice, non-progressive Asp, and so on, though the upshot of the digression will be that the analysis here can accommodate either option.

For example, Legate 2003 argues that there is also a Voice head with unaccusatives. This is reflected in the FSA in Figure 3. I have added the redundant transition labels back in for clarity, except that [PAST] is now an optional non-projecting feature (as in Svenonius 2012). The formative $v_{\emptyset}$ is a Voice head which has no external argument theta role to assign.

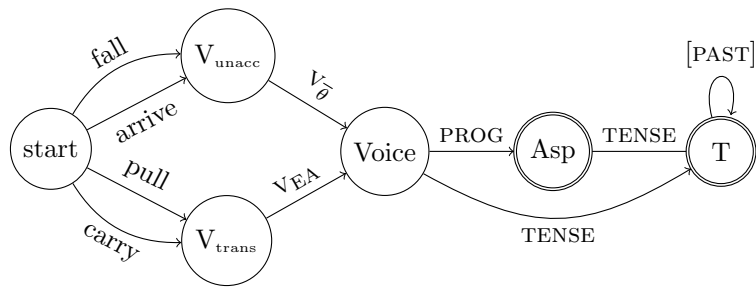


Figure 3: pls provide a caption

If the selection of Voice by verb roots can be relegated to something non-syntactic, as in Borer 2005, then V_{unacc} and V_{trans} can be collapsed into a single category.

Additional states are obviously necessary for the full range of English extended projections, as discussed further in the next section. Although perfect participles and passive participles are morphologically identical, they must be distinct categories, since they have distinct syntactic distributions; for example the passive can be embedded under the perfect but not vice versa.

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- (8) a. The books have been returned. (perfect of the passive *be returned*)
 b. *The books were had returned. (*passive of the perfect *have returned*)

The four nonfinite verbal categories in Table 1 each combine with an auxiliary or modal to form four sets of complex tense-aspect-modality (TAM) forms, namely the passive, the progressive, the perfect, and the modal forms. The progressive can contain the passive, and the perfect can contain the progressive, and the modal can contain the perfect, for a total of sixteen finite TAM forms for transitive dynamic verbs, if we don't distinguish modals and we don't distinguish present from past (fewer for verbs which cannot combine with passive and/or progressive). Since modals and tenses involve distinct formatives, they should be counted as well. Setting aside the question of how compositional the modal-tense combinations are in English, we can distinguish eight finite modals (*can, could, will, would, may, might, must, and should*) and the two tenses, giving 80 forms (16 without modals and 64 forms that include one modal each). With negation there are 160 forms. If the kind of emphatic focus seen in *We DO like it* reflects the presence of a formative Σ , as argued above, which can be combined more generally (e.g., in *We MIGHT like it*), then there are at least 320 forms of the extended projection, up to T, for each verb which can be passivized and can appear in the progressive.

4 The linearization label @

The FSA in Figure 4 generates all and only the 64 forms mentioned above prior to the introduction of Σ (T_{inf} is the bare infinitival which combines with modals; T is finite tense). I have simplified the start state, collapsing V_{trans} and V_{unacc} to a single composite category V with a single Voice for compactness, and giving it a different shape (rounded square) to indicate its status as the start of the FSA. I am omitting the higher Fin projection mentioned in Section 1.

Apart from the composite V, each state represents a distinct category, and each transition corresponds to a set of formatives licensing a projection to that higher category. The states in the form of circles are the inflectional categories from Table 1, and are distinguished by the feature [w] which is the feature designating a node for word-level spell-out – a category which is expressed by word-final morphology has [w] (Marantz 2001, Svenonius 2016, Bye & Svenonius to appear).

Nodes lacking [w] are stems which need to be inflected, and their states have a diamond shape in Figure 4 for visual reference.

Transitions with the symbol @ (“at”) indicate that the category they lead to is a possible linearization point in the extended projection (Brody 2000b, Svenonius

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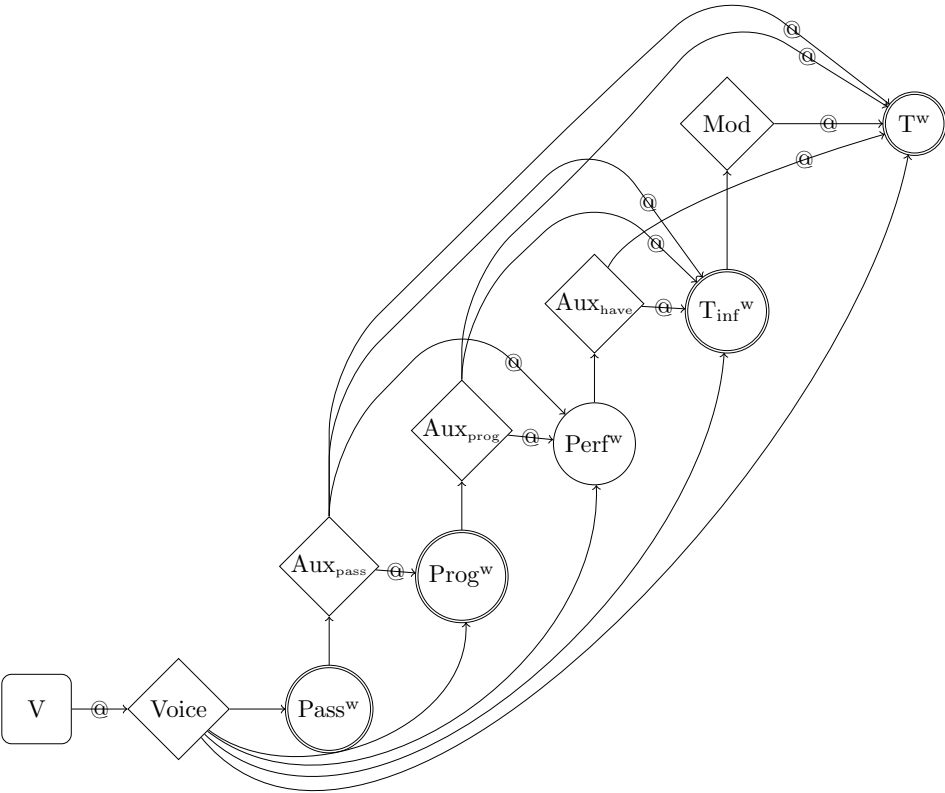


Figure 4: pls provide a caption

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2016). Like [w], this is a parametric setting, set by the learner according to input: if an exponent of a span $X\text{--}Y$ (Y taking X as a complement) precedes an adjunct or specifier of X , then the span must have @ in Y . In previous work, such as Svenonius 2016, I have assumed that @ is fixed for a category, but here I propose that it is relativized to transitions. This allows inflectional heads to have @ when they dominate auxiliaries and not when they dominate lexical verbs.

The placement of @ in Figure 4 on transitions leading from auxiliary stems to inflectional states reflects the Harwood claim that auxiliaries linearize according to their inflections. If the Adger-Bjorkman position instead turns out to be correct, these @s should be moved to the transitions leading to the auxiliary stems from the inflectional states.

The extended projection in a particular phrase marker is gradually built up by successively progressing from state to state in the FSA and the full phrase structure involves importing phrases from other workspaces to Merge in as specifiers and adjuncts (see Svenonius 2024 for details). To implement an FSA in performance, properties of categories or of formatives could be assigned different degrees of activation according to the informational context (this would correspond to part of Chomsky's 1995 Numeration, a selection of formatives which are used to construct a sentence). At each state from which there is more than one possible transition, the transition would be chosen which leads to the state with the greatest degree of activation.

5 Negation

Complex finite TAM forms combine regularly with sentential negation, but the simple finite forms do not; some auxiliary or modal is required, and if no other one is motivated by the TAM context, then the auxiliary *do* is used.

As discussed in Section 1, Laka 1990 proposed the polarity category Sigma (Σ) for the position of negation; the negative particle *n't* is introduced as a specifier of Sigma if there is no other negative licenser available to a feature probe on Σ . The appearance of *do* in negative clauses motivates an additional auxiliary position, Aux_{do} . In Figure 5, I omit the passive and perfect from this diagram for simplicity; they work exactly like the progressive, as seen above in Figure 4.

Now, a lexical infinitive verb can combine either with a modal or with the auxiliary *do*; if it combines with *do*, it must then project to the negative head or the emphatic positive variant of Σ seen in *Marit DOES walk*. For the other auxiliaries and the modal, Σ can be bypassed via a direct transition to T. The transition from Σ to T without @ is the marked option which allows focused auxiliaries to remain in Σ (as seen in (3); otherwise they linearize in T.

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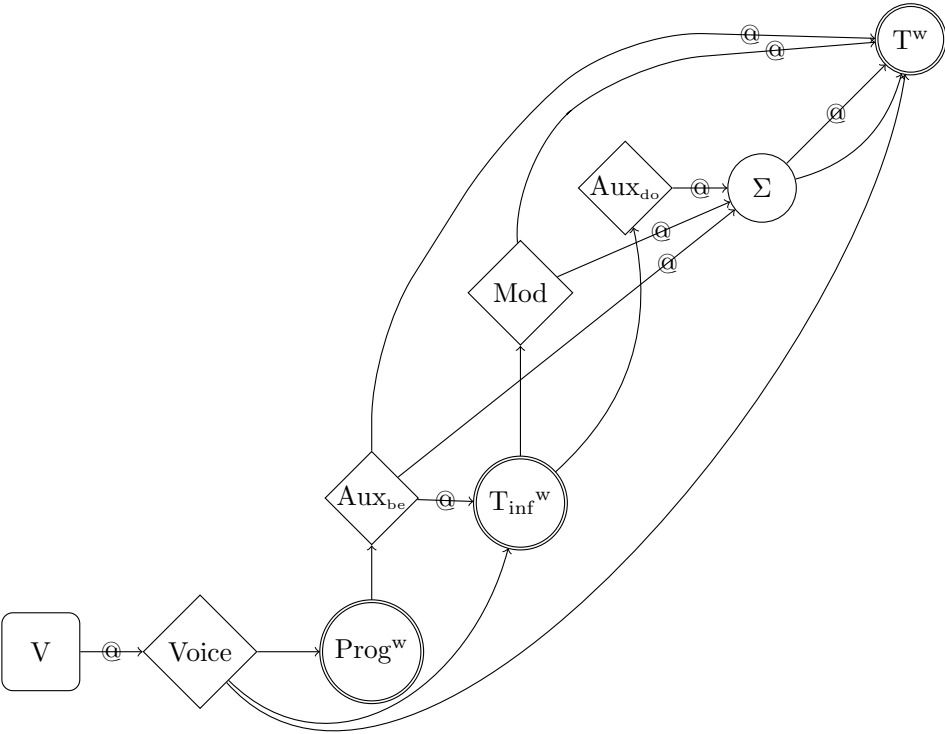


Figure 5: pls provide a caption

The FSA allows auxiliaries to be infinitival, to combine with modals, as in (9a). But in Figure 5, this has the undesired effect that auxiliaries can combine with the auxiliary *do*, which is ungrammatical, as illustrated in (9b).

- (9) a. The kettle should be on the counter.
- b. *The kettle doesn't be on the counter.

This shows that the infinitive of an auxiliary does not have exactly the same distribution as the infinitive of a lexical verb. This could be treated in terms of a feature which distinguishes auxiliaries from lexical verbs, but it would require information to be ‘percolated’ from the category below T_{inf} , allowing a nonlocal

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featural interaction (between the verb and Aux_{do} , across the intervening T_{inf}) which is distinct from Agree.

Feature percolation of some kind is commonly assumed in other frameworks (e.g., HPSG, Pollard & Sag 1994), but a Minimalist Grammar approaches such mechanism with circumspection, in the interest of restrictiveness (see Adger & Svenonius 2011 on the restrictiveness of Minimalist feature theory). In fact, Kobele 2005 demonstrates formally that adding feature percolation to a Minimalist grammar potentially increases its expressive power beyond that which is motivated for natural language, and extended projection generation appears to be subject to strict locality. Features may have nonlocal effects only insofar as they trigger a small set of specific operations such as move (internal Merge) or Agree (cf. Stabler 1997).

This motivates a solution to the puzzle in (9) in the form of a connegative or polar infinitive (T_{pol}), distinct from the modal infinitive (T_{inf}), only for lexical verbs. The revised FSA then appears as in Figure 6.

Now there are two kinds of infinitives for lexical verbs, the polar infinitives, which are like participles in having a dedicated auxiliary, and the regular infinitives, which combine with modals.

In the technical sense of category that I have been using here, these two infinitives are different categories, since they have different distributions (one combines with *do*, the other with modals). But they occupy the same height in the extended projection, which has a dimension diagonally upward to the right in Figure 6. In Grimshaw's (2005) terms, they have the same F-value. Shared F-value is the sense in which a complementizer in German (such as *dass*) is said to be the same category as the position of a verb in a main clause, because they are at the same height in the functional hierarchy; in this broader sense T_{inf} and T_{pol} are the same category. I will not use the term category in this broader sense, so in my terms T_{inf} and T_{pol} are at the same height in the functional hierarchy but are different categories because they do not have identical sets of transitions to other categories. If there are indications that, say, Mod and Aux_{do} have the same F-value, this could be reflected in the FSA by putting them at the same distance from V on the diagonal. In general, extended projections have the monotonic property that they do not decrease in F-value as more categories are added to the lexical head, something which may be due to a constraint on how the learner develops FSAs for extended projection production (Svenonius 2024).

The main innovation in this paper has been to decompose lexical entries so that the property governing word linearization (@) is separated from other properties of the formative, namely by putting @ in the transition and keeping the other properties in the list of formatives that belong to the category identified by

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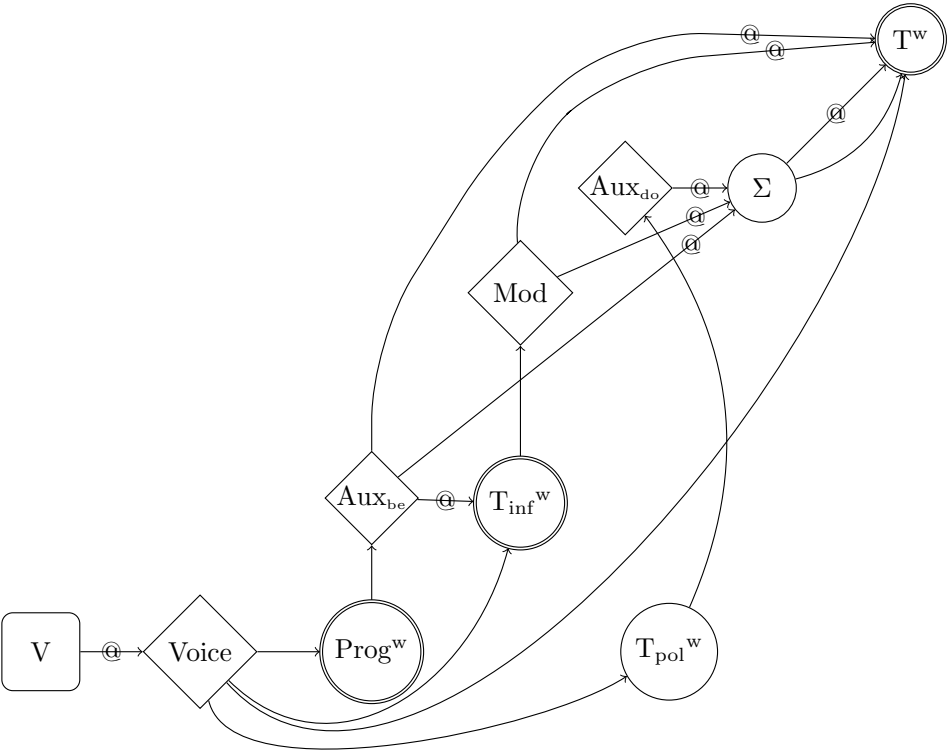


Figure 6: pls provide a caption

the state. A similar decomposition would allow the elimination of the connegative T_{pol} , if the $[w]$ feature designating states as morphological words could also be associated with a transition. In this case, the transition in question would be one from Voice to Aux_{do} . If that transition specifies that Voice is a word, then T_{pol} is not needed. Note that $[w]$ on a transition affects the state that the transition is exiting, while $@$ on a transition affects the state that the transition enters.

In Figure 7, there is no path from auxiliaries to Aux_{do} like the one that caused overgeneration problems in Figure 5, and there is no connegative infinitive duplicating the modal infinitive like the one in Figure 6. Instead, active Voice can be converted into a connegative verb syncategorematically by being embedded under Aux_{do} , as indicated by the transition from Voice to Aux_{do} marked with

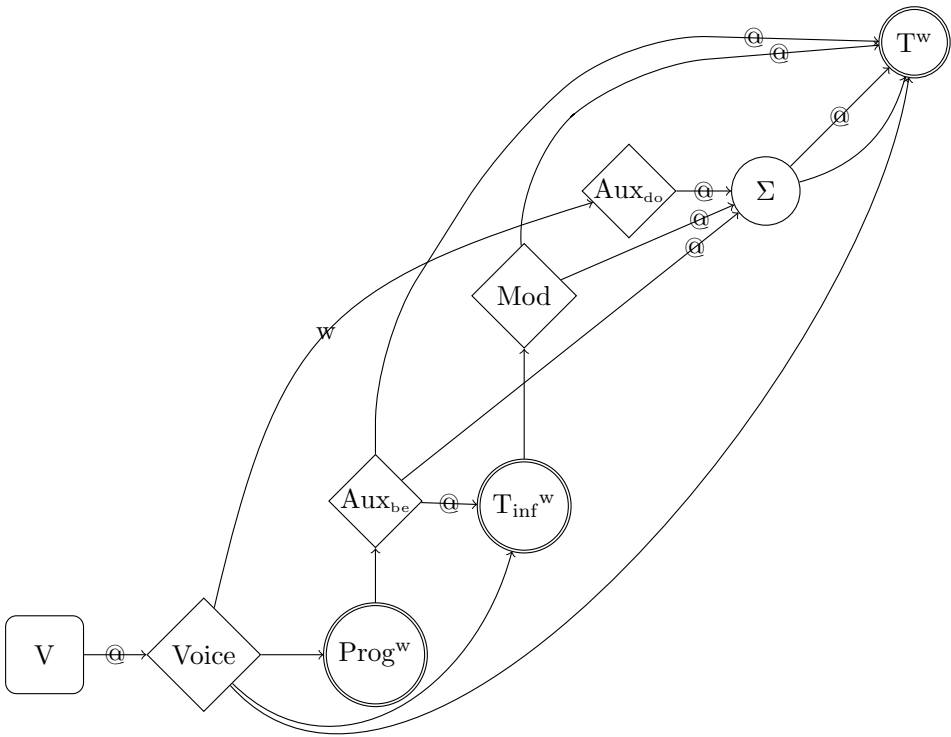


Figure 7: pls provide a caption

[w].

I retain the more conservative treatment of T_{inf} as a formative with its own category because there is a category of infinitive verb phrase which can be embedded without the additional trappings of finite inflection. The syncategorematic treatment of the connegative implies that it cannot be embedded, because only a state can be an accepting state. At the moment this does not seem to be testable since the connegative infinitive generated by Figure 5 did not have properties distinguishing it from T_{inf} . The association of [w] with the transition is somewhat tentative, so I haven't moved the other [w] features in the diagram to transitions.

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6 The copula

The copula in English has the special property that it linearizes like an auxiliary when it is finite, but as a lexical verb when it is nonfinite. This special behavior can be captured if the copula can combine directly with T in neutral affirmative clauses, or with Σ in negative and verum focus clauses. This means that just for that verb, there are transitions directly from V to T and from V to Σ , with @, as depicted in Figure 8. Recall that the rounded box labeled V is a composite category including different classes of verbs which combine with different Voice heads. Now it also includes the category of the copula which has its own transitions to Σ and to T (I am omitting transitions without @ to Prog and T_{inf} for nonfinite copulas).

Restoring the other composite TAM categories which were removed for perspicuity, the full FSA generating all 320 verbal extended projections up to T for each dynamic verb in English with an external argument looks something like Figure 9. This FSA consists of 13 states representing around 19 formatives (depending on specifics of the analysis of the modals) and 32 transitions.

Additional forms can easily be added as needed, for example for expressions like *be going to*, or *ought to*, or *be about to*. The prediction is that insofar as they are grammatically encoded as part of a single extended projection, they should be compatible with the constraints of this simple FSA architecture. If, on the other hand, they involve embedding of one extended projection in another, they will require the additional syntax of phrasal embedding, needed for arguments and adjuncts (see [Svenonius 2024](#) for some discussion).

7 Conclusion

The original puzzle I started with was how to formally capture the observation that English auxiliaries linearize in T while English main verbs don't. One of my background assumptions has been that word linearization is regulated by the property @. Part of my motivation has been to constrain the power of the feature system.

I have argued that the subpart of a standard Minimalist Grammar that generates extended projections constitutes a simple Finite State Automaton, and I have suggested that thinking of it in this way is useful. In particular, I have suggested that the property regulating the linearization of heads be reconceptualized not as a property of formatives but rather of transitions between categories, which is a particular instantiation of Merge.

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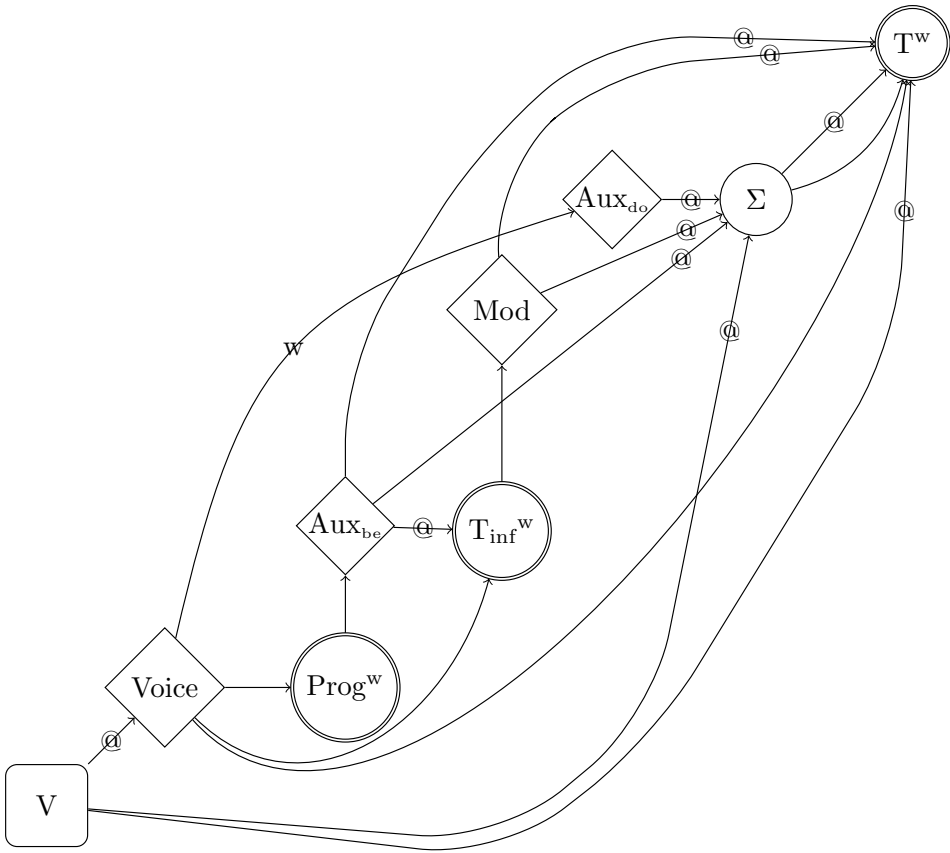


Figure 8: pls provide a caption

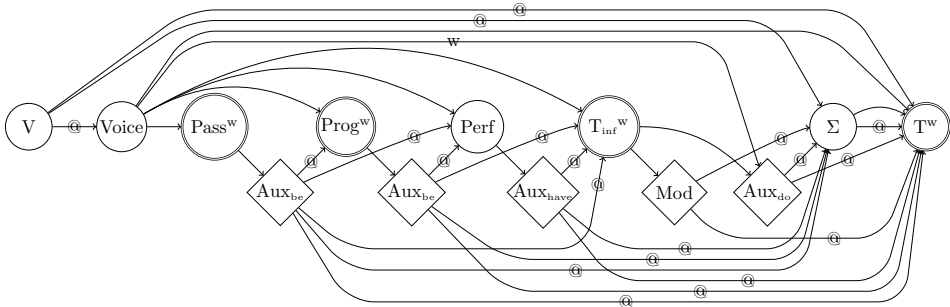


Figure 9: pls provide a caption

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This bears on the Borer-Chomsky conjecture that all linguistic variation is confined to the functional lexicon. The implementation of the Borer-Chomsky conjecture in current work is sometimes procrustean. In that spirit, one could recast the distribution of @ in lexical terms by saying that English Voice has the property of projecting to T, while English Aux has the property of projecting to T@. In this way, the distribution of @ is stated entirely in terms of properties of T and Aux.

This in turn can be further formalized by saying that the property of projecting to T@ can be recognized by learners as a feature [+Aux], and that such arbitrary features can be freely invented to create ad hoc natural classes.

I have suggested instead that certain specifics of the Hierarchy of Projections, in the form of an FSA for each major category, is a cross-linguistically variable part of the grammar, and although inextricably intertwined with the categories of the formatives that it regulates, it is usefully thought of as distinct from them.

Acknowledgements

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References

- Adger, David. 2003. *Core syntax: A minimalist approach*. Oxford: Oxford University Press.
- Adger, David. 2013. *A syntax of substance*. Cambridge: MIT Press. DOI: [10.1017/s0022226714000073](https://doi.org/10.1017/s0022226714000073).
- Adger, David & Peter Svenonius. 2011. Features in minimalist syntax. In Cedric Boeckx (ed.), *The Oxford handbook of linguistic minimalism*, 27–51. Oxford: Oxford University Press.
- Baker, C. L. 1991. The syntax of English *not*: The limits of core grammar. *Linguistic Inquiry* 22(3). 387–429.
- Benmamoun, Elabbas. 2000. *The feature structure of functional categories: A comparative study of Arabic dialects*. New York: Oxford University Press.
- Bjorkman, Bronwyn. 2011. *BE-ing default: The morphosyntax of auxiliaries*. Cambridge: MIT. (Doctoral dissertation).
- Bloomfield, Leonard. 1933. *Language*. New York: Holt Rinehart & Winston.
- Borer, Hagit. 1984. *Parametric syntax*. Dordrecht: Foris.
- Borer, Hagit. 2005. *The normal course of events (Structuring sense, Vol. II)*. Oxford: Oxford University Press.

24 *A spanning approach to categories and their linearization*

- Brody, Michael. 2000a. Mirror Theory: Syntactic representation in Perfect Syntax. *Linguistic Inquiry* 31(1). 29–56.
- Brody, Michael. 2000b. Word order, restructuring, and Mirror Theory. In Peter Svenonius (ed.), *The derivation of VO and OV*, 27–43. Amsterdam: John Benjamins.
- Bye, Patrik & Peter Svenonius. to appear. The phonological word: At the interface of lexical insertion, (morpho)syntax, and pronunciation. In Paul de Lacy & Adam Jardine (eds.), *The Cambridge handbook of phonology*. Cambridge: Cambridge University Press.
- Chomsky, Noam. 1957. *Syntactic structures*. Berlin: Mouton.
- Chomsky, Noam. 1986. *Knowledge of language: Its nature, origins, and use*. New York: Praeger.
- Chomsky, Noam. 1991. Some notes on economy of derivation and representation. In Robert Freidin (ed.), *Principles and parameters in comparative grammar*, 417–454. Cambridge: MIT Press.
- Chomsky, Noam. 1995. *The minimalist program*. Cambridge: MIT Press.
- Chomsky, Noam. 2001. Derivation by phase. In Michael Kenstowicz (ed.), *Ken Hale: A life in language*, 1–52. Cambridge: MIT Press.
- Cinque, Guglielmo. 1999. *Adverbs and functional heads: A cross-linguistic perspective*. New York: Oxford University Press.
- Dékány, Éva. 2018. Approaches to head movement. *Glossa* 3(1). 1–43.
- Emonds, Joseph. 1978. The verbal complex V-V' in French. *Linguistic Inquiry* 9. 151–175.
- Ernst, Thomas. 2002. *The syntax of adjuncts*. Cambridge: Cambridge University Press.
- Grimshaw, Jane. 2005. Extended projection. In Jane Grimshaw (ed.), *Words and structure*, 1–73. revised version of 1991 ms. Stanford: CSLI.
- Halle, Morris & Alec Marantz. 1993. Distributed Morphology and the pieces of inflection. In Kenneth Hale & Samuel Jay Keyser (eds.), *The view from building 20: Essays in linguistics in honor of Sylvain Bromberger*, 111–176. Cambridge: MIT Press.
- Hanson, Kenneth. 2024. Strict locality in syntax. In Kutay Serova & M. K. Snigarsoff (eds.), *CLS 59: Proceedings of the Fifth-ninth Annual Meeting of the Chicago Linguistics Society*, 131–145. Chicago: The Chicago Linguistics Society.
- Harwood, Will. 2014. Rise of the auxiliaries: A case for auxiliary raising vs. affix lowering. *The Linguistic Review* 31(2). 295–362.
- Kobele, Greg. to appear. Minimalist grammars and decomposition. In Evelina Leivada & Kleanthes K. Grohmann (eds.), *The Cambridge Handbook of Minimalism and its Applications*. Cambridge: Cambridge University Press.

Peter Svenonius

- Kobele, Gregory M. 2005. Features moving madly: A formal perspective on feature percolation in the minimalist program. *Research on Language and Computation* 3(2). 391–410.
- Kratzer, Angelika. 1996. Severing the external argument from the verb. In Johann Rooryck & Laurie Zaring (eds.), *Phrase structure and the lexicon*, 109–137. Dordrecht: Kluwer.
- Laka, Itziar. 1990. *Negation in syntax: On the nature of functional categories and projections*. Cambridge: MIT. (Doctoral dissertation).
- Lasnik, Howard. 1995. Case and expletives revisited: On greed and other human failings. *Linguistic Inquiry* 26(4). 615–633.
- Legate, Julie Anne. 2003. Some interface properties of the phase. *Linguistic Inquiry* 34(3). 506–516.
- Marantz, Alec. 2001. Words. ms. MIT.
- Moore, Edward F. 1956. Gedanken-experiments on sequential machines. In Claude E. Shannon & John McCarthy (eds.), *Automata studies*, vol. 34 (Annals of Mathematics Studies), 129–153. Princeton: Princeton University Press.
- Pollard, Carl & Ivan A. Sag. 1994. *Head-driven Phrase Structure Grammar*. Stanford: CSLI & University of Chicago Press.
- Pollock, Jean Yves. 1989. Verb movement, universal grammar, and the structure of IP. *Linguistic Inquiry* 20(3). 365–424.
- Rizzi, Luigi. 2017. On the format and locus of parameters: The role of morphosyntactic features. *Linguistic Analysis* 41(3-4). 159–190.
- Sag, Ivan A., Rui P. Chaves, Anne Abeillé, Bruno Estigarribia, Dan Flickinger, Paul Kay, Laura A. Michaelis, Stefan Müller, Geoffrey K. Pullum & Frank van Eynde. 2020. Lessons from the English auxiliary system. *Journal of Linguistics* 56(1). 87–155. DOI: [10.1017/S002222671800052X](https://doi.org/10.1017/S002222671800052X).
- Stabler, Edward. 1997. Derivational minimalism. In Christian Retoré (ed.), *Logical aspects of computational linguistics*, 68–95. Dordrecht: Springer.
- Stabler, Edward P. 2011. Computational perspectives on minimalism. In Cedric Boeckx (ed.), *Oxford handbook of linguistic minimalism*, 617–642. Oxford: Oxford University Press.
- Svenonius, Peter. 2012. Merge, project, and bundle. Ms. CASTL, University of Tromsø, available at <https://ling.auf.net/lingbuzz/001581>.
- Svenonius, Peter. 2016. Spans and words. In Heidi Harley & Daniel Siddiqi (eds.), *Morphological metatheory*, 199–220. Amsterdam: John Benjamins. DOI: [10.1075/la.229.07sve](https://doi.org/10.1075/la.229.07sve).
- Svenonius, Peter. 2021. Prepositions with CP and their implications for extended projections. *Linguistic Variation* 21(1). 11–45.

24 *A spanning approach to categories and their linearization*

- Svenonius, Peter. 2024. The workspace as a source of hierarchy in extended projections. *Nordlyd* 48(1). 115–132. DOI: doi.org/10.7557/12.7982.
- Svenonius, Peter. to appear. Structural implications of late insertion. In Evelina Leivada & Kleanthes K. Grohmann (eds.), *The Cambridge handbook of minimalism*. Cambridge: Cambridge University Press.
- Westergaard, Marit. 2014. Linguistic variation and micro-cues in first language acquisition. *Linguistic Variation* 14(1). 26–45. DOI: [10.1075/lv.14.1.02wes](https://doi.org/10.1075/lv.14.1.02wes).
- Wilder, Chris. 1997. English finite auxiliaries in syntax and phonology. In James R. Black & Virginia Motapanyane (eds.), *Clitics, pronouns and movement*, 321–362. Amsterdam: John Benjamins.
- Zagona, Karen. 2014. Sequence-of-tense and the feature of finite tenses. *Nordlyd, Tromsø Working Papers in Language & Linguistics* 41(2). available at <http://septentrio.uit.no/index.php/nordlyd>, 261–272.
- Zwicky, Arnold M. & Geoffrey K. Pullum. 1983. Cliticization vs. inflection: English *n't*. *Language* 59(3). 502–513.

Chapter 25

Categorization of novel nouns with ambiguous gender cues: Experimental evidence from Russian

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This article reports on an experimental study investigating the role of probabilistic rules in gender assignment to ambiguously cued novel nouns by adult monolingual speakers of Russian. Unlike previous works on the topic, the study is large in scale (267 participants jointly providing around 35000 gender judgments) and comprehensive in scope (1648 ambiguous monosyllabic stimuli including all possible phonotactically licit combinations of cue-bearing phonological units - codas, nuclei and onsets). The main results indicate that speakers can reliably access probabilistic rules to classify novel nouns into gender classes based on their phonological properties, and the information relevant for gender assignment is generalized at the level of segment classes captured by binary phonological features [$\pm$ sonorant] (codas), [$\pm$ continuant] (onsets), as well as [$\pm$ high] and [$\pm$ low] (nuclei).

1 Introduction

Grammatical gender is an inherent morphosyntactic property of nouns encoded by an abstract feature in the lexical entry of every noun (Chomsky 1965, Carroll 1989). Grammatical gender features categorize nouns into agreement classes, and in some cases correlate with inflectional paradigms the nouns follow (Corbett 1991, Halle 1994). The mechanisms underlying the grammatical gender assignment – i.e. the process by which nouns are paired with abstract gender features in the course of language acquisition – has been subject of much research, both theoretical and experimental. One of the central issues raised in these studies

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is whether gender assignment is rule-governed (Tucker et al. 1977, Harley 1979, Karmiloff-Smith 1979, Pinker 1984, Carroll 1989, Taraban & Kempe 1999, Gagliardi 2012, Slioussar 2018, Mitrofanova et al. 2018). That is, whether a) language learners initially attribute gender to groups of nouns based on their morphophonological properties and later revise that classification in response to positive evidence, or b) acquisition of gender proceeds on noun-to-noun basis without regard to noun-internal gender cues. It is likely that the answer to this question is language-specific, and depends crucially both on how gender is marked in a given language, and how transparent such marking is. For example, Carroll (1989) makes a convincing case for French, where, she argues, gender features are acquired simultaneously with each noun, accounting for the fact that Francophone children make virtually no gender attribution errors. She claims that this learning strategy is enabled by the fact that French marks gender on prenominal articles, which, being clitics, initially get segmented as part of nouns they specify.

Obviously, this strategy is unavailable in languages that lack articles – like, for instance, Russian, where acquisition of grammatical gender is a relatively lengthy (up to 7 years until full mastery) and error-prone process (Gvozdev 1961, Ceitlin 2009, Rodina 2008). Developmental evidence from Russian strongly supports the claim that children form and then revise gender hypothesis for groups of nouns defined through their morphophonological properties (along the lines of Pinker 1984). For instance, all common developmental errors involve overgeneralization in the direction of a more numerous class based on the nominal ending in the citation form (Gvozdev 1961, Ceitlin 2009, Rodina 2008). Thus, consonant-final feminines initially get erroneously grouped with consonant-final masculines, while vowel-final stem-stressed neuters are classified with the feminines of the same shape. Notably, Carroll (1989) maintains that this latter “shape-based” strategy is available to Francophone speakers as well, and is employed when classifying novel morphologically simplex nouns (as, for instance, in a novel noun classification task). The evidence supporting the use of probabilistic “rules of thumb” (as Carroll 1989 calls them) for novel noun classification in Russian comes from non-word experimental studies involving both children (Mitrofanova et al. 2018) and adult speakers (Slioussar 2018). Although preliminary in nature, such studies convincingly demonstrate that speakers have internalized generalizations concerning the phonological shape of nouns belonging to each gender category and are able to access such generalizations for gender attribution of novel nouns. They also show that attribution accuracy is modulated by the transparency of gender marking, such that nouns that contain unambiguous gender cues are classified more readily and elicit fewer errors than those containing ambiguous cues.

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This article continues the experimental investigation into gender assignment to novel nouns, presenting findings from a study with monolingual, non-attributed adult Russian speakers. It will show that speakers make use of probabilistic rules to reliably classify novel nouns into gender classes based on their phonological properties. I will also argue that the information relevant for gender assignment is generalized at the level of segment classes, which can be captured by binary phonological features [$\pm$ sonorant] (codas), [$\pm$ continuant] (onsets), as well as [$\pm$ high] and [$\pm$ low] (nuclei).

This article is organized as follows: the next section provides a brief overview of grammatical gender in Russian; Sections 3 and 4 outline the research questions and study methodology; Section 5 presents the results; Section 6 discusses the findings; and Section 7 offers concluding remarks.

2 Grammatical gender in russian

Russian distinguishes between three grammatical genders: masculine, feminine and neuter. Gender is expressed as a suffix on agreeing adjectives, participles and past-tense verbs, as well as on demonstratives, possessives, and certain numerals and quantifiers. Gender agreement is illustrated in (1).

(1) Grammatical gender agreement on possessive pronouns and adjectives

- a. Mój golub-ój dom
my.M blue-M house.M
'My blue house'
- b. Moj-á golub-áj¹ rek-á
my-F blue-F river.F
'My blue river'
- c. Moj-ó golub-ójə per-ó
my-N blue-N feather.N
'My blue feather'

Masculine nouns are most frequent, constituting about 46% of all nouns in the lexicon, followed by feminine (41%) and neuter nouns (13%) (Corbett 1991: 78), or 48% vs. 35% vs. 17% according to the recent estimation by Slioussar (2018) based

¹Note that since unstressed /a, e, o/ reduce to reduce to [ə] word-finally, FEM vs. NEUT distinction is neutralized in adjectives where the main stress does not fall on the inflection, e.g. /'krasn-aja/ 'red, F' and /'krasn-oje/ 'red, N' are both realized as ['krasnəjə] on the surface.

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on the grammatically disambiguated subcorpus of the Russian National Corpus. Masculine gender is considered to be the grammatical default (Corbett, *ibid*), in that it is typically used for mixed-gender groups (e.g. *nemcy*-M, ‘Germans-M’), and in cases where the gender of the referent is unknown or unclear (*Kto eto sdelal*-M? ‘Who did-M this?’). The status of masculine as the default gender is also corroborated by experimental and observational findings in language acquisition. Thus, it has been shown that both monolingual and bilingual children tend to over-generalize to masculine in case of non-target-like productions (see Rodina & Westergaard 2012, Schwartz et al. 2015, Mitrofanova et al. 2018, Urek et al. 2017 for experimental data, Gvozdev 1961; Ceitlin 2009 for corpus data).

In most instances, the grammatical gender of a noun is transparent, i.e. can be easily determined based on its ending in the citation form. Thus, all (declinable) nouns ending in non-palatal consonants in the nominative singular are masculine, most nouns ending in (stressed) [-á] are feminine, and most nouns ending in (stressed) [-é] and [-ó] are neuter. However, there are two groups of nouns where the relationship between the shape of the noun and its grammatical gender is opaque. In the first group, feminine vs. neuter distinction is obscured by the vowel reduction process, whereby unstressed /a, e, o/ are reduced to [ə] word-finally (see Iosad 2012 for details and the analysis of vowel reduction in Russian). As a result, the surface endings of the feminine *súmk*-[ə] ‘handbag’ and neuter *mýl*-[ə] ‘soap’ become indistinguishable in the citation form.

The second locus of ambiguity (that is the main focus of this article) is the nouns that end in postalveolar/palatalized consonants in the citation form. For example, the grammatical gender of the feminine *te*[nʲ] ‘shadow’ and the masculine *ogo*[nʲ] ‘fire’ cannot be determined based on the nominative singular alone, as both end with a palatalized alveolar nasal stop. Unlike non-end-stressed feminines and neuters, end-palatalized nouns are not usually disambiguated in orthography. The only exception are a subset of them ending in shibilant postalveolars [ʃ, ʃʲ, ʒ, tʃ], where the orthographic convention requires the addition of the word-final softening grapheme in feminines. For example, *ra*[ʒ] ‘rage’ and *ro*[ʒ] ‘rye’ can be correctly identified as respectively masculine and feminine because of their orthographic representations being *раж* and *рожь* (notice the softening grapheme). Nouns ending in other palatalized consonants require a softening sign regardless of gender.

Although some earlier studies suggest that “the distribution of [...] end-palatalized nouns to the feminine and masculine genders is roughly even” and “end-palatalization does not favor either gender” (Taraban & Kempe 1999:124, with a reference to Pulkina 1978), a closer look at the corpus data reveals that surface ambiguity between feminine and masculine palatal-final nouns is not

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absolute. This is especially true if the entire word-final rhyme is taken into the account. This is illustrated in Table 1, which shows the proportion of feminine nouns among frequent palatal-final nouns with singleton non-shibilant codas.

For the table, all noun lemmas ending in a postalveolar/palatalized consonant, along with their grammatical gender specification, were extracted from the morphological dictionary of Russian (Open Corpora project, <http://opencorpora.org>, approx. 5 million morphologically tagged word forms of 390 thousand lemmas). The resulting list consisted of 6309 unique lemmas, of which 2127 (or about 34%) were masculine. The list was then reduced to eliminate nouns where the final syllable contained a word-final coda cluster or shibilant coda. Of the remaining 2345 nouns, those with lemma frequency of less than 1 instance per million words (ipm) were excluded, leaving 678 nouns (about 60% masculine). Lemma frequency information was derived from the Frequency Dictionary of Russian (<http://www.artint.ru/projects/frqlist.php>, Sharoff 2002).

Each cell in Table 1 shows the proportion of feminines among the nouns with a given final coda (rows) and final nucleus (columns). Two final columns indicate the overall proportion of feminines among the nouns ending in a given coda (total instances and percent respectively), while two final rows show the overall proportion of feminines among the nouns with a given final nucleus. Cells shaded in gray indicate that a given nucleus-coda combination is not attested. Coda-nucleus combinations that are feminine in 100% of cases are shaded in green, while combinations that are never associated with feminine gender are shaded in red. Note that the distribution of final onsets is not taken into the account.

Notice that out of 96 possible nucleus-coda combinations (or rhymes), 37 (or 38%) are not attested in the corpus, and especially that [f] never occurs as a singleton coda. Out of remaining 59 rhymes, 30 are associated only with feminine gender, and 7 - only with masculine. This leaves us with 27 rhymes (28%) that are truly ambiguous with respect to grammatical gender.

3 Research questions

Developmental errors whereby children overgeneralize palatal-final feminine nouns to masculine are common, as evidenced by both experimental and observational data (Rodina & Westergaard 2012, Schwartz et al. 2015, Mitrofanova et al. 2018, Urek et al. 2017; Gvozdev 1961, Ceitlin 2009). Less frequent palatal-final nouns are occasionally mis-assigned by adult speakers as well. The overuse of masculine is usually explained as being due to the masculine default.

Table 1: *Proportion of feminine nouns by final coda and nucleus*

	/a/	/e/	/i/	/o/	/u/	/ɨ/	/ju/	/ja/	Total
/b'/				100%	67%	100%		100%	7/8 88%
/v'/	100%			100%	100%			100%	11/11 100%
/d'/	100%	82%		67%	50%			100%	24/28 86%
/z'/	100%	100%	100%	100%				80%	15/17 88%
/l'/	71%	16%	18%	42%	0%	45%	0%	0%	79/343 23%
/m'/		100%							1/1 100%
/n'/	100%	27%	100%	60%	0%	100%	0%	100%	40/89 45%
/p'/		100%	100%	100%	100%	100%			13/13 100%
/r'/	14%	40%	33%	33%	100%	0%			17/88 19%
/s'/	50%	100%	100%	50%	0%	100%			28/32 88%
/t'/	90%	100%	100%	53%	80%	100%		67%	34/44 77%
/f'/									
Total	66/120	86/349	24/35	41/72	13/28	19/41	0/5	20/24	269/674
	55%	25%	69%	57%	46%	46%	0%	83%	

Strikingly, nonce-word studies also find the opposite pattern, i.e. the one where palatal-final nonce nouns are predominantly judged as feminine. For example, [Mitrofanova et al. \(2018\)](#) show that monolingual children are more likely to assign feminine than masculine to a set of palatal-final nonce-word stimuli, while bilingual children with reduced Russian input overuse masculine in the same condition. These results suggest that the overuse of masculine by the bilingual children in this case is the effect of the masculine default, while the overuse of feminine by monolinguals reflects true sensitivity to the corpus distribution.

As [Mitrofanova et al. \(2018\)](#) themselves note, learners appear to draw on phonological information to generalize the typical shapes of nouns in different gender classes, using these generalizations to form probabilistic rules for assigning gender to unfamiliar nouns. However, it remains to be determined what form these rules take, how they are applied, and which aspects of phonological structure are relevant for gender assignment. To investigate these issues, this study addresses the following research questions:

RQ1. What are the rules of gender assignment to morphologically simplex, phonologically ambiguous nouns? More precisely, what are the rules the

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speaker applies for selecting the initial gender hypothesis when presented with a novel noun in the absence of syntactic, morphological and semantic gender cues?

RQ2. Is a gender judgment elicited by a given novel noun consistently linked to this noun?

RQ3. What phonological constituents bear information that is relevant for gender assignment, and what is the nature of such information?²

4 Methodology

4.1 Participants

Participants in this study were 267 native speakers of Russian, born and resident in Russia (112 female). Participants were recruited by a marketing research agency according to the pre-specified criteria and received remuneration for their participation. In terms of age, 49% of the participants fell into the age group 36–45 years, and 38% were aged between 26 and 35 years. 66% of the participants had higher education, while 34% had vocational education or below. Participants were anonymous at all stages of data collection.

4.2 Stimuli

For the test dataset, 1648 novel monosyllables were constructed by combining 12 simplex non-shibilant palatal codas, 8 nuclei and 20 simplex onsets. The set of codas included all palatalized consonants that make phonotactically licit codas ([l', m', n', p', r', s', t', f, b', v', d', z']), except for the palatoalveolars ([ʃ, ʒ, tʃ]). The latter were excluded because the Russian orthographic convention requires the corresponding graphemes to be followed by a softening sign (“ь”) in feminine nouns but not in masculine nouns, thus giving a salient orthographic cue to

²As pointed out in Carroll (1989), “noun ending” that is frequently invoked as a unit that supposedly carries the distributional information relevant for gender assignment, is not a constituent in any linguistically relevant sense, and therefore cannot be targeted by linguistic processes. Since I am concerned here with gender assignment to morphologically simplex nouns, my analysis is based on phonological constituents – and namely rhymes (and their subconstituents, nuclei and codas) and onsets.

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grammatical gender. The set of nuclei included [a, e, i, i̞, o, u, ju, ja]³. The set of onsets included all licit singleton onsets, i.e. [b, v, g, d, ʃ, ʒ, tʃ, z, k, l, m, n, p, r, s, t, f, x, ts].

The care was taken to exclude monosyllables that coincided with existing nouns (but not other parts of speech), as well as combinations violating phonotactics or orthographic conventions. The items were distributed across 24 lists (approx. 75 items per list), in such a way that all nucleus-coda combinations (rhymes) in each list were unique, and all onset-nucleus combinations in each list were unique. Codas, onsets and nuclei were distributed as evenly as possible across the lists (e.g. each of the codas occurred on the average 6 times per list). Each list also contained 80 fillers, 56 of which were existing Russian nouns (top of the frequency list in each of the categories defined below). The order of stimuli in each list was randomized for each participant. Table 2 and 3 give examples of the types of novel and familiar nouns included into the experiment.

Table 2: *Familiar nouns (fillers)*

	End-palatalized	Non-end-palatalized
Monosyllabic	день [denʹ] ‘day’(M), боль [bolʹ] ‘pain’(F)	год [got] ‘year’(M)
Polysyllabic	зритель [zritelʹ] ‘spectator’(M), любовь [ljubovʹ] ‘love’(F)	город [gorət] ‘town’(M), рука [ruka] ‘arm’(F), лицо [litso] ‘face’(N)

4.3 Procedure

Data collection for this study took place online, in 2018. All the tasks were hosted on Ibex Farm platform, with the exception of the receptive vocabulary test that was hosted, administered and processed by its developer (Golovin 2015). In total, participants completed 5 tasks (described in detail in the remainder of this

³For the purposes of this study, I treat surface sequences where a palatalized consonant is followed by a non-palatalizing vowel (e.g. [lʹu], [lʹa], etc.) as derived from underlying representations where plain consonants are followed by diphthongal sequences (i.e. /lju/, /lja/, etc.), rather than postulating underlying palatalized consonants (cf. Iosad & Morén-Duolljá 2010). This has consequences for the phonological segmentation of stimuli, where the list of onsets does not include any palatalized consonants, but the list of nuclei includes diphthongal sequences /ju, ja/ (orthographic ю, я).

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Table 3: Novel nouns (fillers and test items)

	End-palatalized	Non-end-palatalized
Monosyllabic	TEST ITEMS: <i>бонь</i> [bon’], <i>хєб</i> [heb] <i>сатъ</i> [sat’]	
Polysyllabic		<i>пунип</i> [punip], <i>бульга</i> [bul’ga], <i>пруно</i> [pruno]

section): noun classification task (test and retest presentation), receptive vocabulary test, digit span task and pattern detection task. In addition, participants filled out a short background questionnaire, including questions on gender, age group, language skills and education level. The structure of the study is given in (2).

(2) *Structure of the study*

- I. Information and consent form;
- II. Background questionnaire;
- III. **Noun classification: test presentation**
- IV. Digit span task
- V. Pattern detection task
- VI. Receptive vocabulary test
- VII. **Noun classification: retest presentation**

The entire study took approximately 45 minutes to complete. Of 267 participants that started the study, 9 failed to complete the receptive vocabulary test and were excluded. The remaining participants completed all 7 steps of the study.

4.4 Noun classification

Each participant was randomly assigned to the test list. During the test, the participants saw the items from the list in the middle of the screen one by one, accompanied by three gendered 1st person possessive pronouns (*мой* [moj] ‘my’(M), *моя* [maja] ‘my’(F), *моѐ* [majo] ‘my’(N)), as shown below⁴. The participants were instructed to choose one pronoun that goes best with the test item by clicking on

⁴Even though I did not expect participants to choose neuter agreement for any of the stimuli, the neuter pronoun was included as an option to identify cases where participants might respond at random. The analysis of the results showed that neuter agreement was selected extremely rarely, which is consistent with accidental clicks.

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the radio button. The order of radio buttons was randomized for each trial. After selecting the pronoun, the participant was automatically redirected to the next item. Reaction time was registered as the time elapsed from item presentation to the click on the radio button.

(3) Example of stimulus presentation



After completing the noun classification test, the participant was redirected to complete the three background tasks (see below for description). Having completed the tasks, the participant had to once again classify all the nouns in the test list she was assigned to. In other words, each participant got to judge the list she was assigned to twice. Retest presentation was included to analyze whether individual participants judged the same items consistently. In what follows, I will refer to the two presentations as “test presentation” and “retest presentation” respectively. At the retest presentation, all fillers in the list were different, and all items in the list were re-randomized.

4.5 Background measures

In addition to the noun classification task, each participant also participated in three background measures: receptive vocabulary test, digit span task, and pattern detection task.

Online receptive vocabulary test used in this study is a yes/no test with a built-in attention measure, based on computerized adaptive testing technique. The test was developed, hosted and administered by Gregory Golovin (Golovin 2015). The test has previously been validated, and normed with 123 000 respondents (Ashk-inazi & Golovin 2016). In the digit span task followed the standard procedure where the participants were requested to memorize and reproduce sequences of digits appearing on the screen one by one. Digit sequences increased in length from 3 to 7 digits. In total, 15 sequences were used. Pattern detection task was adapted from the collection of psychological tests (Stolyarenko 2006). In the pattern detection task, the participants were requested to deduce which of the given words was encoded by a provided sequence of symbols. In total, 25 symbol sequences were presented, each of them along with 5 answer options (of which

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one or two were correct). The length of symbol sequences increased from 4 to 8 characters.

5 Results

5.1 Test-retest reliability

Let us first consider test-retest reliability, which is the measure of the consistency of participants' responses across repeated presentations of the same noun. The median proportion of test-retest matches on test items across all participants was 68%, with values ranging from 25% to 100% (SD = 0.144). This means that at least half of the participants had a consistent gender judgment rate of 68% or higher between the first and second presentations of each noun. To determine if participants' test-retest reliability was significantly above chance, it was necessary to establish the likelihood of consistent gender judgments by calculating the probability intervals for FEM and MASC responses. Since most participants had a FEM bias, the chance of a test-retest match was calculated specifically for this bias, yielding an interval of 0.557 to 0.588 for chance matches⁵. With this threshold, 75% of participants had test-retest reliability above chance, showing consistent gender judgments beyond random assignment.

In summary, two groups of participants emerged: (1) those with test-retest reliability above chance, and (2) those with reliability at chance level. Participants in the first group likely make consistent judgments due to their sensitivity to certain formal properties of the stimuli (explored in detail in the following sections). In contrast, participants in the second group appear to lack this sensitivity and seem to assign gender judgments at random. For convenience, I will refer to these groups as 'analyzers' and 'guessers.' Two-samples unpaired Wilcoxon test revealed that analyzers outperform guessers on all background measures: analyzers are significantly better than guessers at pattern detection ($W = 6874$, $p = 0.019$), they score significantly higher at digit span task ($W = 6958$, $p = 0.01$), and have larger receptive vocabularies ($W = 6772$, $p = 0.033$).

5.2 Association between the shape of a noun and a gender judgement

In order to investigate the relationship between the likelihood of a feminine judgment and the phonological shape of a noun, a generalized (binomial) mixed-effects model was fitted (lme4 R package; [Bates et al. 2014](#)). The model in-

⁵Please refer to [Urek \(2019\)](#) for the details of this calculation.

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cluded “coda”, “nucleus”, “onset”, and “participant type” as fixed-effect predictors, with random intercepts for “participant” and “item” as grouping factors to account for variability across participants and items. The model also included several interaction terms: *participant_type:coda*, *participant_type:nucleus*, *participant_type:onset* and *coda:nucleus*. Variable “participant type” had two levels, “analyzers” (participants whose proportion of test-retest matches was above chance) and “guessers” (participants whose proportion of test-retest matches was at chance). Only the judgments supplied during the first presentation of stimuli were analyzed.

Likelihood ratio tests of nested models revealed that all fixed effects were significantly associated with the dependent variable: coda ($\chi^2 = 153.78$, $p < 0.0001$), nucleus ($\chi^2 = 166.26$, $p < 0.0001$), onset ($\chi^2 = 114.06$, $p < 0.0001$) and participant type ($\chi^2 = 40.11$, $p < 0.0001$). All interactions except for *nucleus:participant_type* were also significant: *coda:participant_type* ($\chi^2 = 38.62$, $p < 0.0001$), *onset:participant_type* ($\chi^2 = 30.39$, $p = 0.047$) and *coda:nucleus* ($\chi^2 = 173.32$, $p < 0.0001$). Posthoc pairwise comparisons were conducted to compare the likelihood of feminine responses by coda, nucleus and onset for the two participant types (*lsmeans*, Lenth 2016). These will be discussed in detail below.

5.2.1 Codas

Twelve codas were included in the experiment, including four sonorants [l, m, n, r] and eight obstruents [p, s, t, f, b, v, d, z]. Each appeared in an average of 140 unique nouns and received 1500 judgments in the test presentation. All codas showed consistent test-retest judgments, with a median consistency of 67% (range: 63–73%, SD = 3%). The median proportion of feminine judgments was 70% (range: 59–77%, SD = 5%). Posthoc pairwise comparisons assessed the likelihood of feminine responses by coda type for ‘analyzers’ and ‘guessers’ (*lsmeans*, Lenth 2016).

Figure 1 shows a Hasse diagram (R package *hasseDiagram*, Ciomek 2017) summarizing findings for the analyzers group, where arrows indicate statistically significant relations⁶ and should be read as ‘more likely to trigger a feminine response than.’ Codas [-b, -v, -p] (top row) are the most likely to trigger feminine responses; for instance, [-b] elicits more feminine responses than [m, t, f, z, d], and by transitivity, also more than [r, s, l, n]. The coda least associated with

⁶Note that the arrows in the diagram indicate transitivity relationships: if A is more likely to trigger F than B, and B is more likely to trigger F than C, then A is more likely to trigger F than C.

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feminine responses is [n'] (bottom row), with sonorants generally clustering toward the bottom and no sonorant more likely to trigger feminine responses than any obstruent.

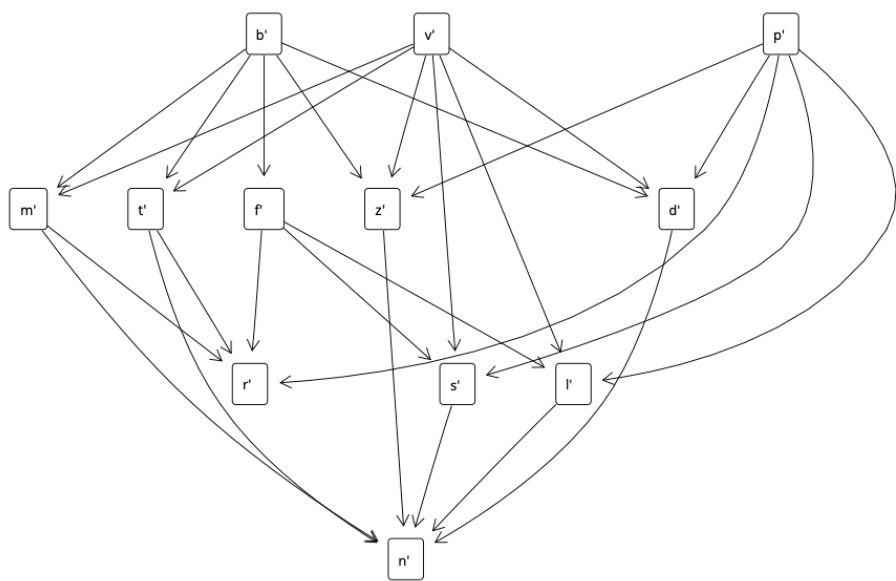


Figure 1: Likelihood of feminine response by coda: “analyzers”

Figure 2 presents a Hasse diagram summarizing findings for the ‘guessers’ group. Unlike ‘analyzers,’ ‘guessers’ make fewer distinctions among coda types, with feminine responses more likely for nouns ending in [-b’, -v’, z’, p’, t’, f’] than [r’], and no other coda differences reaching significance. Notably, while ‘guessers’ make fewer contrasts, their coda relationships never reverse those observed in ‘analyzers.’ The obstruent-sonorant distinction also holds, as five obstruent codas elicit more feminine responses than the sonorant [-r’].

5.2.2 Nuclei

Eight nuclei were included in the experiment, each appearing in an average of 223 unique nouns and receiving 2367 judgments in the test presentation. The median proportion of feminine judgments across nuclei was 69% (range: 63–79%, SD = 6%), and test-retest consistency had a median of 66% (range: 63–73%, SD = 4%). Figures 3 and 4 provide Hasse diagrams summarizing posthoc pairwise comparisons of nuclei for the ‘analyzers’ and ‘guessers’ groups, respectively.

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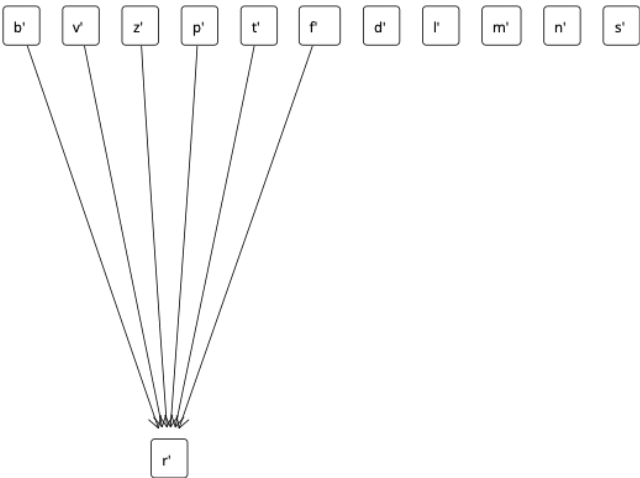


Figure 2: Likelihood of feminine response by coda: “guessers”

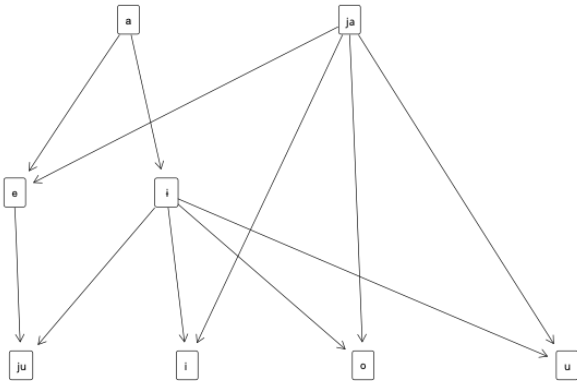


Figure 3: Likelihood of feminine response by nucleus: “analyzers”

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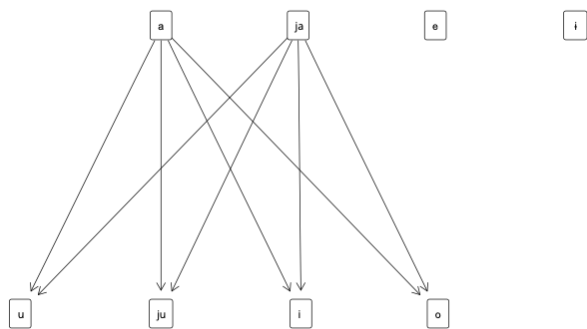


Figure 4: Likelihood of feminine response by nucleus: “guessers”

An important feature of the diagrams in Figures 3 and 4 is that the likelihood of a feminine response aligns with phonological height: lower vowels are more likely to trigger a feminine gender (see [Timberlake 2004](#), for formant frequencies of Russian vowels). Similar to codas, ‘analyzers’ make more distinctions with nuclei, differentiating among three height levels (e.g., low [a], mid [e], high [u]), while ‘guessers’ only make a two-way contrast.

5.2.3 Onsets

There were 20 onsets included in the experiment, each appearing in an average of 91 unique nouns and receiving 969 judgments. The median proportion of feminine judgments for onsets was 70% (range: 64–81%, SD = 4%), with consistent judgments across test and retest (median test-retest match: 67%, range: 64–74%, SD = 3%). Figures 5 and 6 provide Hasse diagrams summarizing posthoc pairwise comparisons of onsets for ‘analyzers’ and ‘guessers’, respectively.

Unlike codas and nuclei, the likelihood of a feminine response for individual onsets does not clearly align with a single phonological property. However, onsets more likely to trigger feminine judgments (upper row) are all [+continuant], while all non-continuants appear in the bottom row. Some [+continuant] segments also appear in the bottom row ([h, v, f, z, l] for ‘analyzers’ and [h] for ‘guessers’). Once again, ‘analyzers’ make more distinctions than ‘guessers.’

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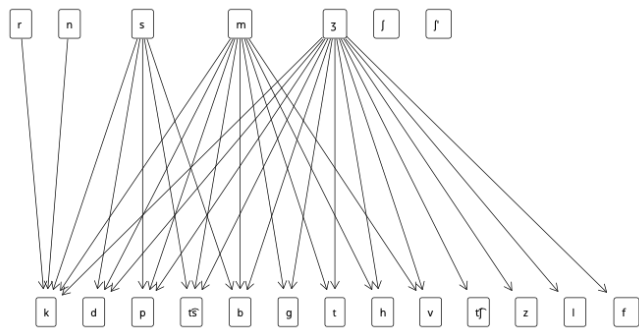


Figure 5: Likelihood of feminine response by onset: “analyzers”

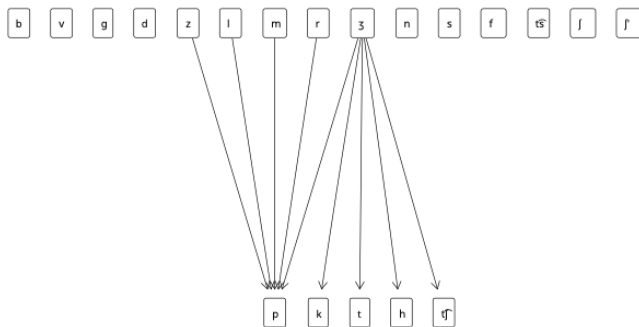


Figure 6: Likelihood of feminine response by onset: “guessers”

5.2.4 Rhymes

There were 96 unique rhymes (i.e. nucleus + coda combinations, e.g. [al', um', etc.]) included in the experiment. Figure 7 illustrates the variability in the proportion of feminine judgments for each rhyme, with some rhymes judged as feminine in around 50% of cases (left), while others assigned feminine in over 90% of cases (right).

The large number of rhymes made pairwise comparisons challenging, so I recoded the data using phonological contrasts previously identified as influencing feminine judgment likelihood. Codas were classified as 'obstruent' or 'sonorant,' nuclei as 'high,' 'mid,' or 'low,' and onsets as 'continuant' or 'non-continuant' (with laterals and nasals as continuants). A generalized mixed-effects model was then used to predict feminine judgment likelihood based on 'coda sonority,' 'nucleus height,' 'onset continuancy,' and 'participant type,' with 'participant' and

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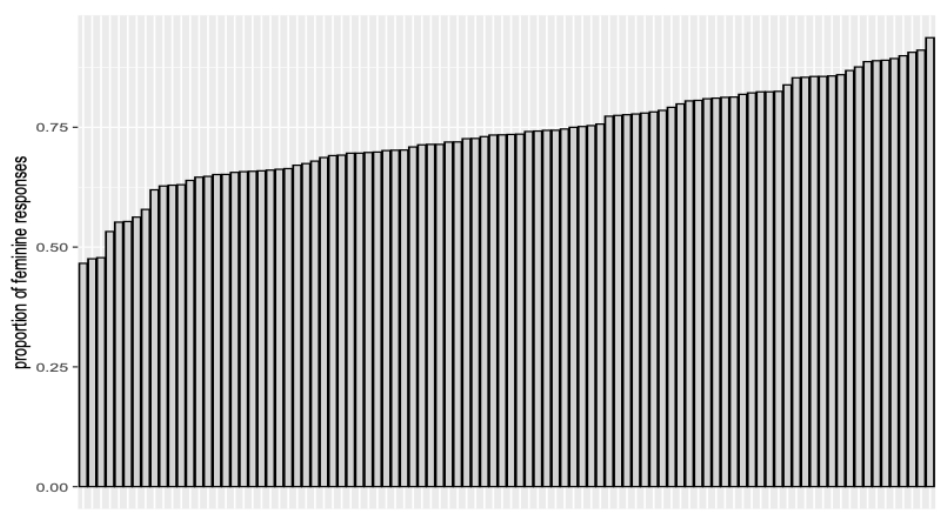


Figure 7: Proportion of feminine judgments by rhyme

'item' as random effects, and interaction terms including `participant_type:coda_sonority`, `participant_type:nucleus_height`, `participant_type:onset_continuancy` and `coda_sonority:nucleus_height`. I then conducted posthoc pairwise comparisons to assess the likelihood of feminine responses across all coda, onset, and nucleus types, as well as their combinations (i.e., all rhymes) for the two participant groups (lsmeans, [Lenth 2016](#)). The analysis confirmed that each identified phonological dimension contributes to feminine gender assignment for both groups.

Let's examine the rhyme results more closely (refer to Hasse diagrams in Figures 8 and 9). As evident from the diagrams, while 'analyzers' and 'guessers' behave similarly, 'guessers' recognize fewer distinctions. For both groups, sonorant-final rhymes are less likely to trigger feminine responses than obstruent-final rhymes with vowels of the same height. Rhymes with low vowels are more likely to elicit feminine responses than those with high or mid vowels with similar codas. Rhymes with mixed likelihoods of feminine (e.g., high vowel and obstruent or low vowel and sonorant) appear in the middle of the diagram, while rhymes where both nucleus and coda increase or decrease feminine likelihood (e.g., low vowel and obstruent, high vowel and sonorant) are at the top and bottom, respectively. This suggests an additive effect: rhymes where both constituents favor feminine responses are more likely to trigger feminine.

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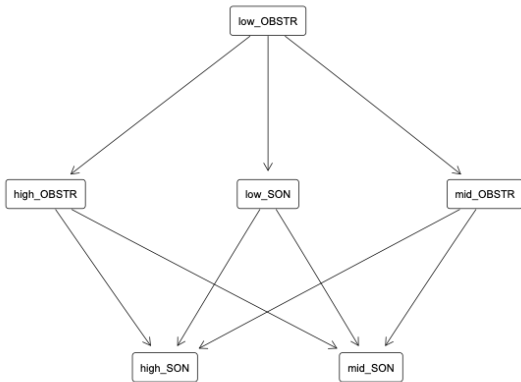


Figure 8: Likelihood of feminine response by rhyme: “analyzers”

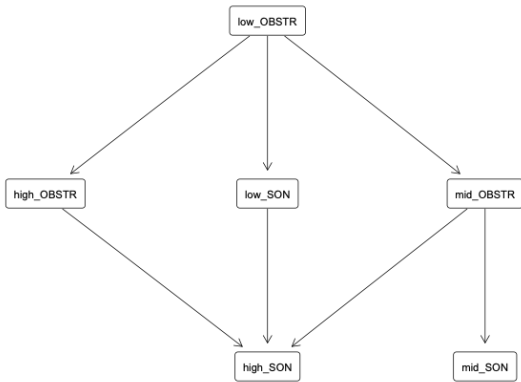


Figure 9: Likelihood of feminine response by rhyme: “guessers”

6 Discussion

The study presented here is among the first to evaluate test-retest reliability in the context of cue-based gender assignment, where high test-retest reliability in gender judgments strongly indicates participants' sensitivity to noun-internal gender cues, while low reliability suggests that specific gender judgments are not consistently triggered by the stimulus. In this sample, 188 participants, or approximately 75%, demonstrated test-retest reliability above chance, suggesting that these participants consistently used phonological cues within nouns to assign gender to novel items. These participants, referred to as 'analyzers,' performed better than participants with chance-level reliability ('guessers') across all background measures, including receptive vocabulary, pattern detection, and digit span. One interpretation of these results is that individuals with better pattern detection and larger receptive vocabularies are more adept at detecting distributional patterns in language input. An alternative explanation is that individuals with stronger memory skills (as indicated by both digit span and vocabulary size) may simply recall their responses from the first presentation of the noun classification task. This, however, seems unlikely, as it would require participants to memorize their binary responses to 75 novel items, near-identical in shape and presented in random order, and to reproduce those judgments after completing 3 unrelated cognitively demanding tasks, with a re-randomized item list.

We have also seen that, despite an overall feminine bias⁷, the likelihood of a feminine response for palatal-final nonce stimuli can be reliably predicted based on the phonological properties of the coda, nucleus, and onset. This finding aligns with earlier experimental results (Slioussar 2018, Mitrofanova et al. 2018) and supports the role of rules in classifying ambiguously marked nouns (Tucker et al. 1977, Harley 1979, Karmiloff-Smith 1979, Pinker 1984). Importantly, distinctions among individual codas, onsets, and nuclei reduce to general phonological dimensions: obstruent codas (e.g., [p', b', v']), low nuclei (e.g., [a, ja]), and continuant onsets (e.g., [s, ʒ]) are significantly more likely to trigger a feminine response than sonorant codas (e.g., [n', l']), high nuclei (e.g., [u, i]), and non-continuant onsets (e.g., [p, d]). When combined into rhymes, the effects of nuclei and codas were shown to be additive. "Analyzers" make more distinctions between different coda, nucleus and onset types than "guessers" do, yet both groups display similar overall tendencies, and relationships observed in 'analyzers' are never

⁷The overall feminine bias in the results likely reflects the fact that most palatal-final monosyllabic nouns in the lexicon are indeed feminine (cf. Table 1). As noted in Section 3, feminine bias has also been found in experiments involving monolingual and bilingual children judging palatal-final nonce nouns (e.g. Mitrofanova et al. 2018).

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reversed by 'guessers.' Notably, neither group defaults to masculine, counter to expectations of an 'emergence-of-the-unmarked' effect.

7 Conclusions

The results presented above strongly suggest that adult monolingual speakers of Russian have internalized a set of probabilistic rules for the classification of nouns into gender classes based on their phonological properties, and can access these rules to categorize morphologically simplex, ambiguously cued novel nouns presented outside of syntactic context. Gender judgments elicited by ambiguously cued novel nouns are reliably linked to individual nouns for 75% of participants (analyzers), indicating sensitivity to the likelihood with which individual phonological constituents occur in nouns of either gender. The responses of the remaining 25% of participants (guessers) indicate that they are sensitive to the general feminine bias in the distribution, but only link it to individual constituents in a small subset of cases. Onsets, nuclei and codas occurring at the right edge of nouns all bear information relevant for gender classification⁸. That information appears to be generalized at the level of segment classes captured by binary phonological features [$\pm$ sonorant] (codas), [$\pm$ continuant] (onsets), as well as [$\pm$ high] and [$\pm$ low] (nuclei).

Acknowledgements

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References

- Ashkinazi, Leonid A. & Grigory V. Golovin. 2016. *Studying passive vocabulary by means of the internet*. Paper presented at the 3rd ISA Forum of Sociology, Vienna, Austria.
- Bates, Douglas, Martin Mächler, Ben Bolker & Steve Walker. 2014. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* 67(1). 1–48.
- Carroll, Susanne. 1989. Second-language acquisition and the computational paradigm. *Language Learning* 39. 535–594.

⁸It remains to be seen to what extent probabilistic phonological cues play a role in acquiring gender of non-transparent nouns in languages like Russian, where deterministic cues from gender agreement are readily available.

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- Ceitlin, Stella. N. 2009. *Očerki po slovoobrazovaniju i formoobrazovaniju v detskoj rechi* [On inflection and derivation in child language]. Moscow: Znak.
- Chomsky, Noam. 1965. *Aspects of the theory of syntax*. Cambridge: The MIT Press.
- Ciomek, K. 2017. *Hassediagram*. R package version 0.1.3.
- Corbett, Greville G. 1991. *Gender*. Cambridge: Cambridge University Press.
- Gagliardi, Annie C. 2012. *Input and intake in language acquisition*. University of Maryland. (Doctoral dissertation).
- Golovin, Grigory V. 2015. Izmerenie passivnogo slovarnogo zapasa riusskogo jazyka [Measuring passive vocabulary in Russian]. *Socio- and psiholingvisticheskie issledovanija* [Studies in socio- and psycholinguistics] 3. 148–159.
- Gvozdev, Alexandr N. 1961. *Voprosy izucheniya detskoy rechi*. Moscow: APN RS-FSR.
- Halle, Morris. 1994. The russian declension: An illustration of the theory of distributed morphology. In J. Cole & C. Kisseberth (eds.), *Perspectives in phonology*, Stanford. Center for the Study of Language & Information: CSLI Publications.
- Harley, Birgit. 1979. French gender “rules” in the speech of English-dominant, French-dominant, and monolingual French-speaking children. *Working Papers on Bilingualism* 19. 129–156.
- Iosad, Pavel. 2012. Vowel reduction in Russian: No phonetics in phonology. *Journal of Linguistics* 48. 521–571.
- Iosad, Pavel & Bruce Morén-Duolljá. 2010. *Russian palatalization: The true(r) story*. Paper presented at Old World Conference in Phonology 7.
- Karmiloff-Smith, Annette. 1979. *A functional approach to child language: A study of determiners and reference*. Cambridge: Cambridge University Press.
- Lenth, Russell V. 2016. Least-squares means: The R package lsmeans. *Journal of Statistical Software* 69(1). 1–33. DOI: [10.18637/jss.v069.i01](https://doi.org/10.18637/jss.v069.i01).
- Mitrofanova, Natalia, Yulia Rodina, Olga Urek & Marit Westergaard. 2018. Bilinguals’ sensitivity to grammatical gender cues in Russian: The role of cumulative input, proficiency, and dominance. *Frontiers in Psychology* 9. 1894. DOI: [10.3389/fpsyg.2018.01894](https://doi.org/10.3389/fpsyg.2018.01894).
- Pinker, S. 1984. *Language learnability and language development*, vol. 7. Cambridge: Harvard University Press.
- Pulkina, I. 1978. *A short Russian reference grammar*. Moscow: Russian Language Publishers.
- Rodina, Yulia. 2008. *Semantics and morphology: The acquisition of grammatical gender in Russian*. University of Tromsø. (Doctoral dissertation).
- Rodina, Yulia & Marit Westergaard. 2012. A cue-based approach to the acquisition of grammatical gender in Russian. *Journal of Child Language* 39(5). 1077–1106.

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- Schwartz, Mila, Miriam Minkov, Elena Dieser, Ekaterina Protassova, Victor Moin & Maria Polinsky. 2015. Acquisition of Russian gender agreement by monolingual and bilingual children. *International Journal of Bilingualism* 19(6). 726–752.
- Sharoff, S. 2002. Meaning as use: Exploitation of aligned corpora for the contrastive study of lexical semantics. In Manuel González Rodríguez & Carmen Paz Suarez Araujo (eds.), *Proceedings of Language Resources and Evaluation Conference (LREC02)*. Las Palmas: ELRA.
- Slioussar, Natalia A. 2018. Gender, declension and stem-final consonants: An experimental study of gender agreement in Russian. In **no editor** (ed.), *Computational linguistics and intellectual technologies: Papers from the Annual International Conference “Dialogue”*, vol. 17, 688–700. **no address: no publisher**.
- Stolyarenko, Lyudmila D. 2006. *Osnovy psihologii: Praktikum [basics of psychology: Practical course]*. Rostov-on-Don: Feniks.
- Taraban, Roman & Vera Kempe. 1999. Gender processing in native and nonnative Russian speakers. *Applied Psycholinguistics* 20(1). 119–148.
- Timberlake, A. 2004. *A reference grammar of Russian*. Cambridge: Cambridge University Press.
- Tucker, G. Richard, Wallace E. Lambert & André G. Rigault. 1977. *The French speaker’s skill with grammatical gender*. The Hague: Mouton.
- Urek, O., M. Westergaard & A. Taurina. 2017. *Acquisition of grammatical gender in Latvian-Russian bilinguals*. Paper presented at the International Symposium on Bilingualism 11.
- Urek, Olga. 2019. *The role of probabilistic phonological cues in gender assignment to novel nouns*. Paper presented at the MultiGender kick-off seminar, Oslo, 23–24 May 2019.

Chapter 26

The nanosyntax of Norwegian *som*

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The Norwegian item *som* is used in a wide range of constructions, ranging from equative and essive constructions to relative clauses and embedded *wh*-questions, and in some varieties it is even used in comparatives and matrix subject *wh*-questions. This paper discusses the various uses in a comparative perspective mainly against other Germanic varieties, arriving at an understanding of the wide lexicalization range as reflexes of historical grammaticalization that involves step-by-step cumulative expansion from smaller syntactic structures to bigger ones. Under a nanosyntactic approach this can in turn be understood as gradual growth regarding the specification of what sub-trees *som* may lexicalize.

1 Introduction

The Norwegian functional item *som* has a range of uses, across all varieties, in which it sometimes corresponds to English *as*, sometimes to *that*, and sometimes lacks a corresponding item in English. The many uses are listed in the dictionary Norsk Ordbok and discussed in Halmøy & Berg-Olsen (2012), and include the following: (1) predicate marker in essive small clauses (English *as*), (2) equative preposition (English *as*), (3) temporal complementizer (archaic use; English *as*), (4) relative complementizer (English *that*; also *who*, *which*), (5) obligatory element in subject clefts (English *that*; also *who*, *which*), (6) obligatory element in subject exclamatives (no English equivalent) (7) complementizer introducing manner exclamatives (no English equivalent), (8) obligatory element in embedded subject *wh*-clauses but ungrammatical in non-subject *wh*-clauses.

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- (1) Jon jobba der som lærer.
Jon worked there SOM teacher
'Jon worked there as a teacher.'
- (2) Jon er like gammel som Per.
Jon is equally old SOM Per
'Jon is as old as Per.'
- (3) Som Jon gikk mot bilen, begynte det å regne.
SOM Jon walked towards car-DEF started it to rain
'As Jon was walking towards the car it started raining.'
- (4) a. Jon liker studenter (*som) møter på forelesninger.
Jon likes students SOM meet to lectures
'Jon like students who/that meet for lectures.'
b. Jon liker studenter (som) han kan stole på.
Jon likes students SOM he can trust on
'Jon likes students that/whom he can trust.'
- (5) a. Det var Jon *(som) kom med et forslag.
it was Jon SOM came with a proposal
'It was Jon that/who made the proposal.'
b. Det var et forslag (som) Jon kom med.
it was a proposal SOM Jon came with
'It was a proposal that/who Jon made.'
- (6) a. Så mye folk *(som) har møtt opp!
so much people SOM have met up
'What a lot of people have turned up!'
b. For en stilig kar *(som) kommer der!
for a stylish guy SOM comes there
'What a stylish guy is coming there!'
- (7) Som dere roper!
SOM you.PL shout
'How you are shouting!'
- (8) a. Jon lurte på hvem *(som) hadde sagt det.
Jon wondered on who SOM had said it
'Jon wondered who had said it.'

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- b. Jon lurte på hva (*som) du hadde sagt.
Jon wondered on what som you had said
'Jon wondered what you had said.'
- c. Jon lurte på hvorfor (*som) du hadde sagt det.
Jon wondered on why som you had said it
'Jon wondered why you had said it.'

Additionally, there are also uses that are restricted to certain varieties of Norwegian. As discussed by [Dancus & Mørck \(2008\)](#), some varieties of Northern Norwegian allow *som* to be used not just in equative constructions, but also as a comparative particle, i.e. corresponding to English *than*, either alone or in addition to the item *enn* used in other varieties of Norwegian.

- (9) Jon e større (enn) som Ola.
Jon is taller (than) som Ola.
'Jon is taller than Ola.'

Furthermore, there are some dialects of Norwegian in which *som* appears in yet two additional contexts: (10) in the embedded left periphery under extraction of an embedded *wh*-subject, hence in a **that*-trace configuration, and (11) after a matrix *wh*-subject (no English equivalent).

- (10) Kem du trur som sa det?
who you think som said it
'Who do you think said so?'
- (11) Kem som sa det?
who som said it
'Who said so?'

A key question to ask is how many *som*'s there are in the Norwegian lexicon: a single lexical entry or several ones? In their discussion of the choices made for the dictionary entry for *som* in Norsk Ordbok, [Halmøy & Berg-Olsen \(2012: 18\)](#) state that their point of departure is that there is only one *som* with a constant, underlying meaning from which all the various meanings are contextual derivatives, and they describe this context independent meaning as "a predication particle or operator, i.e. as a non-verbal copula. In other words: *som* connects two constituents so that one says something about the other – and nothing else" (my translation and underlining, ØAV).

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Although it may be problematic to entertain the idea that *som* itself contains a particular meaning, I will adhere to the view that there is just one *som*, and through comparisons across different grammars I will claim that we can view the particular systems found in Norwegian varieties as a reflex of grammaticalization processes that have followed certain systematic patterns.

If we for the sake of simplicity (i) leave out the rather archaic temporal use of *som*, (ii) group clefts and exclamatives with relatives and similarly (iii) view equatives and essive SCs together, we can summarize the various uses of *som* presented above as in Table 1.¹

Table 1: The lexicalization of various C/P functions in English, Norwegian in general, and dialects of Norwegian

	Matrix <i>wh</i> -subject	Extracted <i>wh</i> -subject	Embedded <i>wh</i> -subject	Relatives	Equative P/ Essive SC	Comparative P
English	Ø	Ø/ <i>that</i>	Ø	<i>that</i>	<i>as</i>	<i>than</i>
Norwegian	Ø	Ø/ <i>at</i>	<i>som</i>	<i>som</i>	<i>som</i>	<i>enn</i>
Dialectal Norw.	<i>som</i>	Ø/ <i>som</i>	<i>som</i>	<i>som</i>	<i>som</i>	<i>som</i>

The comparative use of *som* appears to be the original functional use of the item, which etymologically stems from the same root as the adjective *same* (cf. Latin *simul*-). The relative use of the item developed during the Old Norse period (Stroh-Wollin 2002, Wagener 2017), and Icelandic never developed the use of the corresponding item *sem* in embedded subject *wh*-clauses. Viewed from right to left the various constructions involved are increasingly larger and/or more complex, and from the point of view of grammaticalization and reanalysis this points to a possible development for *som* whereby it has started out as an item used in comparatives and from there gradually extended its use to higher functional heads in the clause. This general observation will be the overarching intuition pursued in this paper.

Evidence from other non-standard varieties of Germanic suggest that the division of labor between items associated with relativization and complementation can vary. Brandner (2020) and Brandner & Bräuning (2013) point to the fact that also the function word *wo* in many German dialects has a diachronic development where it starts as a comparative marker and is later used in relative clauses.

Furthermore, whereas it is fairly well-known that *as* can be used as a relativizer in dialects of English, it can also in some varieties be used as a declarative complementizer, cf. the example in (12) from the Freiburg Corpus of English Dialects (FRED) provided by Kolbe (2003: 138):

¹‘Dialectal Norwegian’ is intentionally vague here and there is not an exact overlap between allowing *som* in matrix subject *wh*-clauses and in the comparative construction.

- (12) When I started to repair them I thought, Well, to think as we drink water as is pumped through these pipes. (FRED, SAL_013)

This latter fact about English adds to the mix that *as* can replace *that* even in regular declarative complements. Axel-Tober & Brandner (To appear) discuss a similar global use of *als/ass* across varieties of German.

Are lexicalization patterns like these random or are they governed by more profound principles? Obviously, the latter alternative is a more exiting one, and the one that I will explore in this paper. In doing so I will view the variation observed as reflexes of grammatical reanalysis where the cline is constrained by syntactic structure. The basic idea will be that a given functional item may only extend its lexicalization range to a functional projection adjacent to one that it already lexicalizes. A theoretical framework that naturally lends itself to this general approach is – in my opinion – Nanosyntax (Caha 2009, Starke 2009, Baunaz & Lander 2018).

2 Nanosyntax – essential parts

A thorough introduction to Nanosyntax would lead too far in this paper – a quick glance at some essential traits will have to suffice. Although Nanosyntax first and foremost is designed to capture morphological patterns, in particular inflection, it is also well suited to deal with variation pertaining to the syntax of functional items (see for instance my own work on *wh*-inventories across varieties of Germanic discussed in Vangsnes 2013).

Nanosyntax shares with other frameworks like Distributed Morphology the view that *morphology is syntax*, or rather that morphology is driven by syntax. Furthermore, it assumes the basic principles of Cartographic Syntax (Cinque & Rizzi 2009), which means that the functional domain of clauses and phrases is fine-grained, consisting of a number of functional projections, ultimately reflecting interpretational properties.

In the nanosyntactic jargon a particular sequence of functional projections is referred to as an *fseq*, and a crucial assumption within the framework is that the ordering of projections within a given *fseq* is the same across languages, hence universal. The main job of phonological material – i.e. word forms – is to spell out *fseqs* according to certain general principles. One of these is the Superset Principle, here formulated as in Caha (2009: 55).

- (13) The Superset Principle (Caha 2009: 55):
A phonological exponent is inserted into a node if its lexical entry has a (sub-) constituent that is identical to the node (ignoring traces).

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Given the abstract fseq in (14), an item X which can lexicalize (or spell out) A can also lexicalize B and C. An item Y fit for lexicalizing B can also lexicalize C, but crucially not A since that is not a subconstituent of B.

$$(14) \quad [A \text{ x } [B \text{ y } [C$$

The Superset Principle is constrained by the ‘Elsewhere Principle’ (Kiparsky 1973), given here as it is formulated by Lander (2015: 99).

(15) The Elsewhere Principle (Lander 2015: 99):
When more than a single L-tree can lexicalize an S-tree, the L-tree with the fewest unused features are chosen.

Again referring to the fseq in (14), the Elsewhere Principle ensures that Y will be chosen over X if the task is to lexicalize just the structure $[B\ [C$ since X would contain unused features (i.e. the $[A$ part).

Nanosyntax furthermore assumes that syntactic trees are built bottom-up with successive matching against the lexicon. The ‘Principle of Cyclic Override’ ensures that the most appropriate exponent gets spelled out at the end of the derivation.

(16) **The Principle of Cyclic Override** (Lander 2015: 102):
Later, higher-level spellouts cancel out previous, lower-level spellouts.

This means that once $[A]$ is merged on top of $[B]$ $[C]$ as a later step in the derivation of the fseq in (14), X will be the best candidate and cancel out the merger of Y .

Equipped with this brief sketch of some essential assumptions underlying the nanosyntactic framework we move on to look more closely at the various uses of *som* in Norwegian, and we start with the ones related to the smaller, non-finite predication structure.

3 A run through the data

3.1 Equative structures and the like

The equative and essive uses of *som* are repeated here.

(1) Jon jobba der som lærer. ESSIVE
Jon worked there SOM teacher
'Jon worked there as a teacher.'

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- (2) Jon er like høy som Per. SIMILATIVE
 Jon is equally old som Per
 ‘Jon is as old as Per.’

Haspelmath & Buchholz (1998) make a finer distinction between equative and *similative* constructions (termed *degree* and *non-degree equatives* by Jäger 2019: 2ff.). Examples of the latter, with both a DP and a clausal predicate, are given in (17), and whereas Norwegian uses *som* also in this construction, we see from the translation that English resorts to the item *like*.

- (17) a. Jon er akkurat som deg. SIMILATIVE
 Jon is just som you
 ‘Jon is just like/*as you.’
 b. Jon snakker akkurat som du snakker.
 Jon speaks just like you speak
 ‘Jon speaks just like you.’

If we now move on to Icelandic, we see a different pattern across the three constructions.

- (18) Jón starfaði þar sem kennari. ESSIVE
 Jón worked there SEM teacher
 ‘Jón worked there as a teacher.’
 (19) a. Jón er á sama aldri og Pétur. SIMILATIVE
 Jón is on same age and Pétur
 ‘Jón is the same age as Pétur.’
 b. Jón er alveg eins og þú. SIMILATIVE
 Jón is just same and you
 ‘Jón is just like you.’

The examples show that Icelandic uses the etymological counterpart of *som* in the essive construction only and rather *og* in the other two constructions (in combination with ‘same’ in the similative): the latter item is otherwise used as a conjunction corresponding to English *and*. This division of labor between one item/expression used in essive constructions and another in equative and similative is also found in for example (Standard) German and Finnish (not illustrated here for space reasons).

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These facts can be summarized as in Table 2, and in order for the nanosyn-tactic puzzle to work, along the lines sketched in Section 3, the ordering of the three constructions viewed as functions in an fseq must be *essive* – *equative* – *similative*.

Table 2: Lexicalizations across the ESSIVE-SIMILATIVE-SIMILATIVE domain

	ESSIVE	SIMILATIVE	SIMILATIVE
Norwegian	<i>som</i>	<i>som</i>	<i>som</i>
English	<i>as</i>	<i>as</i>	<i>like</i>
Icelandic	<i>sem</i>	<i>og</i>	<i>eins og</i>
German	<i>als</i>	<i>wie</i>	<i>wie</i>
Finnish	<i>-na</i> (essive case)	<i>kuin</i>	<i>kuin/kuten</i>

An important assumption pertaining to the methodology pursued here is that translational equivalence amounts to structural and categorial identity across linguistic varieties. Although languages sometimes resort to clearly different constructions to convey a particular state-of-affairs, that does not apply to the constructions considered in the present paper.

Given this (partial) fseq we see that Norwegian *som* lexicalizes all three parts ($[A [B [C$ in (14)), English *as* only the two higher parts ($[A [B ...]$), and Icelandic *sem* only the first one ($[A ...]$).

It would be natural to also address the dialectal use of *som* in comparatives, cf. (9) repeated here.

- (9) Jon e større (enn) som Ola. COMPARATIVE
Jon is taller (than) som Ola.
'Jon is taller than Ola.'

As in the equative and similitive constructions, comparatives may also involve clausal predicates.

- (20) Marit gikk dit raskere enn jeg sykla dit.
Marit walked there faster than I biked there
'Marit walked there faster than I biked there.'

Syncretism between the particle used in equatives/similatives on the one hand and comparatives on the other is by no means uncommon across languages.

Finnish and North Sámi would be cases in point, using the items *kuin* and *go*, respectively, and within Germanic varieties like West Frisian and Low Saxon have *as/als* across the board, including the essive construction. Bavarian has *wie* in all these constructions and also in comparatives where Standard German uses *als*.

The latter fact about Standard German represents a quirk that makes us defer addressing the comparative construction until the discussion in Section 4.2. In the meantime, we move on to uses of *som* in finite constructions.

3.2 Relatives and embedded interrogatives – and clefts and exclamatives

The use of *som* in relative clauses is repeated below. In Norwegian, subject relatives *must* be introduced by *som* whereas it is optional in object relatives (and more generally in other non-subject relatives).

- (4) a. Jon liker studenter *(som) møter på forelesninger.
 Jon likes students SOM meet to lectures
 ‘Jon likes students who show up at lectures.’
 b. Jon liker studenter (som) han kan stole på.
 Jon likes students SOM he can trust on
 ‘Jon likes students that/whom he can trust.’

The same $\pm$ subject asymmetry is found in embedded *wh*-clauses: presence of *som* is obligatory if the *wh*-phrase is the subject of the embedded clause, but, in this case, *som* is disallowed if the *wh*-phrase is the object or an adjunct of the embedded clause, (8b-8c).

- (8) a. Jon lurte på hvem *(som) hadde sagt det.
 Jon wondered on who SOM had said it
 ‘Jon wondered who had said it.’
 b. Jon lurte på hva (*som) du hadde sagt.
 Jon wondered on what SOM you had said
 ‘Jon wondered what you had said.’
 c. Jon lurte på hvorfor (*som) du hadde sagt det.
 Jon wondered on why SOM you had said it
 ‘Jon wondered why you had said it.’

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Furthermore, the $\pm$ subject asymmetry is also found in clefts and exclamatives: *som* is obligatory if the clefted or exclamated phrase is the subject and optional if it has a different syntactic function. We will not discuss this further here, but for the sake of convenience we may assume that the structure of clefts and exclamatives bears on that of relatives.²

In Icelandic, the cognate item *sem* is also used in relatives, but unlike the case of Norwegian *som* there is no subject/object asymmetry associated with it: *sem* is obligatory across the board in relatives.

- (21) a. Jóni líkar vel við nemendur *(sem) mæta á fyrirlestra.
 Jón likes well with students SEM meets on lectures
 ‘Jón likes students who show up at lectures.’
 b. Jóni líkar vel við nemendur *(sem) hann getur treyst.
 Jón likes well with students SEM he kan trust
 ‘Jón likes students that he can trust.’

On the other hand, *sem* is never used in embedded *wh*-clauses, not even when the *wh*-phrase is the subject, as shown in (22).

- (22) Jón langaði að vita hver (*sem) var búin að segja það.
 Jón wanted to know who SEM was done to say it
 ‘Jón wanted to know who had said it.’

What emerges in this domain is thus that (i) Norwegian – and Mainland Scandinavian in general – shows a $\pm$ subject asymmetry in relative clauses insofar that *som* is obligatory in subject relatives but optional in non-subject relatives, and (ii) Norwegian/Mainland Scandinavian obligatorily inserts *som* in embedded subject *wh*-questions. The latter insertion of a relativizer in embedded *wh*-questions is not found in Icelandic and English. These facts can be rendered as in Table 3.

²As noted in the introduction, *som* may also introduce what I termed adverbial exclamatives like the one in (7), *Som dere roper!* ‘How you are shouting!’. English does not have a comparable construction, and the ungrammatical translation is therefore merely some kind of approximation. The corresponding construction in Icelandic suggests that the exclamated constituent also in this case is nominal and that we have a structure akin to relative clauses. As in Norwegian such a sentence may start with *sem*, but it may also be embedded under the pronoun *það* it, that’, or it may rather start with the *wh*-item *hvað* which is otherwise used in degree questions. (Thanks to Thorsteinn Indriðasson for clarifying these facts.)

(i) Æ, {hvað}/{(það) sem} þið öskrið!
 gosh, what that SOM you.PL shout
 ‘Gosh, how you are shouting!’

Table 3: Items introducing embedded

	EMBEDDED <i>wh</i>	RELATIVE
Norwegian SUBJ	<i>wh- som</i>	<i>som</i>
Norwegian OBJ	<i>wh- Ø</i>	<i>(som)</i>
Icelandic SUBJ	<i>wh- Ø</i>	<i>sem</i>
Icelandic OBJ	<i>wh- Ø</i>	<i>sem</i>
English SUBJ	<i>wh- Ø</i>	<i>that/who</i>
English OBJ	<i>wh- Ø</i>	<i>(that/who)</i>

wh-questions and relatives in Norwegian, Icelandic and English

Taking stock, we may now summarize all the information we have gone through so far concerning the subclausal and clausal uses of *som* and its corresponding items in English and Icelandic as in Table 4.

Table 4: The lexicalization of embedded subject

	EMB. <i>Wh</i>	RELATIVE	ESSIVE	EQUATIVE	SIMILATIVE
Norwegian	<i>som</i>	<i>som</i>	<i>som</i>	<i>som</i>	<i>som</i>
English	<i>wh-</i>	<i>that/who</i>	<i>as</i>	<i>as</i>	<i>like</i>
Eng. Dialects	<i>wh-</i>	<i>as</i>	<i>as</i>	<i>as</i>	?
Icelandic	<i>wh-</i>	<i>sem</i>	<i>sem</i>	<i>og</i>	<i>og</i>

wh-questions and relatives, essives, equatives and similatives across Norwegian, English and Icelandic

Next, we turn to consider some of the dialectally restricted uses of *som*.

3.3 Extraction and the matrix use of *som*

In Northern, Central, and Western varieties of Norwegian extraction of an embedded *wh*-subject to the matrix Left Periphery can be accompanied by insertion of *som* at the left edge of the embedded clause from which the *wh*-constituent has moved.

- (10) Kem du trur (som) sa det?
who you think som said it
‘Who do you think said so?’

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A $\pm$ subject asymmetry is at play also here: *som* cannot be inserted if the extracted phrase is not a subject.

- (23) a. Ka du trur (*som) Jon sa?
 what you think som Jon said
 ‘What do you think that Jon said?’
 b. Koffør trur du (*som) Jon sa det?
 why think you som Jon said that
 ‘Why do you think that Jon said it?’

On the other hand, inserting the declarative complementizer *at* in (23) would be fine, and in a largely complementary area (Eastern Norway) speakers to some extent allow insertion of *at* also under extraction of subjects. (See Vangsnes 2019 for details.)

- (24) Hvem tror du (at) sa det?
 who think you that said it
 ‘Who do you think said so?’

Nordgård (1985) makes the observation that dialects which allow non-V2 in matrix *wh*-questions, including inserting *som* in second position after a matrix *wh*-question as in (9), seem to consistently also allow insertion of *som* after an extracted *wh*-subject.

- (9) Kem som (ikkje) sa det?
 who som (not) said it
 ‘Who did (not) say so?’

For the sake of completeness, insertion of *som* is not a viable manifestation of non-V2 with non-subject *wh*-constituents, emphasizing the connection to subjecthood in this domain of grammar.

- (25) Kem (*som) du skulle treffe i morra?
 who som you should meet tomorrow
 ‘Who were you going to meet tomorrow?’

In the next section we will summarize the findings so far and address a remaining quirk: how to fit the comparative use into the picture.

4 Discussion

4.1 Syncretisms and lexicalization ranges

Table 5 sums up what we discussed in the preceding sections and also adds some uncontroversial information about German, notably the fact that relatives are introduced by case inflected *d*-pronouns and that the declarative complementizer *daß* appears under extraction from an embedded clause.³

Table 5: Functional items across varieties of Norwegian and English, Icelandic and German used in a variety of constructions from SIMILATIVE via relatives and *wh*-extraction to matrix *wh*-questions

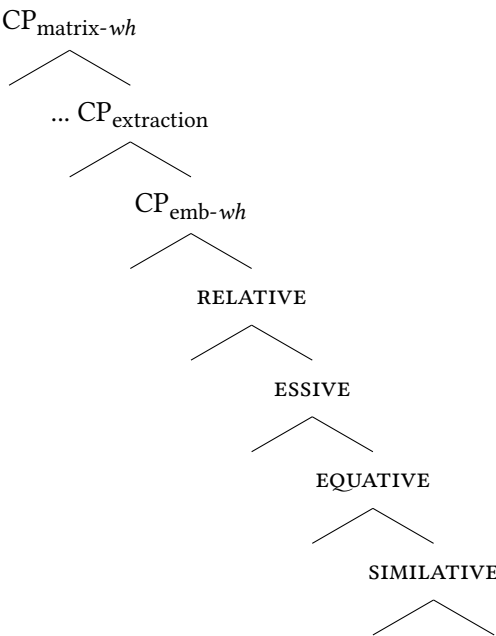
	<i>wh</i> _{SUBJ}	NON-V2	EXTR. SUBJ	EMB. <i>wh</i> _{SUBJ}	RELATIVE	ESSIVE	EQUATIVE	SIMILATIVE
St. Norw.	–	(<i>at</i>)	<i>som</i>	<i>som</i>	<i>som</i>	<i>som</i>	<i>som</i>	<i>som</i>
Dial. Norw.	<i>som</i>	(<i>som</i>)	<i>som</i>	<i>som</i>	<i>som</i>	<i>som</i>	<i>som</i>	<i>som</i>
Icelandic	–	–	–	<i>sem</i>	<i>sem</i>	<i>og</i>	(<i>eins</i>) <i>og</i>	
English	–	(<i>that</i>)	–	<i>that</i>	<i>as</i>	<i>as</i>	<i>like</i>	
Dial. Eng.	–	(<i>that</i>)	–	<i>as</i>	<i>as</i>	<i>as</i>	<i>as</i>	
German	–	<i>daß</i>	–	<i>d-</i>	<i>als</i>	<i>wie</i>	<i>wie</i>	

In this table all the various uses are arranged in such a way that the various items that we have been looking at span contiguous parts. Except for the relative order of the use in matrix non-V2 clauses and *wh*-extraction any reordering of the functions would lead to non-contiguous spans. The nanosyntactic interpretation of this is that the ordering in fact reflects a particular sequence of hierarchically ordered functional projections.

(26) (Preliminary analysis)

³As with ‘dialectal Norwegian’, also ‘dialectal English’ is kept vague on purpose, and whether there is identity between those varieties allowing the relative use of *as* and the simulative one is not known.

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If we assume, as is commonplace, a general principle that functional projections must be ‘identified’ (Vangsnes 1999) by merger of appropriate phonological material (either directly or by movement), and furthermore given the Superset and Elsewhere Principles, we can interpret the patterns in Table 5 as a matter of how the various items can identify various subparts of the tree in (26). We can refer to this as their lexicalization ranges.

Furthermore, the tree in (26) looks like a credible syntactic tree insofar that it goes from smaller sub-clausal structures via embedded finite structures to matrix clauses. We will return to that with some important qualifications below, but let us for now stress that this approach informs us of how items may have expanded their lexicalization ranges stepwise one projection at the time up the tree through reanalysis. Regarding Norwegian *som* for instance, if it started out lexicalizing SIMILATIVE we may conjecture that it expanded to EQUATIVE and ESSIVE and then during the Old Norse period started being used as a relativizer before continuing the expansion to embedded subject *wh*-clauses and, in some dialects, further to extraction cases involving *wh*-subjects and even lexicalizing the matrix C after *wh*-subjects, hence the whole tree.

In Icelandic, on the other hand, which like Norwegian has evolved from Old Norse, *sem* has not expanded further from relatives. On the other hand, a new

item has been introduced for the lowest part of the tree, namely *og*. By the Elsewhere Principle this item is a better fit than *sem* since it carries less irrelevant features than *sem* for the EQUATIVE and SIMILATIVE uses.

4.2 Splitting it in two: The Comparative and Relative Fseqs

At this point it is time to address the quirk represented by Standard German comparatives where the item used is *als* rather than *wie*.

- (27) Der Hans ist größer als/*wie der Peter.
 the Hans is taller ALS/WIE the Peter.
 ‘Hans is taller than Peter.’

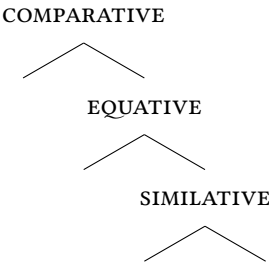
If we concentrate on the lower part of the tree in (26) where we find the functions that allow sub-clausal structures we see that any attempt to fit in the comparative will break up the syncretisms established so far. Placing COMPARATIVE below SIMILATIVE would work fine for Standard Norwegian, Icelandic and English which all have a unique item (*enn/en/than*), but in German *als* in ESSIVE and COMPARATIVE would be intervened by *wie* in EQUATIVE and SIMILATIVE (*als* > *wie* > *wie* > *als*). Placing COMPARATIVE between EQUATIVE and SIMILATIVE would entail that the syncretisms will be broken in all varieties except the Norwegian ones that allow *som* in comparatives. Placing it above EQUATIVE would be in line with Jäger (2019: 7ff.) who makes an argument for merging German comparative *als* in a conjunction head above the *standard-of-comparison* predicate, leaving room for merger of a complementizer or *wh*-item between *als* and the (rest of the) predicate, as is attested in various non-standard varieties of German in COMPARATIVE but not in EQUATIVE and SIMILATIVE.

However, placing COMPARATIVE between ESSIVE and EQUATIVE will break the syncretisms in Standard Norwegian (*som* > *enn* > *som* > *som*) and English (*as* > *than* > *as* > *like*) although it will not in Icelandic (*sem* > *en* > *og* > (*eins*) *og*) and Standard German (*als* > *als* > *wie* > *wie*) (and some dialects of Norwegian). Furthermore, placing COMPARATIVE above ESSIVE will break the syncretism for Norwegian (most varieties) and dialectal English since *som* and *as* is used in relatives.

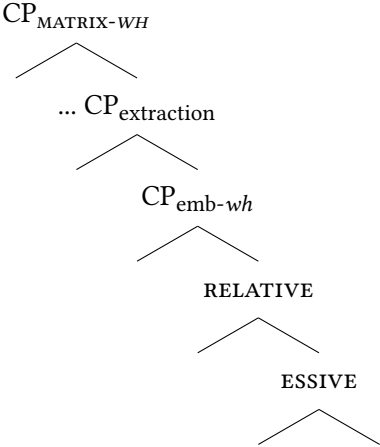
A solution that alleviates these issues is to conjecture that there is in fact two fseqs at play in this domain of the grammar, and that the structure in (26) can be broken up in what we may term The Comparative Fseq and The Relative Fseq, respectively, as given in (28) and (29).

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(28) The Comparative Fseq



(29) The Relative Fseq



By this move, the ESSIVE function heads a distinct fseq and we need not worry about how it structurally relates to the COMPARATIVE, EQUATIVE, and SIMILATIVE functions.

One argument for detaching ESSIVE from the other sub-clausal uses is that the ESSIVE in fact *cannot* involve a finite clause: it is *per se* a non-finite copula sentence (*John worked there as a teacher* = *John worked there and was a teacher*). All the other uses in the Comparative Fseq may involve finite clauses with lexical verbs.

At the same time, we can argue that there is a link between the two fseq's through EQUATIVE~ESSIVE which both essentially carry the semantics 'identity, same'. This is what allows syncretism to obtain *across* the two fseqs: rather than upward expansion through mere syntax it is shared semantics that facilitates it.⁴ One prediction from this is that syncretism should not be found between COM-

⁴Such a link *between* fseqs is akin to what is argued for in Vangsnes (2013) concerning functional expansion of *wh*-items that can lexicalize both manner at the clausal level in (i-a) and type at the DP level in (i-b), the link being the predicative property function in (i-c).

- (i) a. Korsn skal du løse de hær probleme? MANNER
 wh-s-n shall you solve the here problem-def
 'How will you solve this problem?'
- b. Korsn bil har du? TYPE
 wh-s-n car have you
 'What kind of car do you have?'

PARATIVE or SIMILATIVE with ESSIVE (or any node above it) without syncretism also holding with EQUATIVE.

It seems that Icelandic immediately crushes this prediction, but notice that Icelandic used to have *sem* in EQUATIVE, and the later replacement by *og* presumably took place after the use in ESSIVE was established.⁵ The same can be said about (High) German where according to Jäger (2019: 3ff.) *wie* replaces *als* in SIMILATIVE in the 16th century, spreading to EQUATIVE in the 19th century, and on to COMPARATIVE in colloquial German and most High German dialects. The use of *als* in ESSIVE is nevertheless unaffected by this ‘comparative cycle’ (term coined by Jäger) in Standard German. A possible spread of *wie* to ESSIVE should however not surprise anyone.

Let us now end the discussion with a closer look at some issues concerning the Relative Fseq.

4.3 Subject orientation in the Relative Fseq

As noted above in Section 4.2, Icelandic *sem* is obligatory in all relatives whereas Norwegian *som* is only obligatory in subject relatives and optional otherwise. This is a well-known and much discussed fact about Norwegian and the other Mainland Scandinavian languages, and the ±subject asymmetry is even stronger one node up the tree in CP_{emb-wh} function, i.e. embedded *wh*-questions, where Icelandic does not insert *sem*. It will lead too far here to delve into the explanations for the ±subject asymmetry in Norwegian/Mainland Scandinavian versus the lack thereof in Icelandic beyond the fact that it presumably is related to the licensing of the subject position, which is met by overt nominative case marking in Icelandic but not in Norwegian/Mainland Scandinavian. In relatives, Icelandic *sem* is therefore a mere marker of the left edge of any relative whereas Norwegian *som* is primarily a licenser of the subject gap.

In turn, in embedded *wh*-clauses the subject gap is licensed by the *wh*-subject itself in Icelandic but by insertion of *som* in Norwegian. Furthermore, in some

c. Korsn ser han ut? wh-s-n looks he out ‘What does he look like?’	PROPERTY
--	----------

⁵The following examples from J. Fritzner’s Old Norse dictionary suggest that the language did use *sem* in EQUATIVE constructions: *fölr sem gras* ‘pale as grass’, *blár sem hel* ‘blue as hell’. Thorsteinn Indriðason (p.c.) notes that using *sem* is still possible in Modern Icelandic, but that *eins og* would be more natural in these cases. An ESSIVE use of *sem* from the same dictionary is *hann fór sem úsekr maðr* ‘he left as an innocent man’. More detailed investigations of these matters would be ideal.

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dialects of Norwegian this mechanism carries over to cases of extraction of *wh*-subjects. Once that has been established the door is open for extending the mechanism to matrix *wh*-questions, inserting *som* in matrix C whenever the *wh*-constituent is the subject of the clause and leaving movement of the finite verb to second position to meet the V2 requirement obsolete. Whether this last part should be included in the Relative Fseq may be debated: it may very well make sense to keep it separate and assume that the properties of the embedded C may be copied onto the matrix C without assuming a strict sequential relationship between the two.⁶

In any event, under the story developed here this final step represents the end point of a quite elaborate grammaticalization process which started in a very different part of the grammar, presumably the SIMILATIVE function.

5 Conclusion

The item *som* in Norwegian has a wide range of functions which spans from a particle-like use in similative constructions and other subclausal uses via uses in the left periphery of embedded clauses to the matrix left periphery in some non-standard varieties. This wide lexicalization range can be understood as reflexes of historical grammaticalization which involves step-by-step cumulative “expansion” from smaller syntactic structures to bigger ones, but also semantic facilitation in at least one case.

Under a nanosyntactic account the various syntactic functions can be analyzed by means of two functional sequences: the Comparative Fseq involving SIMILATIVE – EQUATIVE – COMPARATIVE and the Relative Fseq involving ESSIVE – RELATIVE – CP_{emb-wh} – CP_{extraction} and possibly also CP_{matrix-wh}, semantically linked through EQUATIVE versus ESSIVE. This allows the expansion of *som* in surface syntax to be viewed as a gradual growth of its L-tree, i.e. the specification of what sub-trees it may lexicalize.

The view advocated by Halmøy & Berg-Olsen (2012) that there is only one lexical entry *som* can be maintained, but rather than attributing a particular meaning to the item itself, on a compositional view of semantics various meanings associated with *som* come from the size of the structures that it licenses.

⁶As noted in Vangsnes (2019), there are varieties of Scandinavian (colloquial Danish, Western Jutlandic, Faroese, Övdalian) that allow the insertion of an item otherwise found in relatives under extraction of *wh*-subjects without also allowing this in matrix subject *wh*-questions, hence the final step of the grammaticalization cline has not happened.

On various points, especially concerning diachronic matters, the story developed above could deserve some more substantiation. The predictions are nevertheless clear and ready to meet scrutiny.

Acknowledgements

The ideas about the item *som* presented here have evolved out of a long-lasting interest in the syntax of matrix *wh*-questions and (lack of) Verb Second across Norwegian dialects, a topic on which I have had a long-standing collaboration with Marit Westergaard. I wish to thank her for all the sparring provided when I have advocated the relevance of *som* for understanding that phenomenon. Thanks go also to Terje Lohndal who joined our collaboration at one point, and who furthermore have given very useful feedback during the writing of this paper. Moreover, I am indebted to two anonymous reviewers as well as Peter Svenonius for detailed comments to an earlier version, and finally I would also acknowledge the very fruitful feedback from the audiences at the workshop Microvariation and microchange in the Scandinavian languages in Kristiansand and the Gramino 4 conference in Sigtuna in May and June 2024, respectively.

References

- Axel-Tober, Katrin & Ellen Brandner. To appear. Complementation and relativization – A new perspective. In Ermenegildo Bidese & Federica Cognola (eds.), *Germanic syntax from a comparative perspective: Diachronic and synchronic issues* (Open Germanic Linguistics). Berlin: Language Science Press.
- Baunaz, Lena & Eric Lander. 2018. Nanosyntax: The basics. In Lena Baunaz, Liliane Haegeman, Karen De Clercq & Eric Lander (eds.), *Exploring nanosyntax* (Oxford Studies in Comparative Syntax), 3–56. New York: Oxford University Press. DOI: [10.1093/oso/9780190876746.003.0001](https://doi.org/10.1093/oso/9780190876746.003.0001).
- Brandner, Ellen. 2020. Re-thinking re-categorization: Is that really a complementizer? In András Bárány, Theresa Biberauer, Jamie Douglas & Sten Vikner (eds.), *Syntactic architecture and its consequences II: Between syntax and morphology*, 259–274. Berlin: Language Science Press. DOI: [10.5281/zenodo.4280651](https://doi.org/10.5281/zenodo.4280651).
- Brandner, Ellen & Iris Bräuning. 2013. Relative *wo* in Alemannic: Only a complementizer? *Linguistische Berichte* 234. 131–169.
- Caha, Pavel. 2009. *The nanosyntax of case*. CASTL, University of Tromsø. (Doctoral dissertation).

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- Cinque, Guglielmo & Luigi Rizzi. 2009. The cartography of syntactic structures. In Bernd Heine & Heiko Narrog (eds.), *The oxford handbook of linguistic analysis*. Oxford: Oxford University Press. DOI: [10.1093/oxfordhb/9780199544004.013.0003](https://doi.org/10.1093/oxfordhb/9780199544004.013.0003).
- Dancus, Adriana & Endre Mørck. 2008. Sammenlikningskonstruksjoner med enn som eller bare som etter komparativ og annen i norske dialekter. *Norsk Lingvistisk Tidsskrift* 26. 13–36.
- Halmøy, Madeleine & Sturla Berg-Olsen. 2012. I leksikografiens periferi – Tydingsbeskriving av som i Norsk Ordbok. *LexicoNordica* 19. 17–38.
- Haspelmath, Martin & Oda Buchholz. 1998. Equative and similitive constructions in the languages of Europe. In Johan van der Auwera (ed.), *Adverbial constructions in the languages of Europe*, 277–334. Berlin: Walter de Gruyter.
- Jäger, Agnes. 2019. The syntax of comparison constructions in diachronic and dialectal perspective. *Glossa: A Journal of General Linguistics* 4(1). 70: 1–51. DOI: [10.5334/gjgl.651](https://doi.org/10.5334/gjgl.651).
- Kiparsky, Paul. 1973. 'elsewhere' in phonology. In S. R. Anderson & P. Kiparsky (eds.), *A Festschrift for Morris Halle*, 93–106. New York: Holt, Rinehart & Winston.
- Kolbe, Daniela. 2003. The semantic and grammatical overlap of as and that: Evidence from non-standard English. In U. Römer & R. Schulze (eds.), *Exploring the lexis–grammar interface*, 137–152. Amsterdam: John Benjamins.
- Lander, Eric. 2015. *The nanosyntax of Northwest Germanic reinforced demonstratives*. University of Ghent. (Doctoral dissertation).
- Nordgård, Torbjørn. 1985. *Word order, binding and the empty category principle*. University of Trondheim. (Cand. Philol. thesis).
- Starke, Michel. 2009. Nanosyntax: A short primer to a new approach to language. *Nordlyd* 36(1). 1–6.
- Stroh-Wollin, Ulla. 2002. *Som-satser med och utan som*. Uppsala University. (Doctoral dissertation).
- Vangsnes, Øystein A. 1999. *The identification of functional architecture*. University of Bergen. (Doctoral dissertation).
- Vangsnes, Øystein A. 2013. Syncretism and functional expansion in Germanic wh-expressions. *Language Sciences* 36. 47–65. DOI: [10.1016/j.langsci.2012.03.019](https://doi.org/10.1016/j.langsci.2012.03.019).
- Vangsnes, Øystein A. 2019. COMP trace effects across North Germanic varieties. In K. R. Christensen, H. Jørgensen & J. Wood (eds.), *The sign of the V – Papers in honour of Sten Vikner*, 657–683. Aarhus: Dept. of English, School of Communication & Culture, Aarhus University. DOI: [10.7146/aul.348.119](https://doi.org/10.7146/aul.348.119).

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Wagener, Terje. 2017. *The history of Nordic relative clauses* (Trends in Linguistics. Studies and Monographs 304). Berlin: De Gruyter Mouton.

Chapter 27

Multilingualism and syntactic change in English 1400–1700

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This paper will bring together recent work on the loss of Verb Second (V2) in English with insights from the literature on the rise of *do*-support and on the auxiliation of modal auxiliaries. It adds to the discussion of the interrelatedness between these changes, which form a key part of the reorganization of the English verbal system from late Middle English (ME) to the end of early Modern English (EME) and beyond. This complex of changes will also be considered from the perspective of increasing language and dialect contact that I will discuss from the point of view of multilingualism.

1 Introduction

This paper will bring together recent work on the loss of Verb Second (V2) in English with insights from the literature on the rise of *do*-support and on the auxiliation of modal auxiliaries, and add to the discussion of the interrelatedness between the changes, which form a key part of the reorganization of the English verbal system in the transition from late Middle English (ME) to early Modern English (EME). I will also consider this complex of changes from the perspective of an increasing scale of dialect and language contact and suggest that the changes may well testify to a more intensive and perhaps disruptive form of language contact that I will call multilingualism. I use this term on the premise that increasing national and international trade brought increasing numbers of dialect speakers and foreign speakers to England, particularly to the larger cities, developing populations with functional command of more than one language/variety. Historians

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document substantial interregional and international migration whose impact, I argue, could be consistent with the effects found in the literature on L2 and L3 learning by adult speakers in the present day and I suggest that it is useful to study the changes from that perspective.¹ Section 2 will discuss some main effects of how dialect contact and migration may have affected the syntax of late ME. Section 3 presents a discussion of the changes, Section 4 concludes.

The argument I present is tentative in the sense that it cannot be “proved” and should be interpreted as an attempt to understand the language dynamics of the time, and the nature and pace of the syntactic changes under study, which were sweeping and had a profound effect on the verbal system of the language in a relatively short space of time, the key part between 1480 and 1570.

2 Multilingualism in late Middle English

Language contact is generally recognized as a major factor in the historical development of English. I refer to [Walkden et al. \(2023\)](#) for a summary of contact-induced syntactic changes, resulting from contact with Celtic, with Scandinavian (see [Warner 2017](#) for an in-depth assessment of the 9th and 10th century Viking invasions and settlements), and with French in the long aftermath of the Norman invasion of 1066 as the language of government and court.

It may seem less obvious to treat the late Middle Ages as an era of language contact, but the increasingly more detailed work on the sociolinguistic situation in late ME indicates much dialect contact and supralocalization (e.g. [Eitler & Westergaard 2014](#), [Auer 2018](#), [Stenroos 2020](#)). This should come as no surprise. The 13th and 14th centuries saw the rise of organised cities attracting trade and migration from a wider area as the feudal system was disintegrating and national and international trade was expanding. The Black Death of the 1340s killed about one third of the population of Europe, leading to massive labour shortage, followed by many bad harvests and famine. One result in England was the Peasant’s revolt (1381). The Wars of the Roses (1455–1487), the dissolution of the monasteries (around 1530) and the mid-17th century civil wars brought further societal upheaval and migration.

¹By “multilingualism” I do not mean the term in the same sense as the recent surge of very interesting historical sociolinguistic work on how communities of writing, from early Middle English onward, practice multilingualism in navigating the contexts in which they use Latin, English and Anglo-Norman, and on the role that these developments play in the later rise of the standard language. For some recent literature and references cited there, see [Pahta et al. \(2017\)](#), [Wright \(2020\)](#), [Räikkönen et al. \(2024\)](#).

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Kermode (2008) and Barron (2008) confirm and partly document that the late 14th and 15th century saw large-scale interregional migration to larger cities, especially to London, including long-distance migration from all parts of England. We may well wonder to what extent these dialects in contact were mutually intelligible. Kermode (2008: 459) also notes long distance trade that brought foreigners to a number of the larger cities in England on a permanent basis (e.g. York and Chester). Barron (2008: 402) lists “alien communities” in London, including Italian and Hanze merchants, and permanent traders from Flanders, Brabant, Holland, Zeeland, Greece and Venice. The total size of these communities is estimated as small (2% of the population), but Barron (2008: 402–403) argues: “This is a tiny percentage and yet by their distinctive appearance, languages and skills, these aliens may have contributed more to the diversity and character of London life than their small number might suggest.” While it is hard to assess the impact of these communities, the combination with large scale interregional migration at least suggests a considerable language dynamic in the larger cities of England.

This historical sketch provides no answer to the question of the nature of the contact situation. Was this a case of contact that involved a substantial scale of adult L2 and L3 acquisition or was there more or less balanced multilingualism involving widespread L1 acquisition? I argue that the nature and speed of the changes could well indicate the former.

2.1 Syntactic effects of dialect and language contact

This section will first consider what kinds of syntactic change might be expected under conditions of substantial contact involving L2 and L3 acquisition, in order to see how the syntactic changes discussed might fit such a scenario.

The first of these is accelerated grammaticalization. Language and dialect contact famously promote the (accelerated) erosion of inflectional morphology and its morphosyntactic counterpart in the shape of grammaticalization of lexical to function words (see, among many others, Traugott & Heine 1991; Roberts & Rousou 2003; van Gelderen 2004). The case discussed here is the range of verbal periphrastic constructions arising from late ME onward (in particular *do*-support as a dummy auxiliary strategy and the reanalysis of pre-modals to function words). This process is interrelated with the loss of finite verb (V_{fin}) movement of lexical finite verbs as evidenced by the loss of two distinct Verb Second (V2) patterns, and V_{fin} movement to the left of negation. I will argue for a clear and detailed causal relationship between the rise of auxiliaries and change as well as loss in some distinct V2 word order patterns.

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The second type of change is the levelling of syntactic and pragmatic distinctions typical of koineization. This term was introduced in the sense understood here by [Kerswill & Williams \(2000\)](#). They report on dialect levelling effects in the new town Milton Keynes in Britain, in which majority forms rather than minority forms in the mix win out. This approach is extended to include morphological and syntactic change by [Howell \(2006\)](#). Simple features are more often adopted than complex ones (I will refine and illustrate this notion in the case studies below). Furthermore, marked regional forms are disfavoured.

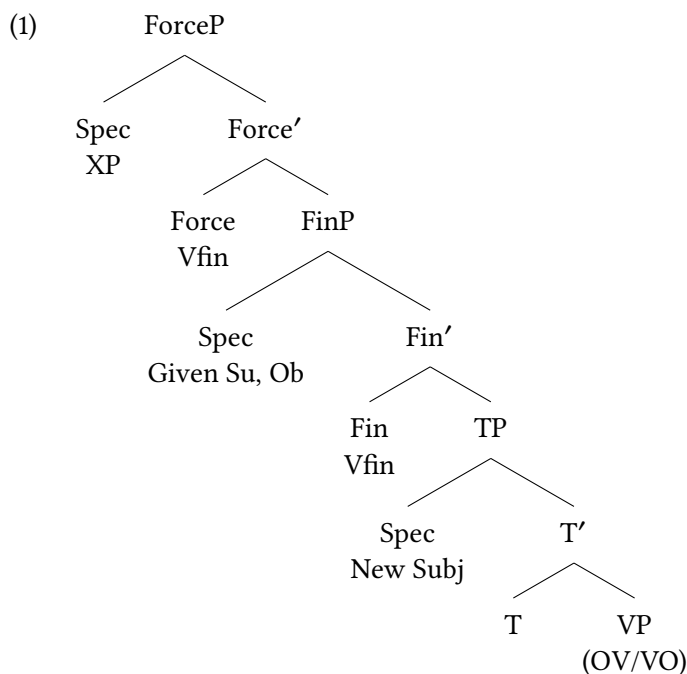
3 Word order and auxiliaries from Late Middle to Early Modern English

This section considers four major syntactic changes that took place over the period between 1480–1570, a timeframe of 90 years, in the following chronology:

- The loss of mixed V2 inversion (Section [3.1](#))
- The rise of *do*-support in negated and adverbial *wh*-questions (Section [3.2](#))
- The reanalysis of auxiliaries to function words (discussed in this section)
- The loss of V2 inversion following clause-initial *then* (Section [3.3](#))

I first consider how these developments may interact with changes in word order, in particular with the dynamics of the loss of different types of V2.

I start with a brief sketch of a formal analysis of the relation between auxiliation and finite (lexical) verb placement, following formal analyses of grammaticalization. I take [Rizzi's \(1997\)](#) clause structure as a starting point in [\(1\)](#), following its application to the history of English by [Roberts \(1996\)](#); [Westergaard \(2009\)](#), among many others.



Early English modal verbs as pre-auxiliaries were a type of main verb that underwent the same Vfin placement rules as other lexical verbs. Thus, any Vfin, could move out of VP by head to head movement to T(ense), Fin(ite), and Force, according to the head to head movement constraint of Travis (1984). Formal models of grammaticalization analyse this as a process whereby a lexical verb undergoing movement to a functional head (T, Fin, Force in (1)), is reanalysed as the fixed functional occupant of this head. The auxiliatation of modal verbs to function words discussed by e.g. Lightfoot (1979); Kroch (1989); Warner 1993) is thus a typical case of grammaticalization. (e.g. van Gelderen 2004; Roberts & Roussou 2003). This perspective on grammaticalization establishes a close relationship between emerging auxiliaries and the position of finite lexical verbs and yields a clear perspective on tracing the processes of change by looking at interrelated changes in Vfin position.

The development of English modals from (more) lexical verb to function word (auxiliary) has been much debated in the literature cited above. Briefly, (see Warner 1993 for detailed discussion), pre-modals had more main verb characteristics in OE. Some of them had non-finite forms, e.g. a verb such as OE *willan*, ME *willen* ('want', the precursor of auxiliary *will*). It could mark future tense, but it could also take a direct object or a *that*-clause complement. A number of

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main verb-like properties were lost over the ME period, and verb endings were reduced more generally over the ME period, not least including infinitival and subjunctive marking, which led to an increased use of finite modals. The loss of such main verb characteristics of pre-modals took several centuries. There is consensus about the dating of the reanalysis as functional heads over the 16th century,

3.1 Mixed V2 inversion and its loss

The clause structure in (1) assumes a more complex left periphery than present-day English, allowing for word order flexibility related to discourse-pragmatic distinctions interacting with V_{fin} placement. I first consider V2 order, which had two distinct types following different pathways of change. I discuss them in the order in which they were lost.

The first type of V2 in fact alternates between V2 and V3, with the choice between them driven by discourse-pragmatic properties of the subject (see [Bech 2001](#)). Examples with initial adverbs are given in (2-4), with the subject in square brackets: pronominal *he* in (2) and nominal *þis dai* in (3) are discourse-given and preverbal in Spec,Fin; nominal *þe mon* in (4) refers generically to how men respond to circumstances and is postverbal in Spec,TP.²

- (2) Cmpeterb. 50.269
 & siððon [he] sende him to Englalande.
 And afterwards he.SBJ.3SG sent.VB.PST.3SG him.OBJ.3SG to England
 ‘and afterwards he sent him to England.’
- (3) Cmlamb. 83.4
 forþi [þis dai] is cleped estrene dai.
 therefore this day.SBJ.3SG is.AUX.3SG called.PTCP Easterday
 ‘This is why this day is called Easter day.’
- (4) Cmlamb. 103.2
 Eaðe mei [þe mon] fundan hu he hine seolfe
 Easily may.AUX.3SG the man.SBJ.3SG find.VB.INF how he himself
 amerre.
 injure.SBJV
 ‘Easily may a man find how he may injure himself’

²The data are drawn from [Kroch & Taylor \(2000\)](#) and from [Kroch et al. \(2004\)](#). The text references are from these corpora.

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This analysis follows van Kemenade & Westergaard (2012); Walkden (2017) and many others. Nominal subjects of passives and unaccusative predicates could also occur in lower positions typical of internal arguments, as in (5-6) (Van Kemenade 1997; Biberauer & van Kemenade 2011).

- (5)

Cmancriw1, II.89.1081

purch blod is in hali boc [sunne] bitacned.
by blood is.AUX.3SG in holy book sin.SBJ.3SG meant.PTCP
'By blood is meant sin in Holy Scripture.'
- (6)

Cmpeterb, 52.304

Ær he wære wel ded, þa wære þær
before he were.PST.SBJV rightly dead, then were.PST.AUX there
coren [twa papes]
chosen.PTCP two popes.SBJ.PL
'He was barely dead before two popes were elected.'

The passivized subject *sunne* in (5) is between an adjunct PP and the nonfinite verb, a low position for bare (i.e. non-referential) subjects in passivized (relic) OV word orders. Discourse-new subjects can also occur clause-finally, as (6) shows for 'two popes'. Such subjects are usually discourse-new and heavy. Summarizing, (2-6) jointly illustrate a range of subject positions in Old and early Middle English which broadly correspond to the NP type and information status (IS) of the subject (cf. Gundel et al. 1993).

- (7)

subjects in clauses with all Vfin types

subjects in passives only

Pronouns > discourse-old NP > discourse-new NP > bare nouns > new/heavy NP

Spec,Fin Spec,Fin/Spec,T Spec,T (or lower) incorporated left of Vnonfin clause-final

The ME development of this type of V2 (van Kemenade 2026) shows that this IS-driven/pragmatic correspondence between subject type and position weakens over the ME period and shows levelling of pragmatic distinctions, in turn leading to loss of structure.

I focus on inversion of nominal subjects following adverbs other than *then*, where the key change takes place.³ Table 1 and Figure 1 show the development

³Three groups of data were left out in Tables 1-3:

1. the M2 period was left out because of the sparsity of text material.

2. Clauses with initial short deictic adverbs like *here*, *there*, *thus*, *now*, *yet*, and *so*, as they increasingly behave like *then*-initial clauses over the ME period (van Kemenade et al. 2023: 13).

3. Clauses with the precursor of *say* as Vfin, as late ME saw the rise of quotative inversion.

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over the ME period:

Table 1: % of inversion of nominal subject following Other Adverbs, per verb type

		M1	M3	M4
		1150–1250	1350–1420	1420–1500
Aux	N	27/41	149/374	68/204
	%	65,9%	39,8%	33,3%
Unacc	N	54/88	230/461	178/380
	%	61,4%	49,9%	46,8%
Tr/Unerg	N	15/37	71/262	35/203
	%	40,5%	27,1%	17,2%

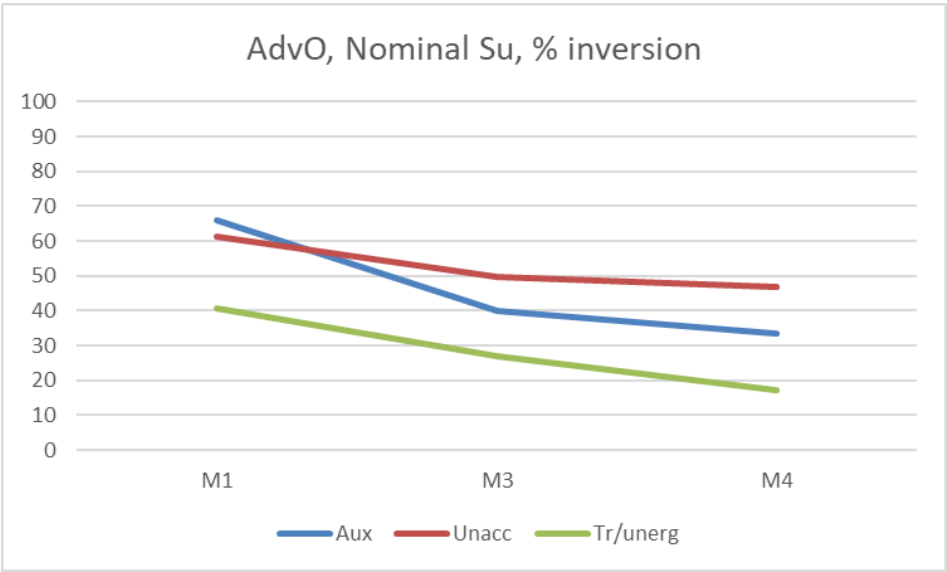


Figure 1: % Inversion of nominal subject following Other Adverbs, per verb type

The figures show diverging rates of inversion for unaccusative and transitive verbs. Both represent V-to-Fin movement, but subjects in unaccusative contexts often occur in a position below Fin, as noted above. Inversion rates are lowest for transitive verbs: agent subjects are more often discourse-given and more often occur in Spec,Fin.

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Tables 2 and 3 present distributional differences between subject type in relation to Vfin type and V2/V3 word order following adverbs other than then (AdvO).⁴ Table 2 lists figures for transitive verbs. Inversion rates (V2) are relatively low in M1, while V3 is relatively robust for transitive verbs. V3 rates rise significantly over the M3 and M4 periods, but we also see increasing rates of V3 for New and Non-referential subjects, showing that these were shifting to the preverbal subject position.

Table 2: IS status of subjects of transitive verbs in V2 and V3 orders after AdvO in ME

	M1 (1150–1250)			M3 (1350–1420)			M4 (1420–1500)		
	V2	V3	%V3	V2	V3	%V3	V2	V3	%V3
Given	16	12	42,9	9	55	85,9	5	96	95,0
New				2	9	81,8	4	8	66,7
Nonref	2	5	71,4	5	17	77,3	0	12	100

Table 3 lists V2 and V3 word orders in unaccusative contexts. V2 order is relatively frequent for all three verb types in M1, showing that unaccusative subjects occur more frequently in a position below Vfin in Fin. New and Non-referential subjects shift to preverbal position over the M3 and M4 periods, with lower rates than for transitive verbs in Table 2.

Table 3: IS status of subjects of unaccusative verbs in V2 and V3 orders after AdvO in ME

	M1 (1150–1250)			M3 (1350–1420)			M4 (1420–1500)		
	V2	V3	%V3	V2	V3	%V3	V2	V3	%V3
Given	27	27	50,0	47	109	69,9	34	87	71,9
New	2	0	0,0	4	10	71,4	11	8	43,1
Nonref	12	2	14,3	26	48	64,9	9	9	50,0

We see that the rate of V3, which in OE and M1 measures the rate of preverbal discourse-given nominal subjects in Spec,Fin, is substantially higher for transitive verbs than it is in unaccusative contexts. However, Tables 2–3 also show a marked increase of V3 for New and Non-referential subjects, although

⁴The data methodology is detailed in [van Kemenade \(2026\)](#)

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V2 word orders were not fully lost in that context by the end of the ME period. This shows that the IS/pragmatic distinction between preverbal (discourse-given) and postverbal (discourse-new) subjects was being neutralized. This part of the change shows levelling of pragmatic distinctions, a type of change that occurs in language and dialect contact. The result of the increase of V3 in unaccusative contexts with New and Non-referential subjects in Spec,Fin eroded the status of Spec,FinP as a host for discourse-given fronted elements, which led to the loss of V to Fin movement, and to the breakdown of this type of V2. [Haeberli & Ihsane \(2016\)](#) present independent evidence for this on the basis of the shifting position of V_{fin} with respect to adverbs and negation in subject-initial clauses. They date the loss of V to Fin movement between 1480 and 1550, much in line with the facts presented here.

3.2 Fast grammaticalization: The rise of *do*-support in questions

We now first turn to the rise of *do*-support. The discussion here is based on [Ellegård's](#) (1953) figure of the development of *do*-support, copied here from [Kauhanen & Walkden \(2017\)](#).

Figure 2 shows that negative questions and affirmative transitive questions lead the rise of *do*-support from 1480 onward. This fast rise falls within the same time frame as the loss of V to Fin movement: between 1480 and 1570. These facts can be accounted for by refining the relationship postulated by [Kroch \(1989\)](#) between the loss of V to T movement and the rise of *do*-support in negation contexts: Kroch analyses *do*-support as a last resort option as V to C/Force movement was partially lost in the second half of the 16th century. I follow the spirit of this account, except that the dating of the loss of V to Fin movement matches the rise of *do*-support in negative and transitive questions much more closely. [Haeberli & Ihsane \(2016\)](#) and [van Kemenade \(2026\)](#), on the basis of distinct and independent evidence, propose compatible datings for the loss of V to Fin movement: between 1480 and 1550. [Haeberli & Ihsane \(2016\)](#) moreover show that V to T movement was lost later. The spirit of the account is the same though: once V can no longer move to Fin, it also cannot move to a higher functional head position Force, according to the head to head movement constraint ([Travis 1984](#)) which requires that movement to Force cannot skip intermediate functional heads. *Do*-support was invoked as a last resort strategy in V to C/Force constructions: questions, focal negative-initial, and *then*-initial constructions.

Last resort auxiliaries are often the result of L2 learning and are thus easily associated with language and dialect contact. A case in point is [van de Craats &](#)

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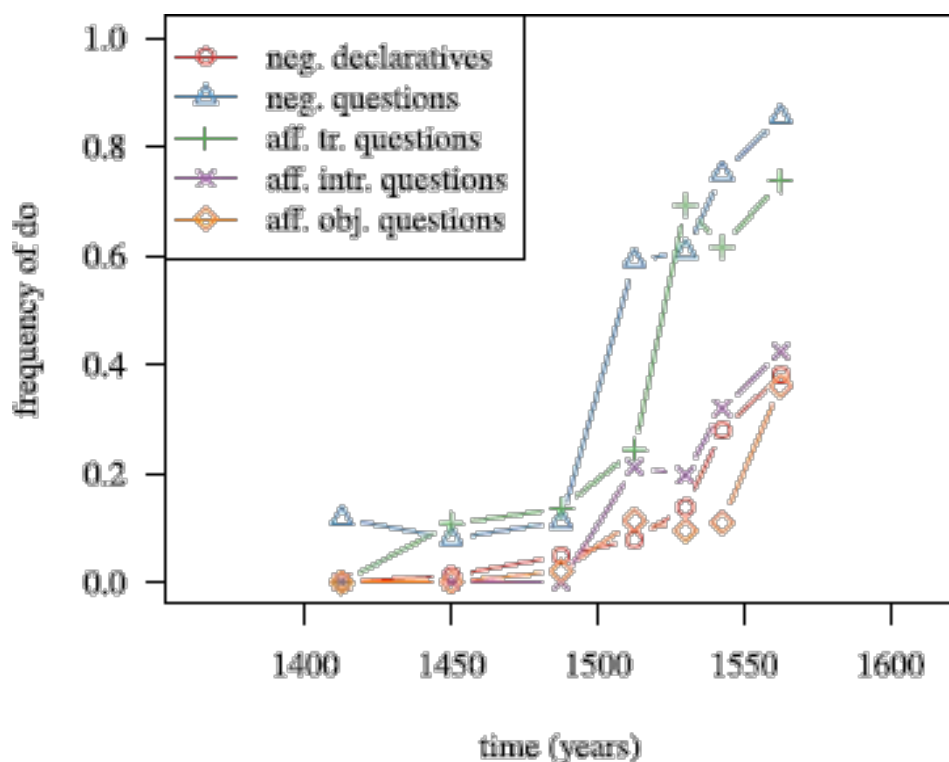


Figure 2: development of *do*-forms in different types of sentence

van Hout (2010). They present an experimental study of second and third generation Moroccan immigrant children in the Netherlands with Moroccan Arabic as their heritage language. Present-day Dutch is an OV language with categorical V to Force movement in main clauses. This compares well with the syntax of V to Force contexts in English. van de Craats & van Hout (2010) report that several of their experimental subjects make no attempt at V to Force movement, but replace lexical V2 verbs with dummy auxiliaries *be* or *go*. It is easy to imagine that the presence of increasing numbers of L2 speakers promoted *do*-support in questions, starting over the 15th century, and accelerating hand in hand with the loss of V to Fin.

3.3 V2 in V to Force constructions and its partial loss

This V2 type was restricted to three specific contexts illustrated in (8-10) with ME examples: questions (8), negative-initial clauses (9), and clauses introduced by the adverb *then* (10)

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- (8) Cmmandev.2.23
Dere god, what loue hadde he to vs his subiettes...?
Dear God, what love.OBJ had.VB.PST.SG he.SBJ.SG for us his subjects
‘Dear God, what love did he have for us, his subjects?’
- (9) Cmearlps.147.6476
ne wil þou nouȝt confounde me.
Not will.AUX.PRS you.SBJ.2SG not confound.VB.INF me
‘You will not confound me.’
- (10) Cmcapchr.101.2151
Tho cam he into Ingland.
Then came.VB.PST.SG he.SBJ.SG to England
‘Then he came to England.’

(8-10) reflect a ME continuation of robust OE patterns. Questions are the most robust and enduring V2 pattern in the history of English, though only with auxiliaries in the present-day.

Negative-initial (9), where initial *ne* was an unstressed element proclitic on *Vfin*, was lost over the ME period. Detailed discussion can be found in Wallage (2017). A negative-initial pattern that does survive to the present day has an initial negated adverb or argument, such as *Never would I do such a thing!* Wallage (2017: 164) argues, extending Nevalainen (1997) that this construction had existed throughout the OE and ME periods. Note, however, that the initial constituent in the surviving pattern is focal and prosodically strong. I come back to this below.

Pattern (10) with initial *then* was lost around 1570. Table 4 and Figure 3 (from van Kemenade et al. 2023) illustrate:

Table 4: Inversion of pronominal subjects following then in Middle and early Modern English

		M1	M3	M4	E1	E2	E3
Aux	N	16/25	74/108	92/153	96/172	21/193	5/115
	%	64,0%	68,5%	60,1%	55,8%	10,9%	4,3%
Unacc	N	44/49	43/98	119/300	45/147	17/249	3/70
	%	89,8%	43,9%	39,7%	30,6%	6,8%	4,3%
Tr/unerg	N	26/33	57/118	82/363	65/178	17/286	0/102
	%	78,8%	48,3%	22,6%	36,5%	5,9%	0,0%

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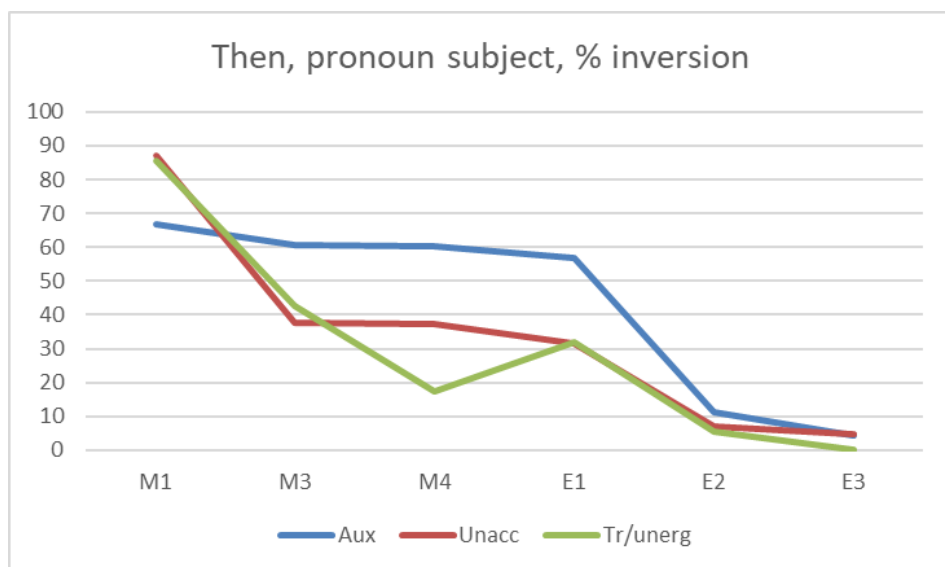


Figure 3: % Inversion of pronoun subjects following *then* in Middle and early Modern English

This is a remarkable change because the syntactic structure of *then*-initial clauses was stably identical to that of questions. This is strikingly confirmed by the fact that the E1 period (1500–1570) features a fair number of examples with *do*-support, e.g. *Than dyd they consulte together* (Mowntayne, 188.250), showing that the pattern was productive.

van Kemenade et al. (2023) revisit the loss of V to Force movement after *then*. The pattern was robust up to 60% with a (pre-)auxiliary (less so with lexical verbs) throughout ME and into the E1 period (1500–1570). Figure 3 shows that it was drastically reduced to around 10% in the transition from E1 to E2. Thus, while V/Aux to Force movement continued robustly in questions and focal negative-initials, it was rapidly lost in *then*-initial V2 clauses. van Kemenade et al. (2023) analyse this as a reflex of the reanalysis of pre-auxiliaries to functional heads and argue that this involved loss of primary stress on the auxiliary. This did not affect the pattern in questions and clauses with initial focal negatives, where the first constituent carries primary stress, allowing it to head the initial prosodic foot. Initial *then* was mostly unstressed, and the loss of stress on the auxiliary, followed by a prosodically light subject pronoun left the initial prosodic foot unheaded, violating basic prosodic requirements.

This account analyses the loss of V2 in *then*-initial clauses as the final stage of the reanalysis of pre-auxiliaries to functional head status, leading to loss of the

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ability to carry primary stress. This narrows down the dating of the final stage of the reanalysis of modals: around 1570.

Haeberli & Ihsane (2016) make a further case that the loss of V to T movement in subject-initial clauses is in fact a two-step development, and that V to T-movement as in Kroch (1989) actually involves two positions: a higher one preceding adverbs (the head of FinP) and a lower one preceding *not* (T). V movement to Fin (to the left of adverbs) was lost between 1480 and 1550; V movement to T (to the left of *not*) was lost over the late 16th century and later. It is the latter that seems to form the counterpart of the rise of *do*-support in negative declarative clauses from 1500 onward.

4 Discussion and conclusion

My account shows that four major changes in the history of English: the loss of mixed V2, the rise of *do*-support in questions, the reanalysis of modals and the partial loss of V to Force movement trace the interplay of the grammaticalization process of auxiliaries and the loss of V movement of lexical verbs in V2 patterns, as in approaches to grammaticalization where functional heads (auxiliaries) and lexical verbs to fronted to functional heads are analysed as two sides of the same coin. The changes show traces of a multilingual language environment:

- the loss of V to Fin movement was at least partially the result of the breakdown of the earlier subtle correspondence between subject position, the IS-status of the subject, and the type of V_{fin}, and may be viewed as a case of levelling of pragmatic distinctions as found in koineization processes.
- The rise of *do*-support in questions follows the time-path of the loss of V to Fin movement, resulting directly in the loss of V to Force movement of lexical V_{fin}. *Do*-support was a last resort strategy in questions and developed quite fast between 1480 and 1550. The fast development of a last resort strategy suggests a contact situation and concomitant L2 learning.
- The partial loss of V to Force movement in *then*-initial clauses reflects the final stage of grammaticalization of auxiliaries: the loss of primary stress (at least in the left periphery). The timing and pace was fast, around 1570.

Space does not permit discussion here of “messy” relics changes typical of multilingualism: one of them would be that modals were all grammaticalized as functional heads, although some of them, e.g. *can*, maintained the semantics of a

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lexical verb. Messy relics that would definitely require micro-cues for language learning in the sense of Westergaard (2021) would include quotative inversion and comparative inversion. I leave this for further research.

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References

- Auer, Anita. 2018. Urban literacies and processes of supralocalisation: A historical sociolinguistic perspective. In Natalie Braber & Sandra Jansen (eds.), *Sociolinguistics in England*, 13–34. London: Palgrave Macmillan.
- Barron, Caroline M. 2008. London 1300–1540. In David M. Palliser (ed.), *The Cambridge urban history of Britain*, vol. 1, 395–440. Cambridge: Cambridge University Press.
- Bech, Kristin. 2001. *Word order patterns in Old and Middle English: A syntactic and pragmatic study*. University of Bergen. (Doctoral dissertation).
- Biberauer, Theresa & Ans van Kemenade. 2011. Subject positions and information-structural diversification in the history of English. *Catalan Journal of Linguistics* 10. 17–94.
- Eitler, Tamas & Marit Westergaard. 2014. Word order variation in late Middle English: The effect of information structure and audience design. In Anneli Meurman-Solin, Maria José López Couso & Bettelou Los (eds.), *Information structure and syntactic change in the history of English*, 87–118. Oxford: Oxford University Press.
- Ellegård, Alvar. 1953. *The auxiliary do: The establishment and regulation of its use in English*. Stockholm: Almqvist & Wiksell.
- Gundel, Jeannette K., Nancy Hedberg & Ron Zacharsky. 1993. Cognitive status and the form of referring expressions in discourse. *Language* 69. 274–307.
- Haeberli, Eric & Tabea Ihsane. 2016. Revisiting the loss of verb movement in the history of English. *Natural Language and Linguistic Theory* 34. 497–542.
- Howell, Robert. 2006. Immigration and koineisation: The formation of early Modern Dutch urban vernaculars. *Transactions of the Philological Society* 104. 207–227.
- Kauhanen, Henri & George Walkden. 2017. Deriving the constant rate effect. *Natural Language and Linguistic Theory* 36. 483–521.

Ans van Kemenade

- Kermode, Jennifer. 2008. The greater towns 1300-1540. In David M. Palliser (ed.), *The Cambridge urban history of Britain*, vol. 1, 441–465. Cambridge: Cambridge University Press.
- Kerswill, Paul & Ann Williams. 2000. Mobility and social class in dialect levelling: Evidence from new and old towns in England. In Klaus Mattheier (ed.), *Dialect and migration in a changing Europe*, 1–13. Frankfurt: Peter Lang.
- Kroch, Anthony. 1989. Reflexes of grammar in patterns of language change. *Language Variation and Change* 1. 199–244. DOI: [10.1017/S0954394500000168](https://doi.org/10.1017/S0954394500000168).
- Kroch, Anthony, Beatrice Santorini & Lauren Delfs. 2004. *Penn Helsinki parsed corpus of early Modern English*. Penn Parsed Corpora of Historical English.
- Kroch, Anthony & Ann Taylor. 2000. *The Penn Helsinki parsed corpus of Middle English*. Penn Parsed Corpora of Historical English.
- Lightfoot, David. 1979. *Principles of diachronic syntax*. Cambridge: Cambridge University Press.
- Nevalainen, Terttu. 1997. Recycling inversion: The case of initial adverbs and negators in early modern English. *Studia Anglica Posnaniensia* XXI. 203–214.
- Pahta, Päivi, Janne Skaffari & Laura Wright. 2017. *Multilingual practices in language history: English and beyond*. Berlin: De Gruyter Mouton.
- Räikkönen, Jenni, Carla Suhr, Minna Palander-Collin, Arja Nurmi, Minna Nevala & Turo Iltunen. 2024. *Multilingualism and language variation in English across genres and registers: A Festschrift in honour of Päivi Pahta*. Helsinki: Société Néophilologique.
- Rizzi, Luigi. 1997. The fine structure of the left periphery. In Liliane Haegeman (ed.), *Elements of grammar: Handbook of generative syntax* (Kluwer International Handbooks of Linguistics), 281–337. Dordrecht: Kluwer. DOI: [10.1007/978-94-011-5420-8_7](https://doi.org/10.1007/978-94-011-5420-8_7).
- Roberts, Ian. 1996. Remarks on the old English C-system and the diachrony of V2. *Linguistische Berichte Sonderheft* 7. 154–168.
- Roberts, Ian & Anna Roussou. 2003. *Syntactic change: A minimalist approach to grammaticalization*. Cambridge: Cambridge University Press.
- Stenroos, Merja. 2020. Regional variation and supralocalization in late medieval English: Comparing administrative and literary texts. In Merja Stenroos & Kjetil Thengs (eds.), *Records of real people: Linguistic variation in Middle English local documents*, 95–128. Amsterdam: John Benjamins.
- Traugott, Elizabeth & Bernd Heine. 1991. *Approaches to grammaticalization*, vol. 2. Amsterdam: John Benjamins.
- Travis, Lisa. 1984. *Parameters and effects of word order variation*. MIT. (Doctoral dissertation).

27 Multilingualism and syntactic change in English 1400–1700

- van Gelderen, Elly. 2004. *Grammaticalization as economy*. Amsterdam: John Benjamins.
- van Kemenade, Ans. 2026. The diachrony of V2/V3 alternations in English: Interaction between syntax and pragmatics. In Jadranka Gvozdanovic (ed.), *Historical linguistics 2023*. Amsterdam: John Benjamins. To appear.
- van Kemenade, Ans, Roland Hinterhölzl & Tara Struik. 2023. Word order change, architecture and interfaces: Evidence from the development of V to C movement in the history of English. *Journal of Historical Syntax* 7(25). 1–55. DOI: [10.18148/hs/2023.v7i25.189](https://doi.org/10.18148/hs/2023.v7i25.189).
- van Kemenade, Ans & Marit Westergaard. 2012. Syntax and information structure: V2 variation in Middle English. In María José López-Couso, Bettelou Los & Anneli Meurmann-Solin (eds.), *Information structure and syntactic change*, 87–120. New York: Oxford University Press.
- van de Craats, Ineke & Roeland van Hout. 2010. Dummy auxiliaries in the second language acquisition of Moroccan learners of Dutch: Form and function. *Second Language Research* 26. 473–500.
- Van Kemenade, Ans. 1997. V2 and embedded topicalisation in Old and Middle English. In Ans van Kemenade & Nigel Vincent (eds.), *Parameters of morphosyntactic change*, 326–352. Cambridge: Cambridge University Press.
- Walkden, George. 2017. Language contact and V3 in Germanic varieties new and old. *Journal of Comparative Germanic Linguistics* 20(1). 49–81.
- Walkden, George, Juhani Klemola & Thomas Rainsford. 2023. An overview of contact-induced morphosyntactic changes in early English. In Sara M. Pons Sanz & Louise Sylvester (eds.), *Medieval English in a multilingual context*, 239–277. London: Palgrave.
- Wallage, Philip. 2017. *Negation in Early English: Grammatical and functional change*. Cambridge: Cambridge University Press.
- Warner, Anthony. 1993. *English auxiliaries: Structure and history* (Cambridge Studies in Linguistics 66). Cambridge: Cambridge University Press.
- Warner, Anthony. 2017. English-Norse contact, simplification, and sociolinguistic typology. *Neuphilologische Mitteilungen* 118. 317–403.
- Westergaard, Marit. 2009. Word order in Old and Middle English: The role of information structure and first language acquisition. *Diachronica* 26(1). 65–102.
- Westergaard, Marit. 2021. Microvariation in multilingual situations: The importance of property-by-property acquisition. *Second Language Research* 37(3). 379–407.
- Wright, Laura. 2020. *The multilingual origins of standard English*. Berlin: Mouton de Gruyter.

Chapter 28

Approaches to L3 phonological acquisition research: challenges, considerations, best practices

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This contribution aims at a comprehensive overview of state-of-the-art approaches to research on L3 phonological acquisition. The analysis involves theoretical and methodological considerations as well as practical solutions to problems pertaining to the production, perception and processing of speech in third and additional languages. A range of research paradigms, elicitation tasks, and instruments are critically evaluated and various approaches to auditory and acoustic analyses are discussed. The overview is illustrated with examples from current research into L3 phonetics and phonology including the findings from the most recent Polish-Norwegian project.

Keywords: L3 phonological acquisition, methodological considerations, theoretical approaches, phonetics and phonology

1 Introduction: Where we stand in L3 speech

An expanding corpus of literature on third language (L3) acquisition highlights the evolving investigations into the speech patterns of multilingual learners. These individuals are distinguished by their heightened metalinguistic awareness, prior linguistic knowledge, and extensive cross-linguistic interactions. The primary objective of research on L3 phonological acquisition is to analyze the distinct characteristics of perception, production, and phonological processing among multilingual speakers. Central to the study of L3 speech learning are

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the nature and origins of cross-linguistic influence, as well as the potential advantages in acquiring subsequent non-native phonological systems. Notable distinctions between L2 and L3 speech learning arise from differences in language learning history and use, an expanded phonetic and phonological repertoire, and heightened perceptual sensitivity. Moreover, specific methodological considerations in L3 speech research include the collection of data across all languages within the multilingual participants' repertoire, the introduction into a particular language mode of data collection, and the complex selection of participant and control groups. Despite the vigorous growth of the L3 speech research agenda, further development is necessary in some areas such as theoretical modeling and L3 speech processing. Quite extensive overviews of previous research on L3 phonological acquisition have been presented in [Cabrelli Amaro \(2012\)](#), [Wrembel \(2012, 2015\)](#) and [Cabrelli Amaro & Wrembel \(2016\)](#) as well as [Wrembel \(2023\)](#).

The present contribution will attempt to build on previous analyses and expand the state-of-the-art approaches by showcasing examples from the most current research into L3 phonetics and phonology stemming from the MultiPHON, CLIMAD and ADIM projects.¹ In the subsequent sections we will review main theoretical and methodological considerations as well as issues related to the production, perception and processing of speech in third and additional languages. The discussion will focus on a variety of research paradigms, elicitation tasks, and instruments while presenting differential approaches to auditory and acoustic analyses.

2 Searching for the explanatory potential: Models of L3 speech

A significant challenge in the research on multilingual speech acquisition is related to the theoretical complexities inherent in the field. In [Wrembel \(2023\)](#) four distinct approaches to the theoretical modeling of third language (L3) phonology were identified, namely: (1) extending conventional second language (L2) speech

¹The acronyms of the projects found in this chapter refer to the following research grants:

ADIM - Across-domain investigations in multilingualism: Modeling L3 acquisition in diverse settings, Norway Grants GRIEG-1 project, AMU Poznań in collaboration with UiT Tromsø & NTNU Trondheim, 2021–2024.

CLIMAD - Cross-linguistic influence in multilingualism across domains: Phonology and syntax (PI), Polish National Science Centre grant OPUS-19-HS, 2021–2026.

MultiPHON - Phonological cross-linguistic influence in young multilinguals, International grant of Polish-German Foundation of Science, AMU Poznań in co-operation with the University of Münster, 2017–2019.

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models to encompass a multilingual framework, (2) modifying L3 morphosyntactic models to address phonological acquisition, (3) refining broader interdisciplinary theories, and/or (4) proposing entirely new models specifically for L3 phonological acquisition.

Over the years, several models of speech acquisition have gained prominence among researchers in second language acquisition (SLA). These include the Speech Learning Model (SLM) proposed by [Flege \(1995\)](#) and its recent revision (SLM-r, [Flege & Bohn 2021](#)); the Perceptual Assimilation Model (PAM and PAM-L2, [Best 1995](#); [Best & Tyler 2007](#)); the Second Language Linguistic Perception (L2LP) model (e.g., [Escudero 2009](#)); the Exemplar-based model ([Pisoni 1996](#)); and the Native Language Magnet model (NLM, [Kuhl & Iverson 1995](#)). In general, most traditional models of L2 speech acquisition emphasize phonetic categories over phonological ones, focusing on context-dependent phonetic segments, exemplars, or prototypes rather than phonemes or features. These models tend not to provide explicit insights into how phonological processes operate in the minds of L2 learners. They typically operate under the assumption of a shared phonological space where the phonological categories of native and non-native languages interact, resulting in mutual (bidirectional) influence between L1 and L2. This interaction can lead to the formation of composite or merged categories (hybrids of L1 and L2) while also fostering divergence and more peripheral trends within the two phonological systems (see [Flege 1995](#)). However, these models do not generate specific predictions regarding the speech development of multilingual individuals (for a detailed discussion, see [Dziubalska-Kołodziejczyk & Wrembel 2022](#)).

Another approach to modeling L3 speech relies on adapting theoretical models designed specifically for multiple language acquisition, yet stemming from the generative paradigm and focused on morphosyntax. They include as follows: the Cumulative-Enhancement Model (CEM) by [Flynn et al. \(2004\)](#); the L2 Status Factor Model (L2SF) by [Bardel & Falk \(2012\)](#); the Typological Primacy Model (TPM) by [Rothman \(2015\)](#); the Linguistic Proximity Model (LPM) by [Westergaard et al. 2017](#); and the Scalpel Model by [Slabakova \(2017\)](#). For a broader discussion on the applicability of the above models to L3 phonological acquisition, see [Wrembel \(2015\)](#) and [Dziubalska-Kołodziejczyk & Wrembel \(2024\)](#).

With the view to the unique characteristics of third language (L3) phonological acquisition, which intertwines phonological representation with neuro-motor articulatory processes, researchers have suggested that a specialized model may be necessary to effectively explain the various and interconnected patterns of cross-linguistic influence (CLI) observed in this area. Therefore, another potential approach involves the development of innovative models specifically addressing

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L3 phonological acquisition. One such model is the Natural Growth Theory of Acquisition (NGTA) (Dziubalska-Kolaczyk & Wrembel 2022, 2024), which offers a more comprehensive explanatory framework that can accommodate the complexities inherent in the multilingual acquisition process. Although NGTA adopts an ecologically valid approach to data collection and analysis, it may initially seem less rigorous than conventional formal theories. Nonetheless, it adheres to a logically defined set of principles and is firmly rooted in the linguistic theory of Natural Phonology (e.g. Donegan & Stampe 2009), while also being enhanced by a dynamic perspective informed by complexity theory (e.g. Kretzschmar 2015). For a detailed discussion of the model as well as its testing against data from L2 morphonotactics and L3 phonology see Dziubalska-Kolaczyk & Wrembel (2022, 2024).

As the authors of particular models discussed above call for further empirical testing of the proposed hypotheses, it seems that such theory verifications should be carried out on a broader set of data and across different domains. This goal has been targeted, among other things, in the ADIM project intended to contribute to modelling of L3 acquisition across language domains (phonology, syntax and semantics) as well as across various acquisition settings (instructed vs. naturalistic).

3 To test or not to test: Methodological considerations in L3 speech

A range of methodological considerations in research on L3 phonological acquisition have emerged in previous metaanalyses (see especially Wrembel 2023 and Gut & Wrembel 2024); however, the present contribution aims to zoom in on the challenges most recently acknowledged.

The current trends shift from product- to process-oriented investigations, focusing not only on particular stages of L3 acquisition in the course of instruction but also by examining the developmental trajectories of foreign language phonologies. Consequently, complex cross-linguistic interactions are now frequently investigated as they evolve over time (e.g., Cal & Wrembel 2023, Weckwerth et al. 2023). However promising, this approach has several inherent challenges related to unpredictability of drop-out rates in longitudinal data collection and, consequently, acquiring incomplete data sets for the analysis. Further, three data collection points typically employed in such investigations may not be sufficient to trace subtle changes in the multilinguals' performance and thus unveil the trajectory of development of L3/Ln phonologies.

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Another methodological challenge is related to testing metalinguistic awareness. New paradigms have been developed in this area and specifically for phonological awareness, ranging from a focus on tacit to overt manifestations thereof. For example, [Kopečková et al. \(2023\)](#) assessed the applicability of the delayed mimicry paradigm for evaluating phonological awareness among young multilinguals. This paradigm consisted of imitating a specific foreign accent when telling a picture story in the native language. The ability to mimic characteristic features of a given phonological system was operationalized as reflective of the degree of phonological awareness. In this study, the accent mimicry task tapping into implicit awareness was supplemented by an oral protocol eliciting explicit responses and comments.

Another consideration is related to measuring cross-linguistic distances between languages in the multilingual repertoire, trying to reconcile purely typological and psychotypological approaches. With a view to the latter, [Nelson et al. \(2021\)](#) proposed a visual psychotypological measure that examines perceived distances among multiple phonological systems. This measure aims to reliably capture the perceived relatedness among the phonologies of all languages within a multilingual's repertoire and thus reflects the perceived psychotypology between them.

The lack of direct comparability between testing two speech modalities, i.e., speech perception and speech production, has been widely discussed as yet another methodological challenge. This has led to a rather limited body of research on the relationship between speech perception and production in multilinguals. Some recent studies (e.g., [Wrembel et al. 2022](#); [Gut et al. 2023](#)) provide new insights into the longstanding question of the perceptuo-productive link in foreign language phonology acquisition. The former exploratory longitudinal study investigates the acquisition of rhotic sounds in English, German, and Polish, examining participants' performance on both a production task (delayed repetition) and a perception task (forced-choice goodness) in their L2 and L3 across several testing intervals. The overall findings indicate a notable disparity between the two modalities, with more established L2 English evidencing patterns of co-involvement, while newly acquired L3 German/L3 Polish pointing to dissociations between perception and production development. Other ongoing investigations are intended to shed more light on the perception and production interface of the same phenomenon in L3 phonology such as voice onset time (VOT) (e.g., [Cal & Wrembel 2023](#)).

An area that has remained largely unexplored in L3/Ln speech research pertains to suprasegmental features, and, in particular, fluency metrics as well as rhythm perception and production. [Sypiańska & Wrembel's](#) () study is among

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the first to pave the way to a deeper understanding of how these aspects of multilinguals' speech differ from what we already know from L2 acquisition. The novelty of Sypiańska & Wrembel's () study lies in analysing both language and task effects on fluency across the three languages of multilinguals in two speech modes. The findings demonstrate a gradient decline in fluency with language status in the expected direction (language effect: $L1 > L2 > L3$). However, diverse fluency profiles were introduced as a way to interpret differential (dis)fluency patterns in the two foreign languages ($L2$ vs. $L3$). Task effects were greater in magnitude than language effects with extemporaneous speech displaying longer pauses, higher pause to speech ratio and lower speech rate compared to text reading. It was concluded that pausing has different purposes depending on the mode of speech with syntax-driven pausing for text reading and cognitively-based pausing for extemporaneous speech. These findings establish the grounds for ongoing suprasegmental analyses of multilingual speech.

Further considerations in multilingual research are related to inherent challenges in the study design and task selection. For instance, care must be exercised in order to introduce a specific language mode in the elicitation of $L1$, $L2$, $L3/Ln$ speech. One such method is through exposure to short films before each language block, with the fragments constituting a whole series presented in turn in different languages (e.g., the Peppa cartoon watching task in Czarnecki-Verner et al. 2025). The induction into the language mode can, at the same time, form a prompt for a speech elicitation task, i.e., video story retelling in particular languages from the participants' multilingual repertoire.

Another consideration concerns the application of new technologies in research such as using AI generated stimuli for text elicitation or online platforms for participant recruitment. Target sentence items generation supported by AI has been tested for $L2$ English and $L3$ Norwegian (Czarnecki-Verner et al. 2025). The experience shows that the human evaluator cannot be fully replaced as yet since the AI generated stimuli had to be hand corrected by humans (native speakers of the respective languages) prior to them being used in the experiment. With respect to participant recruitment, platforms such as Prolific have facilitated the acquisition of larger group samples. However, online recruitment still appears challenging especially for smaller languages (like Norwegian or Polish) with researchers having to rely on more traditional methods of recruitment e.g., via networks of acquaintances. Moreover, following a more nuanced approach to $L2$ data analysis, research into $L3/Ln$ phonological acquisition starts to account more and more frequently for individual differences in the speech performance. Identifying unique trends and developmental paths presents a more ecologically

valid stance which complements the analysis at the group level and further informs theoretical models that endorse such an approach (see e.g., micro & macro variables in NGTA by [Dziubalska-Kořaczyk & Wrembel 2024](#)).

In the following subsections novel avenues in L3 phonological acquisition research will be illustrated based on selected examples of state-of-the-art research from the ADIM project, including (1) a holistic comparison of perceived accentedness and comprehensibility in both L2 and L3, (2) across-domain investigation at the intersection of syntax and phonology, (3) a neurophysiological study of phonetic auditory discrimination in the three languages in trilinguals' repertoire.

3.1 Study 1: Is there a difference? Sounding accented in L2 vs. L3

While foreign-accented speech has been extensively studied from various angles, no research to date has examined the phenomenon across two non-native languages within the same cohort of speakers. [Kędzierska et al.](#)'s (2025) study represents one of the initial efforts to systematically compare the degree of foreign accentedness and comprehensibility in two non-native languages (L2 and L3) among the same group of individuals. The foreign accentedness ratings (FAR) employed in this investigation were novel, not only because they were applied to two distinct non-native languages but also because they incorporated four different conditions: language status (L2 vs. L3/Ln), speaking mode (read vs. extemporaneous speech), grammaticality (error-free vs. erroneous sentences), and acquisition setting (instructed vs. naturalistic learners). The sampling of erroneous and error-free sentences was done by the researcher after the recording of speech samples.

The primary findings indicate that speaking mode significantly influenced comprehensibility ratings in L2 and in L3/Ln with read sentences perceived as generally more comprehensible than spoken sentences. However, the production mode effect was not attested in accentedness ratings. On the other hand, the grammaticality condition was shown to affect both perceived accentedness and comprehensibility in L2, as well as comprehensibility in L3.

Interestingly, in L2 English error-free sentences were rated as significantly less accented and more comprehensible compared to erroneous sentences. In contrast, these effects did not extend to L3/Ln Norwegian, where no significant differences in accentedness ratings were found based on grammaticality. However, error-free sentences were still assessed as more comprehensible than erroneous ones. This suggests that the speakers' L3/Ln Norwegian has not yet developed to a level where nuanced grammatical differences can influence perceived accent-

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edness, likely because any potential impact of grammaticality is overshadowed by a pronounced foreign accent.

Moreover, the acquisition setting impacted comprehensibility judgments in both languages, as the naturalistic language learners were perceived as significantly more comprehensible than instructed learners. Furthermore, we found strong negative correlations between the level of perceived accentedness and comprehensibility in L2 English and in L3/Ln Norwegian, i.e., the stronger the perceived accent, the less comprehensible the speech sample was rated to be.

Additionally, participants had to engage in an L1 identification task, i.e., identify the first language of the speakers based on their L2 English and L3 Norwegian speech samples, which was interpreted as indicative of potential CLI. The findings demonstrated that L1-driven CLI largely prevailed, yet with growing proficiency in L2 English there was a tendency to override it. Results from the L1 identification tasks further indicate that speakers tended to rely more on their L2 English when producing L3/Ln Norwegian, particularly among those producing error-free extemporaneous sentences. This study underscores that perceived accentedness and comprehensibility are influenced by several factors, including language status, speaking mode, the presence of grammatical errors, and the acquisition context. Given the overall higher proficiency of the speakers in L2 English relative to L3/Ln Norwegian, we observed greater cross-linguistic influence from L1 Polish on L3/Ln Norwegian than on L2 English as 47% of native English raters identified the speakers of L2 English samples as Polish or Slavic, whereas such identification for L3 Norwegian samples were significantly higher among Norwegian native raters at 62%. Additionally, naturalistic learners residing in Norway were more frequently perceived as L1 English speakers when producing L3/Ln Norwegian compared to instructed learners. This could be due to their frequent exposure to and use of English while residing in Norway. Although no previous study on foreign-accented speech has directly compared naturalistic and instructed learners, it is reasonable to assume that naturalistic learners would be better positioned to achieve higher L3 Norwegian proficiency given their enhanced exposure to the language of the environment.

The fact that each speaker was represented by samples in three speaking modes (read, extemporaneous error-free and extemporaneous erroneous) made the study design quite complex. Furthermore, control speech samples included native (English or Norwegian) samples as well as learners of L2/L3 Norwegian with L1 backgrounds other than the main experimental group. However, one inherent limitation of the study related to the challenge of ensuring comparable speaker samples across two languages in terms of grammaticality, as the nature of the errors often differed between the languages, particularly due to the

varying proficiency levels of our participants. Moreover, the distinction between instructed and naturalistic learners was not a straightforward one, given that proficiency levels differed between the two groups—most notably, self-assessed proficiency was generally higher in the naturalistic group. Further research is necessary to fully elucidate the impact of the learning context and proficiency levels on language acquisition outcomes.

3.2 Study 2: At the intersection of syntax and phonology: How grammatical do I sound when I am accented?

The second study constitutes a follow-up to the previous one as it takes the idea of foreign accented speech even further by applying it across language domains. To the best of our knowledge, this is the first such study investigating accentedness in relation to grammaticality judgements in multilingual acquisition (Wrembel et al. 2025). To this end, we considered the possible interaction of two domains, i.e., phonology and syntax, by presenting learners of L3 Norwegian who have Polish and English as their previously acquired languages with an Acceptability Judgement Task (AJT) with stimuli where Norwegian is either similar to Polish and different from English or vice versa (see Westergaard et al. 2023 for such a design). The stimuli were spoken with a Polish, an English, or a native Norwegian accent. We controlled for the degree of accentedness by preselecting equally (moderately) accented samples by Polish and English speakers of Norwegian based on a foreign-accent rating pre-task performed on a larger set of non-native samples. The rationale of the study is based on the idea of activation competition, as argued for in the Linguistic Proximity Model (e.g., Westergaard 2021), and thus, that a Polish or English accent may enhance the activation of the relevant language and thus affect the acceptability judgement.

This innovative design involved self-paced listening to sentence items while performing an acceptability judgment task conducted online in Qualtrics. There were three accent blocks, i.e., Polish-accented Norwegian, English-accented Norwegian and native Norwegian. The test items covered eight syntactic properties, selected based on their grammatical similarities or dissimilarities with target Norwegian, mirroring grammatical structures in Polish (3), English (3), or neither (2), the properties being referred to as conditions. The participants included two groups of multilingual learners of L3 Norwegian in a classroom instructed setting as well as in an immersive naturalistic setting.

The results demonstrated significant effects of accent, group and condition/grammaticality, as well as a significant group by condition/grammaticality interaction. Conditions not similar to Polish and English did not exhibit any

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accent effects. We found some relationships between condition and accent in the English-like conditions. For instance, unacceptable Norwegian sentences without a definite article were rated lower when they were English-accented as opposed to Polish-accented, in the naturalistic learner group. Ditransitives were similarly sensitive to the English accent, but only among the instructed learner group. These findings suggest that the differences in ratings based on accentedness could be linked to structural similarity but also to exposure conditions (i.e., instructed vs. immersive setting).

Interestingly, the study also suggests that accent may activate linguistic representations together with other factors that affect crosslinguistic influence such as variable acceptability, input frequency or language proficiency. The originality of the investigation rested on the idea that language users provide different judgements for the same morphosyntactic phenomena depending on whether the auditory input presented was foreign accented or not. All in all, combining syntax and phonology in L3 acquisition research is innovative, as it constitutes one of very few investigations into L3 acquisition across language domains. Furthermore, it is of high relevance since foreign-accented speech has significant social and pedagogical implications.

3.3 Study 3: Can the brain tell us more? Neural sound processing in native vs. non-native languages

Auditory discrimination is generally considered to be less precise in non-native languages compared to native languages. To date, there has been a dearth of studies investigating sound processing in the third language with the application of EEG, with previous related studies focusing mostly on bilinguals (e.g., [Jakoby et al. 2011](#); [Liang & Chen 2022](#)). To bridge this gap, we have conducted a series of studies that focused on electrophysiological markers of speech sound discrimination in multilingual speakers ([Kędzierska et al. 2023, 2024](#), [Under review](#)). We explored whether pre-attentive phonological processing differences arise between the first foreign language (L2) and subsequent languages (L3), and whether these differences are influenced by various learner-related factors such as e.g., the acquisition context.

Specifically, we conducted event-related brain potentials (ERP) oddball studies involving native Polish speakers learning English (L2) and Norwegian (L3/Ln). We aimed to explore the mismatch negativity (MMN), which is the component typically associated with listeners' sensitivity to discernible phonemic changes within a sequence of sounds (e.g., [Näätänen et al. 1997](#)). This component is generally elicited during oddball tasks, in which a sequence of standard stimuli (e.g.,

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identical sounds) is intermittently interrupted by a deviant stimulus (e.g., a different sound). For two out of three vowel pair discrimination EEG studies, our selection of standard stimuli was guided by the high degree of cross-linguistic similarity among the three standard sounds: Polish /i/, English /ɪ/, and Norwegian /i/. Conversely, the deviants were chosen based on systematic differences among the three studied languages, with the added requirement that the standard and deviant sounds within each language be comparably distanced from one another. Consequently, experimental stimuli included three pairs of phonemes: the Polish /i/-/ɛ/ contrast, the English /ɪ/-/ʊ/ contrast and the Norwegian /i/-/y/ contrast. In turn, in the most recent study (Kędzierska et al. *Under review*) we investigated potential phoneme perception asymmetries in the multilinguals with the application of the ‘flip-flop’ design, in which we selected different phonemes and each member of the phoneme pair served as a deviant or standard an equal number of times.

To effectively control the experimental conditions, it would have been ideal to utilize the same standard across all three languages, along with deviants that are equidistant and consistent in their phonetic features. This would allow for a clearer examination of the impact of language status on processing. However, such ideal conditions are not feasible within natural languages, necessitating the exploration of various scenarios, as each scenario provides insights into the multitude of variables influencing non-native speech acquisition. During the experimental procedure, participants viewed a silent cartoon and were explicitly instructed to focus their attention on the cartoon rather than the auditory stimuli. In the reported studies, we establish a quasi-common standard across all three languages (i.e., i-like sounds), with distinct, non-overlapping deviants (i.e., Polish /ɛ/, English /ʊ/, and Norwegian /y/) varying in their phonetic features, however much we try to control for the Euclidean distances.

Our findings indicated a graded amplitude of the mismatch negativity (MMN) effect, with the largest amplitude observed for L1, a more reduced one for L2, and the smallest for L3. By comparing parallel EEG research conducted in two different groups of learners with the same linguistic repertoire, we concluded that the acquisition setting significantly affects the perception of phonemic contrasts. In the naturalistic group, no notable differences were found between L1 and L2; however, the instructed group demonstrated more nuanced distinctions among all languages tested. Another ERP component that we decided to enter into the analysis involved the late discriminative negativity (LDN) component as it is typically associated with pre-attentive cognitive evaluation of the stimulus and a successful formation of memory traces associated with a specific phonemic

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representation (e.g., [Jakoby et al. 2011](#)). Interestingly, we did not find any significant difference between L2 and L3/Ln in terms of the LDN amplitude and the effect was smaller for both foreign languages when compared with L1 ([Kędzierska et al. 2023](#)). Consequently, the two components (MMN and LDN) were not found to align with each other, providing further evidence against the disputed interpretation of the late discrimination negativity as extending beyond a mere continuation of the earlier effect associated with the mismatch negativity. Instead, the LDN appeared to be better characterized as an indicator of successful phonological representation formation, which, in our study, was unaffected by the foreign language status (L2 vs. L3/Ln) or the learning context (instructed vs. naturalistic). This suggests that processing differences between the two non-native languages are more effectively captured at the pre-attentional level within the MMN time window, which, in line with previous findings, may be assumed to index auditory discrimination. In terms of the differences between L2 English and L3/Ln Norwegian, a weaker MMN effect in the L3/Ln also signals the importance of language dominance as a factor contributing to effective phoneme discrimination. This task appears to be facilitated by early age of acquisition and higher language proficiency levels. All in all, our findings support the idea that foreign language status (i.e., L2 vs. L3/Ln influences early auditory processing as demonstrated by MMN patterns).

To the best of our knowledge, this is the first such ERP study focused on auditory discrimination in three languages. The novelty of our investigations stems primarily from its inclusion of diverse learner groups, as all prior research has typically examined non-native phoneme perception from the perspective of a single acquisition context — either naturalistic or instructed. The observed advantage for the naturalistic group, who seemed to process L1 and L2 sounds with equal efficacy, reinforces the idea that non-native phonemic processing benefits from frequent use in authentic communicative contexts. Undoubtedly, phonemic processing and, consequently, effective phonemic production represent critical skills that language learners strive to master; thus, these findings have the potential to inform the educational decisions of relevant stakeholders.

4 Final recommendations

In conclusion, in order to address the challenges facing research into L3 speech, let us present some recommendations, thus assuming a forward-looking perspective. As already emphasised in [Wrembel \(2023\)](#), a comprehensive range of contemporary behavioral and online processing techniques should be applied to

achieve robust triangulation across methodologies. Research on third-language speech production and perception should be advanced by incorporating larger participant samples, a variety of language pairings, and distinct phonological features that exhibit unique cross-linguistic patterns. Systematic comparisons of multilingual speech acquired across diverse learning environments remain largely unexplored, as prior studies have predominantly focused on either naturalistic acquisition or formal settings, which has led to somewhat divergent patterns of CLI reported in the studies. A key recommendation formulated in the field has been to adopt a more interdisciplinary approach to L3 phonology research, broadening studies across different linguistic domains to determine whether patterns of cross-linguistic influence (CLI) in L3 acquisition emerge as holistic phenomena or remain specific to individual domains.

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References

- Bardel, Cecilia & Ylva Falk. 2012. The L2 status factor and the declarative/procedural distinction. In Jennifer Cabrelli Amaro, Susan Flynn & Jason Rothman (eds.), *Third language acquisition in adulthood*, 61–78. Philadelphia: John Benjamins.
- Best, Catherine. 1995. A direct realist view of cross-language speech perception. In Winifred Strange (ed.), *Speech perception and linguistic experience: Issues in cross-language research*, 171–204. Timonium: York Press.
- Best, Catherine T. & Michael D. Tyler. 2007. Nonnative and second-language speech perception: Commonalities and complementarities. In O.-S. Bohn & M. J. Munro (eds.), *Language experience in second language speech learning: In honor of James Emil Flege*, 13–34. Amsterdam: John Benjamins.
- Cabrelli Amaro, Jennifer. 2012. L3 phonology: An understudied domain. In Jennifer Cabrelli Amaro, Susanne Flynn & Jason Rothman (eds.), *Third language acquisition in adulthood*, 33–60. Philadelphia: John Benjamins.
- Cabrelli Amaro, Jennifer & Magdalena Wrembel. 2016. Investigating the acquisition of phonology in a third language—A state of the science and an outlook for the future. *International Journal of Multilingualism* 13. 395–409.

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- Cal, Zuzanna & Magdalena Wrembel. 2023. Developmental acquisition of stops by multilingual speakers. *International Journal of Multilingualism* 22(1). 13–35. DOI: [10.1080/14790718.2024.2363777](https://doi.org/10.1080/14790718.2024.2363777).
- Czarnecki-Verner, Tristan, Jarosław Weckwerth, Kamil Kaźmierski & Magdalena Wrembel. 2025. Cross-linguistic interactions in voiceless sibilant production in L1 Polish, L2 English, and L3 Norwegian. *International Journal of Multilingualism*. 1–26. DOI: [10.1080/14790718.2025.2577835](https://doi.org/10.1080/14790718.2025.2577835).
- Donegan, Patricia J. & David Stampe. 2009. Hypotheses of natural phonology. *Poznan Studies in Contemporary Linguistics* 45(1). 3–31.
- Dziubalska-Kończyk, Katarzyna & Magdalena Wrembel. 2022. Natural Growth Theory of Acquisition (NGTA): Evidence from (mor)phonotactics. In Veronica Sardegna & Anna Jarosz (eds.), *Theoretical and practical developments in English speech assessment, research and training*, 281–298. Cham: Springer.
- Dziubalska-Kończyk, Katarzyna & Magdalena Wrembel. 2024. A revised natural growth theory of acquisition: Evidence from L3 phonology. In Elena Babatsouli (ed.), *Multilingual acquisition and learning: An ecosystemic view to diversity*, 426–449. Amsterdam: John Benjamins. DOI: [10.1075/sibil.67.16dzi](https://doi.org/10.1075/sibil.67.16dzi).
- Escudero, Paula. 2009. Linguistic perception of “similar” L2 sounds. In Paul Boersma & Silke Hamann (eds.), *Phonology in perception*, 151–190. Berlin: Mouton de Gruyter.
- Flege, James E. 1995. Second-language speech learning: Theory, findings, and problems. In Winifred Strange (ed.), *Speech perception and linguistic experience: Issues in cross-language research*, 233–272. Timonium: York Press.
- Flege, James E. & Ocke-Schwen Bohn. 2021. The revised speech learning model (SLM-r). In Ratreë Wayland (ed.), *Second language speech learning*, 3–83. Cambridge: Cambridge University Press. DOI: [10.1017/9781108886901.002](https://doi.org/10.1017/9781108886901.002).
- Flynn, Suzanne, Claire Foley & Inna Vinnitskaya. 2004. The cumulative-enhancement model for language acquisition: Comparing adults’ and children’s patterns of development in L1, L2 and L3 acquisition of relative clauses. *The International Journal of Multilingualism* 1(1). 3–16.
- Gut, Ulrike, Romana Kopečková & Christina Nelson. 2023. *Phonetics and phonology in multilingual language development*. Cambridge: Cambridge University Press. DOI: [10.1017/9781108992527](https://doi.org/10.1017/9781108992527).
- Gut, Ulrike & Magdalena Wrembel. 2024. Comparing bilingual and trilingual phonetics and phonology. In Mark Amenguel (ed.), *Handbook of bilingual phonetics and phonology*, 631–652. Cambridge: Cambridge University Press.
- Jakoby, Hilla, Abraham Goldstein & Miriam Faust. 2011. Electrophysiological correlates of speech perception mechanisms and individual differences in second

28 Approaches to L3 phonological acquisition research

- language attainment. *Psychophysiology* 48. 1516–1530. DOI: [10.1111/j.1469-8986.2011.01227.x](https://doi.org/10.1111/j.1469-8986.2011.01227.x).
- Kędzierska, Hanna, Krzysztof Hwaszcz & Magdalena Wrembel. 2025. Are we equally accented in L2 and L3? Perception of foreign accentedness and comprehensibility in multilingual speakers. *International Journal of Bilingualism* 30(2). 339–357. DOI: [10.1177/13670069251324466](https://doi.org/10.1177/13670069251324466).
- Kędzierska, Hanna, Karolina Rataj, Anna Balas & Magdalena Wrembel. 2023. Vowel perception in multilingual speakers: ERP evidence from Polish, English and Norwegian. *Frontiers in Psychology* 14. DOI: [10.3389/fpsyg.2023.1270743](https://doi.org/10.3389/fpsyg.2023.1270743).
- Kędzierska, Hanna, Karolina Rataj, Anna Balas & Magdalena Wrembel. 2024. The neurophysiology of phonemic contrasts perception in L2/L3 learners: The role of acquisition setting. *Bilingualism: Language and Cognition* 28(2). 550–562. DOI: [10.1017/S136672892400066X](https://doi.org/10.1017/S136672892400066X).
- Kędzierska, Hanna, Karolina Rataj, Anna Balas & Magdalena Wrembel. Under review. MMN asymmetries in trilingual listeners. *no journal no volume. no pages*.
- Kopečková, Romana, Magdalena Wrembel, Ulrike Gut & Anna Balas. 2023. Differences in phonological awareness of young L3 learners: An accent mimicry study. *International Journal of Multilingualism* 20(2). 408–424. DOI: [10.1080/14790718.2021.1897127](https://doi.org/10.1080/14790718.2021.1897127).
- Kretzschmar, William A., Jr. 2015. *Language and complex systems*. Cambridge: Cambridge University Press.
- Kuhl, Patricia K. & Paul Iverson. 1995. Linguistic experience and the “perceptual magnet effect.” In Winifred Strange (ed.), *Speech perception and linguistic experience: Issues in cross-language research*, 121–154. Baltimore: York Press.
- Liang, Lijuan & Baoguo Chen. 2022. The non-native phonetic perception mechanism utilized by bilinguals with different L2 proficiency levels. *International Journal of Bilingualism* 26(3). 368–386. DOI: [10.1177/13670069211058275](https://doi.org/10.1177/13670069211058275).
- Näätänen, R., A. Lehtokoski, M. Lennes, M. Cheour, M. Houtilainen, A. Iivonen & E. Aronen. 1997. Language-specific phoneme representations revealed by electric and magnetic brain responses. *Nature* 385. 432–434. DOI: [10.1038/385432a0](https://doi.org/10.1038/385432a0).
- Nelson, Christina, Iga Krzysik, Halina Lewandowska & Magdalena Wrembel. 2021. Multilingual learners’ perceptions of cross-linguistic distances: A proposal for a visual psychotypological measure. *Language Awareness* 30(2). 176–194. DOI: [10.1080/09658416.2021.1897132](https://doi.org/10.1080/09658416.2021.1897132).
- Pisoni, David. 1996. Some thoughts on normalization in speech perception. In Keith Johnson & John W. Mullennix (eds.), *Talker variability in speech processing*, no pages. San Diego: Academic Press.

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- Rothman, Jason. 2015. Linguistic and cognitive motivations for the typological primacy model (TPM) of third language (L3) transfer: Timing of acquisition and proficiency considered. *Bilingualism: Language and Cognition* 18(2). 179–190.
- Slabakova, Roumyana. 2017. The scalpel model of third language acquisition. *International Journal of Bilingualism* 21(6). 651–665.
- Sypiańska, Jolanta & Magdalena Wrembel. Under review. Fluency profiles in multilingual speech: Task effects outweigh language status. *International Journal of Multilingualism*.
- Weckwerth, Jarosław, Magdalena Wrembel, Kamil Kaźmierski & Anna Balas. 2023. L3 Norwegian /ɐ:/ and /u:/ in L1 Polish/L2 English learners: Different patterns of cross-linguistic interactions. In *ICPhS Proceedings 2023, Prague*, 2666–2670. Guarant International. <https://guarant.cz/icphs2023/775.pdf>.
- Westergaard, Marit. 2021. Microvariation in multilingual situations: The importance of property-by-property acquisition. *Second Language Research* 37(3). 379–407.
- Westergaard, Marit, Natalia Mitrofanova, Roksolana Mykhaylyk & Yulia Rodina. 2017. Crosslinguistic influence in the acquisition of a third language: The linguistic proximity model. *International Journal of Bilingualism* 21(6). 666–682.
- Westergaard, Marit, Natalia Mitrofanova, Yulia Rodina & Roumyana Slabakova. 2023. Full transfer potential in L3/Ln acquisition: Crosslinguistic influence as a property-by-property process. In Jennifer Cabrelli, Adel Chaouch-Orozco, Jorge González Alonso, Sergio Miguel Pereira Soares, Eloi Puig-Mayenco & Jason Rothman (eds.), *The Cambridge handbook of third language acquisition*, chap. 9, 219–242. Cambridge: Cambridge University Press.
- Wrembel, Magdalena. 2012. Foreign accentedness in third language acquisition: The case of L3 English. In Jennifer Cabrelli Amaro, Suzanne Flynn & Jason Rothman (eds.), *Third language acquisition in adulthood*, 281–309. Amsterdam: John Benjamins.
- Wrembel, Magdalena. 2015. *In search of a new perspective: Cross-linguistic influence in the acquisition of third language phonology*. Poznań: Wydawnictwo Naukowe UAM.
- Wrembel, Magdalena. 2023. Exploring the acquisition of L3 phonology: Challenges, new insights, and future directions. In Jennifer Cabrelli, Adel Chaouch-Orozco, Jorge González Alonso, Sergio M. Pereira Soares, Eloi Puig-Mayenco & Jason Rothman (eds.), *The Cambridge handbook of third language acquisition and processing*, 115–141. Cambridge: Cambridge University Press. DOI: [10.1017/9781108957823](https://doi.org/10.1017/9781108957823).

28 *Approaches to L3 phonological acquisition research*

- Wrembel, Magdalena, Chloe Castle, Justyna Gruszecka, Roumyana Slabakova, Marta Velnić & Marit Westergaard. 2025. The role of accentedness in acceptability judgements in L3 Norwegian: An across-domain investigation. *International Journal of Multilingualism* 23(1). 655–677. DOI: [10.1080/14790718.2025.2533470](https://doi.org/10.1080/14790718.2025.2533470).
- Wrembel, Magdalena, Ulrike Gut, Romana Kopečková & Anna Balas. 2022. The relationship between the perception and production of L2 and L3 rhotics in young multilinguals; an exploratory cross-linguistic study. *International Journal of Multilingualism* 21(1). 92–111. DOI: [10.1080/14790718.2022.2036158](https://doi.org/10.1080/14790718.2022.2036158).

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